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出 版 说 明

本书是陈希孺院士的自选集，包含了他自20世纪60年代以来所发表的百余篇论文中的25篇，内容涉及参数估计、线性统计模型（LS估计、M估计等）、非参数统计（U-统计量、线性秩统计量、非参数回归和密度估计等）等领域，可以反映他在这些领域工作的基本成果.

本书的出版得到了中国科学技术大学科技处、出版社和统计与金融系等有关方面的大力支持和协助，具体工作主要是由目前在中国科学技术大学工作的陈希孺院士的学生们完成的.这件事自2001年开始酝酿，历时两年多，在上述各方面通力合作之下，得以顺利完成，特书此以志其事.

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1

On Minimax Invariant Estimation of Scale and Location Parameters

1 Introduction

Let X be a random variable whose distribution function $F(x)$ is known. By changing the origin and scale we modify linearly the results of measurements. This leads us to consider a family of distributions $\left\{F\left(\frac{x-b}{a}\right)\right\}$, where $a > 0$ and b are parameters — the so-called scale and location parameters.

The problem of estimating these parameters has been investigated by many authors since 1937. Roughly Speaking, up to now the efforts have been centred on two points: the seeking of minimax invariant estimators (MIE) and the admissibility of such estimators. The second point has been investigated more successfully in the case when only a location parameter is present (see [4],[5],[6],[10],[13], [14], and [15]). As for the problem of existence of MIE, the present situation is that although a number of results in various particular cases were obtained (see [3],[4],[6],[7],[8], and [12]), the general case that two parameters are present simultaneously has not yet been worked out.

In this section an important paper by J. Kiefer should be here mentioned. In [11] Kiefer developed the general theory of invariance principle in the nonsequential case. His result is so general that the problem of existence of MIE may be considered as a solved one, at least partially. However, the solution by his theory, it seems, cannot be considered a rigorous and complete one, for, first, the main theorem of [11] states only that the restriction to invariant estimators does not enlarge the maximum risk and the problem of existence of MIE is still left open; second, little attention was paid in the paper to the measurability questions. So, although it may happen that in some cases MIE can be formally constructed, a verification is needed to insure the measurability of the estimator obtained. It goes without saying that from the rigorous mathematical viewpoint, it is very desirable to have these problems clarified. Third, Kiefer's theory does not apply to discrete distribution case. Owing to these

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reasons, it seems worth while to give a rigorous treatment of this problem.

The purpose of this paper is to prove the existence of MIE of scale and location parameters. For simplifying the wording, we shall discuss in detail only the one-dimensional case. The extension to multidimensional case will be indicated in the last section.

The case where the location parameter is the only one present may be treated in a somewhat similar manner (see in this connexion an article published by the author in *Acta Mathematica Sinica*, Vol. 14, No. 2, pp. 276~290).

2 The Main Results

Let $X = (X_1, \cdots, X_N)$ be a random vector with a known distribution F, and let $\varepsilon = (1, \cdots, 1)$. Then for any $a > 0$ and b, $aX + b\varepsilon$ has a distribution function

$$F(x \mid a, b) = F\left(\frac{x - b\varepsilon}{a}\right).$$

Now we draw a sample x from the population $F(x|a, b)$, and want to make some estimation of the parameter vector (a, b).

Let $V(a, b; d_1, d_2)$ be the loss function. We impose the natural condition $d_1 \geqslant 0$, and suppose that V is invariant, i. e., for any $c_1 > 0$ and c_2, we have

$$V(ac_1, bc_1 + c_2; d_1c_1, d_2c_1 + c_2) = V(a, b; d_1, d_2).$$

This is equivalent to the condition that there exists a function $W = W(t_1, t_2)$ defined on set $\Sigma = \{(t_1, t_2) \mid t_1 \geqslant 0\}$ such that

$$V(a, b; d_1, d_2) = W\left(\frac{d_1}{a}, \frac{d_2 - b}{a}\right).$$

In what follows the term "loss function" will be understood as the function W, and not V itself.

We call "estimator" any Borel measurable function defined on R_N with range Σ. The class of all estimators will be denoted by μ. An estimator (u, v) is called invariant if for $(c, d) \in \Sigma^0$ (Σ^0— interior of Σ) we have

$$u(cx + d\varepsilon) = cu(x), \ v(cx + d\varepsilon) = cv(x) + d.$$

Let (u_0, v_0), where $u_0(x) > 0$, be a fixed invariant estimator. For any $(u, v) \in \mu$, we define

$$s(x) = \frac{u(x)}{u_0(x)}, \quad r(x) = \frac{v(x) - v_0(x)}{u_0(x)};$$

then $(s, r) \in \mu$, and the correspondence between (u, v) and (s, r) is one-to-one. A necessary and sufficient condition that (u, v) is invariant is that

$$s(cx + d\varepsilon) = s(x), \quad r(cx + d\varepsilon) = r(x) \tag{1}$$

for any $(c, d) \in \Sigma^0$. In what follows we shall often use (s, r) instead of (u, v), and the subclass of all functions in μ satisfying (1) will be denoted by ϑ.

The risk function $R(a, b; u, v)$ may conveniently be expressed as a functional ρ of (a, b) and (s, r). Thus

$$\begin{aligned}\rho(s, r; a, b) &= \rho(s, r; a, b; F, W)\\ &= \int W\left(\frac{u(x)}{a}, \frac{v(x)-b}{a}\right)\mathrm{d}F\left(\frac{x-b}{a}\right)\\ &= \int W[s(ay+b\varepsilon)u_0(y), r(ay+b\varepsilon)u_0(y)+v_0(y)]\mathrm{d}F(y).\end{aligned}$$

Notice that if $(s, r)\in\vartheta$, $\rho(s, r; a, b)$ would be independent of (a, b) and might justifiably be written as $\rho(s, r)$. Let

$$V = V(F, W) = \inf_{(s, r)\in\vartheta}\sup_{(a, b)\in\Sigma^0}\rho(s, r; a, b; F, W), \qquad (*)$$

$$\overline{V} = \overline{V}(F, W) = \inf_{(s, r)\in\mu}\sup_{(a, b)\in\Sigma^0}\rho(s, r; a, b; F, W),$$

$$\underline{V} = \underline{V}(F, W) = \sup_{G}\inf_{(s, r)\in\vartheta}\int\rho(s, r; a, b; F, W)\mathrm{d}G,$$

where G is an *a priori* distribution on Σ^0. We remark that in the foregoing expressions it is unnecessary to require that F is a probability measure.

It is easily seen that $V\geqslant\overline{V}\geqslant\underline{V}$. What we want to prove is $V=\overline{V}$ and the inf standing on the right hand of ($*$) is attainable. This will be proved in this paper under mild restrictions. We now formulate the conditions imposed on the loss function W:

1°. W is continuous on Σ^0.

2°. $W(1, 0)=0$.

3°. For any $t_1\geqslant 0$ fixed, W is nonincreasing in t_2 for $t_2\leqslant 0$ and nondecreasing in t_2 for $t_2\geqslant 0$. For any t_2 fixed, $W(t_1, t_2)$ viewed as a function of t_1 nonincreases for $0\leqslant t_1\leqslant 1$ and nondecreases for $t_1\geqslant 1$. Notice that from 2°, 3° it follows that W is nonnegative on Σ.

4°_1. The relation

$$W(t_1, t_2)\to\infty$$

holds uniformly as $\frac{1}{t_1}+t_1+|t_2|\to\infty$. This in conjunction with 3° implies that $W(t_1, t_2)=\infty$ when $t_1=0$.

In many cases it is reasonable to assume that the loss remains bound as the error of estimation remains bound. Accordingly we impose the following conditions.

4°_2. W is continuous on Σ and $W(t_1, t_2)\to\infty$ holds uniformly as $t_1+|t_2|\to\infty$. The important quadratic loss function

$$W(t_1, t_2) = (t_1-1)^2+t_2^2$$

satisfies this condition.

4°_3. W is continuous on Σ, and

$$W(t_1, t_2) \to \alpha < \infty$$

uniformly as $t_1 + |t_2| \to \infty$. An example of such a loss function is furnished by $W(t_1, t_2) = \frac{|1-t_1|+|t_2|}{1+t_1+|t_2|}$ with $\alpha = 1$.

Now we turn to the distribution F. We assume that it satisfies the condition:

$$F(A) = 0, \tag{2}$$

where

$$A = \{x \mid x_1 = \cdots = x_N\}. \tag{3}$$

It is easily seen that $aA + b\varepsilon = \{(ax_1 + b, \cdots, ax_N + b) \mid (x_1, \cdots, x_N) \in A\} = A$ for any $(a, b) \in \Sigma^0$, and so $P\{(ax + b\varepsilon) \in A\} = F(A) = 0$. This being the case, the invariant estimator (u_0, v_0) with $u_0 > 0$ exists and may be taken as

$$u_0(x) = \max_{1 \leqslant i < j \leqslant N} |x_i - x_j|, \quad v_0(x) = x_1.$$

The following may be said about the condition (2). Suppose that (u, v) is an invariant estimator and $x = (x_1, \cdots, x_1)$, then it is easily seen that

$$u(x) = 0, \ v(x) = x_1.$$

But under condition 4_1°, the risk of any such estimator is ∞ if $F(A) > 0$. Therefore, in this case, the condition (2) may be considered as necessary. The situation becomes more complicated when 4_2° or 4_3° holds. Fortunately the most important case where the condition (2) fails is that F is a discrete distribution, and we shall prove that in this case, the MIE exists under 4_2°.

The main results of this paper may be summarized in the following two theorems:

Theorem 1 If the loss function W satisfies conditions 1°, 2°, 3° and one of 4_1°, 4_2°, 4_3°, and if the distribution F satisfies (2), then there exists an MIE of the scale and location parameters.

Theorem 2 If W satisfies 1°, 2°, 3°, 4_2° and if F is discrete on A, i. e., there exists a denumerable set $A_1 \subset A$ such that $F(A_1) = F(A)$, then the conclusion of Theorem 1 still holds true.

In practical applications, the case most commonly dealt with is that F is the direct product of N one-dimensional distributions, all of which belong to the continuous of discrete type. In this case, the condition of F in the above theorems is satisfied.

3 Some Lemmas

The proof of the above theorems are preceeded by several lemmas, which are discussed in the present section. Some of them are of general interest.

Lemma 1 Suppose that the probability distribution F satisfies (2) and the condition

$$F(\{x \mid (u_0(x), v_0(x)) \in L_m\}) = 1, \tag{4}$$

where (u_0, v_0) is a fixed invariant estimation with $u_0(x) > 0$ for all x, and

$$L_m = \left\{(t_1, t_2) \,\middle|\, \frac{1}{m} \leqslant t_1 \leqslant m, \ |t_2| \leqslant m\right\}.$$

Furthermore, we assume that one of the following conditions is satisfied by the loss function W:

(Ⅰ) W satisfies 2°, 3°, 4_1° and is finite on Σ^0;

(Ⅱ) W satisfies 2°, 3°, finite on Σ, and $\lim W(t_1, t_2) = \infty$ uniformly, as $t_1 + |t_2| \to \infty$;

(Ⅲ) W is a bound and Borel measurable function on Σ.

Then we have

$$V(F, W) = \underline{V}(F, W).$$

This lemma① may be considered as an extension of the one proved by Girshick and Savage (see lemma 3.2 in [3]).

Proof For any $q > 1$, $(a, b) \in \Sigma - L_q$, we define

$$(a, b)^* = (a^*, b^*) \tag{5}$$

as the point on L_q with shortest distance to (a, b). For any $(a, b) \overline{\in} L_{2m^2}$ and $(c, d) \in L_m$, we have

$$W(ac, bc + d) \geqslant W(a^* c, b^* c + d),$$

where (a^*, b^*) is determined as above with $q = 2m^2$.

Now define

$$W^*(t_1, t_2) = \begin{cases} W(t_1, t_2), & \text{for } (t_1, t_2) \in L_{m(2m^2+1)}, \\ W(t_1^*, t_2^*), & \text{for } (t_1, t_2) \in \Sigma - L_{m(2m^2+1)}, \end{cases}$$

where (t_1^*, t_2^*) is determined by (5) with $q = m(2m^2 + 1)$. From the conditions imposed on W, it follows easily that W^* is a Borel measurable bound function, and $W^* \leqslant W$.

By the definition of $\underline{V}$, for any $\varepsilon > 0$ and $T > 1$, there exists $(s, r) \in \mu$ such that

$$\begin{aligned} &\underline{V}(F, W^*) + \varepsilon \\ &\geqslant \frac{1}{\varphi(T)} \int_{L_T} \left[\int W^*[s(ay + b\varepsilon)u_0(y), r(ay + b\varepsilon)u_0(y) + v_0(y)] \mathrm{d}F(y)\right] \frac{1}{a} \mathrm{d}a\mathrm{d}b \\ &= \frac{1}{\varphi(T)} \int \mathrm{d}F(y) \int_{L_T} W^*[s(ay + b\varepsilon)u_0(y), r(ay + b\varepsilon)u_0(y) + v_0(y)] \frac{1}{a} \mathrm{d}a\mathrm{d}b. \end{aligned} \tag{6}$$

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① The author is indebted to Comrades Cheng Ping, Chien T T, and Tao Po for their assistance in the proof of this lemma.

Here $\varphi(T)=\int_{L_T}\frac{1}{a}\mathrm{d}a\mathrm{d}b=4T\log T$. But

$$\int_{L_T}W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\frac{\mathrm{d}a\mathrm{d}b}{a}$$
$$\geqslant\int_{1/T}^{T}\frac{1}{a}\mathrm{d}a\int_{-T-av_0(y)}^{T-av_0(y)}W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\mathrm{d}b$$
$$-2\mid v_0(y)\mid TM,\tag{7}$$

where

$$M=\sup_{(t_1,t_2)\in\Sigma}W^*(t_1,t_2)=\sup\{W(t_1,t_2)\mid(t_1,t_2)\in L_{m(2m^2+1)}\}<\infty.$$

Also, because of condition (4), we may restrict ourselves to those values of y, for which $(u_0(y),\ v_0(y))\in L_m$. Thus from (7) we have

$$\int_{L_T}W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\frac{\mathrm{d}a\mathrm{d}b}{a}$$
$$\geqslant\int_{[u_0(y)T]^{-1}}^{u_0^{-1}(y)T}\frac{\mathrm{d}a}{a}\int_{-T-av_0(y)}^{T-av_0(y)}W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\mathrm{d}b$$
$$-2mTM-2TM\int_T^{mT}\frac{\mathrm{d}a}{a}-2TM\int_{1/mT}^{1/T}\frac{\mathrm{d}a}{a}$$
$$=\int_E W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\frac{\mathrm{d}a\mathrm{d}b}{a}$$
$$-2mTM(1+2\log m),\tag{8}$$

where $E=\{(a,b)\mid(au_0(y),\ b+av_0(y))\in L_T\}$. Making the transform $a'=au_0(y)$, $b'=av_0(y)+b$ and rewriting (a,b) instead of (a',b'), we obtain

$$\int_E W^*[s(ay+b\varepsilon)u_0(y),\ r(ay+b\varepsilon)u_0(y)+v_0(y)]\frac{1}{a}\mathrm{d}a\mathrm{d}b$$
$$=\int_{L_T}W^*\Big[s\Big\{\frac{a}{u_0(y)}y+\Big(b-\frac{av_0(y)}{u_0(y)}\Big)\varepsilon\Big\}u_0(y),$$
$$r\Big\{\frac{a}{u_0(y)}y+\Big(b-\frac{av_0(y)}{u_0(y)}\Big)\varepsilon\Big\}u_0(y)+v_0(y)\Big]\frac{1}{a}\mathrm{d}a\mathrm{d}b.\tag{9}$$

Define

$$\begin{aligned}\bar{s}_{ab}(x)&=s\Big\{\frac{a}{u_0(x)}x+\Big(b-\frac{av_0(x)}{u_0(x)}\Big)\varepsilon\Big\},\\ \bar{r}_{ab}(x)&=r\Big\{\frac{a}{u_0(x)}x+\Big(b-\frac{av_0(x)}{u_0(x)}\Big)\varepsilon\Big\}.\end{aligned}\tag{10}$$

Then it is easy to verify that $(\bar{s}_{ab},\ \bar{r}_{ab})\in\vartheta$. Therefore, from (6), (7), (8), (9), and (10), it follows that

$$\underline{V}(F, W^*) + \varepsilon \geqslant \frac{1}{\varphi(T)} \inf_{(a,b)\in\Sigma''} \int W^*[\bar{s}_{ab}(y)u_0(y), \bar{r}_{ab}(y)u_0(y) + v_0(y)]\mathrm{d}F(y)\int_{L_T} \frac{1}{a}\mathrm{d}a\mathrm{d}b$$
$$- \frac{1}{\varphi(T)} 2mTM(1 + 2\log m)$$
$$\geqslant \inf_{(s,r)\in\vartheta} \int W^*[s(y)u_0(y), r(y)u_0(y) + v_0(y)]\mathrm{d}F(y) - \varepsilon(T). \tag{11}$$

where

$$\lim_{T\to\infty} \varepsilon(T) = \lim_{T\to\infty} \frac{mM}{2\log T}(1 + 2\log m) = 0. \tag{12}$$

But it is not difficult to show that

$$\inf_{(s,r)\in\vartheta} \int W^*[s(y)u_0(y), r(y)u_0(y) + v_0(y)]\mathrm{d}F(y)$$
$$= \inf\left\{ \int W^*[s(y)u_0(y), r(y)u_0(y) + v_0(y)]\mathrm{d}F(y) \mid (s, r) \in \vartheta;\right.$$
$$\left.\text{for any } y, (s(y), r(y)) \in L_{2m^2} \right\}. \tag{13}$$

In fact, for any $(s, r) \in \vartheta$, let us define a pair of functions s_1, r_1 in the following manner:

$$(s_1(x), r_1(x)) = \begin{cases} (s(x), r(x)), & \text{for } (s(x), r(x)) \in L_{2m^2}; \\ ((s(x))^*, (r(x))^*), & \text{for } (s(x), r(x)) \overline{\in} L_{2m^2}. \end{cases} \tag{14}$$

Here $((s(x))^*, (r(x))^*)$ is determined from $(s(x), r(x))$ by means of (5), with $q = 2m^2$. Clearly we have $(s_1, r_1) \in \vartheta$, $(s_1(x), r_1(x)) \in L_{2m^2}$ for every x, and

$$W^*[s_1(x)u_0(x), r_1(x)u_0(x) + v_0(x)] \leqslant W^*[s(x)u_0(x), r(x)u_0(x) + v_0(x)].$$

Therefore

$$\int W^*[s_1(y)u_0(y), r_1(y)u_0(y) + v_0(y)]\mathrm{d}F(y)$$
$$\leqslant \int W^*[s(y)u_0(y), r(y)u_0(y) + v_0(y)]\mathrm{d}F(y),$$

and (13) is proved. But we have $(s_1(x)u_0(x), r_1(x)u_0(x) + v_0(x)) \in L_{m(2m^2+1)}$ when $(s_1(x), r_1(x)) \in L_{2m^2}$, and $(u_0(x), v_0(x)) \in L_m$. So, from the definition of W it follows at once

"The right hand side of (11)"

$$\geqslant \inf_{(s,r)\in\vartheta} \int W[s(y)u_0(y), r(y)u_0(y) + v_0(y)]\mathrm{d}F(y) - \varepsilon(T)$$
$$= V(F, W) - \varepsilon(T). \tag{15}$$

From (11) and (15) it follows that

$$\underline{V}(F, W^*) + \varepsilon \geqslant V(F, W) - \varepsilon(T).$$

But obviously $\underline{V}(F, W) \geqslant \underline{V}(F, W^*)$. Inserting this into the above inequality and letting

$T \to \infty$ and then $\varepsilon \to 0$, we obtain $\underline{V}(F, W) \geqslant V(F, W)$. As the reverse inequality holds true in any case $V(F, W) = \underline{V}(F, W)$ follows and the proof is terminated.

In case W is bound, it follows by the same argument as used in [3] that $V(F, W) = \underline{V}(F, W)$, provided F satisfies (4). Nevertheles, it seems impossible to prove the existence of MIE without some additional assumptions.

Lemma 2 (A) Suppose that $W \geqslant 0$ is continuous on Σ^0, $W(0, t_2) = \infty$ and $W(t_1, t_2) \to \infty$ uniformly as $\frac{1}{t_1} + t_1 + |t_2| \to \infty$. Let μ be a measure defined on the σ-field $\mathscr{F}$ of all Borel sets in Σ^0 such that $\mu(B) < \infty$ for any closed bound set B, then

$$\lim_{m \to \infty} \inf_{(a, b) \in \Sigma^0} \int_{L_m} W(at_1, bt_1 + t_2) \mathrm{d}\mu = \inf_{(a, b) \in \Sigma^0} \int_{\Sigma^0} W(at_1, bt_1 + t_2) \mathrm{d}\mu. \tag{16}$$

(B) Suppose that $W \geqslant 0$ is continuous on Σ, $W(t_1, t_2) \to \alpha \leqslant \infty$ uniformly as $t_1 + |t_2| \to \infty$, and μ is a measure on $\mathscr{F}$ such that $\mu(C) < \infty$ for any bound Borel set C, then (16) holds true, but the symbol "$\inf\limits_{(a, b) \in \Sigma^0}$" should be replaced by "$\inf\limits_{(a, b) \in \Sigma}$".

Proof We shall prove only (A), for the proof of (B) is essentially the same. Without loss of generality we can assume that m takes only positive integer value. Write

$$h_m(a, b) = \int_{L_m} W(at_1, bt_1 + t_2) \mathrm{d}\mu.$$

It is easily seen that h_m is continuous on Σ^0. Clearly we may assume that $\mu(\Sigma^0) > 0$, so $\mu(L_m) > 0$ for the sufficiently large m. For such an m, $W(t_1, t_2) \to \infty$ uniformly as $\frac{1}{t_1} + t_1 + |t_2| \to \infty$ and h_m attains its minimum at some finite point $(a_m, b_m) \in \Sigma^0$. Writing $c_m = h_m(a_m, b_m)$, we have $c_1 \leqslant c_2 \leqslant \cdots$ and $\lim\limits_{m \to \infty} c_m = c \leqslant \infty$ exists. Two cases may occur: (Ⅰ) $h(a, b) = \int_{\Sigma^0} W(at_1, bt_1 + t_2) \mathrm{d}\mu \equiv \infty$, and (Ⅱ) $h \not\equiv \infty$. We consider only case (Ⅰ) as case (Ⅱ) may be discussed in a similar way.

We have to show that $c = \infty$. Suppose, on the contrary, that $c < \infty$; then the sequence $\left\{\left(\frac{1}{a_m} + a_m + |b_m|\right)\right\}$ must be bound, for otherwise we may assume that $\lim\limits_{m \to \infty}\left(\frac{1}{a_m} + a_m + |b_m|\right) = \infty$. Then for T sufficiently large (so as to make $\mu(L_T) > 0$), one has

$$\begin{aligned} \infty > c = \lim_{m \to \infty} c_m &= \lim_{m \to \infty} \int_{L_T} W(a_m t_1, b_m t_1 + t_2) \mathrm{d}\mu \\ &\geqslant \int_{L_T} \liminf_{m \to \infty} W(a_m t_1, b_m t_1 + t_2) \mathrm{d}\mu = \infty. \end{aligned}$$

This contradiction proves the boundness of $\left\{\left(\frac{1}{a_m} + a_m + |b_m|\right)\right\}$. Therefore without loss of generality we may assume that $(a_m, b_m) \to (a_0, b_0) \in \Sigma^0$. For any fixed T and $m \geqslant T$, we have

$$\int_{L_T} W(a_m t_1, b_m t_1 + t_2)\mathrm{d}\mu \leqslant \int_{L_m} W(a_m t_1, b_m t_1 + t_2)\mathrm{d}\mu = c_m \leqslant c < \infty.$$

Let $m \to \infty$ and then $T \to \infty$. Then it follows that $h(a_0, b_0) \leqslant c < \infty$, Which contradicts $h \equiv \infty$, and the lemma is proved.

Remark 1 Suppose that W is a strictly convex function on Σ and μ is a nonzero measure on $\mathscr{F}$. Then it is easily seen that $h(a, b) = \int_{\Sigma^0} W(at_1, bt_1 + t_2)\mathrm{d}\mu$ is strictly convex on the convex set $H = \{(a, b) \mid h(a, b) < \infty\}$. In case H is not empty, it follows that the point at which h attains its minimum is unique.

Before formulating Lemma 3, it is convenient to introduce the following notion: Let $A \subset R_k$ be a bound set. Define

$$\bar{a}_1 = \sup\{a_1 \mid a = (a_1, \cdots, a_k) \in A\}.$$

The intersection of $\bar{A}$(closure of A) and the hyperplane $x_1 = a_1$ is a bound set A_1. Let

$$\bar{a}_2 = \sup\{a_2 \mid a = (a_1, \cdots, a_k) \in A_1\}.$$

Suppose now that the numbers $\bar{a}_1, \cdots, \bar{a}_p (p \leqslant k-1)$ have already been determined. Let $A_p = \bar{A} \cap \{(a_1, \cdots, a_k) \mid a_i = \bar{a}_i, i \leqslant p\}$ and $\bar{a}_{p+1} = \sup\{a_{p+1} \mid a = (a_1, \cdots, a_k) \in A_p\}$. The point $\bar{a} = (\bar{a}_1, \cdots, \bar{a}_k)$ thus obtained clearly belongs to $\bar{A}$ and will be denoted by

$$\bar{a} = \sup a\{A\}.$$

Lemma 3 Let $\varphi^{(i)}(y) = (\varphi_1^{(i)}(y), \cdots, \varphi_k^{(i)}(y))$, $i = 1, 2, \cdots$ be a sequence of Borel measurable functions defined on a Borel set $E \subset R_m$. Suppose that for every $y \in E$ the set $A_y = \{\varphi^{(i)}(y), i = 1, 2, \cdots\}$ is bound; then the function f defined on E by the formula $f(y) = (f_1(y), \cdots, f_k(y)) = \sup a\{A_y\}$ is Borel measurable.

Proof The case $k = 1$ being a well-known result, we proceed to prove the general case by the method of induction. We first show that for any real u

$$\begin{aligned} &\{y \mid y \in E; f_k(y) > u\} \\ &= \sum_{n=1}^{\infty} \prod_{m=1}^{\infty} \sum_{i=1}^{\infty} \left\{ y \,\middle|\, y \in E;\ \varphi_d^{(i)}(y) > f_d(y) - \frac{1}{m},\ d = 1, \cdots, k-1;\ \varphi_k^{(i)}(y) \geqslant u + \frac{1}{n} \right\}. \end{aligned} \tag{17}$$

In fact, assume that y_0 belongs to the left hand of (17); then $f_k(y_0) - u = \delta > \frac{1}{n}$ for n_0 sufficiently large. Let m be arbitrarily fixed. Since $f(y_0) \in \bar{A}_{y_0}$, an integer i_m may be found such that $\varphi^{(i_m)}(y_0) \in V_\varepsilon$ where V_ε is a sphere with a centre $f(y_0)$ and a radius $\varepsilon < \min\left(\frac{1}{m}, \delta - \frac{1}{n_0}\right)$. Consequently we have shown that there exists an integer n such that to every m there corresponds an integer i_m such that

$$\varphi_d^{(i_m)}(y_0) > f_d(y_0) - \frac{1}{m},\ d = 1, \cdots, k-1,\ \varphi_k^{(i_m)}(y_0) \geqslant u + \frac{1}{n_0}, \tag{18}$$

where y_0 belongs to the right hand of (17). Conversely, if y_0 belongs to the right hand of (17), then an integer n_0 exists such that for any m, an integer i_m satisfying (18) may be found. The sequence $\{\varphi^{(i_m)}(y_0)\}$ being bound, there exists a subsequence $\{\varphi^{(i_{m_r})}(y)\}$ converging to a point $a = (a_1, \cdots, a_k)$. Clearly

$$a_d \geqslant f_d(y_0),\ d = 1, \cdots, k-1;\ a_k \geqslant u + \frac{1}{n_0}.$$

As the reverse inequality $a_d \leqslant f_d(y_0)$, $d = 1, \cdots, k-1$ follows easily from the definition of "sup a", we have $a_d = f_d(y_0)$ for $d = 1, \cdots, k-1$. Therefore $f_k(y_0) \geqslant a_k \geqslant u + 1/n_0 > u$ by the very definition of "sup a". This shows that y_0 belongs to the left hand of (17) and concludes the proof of (17).

Suppose now that Lemma 3 holds true for $k-1$ instead of k; then the right hand of (17) is a Borel set, and this proves the measurability of f_k. The truth of Lemma 3 for k follows, and the proof is terminated.

Lemma 4 Suppose that a sequence of functions $\{\varphi^{(i)}(y) = (\varphi_1^{(i)}(y), \cdots, \varphi_k^{(i)}(y)),\ i = 1, 2, \cdots\}$ defined on a Borel set $E \subset R_m$ satisfies the condition of Lemma 3. Then there exists a Borel measurable function $\varphi(y) = (\varphi_1(y), \cdots, \varphi_k(y))$ defined on E such that for any $y \in E$, a subsequence $\{i_r\}$ (depending, in general, on y) of positive integers may be found such that $\varphi(y) = \lim_{r\to\infty} \varphi^{(i_r)}(y)$.

Proof For $k = 1$ the lemma follows by taking $\varphi(y) = \limsup_{n\to\infty} \varphi^{(n)}(y)$. The general case may again be treated by the method of induction.

For any $y \in E$, let $A_{y_n} = \{\varphi^{(i)}(y),\ i = n, n+1, \cdots\}$, $n = 1, 2, \cdots$. By Lemma 3, the functions $f^{(n)}(y) = (f_1^{(n)}(y), \cdots, f_k^{(n)}(y)) = \sup a\{A_{y_n}\}$, $n = 1, 2, \cdots$ are Borel measurable on E. Clearly, for any $y \in E$, $\lim_{n\to\infty} f_1^{(n)}(y)$ exists. Consider the sequence of functions defined on E by $g^{(n)}(y) = (f_2^{(n)}(y), \cdots, f_k^{(n)}(y))$, $n = 1, 2, \cdots$. This sequence satisfies all conditions of Lemma 4. Therefore by induction hypothesis there exists on E a Borel measurable function $g(y) = (g_2(y), \cdots, g_k(y))$ such that for any $y \in E$, a subsequence $\{g^{(n_s)}(y)\}$ tending to $g(y)$ exists. Now define

$$\varphi(y) = (\varphi_1(y), \cdots, \varphi_k(y)) = (\lim_{n\to\infty} f_1^{(n)}(y), g_2(y), \cdots, g_k(y))$$

and proceed to show that it satisfies all conditions of Lemma 4. Since the measurability of φ follows from g, we have only to show that for any $y \in E$, there exists a subsequence $\{\varphi^{(i_r)}(y)\}$ tending to $\varphi(y)$. Suppose on the contrary that for some $\varepsilon > 0$, $\varphi^{(i)}(y) \overline{\in} V$ as $i \geqslant n_0$, where V is a sphere with a radius ε and a centre $\varphi(y)$. From the definition of "sup a" it then follows that $f^{(n)}(y) = \sup a\{A_{y_n}\} \overline{\in} V$ for $n \geqslant n_0$. This is a contradiction since for any y we have $g^{(n_s)}(y) \to g(y)$, and thus $f^{(n_s)}(y) \to \varphi(y)$ as $s \to \infty$. The proof of Lemma 4 is concluded.

Lemma 5 Let $P(Z, B)$ be a Borel measurable probability defined on $D \times \mathscr{F}$, where $D \subset R_n$ is a Borel set. Suppose further that the loss function W satisfies the conditions of

Lemma 2(A); then for any m, $0 < m < \infty$, there exists a Borel measurable function (a_m, b_m) mapping D into Σ^0 such that

$$\int_{L_m} W(a_m(z)t_1, b_m(z)t_1 + t_2)P(z, \mathrm{d}\omega) = \inf_{(a, b)\in\Sigma^0}\int_{L_m} W(at_1, bt_1 + t_2)P(x, \mathrm{d}\omega), \quad (19)$$

where $\mathrm{d}\omega = d(t_1, t_2)$. There exists also a Borel measurable function (a, b) mapping D into Σ^0 such that

$$\int_{\Sigma^0} W(a(z)t_1, b(z)t_1 + t_2)P(z, \mathrm{d}\omega) = \inf_{(a, b)\in\Sigma^0}\int_{\Sigma^0} W(at_1, bt_1 + t_2)P(z, \mathrm{d}\omega). \quad (20)$$

If the function W satisfies the conditions of Lemma 2 (B), then the above conclusion remains true, only that the symbol " $\inf\limits_{(a, b)\in\Sigma^0}$ " should be replaced by " $\inf\limits_{(a, b)\in\Sigma}$ ", and the functions (a_m, b_m), (a, b) will map D into Σ.

Proof We shall treat only the first case for in the second case the proof remains essentially unchanged.

The set $B_m = \{z \mid z \in D, P(z, L_m) = 0\}$ is a Borel set. If $z_0 \in B_m$, then any point in Σ^0 may be chosen for $(a_m(z_0), b_m(z_0))$ in (19). So without loss of generality we can assume $B_m = \varnothing$. Define

$$H_m(a, b; z) = \int_{L_m} W(at_1, bt_1 + t_2)P(z, \mathrm{d}\omega).$$

By a well-known fact of measure theory, $H_m(a, b; z)$ is Borel measurable on D for any fixed $(a, b) \in \Sigma^0$. The continuity of W implies the continuity of $H_m(a, b; z)$ in (a, b) for any fixed $z \in D$, and we have $\inf\limits_{(a, b)\in\Sigma^0} H_m(a, b; z) = \inf\limits_n H_m(a_n, b_n, z)$, where $\{(a_n, b_n)\}$ is the sequence of all rational points in Σ^0. In this way, we have proved that the right hand of (19) is Borel measurable. Now let $\varepsilon > 0$ be given, and define

$$E_{n\varepsilon} = \{z \mid z \in D, H_m(a_n, b_n, z) \leqslant \inf_{(a, b)\in\Sigma^0} H_m(a, b, z) + \varepsilon\}, \ n = 1, 2, \cdots$$

Clearly, $\{E_{n\varepsilon}\}$ is a sequence of the Borel set and $D = \sum\limits_{n=1}^{\infty} E_{n\varepsilon}$. Write $E^0_{1\varepsilon} = E_{1\varepsilon}, \cdots, E^0_{n\varepsilon} = E_{n\varepsilon} - (E_{1\varepsilon} + \cdots + E_{n-1\varepsilon}), \cdots$; then the sets of $\{E^0_{n\varepsilon}\}$ are pairwise disjoint. Now define a sequence of functions on D as follows:

$$(a_{m\varepsilon}(z), b_{m\varepsilon}(z)) = (a_n, b_n), \text{ for } z \in E^0_{n\varepsilon}, \ n = 1, 2, \cdots$$

These functions are Borel measurable on D. If we choose a sequence $\{\varepsilon_s\}$, $\varepsilon_s \downarrow 0$, and if for every s we define a function $(a_{m\varepsilon_s}, b_{m\varepsilon_s})$ as above, then for any $z \in D$ we have

$$\inf_{(a, b)\in\Sigma^0} H_m(a, b; z) \leqslant H_m(a_{m\varepsilon_s}(z), b_{m\varepsilon_s}(z); z) \leqslant \inf_{(a, b)\in\Sigma^0} H_m(a, b; z) + \varepsilon_s. \quad (21)$$

Setting $s \to \infty$, we obtain

$$\lim_{s\to\infty} H_m(a_{m\epsilon_s}(z),\ b_{m\epsilon_s}(z);\ z) = \inf_{(a,\ b)\in\Sigma^0} H_m(a,\ b;\ z). \tag{22}$$

Since the right hand of (19) is finite and since by assumption $P(z,\ L_m) > 0$, by an argument similar to what was used in the proof of Lemma 2, for any fixed $z \in D$, the sequence $\{(a_{m\epsilon_s}^{-1}(z) + a_{m\epsilon_s}(z) + |b_{m\epsilon_s}(z)|)\}_{s=1}^{\infty}$ is bound, and therefore there exists a compact set $J_z \subset \Sigma^0$ such that $(a_{m\epsilon_s}(z),\ b_{m\epsilon_s}(z)) \in J_z$ for all s. By Lemma 4, there exists a function on D such that for any $z \in D$ a subsequence $\{s_i\}$ can be found to satisfy the relation

$$\lim_{i\to\infty}(a_{m\epsilon_{s_i}}(z),\ b_{m\epsilon_{s_i}}(z)) = (a_m(z),\ b_m(z)).$$

Clearly $a_m(z) > 0$ for all z. Owing to the continuity of $H_m(a,\ b;\ z)$ in $(a,\ b)$, from (21) and (22) it follows at once that $H_m(a_m(z),\ b_m(z);\ z) = \inf_{(a,\ b)\in\Sigma^0} H_m(a,\ b;\ z)$. This proves (19).

We now proceed to prove (20). Write

$$H(a,\ b;\ z) = \int_{\Sigma^0} W(at_1,\ bt_1 + t_2)P(z,\ d\omega).$$

By (16) and the Borel measurability of $\inf_{(a,\ b)\in\Sigma^0} H_m(a,\ b;\ z)$ it follows that $\inf_{(a,\ b)\in\Sigma^0}\int_{\Sigma^0} W(at_1,\ bt_1 + t_2)P(z,\ d\omega)$ is Borel measurable. As before we may assume that $B = \{z \mid z \in D,\ H(a,\ b;\ z) \equiv \infty\}$ is empty.

Now let $(a_m,\ b_m)$, $m = 1,\ 2,\ \cdots$ be the functions satisfying (19) found above. For any fixed b' and $m \geqslant b'$,

$$\begin{aligned}
&\int_{L_{b'}} W(a_m(z)t_1,\ b_m(z)t_1 + t_2)P(z,\ d\omega)\\
&\leqslant \int_{L_m} W(a_m(z)t_1,\ b_m(z)t_1 + t_2)P(z,\ d\omega)\\
&\leqslant \inf_{(a,\ b)\in\Sigma^0} H(a,\ b;\ z) < \infty,
\end{aligned} \tag{23}$$

and as before for any fixed $z \in D$ there exists a compact set $J_z \subset \Sigma^0$ such that $(a_m(z),\ b_m(z)) \in J_z$ for all m. By Lemma 4 there exists a Borel measurable function $(a,\ b)$ such that for any $z \in D$ a subsequence $\{m_i\}$ can be found to satisfy

$$\lim_{i\to\infty}(a_{m_i}(z),\ b_{m_i}(z)) = (a(z),\ b(z));$$

clearly, $a(z) > 0$. From (23) it follows that

$$\begin{aligned}
&\int_{L_{b'}} W(a(z)t_1,\ b(z)t_1 + t_2)P(z,\ d\omega)\\
&= \lim_{i\to\infty}\int_{L_{b'}} W(a_{m_i}(z)t_1,\ b_{m_i}(z)t_1 + t_2)P(z,\ d\omega)\\
&\leqslant \inf_{(a,\ b)\in\Sigma^0} H(a,\ b;\ z).
\end{aligned} \tag{24}$$

Setting $b' \to \infty$, we see that (20) follows from (24). Lemma 5 is thus proved.

Remark 2 If W is strictly convex and $B = \varnothing$, then it follows from Remark 1 and

Lemma 5 that the function (a, b) found above to satisfy (20) is unique.

4 Proof of Theorem 1

Define a function z on $R_N - A$:

$$z(x) = \left(i_x,\ \mathrm{sign}(x_{i_x} - x_1),\ \frac{x_{i_x} - x_j}{x_{i_x} - x_1},\ j = 2, \cdots, N\right),$$

where $i_x = \min\{j \mid x_j \neq x_1,\ j = 2, \cdots, N\}$. It is readily seen that z is a maximal invariant, and therefore the condition $(s, r) \in \vartheta$ is equivalent to the condition that (s, r) depends only on z. In such a case, we shall write $s(x) = s(z(x))$ for convenience. It is easily seen that z is a Borel measurable function.

Let $P(z, B)$ be the mixed conditional distribution of (u_0, v_0) (see Section 2) given $z(X)$, the underlying distribution being F (see, for example, p. 361 in [16]). Suppose that $(s, r) \in \vartheta$; we have

$$\begin{aligned} &\mathrm{E}_F\{W[s(X)u_0(X),\ r(X)u_0(X) + v_0(X)] \mid Z\} \\ &= \mathrm{E}_F\{W[s(Z)u_0(X),\ r(Z)u_0(X) + v_0(X)] \mid Z\} \\ &= \int_{\Sigma^0} W[s(Z)t_1,\ r(Z)t_1 + t_2]P(Z,\ \mathrm{d}\omega). \end{aligned}$$

By Lemma 5, there exists a Borel measurable function (s_0, r_0) such that

$$\begin{aligned} w(Z) &= \mathrm{E}_F\{W[s_0(Z)u_0(X),\ r_0(Z)u_0(X) + v_0(X)] \mid Z\} \\ &= \int_{\Sigma^0} W[s_0(Z)t_1,\ r_0(Z)t_1 + t_2]P(Z,\ \mathrm{d}\omega) \\ &= \inf_{(a,\ b) \in \Sigma^0} \int_{\Sigma^0} W[at_1,\ bt_1 + t_2]P(Z,\ \mathrm{d}\omega). \end{aligned}$$

Therefore, for any $(s, r) \in \vartheta$ we have

$$\begin{aligned} \rho(s, r) &= \mathrm{E}_F\left\{\int_{\Sigma^0} W[s(Z)t_1,\ r(Z)t_1 + t_2]P(Z,\ \mathrm{d}\omega)\right\} \\ &\geqslant \mathrm{E}_F\left\{\int_{\Sigma^0} W[s_0(Z)t_1,\ r_0(Z)t_1 + t_2]P(Z,\ \mathrm{d}\omega)\right\} \\ &= \rho(s_0, r_0). \end{aligned} \tag{25}$$

This leads to

$$V = V(F, W) = \inf_{(s,\ r) \in \vartheta} \rho(s, r) = \rho(s_0, r_0) = \mathrm{E}_F\{w(Z)\}.$$

Now let m be a positive integer, define a measure F_m:

$$F_m(C) = F(C \cap \{x : (u_0(x),\ v_0(x)) \in L_m\}),$$

and let $h(x, m)$ be the characteristic function of the set $\{x \mid (u_0(x), v_0(x)) \in L_m\}$; then for any $(s, r) \in \vartheta$, we have

$$\begin{aligned}&\mathrm{E}_F\{h(X, m)W[s(Z)u_0(X), r(Z)u_0(X)+v_0(X)] \mid Z\}\\&\quad=\int_{L_m} W[s(Z)t_1, r(Z)t_1+t_2]P(Z, \mathrm{d}\omega).\end{aligned}$$

By Lemma 5 there exists a Borel measurable function (s_m, r_m) such that

$$\begin{aligned}w_m(Z) &= \mathrm{E}_F\{h(X, m)W[s_m(Z)u_0(X), r_m(Z)u_0(X)+v_0(X)] \mid Z\}\\&=\int_{L_m} W[s_m(Z)t_1, r_m(Z)t_1+t_2]P(Z, \mathrm{d}\omega)\\&=\inf_{(a, b)\in\Sigma^0}\int_{L_m} W(at_1, bt_1+t_2)P(Z, \mathrm{d}\omega).\end{aligned}$$

By an argument similar to that employed above, we obtain

$$V_m = V(F_m, W) = \rho(s_m, r_m; F_m, W) = \mathrm{E}_F\{w_m(Z)\}.$$

Now, by Lemma 2, $\lim\limits_{m\to\infty} w_m(z) = w(z)$, and $w_m(z)$ is nonnegative and nondecreasing in m. By the theorem of Lebesgue we have $\lim\limits_{m\to\infty} V(F_m, W) = V(F, W)$. But we have $V(F_m, W) = \underline{V}(F_m, W) \leqslant \underline{V}(F, W)$ by Lemma 1; therefore $\underline{V} \geqslant V$ and $\underline{V} = V$. Relation (25) shows that

$$(u(x), v(x)) = (u_0(x)s_0(x), u_0(x)r_0(x)+v_0(x)) \tag{26}$$

is a MIE. Theorem 1 is proved.

Remark 3 Suppose that there exists $(s', r') \in \vartheta$ such that $\rho(s', r') < \infty$, and W is strictly convex; then it is easily seen from the above discussion and Remark 2 that the function $(s_0, r_0) \in \vartheta$ found above is essentially unique in the class of all invariant estimators.

In most cases of practical importance, $(X_1, \cdots, X_N)$ is usually a random sample taken from a univariate distribution possessing a density function f. Under this assumption explicit formulae of the MIE analogous to the well-known Pitman estimate in translate parameter case can be derived. By means of these formulae in some simple cases, elegant expressions of the MIE may be established. For example, in the case $W(t_1, t_2) = (t_1-1)^2 + t_2^2$, when a^2 and b are variance and expectation of a univariate normal variable, the above method yields $\left(\dfrac{\Gamma\left(\frac{N}{2}\right)}{\Gamma\left(\frac{N-1}{2}\right)}\sqrt{\dfrac{2}{N-1}}s, \bar{x}\right)$ as the MIE of (a, b), where $\bar{x} = \dfrac{1}{N}\sum\limits_{i=1}^{N} x_i$ and $s^2 = \dfrac{1}{N-1}\sum\limits_{i=1}^{N}(x_i - \bar{x})^2$. It seems that this result has not been proved or explicitly stated elsewhere, and we should like to give another proof based on Theorem 1, which seems to be the simplist one.

By Theorem 1, the MIE(u, v) exists. Since $(s, \bar{x})$ is a sufficient statistic, we can write $u(x) = f(s, \bar{x})$, $v(x) = g(s, \bar{x})$①. Invariance property yields $f(s, \bar{x}+c) = f(s, \bar{x})$ and $g(s, \bar{x}+c) = g(s, \bar{x})+c$, so that $f(s, \bar{x}) = f_1(s)$, $g(s, \bar{x}) = \bar{x}+g_1(s)$. By invariance property again we have $f_1(ks) = kf_1(s)$ for $k>0$, and $f_1(s) = ks$ for some $k>0$.

Now the risk of $(f, g) = (ks, \bar{x}+g_1(s))$ is

① By the convexity of the function W and the famous theorem of Rao and Blackwell.

$$\mathrm{E}_{ab}\left[\frac{(\bar{x}-b+g_1(s))^2}{a^2}\right]+\mathrm{E}_{ab}\left[\frac{(ks-a)^2}{a^2}\right]$$

$$=\frac{1}{N}+\mathrm{E}_{ab}(g_1^2)/a^2+\left[k^2+1-2k\sqrt{\frac{2}{N-1}}\frac{\Gamma(N/2)}{\Gamma\left(\frac{N-1}{2}\right)}\right]$$

and its minimum is attained when $g_1 \equiv 0$, $k=\dfrac{\Gamma(N/2)}{\Gamma\left(\frac{N-1}{2}\right)}\sqrt{\dfrac{2}{N-1}}$. This proves the result stated above. It is easily seen that the result can be extended to the case where W is a convex function and $W(t_1, t_2)=W(t_1, |t_2|)$. The MIE of (a, b) would be $(cs, \bar{x})$, where c is determined by the condition $\mathrm{E}_{0,1}[W(cs, \bar{x})]=\inf\limits_k \mathrm{E}_{0,1}[W(ks, \bar{x})]$.

In [11] Kiefer proved a theorem, asserting that under certain conditions the sequential MIE reduces to a fixed sample size one. These conditions are easily verified in our problem, and thus we obtain the following result:

If the loss function is $(t_1-1)^2+t_2^2$ and the cost function $c(n)$ is nondecreasing in n, then the sequential MIE of (a, b) reduces to a fixed sample size one found above and the sample size N_0 is determined by the equation

$$c(N_0)+\frac{1}{N_0}+1-\frac{\Gamma^2(N_0/2)}{\Gamma^2\left(\frac{N_0-1}{2}\right)}\frac{2}{N_0-1}$$

$$=\min_N\left\{c(N)+\frac{1}{N}+1-\frac{\Gamma^2(N/2)}{\Gamma^2\left(\frac{N-1}{2}\right)}\frac{2}{N-1}\right\}.$$

We should like to point out that Chen Pin, in a recent paper[18], has obtained the minimax estimator of (a^2, b) by the method of Baye's estimation in the nonsequential case.

5 Proof of Theorem 2

Suppose that $F(A_1)=F(A)=c>0$, where A is defined by (3), $A_1=\{\alpha^{(n)}=(\alpha_n, \cdots, \alpha_n), n=1, 2, \cdots\}\subset A$ is a denumerable set. Let $c_n=F(\{\alpha^{(n)}\})>0$, $n=1, 2, \cdots$. Since A and R_N-A and invariant under the transformation $y=ax+b\varepsilon\,(a>0)$, we have $F_{a,b}(A)=F(A)=c$, $F_{a,b}(R_N-A)=1-c$, where $F_{a,b}$ is the measure induced by the random vector $aX+b\varepsilon$. Thus, our original problem may be considered as a mixture of two problems with probability $1-c$ and c respectively. The underlying probability distribution of these problems is defined by $F_1(B)=\frac{1}{1-c}F(B\cap(R_N-A))$ and $F_2(B)=\frac{1}{c}F(B\cap A)$ respectively.

We shall employ the symbols R, R_1, R_2 to denote the risk function of the original, the first and the second problems. Clearly

$$R(u, v;\, a, b)=(1-c)R_1(u, v;\, a, b)+cR_2(u, v;\, a, b).$$

Since F_1 satisfies (2), by Theorem 1 there exists an invariant estimator (u_0', v_0') such that

$$\rho(u_0', v_0') \equiv R_1(u_0', v_0'; a, b) \leqslant \sup_{(a, b) \in \Sigma^0} R_1(u, v; a, b)$$

for any estimator (u, v), invariant or not.

On the other hand, we know that the only invariant estimator of the second problem is $(u_0''(x), v_0''(x)) = (0, x_1)$, for $x = (x_1, \cdots, x_1)((u_0'', v_0'')$ may be defined in any way on $R_N - A)$. It is easily seen that

$$\rho_2(u_0'', v_0'') = R_2(u_0'', v_0''; a, b) = \int W\left(0, \frac{v_0''(x) - b}{a}\right) dF_2\left(\frac{x - b\varepsilon}{a}\right)$$
$$= \int W(0, y) dF_2(y) = \frac{1}{c}\sum_{n=1}^{\infty} c_n W(0, \alpha_n).$$

Therefore, if we take

$$(u_0(x), v_0(x)) = \begin{cases} (u_0'(x), v_0'(x)), \text{ for } x \in R_N - A, \\ (u_0''(x), v_0''(x)), \text{ for } x \in A, \end{cases}$$

then (u_0, v_0) is an invariant estimator, and

$$\rho(u_0, v_0) \equiv R(u_0, v_0; a, b) = (1 - c)\rho_1(u_0', v_0') + c\rho_2(u_0'', v_0'').$$

Now let (u, v) be any estimator. If a point $x^{(0)} = (x_0, \cdots, x_0)$ exists such that either $u(x^{(0)}) > 0$ or $v(x^{(0)}) \neq x_0$, then by the formula,

$$R(u, v;, a, b) = (1 - c)R_1(u, v; a, b) + cR_2(u, v; a, b)$$
$$\geqslant cR_2(u, v; a, b)$$
$$= \sum_{n=1}^{\infty} c_n W\left[\frac{u(a\alpha^{(n)} + b\varepsilon)}{a}, \frac{v(a\alpha^{(n)} + b\varepsilon) - b}{a}\right],$$

we have

$$R\left(u, v; \frac{1}{n}, x_0 - \frac{\alpha_1}{n}\right) \geqslant c_1 W[nu(x^{(0)}), n[v(x^{(0)}) - x_0] + \alpha_1]. \tag{27}$$

The right hand of (27) tends to ∞ as $n \to \infty$, so that

$$\sup_{(a, b) \in \Sigma^0} R(u, v; a, b) = \infty \geqslant \rho(u_0, v_0).$$

If $(u(x), v(x)) = (u_0''(x), v_0''(x))$ for $\alpha \in A$, then

$$\sup_{(a, b) \in \Sigma^0} R(u, v; a, b) = (1 - c) \sup_{(a, b) \in \Sigma^0} R_1(u, v; a, b) + c\rho_2(u_0'', v_0'')$$
$$\geqslant (1 - c)\rho_1(u_0', v_0') + c\rho_2(u_0'', v_0'')$$
$$= \rho(u_0, v_0).$$

This shows that (u_0, v_0) is an MIE, and Theorem 2 is proved.

Remark 4 If all conditions of Remark 3 are satisfied, the above arguments show again that the MIE is essentially unique in the class of all invariant estimators.

Remark 5 We may ask whether Theorem 2 remains true when W satisfies condition 4_3° of Section 2. It seems reasonable to conjecture that the answer is in the negative. While the author has not yet succeeded in constructing a counter example, the following example points out that in this case the MIE may be inadmissible.

Example $F((n, \cdots, n)) = \frac{1}{2^n}$, $n = 1, 2, \cdots$, and

$$W(t_1, t_2) = \frac{|1 - t_1| + |t_2|}{1 + t_1 + |t_2|}.$$

Here W satisfies conditions $1°$, $2°$, $3°$, and 4_3° of Section 2. The risk of the invariant estimator (u_0, v_0) is a constant:

$$\rho(u_0, v_0) = \sum_{n=1}^{\infty} \frac{|1 - 0| + n}{1 - 0 + n} \frac{1}{2^n} = 1.$$

However, the risk function of the estimator $(1, v_0)$ is

$$\rho(1, v_0; a, b) = \sum_{n=1}^{\infty} \frac{|a-1| + an}{a + 1 + an} \cdot \frac{1}{2^n} < \sum_{n=1}^{\infty} \frac{1}{2^n} = 1,$$

and the inadmissibility of (u_0, v_0) follows. Generally speaking, since on A the invariant estimator is unique, it is natural that in many cases the use of such an estimator would be unwise.

Remark 6 In above discussions we have assumed that the parameter vector (a, b) varies in the whole right open half-plane Σ^0. A close inspection of the foregoing proofs shows that this condition may be slightly relaxed. For example, the theorems remain true when any one of the following set is taken as the parameter space instead of Σ^0:

$$\Sigma_1^0 = \{(a, b) \mid a \geqslant a_0 > 0\}, \qquad \Sigma_2^0 = \{(a, b) \mid b \geqslant b_0\},$$
$$\Sigma_3^0 = \{(a, b) \mid b \leqslant b_0\}, \qquad \Sigma_4^0 = \{(a, b) \mid a \geqslant a_0 > 0, b \geqslant b_0\},$$
$$\Sigma_5^0 = \{(a, b) \mid a \geqslant a_0 > 0, b \leqslant b_0\}, \qquad \Sigma_6^0 = \{(a, b) \mid 0 < a \leqslant a_0, b_1 \leqslant b \leqslant b_2\}.$$

But the conclusion of the theorems becomes incorrect when Σ^0 is replaced by any bound subset of Σ^0.

6 Extension to Multidimensional Case

The results obtained above may be extended to the multidimensional case. We shall restrict ourselves to the formulation of the results without proof.

Let $X = (X_1, \cdots, X_N)$, where $X_i = (X_{i1}, \cdots, X_{ik})$, be a kN-dimensional random vector with a known distribution $F(x) = F(x_1, \cdots, x_N) = F(x_{11}, \cdots, x_{1k}, \cdots, x_{N1}, \cdots, x_{Nk})$. Let $a_1, \cdots, a_k; b_1, \cdots, b_k$ be $2k$ real numbers, where $a_i > 0$, $i = 1, \cdots, k$. The random vector

$$(a_1 X_{11} + b_1, \cdots, a_k X_{1k} + b_k; \cdots; a_1 X_{N1} + b_1, \cdots, a_k X_{Nk} + b_k)$$

possesses the distribution function

$$F(x \mid a_1, \cdots, a_k; b_1, \cdots, b_k) = F\left(\frac{x_{11} - b_1}{a_1}, \cdots, \frac{x_{1k} - b_k}{a_k}; \cdots; \frac{x_{N1} - b_1}{a_1}, \cdots, \frac{x_{Nk} - b_k}{a_k}\right). \tag{28}$$

The vector $(a_1, \cdots, a_k; b_1, \cdots, b_k)$ is naturally called the scale-location parameter vector. We want some estimation of this vector, on the basis of a sample drawn from population (28).

The invariance property of the loss function and the estimator may be defined analogously as in the one-dimensional case. For loss function V, this reduces to the existence of a function W defined on $R_k^+ \times R_k$ such that

$$V(a_1, \cdots, a_k; b_1, \cdots, b_k; c_1, \cdots, c_k; d_1, \cdots, d_k)$$
$$= W\left(\frac{c_1 - a_1}{a_1}, \cdots, \frac{c_k - a_k}{a_k}; \frac{d_1 - b_1}{a_1}, \cdots, \frac{d_k - b_k}{a_k}\right),$$

where $R_k^+ = \{\alpha = (\alpha_1, \cdots, \alpha_k) \mid \alpha_i \geqslant 0, i = 1, \cdots, k\}$. The conditions imposed on W are as follows:

1°. W is continuous at any interior point of $R_k^+ \times R_k$.

2°. $W(1, \cdots, 1; 0, \cdots, 0) = 0$.

3°. For any i, $1 \leqslant i \leqslant k$; $W = W(t_1^{(1)}, \cdots, t_k^{(1)}; t_1^{(2)}, \cdots, t_k^{(2)})$, viewed as a function of $t_i^{(1)}$ and $t_i^{(2)}$, satisfies condition 3° of Section 2.

4_1°. $W(t_1^{(1)}, \cdots, t_k^{(1)}; t_1^{(2)}, \cdots, t_k^{(2)}) \to \infty$ uniformly as

$$\sum_{i=1}^{k} [(t_i^{(1)})^{-1} + t_i^{(1)} + | t_i^{(2)} |] \to \infty.$$

4_2°. W is continuous on $R_k^+ \times R_k$ and $W(t_1^{(1)}, \cdots, t_k^{(1)}; t_1^{(2)}, \cdots, t_k^{(2)}) \to \infty$ uniformly as $\sum_{i=1}^{k} (t_i^{(1)} + | t_i^{(2)} |) \to \infty$.

4_3°. W is continuous on $R_k^+ \times R_k$ and $W(t_1^{(1)}, \cdots, t_k^{(1)}; t_1^{(2)}, \cdots, t_k^{(2)}) \to \alpha < \infty$ uniformly as $\sum_{i=1}^{k} (t_i^{(1)} + | t_i^{(2)} |) \to \infty$.

The following theorem holds true:

Theorem 1′ Suppose that W satisfies conditions 1°, 2°, 3°, and one of 4_1°, 4_2°, and 4_3° of Section 6, and that the distribution F satisfies $F(A^*) = 0$, where

$$A^* = \{x = (x_{11}, \cdots, x_{1k}; \cdots; x_{N1}, \cdots, x_{Nk}) \mid \text{there exists } i \text{ such that } x_{1i} = \cdots = x_{Ni}\}.$$

Then an MIE of $(a_1, \cdots, a_k; b_1, \cdots, b_k)$ exists.

It should be remarked that when $F(A^*) \neq 0$. Theorem 2 of Section 2 cannot be extended in a straight-forward way. The truth is that when we consider the original problem as a probability mixture of several problems, the sample spaces of these problems are of the following form:

$$A_{i_1 \cdots i_r}^* = \{x = (x_{11}, \cdots, x_{1k}; \cdots; x_{N1}, \cdots, x_{Nk}) \mid x_{1i_s} = \cdots = x_{Ni_s} \text{ for } s = 1, \cdots, r, \text{ and } \max_{1 \leqslant p < q \leqslant n} | x_{pt} - x_{qt} | > 0 \text{ for } t \neq i_s, s = 1, \cdots, r\}.$$

But as long as $r \neq N$, the invariant estimator on $A_{i_1 \cdots i_r}^*$ would not be uniquely determined, and the argument employed in Section 5 fails in the present case. It seems that, in general, Theorem 2 of Section 2 cannot be extended to the multidimensional case.

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2

拟秩统计量的一极限定理

1 摘要

在本文中我们将证明下面的结果.

定理 1 设 $X_1, \cdots, X_m$; $Y_1, \cdots, Y_n$; $Z_1, \cdots, Z_l$ 是从具有连续分布函数 F 的总体中抽出的独立样本,此处 F 满足以下三个条件:

1° F 为关于原点对称,并具密度函数 f 的分布;

2° 存在一有限常数 K,使得对任何 x 及 $h \neq 0$ 有

$$\left|\frac{F(x+h)-F(x)}{h}\right| \leqslant K;$$

3° $\int_{-\infty}^{\infty} |x|^3 \mathrm{d}F(x) < \infty$,

以 $\overline{X}$, $\overline{Y}$, $\overline{Z}$ 分别记 $\frac{1}{m}\sum_1^m X_i$, $\frac{1}{n}\sum_1^n Y_i$, $\frac{1}{l}\sum_1^l Z_k$. 定义

$$X_i^* = X_i - \overline{X} + \overline{Z},\ i = 1, \cdots, m,$$

以 r_i 记 X_i^* 在 $\{X_1^*, \cdots, X_m^*, Y_1, \cdots, Y_n\}$ 中的秩, $i = 1, \cdots, m$. 引进统计量

$$M^* = \sum_{i=1}^m \left(r_i - \frac{N+1}{2}\right)^2,$$

此处 $N = m+n$, 则当 m, n, l 皆$\to\infty$而

$$\lim \lambda_N = \lim \frac{m}{N} = \lambda,\ 0 < \lambda < 1,\ \liminf \frac{l}{N} > 0 \tag{1}$$

成立时,对一切 x 必有

$$P\left\{\frac{\sqrt{180}\left(M^* - \frac{m(N^2-1)}{12}\right)}{\sqrt{mn(N+1)(N^2-4)}} < x\right\} \to \Phi(x), \tag{2}$$

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此处 Φ 为 $N(0,1)$的分布函数.

在第 2 节中,我们指出这一结果在非参数检验理论中的意义. 在第 4 节中,我们指出类似的定理对其他某些统计量也成立,其中包括 Ansari 和 Bradley 的统计量 W(见[1]).

2 引言

在关于刻度参数(scale parameter)的非参数两样本问题中,下面两种情况有着很大的差别:一种是两总体的某种转移参数皆已知或至少其差为已知. 在这种情况下,经过适当的平移变换,可假定二者有相同的转移参数,从而可以引进具有一定良好性质的秩检验. 这方面已有了一些工作,在此我们提到 Mood[2] 与 Ansari 和 Bradley[1].

另外一种,也是在实际中最有兴趣的情况,是两总体的转移参数皆全然未知. 在这种情况下,如 Fraser[3],Moses[4] 等人的工作所指出的,除非对所涉及的分布族作重大的限制,否则不可能存在性质足够良好的秩检验. 解决这一问题的办法是引进所谓拟秩检验(Rank-like Test),其具体途径之一是"修正"有关的秩检验统计量. 例如,在 Mood 检验(见[2])的情况,若以 $\hat{\xi}(X_1,\cdots,X_m)$及 $\hat{\eta}(Y_1,\cdots,Y_n)$分别记第一、二总体的转移参数的某种估计值,令

$$X_i^* = X_i - \hat{\xi},\ i=1,\cdots,m,\ Y_j^* = Y_j - \hat{\eta},\ j=1,\cdots,n.$$

以 r_i^* 记 X_i^* 在"中心化合样本"$\{X_1^*,\cdots,X_m^*;Y_1^*,\cdots,Y_n^*\}$中的秩而作统计量

$$M^* = \sum_{i=1}^{m}\left(r_i^* - \frac{m+n+1}{2}\right)^2,$$

则"修正的 Mood 检验"可基于"修正的 Mood 统计量"M^* 而引进.

这一方法直观看来很合理,也很自然,但在理论上颇有困难,主要在于有关的分布理论复杂化了. 为了这种修正至少能部分地保持秩检验的优良性质,最好修正后的统计量在原假设下是分布无关的,或最低限度是渐近分布无关的(asymptotic distribution free).

Sukhatme[5] 是这一领域内的唯一文献. 在该文中,他处理了基于 U 统计量的检验,其中包括他自己在[6]中提出的检验. 在文章的末尾,他证明了 Mood 统计量 M 不论作何种修正(但应满足该文中提出的条件)恒不能渐近分布无关. 由于 Mood 统计量的简要,这一结论颇出人意外,从而对这种修正的途径能否在一定程度上获得成功引起了疑问. 本文的结果显示出,若对通常所了解的"修正"手续作一些不大的改变,则对若干统计量来说,在对称分布族中渐近分布无关性仍能保持.

本文方法对于推导在对立假设下修正秩统计量的极限分布也能应用. 因此,可以将这种检验与由另一途径(见[4])得到的拟秩检验加以比较,这方面有关的结果将另文叙述.

3 若干引理

本节的目的是为证明第 1 节中的定理做准备.

引理 1 设 $g(t_1,t_2)$为实变数 t_1,t_2 的实函数,满足以下两条件:

1° 存在常数 $K<\infty$,对一切(t_1,t_2),致 $|g(t_1,t_2)|\leqslant K$;

2° 存在 Lebesgue 零测度集 A,使当 $t_1 \overline{\in} A$ 时有 $\lim\limits_{t_2 \to 0} g(t_1, t_2) = 0$.

又设 $X_1, X_2, \cdots$ 为一串相互独立、相同分布的随机变量,X_1 有密度 f, $Y_1, Y_2, \cdots$ 为一串依概率收敛于 0 的随机变量,则当 $n \to \infty$ 时

$$Z_n = \frac{1}{n} \sum_{i=1}^{n} g(X_i, Y_n) = o_p(1).$$

证 任意取定 $\varepsilon > 0$, 定义一串集合

$$B_n = \left\{ t_1 : |g(t_1, t_2)| < \varepsilon, \text{当} |t_2| \leqslant \frac{1}{n} \right\}, n = 1, 2, \cdots,$$

则显然 $B_1 \subset B_2 \subset B_3 \subset \cdots$, $\sum\limits_{n=1}^{\infty} B_n = A^c$ (此处 A^c 记 A 的余集). 取 N_1 充分大,致

$$\int_{B_{N_1}} f(x) \mathrm{d}x > 1 - \varepsilon,$$

取 N_2 充分大,使当 $n \geqslant N_2$ 时有

$$P\left\{ |Y_n| \leqslant \frac{1}{N_1} \right\} > 1 - \frac{\varepsilon}{2}.$$

由大数定律可知,存在 N_3 充分大,使当 $n \geqslant N_3$ 时有

$$P(D) > 1 - \frac{\varepsilon}{2},$$

此处

$$D = \{X_1, \cdots, X_n \text{ 落在集合 } B_{N_1} \text{ 中的个数} \geqslant n(1 - 2\varepsilon)\}.$$

记 $N = \max\{N_2, N_3\}$, 则当 $n \geqslant N$ 时有

$$P(E) = P\left\{ \left[|Y_n| < \frac{1}{N_1} \right] \cap D \right\} > 1 - \varepsilon.$$

但易见当事件 E 发生时有

$$|Z_n| \leqslant \frac{1}{n} \sum_{i=1}^{n} |g(X_i, Y_n)| \leqslant \frac{1}{n} [\varepsilon n(1 - 2\varepsilon) + n2\varepsilon K] < \varepsilon(1 + 2K).$$

换句话说,对任给的 $\varepsilon > 0$, 存在 $N(\varepsilon)$,使当 $n \geqslant N(\varepsilon)$ 时有

$$P\{|Z_n| < \varepsilon(1 + 2K)\} > 1 - \varepsilon.$$

这就证明了所要的结果.

引理 2 采用定理 1 的记号. 设定理 1 中的 2°,3°两条件都满足且 $\int_{-\infty}^{\infty} x \mathrm{d}F(x) = 0$, 又当 $N \to \infty$ 时(1)式成立,则若以 F_m 及 F_m^* 分别记 $X_1, \cdots, X_m$ 及 $X_1^*, \cdots, X_m^*$ 的经验分布函数,则

$$\mathrm{E}\{(F_m^*(x) - F_m(x))^2\} = o\left(\frac{1}{\sqrt{N}}\right)$$

且对 x 是一致的.

证 在此我们要用到极限理论中的下述结果(参看[7]):设 ξ_1, ξ_2, …为一串相互独立、相同分布的随机变量序列,其三阶矩存在、有限且满足条件

$$\limsup_{|t|\to\infty} |\varphi(t)| < 1, \tag{3}$$

此处 φ 为 ξ_1 的特征函数,则一致地对于 x 有

$$F_n(x) - \Phi(x) = \frac{1}{\sqrt{2\pi n}} e^{-x^2/2} Q(x) + o\left(\frac{1}{\sqrt{n}}\right),$$

这里 F_n 为 $\sum_{i=1}^{n}(\xi_i - E\xi_i)/\sqrt{n \cdot D\xi_1}$ 的分布函数,而 Q 为一多项式,其系数只与 ξ_1 的前三阶矩有关.

因为当密度存在时(3)式必然成立,在引理的假定下,上述定理的条件全部满足. 现在取定 $\delta_1 > 0$, $\delta_2 > 0$ 且 $\delta_1 + \delta_2 < \frac{1}{4}$. 记 $d^2 = \int_{-\infty}^{\infty} x^2 \mathrm{d}F(x)$, 由上面所引的定理有

$$\begin{aligned} P\{|\overline{X}| \geqslant N^{-\frac{1}{2}+\delta_1}\} &\leqslant P\left\{\frac{\sqrt{m}\,|\overline{X}|}{d} \geqslant hN^{\delta_1}\right\} \\ &= 2[1 - \Phi(hN^{\delta_1})] + \frac{1}{\sqrt{2\pi m}} e^{-h^2 N^{2\delta_1}/2} Q(hN^{\delta_1}) \\ &\quad - \frac{1}{\sqrt{2\pi m}} e^{-h^2 N^{2\delta_1}/2} Q(-hN^{\delta_1}) + o\left(\frac{1}{\sqrt{N}}\right), \end{aligned} \tag{4}$$

这里 h 为某一个大于 0 的常数. 由于

$$1 - \Phi(hN^{\delta_1}) \approx \frac{1}{\sqrt{2\pi} hN^{\delta_1}} e^{-h^2 N^{2\delta_1}/2} = o\left(\frac{1}{\sqrt{N}}\right),$$

由(4)可知

$$P\{|\overline{X}| \geqslant N^{-\frac{1}{2}+\delta_1}\} = o\left(\frac{1}{\sqrt{N}}\right).$$

因为由关系式(1)知,当 N 充分大时,对某一个固定的 $u > 0$, $l \geqslant uN$, 用上述同样的推理知

$$P\{|\overline{Z}| \geqslant N^{-\frac{1}{2}+\delta_1}\} = o\left(\frac{1}{\sqrt{N}}\right).$$

固定一个 x,记 $I_N = [x - 2N^{-\frac{1}{2}+\delta_1}, x + 2N^{-\frac{1}{2}+\delta_1}]$, 且定义事件 $B_N = \{X_1, \cdots, X_m$ 落在 I_N 中的个数 $\geqslant N^{\frac{3}{4}-\delta_2}\}$, 则

$$P(B_N) = \sum_{k \geqslant N^{\frac{3}{4}-\delta_2}} b(k; m, p_N), \tag{5}$$

这里 $p_N = F(x + 2N^{-\frac{1}{2}+\delta_1}) - F(x - 2N^{-\frac{1}{2}+\delta_1})$. 由定理 1 条件 2°得知,存在常数 q 致 $p_N \leqslant qN^{-\frac{1}{2}+\delta_1} = p'_N$. 又众所周知,(5)式右边为 p_N 的非降函数,故

$$P(B_N) \leqslant \sum_{k \geqslant N^{\frac{3}{4}-\delta_2}} b(k;\ m,\ p'_N).$$

由 $\delta_1 + \delta_2 < \frac{1}{4}$ 可知,当 N 充分大时有 $N^{\frac{3}{4}-\delta_2} > mp'_N + 1$. 故有(此结果见[8],第 6 章)

$$P(B_N) \leqslant \frac{m}{[N^{\frac{3}{4}-\delta_2}] - mp'_N} b([N^{\frac{3}{4}-\delta_2}];\ m,\ p'_N),$$

此处$[x]$记不超过 x 的最大整数,但是由 Stirling 公式($\sim$表示当 $N\to\infty$时两边的比$\to 1$),有

$$b([N^{\frac{3}{4}-\delta_2}];\ m,\ p'_N) \sim \left\{\frac{m}{2\pi[N^{\frac{3}{4}-\delta_2}](m-[N^{\frac{3}{4}-\delta_2}])}\right\}^{\frac{1}{2}}$$
$$\times \left(\frac{mp'_N}{[N^{\frac{3}{4}-\delta_2}]}\right)^{[N^{\frac{3}{4}-\delta_2}]} \left(\frac{m(1-p'_N)}{m-[N^{\frac{3}{4}-\delta_2}]}\right)^{m-[N^{\frac{3}{4}-\delta_2}]},$$

右边第一因子当 $N\to\infty$时$\to 0$. 取对数,并注意 $\delta_1 + \delta_2 < \frac{1}{4}$, 易知后两因子之积有数量级 $O(N^{-(N^{\frac{3}{4}-\delta_2})})$. 由此推出

$$P(B_N) = o\left(\frac{1}{\sqrt{N}}\right),$$

且由上面的讨论知,右边对 x 为一致的. 故若记

$$C_N = \{|\overline{X}| \geqslant N^{-\frac{1}{2}+\delta_1}\},\ D_N = \{|\overline{Z}| \geqslant N^{-\frac{1}{2}+\delta_1}\},$$

则由以上讨论知

$$P(B_N + C_N + D_N) = o\left(\frac{1}{\sqrt{N}}\right),$$

且右边对 x 一致,但不难看出,当 $B_N + C_N + D_N$ 不发生时有

$$|F^*_m(x) - F_m(x)| \leqslant N^{\frac{3}{4}-\delta_2}/m \leqslant h' N^{-\frac{1}{4}-\delta_2},$$

此处 $h' > 0$ 为一常数. 因此

$$\mathrm{E}\{(F^*_m(x) - F_m(x))^2\} \leqslant 1 \cdot o\left(\frac{1}{\sqrt{N}}\right) + (h' N^{-\frac{1}{4}-\delta_2})^2 = o\left(\frac{1}{\sqrt{N}}\right),$$

且右边对 x 一致. 这就证明了引理 2.

引理 3 在引理 2 的条件下,若记

$$F^{(m)}(x) = \mathrm{E}(F^*_m(x)),\ T^{(m)}(x,\ y) = \mathrm{E}(F^*_m(x)F^*_m(y)),$$

则 $F^{(m)}$ 及 $T^{(m)}$ 分别为一维及二维的连续分布函数,且

$$\lim_{N\to\infty} F^{(m)}(x) = F(x) \text{ 对一切 } x, \tag{6}$$

$$\lim_{N\to\infty} T^{(m)}(x,\ y) = F(x)F(y) \text{ 对一切 } x,\ y. \tag{7}$$

证 $F^{(m)}$的非降性显然,现证其连续性.注意

$$F^{(m)}(x+h)-F^{(m)}(x)=\mathrm{E}\{F_m^*(x+h)-F_m^*(x)\}$$
$$=\mathrm{E}\{\mathrm{E}\{(F_m^*(x+h)-F_m^*(x))\mid\overline{Z}\}\},$$

记

$$\mathrm{E}\{(F_m^*(x+h)-F_m^*(x))\mid\overline{Z}\}=g(\overline{Z},x,h),$$

$$\mathrm{E}\{(F_m^*(x+h)-F_m^*(x))\mid\overline{Z};\ \overline{x}=y\}=\tilde{g}(\overline{Z},y,x,h),$$

则有

$$g(\overline{Z},x,h)=\int_{-\infty}^{\infty}\tilde{g}(\overline{Z},y,x,h)\mathrm{d}G_1(y),$$

此处G_1为$\overline{X}$的分布函数.由引理的假定易知,当给定$\overline{X}=y$时,$(X_1,\cdots,X_m)$在超平面$\overline{X}=y$上的条件分布将有密度$f(x_1,\cdots,x_m\mid y)$.由$(X_1,\cdots,X_m)$与$(Z_1,\cdots,Z_l)$的独立性易知

$$\mid\tilde{g}(\overline{Z},y,x,h)\mid\leqslant\sum_{i=1}^{m}\int_{D_i}f(x_1,\cdots,x_m\mid y)\mathrm{d}s,$$

此处$\mathrm{d}s$为平面$\overline{X}=y$上的面积元素,而

$$D_i=\{(x_1,\cdots,x_m):\overline{X}=y,\ x+y-\overline{Z}\leqslant x_i\leqslant x+h+y-\overline{Z}\}.$$

由此可知,当$\overline{Z}$, y, x固定时有$\lim\limits_{h\to0}\tilde{g}(\overline{Z},y,x,h)=0$.再由$\tilde{g}$的有界性,用控制收敛定理即得:对任何固定的$\overline{Z}$, x,

$$\lim_{h\to0}g(\overline{Z},x,h)=0.$$

但

$$\mid F^{(m)}(x+h)-F^{(m)}(x)\mid\leqslant\mathrm{E}\{\mid g(\overline{Z},x,h)\mid\}$$

而g有界,故当$h\to0$时$\mid F^{(m)}(x+h)-F^{(m)}(x)\mid\to0$.这就证明了$F^{(m)}$的连续性.现在证明

$$\lim_{x\to\infty}F^{(m)}(x)=1,\quad\lim_{x\to-\infty}F^{(m)}(x)=0.\tag{8}$$

此两式的证明相似,故只证第一式.任给$\varepsilon>0$,找出$A>0$充分大致

$$P\left\{\mid\overline{X}\mid\geqslant\frac{A}{2}\right\}<\frac{\varepsilon}{m+2},\ P\left\{\mid\overline{Z}\mid\geqslant\frac{A}{2}\right\}<\frac{\varepsilon}{m+2}.$$

取x_0充分大致$F(x_0)>1-\dfrac{\varepsilon}{m+2}$,令$x_0'=x_0+A$,定义事件$B=\{\max\limits_{1\leqslant i\leqslant m}x_i<x_0\}$,则$P(B)\geqslant\left(1-\dfrac{\varepsilon}{m+2}\right)^m$.故

$$\begin{aligned}&P\left\{B\cap\left[\mid\overline{X}\mid\leqslant\frac{A}{2}\right]\cap\left[\mid\overline{Z}\mid\leqslant\frac{A}{2}\right]\right\}\\&\geqslant\left(1-\frac{\varepsilon}{m+2}\right)^m-2\frac{\varepsilon}{m+2}\\&\geqslant1-m\frac{\varepsilon}{m+2}-2\frac{\varepsilon}{m+2}\\&=1-\varepsilon.\end{aligned}$$

但当 $B\cap\left\{|\overline{X}|\leqslant\frac{A}{2}\right\}\cap\left\{|\overline{Z}|\leqslant\frac{A}{2}\right\}$ 发生时不难看出有 $F_m^*(x_0')=1$，因此

$$F^{(m)}(x_0')\geqslant 1-\varepsilon.$$

这就证明了(8)的第一式. 为了证明(6)，注意

$$(F_m^*(x)-F(x))^2\leqslant 2(F_m^*(x)-F_m(x))^2+2(F_m(x)-F(x))^2,$$

故由引理 2 有

$$\begin{aligned}\mathrm{E}\{(F_m^*(x)-F'(x))^2\}&\leqslant o\left(\frac{1}{\sqrt{N}}\right)+2\mathrm{E}\{(F_m(x)-F(x))^2\}\\&=o\left(\frac{1}{\sqrt{N}}\right)+o\left(\frac{1}{N}\right)\\&=o\left(\frac{1}{\sqrt{N}}\right).\end{aligned}$$

因此，当 $N\to\infty$ 时有

$$\begin{aligned}|F^{(m)}(x)-F(x)|&\leqslant \mathrm{E}\{|F_m^*(x)-F(x)|\}\\&\leqslant\{\mathrm{E}\{(F_m^*(x)-F(x))^2\}\}^{1/2}\\&\to 0.\end{aligned}$$

$T^{(m)}(x,y)$ 为二维连续分布函数以及(7)的证明两者可类似地作出，引理 3 证毕.

4 定理 1 的证明

我们分以下几段来进行.

1° 引进下面的一些记号:

G_n: $\{y_1,\cdots,y_n\}$ 的经验分布函数，

H_N: $\{X_1^*,\cdots,X_m^*,y_1,\cdots,y_n\}$ 的经验分布函数，

F_m，F_m^*，$F^{(m)}$ 的意义同第 3 节，

$$\lambda_N=\frac{m}{N},\ J(x)=\left(x-\frac{1}{2}\right)^2\quad(\text{当 } 0<x<1).$$

根据 Chernoff-Savage[9] 的理论，经过一些简单的论证，得知定理 1 的证明可转化为以下两点事实的证明:

(a) $\sqrt{N}\{(B_{1N}^*+B_{2N}^*)-(B_{1N}+B_{2N})\}=o_p(1)$，

(b) $C_{iN}^*=o_p\left(\frac{1}{\sqrt{N}}\right)$，$i=1,\cdots,5$，这里

$$C_{1N}^*=\lambda_N\int_{-\infty}^{\infty}[F_m^*(x)-F(x)]J'[F(x)]\mathrm{d}[F_m^*(x)-F(x)],$$

$$C_{2N}^*=(1-\lambda_N)\int_{-\infty}^{\infty}[G_n(x)-F(x)]J'(F(x))\mathrm{d}[F_m^*(x)-F(x)],$$

$$C_{3N}^{*}=\int_{I_N}\frac{[H_N(x)-F(x)]^2}{2}J''[\varphi H_N(x)+(1-\varphi)F(x)]\mathrm{d}F_m^{*}(x),$$

$$C_{4N}^{*}=\int_{H_N=1}\{-J(F(x))-[H_N(x)-F(x)]\}J'(F(x))\mathrm{d}F_m^{*}(x),$$

$$C_{5N}^{*}=\int_{H_N=1}J[H_N(x)]\mathrm{d}F_m^{*}(x),$$

其中 $I_N=\{x:0<H_N(x)<1\}$ 而 $0\leqslant\varphi\leqslant1$($\varphi$ 与 x, X_i, Y_j, Z_k 有关). 又

$$B_{1N}+B_{2N}=(1-\lambda_N)\left\{\frac{1}{m}\sum_{i=1}^{m}[B(X_i)-\mathrm{E}B(X_i)]-\frac{1}{n}\sum_{j=1}^{n}[B(Y_j)-\mathrm{E}B(Y_j)]\right\},$$

$$B_{1N}^{*}+B_{2N}^{*}=(1-\lambda_N)\left\{\frac{1}{m}\sum_{i=1}^{m}[B(X_i^{*})-\mathrm{E}B(X_i)]-\frac{1}{n}\sum_{j=1}^{n}[B(Y_j)-\mathrm{E}B(Y_j)]\right\},$$

而

$$B(x)=\int_0^x J'[F(x)]\mathrm{d}F(x)=\left(F(x)-\frac{1}{2}\right)^2.$$

2° 现在我们来证明(a). 因为

$$B(X_i^{*})-B(X_i)=(F(X_i^{*})-F(X_i))[F(X_i^{*})+F(X_i)-1],$$

由定理的假定不难推得

$$F(X_i^{*})=F(X_i)+\hat{\xi}[f(X_i)+g(X_i,\hat{\xi})],$$

此处 $\hat{\xi}=\overline{Z}-\overline{X}$, 而 g 满足引理 1 的条件. 因此

$$\begin{aligned}B(X_i^{*})-B(X_i)=&\ \hat{\xi}f(X_i)[2F(X_i)-1]+\hat{\xi}^2f(X_i)\\&+\hat{\xi}g(X_i,\hat{\xi})[2F(X_i)-1]+2\hat{\xi}^2f(X_i)g(X_i,\hat{\xi}),\end{aligned}$$

因而我们得到

$$\begin{aligned}\sqrt{N}\{(B_{1N}^{*}+B_{2N}^{*})-(B_{1N}+B_{2N})\}=&\ (1-\lambda_N)\sqrt{N}\frac{1}{m}\sum_{i=1}^{m}\hat{\xi}f(X_i)[2F(X_i)-1]\\&+(1-\lambda_N)\sqrt{N}\frac{1}{m}\sum_{i=1}^{m}\hat{\xi}^2f^2(X_i)\\&+(1-\lambda_N)\sqrt{N}\frac{1}{m}\sum_{i=1}^{m}\hat{\xi}g(X_i,\hat{\xi})[2F(X_i)-1]\\&+2(1-\lambda_N)\sqrt{N}\cdot\frac{1}{m}\sum_{i=1}^{m}\hat{\xi}^2f(X_i)g(X_i,\hat{\xi})\\=&\ I_1+I_2+I_3+I_4.\end{aligned}\tag{9}$$

因为 $\sqrt{N}\hat{\xi}=O_p(1)$, 由定理假定知 $\int_{-\infty}^{\infty}f^2(x)\mathrm{d}x<\infty$, 故由 F 的对称性知 $\int_{-\infty}^{\infty}f^2(x)[2F(x)-1]\mathrm{d}x=0$, 因而由大数定律知

$$I_1=o_p(1).\tag{10}$$

对 I_2,注意因 $\sqrt{N}\,\hat{\xi}=O_p(1)$, $\hat{\xi}=o_p(1)$ 及$\frac{1}{m}\sum_{i=1}^{m}f^2(X_i)=O_p(1)$, 故

$$I_2=o_p(1). \tag{11}$$

现在注意

$$|I_3|\leqslant|\sqrt{N}\,\hat{\xi}\,|\cdot\frac{1}{m}\sum_{i=1}^{m}|g(X_i,\ \hat{\xi})|.$$

由引理 1 知 $\frac{1}{m}\sum_{i=1}^{m}|g(X_i,\ \hat{\xi})|=o_p(1)$, 故

$$I_3=o_p(1). \tag{12}$$

同理推出

$$I_4=o_p(1). \tag{13}$$

由(10)~(13)即得(a).

3° 现在我们证明,诸 C_{iN}^{*} 都是 $o_p\left(\frac{1}{\sqrt{N}}\right)$. 先考虑 C_{1N}^{*}. 由[9]知

$$C_{1N}^{*}=-\frac{\lambda_N}{2}(C_{11N}^{*}-C_{12N}^{*}),$$

其中

$$C_{11N}^{*}=\int_{-\infty}^{\infty}[F_m^{*}(x)-F(x)]^2J''[F(x)]\mathrm{d}F(x),$$

$$C_{12N}^{*}=\frac{1}{m}\int_{-\infty}^{\infty}J'(F(x))\mathrm{d}F_m^{*}(x).$$

因为 $|J''(x)|=2$, 由引理 2 可知

$$\begin{aligned}
\mathrm{E}\,|C_{11N}^{*}|&\leqslant 2\int_{-\infty}^{\infty}\mathrm{E}[F_m^{*}(x)-F(x)]^2\mathrm{d}F(x)\\
&\leqslant 4\int_{-\infty}^{\infty}\mathrm{E}[F_m^{*}(x)-F_m(x)]^2\mathrm{d}F(x)\\
&\quad+4\int_{-\infty}^{\infty}\mathrm{E}[F_m(x)-F(x)]^2\mathrm{d}F(x)\\
&=o\left(\frac{1}{\sqrt{N}}\right)+o\left(\frac{1}{N}\right)\\
&=o\left(\frac{1}{\sqrt{N}}\right).
\end{aligned}$$

因此,对任给的 $\varepsilon>0$, 有

$$P\{|C_{11N}^{*}|\geqslant\varepsilon/\sqrt{N}\}\leqslant\frac{\sqrt{N}}{\varepsilon}\mathrm{E}\,|C_{11N}^{*}|=\frac{1}{\varepsilon}o(1).$$

从而 $C_{11N}^{*}=o_p\left(\frac{1}{\sqrt{N}}\right)$, $C_{12N}^{*}=o_p\left(\frac{1}{\sqrt{N}}\right)$ 由 J'的有界性是显然的,故 $C_{1N}^{*}=o_p\left(\frac{1}{\sqrt{N}}\right)$.

现在考虑 C_{2N}^{*}. 我们有

$$(C_{2N}^{*})^2 = (1-\lambda_N)\iint_{-\infty}^{\infty} [G_n(x)-F(x)][G_n(y)-F(y)]J'[F(x)]J'[F(y)]$$
$$\cdot \mathrm{d}[F_m^{*}(x)-F(x)]\mathrm{d}[F_m^{*}(y)-F(y)]$$
$$=\Big\{2\iint_{x<y}[G_n(x)-F(x)][G_n(y)-F(y)]$$
$$\cdot J'[F(x)]J'[F(y)]\mathrm{d}[F_m^{*}(x)-F(x)]\mathrm{d}[F_m^{*}(y)-F(y)]$$
$$+\frac{1}{m}\int_{-\infty}^{\infty}[G_n(x)-F(x)]^2[J'[F(x)]]^2\mathrm{d}F_m^{*}(x)\Big\}(1-\lambda_N).$$

从而

$$\mathrm{E}\left\{\frac{(C_{2N}^{*})^2}{1-\lambda_N}\,\bigg|\,X_1,\ \cdots,\ X_m;\ Z_1,\ \cdots,\ Z_l\right\}=C_{21N}^{*}+C_{22N}^{*}, \tag{14}$$

其中

$$C_{21N}^{*}=\frac{2}{n}\iint_{\{x<y\}}G(x)[1-G(y)]J'[F(x)]J'[F(y)]\mathrm{d}[F_m^{*}(x)-F(x)]\mathrm{d}[F_m^{*}(x)-F(y)],$$

$$C_{22N}^{*}=\frac{1}{mn}\iint_{\{x<y\}}G(x)[1-G(x)][J'[F(x)]]^2\mathrm{d}F_m^{*}(x).$$

由 J'的有界性知, $C_{22N}^{*}=O\left(\frac{1}{N^2}\right)=o\left(\frac{1}{N}\right)$. 现在考察 C_{21N}^{*}. 我们有

$$n\mathrm{E}C_{21N}^{*}=\iint_{\{x<y\}}U(x,\ y)\mathrm{d}T^{(m)}(x,\ y)-\iint_{\{x<y\}}U(x,\ y)\mathrm{d}F^{(m)}(x)\mathrm{d}F(y)$$
$$-\iint_{\{x<y\}}U(x,\ y)\mathrm{d}F(x)\mathrm{d}F^{(m)}(y)+\iint_{\{x<y\}}U(x,\ y)\mathrm{d}F(x)\mathrm{d}F(y),$$

此处

$$U(x,\ y)=2G(x)[1-G(y)]J'[F(x)]J'[F(y)]$$

为一有界连续函数,故由引理 3 立即推出

$$\lim_{N\to\infty} n\mathrm{E}C_{21N}^{*}=0,$$

即 $\mathrm{E}C_{21N}^{*}=o\left(\frac{1}{N}\right)$. 因此,由(14)式知 $\mathrm{E}[(C_{2N}^{*})^2]=o\left(\frac{1}{N}\right)$. 再由 Чебышев 不等式即得 $C_{2N}^{*}=o_p\left(\frac{1}{\sqrt{N}}\right)$.

现在考虑 C_{3N}^{*}. 由

$$|C_{3N}^{*}|\leqslant\int_{-\infty}^{\infty}[H_N(x)-F(x)]^2\mathrm{d}F_m^{*}(x)$$
$$\leqslant 2\int_{-\infty}^{\infty}[G_n(x)-F(x)]^2\mathrm{d}F_m^{*}(x)+2\int_{-\infty}^{\infty}[F_m^{*}(x)-F(x)]^2\mathrm{d}F_m^{*}(x)$$

$$=2\int_{-\infty}^{\infty}[G_n(x)-F(x)]^2\mathrm{d}F_m^*(x)+2\int_{-\infty}^{\infty}[F_m^*(x)-F(x)]^2\mathrm{d}F(x)$$

$$+2\int_{-\infty}^{\infty}[F_m^*(x)-F(x)]^2\mathrm{d}[F_m^*(x)-F(x)]$$

$$=C_{31N}^*+C_{32N}^*+C_{33N}^*.$$

易见

$$\mathrm{E}\{C_{31N}^* \mid X_1, \cdots, X_m;\ Z_1, \cdots, Z_l\} = 2\int_{-\infty}^{\infty}\mathrm{E}[G_n(x)-F(x)]^2\mathrm{d}F_m^*(x) = O\left(\frac{1}{N}\right),$$

且右边对 $X_1, \cdots, X_m;\ Z_1, \cdots, Z_l$ 为一致的. 故 $C_{31N}^* = O\left(\frac{1}{N}\right)$. 又由引理 2 知

$$\mathrm{E}C_{32N}^* = 2\int_{-\infty}^{\infty}\mathrm{E}[F_m^*(x)-F(x)]^2\mathrm{d}F(x) = o\left(\frac{1}{\sqrt{N}}\right),$$

从而 $C_{32N}^* = o_p\left(\frac{1}{\sqrt{N}}\right)$. 最后考虑 C_{33N}^*. 为此,我们证明关系式

$$\int_{-\infty}^{\infty}[F_m^*(x)-F(x)]^2\mathrm{d}[F_m^*(x)-F(x)] = \frac{1}{3}\sum_{i=1}^{m}\frac{3i-1}{m^3}+\frac{1}{m^2}\sum_{i=1}^{m}F(X_i^*). \tag{15}$$

由此式立得 $C_{33N}^* = O\left(\frac{1}{N}\right)$, 从而证明了 $C_{3N}^* = o_p\left(\frac{1}{\sqrt{N}}\right)$.

为了证明(15)式,以 R 记集合 $\{X_1^*, \cdots, X_m^*\}$,则有

$$\begin{aligned}0 &= \int_{-\infty}^{\infty}\mathrm{d}[F_m^*(x)-F(x)]^3\\ &= \int_{R^c}\mathrm{d}[F_m^*(x)-F(x)]^3+\int_R\mathrm{d}[F_m^*(x)-F(x)]^3\\ &= 3\int_{R^c}[F_m^*(x)-F(x)]^2\mathrm{d}[F_m^*(x)-F(x)]\\ &\quad+\sum_{i=1}^{m}\left[\left(\frac{i}{m}-F(X_i^*)\right)^3-\left(\frac{i-1}{m}-F(X_i^*)\right)^3\right]\\ &= 3\int_{R^c}[F_m^*(x)-F(x)]^2\mathrm{d}[F_m^*(x)-F(x)]\\ &\quad+\sum_{i=1}^{m}\left[\frac{3i^2-3i+1}{m^3}-\frac{6i+3}{m^2}F(X_i^*)+\frac{1}{m}F^2(X_i^*)\right].\end{aligned} \tag{16}$$

又

$$\begin{aligned}&3\int_R[F_m^*(x)-F(x)]^2\mathrm{d}[F_m^*(x)-F(x)]\\ &\quad= 3\,\frac{1}{m}\sum_{i=1}^{m}\left[\frac{i}{m}-F(X_i^*)\right]^2\\ &\quad= \frac{3}{m}\sum_{i=1}^{m}\left[\frac{i^2}{m^2}-2\,\frac{i}{m}F(X_i^*)+F^2(X_i^*)\right],\end{aligned} \tag{17}$$

(16)、(17)两式相加并整理之，即得(15)式.

由 J 及 J' 的有界性立即推得 $C_{4N}^{*}=O\left(\frac{1}{N}\right)$，$C_{5N}^{*}=O\left(\frac{1}{N}\right)$. 这就完全证明了(b).

定理 1 证毕.

注 1. 仔细检查定理证明的过程即可发现，当对称性的要求不成立时，(2)式中的表达式仍有正态的极限分布. 但此极限分布的方差将与 f 有关.

2. 证明定理 1 所用的方法不难推广到其他统计量. 因为由证明过程看出，只须该统计量为 Chernoff-Savage 型且对应的函数 J 具有以下两点性质：

(i) J, J', J''在(0, 1)内有界(J', J''可在个别点不存在)；

(ii) $J\left(\frac{1}{2}-x\right)=J\left(\frac{1}{2}+x\right)$，当 $|x|<\frac{1}{2}$.

在此情形下，上述证明全部有效. 将这一事实用到 Ansari 和 Bradley 的统计量 W，我们将得出如下的定理：

定理 2 设 X_i, Y_j, Z_k 及 X_i^* 的意义同定理 1，F 满足定理 1 的条件. 定义统计量

$$W^{*}=W(X_1^{*},\cdots,X_m^{*};Y_1,\cdots,Y_n),$$

上式右边表示从“修正样本”$\{X_1^{*},\cdots,X_m^{*};Y_1,\cdots,Y_n\}$算出的 Ansari-Bradley 统计量 W 的值，则当 l, m, n 都$\to\infty$而(1)式成立时，对一切 x 有

$$P\left\{\frac{\sqrt{48}\left(W^{*}-\frac{1}{4}mN\right)}{\sqrt{mnN}}<x\right\}\to\Phi(x).$$

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3

具任给精确度的区间估计的存在问题

1　问题和结果

设$(\Omega, \mathscr{F})$为一可测空间,$\mathscr{P}$为定义于其上的一族概率测度. 设 X_1, X_2, $\cdots$为定义于 Ω 而取值于R_k 内的一串随机变量,对任何 $P \in \mathscr{P}$它们是独立同分布的. X_1 在$(R_k, \mathscr{B}_k)$上的分布及分布函数都记为 F_P. 设 $h(P)$为定义在 $\mathscr{P}$上的一有限实值函数,通常 $\mathscr{P}$中的分布由某一距离空间 Θ 上的点θ 确定(不同的 θ 对应不同的 P). 这时我们用 F_θ 及 $h(\theta)$分别记 F_P 及 $h(P)$,设 ε 为一基于$\{X_i\}$的、$h(P)$的一区间估计类. 若对任给 $\delta > 0$ 及 $\alpha > 0$ 存在 ε 中之一估计,致其长度不超过 δ 而置信系数不小于 $1-\alpha$, 则称 $h(P)$在 ε 中可精确估计且记为 $h \in L(\varepsilon)$. 在各种情况下寻找后一事实成立的相应条件是一重要而有趣的问题. Singh 在文献[1]中得出了一些必要条件,在充分条件方面至今只有零散的结果. 本文对于三类最重要的 ε 得出了 $h \in L(\varepsilon)$ 的充分条件,这些足够宽广的条件在许多常见的情况下甚至是充分必要的.

为了叙述我们的结果,先引进如下的定义:

设 $P_i \in \mathscr{P}$, $i = 1, 2$. 令

$$D(P_1, P_2) = \sup_{x \in R_k} | F_{P_1}(x) - F_{P_2}(x) |$$

$$d(P_1, P_2) = \sup_{B \in \mathscr{B}_k} | F_{P_1}(B) - F_{P_2}(B) |$$

这些可看作 P_1, P_2 之间的某种距离. 定义在 $\mathscr{P}$上的函数在这些距离下的连续性及一致连续性的意义是明显的. 由于 $D(P_1, P_2) \leqslant d(P_1, P_2)$, 由在 D 下的连续性推出在 d 下的连续性,反过来不必对.

以 ε_1, ε_2, ε_3 分别记一切基于固定样本、纯序贯样本及二阶段样本的估计的类,则有

定理 1　$h \in L(\varepsilon_1)$ 的充分条件为 h 为 D 一致连续的.

定理 2　$h \in L(\varepsilon_2)$ 的充分条件为 h 为 D 连续的.

为了叙述定理 3,还要引进一记号. 设 h 为 D 连续于 $\mathscr{P}$上,则对任何 $P \in \mathscr{P}$及 $\varepsilon > 0$ 存在 $\eta > 0$ 致当 $D(P, P') \leqslant \eta$时有 $| h(P) - h(P') | \leqslant \varepsilon$. 一切满足这一条件的 η 的上确界记为 $\gamma(P, \varepsilon)$.

定理 3　$h \in L(\varepsilon_3)$ 的充分条件为存在 $c > 0$ 使对 $\mathscr{P}$之任何子集 $\mathscr{P}^*$ 其在距离 D 意义下的

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直径不超过 c 者,恒有 $\inf\limits_{P\in\mathscr{P}^*}\gamma(P,\epsilon)>0$.

在文献[2]中曾指出一类重要的情况,在其中 D, d 两种连续性等价(但在文献[2]中并未提及 D, d 连续性的概念). 将这一点与文献[1]中的结果结合即得出:在这些重要情况下,以上定理 1 和定理 2 中的条件都是充分且必要的(但同样结论对定理 3 不适用).

2 定理的证明

作为例子,我们在此给出定理 2 的证明. 对任何 n,以 S_n 记 $X_1,\cdots,X_n$ 的经验分布函数. 设 h 在 $\mathscr{P}$ 上 D 连续而要得出其置信系数不小于 $1-\alpha$、而长不超过 δ 的序贯区间估计. 取一串数 $\{\alpha_i\}_{i=1}^{\infty}$, 致 $\alpha_i>0$ 且 $\sum\limits_{i=1}^{\infty}\alpha_i\leqslant\alpha$. 由文献[3]中的定理可知,存在一串 $\{c_i\}$, $c_1>c_2>\cdots\downarrow 0$, 使对任何 i 存在 n_i 充分大,致(对一切 $P\in\mathscr{P}$ 同时成立)

$$P_P\{D(S_{n_i},F_P)>c_{n_i}\}\leqslant\alpha_i,\ i=1,2,\cdots$$

且显然可令 $n_1<n_2<n_3<\cdots$.

现在定义一抽样程序 S 和一区间估计 $[T_1,T_2]$ 如下:首先观测 $X_1,\cdots,X_{n_1}$,考虑集合

$$A(X_1,\cdots,X_{n_1})=\{P:P\in\mathscr{P},D(S_{n_1},F_P)\leqslant c_{n_1}\}$$

$$A^*(X_1,\cdots,X_{n_1})=\{h(P):P\in A(X_1,\cdots,X_{n_1})\}$$

且以 $\overline{A^*}(X_1,\cdots,X_{n_1})$ 记包含 $A^*(X_1,\cdots,X_{n_1})$ 的最小闭区间. 若 $A(X_1,\cdots,X_{n_1})\neq\varnothing$ ($\varnothing$ 表示空集,下同)且 $\overline{A^*}(X_1,\cdots,X_{n_1})$ 之长 $\leqslant\delta$, 则停止抽样而取 $[T_1(x),T_2(x)]=\overline{A^*}(X_1,\cdots,X_{n_1})$. 若不然,则继续观测 $X_{n_1+1},\cdots,X_{n_2}$. 一般地,若在观测到 X_{n_i} 为止时仍不能停止,则继续观测 $X_{n_i+1},\cdots,X_{n_{i+1}}$ 而考虑集合 $A(X_1,\cdots,X_{n_{i+1}})$, $A^*(X_1,\cdots,X_{n_{i+1}})$ 及 $\overline{A^*}(X_1,\cdots,X_{n_{i+1}})$ (定义与上类似). 若 $A(X_1,\cdots,X_{n_{i+1}})\neq\varnothing$ 且 $\overline{A^*}(X_1,\cdots,X_{n_{i+1}})$ 之长 $\leqslant\delta$, 则停止抽样而取 $[T_1(x),T_2(x)]=\overline{A^*}(X_1,\cdots,X_{n_{i+1}})$, 否则继续观测 $X_{n_{i+1}+1},\cdots,X_{n_{i+2}}$. 我们证明,这样作出的序贯区间估计满足所要的条件.

首先必须证明序贯抽样程序 S 是封闭的,换句话说,若以 N 记样本大小函数,要证明对任何 $P_0\in\mathscr{P}$ 有 $\lim\limits_{n\to\infty}P_{P_0}\{N>n\}=0$. 由于 h 为 D 连续,对给定的 $\delta>0$ 存在(与 P_0 有关的) $\eta>0$ 致当 $D(P,P_0)\leqslant\eta$ 时有 $|h(P)-h(P_0)|\leqslant\delta/2$. 取 i_0 充分大致 $c_{n_i}\leqslant\eta$ 当 $i\geqslant i_0$, 则当 $i\geqslant i_0$ 时若 $P_0\in A(X_1,\cdots,X_{n_i})$, 其中任一 P 与 P_0 的 D 距离必 $\leqslant\eta$. 因而由 η 的定义知 $\overline{A^*}(X_1,\cdots,X_{n_i})$ 之长必 $\leqslant\delta$. 因此当 $i\geqslant i_0$ 时有

$$\{N>n_i\}\subset\{P_0\overline{\in}A(X_1,\cdots,X_{n_i})\}=\{D(S_{n_i},P_0)>c_{n_i}\}$$

从而 $P_{P_0}\{N>n_i\}\leqslant P_{P_0}\{D(S_{n_i},P_0)>c_{n_i}\}\leqslant\alpha_i\to 0$ 当 $i\to\infty$, 这就证明了程序 S 的封闭性.

现在来考虑这一区间估计的置信系数. 由程序 S 的作法可知

$$\{h(P_0)\in[T_1,T_2]\}\supset\{N<\infty\}\prod_{i=1}^{\infty}\{D(S_{n_i},F_{P_0})\leqslant c_{n_i}\}$$

因此由上面已证的 $P_{P_0}(N<\infty)=1$ 可知

$$\begin{aligned}P_{P_0}\{h(P_0)\in[T_1,T_2]\} &= P_{P_0}\left\{\prod_{i=1}^{\infty}D(S_{n_i},F_{P_0})\leqslant c_{n_i}\right\}\\ &\geqslant 1-\sum_{i=1}^{\infty}P_{P_0}\{D(S_{n_i},F_{P_0})>c_{n_i}\}\\ &\geqslant 1-\sum_{i=1}^{\infty}\alpha_i\\ &\geqslant 1-\alpha\end{aligned}$$

对任何 $P_0\in\mathscr{P}$, 因而此区间估计的置信系数不小于 $1-\alpha$. 定理完全证明了.

3 应用举例

本节的目的是举例说明上述诸定理的应用. 为节省篇幅计,在各例中关于相应条件的确适合的验证(在有些场合下颇复杂)概行略去.

例 1 Poisson 分布族 $\mathscr{P}=\{P_\lambda,\lambda>0\}$. 用文献[1]中的结果易证函数 $h_1(\lambda)\equiv\lambda$ 不属于 $L(\varepsilon_1)$. 若令 $h_2(\lambda)\equiv\log\lambda$, 则根据定理 1 可证明 $h_2\in L(\varepsilon_1)$. 对 $h_3(\lambda)\equiv \mathrm{e}^{-\lambda}\sin\mathrm{e}^{\lambda}$ 来说有同样的结果. 后一情况尤为有趣,因为易见对任何自然数 n,基于 $X_1,\cdots,X_n$ 的 h_3 的无偏估计必存在,但任何无偏估计的方差不能为 λ 的有界函数,盖由 Cramer-Rao 不等式可知,h_3 的任何无偏估计(基于 $X_1,\cdots,X_n$)的方差 $\mathrm{D}(\lambda)\geqslant\frac{\lambda}{n}[h_3'(\lambda)]^2=\frac{\lambda}{n}(\cos\mathrm{e}^{\lambda}-\mathrm{e}^{-\lambda}\sin\mathrm{e}^{\lambda})^2$. 因此,一估计问题可能在区间估计范围内有满意的解决,而在点估计范围内则否.

例 2 设 $0<p<1$, p 为一已知数,而 $\mathscr{P}$ 为一切其 p 分位点唯一的一维分布族,而 $h(P)=P$ 的 p 分位点($P\in\mathscr{P}$). 容易证明 h 在 $\mathscr{P}$ 上为 D 连续的,因此根据定理 2 得知 $h\in L(\varepsilon_2)$. 这一结果首先由 Farrell 在文献[4]中证明过.

例 3 以 $\mathscr{P}$ 记一切其均值存在有限的一维分布族,而 $h(P)=P$ 的均值(当 $P\in\mathscr{P}$). 已经证明(见文献[5]): $h\,\overline{\in}\,L(\varepsilon_2)$, 但若对 $\mathscr{P}$ 作一些限制,例如,取

$$A_{cK\gamma}=\{\text{一维分布 }F:1-F(x)+F(-x)\leqslant c\cdot x^{-\gamma}\text{ 当 }x>K\}$$

其中 $c>0$, $K>0$, $\gamma>1$ 皆为常数,而定义

$$\mathscr{P}_{cK\gamma}=\{\text{一维分布 }F:F(x)\equiv\widetilde{F}(x+a),\text{其中 }\widetilde{F}\in A_{cK\gamma},\text{而 }-\infty<a<\infty\}$$

则易见 h 在 $\mathscr{P}_{cK\gamma}$ 上为 D 连续的,因此由定理 2 可知这时 $h\in L(\varepsilon_2)$.

又若 g 为 $(-\infty,\infty)$ 上任一有界实连续函数,$\mathscr{P}$ 为一切一维分布的族,而 $h(P)=\int_{-\infty}^{\infty}g(x)\mathrm{d}P(x)$. 这时由定理 2 易证 $h\in L(\varepsilon_2)$.

例 4 设 $\mathscr{P}=\{P_\theta,0<\theta<1\}$, 而 $P_\theta\{X_1=1\}=\theta$, $P_\theta\{X_1=0\}=1-\theta$, $h(\theta)=\log\theta$. 在文献[1]中曾证明对任何 m,在 m 阶段抽样下 $h(\theta)$ 的具任给精确度的区间估计不存在. 但易证 $h(\theta)$ 为 D 连续,因此由定理 2 知 $h\in L(\varepsilon_2)$.

例 5 设 F 为一已知的一维连续分布,而 $\mathscr{P}$ 为转移一刻度分布族 $\mathscr{P}=\left\{F\left(\frac{x-\theta}{\sigma}\right),\sigma>0,-\infty<\theta<\infty\right\}$, $h(P)=g(\theta,\sigma)$, 其中 g 为 $\{-\infty<\theta<\infty,\sigma>0\}$ 上的实连续函数. 可以

验证定理 3 的条件在此成立,因而由定理 3 知 $h \in L(\varepsilon_3)$. 作为特例,这一事实包括和推广了许多目前已知的结果. 又本文作者在另一工作中证明了:在本例中第一阶段的样本大小甚至可取为 1.

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4

线性模型中误差方差估计的 Berry-Esseen 界限

摘要 在线性模型中，通常用残差平方和（除以适当的自由度）来估计模型中随机误差的方差，本文在误差独立但不必同分布的情况下，证明了在一定条件下，这种估计量经过规则化后，其分布以 $O\left(\frac{1}{\sqrt{n}}\right)$ 的理想速度收敛于标准正态分布.

1 引言和主要结果

早在 1945 年，即 Berry-Esseen 理论刚建立不久，许宝騄教授在文献[1]中证明了下面的结果：设 $X_1, X_2, \cdots, X_n, \cdots$ 为取自某总体的独立随机样本，假定

$$0 < \mathrm{Var}(X_1) = \sigma^2 < \infty,\ \mathrm{E}(X_1^6) < \infty,\ \mathrm{Var}\{(X - \mathrm{E}X)^2\} > 0. \tag{1}$$

记 $s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$, $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n}X_i$，并以 $G_n(x)$ 和 $\Phi(x)$ 分别记

$$(s_n^2 - \sigma^2)/\sqrt{\mathrm{Var}(s_n^2)}$$

和 $N(0, 1)$ 的分布函数，则存在与 n 无关的常数 C，致

$$\sup_x | G_n(x) - \Phi(x) | \leqslant C/\sqrt{n}.$$

这是非独立和形式的统计量的收敛速度最早的一个重要结果. 众所周知，虽则常见的重要统计量大都有渐近正态性，但在非独立和形式的场合，要建立其收敛速度的 Berry-Esseen 界限，则并非易事，近年来在这方面颇有些进展.

许宝騄教授考虑的模型是一般线性回归模型的一个很特殊的例子. 本文将在两个方面推广和改进文献[1]中的结果. 一是我们将讨论一般线性模型，另一是观察值同分布的假定可以免除（这在文献[1]中是一个实质性条件）. 因此在一定意义上说，本文可视为许教授上述工作的继续，谨以此表达作者对我国这位杰出的数理统计学者的纪念.

考虑线性回归模型

$$Y_i = x_i'\beta + e_i,\ i = 1, \cdots, n, \cdots. \tag{2}$$

试验点列 $\{x_i\}$ 是一串已知的 p 维向量，β 是未知的回归系数向量，$\{e_i\}$ 是一串独立的试验误差，

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满足条件：

$$\mathrm{E}e_i = 0,\ \mathrm{Var}(e_i) = \sigma^2,\ 0 < \sigma^2 < \infty,\ i = 1, 2, \cdots, \tag{3}$$

误差方差 σ^2 是模型的一个重要未知参数，若记

$$X_n = (x_1 \mid \cdots \mid x_n),\ r_n = rk(X_n),\ Y_{(n)} = (Y_1, \cdots, Y_n)',\ e_{(n)} = (e_1, \cdots, e_n)',$$

则在(2)式的前 n 次试验结果的基础上，最小二乘法规定以

$$\begin{aligned}\hat{\sigma}_n^2 &= \frac{1}{n-r_n} Y'_{(n)}(I - X_n'(X_n X_n')^- X_n) Y_{(n)} \\ &= \frac{1}{n-r_n} e'_{(n)}(I - X_n'(X_n X_n')^- X_n) e_{(n)}\end{aligned}$$

去估计 σ^2. 有 $\mathrm{E}(\hat{\sigma}_n^2) = \sigma^2$. 如所周知，方阵 $X_n'(X_n X_n')^- X_n$ 为对称、幂等且有秩 r_n，故上式可写为

$$\hat{\sigma}_n^2 = \frac{1}{n-r_n}\left\{\sum_{i=1}^n e_i^2 - \sum_{j=1}^{r_n}\left(\sum_{k=1}^n a_{njk} e_k\right)^2\right\}. \tag{4}$$

这里 $\sum_{k=1}^n a_{njk}^2 = 1,\ j = 1, \cdots, r_n$. 当 n 充分大时 r_n 稳定到某个值 $r \leqslant p$. 因此，以后写 r 代替(4)式中的 r_n.

从(4)式出发，只要对 e_i 的矩作适当的假定，就不难证明 $\hat{\sigma}_n^2$ 的渐近正态性. 本文的目的是建立 $\hat{\sigma}_n^2$(经过规则化)的分布收敛于 Φ 的速度的 Berry-Esseen 界限. 主要结果为：

定理 1 在上述记号下，设 $e_1, e_2, \cdots$ 相互独立，满足条件(3)式，且存在常数 $D_1 < \infty$, $D_2 > 0$, $\epsilon > 0$, 使

$$\frac{1}{n}\sum_{i=1}^n \mathrm{E}e_i^6 \leqslant D_1, \tag{5}$$

$$\frac{1}{n}\sum_{i=1}^n d_i^2 \geqslant D_2,\ d_i^2 = \mathrm{Var}(e_i^2) = \mathrm{E}e_i^4 - \sigma^4, \tag{6}$$

$$\max_{1 \leqslant i \leqslant n} \int_{|x| \leqslant n^{1/4}} x^6 \mathrm{d}F_i = O(n^{1/4-\epsilon}),\ F_i \text{ 为 } e_i \text{ 的分布}. \tag{7}$$

以 G_n 记 $(\hat{\sigma}_2^n - \sigma^2)/\sqrt{\mathrm{Var}(\hat{\sigma}_n^2)}$ 的分布，则存在与 n 无关的常数 C，致

$$\| G_n - \Phi \| = \sup_x | G_n(x) - \Phi(x) | \leqslant C/\sqrt{n}. \tag{8}$$

特别，当 $e_1, e_2, \cdots$ 独立同分布，而 $\mathrm{E}e_1^6 < \infty$, $\mathrm{Var}(e_1^2) > 0$ 时，(5)～(7)式成立. 条件(5), (6)在某种程度上是必要的. 作者猜测，去掉条件(7)后，定理结论应当仍成立，但迄今未能证明这一点. 显然，若

$$\mathrm{E}e_n^6 = O(n^{1/4-\epsilon}), \tag{7'}$$

则(7)式成立. 但很容易举出满足(7)式而并非(7′)式的例子. 另一方面，我们注意到定理 1 的条件与试验点列完全无关. 如果对它作一定的要求，就可免除条件(7)式. 结果如下：

定理 2 假设条件(5),(6)式成立,且存在 $\epsilon > 0$, 致

$$\max_{1\leqslant k\leqslant n} | a_{njk} | = O(n^{-3/20-\epsilon}), \ 1\leqslant j\leqslant r, \tag{9}$$

则(8)式成立.

特别,对许教授研究的情况,有 $r=1$, $a_{n1k}=\frac{1}{\sqrt{n}}$, $1\leqslant k\leqslant n$, 这时(9)式成立.这样,可将许教授的条件(1)式削弱为:X_1, X_2, …独立,

$$\sum_{i=1}^n \mathrm{E}(X_i-\mathrm{E}X_i)^6 \leqslant nD_1, \ \sum_{i=1}^n \mathrm{Var}\{(X_i-\mathrm{E}X_i)^2\} \geqslant nD_2,$$

此处 $D_1<\infty$, $D_2>0$. 还可举出某些其他适合(9)式的重要特例.特别,若假定试验区域有界,则只需

$$\left(\sum_{i=1}^n x_i x_i'\right)^{-1} = O(n^{-3/20-\epsilon}) \tag{9'}$$

就可推出(9)式.为了使回归系数向量 β 的最小二乘估计有良好的大样本性质,需要 $\left(\sum_{i=1}^n x_i x_i'\right)^{-1}$ 趋于 0 的速度有一定的数量级,且这数量级愈高愈好.这时(9′)式一般都能满足.比方说,对逼近于所谓"D 最优设计"的试验点序列$\{x_i\}$而言,有 $\left(\sum_{i=1}^n x_i x_i'\right)^{-1} = O\left(\frac{1}{n}\right)$, 这时满足条件(9′)式,因而(9)式当然满足.然而,从纯理论的角度看,是否能够和怎样在没有附加条件(7)或(9)式之下去证明(8)式,是一个很有兴趣的问题.

2 若干引理

由于本文定理的证明过程很复杂,我们将证明中一些主要之点提出来先加以讨论.

引理 1 设 e_1, e_2, …相互独立,满足条件(3), (5), (6)式.定义

$$\hat{e}_j = \begin{cases} e_j, & \text{当 } |e_j| \leqslant n^{1/4}, \\ 0, & \text{当 } |e_j| > n^{1/4}, \end{cases} \tag{10}$$

$$\xi_j = \hat{e}_j^2 - \mathrm{E}(\hat{e}_j^2), \ j=1, \cdots, n$$

($\hat{e}_j$, ξ_j 实际上与 n 有关.为简化书写,记号中未反映这一点).又记

$$nB_n^2 = \sum_{j=1}^n d_j^2, \ \sigma_{nj}^2 = \mathrm{Var}(\xi_j), \ \alpha_{nj} = \mathrm{E}|\xi_j|^3, \ 1\leqslant j\leqslant n,$$

$$n\widetilde{B}_n^2 = \sum_{j=1}^n \sigma_{nj}^2, \ n\alpha_n = \sum_{j=1}^n \alpha_{nj}, \tag{11}$$

则当 n 充分大时,有:

(1) 存在与 n 无关的 $D_3<\infty$, 使

$$B_n^2 \leqslant D_3, \ n=1, 2, \cdots. \tag{12}$$

(2) 存在与 n 无关的常数 C,使

$$|B_n^2-\widetilde{B}_n^2|\leqslant C/\sqrt{n}. \tag{13}$$

(3) 存在与 n 无关的常数 $D_1^\circ<\infty$, $D_2^\circ>0$, 使

$$\widetilde{B}_n^2\geqslant D_2^\circ,\ \alpha_n\leqslant D_1^\circ. \tag{14}$$

(4) 存在与 n 无关的常数 $b_i>0$, $i=1,2,3$, 使集合

$$A_n=\{j:\sigma_{nj}^2\geqslant b_2,\ \alpha_{nj}\leqslant b_3,\ 1\leqslant j\leqslant n\} \tag{15}$$

包含的元素个数不少于 b_1n.

证 (1)~(3)证明很容易. 首先

$$nB_n^2\leqslant\sum_{j=1}^n \mathrm{E}e_j^4\leqslant\sum_{j=1}^n \mathrm{E}^{2/3}(e_j^6)\leqslant\sum_{j=1}^n\{1+\mathrm{E}e_j^6\}\leqslant n(1+D_1),$$

即(12)式,其次当 n 充分大时,

$$\begin{aligned}|nB_n^2-n\widetilde{B}_n^2|&\leqslant\sum_{j=1}^n|\mathrm{E}e_j^4-\mathrm{E}\,\hat{e}_j^4|+\sum_{j=1}^n|\sigma^4-\mathrm{E}^2(\hat{e}_j^2)|\\&\leqslant\sum_{j=1}^n\int_{|x|\geqslant n^{1/4}}x^4\mathrm{d}F_j+2\sigma^2\sum_{j=1}^n\int_{|x|\geqslant n^{1/4}}x^2\mathrm{d}F_j,\\&\leqslant\frac{1}{\sqrt{n}}\sum_{j=1}^n\mathrm{E}e_j^6+\frac{2\sigma^2}{n}\sum_{j=1}^n\mathrm{E}e_j^6\\&\leqslant\sqrt{n}D_1+2\sigma^2D_1\leqslant2\sqrt{n}D_1,\end{aligned}$$

即为(13)式. (14)式前一式由(13)式推出,后一式由(5)式及 $\alpha_{nj}\leqslant 8\mathrm{E}e_j^6$ 推出.

现证(4). 先证存在 $\varepsilon_1>0$, 使当 n 充分大时,集合 $A_{n1}=\{j:1\leqslant j\leqslant n,\ \sigma_{nj}^2\geqslant D_2^\circ/2\}$ 的元素个数不少于 $\varepsilon_1 n$. 事实上,若此事不真,则将存在 $1\leqslant n_1<n_2<\cdots$, 使集合 A_{n_k1} 的元素个数 $m_k\leqslant n_k/k$. 由(14)式,有 $\sum\limits_{j\in A_{n_k1}}\sigma_{n_kj}^2\geqslant n_kD_2^\circ/2$. 又因 $\alpha_{nj}\geqslant\sigma_{nj}^3$, 有

$$\sum_{j\in A_{n_k1}}\alpha_{n_kj}\geqslant\sum_{j\in A_{n_k1}}\sigma_{n_kj}^3\geqslant\Big(\sum_{j\in A_{n_k1}}\sigma_{n_kj}^2\Big)^{3/2}\Big/\sqrt{m_k}\geqslant(D_2^\circ/2)^{3/2}n_k^{3/2}/\sqrt{m_k}.$$

于是 $\alpha_{n_k}\geqslant\dfrac{1}{n_k}\sum\limits_{j\in A_{n_k1}}\alpha_{n_kj}\geqslant(D_2^\circ/2)^{3/2}\sqrt{n_k/m_k}\geqslant(D_2^\circ/2)^{3/2}\sqrt{k}\to\infty$. 这与(14)式矛盾,因而证明了所述事实. 另一方面,由(14)式知,对任给 $\varepsilon_2>0$, 当 n 充分大时集合 $A_{n2}=\{j:1\leqslant j\leqslant n,\ \alpha_{nj}\leqslant D_1^\circ/\varepsilon_2\}$ 的元素个数不少于 $(1-\varepsilon_2)n$. 取 $\varepsilon_2=\varepsilon_1/3$, 即知当 n 充分大时,集合 $A_{n1}\cap A_{n2}$ 的元素个数不少于 $\varepsilon_1n/2$. 取 $b_1=\varepsilon_1/2$, $b_2=D_2^\circ/2$, $b_3=3D_1^\circ/\varepsilon_1$, 即得(4)证明.

下面的引理是文献[2]中定理 2 的显然推广,故证明从略.

引理 2 设 $\mathscr{F}$ 为一族随机变量,存在常数 $a>0$ 及 $b<\infty$, 使对任何 $X\in\mathscr{F}$, 有 $\mathrm{Var}(X)\geqslant a$, $\mathrm{E}|X|^3\leqslant b$, 则存在 $\eta>0$, $\delta>0$. 若 $f(t)$ 为 $\mathscr{F}$ 中任一变量 X 的特征函数,则有

$$|f(t)|\leqslant1-\delta t^2,\ 当\ |t|\leqslant\eta. \tag{16}$$

在以下,记号 i 专用于表示 $\sqrt{-1}$.

引理 3 设 $e_1, \cdots, e_n, \cdots$ 独立,满足条件(3),(5),(6)式,并令

$$e_j^* = \hat{e}_j - \mathrm{E}(\hat{e}_j),\ 1 \leqslant j \leqslant n,\ S_n = \sum_{j=1}^{n} \xi_j / \sqrt{n} B_n. \tag{17}$$

$\hat{e}_j$, ξ_j 的定义见(10)式. 设 a_{nj}, b_{nj}, $1 \leqslant j \leqslant n$, 都是常数,满足条件

$$\sum_{j=1}^{n} a_{nj}^2 = \sum_{j=1}^{n} b_{nj}^2 = 1.$$

又设 $1 \leqslant j_{n1} < \cdots < j_{nm_n} \leqslant n$, 并用 C 表示任一与 n, a_{nj}, b_{nj}, j_{ns} 等无关的常数,每次出现时其值可不同,则存在与 n, a_{nj}, b_{nj}, j_{ns} 等无关的 $\eta > 0$, $\sigma > 0$. 当 n 充分大,而

$$|t| \leqslant \sqrt{n}\eta$$

时,有

$$\left| \mathrm{E}\left\{ \mathrm{e}^{\mathrm{i}tS_n} \left(\sum_{s=1}^{m_n} a_{nj_{ns}} e_{j_{ns}}^* \right)^2 \right\} \right| \leqslant C(1+t^2)(1-\delta t^2/n)^{[b_1 n]-2}, \tag{18}$$

$$\left| \mathrm{E}\left\{ \mathrm{e}^{\mathrm{i}tS_n} \left(\sum_{s=1}^{m_n} a_{nj_{ns}} e_{j_{ns}}^* \right)^2 \left(\sum_{s=1}^{m_n} b_{nj_{ns}} e_{j_{ns}}^* \right)^2 \right\} \right| \leqslant C \sum_{j=0}^{4} |t|^j (1-\delta t^2/n)^{[b_1 n]-4}. \tag{19}$$

证 为简化起见,设 $m_n = n$ (这时 $j_{ns} = s$),检查以下的证明会看出这无损于普遍性. 记

$$g_j = \mathrm{E}\{\exp(\mathrm{i}t\xi_j / \sqrt{n} B_n) e_j^*\},\ 1 \leqslant j \leqslant n.$$

因为 $\mathrm{E}e_j^* = 0$, 有

$$g_j = \mathrm{E}\{(\exp(\mathrm{i}t\xi_j / \sqrt{n} B_n) - 1) e_j^*\} = \frac{\mathrm{i}t}{\sqrt{n} B_n} \theta E(|\xi_j e_j^*|), \tag{20}$$

此处 $|\theta| \leqslant 1$. 但

$$\begin{aligned} \mathrm{E}^2(|\xi_j e_j^*|) &\leqslant \mathrm{E}\xi_j^2 \mathrm{E}(e_j^*)^2 \leqslant \sigma^2 \sigma_{nj}^2 \leqslant \sigma^2 \left(d_j^2 + 2\sigma^2 \int_{|x| \geqslant n^{1/4}} x^2 \mathrm{d}F_j \right) \\ &\leqslant \sigma^2 \left(d_j^2 + 2\sigma^2 \frac{1}{n} E e_j^6 \right) \leqslant \sigma^2 (d_j^2 + 2\sigma^2 D_1) \\ &\leqslant \sigma^2 (d_j + \rho)^2, \quad \rho = \sqrt{2\sigma^2 D_1}. \end{aligned}$$

由此式及(20)式,得

$$|g_j| \leqslant |t| \sigma(d_j + \rho) / \sqrt{n} B_n,\ 1 \leqslant j \leqslant n. \tag{21}$$

以 $f_{nj}(t)$ 记 ξ_j 的特征函数,则有

$$|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n} e_j^* e_h^*)| \leqslant \prod_{\substack{u=1 \\ u \neq j,h}}^{n} |f_{nu}(t/\sqrt{n} B_n)| \cdot |g_j g_h|,\ 1 \leqslant j \neq h \leqslant n. \tag{22}$$

依引理 1,当 n 充分大时,由(15)式所定义的集 A_n 至少有 $[b_1 n]$ 个元素. 依引理 2,存在 $\eta_1 > 0$,

$\delta_1 > 0$，使当 n 充分大时，

$$| f_{nj}(t) | \leqslant 1 - \delta_1 t^2,\ | t | \leqslant \eta_1,\ j \in A_n. \tag{23}$$

由(6)，(12)式，$D_2 \leqslant B_n^2 \leqslant D_3$. 取 $\eta = \eta_1 \sqrt{D_2}$，$\delta = \delta_1 / D_3$，将有

$$| f_{nj}(t/\sqrt{n}B_n) | \leqslant 1 - \delta t^2/n,\ | t | \leqslant \sqrt{n}\eta,\ j \in A_n. \tag{24}$$

由于 A_n 中至少有$[b_1 n]$个元素，由(22)，(24)式得

$$| \mathrm{E}(\mathrm{e}^{itS_n} e_j^* e_h^*) | \leqslant (1 - \delta t^2/n)^{[b_1 n]-2} | g_j g_h |,\ 1 \leqslant j \neq h \leqslant n. \tag{25}$$

又显然

$$| \mathrm{E}(\mathrm{e}^{itS_n} e_j^{*2}) | \leqslant (1 - \delta t^2/n)^{[b_1 n]-1} \sigma^2 \leqslant (1 - \delta t^2/n)^{[b_1 n]-2} \sigma^2,\ 1 \leqslant j \leqslant n, \tag{26}$$

(25)，(26)式都在 n 充分大而 $| t | \leqslant \sqrt{n}\eta$ 时成立. 现有

$$\begin{aligned}
&\left| \mathrm{E}\left\{ \mathrm{e}^{itS_n} \Big(\sum_{j=1}^n a_{nj} e_j^* \Big)^2 \right\} \right| \\
&\leqslant \sum_{j=1}^n a_{nj}^2 | \mathrm{E}(\mathrm{e}^{itS_n} e_j^{*2}) | + \sum_{\substack{u \neq v \\ 1}}^n | a_{nu} a_{nv} | | \mathrm{E}(\mathrm{e}^{itS_n} e_u^* e_v^*) | \\
&\leqslant \sigma^2 (1 - \delta t^2/n)^{[b_1 n]-2} + (1 - \delta t^2/n)^{[b_1 n]-2} \Big(\sum_{j=1}^n | a_{nj} g_j | \Big)^2.
\end{aligned} \tag{27}$$

由(21)式，

$$\begin{aligned}
\Big(\sum_{j=1}^n | a_{nj} g_j | \Big)^2 &\leqslant \sum_{j=1}^n a_{nj}^2 \sum_{j=1}^n | g_j^2 | = \sum_{j=1}^n | g_j^2 | \leqslant \frac{\sigma^2 t^2}{n B_n^2} \sum_{j=1}^n 2(d_j^2 + \rho^2) \\
&= \frac{\sigma^2 t^2}{n B_n^2} (2n B_n^2 + n\rho^2) = \sigma^2 t^2 (1 + \rho^2 / B_n^2) \\
&\leqslant \sigma^2 (1 + \rho^2 / D_2) t^2 \leqslant C t^2,
\end{aligned}$$

代入(27)式得到(18)式. (19)式的证明可用类似的办法得到，只需用上述办法处理

$$\mathrm{E}(\mathrm{e}^{itS_n} e_{j_1}^{*4}),\ \mathrm{E}(\mathrm{e}^{itS_n} e_{j_1}^{*3} e_{j_2}^*),\ \mathrm{E}(\mathrm{e}^{itS_n} e_{j_1}^{*2} e_{j_2}^{*2}),\ \mathrm{E}(\mathrm{e}^{itS_n} e_{j_1}^{*2} e_{j_2}^* e_{j_3}^*),\ \mathrm{E}(\mathrm{e}^{itS_n} e_{j_1}^* e_{j_2}^* e_{j_3}^* e_{j_4}^*)$$

这五种类型的项即可. 引理 3 证毕.

引理 4 设 $e_1, e_2, \cdots$独立，满足条件(3)式，e_j 的分布为 F_j. 又$\{a_{nj}\}$为常数，

$$\sum_{j=1}^n a_{nj}^2 \leqslant 1,\ n = 1, 2, \cdots$$

设条件(7)式满足，或条件(5)式满足，且存在 $\varepsilon_1 > 0$，使

$$\max_{1 \leqslant j \leqslant n} | a_{nj} | = O(n^{-3/20-\varepsilon_1}), \tag{28}$$

则存在常数 $\varepsilon_2 > 0$，使

$$\mathrm{E}\Big(\sum_{j=1}^n a_{nj} e_j^* \Big)^6 = O(n^{1/4-\varepsilon_2}). \tag{29}$$

证 先设条件(7)式成立.与 $\mathrm{E}\mid e_j^*\mid^m\leqslant 2^m\mathrm{E}\mid \hat{e}_j\mid^m$ 结合,知 $\sup\limits_{1\leqslant j\leqslant n}\mathrm{E}\mid e_j^*\mid^m=O(n^{(1/4-\varepsilon)m/6})$, $m=3,4,6$. 再注意到 $\mathrm{E}e_j^*=0$, $\mathrm{E}e_j^{*2}\leqslant\sigma^2$, $\sum\limits_{j=1}^n\mid a_{nj}\mid^l\leqslant 1$,当 $l\geqslant 2$,以及

$$
\begin{aligned}
\mathrm{E}\Big(\sum_{j=1}^n a_{nj}e_j^*\Big)^6 &\leqslant \sum_{j=1}^n a_{nj}^6\mathrm{E}e_j^{*6}+15\sum_{j=1}^n a_{nj}^4\mathrm{E}e_j^{*4}\sum_{j=1}^n a_{nj}^2\mathrm{E}e_j^{*2}\\
&\quad+20\Big(\sum_{j=1}^n a_{nj}^3\mathrm{E}e_j^{*3}\Big)^2+90\Big(\sum_{j=1}^n a_{nj}^2\mathrm{E}e_j^{*2}\Big)^3\\
&=I_1+I_2+I_3+I_4,
\end{aligned}\tag{30}
$$

便不难得到(29)式,其中 $\varepsilon_2=\varepsilon$.

现设条件(5)和(28)式成立.由(5)式知, $\sum\limits_{j=1}^n(\mathrm{E}e_j^{*6})^2=O(n^2)$. 故

$$
\begin{aligned}
I_1^2 &\leqslant \sum_{j=1}^n a_{nj}^{12}\sum_{j=1}^n(\mathrm{E}e_j^{*6})^2\leqslant\max_{1\leqslant j\leqslant n}\mid a_{nj}\mid^{10}O(n^2)\\
&=O(n^{-3/2-10\varepsilon_1})O(n^2)=O(n^{1/2-10\varepsilon_1}),
\end{aligned}
$$

故 $I_1=O(n^{1/4-5\varepsilon_1})$. 又因 $(\mathrm{E}e_j^{*4})^2\leqslant\mathrm{E}e_j^{*2}\mathrm{E}e_j^{*6}\leqslant\sigma^2\mathrm{E}e_j^{*6}$,有

$$
\begin{aligned}
I_2^2 &\leqslant \sum_{j=1}^n a_{nj}^8\sum_{j=1}^n(\mathrm{E}e_j^{*4})^2(15\sigma^2)^2\\
&\leqslant 225\sigma^6\max_{1\leqslant j\leqslant n}\mid a_{nj}\mid^6\sum_{j=1}^n\mathrm{E}e_j^{*6}\\
&=O(n^{-18/20-6\varepsilon_1})O(n)=O(n^{1/10-6\varepsilon_1}).
\end{aligned}
$$

故 $I_2=O(n^{1/20-3\varepsilon_1})$. 对 I_3,则注意 $\mid e_j^*\mid\leqslant 2n^{1/4}$ 且 $\mathrm{E}e_j^{*2}\leqslant\sigma^2$,知 $\max\limits_{1\leqslant j\leqslant n}\mid\mathrm{E}e_j^{*3}\mid=O(n^{1/4})$,于是

$$
I_3=O(n^{1/2})\big(\max_{1\leqslant j\leqslant n}\mid a_{nj}\mid^2\big)=O(n^{1/2-3/10-2\varepsilon_1})=O(n^{1/5-2\varepsilon_1}).
$$

最后, $I_4=O(1)$,综合以上结果即得(29)式,其中 ε_2 可取为 ε_1. 引理4证毕.

引理5 在引理1,3的记号和假定下,以 $\phi_n(t)$ 记 S_n 的特征函数,则存在与 n 无关的 $\varepsilon>0$, $C<\infty$,使当 n 充分大时

$$
\int_{|t|\leqslant\sqrt{n}\varepsilon}\left|\frac{\phi_n(t)-\exp(-t^2/2)}{t}\right|\mathrm{d}t\leqslant\frac{C}{\sqrt{n}}.\tag{31}
$$

证 以 C 记任一与 n 无关的常数,每次出现时其值不必相同.由(13)式结合 $B_n^2\geqslant D_2>0$,知 $\mid 1-(\tilde{B}_n/B_n)\mid\leqslant C/\sqrt{n}$. 又我们有 $\sum\limits_{j=1}^n\mathrm{E}\mid\xi_j\mid^3\leqslant 8Dn$. 于是根据文献[2]的引理1,知存在 $\varepsilon_1>0$ 与 n 无关,使

$$
\mid\phi_n(B_nt/\tilde{B}_n)-\mathrm{e}^{-t^2/2}\mid\leqslant C\mid t\mid^3\mathrm{e}^{-t^2/3}/\sqrt{n},\quad\mid t\mid\leqslant\sqrt{n}\varepsilon_1.\tag{32}
$$

又

$$
\begin{aligned}
\mid\phi_n(t)-\mathrm{e}^{-t^2/2}\mid &\leqslant\mid\phi_n(t)-\exp(-\tilde{B}_nt^2/2B_n^2)\mid+\mid\exp(-\tilde{B}_nt^2/2B_n^2)-\mathrm{e}^{-t^2/2}\mid\\
&=I_1+I_2.
\end{aligned}\tag{33}
$$

当 n 充分大时，有 $\frac{1}{2} \leqslant \tilde{B}_n / B_n \leqslant 2$，故由(32)式有

$$I_1 \leqslant C \mid t \mid^3 e^{-t^2/3} / \sqrt{n}, \quad \mid t \mid \leqslant \sqrt{n}\varepsilon_1 / 2 = \sqrt{n}\varepsilon. \tag{34}$$

由中值定理得

$$I_2 \leqslant C \mid t \mid e^{-t^2/8} \mid \tilde{B}_n / B_n - 1 \mid \mid t \mid \leqslant C t^2 e^{-t^2/12} / \sqrt{n}. \tag{35}$$

由(33)～(35)式知，当 n 充分大且 $\mid t \mid \leqslant \sqrt{n}\varepsilon$ 时，有

$$\mid \phi_n(t) - e^{-t^2/2} \mid \leqslant C(t^2 + \mid t \mid^3) e^{-t^2/12} / \sqrt{n}, \tag{36}$$

由(36)式即得(31)式. 引理 5 证毕.

以 $S_n^*(x)$ 记 S_n 的分布函数，由(31)式，根据 Berry-Esseen 的基本不等式，得到

$$\| S_n^* - \Phi \| = \sup_x \mid S_n^*(x) - \Phi(x) \mid \leqslant C / \sqrt{n}. \tag{37}$$

引理 6 设 $e_1, e_2, \cdots$ 满足定理 1 或定理 2 的条件，e_j^*，S_n，$\phi_n(t)$ 等的意义同前. 又设 $\{a_{nuv}, v = 1, \cdots, n, u = 1, \cdots, r, n = 1, 2, \cdots\}$ 满足条件

$$\sum_{v=1}^{n} a_{nuv}^2 = 1, \ u = 1, \cdots, r, \ n = 1, 2, \cdots. \tag{38}$$

令 $m_n = n - [\sqrt{n}]$. 对每个 n，取足标 $1 \leqslant v_{n1} < \cdots < v_{nm_n} \leqslant n$，使剩下的 $[\sqrt{n}]$ 个足标都在(15)式定义的集 A_n 内. 令

$$T_n = S_n - \frac{4}{\sqrt{n} B_n} \sum_{u=1}^{r} \left(\sum_{s=1}^{m_n} a_{nuv_{ns}} e_{v_{ns}}^* \right)^2 = S_n + \Delta_n, \tag{39}$$

以 $T_n^*(x)$ 记 T_n 的分布函数，则存在与 n，$\{a_{nuv}\}$，$\{v_{ns}\}$ 都无关的常数 C，使

$$\| T_n^* - \Phi \| \leqslant C / \sqrt{n}. \tag{40}$$

证 仍以 C 记任一具有引理中所述性质的数，每次出现时取值可不同. 为简化起见，设 $v_{ns} = S$，$1 \leqslant S \leqslant m_n$，并将 m_n 和 a_{nuv} 分别写为 m 和 a_{uv}. 检查以下的证明将看出，v_{ns} 的这个特殊取法对证明的普遍性没有损害.

以 $\psi_n(t)$ 记 T_n 的特征函数，由 Berry-Esseen 的基本不等式，欲证(40)式，只需证明存在 $\mu > 0$ 与 n 无关，使

$$\int_{|t| \leqslant \sqrt{n}\mu} \left| \frac{\psi_n(t) - \exp(-t^2/2)}{t} \right| dt \leqslant C / \sqrt{n}. \tag{41}$$

而据引理 5，欲证(41)式，只需证

$$\int_{|t| \leqslant \sqrt{n}\mu} \left| \frac{\psi_n(t) - \phi_n(t)}{t} \right| dt \leqslant C / \sqrt{n}. \tag{42}$$

为此注意

$$|\psi_n(t)-\phi_n(t)|=|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}(\mathrm{e}^{\mathrm{i}t\Delta_n}-1))|$$
$$\leqslant|t||\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\Delta_n)|+t^2|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\Delta_n^2)|+|t|^3|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\theta\Delta_n^3)|. \tag{43}$$

此处 $|\theta|\leqslant 1$, θ 为随机变量,由引理 3 知,存在 $\eta>0$, $\delta>0$, $b_1>0$, 使(18), (19)式成立. 由此出发不难知道,当 n 充分大而 $|t|\leqslant\sqrt{n}\eta$ 时,有

$$|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\Delta_n)|\leqslant C(1+t^2)(1-\delta t^2/n)^{[b_1n]-2}/\sqrt{n}, \tag{44}$$

$$|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\Delta_n^2)|\leqslant C\sum_{j=0}^{4}|t|^j(1-\delta t^2/n)^{[b_1n]-4}/n. \tag{45}$$

又当 n 充分大而 $|t|\leqslant\sqrt{n}\eta$ 时,有

$$(1-\delta t^2/n)^{[b_1n]-2h}\leqslant\exp(-\delta b_1t^2/2),\ h=1,\ 2 \tag{46}$$

于是由(44)~(46)式,得

$$\int_{|t|\leqslant\sqrt{n}\eta}\sum_{h=1}^{2}|t|^h|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\Delta_n^h)|/|t|\,\mathrm{d}t\leqslant C/\sqrt{n}. \tag{47}$$

为处理(43)式右边第三项,分 $|t|\leqslant n^{1/4+\alpha}$ 和 $n^{1/4+\alpha}<|t|\leqslant\sqrt{n}\eta$ 两段来考虑. $\alpha>0$ 充分小,其具体值将在后面指定. 注意(43)式中的 θ 满足 $|\theta|\leqslant 1$, 且 θ 只依赖 e_j^*, $1\leqslant j\leqslant m$, 故与 e_{m+1}^*, …, e_n^* 独立. 又因足标 $m+1$, …, n (共$[\sqrt{n}]$个)都在集 A_n 内,有

$$|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\theta\Delta_n^3)|\leqslant(1-\delta t^2/n)^{[\sqrt{n}]}\mathrm{E}|\Delta_n|^3. \tag{48}$$

由 Δ_n 的定义及 $B_n^2\geqslant D_2>0$, 易见

$$\mathrm{E}|\Delta_n^3|\leqslant Cn^{-3/2}\sum_{u=1}^{r}\mathrm{E}\Big(\sum_{v=1}^{m}a_{uv}e_v^*\Big)^6.$$

于是由引理 4,存在 $\varepsilon_2>0$, 使

$$\mathrm{E}|\Delta_n^3|\leqslant Cn^{-5/4-\varepsilon_2}. \tag{49}$$

取 $\alpha=\varepsilon_2/4$, 则由(48), (49)式,知

$$\begin{aligned}&\int_{|t|\leqslant n^{1/4+\alpha}}\frac{1}{|t|}|t|^3|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\theta\Delta_n^3)|\,\mathrm{d}t\\&\leqslant Cn^{-5/4-\varepsilon_2}\int_{|t|\leqslant n^{1/4+\alpha}}t^2\,\mathrm{d}t\leqslant Cn^{-5/4-\varepsilon_2}n^{3(1/4+\alpha)}\\&=Cn^{-1/2-\alpha}\leqslant C/\sqrt{n}.\end{aligned} \tag{50}$$

另一方面,在 $n^{1/4+\alpha}<|t|\leqslant\sqrt{n}\eta$ 内,有

$$(1-\delta t^2/n)^{[\sqrt{n}]}\leqslant(1-\delta n^{-1/2+2\alpha})^{[\sqrt{n}]}\leqslant\exp(-\delta n^{\alpha}). \tag{51}$$

由(48), (49), (51)式,得 $|\mathrm{E}(\mathrm{e}^{\mathrm{i}tS_n}\theta\Delta_n^3)|\leqslant C\exp(-\delta n^{\alpha})$, 因而

$$\int_{n^{1/4+\alpha}\leqslant |t|\leqslant \sqrt{n}\eta} \frac{1}{|t|}\,|t|^3\,|\mathrm{E}(\mathrm{e}^{itS_n}\theta\Delta_n^3)|\,\mathrm{d}t \leqslant Cn^{3/2}\exp(-\delta n^{\alpha}) \leqslant C/\sqrt{n} \quad (n\ \text{充分大}), \tag{52}$$

结合(47), (50), (52)式,即得(42)式,其中 $\mu = \eta$, 引理 6 证毕.

引理 7 沿用引理 3 的假定和记号,又设 $\{a_{nuv}\}$ 满足(38)式,则存在与 n, $\{a_{nuv}\}$ 无关的常数 L,使当 n 充分大时,必存在$[\sqrt{n}]$个足标 $v_{n1}, \cdots, v_{n[\sqrt{n}]}$,使

(1) $v_{ns} \in A_n$, $s = 1, \cdots, [\sqrt{n}]$, A_n 的定义见(15)式.

(2) $a_{nuv_{ns}}^2 \leqslant L/n$, $s = 1, \cdots, [\sqrt{n}]$, $u = 1, \cdots, r$.

证 因 $\sum_{v=1}^{n} a_{nuv}^2 = 1$, 对任何 $R > 0$ 和 u,使 $a_{nuv}^2 \geqslant R/n$ 的足标 v 的个数,不超过 n/R,故集合

$$H_{nR} = \left\{v: 1 \leqslant v \leqslant n, \text{存在 } u, 1 \leqslant u \leqslant r, \text{使 } a_{nuv}^2 \geqslant \frac{R}{n}\right\}$$

中的元素个数不超过 rn/R,取 $R = 2r/b_1 = L$ (b_1 的意义见引理 3),则集合

$$A_n \cap \{v: 1 \leqslant v \leqslant n,\ a_{nuv}^2 \leqslant L/n,\ u = 1, \cdots, r\} \tag{53}$$

所包含的元素个数不少于 $b_1 n/2 > \sqrt{n}$, 当 n 充分大. 在集(53)式中任取$[\sqrt{n}]$个足标作为 $v_{n1}, \cdots, v_{n[\sqrt{n}]}$ 即可. 引理 7 证毕.

引理 8 设 $e_1, e_2, \cdots$相互独立,并满足条件(3), (5), (6)式,则存在与 n 无关的 C,使当 $n(> r)$ 充分大时,有

$$\left|\frac{\sqrt{n}B_n}{\sqrt{\mathrm{Var}\{(n-r)\ \hat{\sigma}_n^2\}}} - 1\right| \leqslant C/\sqrt{n}. \tag{54}$$

证 记 $M_n = (m_{nuv})_{u,v=1,\cdots,n} = I - X_n'(X_nX_n')^{-}X_n$, $\mathrm{E}e_j^4 = \beta_j\sigma^4$, 则由周知的公式

$$\begin{aligned}\mathrm{Var}\{(n-r)\ \hat{\sigma}_n^2\} &= \mathrm{Var}(e_{(n)}'M_ne_{(n)}) \\ &= \sigma^4\left\{\sum_{j=1}^{n}(\beta_j - 3)m_{njj} + 2\sum_{j=1}^{n}\sum_{k=1}^{n}m_{njk}^2\right\},\end{aligned} \tag{55}$$

因 M_n 为对称的幂等阵,有

$$\sum_{j=1}^{n}\sum_{k=1}^{n}m_{njk}^2 = \mathrm{tr}(M_nM_n') = \mathrm{tr}(M_n) = rk(M_n) = n - r. \tag{56}$$

取(4)式中的 a_{njk},有 $m_{njj} = 1 - \sum_{u=1}^{r}a_{nuj}^2$, 故

$$m_{njj}^2 = 1 - 2\sum_{u=1}^{r}a_{nuj}^2 + \left(\sum_{u=1}^{r}a_{nuj}^2\right)^2, \tag{57}$$

$$1 - 2\sum_{u=1}^{r}a_{nuj}^2 \leqslant m_{njj}^2 \leqslant 1 + r\sum_{u=1}^{r}a_{nuj}^2. \tag{58}$$

于是由(56)~(58)式,得到

$$\sigma^4\left\{\sum_{j=1}^n(\beta_j-3)m_{njj}^2+2\sum_{j=1}^n\sum_{k=1}^n m_{njk}^2\right\}$$
$$=\sigma^4\left\{\sum_{j=1}^n(\beta_j-1)m_{njj}^2-2\sum_{j=1}^n m_{njj}^2+2(n-r)\right\}$$
$$\geqslant\sigma^4\left\{\sum_{j=1}^n(\beta_j-1)-2\sum_{j=1}^n(\beta_j-1)\sum_{u=1}^r a_{nuj}^2\right.$$
$$\left.-2n-2r\sum_{j=1}^n\sum_{u=1}^r a_{nuj}^2+2(n-r)\right\}.$$

注意到 $(\beta_j-1)\sigma^4=\mathrm{Var}(e_j^2)$, $\beta_j\sigma^4=\mathrm{E}(e_j^4)\leqslant\mathrm{E}^{1/2}(e_j^2)\mathrm{E}^{1/2}(e_j^6)\leqslant\sigma\sqrt{D_1 n}$, 而

$$\sum_{j=1}^n\sum_{u=1}^r a_{nuj}^2=r,$$

得

$$\mathrm{Var}\{(n-r)\ \hat{\sigma}_n^2\}\geqslant nB_n^2-2r\sigma\sqrt{D_1}\sqrt{n}-2r(r+1).\tag{59}$$

用完全类似的方式可得到

$$\mathrm{Var}\{(n-r)\ \hat{\sigma}_n^2\}\leqslant nB_n^2+r^2\sigma\sqrt{D_1}\sqrt{n}+2r.\tag{60}$$

由 $B_n^2\geqslant D_2>0$, 从(59), (60)两式不难得到(54)式. 引理 8 证毕.

引理 9 设 $X_n=X_{n1}+X_{n2}$, $n=1,2,\cdots$ 为一串随机变量, X_n, X_{n1}的分布记为 $F_n(x)$, $F_{n1}(x)$. 设存在与 n 无关的正常数 C_1, C_2, C_3, 使

$$\|F_{n1}-\Phi\|\leqslant C_1/\sqrt{n},\ P\left(|X_{n2}|\geqslant\frac{C_2}{\sqrt{n}}\right)\leqslant\frac{C_3}{\sqrt{n}},\ n=1,2,\cdots,$$

则存在与 n 无关的常数 C, 使

$$\|F_n-\Phi\|\leqslant C/\sqrt{n}.$$

证 有显然的不等式

$$F_{n1}\left(x-\frac{C_2}{\sqrt{n}}\right)-P\left(|X_{n2}|\geqslant\frac{C_2}{\sqrt{n}}\right)\leqslant F_n(x)\leqslant F_{n1}\left(x+\frac{C_2}{\sqrt{n}}\right)+P\left(|X_{n2}|\geqslant\frac{C_2}{\sqrt{n}}\right),$$

再注意 $\left|\Phi\left(x\pm\frac{C_2}{\sqrt{n}}\right)-\Phi(x)\right|\leqslant C_2/\sqrt{n}$ 即可.

引理 10 设 Z_n 为随机变量, $a_n>0$, b_n 都是常数, $X_n=a_nZ_n+b_n$, $n=1,2,\cdots$. 以 $F_n(x)$和 $G_n(x)$分别记 X_n 和 Z_n 的分布, 设存在与 n 无关的常数 C_1, 使

$$|a_n-1|\leqslant C_1/\sqrt{n},\ |b_n|\leqslant C_1/\sqrt{n},\ \|G_n-\Phi\|\leqslant C_1/\sqrt{n},\ n=1,2,\cdots,$$

则存在与 n 无关的常数 C, 使

$$\|F_n-\Phi\|\leqslant C/\sqrt{n}.$$

证明从略.

3 定理的证明

在作了上述准备以后,现在已不难证明定理 1 和定理 2. 我们沿用前面的记号,首先考虑

$$Q_n^{(1)} = S_n - \frac{2}{\sqrt{n}B_n}\sum_{u=1}^{r}\Big(\sum_{v=1}^{n}a_{muv}e_v^*\Big)^2, \tag{61}$$

r, a_{muv} 的意义见第 1 节, B_n, S_n, e_v^* 见(11)和(17)式.

依引理 7,存在常数 L,使当 n 充分大时,对每个 n,可选出$[\sqrt{n}]$个足标 v_{ns}, $s=1, \cdots, [\sqrt{n}]$, 满足引理 7 的条件(1),(2). 为书写简便,且不失普遍性,不妨设这$[\sqrt{n}]$个足标为 $m+1$, $m+2$, $\cdots$, $n(m=n-[\sqrt{n}])$, 由(61)式得

$$S_n - \frac{4}{\sqrt{n}B_n}\sum_{u=1}^{r}\Big(\sum_{v=1}^{m}a_{muv}e_v^*\Big)^2 - \frac{4}{\sqrt{n}B_n}\sum_{u=1}^{r}\Big(\sum_{v=m+1}^{n}a_{muv}e_v^*\Big)^2 \leqslant Q_n^{(1)} \leqslant S_n. \tag{62}$$

记 $T_n' = \frac{4}{\sqrt{n}B_n}\sum_{u=1}^{r}\Big(\sum_{v=m+1}^{n}a_{muv}e_v^*\Big)^2$, 将上式写为

$$T_n - T_n' \leqslant Q_n^{(1)} \leqslant S_n, \tag{63}$$

T_n 由(39)式定义. 根据引理 6,(40)式成立,其中 $T_n^*(x)$为 T_n 的分布. 另一方面

$$P\Big(|-T_n'|\geqslant\frac{1}{\sqrt{n}}\Big)\leqslant\frac{4}{B_n}\sum_{u=1}^{r}\mathrm{E}\Big(\sum_{v=m+1}^{n}a_{muv}e_v^*\Big)^2\leqslant\frac{4\sigma^2}{B_n}\sum_{u=1}^{r}\sum_{v=m+1}^{n}a_{muv}^2,$$

由足标 $m+1$, $\cdots$, n 的选法,有 $a_{muv}^2 \leqslant L/n$, $v=m+1$, $\cdots$, n, $u=1$, $\cdots$, r. 于是,再注意 $B_n \geqslant \sqrt{D_2} > 0$, 得

$$P\Big(|T_n'|\geqslant\frac{1}{\sqrt{n}}\Big)\leqslant\frac{4\sigma^2}{B_n}r\sqrt{n}L/n\leqslant C/\sqrt{n},$$

$C<\infty$ 与 n 无关. 由(40)式结合引理 9 即知,若以 $H_n^{(1)}(x)$记 $T_n - T_n'$ 的分布函数,则存在与 n 无关的 C,使

$$\| H_n^{(1)} - \Phi \| \leqslant C/\sqrt{n}. \tag{64}$$

另一方面,S_n 的分布函数 $S_n^*(x)$,根据引理 5,满足(37)式. 由(63), (64), (37)式,知若以 $R_n^{(1)}(x)$记 $Q_n^{(1)}$ 的分布,则存在 $C<\infty$ 与 n 无关,使

$$\| R_n^{(1)} - \Phi \| \leqslant C/\sqrt{n}. \tag{65}$$

其次,定义

$$Q_n^{(2)} = \Big\{\sum_{j=1}^{n}\hat{e}_j^2 - (n-r)\sigma^2\Big\}\Big/\sqrt{n}B_n - \frac{1}{\sqrt{n}B_n}\sum_{u=1}^{r}\Big(\sum_{v=1}^{n}a_{muv}\hat{e}_v\Big)^2, \tag{66}$$

则

$$\left|\left\{\sum_{j=1}^{n}\hat{e}_j^2-(n-r)\sigma^2\right\}\Big/\sqrt{n}B_n-S_n\right|\leqslant\frac{1}{\sqrt{n}B_n}\left(\sum_{j=1}^{n}(\sigma^2-\mathrm{E}(\hat{e}_j^2))+r\sigma^2\right)$$
$$=\frac{1}{\sqrt{n}B_n}\left(\sum_{j=1}^{n}\int_{|x|\geqslant n^{1/4}}x^2\mathrm{d}F_j+r\sigma^2\right)$$
$$\leqslant\frac{1}{\sqrt{n}B_n}\left(\frac{1}{n}\sum_{j=1}^{n}\mathrm{E}e_j^6+r\sigma^2\right)$$
$$\leqslant\frac{D_1+r\sigma^2}{\sqrt{n}B_n}. \tag{67}$$

又因 $|\mathrm{E}\hat{e}_j|=\left|\int_{|x|\geqslant n^{1/4}}x\mathrm{d}F_j\right|\leqslant\frac{1}{\sqrt{n}}\mathrm{E}|e_j|^3$，有

$$\left(\sum_{v=1}^{n}a_{nuv}\hat{e}_v\right)^2=\left(\sum_{v=1}^{n}a_{nuv}e_v^*+\sum_{v=1}^{n}a_{nuv}\mathrm{E}\hat{e}_v\right)^2$$
$$\leqslant 2\left(\sum_{v=1}^{n}a_{nuv}e_v^*\right)^2+2\left(\sum_{v=1}^{n}a_{nuv}\mathrm{E}\hat{e}_v\right)^2$$
$$\leqslant 2\left(\sum_{v=1}^{n}a_{nuv}e_v^*\right)^2+2\sum_{v=1}^{n}(\mathrm{E}\hat{e}_v)^2$$
$$\leqslant 2\left(\sum_{v=1}^{n}a_{nuv}e_v^*\right)^2+\frac{2}{n}\sum_{v=1}^{n}(\mathrm{E}|e_v|^3)^2$$
$$\leqslant 2\left(\sum_{v=1}^{n}a_{nuv}e_v^*\right)^2+\frac{2}{n}\sum_{v=1}^{n}\mathrm{E}e_v^6$$
$$\leqslant 2\left(\sum_{v=1}^{n}a_{nuv}e_v^*\right)^2+2D_1. \tag{68}$$

由(66)～(68)式，并注意 $B_n\geqslant\sqrt{D_2}>0$，知存在与 n 无关的常数 C，使

$$Q_n^{(1)}-\frac{C}{\sqrt{n}}\leqslant Q_n^{(2)}\leqslant S_n+\frac{C}{\sqrt{n}}. \tag{69}$$

根据(65)和(40)式，(69)左右两端的变量的分布，都以 $O\left(\frac{1}{\sqrt{n}}\right)$ 的速度逼近于 Φ，故 $Q_n^{(2)}$ 的分布 $R_n^{(2)}(x)$ 亦然，即存在与 n 无关的 C，使

$$\|R_n^{(1)}-\Phi\|\leqslant C/\sqrt{n}. \tag{70}$$

现在考虑

$$Q_n^{(3)}=(n-r)(\hat{\sigma}_n^2-\sigma^2)/\sqrt{n}B_n$$
$$=\left\{\sum_{j=1}^{n}e_j^2-(n-r)\sigma^2\right\}\Big/\sqrt{n}B_n-\frac{1}{\sqrt{n}B_n}\sum_{u=1}^{r}\left(\sum_{v=1}^{n}a_{nuv}e_v\right)^2, \tag{71}$$

比较(66)与(71)式，看出 $Q_n^{(3)}$ 与 $Q_n^{(2)}$ 之差别，仅在于 e_j 换为 $\hat{e}_j$. 因此，显然，若以 $R_n^{(3)}(x)$ 记 $Q_n^{(3)}$ 的分布，则对任何 x，有

$$
\begin{aligned}
|R_n^{(3)}(x)-R_n^{(2)}(x)| &\leqslant \sum_{j=1}^{n} P(e_j \neq \hat{e}_j) \\
&= \sum_{j=1}^{n} \int_{|x| \geqslant n^{1/4}} \mathrm{d}F_j \leqslant n^{-3/2} \sum_{j=1}^{n} \mathrm{E}e_j^6 \\
&\leqslant n^{-3/2} D_1 n = D_1/\sqrt{n}.
\end{aligned} \tag{72}
$$

(70)与(72)式结合,知存在与 n 无关的 C,使

$$
\| R_n^{(3)} - \Phi \| \leqslant C/\sqrt{n}. \tag{73}
$$

最后,有

$$
Q_n = (\hat{\sigma}_n^2 - \sigma^2)/\sqrt{\mathrm{Var}(\hat{\sigma}_n^2)} = \frac{\sqrt{n}B_n}{(n-r)\sqrt{\mathrm{Var}(\hat{\sigma}_n^2)}} Q_n^{(3)}.
$$

根据引理 8,有(54)式. 于是,根据(73)式和引理 10,若以 G_n 记 Q_n 的分布,则得知存在与 n 无关的 C,使

$$
\| G_n - \Phi \| \leqslant C/\sqrt{n}.
$$

即为(8)式. 这就完成了定理 1,2 的证明.

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5

线性估计弱相合性的一个问题

1 引言和主要结果

设有线性回归模型

$$Y_i = x_i'\beta + e_i \quad (i = 1, \cdots, n, \cdots), \tag{1}$$

其中 $x_i = (x_{i1}, \cdots, x_{ip})'$ $(i = 1, 2, \cdots)$ 为一串已知向量(试验点列)，$\beta = (\beta_1, \cdots, \beta_p)'$ 为未知的回归系数向量，$\{e_i\}$ 为随机误差序列. 我们称 $\{e_i\}$ 满足 Gauss-Markov 条件(简称 GM 条件)，如果

$$E(e_i) = 0,\ E(e_ie_j) = \sigma^2\delta_{ij} \quad (i, j = 1, 2, \cdots),\ 0 < \sigma^2 < \infty. \tag{2}$$

此处 $\delta_{ij} = 1$ 或 0，视 $i = j$ 或否而定. 若除(2)外还假定 e_i 都服从正态分布，则称 $\{e_i\}$ 满足正态条件.

记 $X_n = (x_1, \cdots, x_n)'$. 若 X_n 的秩为 p，则在(1)的前 n 次试验的基础上，可求出 β 的最小二乘(LS)估计为

$$\hat{\beta}(n) = (\hat{\beta}_1(n), \cdots, \hat{\beta}_p(n))' = S_n^{-1}X_n'(Y_1, \cdots, Y_n)'.$$

这里 $S_n = X_n'X_n$. 在 GM 条件下，$\hat{\beta}(n)$是 β 的弱相合估计(简记为 WCE)的充要条件为 $S_n^{-1} \to 0$(见[1]). 这一结果可以表述为更一般的形式：若当 n 充分大时，线性函数 $c'\beta$ 在(1)的前 n 次试验的基础上可估，则 $c'\beta$ 的 Gauss-Markov (GM)估计为弱相合的充要条件，是它的方差趋于 0，这说明：在 GM 条件下，可估函数的 GM 估计的弱相合性与均方相合性等价.

形如

$$T_n = c_{n0} + \sum_{i=1}^{n} c_{ni}Y_i \quad (n = 1, 2, \cdots) \tag{3}$$

的估计，称为线性估计. 在 GM 条件下，甚至进一步要求 $e_1, e_2, \cdots$同分布或者独立，也不能排斥如下的可能性：$c'\beta$ 的 GM 估计不是 WCE，但是存在着其他线性的 WCE. 在[1]中我们证明了：在$\{e_i\}$满足正态条件时，这种可能性不存在. 本文将正态条件削弱为只要求 $e_1, e_2, \cdots$相互独立，得到了比较彻底的结果.

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定理 设$\{e_i\}$为满足 GM 条件及下述条件(A)的独立随机变量序列.

(A) 不存在常数 a 及自然数子序列$\{n_k\}$,使

$$e_{n_k} \xrightarrow{P} a \text{,当 } k \to \infty \text{ 时,}$$

则当可估函数 $c'\beta$ 的 GM 估计不为 $c'\beta$ 的 WCE 时,任何形如(3)的线性估计都不是 $c'\beta$ 的 WCE.

反过来,若条件(A)不满足,则可以找到试验点列$\{x_i\}$及可估函数 $c'\beta$,使 $c'\beta$ 的 GM 估计不为 WCE,但 $c'\beta$ 的其他某个线性估计却为其 WCE.

在研究这个问题的过程中,我们附带证明了以下的有趣结果:在$\{e_i\}$满足 GM 条件下,若可估函数 $c'\beta$ 的 GM 估计不是 $c'\beta$ 的均方相合估计,则 $c'\beta$ 的线性均方相合估计根本不存在. 为了避免误解,此处及下文都约定称 $c'\beta$ 可估,是指它在(1)的前 n 次试验的基础上可估,当 n 充分大时.

2 两个引理

本文主要定理的证明依赖于两个有独立兴趣的引理. 其中之一,即引理 1,证明了第 1 节末尾处提到的关于均方相合的事实,又以下沿用第 1 节的记号.

引理 1 设$\{e_i\}$满足 GM 条件而 β_1 可估,则由(3)给出的线性估计 T_n 为 β_1 的均方相合估计的充要条件为

$$\lim_{n\to\infty} c_{n0} = 0, \tag{4}$$

$$\lim_{n\to\infty} \sum_{i=1}^{n} c_{ni} x_{i1} = 1, \tag{5}$$

$$\lim_{n\to\infty} \sum_{i=1}^{n} c_{ni} x_{ij} = 0 \ (j = 2, \cdots, p), \tag{6}$$

$$\lim_{n\to\infty} \sum_{i=1}^{n} c_{ni}^2 = 0. \tag{7}$$

又若 $c'\beta$ 为可估,则当其 GM 估计不为均方相合时,其任何线性估计决非均方相合.

证 由定义,T_n 为 β_1 的均方相合估计的充要条件为 $\lim\limits_{n\to\infty} \mathrm{E}[(T_n-\beta_1)^2]=0$. 但

$$T_n - \beta_1 = c_{n0} + \Big(\sum_{i=1}^{n} c_{ni} x_{i1} - 1\Big)\beta_1 + \sum_{j=2}^{p} \sum_{i=1}^{n} c_{ni} x_{ij} \beta_j + \sum_{j=1}^{n} c_{ni} e_i,$$

因而由$\{e_i\}$满足 GM 条件知

$$\mathrm{E}[(T_n-\beta_1)^2] = \Big[c_{n0} + \Big(\sum_{i=1}^{n} c_{ni} x_{i1} - 1\Big)\beta_1 + \sum_{j=2}^{p} \sum_{i=1}^{n} c_{ni} x_{ij} \beta_j\Big]^2 + \sigma^2 \sum_{i=1}^{n} c_{ni}^2.$$

为了 $\lim\limits_{n\to\infty} \mathrm{E}[(T_n-\beta_1)^2]=0$ 对任何 $\beta_1, \cdots, \beta_p$ 都成立,显然必须只须(4)~(7)同时成立,这证明了引理的第一部分. 为证后一部分,注意通过在必要时改换参数,不失普遍性,可设 $c'\beta = \beta_1$. 记

$$\xi_{jn}=(x_{1j},\cdots,x_{nj})'\ (j=1,\cdots,p),$$

必要时适当改换参数,不失普遍性,可设当 n 充分大时 $\xi_{1n},\cdots,\xi_{pn}$ 线性无关,以 η_n 记 ξ_{1n} 在由 $\xi_{2n},\cdots,\xi_{pn}$ 所张成的线性子空间 $\mathscr{M}_n$ 内的投影,则 η_n 可唯一表示为

$$\eta_n=d_{2n}\xi_{2n}+\cdots+d_{pn}\xi_{pn},\ n\text{ 充分大}. \tag{8}$$

设 β_1 的 LS 估计 $\hat\beta_1(n)$ 不为均方相合,则 $\mathrm{Var}(\hat\beta_1(n))\downarrow v>0$ 当 $n\to\infty$ 时. 后面我们将证明

$$\mathrm{Var}(\hat\beta_1(n))=\sigma^2/\|\zeta_n\|^2,\ \zeta_n=\xi_{1n}-\eta_n, \tag{9}$$

因而存在常数 $R<\infty$, 致

$$\|\zeta_n\|\leqslant R. \tag{10}$$

现在再证存在 $K<\infty$,使 $|d_{in}|\leqslant K$ 当 n 充分大时 $(i=2,\cdots,p)$, 事实上,若这种 K 不存在,则为确定计且不失普遍性,可找到自然数子序列 $\{n_r\}$,致

$$\lim_{r\to\infty}|d_{2n_r}|=\infty, \tag{11}$$

在(8)中改 n 为 n_r,并记 $f_{ir}=d_{in_r}/d_{2n_r}$, 有

$$\eta_{n_r}=d_{2n_r}(\xi_{2n_r}+\sum_{i=3}^{p}f_{ir}\xi_{in_r}). \tag{12}$$

因为当 n 充分大时 $\xi_{2n},\cdots,\xi_{pn}$ 线性无关,存在自然数 d,致

$$u_r=\xi_{2n_r}+\sum_{i=3}^{p}f_{ir}\xi_{in_r}\neq 0,\text{当 } r\geqslant d\text{ 时}, \tag{13}$$

以 α 记 ξ_{2n_d} 在 $\{\xi_{jn_d},j=3,\cdots,p\}$ 所张成的线性子空间上的投影,而 $\delta=\xi_{2n_d}-\alpha$, 则 $\delta\neq 0$. 对任何 $r\geqslant d$, 以 b_r 记 u_r 的前 n_d 个分量构成的列向量,则由 u_r 的定义(13)式及 δ 的定义,知

$$\|b_r\|\geqslant\|\delta\|,\ r\geqslant d. \tag{14}$$

以 θ_r 记 η_{n_r} 的前 n_d 个分量构成的列向量. 由(10)～(14),有

$$\begin{aligned}R&\geqslant\|\zeta_{n_r}\|=\|\xi_{1n_r}-\eta_{n_r}\|\\&\geqslant\|\xi_{1n_d}-\theta_r\|\geqslant\|\theta_r\|-\|\xi_{1n_d}\|\\&=\|d_{2n_r}b_r\|-\|\xi_{1n_d}\|\geqslant|d_{2n_r}|\,\|\delta\|-\|\xi_{1n_d}\|\\&\to\infty.\end{aligned}$$

这与 $R<\infty$ 矛盾,因而证明了所述 K 的存在性.

现设 β_1 有线性均方相合估计(3). 由本引理已证部分,知(4)～(7)成立. 上面已证明了 $\{d_{jn}\ (j=2,\cdots,p),n\text{ 充分大}\}$ 的有界性,故可以用 $-d_{jn}$ 乘(6)式,并与(5)式相加 $(j=2,\cdots,p)$, 得

$$1=\lim_{n\to\infty}\sum_{i=1}^{n}c_{ni}(x_{i1}-\sum_{j=2}^{p}d_{jn}x_{ij})=\lim_{n\to\infty}c_n'\zeta_n, \tag{15}$$

此处 $c_n'=(c_{n1},\cdots,c_{nn})$. 另一方面,由(7)和(10)得

$$(c_n'\zeta_n)^2\leqslant\|c_n\|^2\|\zeta_n\|^2\leqslant R^2\|c_n\|^2\to 0,$$

当 $n \to \infty$ 时. 这与(15)矛盾,因而(3)不能是 β_1 的均方相合估计. 现往证(9)式. 在 GM 条件下,如所周知,若以 g 记 S_n^{-1} 的(1, 1)元,则 $\mathrm{Var}(\hat{\beta}_1(n)) = \sigma^2 g$. 记 $B = (\xi_{2n}, \cdots, \xi_{pn})$, 则

$$S_n = \begin{pmatrix} \|\xi_{1n}\|^2 & \xi'_{1n}B \\ B'\xi_{1n} & B'B \end{pmatrix},$$

由

$$\begin{aligned} g^{-1} &= \|\xi_{1n}\|^2 - \xi'_{1n}B(B'B)^{-1}B'\xi_{1n} \\ &= \|\xi_{1n} - B(B'B)^{-1}B'\xi_{1n}\|^2 \\ &= \|\xi_{1n} - \eta_n\|^2 \\ &= \|\zeta_n\|^2. \end{aligned}$$

这证明了(9)式,因而完成了引理 1 的证明(此处利用了 $B(B'B)^{-1}B'$为向 $\mathscr{M}_n$ 投影的算子这个事实).

引理 2 设$\{e_i\}$为独立随机变量序列,满足如下的条件(A′):

(A′) 不存在常数序列$\{a_k\}$及自然数子序列

$$\{n_k\},致\ e_{n_k} - a_k \xrightarrow{P} 0\ 当\ k \to \infty\ 时.$$

又设 $c_{n1}, \cdots, c_{nn}, b_n$ $(n = 1, 2, \cdots)$ 为给定的常数,致

$$Z_n = \sum_{i=1}^{n} c_{ni}e_i - b_n \xrightarrow{P} 0, \tag{16}$$

则有

$$d_n^2 = \sum_{i=1}^{n} c_{ni}^2 \to 0, 当\ n \to \infty\ 时, \tag{17}$$

证 作随机变量 $e_1^*, e_2^*, \cdots$,致 $e_1, e_1^*, e_2, e_2^*, \cdots$相互独立,且对每个 i, e_i 与 e_i^* 同分布,记 $e_i^s = e_i - e_i^*$. 由(16)知,当 $n \to \infty$ 时, $Z_n^* = \sum_{i=1}^{n} c_{ni}e_i^* - b_n \xrightarrow{P} 0$, 故

$$Z_n^s = Z_n - Z_n^* = \sum_{i=1}^{n} c_{ni}e_i^s \xrightarrow{P} 0. \tag{18}$$

现证在引理条件下,存在 $m > 0$, 使

$$P(|e_i^s| \geqslant m) \geqslant m\ (i = 1, 2, \cdots). \tag{19}$$

事实上,若这样的 m 不存在,则可找到自然数子序列$\{n_k\}$,致 $e_{n_k}^s \xrightarrow{P} 0$ 当 $k \to \infty$时,以 μ_i 记 e_i 的中位数,则对任给 $\varepsilon > 0$ 有(见[2], p. 245)

$$P(|e_{n_k} - \mu_{n_k}| \geqslant \varepsilon) \leqslant 2P(|e_{n_k}^s| \geqslant \varepsilon) \to 0,\ 当\ k \to \infty 时,$$

即 $e_{n_k} - \mu_{n_k} \xrightarrow{P} 0$, 这与$\{e_i\}$满足条件(A′)矛盾.

现作如下的补充假定:存在 $M < \infty$, 使

$$P(|e_i^s| > M) = 0\ (i = 1, 2, \cdots), \tag{20}$$

则将有 $\mathrm{E}(e_i^s) = 0$ $(i = 1, 2, \cdots)$. 又由(19), (20),有

$$\mathrm{E}[(e_i^s)^2] \geqslant m^3,\ \mathrm{E}(|e_i^s|^3) \leqslant M^3.$$

故若以 $f_i(t)$ 记 e_i^s 的特征函数,将有

$$|f_i(t)| \leqslant 1 - t^2 m^3/2 + |t^3| M^3/6\ (i = 1, 2, \cdots),$$

因而存在与 i 无关的常数 $\eta > 0$, $\delta > 0$, 致

$$|f_i(t)| \leqslant 1 - \delta t^2, \text{当} |t| \leqslant \eta \text{时}\ (i = 1, 2, \cdots), \tag{21}$$

显然可设 $\delta\eta^2 < 1$. 若(17)式不对,则存在常数 $A > 0$ 及自然数子序列 $\{n_k\}$,致 $d_{n_k}^2 \geqslant A$ $(k = 1, 2, \cdots)$,记 $c'_{ki} = c_{n_k i}/d_{n_k}$ $(i = 1, \cdots, n_k)$, 有

$$\sum_{i=1}^{n_k} c'^2_{ki} = 1, \tag{22}$$

由 $d_{n_k}^2 \geqslant A > 0$ 及(18),得

$$\boldsymbol{W}_k = \sum_{i=1}^{n_k} c'_{ki} e_i^s \xrightarrow{P} 0. \tag{23}$$

$\boldsymbol{W}_k$ 的特征函数为

$$\varphi_k(t) = \prod_{i=1}^{n_k} f_i(c'_{ki} t). \tag{24}$$

由于 $|c'_{ki}| \leqslant 1$, 由(21)有

$$|f_i(c'_{ki} t)| \leqslant 1 - \delta c'^2_{ki} t^2, \text{当} |t| \leqslant \eta \text{时}\ (i = 1, \cdots, n_k). \tag{25}$$

由(24), (25), $\delta\eta^2 < 1$, 以及 $\log(1 - x) < -x$ 当 $0 < x < 1$ 时,有

$$\log|\varphi_k(t)| = \sum_{i=1}^{n_k} \log|f_i(c'_{ki} t)| \leqslant -\delta t^2 \sum_{i=1}^{n_k} c'^2_{ki} = -\delta t^2 \nrightarrow 0,$$

这与(23)矛盾,因而在补充条件(20)之下证明了(17).

现考虑一般情况,仍用反证法,若(17)式不对,则如前,有 A, $\{n_k\}$, c'_{ki},及(22),(23)式,现再证

$$\{c'_{ki} e_1^s, \cdots, c'_{kn_k} e_{n_k}^s\}\ (k = 1, 2, \cdots).$$

适合 uan 条件(见[2],p. 302). 事实上,若以 $f_{ki}(t)$ 记为 $c'_{ki} e_i^s$ 的特征函数,则因 e_i^s 对称,有 $f_{ki}(t) \geqslant 0$. 又由(23)知 $\lim\limits_{k\to\infty} \prod\limits_{i=1}^{n_k} f_{ki}(t) = 1$ 对任何 t,故有

$$\max_{1\leqslant i\leqslant n_k} |1 - f_{ki}(t)| \leqslant 1 - \prod_{i=1}^{n_k} f_{ki}(t) \to 0.$$

这证明了所要的结论(参看[2],p. 302～303). 这与(23)结合,根据[2],p. 317,并利用 e_i^s 的对称性,知对任给 $\epsilon > 0$ 有

$$\lim_{k\to\infty} \sum_{i=1}^{n_k} \int_{|x|\geqslant\epsilon} \mathrm{d}F_{ki}(x) = 0, \tag{26}$$

$$\lim_{k\to\infty} \sum_{i=1}^{n_k} \int_{|x|<\epsilon} x^2 \mathrm{d}F_{ki}(x) = 0, \tag{27}$$

这里 $F_{ki}(x)$ 为 $c'_{ki}e^s_i$ 的分布函数. 现任取 $M > m$ (m 的意义见(19)式),并定义

$$\hat{e}_i = \min(M,\ \max(-M,\ e^s_i))\ (i = 1,\ 2,\ \cdots),$$

则 $\{\hat{e}_i\}$ 为一相互独立的对称随机变量序列,满足条件

$$P(|\hat{e}_i| \geqslant m) \geqslant m,\ P(|\hat{e}_i| > M) = 0\ (i = 1,\ 2,\ \cdots). \tag{28}$$

又若以 $\hat{F}_{ki}$ 记 $c'_{ki}\hat{e}_i$ 的分布函数,则对任给 $\varepsilon > 0$ 有

$$\int_{|x| \geqslant \varepsilon} \mathrm{d}\hat{F}_{ki}(x) \leqslant \int_{|x| \geqslant \varepsilon} \mathrm{d}F_{ki}(x),$$

$$\int_{|x| < \varepsilon} x^2 \mathrm{d}\hat{F}_{ki}(x) \leqslant \int_{|x| < \varepsilon} x^2 \mathrm{d}F_{ki}(x) + c'^2_{ki}M^2 \int_{|x| > \varepsilon} \mathrm{d}F_{ki}(x).$$

由此及(22),(26),(27),得知(26), (27)式将 $F_{ki}(x)$ 改为 $\hat{F}_{ki}(x)$ 时,仍保持成立. 再利用[2], p. 317, 3°,即得

$$\sum_{i=1}^{n_k} c'_{ki}\hat{e}_i \xrightarrow{P} 0, \text{当 } k \to \infty \text{ 时}, \tag{29}$$

因为 $\{\hat{e}_i\}$ 满足(28),由已证部分,由(29)将得

$$\lim_{k\to\infty} \sum_{i=1}^{n_k} c'^2_{ki} = 0,$$

这与(22)矛盾,因而证明了本引理.

3 定理的证明

现在来证明第 1 节中提出的定理. 先证,若 $\{e_i\}$ 满足定理的条件,则必满足引理 2 的条件(A′).

事实上,若条件(A′)不满足,则存在常数序列 $\{a_k\}$ 及自然数子序列 $\{n_k\}$,使

$$e_{n_k} - a_k \xrightarrow{P} 0, \text{当 } k \to \infty \text{ 时}, \tag{30}$$

易见 $\{a_k\}$ 有界. 不然的话,必要时抽出子序列,不失普遍性可设 $\lim\limits_{k\to\infty} a_k = \infty$. 由(30)知

$$\lim_{k\to\infty} P(|e_{n_k}| \geqslant a_k - 1) = 1. \tag{31}$$

另一方面,由 $\mathrm{E}(e^2_{n_k}) = \sigma^2$ 知 $P(|e_{n_k}| \geqslant a_k - 1) \leqslant \sigma^2/(a_k - 1)^2 \to 0$ 当 $k \to \infty$ 时,与(31)矛盾,因而证明了 $\{a_k\}$ 的有界性. 由此可从 $\{a_k\}$ 中取出一个收敛于有限极限 a 的子序列,为方便计且不失普遍性,设 $\lim\limits_{k\to\infty} a_k = a$, 此与(30)结合将给出 $e_{n_k} - a \xrightarrow{P} 0$, 与 $\{e_i\}$ 满足条件(A)矛盾,这证明了 $\{e_i\}$ 满足条件(A′).

现设 $c'\beta$ 可估,但其 GM 估计不为弱相合,不失普遍性,不妨假定 $c'\beta = \beta_1$ (这可通过适当改换参数做到). 若由(3)确定的 T_n 为 β_1 的线性 WCE,则由

$$T_n-\beta_1=\left(\sum_{i=1}^{n}c_{n_i}x_{i1}-1\right)\beta_1+\sum_{j=2}^{p}\sum_{i=1}^{n}c_{ni}x_{ij}\beta_j+\left(c_{n0}+\sum_{i=1}^{n}c_{ni}e_i\right),$$

而且这个表达式对 $\beta_1,\cdots,\beta_p$ 的任何值都依概率收敛于 0,知(5),(6)两式成立,且

$$c_{n0}+\sum_{i=1}^{n}c_{ni}e_i\xrightarrow{P}0,\text{当 } n\to\infty \text{ 时}. \tag{32}$$

根据定理的假定及上文的论证,$\{e_i\}$满足引理 2 的一切条件,故由引理 2 知(7)式成立. 因此

$$\lim_{n\to\infty}E\left[\left(\sum_{i=1}^{n}c_{ni}e_i\right)^2\right]=\lim_{n\to\infty}\sum_{i=1}^{n}c_{ni}^2\sigma^2=0.$$

这推出 $\sum_{i=1}^{n}c_{ni}e_i\xrightarrow{P}0$, 当 $n\to\infty$时. 与(32)结合,即知(4)式成立,这样,(4)~(7)全成立. 由引理 1,知 T_n 为β_1 的均方相合估计. 再由引理 1,知 β_1 的 LS 估计为均方相合,与假设矛盾,这证明了定理的前一部分.

现设$\{e_i\}$不满足条件(A),即存在自然数子序列$\{n_k\}$及常数 a,使 $e_{n_k}-a\xrightarrow{P}0$ 当 $k\to\infty$时,记 $\tilde{e}_k=e_{n_k}-a$. 找自然数子序列$\{k_j\}$,致

$$P(|\tilde{e}_{k_j}-a|\geqslant 1/j^2)\leqslant 1/j\ (j=1,2,\cdots), \tag{33}$$

任取 p 维向量 d. 取试验点列$\{x_i\}$如下:

$$x_i=\begin{cases}d/j, & \text{当 } i=n_{k_j} \text{ 时 } (j=1,2,\cdots),\\ 0, & \text{其他 } i.\end{cases}$$

则显然 $d'\beta$ 为可估的. 对充分大的 n,取 $d'\beta$ 的线性估计 $T_n=c_{n0}+\sum_{i=1}^{n}c_{ni}Y_i$ 如下:先找出最大的自然数 j,满足条件 $n_{k_j}\leqslant n$, 然后令

$$c_{n0}=-ja,\begin{cases}c_{ni}=j, & \text{当 } i=n_{k_j} \text{ 时},\\ c_{ni}=0, & \text{对其他 } i.\end{cases}$$

这时易见 $T_n=d'\beta+j\tilde{e}_{k_j}$, 因而由(33)得

$$P(|T_n-d'\beta|\geqslant 1/j)\leqslant 1/j.$$

因为当 $n\to\infty$ 时有 $j\to\infty$,得知 T_n 为 $d'\beta$ 的 WCE. 但易见 $d'\beta$ 的 GM 估计为 $\psi_n=\sum_{i=1}^{j}\frac{1}{i}Y_{m_i}\Big/\sum_{i=1}^{j}\frac{1}{i^2}$, 此处 $m_i=n_{k_i}$, j 与 n 的关系同上. 有

$$\mathrm{Var}(\psi_n)=\sigma^2\left(\sum_{i=1}^{j}1/i^2\right)^{-1}.$$

它当 $n\to\infty$时不趋于 0,因而 ψ_n 不为 $d'\beta$ 的 WCE.

从本文结果及其证明过程,可以提供以下两点有意义的解释.

首先,我们前已指出,在 GM 条件下,可估函数的 GM 估计的弱相合性与均方相合性等价,在证明本文定理的过程中,我们得到:若$\{e_i\}$满足本文定理的条件,则对一般线性估计而

言,上述等价性也成立;而当$\{e_i\}$不满足定理中的条件(A)时则不然.

其次,设某个可估函数$c'\beta$,为方便计且不失普遍性,就设$c'\beta=\beta_1$,其 LS 估计为弱相合.这时,若由(3)式所定义的线性估计T_n也是β_1的 WCE,且$\{e_i\}$满足本文定理的条件,则如上所述,T_n也是β_1的均方相合估计.有$\mathrm{Var}(T_n)=\sigma^2\sum\limits_{i=1}^{n}c_{in}^2$.但由引理 1 的证明看出(见(15)式),这时有$\lim\limits_{n\to\infty}c_n'\zeta_n=1$(记号意义见引理 1).因而由

$$\mathrm{Var}(T_n)/\mathrm{Var}(\hat{\beta}_1(n))=\|c_n\|^2\|\zeta_n\|^2,$$

以及$\mathrm{Var}(\hat{\beta}_1(n))=\mathrm{E}[(\hat{\beta}_1(n)-\beta_1)^2]$知

$$\begin{aligned}&\liminf_{n\to\infty}[\mathrm{E}(T_n-\beta_1)^2/\mathrm{E}(\hat{\beta}_1(n)-\beta_1)^2]\\&\quad\geqslant\liminf_{n\to\infty}[\mathrm{Var}(T_n)/\mathrm{Var}(\hat{\beta}_1(n))]\\&\quad\geqslant\liminf_{n\to\infty}\|c_n\|^2\|\zeta_n\|^2\\&\quad\geqslant\lim_{n\to\infty}(c_n'\zeta_n)^2\\&\quad=1.\end{aligned}$$

这说明:在平方损失函数之下,若一可估函数$c'\beta$的 GM 估计ψ_n为弱相合的,则在$c'\beta$的一切弱相合线性估计的类中,ψ_n具有一致最小的渐近风险.因而,不可能存在这样的线性估计(不论是否弱相合)T_n,使

$$\liminf_{n\to\infty}[\mathrm{E}(T_n-c'\beta)^2/\mathrm{E}(\psi_n-c'\beta)^2]\leqslant 1.$$

对一切β,且不等号至少对一个β成立.这可以解释为$c'\beta$的 GM 估计ψ_n为"渐近可容许"的.当样本大小n固定时,关于线性估计在全体线性估计类中的可容许性问题(在平方损失下)有深入的研究(见[3]),特别是证明了:可估函数的 GM 估计在全部线性估计的类中是可容许的.上面指出的"渐近可容许性"从大样本的角度补充和加强了这个结论.

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A Problem of Weakly Consistent Linear Estimate

Chen Xiru

Abstract

Suppose we have a linear regression model $Y_i=x_i'\beta+e_i$ $(i=1,\cdots,n,\cdots)$, $e_1,e_2,\cdots$ independent, $\mathrm{E}(e_i)=0$, $\mathrm{Var}(e_i)=\sigma^2$ $(i=1,2,\cdots)$, and there is no such subsequence of

$\{e_i\}$ which converges in probability to some constant, then when the Gauss-Markov estimate $c'\beta(n)$ of a linear estimable function $c'\beta$ is not a weakly consistent estimate, there exists no weakly consistent linear estimate of $c'\beta$. The final condition imposed on $\{e_i\}$ is necessary in some meaning. This greatly improved the related result in [1].

6

最近邻密度估计的收敛速度

摘要 Loftsgarden 和 Quesenberry 在文献[1]中提出了概率密度函数 f 的最近邻估计 $\hat{f}_n$. 在本文中,我们得到了 (1) $\hat{f}_n(x)-f(x)$ 当 x 固定时的 a. s. 收敛速度;(2) $\sup\limits_x|\hat{f}_n(x)-f(x)|$ 的一致收敛速度;(3) $\hat{f}_n$ 的 a. s. L_r-相合性. 我们也证明了 $\hat{f}_n(x)$ 在 x 固定时的渐近正态性,以及下述结果:若除了 f 在 R_1 上一致连续外无其他假定,则 $\sup\limits_x|\hat{f}_n(x)-f(x)|$ 的收敛速度可任意慢.

设 $x_1, \cdots, x_n$ 是从某个具有分布 F 和密度 f 的一维总体中抽出的 iid. 样本. 为估计 f, Loftsgarden 和 Quesenberry[1] 提出了下面的方法:选定一个与 n 有关的自然数 $k(n)$. 找最小的 $a_n(x)$,使区间 $[x-a_n(x),\ x+a_n(x)]$ 内所包含的样本点 $x_1, \cdots, x_n$ 的个数不小于 $k(n)$,然后以

$$\hat{f}_n(x)=(k(n)-1)/(2na_n(x)) \tag{1}$$

作为 $f(x)$ 的估计. 这在文献中常称为最近邻估计. 这种估计引起了一些作者的兴趣,特别在其相合性方面,取得了不少结果. 例如,文献[1]中证明了:若 $f(x_0)>0$, f 在 x_0 点连续,而 $k(n)\to\infty$, $k(n)/n\to 0$, 则 $\hat{f}_n(x_0)\xrightarrow{P}f(x_0)$. Moore 等在文献[2]中也证明了:若 f 在 R_1 一致连续且处处大于 0, 且 $k(n)/n\to 0$, $\log n/k(n)\to 0$, 则

$$\sup_x|\hat{f}_n(x)-f(x)|\xrightarrow{P}0.$$

在强收敛方面, Wagner[3] 证明了:若

$$k(n)/n\to 0,\ \sum_{n=1}^{\infty}\mathrm{e}^{-\alpha k(n)}<\infty, \text{对任何 } \alpha>0,$$

且 f 在 x_0 点连续,则 $\hat{f}_n(x_0)\to f(x_0)$ a. s.. 1977 年, Devroye 等在文献[4]中证明:若 $k(n)/n\to 0$, $\log n/k(n)\to 0$ 而 f 在 R_1 一致连续,则

$$\sup_x|\hat{f}_n(x)-f(x)|\to 0\quad \text{a. s.}. \tag{2}$$

本文的目的是研究 $\hat{f}_n(x)$ 及 $\sup\limits_x|\hat{f}_n(x)-f(x)|$ 的收敛速度. 就前者而言,它依赖于 f 在指定点 x 附近的局部性质,更确切地说,依赖于 f 在 x 点处的(二阶或高阶)可微性. 如果知

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道了这种性质,也就能知道在可能的$k(n)$的选择下,收敛的阶的上限;也可以知道,在$k(n)$的某种选择下,收敛的阶能达到多少.本文还讨论了(2)式的收敛速度,这自然会与f在R_1上的整体性质有关系.最后我们将证明:若除一致连续外对f不作任何假定,则(2)式的收敛速度可任意慢.

在此需要提到,Loftsgarden等提出的估计$\hat{f}_n$,以及上文所引的一些与此估计有关的工作,多是对一般的高维密度讨论的.就本文结果而言,对高维密度也成立,但第2节的证法则只对一维有效.为行文简单计,我们全部局限于一维的情况来讨论.本文常以k简记$k(n)$,且将以较简单的形式

$$\hat{f}_n(x) = k/(2na_n(x)) \tag{1'}$$

取代(1)式,这对于本文的结果毫无影响.

1 $\hat{f}_n(x)$的收敛速度

在本节中,x总是固定的.我们想弄清楚$\hat{f}_n(x)-f(x)$能以多快的速度强收敛于0.例如,若存在$a>0$,使$n^a(\hat{f}_n(x)-f(x))\to 0$ a.s.,或写为$\hat{f}_n(x)-f(x)=0(n^{-a})$ a.s.,则这反映收敛速度的一种估计.

解决这个问题分两个步骤:定理1给出收敛速度不可逾越的上限,我们(在一定条件下)附带证明了$\hat{f}_n(x)$的渐近正态性,这个结果有其独立的意义.定理2则指出:适当选择$k(n)$,收敛速度确可以充分接近这个上限.

定理1 设$f(x)>0$,且存在自然数$r\geqslant 2$,使$f^{(r)}(x)$存在,而$f^{(2)}(x)=f^{(3)}(x)=\cdots=f^{(r-1)}(x)=0$(若$r=2$,后一条件消失).则

(1) 若$k\to\infty$,且$k=o(n^{2r/(2r+1)})$,则

$$\sqrt{k}(\hat{f}_n(x)-f(x))/f(x)\xrightarrow{c}N(0,1), \tag{3}$$

(2) 若存在常数$c>0$,致

$$\lim_{n\to\infty} cn^{2r/(2r+1)}/k = 1, \tag{4}$$

则

$$\sqrt{k}(\hat{f}_n(x)-f(x))/f(x)-\frac{c^{(2r+1)/2}f^{(r)}(x)}{(r+1)!2^r(f(x))^{r+1}}\xrightarrow{c}N(0,1). \tag{5}$$

(3) 若$f^{(r)}(x)\neq 0$,则在$k=o(n)$的条件下,变量

$$Z_n = n^{r/(2r+1)}\mid \hat{f}_n(x)-f(x)\mid. \tag{6}$$

当$n\to\infty$时,不依概率收敛于0.

(4) 若$f^{(r)}(x)\neq 0$, $\lim\limits_{n\to\infty}n^a/k=c$, $0<c<\infty$,对某个a,$\frac{2r}{2r+1}<a<1$,则当$b>r(1-a)$时,变量

$$W_n = n^b(\hat{f}_n(x)-f(x)),$$

不依概率收敛于0.

证 任取y,有

$$P(\sqrt{k}(\hat{f}_n(x)-f(x))/f(x)<y)=P(a_n(x)>b_n),\tag{7}$$

此处

$$b_n=k/(2nf(x)(1+y/\sqrt{k}))=O(k/n).\tag{8}$$

在(1)～(4)的条件下，都有 $b_n\to 0$. 又记

$$p_n=\int_{x-b_n}^{x+b_n}f(t)\mathrm{d}t,\ q_n=1-p_n.\tag{9}$$

引进 n 个独立同分布变量 $\xi_{n1},\cdots,\xi_{nn},\xi_{n1}$ 有分布

$$P(\xi_{n1}=1)=1-P(\xi_{n1}=0)=p_n,$$

并定义

$$\xi'_{ni}=(\xi_{ni}-p_n)/\sqrt{np_nq_n},\ i=1,\cdots,n.\tag{10}$$

由 $a_n(x)$的定义，得

$$P(a_n(x)>b_n)=P\Big(\sum_{i=1}^{n}\xi_{ni}<k\Big)=P\Big(\sum_{i=1}^{n}\xi'_{ni}<(k-np_n)/\sqrt{np_nq_n}\Big).\tag{11}$$

由于 $f(x)>0$, f 在 x 点连续，以及 $k\to\infty$, $k=o(n)$, 由(9)式不难验证 $p_n\to 0$, 以及

$$np_nq_n=k(1+o(1))\to\infty.\tag{12}$$

于是不难验证，$\{\xi'_{ni}\}$满足文献[5]中所谓“正态收敛准则”，因而

$$\sum_{i=1}^{n}\xi'_{ni}\xrightarrow{\mathcal{L}}N(0,1).\tag{13}$$

又由 $b_n\to 0$ 及 $f^{(2)}(x)=\cdots=f^{(r-1)}(x)=0$, $f^{(r)}(x)$ 存在，由(9)式，易见

$$p_n=2f(x)b_n+\frac{2}{(r+1)!}(f^{(r)}(x)+o(1))b_n^{r+1},\tag{14}$$

此处当 $n\to\infty$ 时，$o(1)\to 0$. 先设 $k=o(n^{2r/(2r+1)})$, 有

$$\begin{aligned}k-np_n&=k-k/(1+y/\sqrt{k})+O(k^{r+1}/n^r)\\&=yk/(\sqrt{k}+y)+\sqrt{k}O(k^{r-\frac{1}{2}}/n^r)\\&=y\sqrt{k}+o(\sqrt{k}).\end{aligned}$$

由此及(12)式，得 $\lim\limits_{n\to\infty}(k-np_n)/\sqrt{np_nq_n}=y$. 再利用(7), (11), (13)式，即得

$$\lim_{n\to\infty}P(\sqrt{k}(\hat{f}_n(x)-f(x))/f(x)<y)=\Phi(y)$$

对任何 y, Φ 为 $N(0,1)$的分布函数，这证明了(3)式.

再设 $k=(c+o(1))n^{2r/(2r+1)}$, 这时简单计算引出

$$\begin{aligned}k-np_n&=yk/(\sqrt{k}+y)-\frac{f^{(r)}(x)}{(r+1)!2^r(f(x))^{r+1}}\frac{k^{r+1}}{n^r}[1+o(1)]\\&=\sqrt{k}\Big(y-\frac{c^{(2r+1)/2}f^{(r)}(x)}{(r+1)!2^r(f(x))^{r+1}}+o(1)\Big).\end{aligned}$$

由此式及(12)式,并利用(7),(11),(13)式,即得(5)式.

现证(3).不难看出,通过在必要时取 n 的子序列的方法,可以归结为以下四种情况来讨论:

(a) $k \to \infty$, $k = o(n^{2r/(2r+1)})$.

(b) 存在常数 $c > 0$, 使(4)式成立.

(c) $\lim\limits_{n\to\infty} n^{2r/(2r+1)}/k = 0$,但 $k = o(n)$.

(d) k 与 n 无关.

先讨论(a). 由于 $f(x) > 0$, 有

$$P(n^{r/(2r+1)} \mid \hat{f}_n(x) - f(x) \mid < y) = P(\sqrt{k} \mid \hat{f}_n(x) - f(x) \mid / f(x) < y\sqrt{k/n^{2r/(2r+1)}}/f(x)).$$

故由 $k = o(n^{2r/(2r+1)})$ 及(3)式,知对任何 y,有

$$\lim_{n\to\infty} P(n^{r/(2r+1)} \mid \hat{f}_n(x) - f(x) \mid < y) = 0 \neq 1.$$

这证明了所要的结论.情况(b)的处理完全类似,只需用(5)式代替(3)式.现考虑情况(c).为确定起见,设 $f^{(r)}(x) = d > 0$, 仿照本定理前半部的方法,易得对任何 y,有

$$P(n^{r/(2r+1)} (\hat{f}_n(x) - f(x)) < y) = P\Big(\sum_{i=1}^{n} \xi'_{ni} < (k - np'_n)/\sqrt{np'_nq'_n}\Big). \tag{15}$$

此处 ξ'_{ni} 的定义过程与前面完全类似,唯一不同之处是,前面的 p_n 要以

$$p'_n = \int_{x-b'_n}^{x+b'_n} f(t)\mathrm{d}t \tag{16}$$

来代替,而 $q'_n = 1 - p'_n$, 又

$$b'_n = k/(2n(f(x) + yn^{-r/(2r+1)})). \tag{17}$$

仍有 $b'_n \to 0$,因而 $p'_n \to 0$,以及 $np'_nq'_n = k(1 + o(1))$. 有

$$k - np'_n = \frac{ykn^{-r/(2r+1)}}{f(x) + yn^{-r/(2r+1)}} - \frac{d + o(1)}{(r+1)!2^r(f(x))^{r-1}} \frac{k^{r+1}}{n^r}, \tag{18}$$

由于 $d > 0$, 以及

$$k^{r+\frac{1}{2}}/n^r = (\sqrt{k}/n^{r/(2r+1)})^{2r+1}, \sqrt{k}/n^{r/(2r+1)} \to \infty,$$

由(18)式得 $\lim\limits_{n\to\infty}(k - np'_n)/\sqrt{np'_nq'_n} = -\infty$. 再注意到对此处的 ξ'_{ni} 而言,(13)式仍成立.故由此式及(15)式,对任何 y 有

$$\lim_{n\to\infty} P(n^{r/(2r+1)}(\hat{f}_n(x) - f(x)) < y) = 0,$$

显然,这证明了本定理(3)的结论.

最后考虑情况(d).仍如在情况(c)那样定义 $\xi'_{n1}, \cdots, \xi'_{nn}$,即 $P(\xi'_{n1} = 1) = 1 - P(\xi'_{n1} = 0) = p'_n$, p'_n和b'_n 分别由(16),(17)式所定义,有

$$P(n^{r/(2r+1)}(\hat{f}_n(x)-f(x))<y)=P(\sum_{i=1}^{n}\xi'_{ni}<k). \tag{19}$$

由于 k 与 n 无关,依(16),(17)式及 f 在 x 点的连续性,有 $\lim\limits_{n\to\infty} np'_n=k$. 于是,按二项分布的 Poisson 逼近,得

$$\lim_{n\to\infty} P(\sum_{i=1}^{n}\xi'_{ni}<k)=\sum_{i=0}^{k-1}e^{-k}k^i/i! \tag{20}$$

与 y 无关. 显然,由(19),(20)式即得:(6)式定义的 Z_n 不能依概率收敛于 0,故证明了(3). (4)的证明与(3)类似. 定理 1 证毕.

注 1 (3)式可直接用于作 $f(x)$ 的大样本置信区间. 这有一个如何选择 k 的问题. 一般说来,由于 f 是未知的,对选定的点 x 无法知道满足条件

$$f^{(2)}(x)=\cdots=f^{(r-1)}(x)=0,\ f^{(r)}(x)\neq 0$$

的自然数 $r\geqslant 2$. 但如果假定 f 在 x 点有二阶导数,则总可取 $r=2$, 因此,可选满足以下条件的 k:

$$k=o(n^{\frac{4}{5}}),\ k/n^{\frac{4}{5}-\varepsilon}\to\infty \text{ 对任给 } \varepsilon>0. \tag{21}$$

如果 $f^{(2)}(x)$不存在,则可以证明:只要 $f(x)>0$, 而

$$\limsup_{h\to 0} |f(x+h)-f(x)|/|h|<\infty, \tag{22}$$

则在 $k\to\infty$, $k=o(n^{\frac{2}{3}})$时,(3)式仍成立.

先将下述已知结果表为引理 1.

引理 1 设变量 Y 服从二项分布 $B(n,p)$,则对 $\varepsilon>0$, 有

$$P(|Y/n-p|\geqslant\varepsilon)\leqslant 2\exp\left(-\frac{n\varepsilon^2}{2p+\varepsilon}\right). \tag{23}$$

证明不难由文献[6]的定理 3 得出.

定理 2 设 $f(x)>0$, 且

$$f^{(2)}(x)=\cdots=f^{(r-1)}(x)=0,\ f^{(r)}(x)\neq 0, \tag{24}$$

而

$$k=o(n), \tag{25}$$

则不论如何选择 k, $\hat{f}_n(x)-f(x)$ 收敛于 0 的速度达不到 $O(n^{-r/(2r+1)})$. 但对任何 $\alpha<r/(2r+1)$,适当选择 k, $\hat{f}_n(x)-f(x)$ 的收敛速度可达到 $O(n^{-\alpha})$. 更精密地,对任一串数 $c_n\to\infty$, 可适当选择 k,使

$$\hat{f}_n(x)-f(x)=O(c_n n^{-r/(2r+1)}\sqrt{\log n}) \text{ a.s.}, \tag{26}$$

又若(24)式成立,且对某个 a, $0<a<1$, 有

$$\lim_{n\to\infty} n^a/k=c \text{ 非零有限},$$

则当 $a\leqslant 2r/(2r+1)$ 时, $\hat{f}_n(x)-f(x)=O(n^{-a/2+\varepsilon})$a. s. ,对任给 $\varepsilon>0$,但$|\hat{f}_n(x)-f(x)=$

$O(n^{-a/2})$a. s. } 不成立. 若 $a > 2r/(2r+1)$, 则对任给 $\varepsilon > 0$ 有 $\hat{f}_n(x) - f(x) = O(n^{-r(1-a)+\varepsilon})$ a. s., 但 { $\hat{f}_n(x) - f(x) = O(n^{-r(1-a)-\varepsilon})$ a. s. } 不成立.

证 关于在条件(24), (25)之下, $\hat{f}_n(x) - f(x)$ 达不到 $O(n^{-r/(2r+1)})$ 这个断言, 是定理 1 的(3)的直接推论. 现设 $c_n \to \infty$, 任给 $\varepsilon > 0$, 考虑事件

$$T_n = \left\{\frac{1}{c_n\sqrt{\log n}} n^{r/(2r+1)}(\hat{f}_n(x) - f(x)) \geqslant \varepsilon\right\}, \tag{27}$$

$\hat{f}_n(x)$定义中的 k 尚待选定. 有

$$P(T_n) = P(a_n(x) \leqslant d_n) = P(Y_n \geqslant k), \tag{28}$$

其中

$$d_n = k/(2n(f(x) + \varepsilon c_n n^{-r/(2r+1)}\sqrt{\log n})). \tag{29}$$

Y_n 服从二项分布 $B(n, p_n)$, 而

$$p_n = \int_{x-d_n}^{x+d_n} f(t)\mathrm{d}t \tag{30}$$

在条件(24), (25)之下有

$$p_n = kf(x)/(n(f(x) + \varepsilon c_n n^{-r/(2r+1)}\sqrt{\log n})) + O(k^{r+1}/n^{r+1}).$$

若取

$$k = [n^{2r/(2r+1)}],$$

则易见

$$\frac{k}{n} - p_n = (1 + o(1))\frac{\varepsilon c_n\sqrt{\log n}}{f(x)n^{(r+1)/(2r+1)}}.$$

故若记

$$\eta_n = \varepsilon' c_n n^{-(r+1)/(2r+1)}\sqrt{\log n} \quad (\varepsilon' = \varepsilon/(2f(x))),$$

依引理 1, 当 n 充分大时有

$$P(Y_n \geqslant k) \leqslant P(|Y_n/n - p_n| \geqslant \eta_n) \leqslant 2\exp(-n\eta_n^2/(2k/n + \eta_n)),$$

不失普遍性, 可设 $c_n \leqslant n^{r/(2r+2)}$. 当 n 充分大时, $\eta_n < k/n$, 故

$$P(Y_n \geqslant k) \leqslant 2\exp(-n\eta_n^2 n/(3k)) \leqslant 2\exp(-\varepsilon'^2 c_n^2 \log n/4).$$

由于 $c_n \to \infty$, 当 n 充分大时有

$$P(Y_n \geqslant k) \leqslant n^{-2}. \tag{31}$$

由(27), (28), (31)式得

$$\sum_{n=2}^{\infty} P\left(\frac{1}{c_n\sqrt{\log n}} n^{r/(2r+1)}(\hat{f}_n(x) - f(x)) \geqslant \varepsilon\right) < \infty. \tag{32}$$

同样可证明对任给 $\varepsilon > 0$ 有

$$\sum_{n=2}^{\infty} P\left(\frac{1}{c_n\sqrt{\log n}} n^{r/(2r+1)}(\hat{f}_n(x) - f(x)) \leqslant -\varepsilon\right) < \infty. \tag{33}$$

由(32), (33)式得知,对任给 $\varepsilon > 0$, 有

$$\sum_{n=2}^{\infty} P\Big(\frac{1}{c_n\sqrt{\log n}} n^{r/(2r-1)} \mid \hat{f}_n(x) - f(x) \mid \geqslant \varepsilon\Big) < \infty.$$

故由 Borel-Cantelli 引理知(26)式成立,这证明了定理 2 的前半部. 定理 2 的后半部的证明是根据定理 1 的(4)和引理 1,仿照前半部的证明而得,故细节从略.

注 2 当然,本定理的主要意义在于指明在一定条件下,在 k 的最佳选择下, $\hat{f}_n(x) - f(x)$ 收敛于 0 的速度的界限,而不在于用它决定 k 的选择. 因为,对一个具体的 x,适合条件(24)式的 $r \geqslant 2$ 无法知道. 但如果有理由认为 $f^{(2)}$ 存在,且 $f^{(2)}(x) = 0$ 是一个例外,而非经常出现的,则本定理指出,最好选择 $k = [n^{\frac{4}{5}}]$. 因若选择 $k = [n^a]$, $0 < a < 1$,则当 $a < 4/5$ 时,收敛速度达不到 $O(n^{-a/2})$;而当 $a > 4/5$ 时,收敛速度也达不到 $O(n^{-2(1-a)})$,这都比选择 $k = [n^{\frac{4}{5}}]$时所能达到的 $O(n^{-\frac{2}{5}}(\log n)^{1-\varepsilon})$ 为低.

2 $\hat{f}_n(x)$一致收敛的速度

前面已提到 Devroye 等的结果(见(2)式). 本节的目的是在 f 满足进一步的条件下,来研究这个收敛的速度. 在此需要 Finkelstein 的一个结果,我们将其列为引理 2.

引理 2 设 x_1, …, x_n 是从具有连续分布 F 的一维总体中抽出的 iid. 样本,而 F_n 为其经验分布,则

$$\limsup_{n\to\infty}[\sup_x \sqrt{n} \mid F_n(x) - F(x) \mid / \sqrt{2\log\log n}] = 1 \text{ a. s.} \tag{34}$$

证明见文献[7].

定理 3 设 f 在 R_1 上满足 Lipschitz 条件. 若取 k,使

$$\lim_{n\to\infty} n^{\frac{2}{3}}(\log\log n)^{\frac{1}{3}}/k \text{ 非零有限}, \tag{35}$$

则

$$\sup_x \mid \hat{f}_n(x) - f(x) \mid = O(n^{-1/6}(\log\log n)^{1/6}) \text{ a. s.} \tag{36}$$

证 令 $N_n(a, b) = \#\{i: 1 \leqslant i \leqslant n, a \leqslant x_i \leqslant b\}$, $A_n = 2\sqrt{\log\log n/n}$, 则由引理 2 知,以概率为 1,当 n 充分大时,对一切 $a < b$ 同时有

$$n(F(b) - F(a) - A_n) \leqslant N_n(a, b) \leqslant n(F(b) - F(a) + A_n), \tag{37}$$

$F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t$. 存在常数 R,使 $\mid f(x) - f(y) \mid \leqslant R \mid x - y \mid$. 记

$$B = \{x: f(x) > (R+2)n^{-1/6}(\log\log n)^{1/6}\},$$

分别考虑 $x \in B$ 及 $x \overline{\in} B$ 两种情况.

(1) $x \in B$,因为对任何 $d > 0$ 有

$$2f(x)d - Rd^2 \leqslant \int_{x-d}^{x+d} f(t)\mathrm{d}t \leqslant 2f(x)d + Rd^2, \tag{38}$$

故由此及(37)式即得 $d_1 \leqslant a_n(x) \leqslant d_2$，其中 d_1，d_2 分别由

$$Rd_1^2 + 2f(x)d_1 = k/n - A_n, \tag{39}$$

$$-Rd_2^2 + 2f(x)d_2 = k/n + A_n \tag{40}$$

所决定.(39)式给出

$$d_1 = \frac{1}{R}\left[-f(x) + f(x)\left(1 + \frac{R(k/n - A_n)}{f^2(x)}\right)^{\frac{1}{2}}\right]. \tag{41}$$

因为当 $x \in B$ 时，

$$f^2(x) > (R+2)^2 n^{-\frac{1}{3}} (\log\log n)^{\frac{1}{3}} \geqslant (R+2)^2 k/n \geqslant 8R(k/n - A_n),$$

且对 $0 < x < 1$ 有 $\sqrt{1+x} > 1 + x/2 - x^2$，故由(41)式得

$$d_1 \geqslant \frac{(k/n) - A_n}{2f(x)} - \frac{R((k/n) - A_n)^2}{f^3(x)} = \frac{(k/n) - A_n}{2f(x)}\left[1 - \frac{2R((k/n) - A_n)}{f^2(x)}\right]. \tag{42}$$

由(1′)，(42)式及 $a_n(x) \geqslant d_1$，得

$$\hat{f}_n(x) \leqslant \frac{k}{n}\,\frac{f(x)}{(k/n) - A_n}\left[1 + \sum_{r=1}^{\infty}\left(\frac{2R((k/n) - A_n)}{f^2(x)}\right)^r\right].$$

因而(仍利用 $x \in B$)

$$\begin{aligned}\hat{f}_n(x) - f(x) &\leqslant \frac{A_n f(x)}{(k/n) - A_n} + \frac{2R}{f(x)}\,\frac{k}{n}\left[1 - \frac{2R((k/n) - A_n)}{f^2(x)}\right]^{-1} \\ &\leqslant \frac{nA_n}{k - nA_n} f(x) + \frac{4Rk}{nf(x)}.\end{aligned} \tag{43}$$

由假定易知

$$\sup_x f(x) = M < \infty. \tag{44}$$

又当 n 充分大时，$A_n \leqslant k/(2n)$，故由(43)式，有

$$\hat{f}_n(x) - f(x) \leqslant (2M+4) n^{-1/6} (\log\log n)^{1/6} \quad (n \text{ 充分大}). \tag{45}$$

同样.由(40)式解出 d_2，利用 $\sqrt{1-x} > 1 - \frac{x}{2} - x^2$（当 $0 < x < 1$ 时），可以证得

$$\hat{f}_n(x) - f(x) \geqslant -(M+2) n^{-1/6} (\log\log n)^{1/6} \quad (n \text{ 充分大}). \tag{46}$$

(2) $x \overline{\in} B$. 由(38)式可知，要 $\int_{x-d}^{x+d} f(t)\mathrm{d}t \geqslant k/n - A_n$，必有 $d \geqslant n^{-1/6}(\log\log n)^{1/6}/[4(R+2)]$. 于是由(37)式易得

$$\hat{f}_n(x) \leqslant (k/n) 2(R+2) n^{1/6} (\log\log n)^{-1/6} \leqslant 2(R+2) n^{-1/6} (\log\log n)^{1/6}.$$

由此式及 $x \overline{\in} B$，得

$$|\hat{f}_n(x) - f(x)| \leqslant 2(R+2) n^{-1/6} (\log\log n)^{1/6} \quad (n \text{ 充分大}), \tag{47}$$

由(45)~(47)式即得(36)式. 定理 3 证毕.

注 3 本定理所确定的 $\sup\limits_x |\hat{f}_n(x)-f(x)|$ 的收敛于 0 的阶,当然有可能被改进. 但是,若只假定 f 满足 Lipschitz 条件,这个阶达不到 $O(n^{-1/4}(\log\log n)^{-1/4})$. 事实上,取一密度函数 f,满足条件

(a) $f(x)=|x|$, 当 $|x|$ 充分小时.

(b) f 在 R_1 上满足 Lipschitz 条件.

(c) 至少存在一点 x_0,使 $f(x_0)>0$ 且 $f''(x_0)$ 存在有限.

通过在必要时取子序列的方法,可设 k 满足下述两条件之一: $k/\sqrt{n\log\log n}\to\infty$ 或 $k=O(\sqrt{n\log\log n})$. 对前者而言. 由(37)式有 $a_n(0)\leqslant\sqrt{(k/n)+A_n}\leqslant 2\sqrt{k/n}$ (当 n 充分大时),故 $\hat{f}_n(0)-f(0)=\hat{f}_n(0)\geqslant\frac{1}{4}\sqrt{k/n}\geqslant\frac{1}{4}(\log\log n/n)^{1/4}$. 对后者而言,依定理 1 的(1)(取 $r=2$),知 $\hat{f}_n(x_0)-f(x_0)$ 收敛于 0 的速度达不到 $O(1/\sqrt{k})$ 即 $O((n\log\log n)^{-1/4})$. 从而证明了所要的结果.

由定理 3 不难得到下面的结果:

定理 4 设 $-\infty=a_0<a_1<\cdots<a_m<a_{m+1}=\infty$,在每个开区间 $(a_i,\ a_{i+1})$ 内,f 满足 Lipshitz 条件,且 f 的方差有限,即

$$\int_{-\infty}^{\infty}x^2f(x)\mathrm{d}x<\infty. \tag{48}$$

又 k 满足条件(35),则对任何 $r>1$ 有

$$\hat{f}_n\xrightarrow{L_r}f\left(\text{即}\int_{-\infty}^{\infty}|\hat{f}_n(x)-f(x)|^r\mathrm{d}x\to 0\right)\ \text{a. s.} \tag{49}$$

证 首先,不难看出,在本定理条件下,用定理 3 的证法不难证明:对任给 $\varepsilon>0$, 有

$$\sup\{|\hat{f}_n(x)-f(x)|:\ x\in B_\varepsilon\}=O(n^{-1/6}(\log\log n)^{1/6})\ \text{a. s.} \tag{50}$$

此处 $B_\varepsilon=(-\infty,\ \infty)-\bigcup\limits_{i=1}^{m}(a_i-\varepsilon,\ a_i+\varepsilon)$. 由本定理条件知(44)式成立. 再由(37), $A_n=o(k/n)$ 及 $\hat{f}_n$的定义,易见

$$\sup_x\hat{f}_n(x)\leqslant 2M,\text{当 } n \text{ 充分大 a. s.} \tag{51}$$

又由(48)式有 $1-F(x)=\int_x^{\infty}f(t)\mathrm{d}t=o(x^{-2})$ (当 $x\to\infty$ 时),故为要 $1-F(x)\geqslant k/n-A_n$, 必须

$$x\leqslant\sqrt{n/(2k)}<n^{1/6}(\log\log n)^{-1/6}\quad(n\text{ 充分大时}).$$

由此可知,若记 $c_n=2n^{1/6}(\log\log n)^{-1/6}$, 则

$$x\geqslant c_n\Rightarrow a_n(x)\geqslant x/2\Rightarrow\hat{f}_n(x)\leqslant 1/x\ \text{a. s.},$$

故有

$$\begin{aligned}\int_{c_n}^{\infty}|\hat{f}_n(x)-f(x)|^r\mathrm{d}x&\leqslant 2^{r-1}\left(\int_{c_n}^{\infty}x^{-r}\mathrm{d}x+\int_{c_n}^{\infty}f^r(x)\mathrm{d}x\right)\\&\leqslant 2^{r-1}(c_n^{1-r}/(r-1)+M^{r-1}(1-F(c_n)))\\&\to 0\ \text{a. s.}\end{aligned} \tag{52}$$

同样得出

$$\int_{-\infty}^{-c_n} | \hat{f}_n(x) - f(x) |^r \mathrm{d}x \to 0 \text{ a. s.} \tag{53}$$

任给 $\varepsilon > 0$,取 $B_{n,\varepsilon} = (-c_n, c_n) - \bigcup_{i=1}^{m}(a_i - \varepsilon, a_i + \varepsilon)$,则由(50)式,得

$$\int_{B_{n,\varepsilon}} | \hat{f}_n(x) - f(x) |^r \mathrm{d}x = O(c_n n^{-r/6} (\log\log n)^{r/6}) \to 0 \text{ a. s.} \tag{54}$$

另一方面,由(44),(51)式知,当 n 充分大时,有 $\sup_x | \hat{f}_n(x) - f(x) | \leqslant 2M$,故

$$\sum_{i=1}^{m} \int_{a_i-\varepsilon}^{a_i+\varepsilon} | \hat{f}_n(x) - f(x) |^r \mathrm{d}x \leqslant 2m\varepsilon (2M)^r \text{ a. s.} \tag{55}$$

由 $\varepsilon > 0$ 的任意性,从(52)~(55)式即得(49)式.定理 4 证毕.

注 4 不论怎样选择 $k = o(n)$,则由强大数律易见对充分大的 n 有 $\hat{f}_n(x) \geqslant k/(4nx)$ a. s.(当 x 充分大时),故当 n 充分大时,

$$\int_{-\infty}^{\infty} \hat{f}_n(x) \mathrm{d}x = \infty \text{ a. s.}$$

因而定理 4 当 $r = 1$ 时不再成立.

最后我们证明:若除在 R_1 一致连续外不对 f 作任何进一步的假定,则(2)式的收敛速度可任意慢.更确切地,有

定理 5 给定任意趋于∞的常数序列 c_n,以及 $k(n)$满足条件

$$k(n)/n \to 0,\ k(n) > (2+\delta)\log n, \tag{56}$$

对某个 $\delta > 0$,则存在 R_1 上一致连续的密度 f,使

$$\sup_x | \hat{f}_n(x) - f(x) | \geqslant \frac{1}{c_n} \text{ (当 } n \text{ 充分大) a. s.} \tag{57}$$

证 对给定的常数 a, b, d, $a > 0$, $b > 0$, $0 < d < 2ab$,选择一个定义于 $|x| \leqslant b$ 的函数 $g(a, b, d, x)$,满足条件

(1) $g(a, b, d, 0) = a$, $g(a, b, d, \pm b) = 0$.

(2) $0 \leqslant g(a, b, d, x) \leqslant a$ (当 $|x| \leqslant b$ 时).

(3) $g(a, b, d, x)$在$[-b, b]$连续.

(4) $\int_{-b}^{b} g(a, b, d, x)\mathrm{d}x = d$.

这种函数的存在是显然的.定义

$$a_n = 3/(2c_n),\ b_n = c_n k(n)/n,\ d_n = \frac{1}{2^n},\ n \geqslant 1$$

$$e_1 = c_1 k(1),\ e_n = 2\sum_{i=1}^{n-1} c_i k(i)/i + c_n k(n)/n,\ n \geqslant 2.$$

引进定义于 R_1 的函数 f 如下:

$$f(x) = \begin{cases} g(a_n, b_n, d_n, x - e_n), & \text{当} \mid x - e_n \mid \leqslant b_n, \ n = 1, 2, \cdots, \\ 0, & \text{当} \ x < 0. \end{cases}$$

易见 f 为一在 R_1 上一致连续的密度. 现证(57)式. 定义 $\xi_n = N_n(e_n - b_n, e_n + b_n)$, 则由引理 1 及 f 的定义,即得

$$P(\mid \xi_n/n - d_n \mid \geqslant k(n)/(2n)) \leqslant 2\exp(-k(n)/2). \tag{58}$$

定义事件 $T_n = \{\mid \xi_n/n - d_n \mid \geqslant k(n)/(2n)\}$. 由(56), (58)式,依 Borel-Cantelli 引理,即得

$$P(\limsup T_n) = 0. \tag{59}$$

现任取 $\omega = (x_1, x_2, \cdots) \overline{\in} \limsup T_n$,则当 n 充分大时有 $\mid \xi_n/n - d_n \mid < k(n)/(2n)$. 由于 $d_n \leqslant k(n)/(2n)$,有 $\xi_n < k(n)$, 当 n 充分大,再由 $a_n(x)$的定义,有

$$a_n(e_n) > b_n \quad (\text{当 } n \text{ 充分大}),$$

因而 $\hat{f}_n(e_n) = k(n)/(2na_n(e_n)) < k(n)/(2nb_n) = 1/(2c_n)$, 而当 n 充分大时,

$$\mid \hat{f}_n(e_n) - f(e_n) \mid \geqslant 3/(2c_n) - 1/(2c_n) = 1/c_n,$$

这完成了定理 5 的证明.

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7

U-统计量的分布的非一致性收敛速度

摘要 Callaert 和 Janssen[1]在只假定核的三阶矩有限的条件下,得到了 U-统计量的分布的最佳一致性收敛速度 $O(n^{-1/2})$. 本文在同样条件下,得到了理想的非一致性收敛速度 $O(n^{-1/2}(1+|x|)^{-3})$.

1 引言

设 $\{X_n\}$ 为一串 iid. 随机变量,$h(x, y)$ 为两个变元 x, y 的对称函数,则称

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leqslant j < k \leqslant n} h(X_j, X_k) \tag{1}$$

为以 $h(x, y)$ 为核的 U-统计量. 为方便计,以下设 $\mathrm{E}h(X_1, X_2) = 0$. 记

$$\sigma_n^2 = \mathrm{Var}(U_n),\ \sigma_g^2 = \mathrm{E}g^2(X_1),\ \Phi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} \mathrm{e}^{-u^2/2}\mathrm{d}u,$$

其中, $g(x) = \mathrm{E}h(x, X_2)$. 1978 年,Callaert 和 Janssen 在文献[1]中证得,若 $\mathrm{E}|h(X_1, X_2)|^3 < \infty$,且 $\sigma_g^2 > 0$, 则

$$\sup_x |P(U_n/\sigma_n \leqslant x) - \Phi(x)| = O(n^{-1/2}). \tag{2}$$

这一结果后来为文献[2]所改进,该文在 $\mathrm{E}|h(X_1, X_2)|^{2+\delta} < \infty$ $(0 < \delta \leqslant 1)$ 的条件下,得到了理想的收敛速度 $O(n^{-\delta/2})$.

这些结果平行于独立随机变量和的经典理论. 近年来,在独立和的情形下发展了非一致性收敛速度的深刻理论,其难度远远超出了"一致性"的理论. 将 U-统计量与独立和类比,自然会提出这样的问题:对独立和建立的非一致性收敛速度,是否可以推广到 U-统计量? 作为这个方向上的一个初步结果,赵林城① 证得:在 $\mathrm{E}|h(X_1, X_2)|^3 < \infty$, $\sigma_g^2 > 0$ 的条件下,存在常数 C,使对所有 x 和充分大的 n,有

$$|P(U_n/\sigma_n \leqslant x) - \Phi(x)| \leqslant Cn^{-1/2}(1+|x|)^{-2}. \tag{3}$$

本文将(3)式右方改进为 $Cn^{-1/2}(1+|x|)^{-3}$, 从而得到了下面的结果.

本文 1982 年发表于《中国科学》(A 辑)第 12 卷第 12 期,合作者为赵林城.

① 见赵林城《U-统计量的非一致性界限》一文.

定理 1 设 $\mathrm{E}\mid h(X_1, X_2)\mid^3<\infty$,且 $\sigma_g^2=\mathrm{E}g^2(X_1)>0$, 则存在常数 C,使对所有 x 和 n,有

$$\mid P(U_n/\sigma_n\leqslant x)-\Phi(x)\mid\leqslant Cn^{-1/2}(1+\mid x\mid)^{-3}. \tag{4}$$

注意,此处的 C 不依赖于 n 和 x,它只依赖于核函数 h,以及 X_1 的分布.

2 若干引理

为证明定理 1,我们在本节证明若干引理.为书写简便,特作如下约定:

(1) 用 C 记一与 n 和 x 都无关的常数,每次出现,即使在同一式内,都可取不同之值.

(2) 经常出现的句式"存在与 n 无关的 $\eta>0$,使当 $\mid t\mid\leqslant\sqrt{n}\eta$ 且 n 充分大时,有……"和"存在与 n 无关的 $\eta>0$, $\mu>0$,使当 $\mid t\mid\leqslant\sqrt{n}\eta$ 且 n 充分大时,有……"分别记为"$\exists\,\eta$, s. t."和"$\exists\,\eta, \mu$, s. t.". 此外,"对充分大的 n"一语常常略去.

(3) 设 $g_1(t)$, $g_2(t)$ 为定义于 R^1 的两个函数(可以依赖于 n),称 $g_1\sim g_2$,若存在与 n 无关的常数 $\eta>0$, 使

$$\int_{|t|\leqslant\sqrt{n}\eta}\mid t\mid^{-1}\mid g_1(t)-g_2(t)\mid\mathrm{d}t\leqslant C/\sqrt{n}.$$

(4) 令 $\lambda_n=n-[n^{2/3}]$. $\sum{}'$ 表示求和范围为:每一指标从 1 到 λ_n,且当涉及的指标多于 1 个时,每一指标均小于后一指标. 例如,$\sum{}'\mathrm{E}(\zeta_1^2Y_{1k}^{*2})=\sum\limits_{k=2}^{\lambda_n}\mathrm{E}(\zeta_1^2Y_{1k}^{*2})$.

又,集 A 的指示函数记为 $I(A)$, $\#(A)$表示集 A 中不同元素的个数. i 总表示 $\sqrt{-1}$.

简记 $h(X_j, X_k)=h_{jk}$, $g(X_j)=g_j$, 令

$$S_n=\frac{1}{\sqrt{n}\sigma_g}\sum_{j=1}^n g_j\triangleq\frac{1}{\sqrt{n}}\sum_{j=1}^n\zeta_j,\ S_n'=\frac{1}{\sqrt{n}}\sum{}'\zeta_j,\ S_n''=S_n-S_n'.$$

$$Y_{jk}=h_{jk}-g_j-g_k,\ g_j=\mathrm{E}(h_{jk}\mid X_j).$$

$$\hat{h}_{jk}=h_{jk}I(\mid h_{jk}\mid\leqslant\sqrt{n}),\ h_{jk}^*=\hat{h}_{jk}-E\hat{h}_{jk}.$$

$$Y_{jk}^*=h_{jk}^*-g_j^*-g_k^*,\ g_j^*=\mathrm{E}(h_{jk}^*\mid X_j).$$

$$\tilde{h}_{jk}=h_{jk}I(\mid h_{jk}\mid\leqslant\sqrt{n}(1+\mid x\mid)),\ \tilde{h}_{jk}^*=\tilde{h}_{jk}-E\tilde{h}_{jk}.$$

$$\widetilde{Y}_{jk}^*=\tilde{h}_{jk}^*-\tilde{g}_j^*-\tilde{g}_k^*,\ \tilde{g}_j^*=\mathrm{E}(\tilde{h}_{jk}^*\mid X_j).$$

$$\Delta_n=\frac{\sqrt{n}}{2\sigma_g}\binom{n}{2}^{-1}\sum_{1\leqslant j<k\leqslant n}Y_{jk}\triangleq b_n\sum Y_{jk},\ \Delta_n^*=b_n\sum Y_{jk}^*,\ \widetilde{\Delta}_n^*=b_n\sum\widetilde{Y}_{jk}^*.$$

$$\Delta_{n1}^*=b_n\sum{}'Y_{jk}^*,\ \Delta_{n2}^*=\Delta_n^*-\Delta_{n1}^*=b_n\sum_{k=\lambda_n+1}^n\sum_{j=1}^{k-1}Y_{jk}^*.$$

易见

$$\frac{\sqrt{n}}{2\sigma_g}U_n=S_n+\Delta_n, \tag{5}$$

$$\mathrm{E}U_n^2=\binom{n}{2}^{-2}\left\{\binom{n}{2}\mathrm{E}h_{12}^2+n(n-1)(n-2)\sigma_g^2\right\}, \tag{6}$$

且有下述重要性质:

$$\begin{cases} Eg_j = Eg_j^* = E\,\tilde{g}_j^* = 0, \\ \{j,\ k\} \nsubseteq \{j_1,\ \cdots,\ j_l\} \Rightarrow E(Y_{jk}\varphi_{j_1\cdots j_l}) = E(Y_{jk}^*\varphi_{j_1\cdots j_l}) = E(\widetilde{Y}_{jk}^*\varphi_{j_1\cdots j_l}) = 0. \end{cases} \tag{*}$$

此处 $\varphi_{j_1\cdots j_l} = \varphi(X_{j_1},\ \cdots,\ X_{j_l})$ 为使上述期望存在的任一 Borel 可测函数.

引理 1 $E|S_n'|^3 \leqslant C$, $E|\Delta_{n2}^*|^3 \leqslant Cn^{-2}$, $E(S_n'\Delta_{n1}^*)^2 \leqslant Cn^{-1}$, $E|\Delta_{n1}^*|^m \leqslant Cn^{-m/2}$, $m=1, 2, 3, 4$.

证 前三个式子的证明见赵林城文(注 1). 为证 $E\Delta_{n1}^{*4} \leqslant Cn^{-2}$, 考虑

$$E\Delta_{n1}^{*4} \leqslant Cn^{-6}E\left(\sum{}'Y_{jk}^*\right)^4 \tag{7}$$

右方的展开式. 对于每一项 $Cn^{-6}E\prod_{l=1}^{4}{}'Y_{j_lk_l}^*$, 记 $A = \{j_l,\ k_l,\ l=1,\ \cdots,\ 4\}$. 若 A 中有一指标, 例如 j_1, 使 $j_1 \neq j_l$ 对 $l \neq 1$, 且 $j_1 \neq k_l$ 对任何 l, 则由($*$)有 $E\prod_{l=1}^{4}{}'Y_{j_lk_l}^* = 0$. 因此, 展开式中的非零项只有以下两种情形:

(1) $\#(A) \leqslant 3$. 因为这些项总数不超过 Cn^3, 因而它们的和的绝对值不超过

$$Cn^{-6}Cn^3EY_{12}^{*4} \leqslant Cn^{-3}EY_{12}^{*4} \leqslant Cn^{-3}\cdot C\sqrt{n} \leqslant Cn^{-5/2}.$$

(2) A 中恰有四个不同的指标, 每个指标都出现两次. 这些项总数不超过 Cn^4, 因此它们的和的绝对值不超过

$$\begin{aligned} &Cn^{-6}Cn^4(EY_{12}^{*2}Y_{34}^{*2} + E|Y_{12}^*Y_{23}^*Y_{34}^*Y_{14}^*|) \\ &\quad\leqslant Cn^{-2}[(EY_{12}^{*2})^2 + (EY_{12}^{*2}Y_{34}^{*2})^{1/2}(EY_{23}^{*2}Y_{14}^{*2})^{1/2}] \\ &\quad\leqslant Cn^{-2}(EY_{12}^{*2})^2 \\ &\quad\leqslant Cn^{-2}. \end{aligned}$$

由此即得 $E\Delta_{n1}^{*4} \leqslant Cn^{-2}$, 及 $E|\Delta_{n1}^*|^m \leqslant (E\Delta_{n1}^{*4})^{m/4} \leqslant Cn^{-m/2}$, $m = 1, 2, 3, 4$. 引理 1 证毕.

引理 2 $E(\widetilde{\Delta}_n^* - \Delta_n^*)^4 \leqslant Cn^{-5/2}(1+|x|)$, $E(\Delta_n - \widetilde{\Delta}_n^*)^2 \leqslant Cn^{-3/2}(1+|x|)^{-1}$.

证 令 $Z_{jk} = \widetilde{Y}_{jk}^* - Y_{jk}^*$, 则

$$\begin{aligned} EZ_{12}^4 &\leqslant C\sqrt{n}(1+|x|), \\ EZ_{12}^2 &\leqslant 3\{E(\tilde{h}_{12}^* - h_{12}^*)^2 + 2E(E[(\tilde{h}_{12}^* - h_{12}^*)\mid X_1])^2\} \\ &\leqslant 3\{E(\tilde{h}_{12}^* - h_{12}^*)^2 + 2E(E[(\tilde{h}_{12}^* - h_{12}^*)^2\mid X_1])\} \\ &\leqslant 9E(\tilde{h}_{12}^* - h_{12}^*)^2 \\ &\leqslant 9Eh_{12}^2I(\sqrt{n} < |h_{12}| \leqslant \sqrt{n}(1+|x|)) \\ &\leqslant 9n^{-1/2}E|h_{12}|^3 \\ &\leqslant Cn^{-1/2}, \end{aligned} \tag{8}$$

类似于(7)式, 有

$$\begin{aligned} E(\widetilde{\Delta}_n^* - \Delta_n^*)^4 &\leqslant Cn^{-6}E\left(\sum_{1\leqslant j<k\leqslant n}Z_{jk}\right)^4 \leqslant Cn^{-3}EZ_{12}^4 + Cn^{-2}(EZ_{12}^2)^2 \\ &\leqslant Cn^{-3}\cdot C\sqrt{n}(1+|x|) + Cn^{-2}\cdot(Cn^{-1/2})^2 \\ &\leqslant Cn^{-5/2}(1+|x|). \end{aligned}$$

另一方面，若令 $W_{jk}=Y_{jk}-\widetilde{Y}^{*}_{jk}$，则由(6)式，及类似于(8)式的论证，有

$$
\begin{aligned}
\mathrm{E}(\Delta_n-\Delta_n^{*})^2 &= \frac{n}{4\sigma_g^2}\binom{n}{2}^{-1}\mathrm{E}W_{12}^2\leqslant Cn^{-1}\mathrm{E}W_{12}^2\\
&\leqslant Cn^{-1}\cdot 9\mathrm{E}h_{12}^2I(|h_{12}|>\sqrt{n}(1+|x|))\\
&\leqslant Cn^{-3/2}(1+|x|)^{-1}\mathrm{E}|h_{12}|^3\\
&\leqslant Cn^{-3/2}(1+|x|)^{-1}.
\end{aligned}
$$

这就完成了引理 2 的证明.

引理 3 (1) 设 $W_n=W_{n1}+W_{n2}$，$n=1,2,\cdots$，为一串随机变量. 分别用 F_n，F_{n1} 记 W_n，W_{n1} 的分布函数. 设

$$|F_{n1}(x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}, \tag{9}$$

且对 $|x|\geqslant 1$，

$$P(|W_{n2}|\geqslant C|x|/\sqrt{n})\leqslant Cn^{-1/2}(1+|x|)^{-3},$$

则对充分大的 n，我们有

$$|F_n(x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}\ \text{对所有}\ x. \tag{10}$$

(2) 设 $W_n=a_nW_{n1}+b_n$，$n=1,2,\cdots$. $\{a_n\}$，$\{b_n\}$ 为实数序列，$|a_n-1|\leqslant C/\sqrt{n}$，$|b_n|\leqslant C/\sqrt{n}$，则当(9)式成立时(10)式也成立.

证 见文献[3]中引理 1.

引理 4 设 $G_n(x)$为非降函数，$H_n(x)$在 R^1 上有有界变差，它们的 F.S. 变换分别记为 g_n 和 h_n. 假定 $G_n(\pm\infty)=H_n(\pm\infty)$，$\int_{-\infty}^{\infty}|x|^3|\mathrm{d}(G_n(x)-H_n(x))|<\infty$，且

$$|H_n(x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}, \tag{11}$$

则对 $T_n=\sqrt{n}\eta$ ($\eta>0$ 与 n 及 x 无关)，有

$$
\begin{aligned}
|G_n(x)-H_n(x)|\leqslant C(1+|x|)^{-3}\Big\{&\int_{|t|\leqslant T_n}|t|^{-1}\cdot|g_n(t)-h_n(t)|\,\mathrm{d}t\\
&+\int_{|t|\leqslant T_n}|t|^{-1}\cdot|\delta_3^{(n)}(t)|\,\mathrm{d}t+C/\sqrt{n}\Big\}，\text{对充分大的}\ n\ \text{和所有}\ x，
\end{aligned}
\tag{12}
$$

此处，

$$\delta_3^{(n)}(t)=\int_{-\infty}^{\infty}\mathrm{e}^{itx}\mathrm{d}\{x^3(G_n(x)-H_n(x))\}. \tag{13}$$

证 由文献[4]，p. 157，引理 7，函数 $\Lambda_3^{(n)}(x)=x^3(G_n(x)-H_n(x))$ 在 R^1 上有有界变差. 为证(12)式，只需证明

$$
\begin{aligned}
\|\Lambda_3^{(n)}\|=\sup_x|\Lambda_3^{(n)}(x)|\leqslant C\Big\{&\int_{|t|\leqslant T_n}|t|^{-1}\cdot|g_n(t)-h_n(t)|\,\mathrm{d}t\\
&+\int_{|t|\leqslant T_n}|t|^{-1}\cdot|\delta_3^{(n)}(t)|\,\mathrm{d}t+C/\sqrt{n}\Big\}.
\end{aligned}
\tag{14}
$$

令 $\alpha = 80/\pi$,选取 y_n,使 $|\Lambda_3^{(n)}(y_n)| \geqslant \frac{1}{2}\|\Lambda_3^{(n)}\|$,且 $y_n \pm \alpha/(2T_n)$ 为 $G_n - H_n$ 的连续点.若 $|y_n| \leqslant 6\alpha$, 则由文献[4], p. 104,定理 1,注意到(11)式,有

$$\begin{aligned}\|\Lambda_3^{(n)}\| &\leqslant 2|\Lambda_3^{(n)}(y_n)| \leqslant 2(6\alpha)^3 |G_n(y_n) - H_n(y_n)| \\ &\leqslant C\int_{|t|\leqslant T_n} |t|^{-1}\cdot|g_n(t) - h_n(t)|\,\mathrm{d}t \\ &\quad + CT_n \sup_x \int_{|y|\leqslant C/T_n} |H_n(x+y) - H_n(x)|\,\mathrm{d}y \\ &\leqslant C\int_{|t|\leqslant T_n} |t|^{-1}\cdot|g_n(t) - h_n(t)|\,\mathrm{d}t \\ &\quad + CT_n \int_{|y|\leqslant C/T_n} 2\sup_x |H_n(x) - \Phi(x)|\,\mathrm{d}y \\ &\quad + CT_n \sup_x \int_{|y|\leqslant C/T_n} |\Phi(x+y) - \Phi(x)|\,\mathrm{d}y \\ &\leqslant C\int_{|t|\leqslant T_n} |t|^{-1}\cdot|g_n(t) - h_n(t)|\,\mathrm{d}t + C/\sqrt{n},\end{aligned}$$

这就证明了(14)式.

今设 $|y_n| > 6\alpha$, $\int_{|t|\leqslant T_n} |t|^{-1}\cdot|\delta_3^{(n)}(t)|\,\mathrm{d}t < \infty$. 由(11)式,对 $x \in [y_n - \alpha/T_n,\ y_n]$ 及充分大的 n, 有

$$\begin{aligned}|H_n(y_n) - H_n(x)| &\leqslant |H_n(y_n) - \Phi(y_n)| + |H_n(x) - \Phi(x)| + |\Phi(y_n) - \Phi(x)| \\ &\leqslant Cn^{-1/2}(1 + |y_n|)^{-3},\end{aligned} \tag{15}$$

由 $|y_n| > 6\alpha$ 及 $T_n > 1$,有

$$\frac{1}{2}|y_n| \leqslant |y_n| + 1 - \alpha/T_n \leqslant 1 + |y_n|.$$

若 $G_n(y_n) - H_n(y_n) < 0$ 且 $x \in [y_n - \alpha/T_n,\ y_n]$, 则由(15)式,有

$$\begin{aligned}G_n(x) - H_n(x) &\leqslant G_n(y_n) - H_n(y_n) + (H_n(y_n) - H_n(x)) \\ &\leqslant G_n(y_n) - H_n(y_n) + Kn^{-1/2}|y_n|^{-3},\end{aligned}$$

此处 $K > 0$ 为与 n, y_n 及 x 都无关的常数. 若 $|\Lambda_3^{(n)}(y_n)| \leqslant 2K/\sqrt{n}$,则(14)式显然成立,故不妨设 $|\Lambda_3^{(n)}(y_n)| > 2K/\sqrt{n}$,此时 $Kn^{-1/2}|y_n|^{-3} < \frac{1}{2}|G_n(y_n) - H_n(y_n)|$. 因此,

$$G_n(x) - H_n(x) < -\frac{1}{2}|G_n(y_n) - H_n(y_n)|.$$

若 $G_n(y_n) - H_n(y_n) > 0$,且 $x \in [y_n,\ y_n + \alpha/T_n]$, 则同样有

$$G_n(x) - H_n(x) > \frac{1}{2}|G_n(y_n) - H_n(y_n)|.$$

令 $\beta_n = y_n - \alpha/(2T_n)$ 或 $y_n + \alpha/(2T_n)$,视 $G_n(y_n) - H_n(y_n) < 0$ 或 > 0 而定,则对 $x \in [\beta_n - \alpha/(2T_n),\ \beta_n + \alpha/(2T_n)]$, 有

$$|x|^3 \geqslant ||y_n| - \alpha/T_n|^3 \geqslant |y_n|^3\left(1-\frac{1}{6}\right)^3 \geqslant \frac{1}{2}|y_n|^3,$$

故如上所述，$\Lambda_3^{(n)}(x)$有确定的符号，且

$$|\Lambda_3^{(n)}(x)| \geqslant \frac{1}{4}|\Lambda_3^{(n)}(y_n)|; \text{当 } |x-\beta_n| \leqslant \alpha/(2T_n). \tag{16}$$

如所周知，$q_n(x) = \dfrac{1-\cos T_n x^2}{\pi T_n x^2}$ 为一概率密度，其 c.f. 为 $v_n(t) = \left(1-\dfrac{|t|}{T_n}\right) I\ (|t| \leqslant T_n)$，因此，$\delta_3^{(n)}(t)v_n(t)$为

$$\Lambda_3^{*(n)}(x) = \int_{-\infty}^{\infty} \Lambda_3^{(n)}(x-u)q_n(u)\mathrm{d}u \tag{17}$$

的 F. S. 变换. 由反转公式，我们有

$$\Lambda_3^{*(n)}(\beta_n) - \Lambda_3^{*(n)}(\gamma) = \frac{1}{2\pi}\int_{-T_n}^{T_n} \frac{\mathrm{e}^{-\mathrm{i}t\beta_n} - \mathrm{e}^{-\mathrm{i}t\gamma}}{-\mathrm{i}t}\delta_3^{(n)}(t)v_n(t)\mathrm{d}t. \tag{18}$$

设 $x>0$，则

$$\begin{aligned} |G_n(x) - H_n(x)| &= \left|\int_x^{\infty} \mathrm{d}(G_n(y) - H_n(y))\right| \\ &\leqslant x^{-3}\int_x^{\infty} y^3\,|\mathrm{d}(G_n(y) - H_n(y))|, \end{aligned}$$

因此，$\lim\limits_{x\to\infty} x^3(G_n(x) - H_n(x)) = 0$，同理 $\lim\limits_{x\to-\infty} \Lambda_3^{(n)}(x) = 0$. 因为 $|\Lambda_3^{(n)}(x-u)q_n(u)| \leqslant Cq_n(u)$，且$\int_{-\infty}^{\infty} q_n(u)\mathrm{d}u = 1$，故由控制收敛定理，若在(17) 式中令 $x \to -\infty$，有

$$\lim_{\gamma\to-\infty} \Lambda_3^{*(n)}(\gamma) = 0, \tag{19}$$

由 Riemann-Lebesgue 引理，在(18)式中令 $\gamma \to -\infty$，有

$$\Lambda_3^{*(n)}(\beta_n) = \frac{1}{2\pi}\int_{|t|\leqslant T_n} \frac{\mathrm{e}^{-\mathrm{i}t\beta_n}}{-\mathrm{i}t}\delta_3^{(n)}(t)v_n(t)\mathrm{d}t. \tag{20}$$

由(16)，(17)式，并注意到当 $|x-\beta_n| \leqslant \alpha/(2T_n)$ 时 $\Lambda_3^{(n)}(x)$ 保持确定的符号，我们有

$$\begin{aligned} |\Lambda_3^{*(n)}(\beta_n)| &= \left|\int_{-\infty}^{\infty} \Lambda_3^{(n)}(\beta_n - u)q_n(u)\mathrm{d}u\right| \\ &\geqslant \frac{1}{4}|\Lambda_3^{(n)}(y_n)|\int_{|u|\leqslant\alpha/(2T_n)} q_n(u)\mathrm{d}u - \|\Lambda_3^{(n)}\|\int_{|u|>\alpha/(2T_n)} q_n(u)\mathrm{d}u \\ &\geqslant \|\Lambda_3^{(n)}\|\left(\frac{1}{8}\int_{|u|\leqslant\alpha/(2T_n)} q_n(u)\mathrm{d}u - \int_{|u|>\alpha/(2T_n)} q_n(u)\mathrm{d}u\right) \\ &= \|\Lambda_3^{(n)}\|\left(\frac{1}{8} - \frac{9}{8}\int_{|u|>\alpha/(2T_n)} q_n(u)\mathrm{d}u\right) \\ &\geqslant \|\Lambda_3^{(n)}\|\left(\frac{1}{8} - \frac{9}{8}\cdot\frac{2}{\pi T_n}\int_{|u|>\alpha/(2T_n)} u^{-2}\mathrm{d}u\right) \\ &= \|\Lambda_3^{(n)}\|\left(\frac{1}{8} - \frac{9}{\pi\alpha}\right) \end{aligned}$$

$$=\frac{1}{80}\|\Lambda_3^{(n)}\|. \tag{21}$$

由(20)及(21)式即得(14)式,因而引理4得证.

引理5 设$\varphi(t)$在$|t|\leqslant T$内有三阶连续导数$\varphi^{(3)}(t)$,且$\varphi^{(j)}(0)=0$, $j=0,1,2$,则

$$\int_{-T}^{T}|t|^{j-4}\cdot|\varphi^{(j)}(t)|\,\mathrm{d}t\leqslant\int_{-T}^{T}|t|^{-1}\cdot|\varphi^{(3)}(t)|\,\mathrm{d}t,\ j=0,1,2.$$

证 见文献[3],引理2.

引理6 $\exists\,\eta,\mu$, s.t.

$$|\mathrm{E}\mathrm{e}^{itS_n''}|\leqslant\exp(-\mu n^{-1/3}t^2),$$
$$|\mathrm{E}S_n''\mathrm{e}^{itS_n''}|\leqslant Cn^{-1/3}|t|\exp(-\mu n^{-1/3}t^2),$$
$$|\mathrm{E}(S_n'')^2\mathrm{e}^{itS_n''}|\leqslant Cn^{-1/3}(1+t^2)\exp(-\mu n^{-1/3}t^2),$$
$$|\mathrm{E}(S_n'')^3\mathrm{e}^{itS_n''}|\leqslant Cn^{-2/3}(1+|t|^3)\exp(-\mu n^{-1/3}t^2).$$

证 见文献[3],引理6(注意:证明中未用到$|\zeta_j|\leqslant C\sqrt{n}$).

引理7 记$\varepsilon_{1j}=\begin{cases}0, j=1\\1, j\neq 1,\end{cases}$ 则$\exists\,\eta,\mu$, s.t.

$$|\mathrm{E}\Delta_{n1}^{*j}\mathrm{e}^{itS_n'}|\leqslant Cn^{-1/2}(\varepsilon_{1j}+|t|^{j+1})\mathrm{e}^{-\mu t^2},\ j=1,2,3, \tag{22}$$

$$|\mathrm{E}\mathrm{e}^{itS_n'}(\mathrm{e}^{it\Delta_{n1}^*}-1)|\leqslant Cn^{-1/2}(t^2+|t|^7)\mathrm{e}^{-\mu t^2}+Cn^{-2}t^4. \tag{23}$$

证 为证(22)式,我们只考虑$j=3$的情形,其他情形证明类似.记

$$\varphi_n(t)=\mathrm{E}\exp(it\zeta_1/\sqrt{n}),$$

由文献[5]引理2,$\exists\,\eta,\mu$, s.t.

$$|\varphi_n(t)|^{\lambda_n-10}\leqslant\mathrm{e}^{-\mu t^2}. \tag{24}$$

对于

$$|\mathrm{E}\Delta_{n1}^{*3}\mathrm{e}^{itS_n'}|\leqslant Cn^{-9/2}|\mathrm{E}(\textstyle\sum{}'Y_{jk}^*)^3\mathrm{e}^{itS_n'}| \tag{25}$$

右方展式中的每一项$Cn^{-9/2}\mathrm{E}(\prod_{l=1}^{3}{}'Y_{j_lk_l}^*)\mathrm{e}^{itS_n'}$,记$A=\{j_l,k_l,l=1,2,3\}$.对于$\#(A)\leqslant 4$的那些项,总数不超过$Cn^4$,故这些项的和的绝对值不超过

$$Cn^4\cdot Cn^{-9/2}|\varphi_n(t)|^{\lambda_n-4}\leqslant Cn^{-1/2}\mathrm{e}^{-\mu t^2},\text{当}|t|\leqslant\sqrt{n}\eta\text{时}. \tag{26}$$

对于$\#(A)=5$的那些项,总数不超过Cn^5.由(*)和(24)式,这些项的绝对值不超过

$$\begin{aligned}&Cn^5\cdot Cn^{-9/2}|\mathrm{E}Y_{12}^*Y_{13}^*Y_{45}^*\mathrm{e}^{itS_n'}|\\&\leqslant Cn^{1/2}\mathrm{E}|Y_{12}^*Y_{13}^*|\cdot|\mathrm{E}Y_{45}^*(\mathrm{e}^{it\zeta_4/\sqrt{n}}-1)(\mathrm{e}^{it\zeta_5/\sqrt{n}}-1)|\cdot|\varphi_n(t)|^{\lambda_n-5}\\&\leqslant Cn^{1/2}\cdot Cn^{-1}t^2\mathrm{E}|Y_{45}^*\zeta_4\zeta_5|\mathrm{e}^{-\mu t^2}\\&\leqslant Cn^{-1/2}t^2\mathrm{e}^{-\mu t^2},\text{当}|t|\leqslant\sqrt{n}\eta\text{时}.\end{aligned} \tag{27}$$

同理,$\#(A)=6$的那些项的和的绝对值不超过

$$Cn^6 \cdot Cn^{-9/2} \mid EY_{12}^* Y_{34}^* Y_{56}^* e^{itS_n'} \mid \leqslant Cn^{-1/2} t^4 e^{-\mu t^2}, \text{当} \mid t \mid \leqslant \sqrt{n}\eta \text{时}. \tag{28}$$

由(25)～(28)式即得(22)式 ($j = 3$). 最后,由引理 1 及(22)式,以及

$$\mid Ee^{itS_n'}(e^{it\Delta_{n1}^*} - 1) \mid \leqslant \sum_{j=1}^{3} \mid t \mid^j \cdot \mid E\Delta_{n1}^{*j} e^{itS_n'} \mid + t^4 E\Delta_{n1}^{*4},$$

即得(23)式. 引理 7 证毕.

引理 8 令 $W = W(X_1, X_2) = \zeta_1^2\zeta_2$ 或 $\zeta_1 Y_{12}^*$, $\overline{S}_n = S_n'$, $\overline{U}_n = \Delta_{n1}^*$ 或 $b_n \sum_{j\geqslant 2}{}' Y_{jk}^*$, 则 $\exists\, \eta, \mu$, s. t.

$$\mid EWe^{it\overline{S}_n}(e^{it\overline{U}_n} - 1) \mid \leqslant Cn^{-1}(\mid t \mid + t^8)e^{-\mu t^2} + Cn^{-2}t^4.$$

证 不失一般性,我们只需考虑 $\overline{U}_n = \Delta_{n1}^*$ 的情形. 当$\overline{U}_n = b_n \sum_{j\geqslant 2}{}' Y_{jk}^*$ 时,下面各式中的 $\sum{}' Y_{1k}^*$, $\sum_{k\geqslant 3}{}' Y_{1k}^*$, Y_{13}^* 等不会出现,因而证明只会简单些. 易见,

$$\begin{aligned} \mid EWe^{itS_n'}(e^{it\Delta_{n1}^*} - 1) \mid &\leqslant \sum_{j=1}^{3} \mid t \mid^j \cdot \mid EW\Delta_{n1}^{*j} e^{itS_n'} \mid + t^4 E \mid W\Delta_{n1}^{*4} \mid \\ &\triangleq \sum_{j=1}^{4} J_j(t). \end{aligned} \tag{29}$$

待证, $\exists\, \eta, \mu$, s. t.

$$J_j(t) \leqslant Cn^{-1}(\mid t \mid^j + t^{2+2j})e^{-\mu t^2},\ j = 1, 2, 3. \tag{30}$$

$$J_4(t) \leqslant Cn^{-2}t^4. \tag{31}$$

考虑到(*)及 $\mid Y_{jk}^* \mid \leqslant C\sqrt{n}$, 我们有

$$\begin{aligned} E \mid W(\textstyle\sum{}' Y_{1k}^*)^4 \mid &\leqslant 8E \mid W \mid Y_{12}^{*4} + 8E \mid W \mid (\sum_{k\geqslant 3}{}' Y_{1k}^*)^4 \\ &\leqslant 8E \mid W \mid Y_{12}^{*4} + 8nE \mid W \mid Y_{13}^{*4} + 8n^2 E \mid W \mid Y_{13}^{*2} Y_{14}^{*2} \\ &\leqslant Cn^2 + Cn^3 + 8n^2 \max\{E\zeta_1^2 Y_{13}^{*2} Y_{14}^{*2} E \mid \zeta_2 \mid, E \mid \zeta_1 Y_{12}^* \mid Y_{13}^{*2} Y_{14}^{*2}\} \\ &\leqslant Cn^{7/2}, \end{aligned}$$

同样, $E \mid W \mid (\sum{}' Y_{2k}^*)^4 \leqslant Cn^{7/2}$. 因此,由引理 1,

$$\begin{aligned} J_4(t) &\leqslant Cn^{-6}t^4\{E \mid W \mid (\textstyle\sum{}' Y_{1k}^*)^4 + E \mid W \mid (\sum{}' Y_{2k}^*)^4 + E \mid W \mid E(\sum_{j\geqslant 3}{}' Y_{jk}^*)^4\} \\ &\leqslant Cn^{-6}t^4\{2Cn^{7/2} + Cn^4\} \\ &\leqslant Cn^{-2}t^4, \end{aligned}$$

此即(31)式. 由(*),有

$$\begin{aligned} \mid EW\exp[it(\zeta_1 + \zeta_2)/\sqrt{n}] \mid &\leqslant E \mid W(e^{it\zeta_2/\sqrt{n}} - 1) \mid \\ &\leqslant \frac{\mid t \mid}{\sqrt{n}} E \mid W\zeta_2 \mid \\ &\leqslant C \mid t \mid / \sqrt{n}, \end{aligned} \tag{32}$$

又

$$\mathrm{E}\mid\zeta_1^2 Y_{13}^*\zeta_3\mid\leqslant(\mathrm{E}\mid\zeta_1^2\zeta_3\mid^{3/2})^{2/3}(\mathrm{E}\mid Y_{13}^*\mid^3)^{1/3}\leqslant C,$$

$$\mathrm{E}\mid\zeta_1 Y_{12}^* Y_{13}^*\zeta_3\mid\leqslant(\mathrm{E}\mid\zeta_1\zeta_3\mid^3)^{1/3}(\mathrm{E}\mid Y_{12}^*\mid^3)^{1/3}(\mathrm{E}\mid Y_{13}^*\mid^3)^{1/3}\leqslant C,$$

故

$$\mathrm{E}\mid WY_{13}^*\zeta_3\mid\leqslant C, \tag{33}$$

同理

$$\mathrm{E}\mid WY_{23}^*\zeta_3\mid\leqslant C. \tag{34}$$

为证(30)式,我们仍以 $j=3$ 为例,有

$$J_3(t)\leqslant Cn^{-9/2}\mid t\mid^3\cdot\mid\mathrm{E}W(\sum{}'Y_{jk}^*)^3\mathrm{e}^{\mathrm{i}tS_n'}\mid. \tag{35}$$

对于每一项 $Cn^{-9/2}\mid t\mid^3\mathrm{E}(W\prod\limits_{l=1}^{3}{}'Y_{j_lk_l}^*)\mathrm{e}^{\mathrm{i}tS_n'}$,记 $A=\{1,\ 2,\ j_l,\ k_l,\ l=1,\ 2,\ 3\}$. 容易验证,$\#(A)\leqslant 5$ 的那些项的和的绝对值不超过 $Cn^{-1}(\mid t\mid^3+t^4)\mathrm{e}^{-\mu t^2}$,当 $\mid t\mid\leqslant\sqrt{n}\eta$ 时. 又,$\#(A)=6$ 的那些项,其项数不超过 Cn^4,当 $\mid t\mid\leqslant\sqrt{n}\eta$ 时,这些项的和的绝对值不超过

$$\begin{aligned}
&Cn^{-9/2}\mid t\mid^3\cdot Cn^4\{\mid\mathrm{E}WY_{12}^*Y_{34}^*Y_{56}^*\mathrm{e}^{\mathrm{i}tS_n'}\mid+\mid\mathrm{E}W(Y_{13}^*+Y_{23}^*)Y_{34}^*Y_{56}^*\mathrm{e}^{\mathrm{i}tS_n'}\mid\\
&\quad+\mid\mathrm{E}W(Y_{13}^*+Y_{23}^*)Y_{45}^*Y_{56}^*\mathrm{e}^{\mathrm{i}tS_n'}\mid+\mid\mathrm{E}W(Y_{34}^*Y_{35}^*Y_{36}^*+Y_{34}^{*2}Y_{56}^*+Y_{34}^*Y_{35}^*Y_{46}^*)\mathrm{e}^{\mathrm{i}tS_n'}\mid\\
&\leqslant Cn^{-1/2}\mid t\mid^3\Big\{2Cn^{1/2}\mathrm{E}\mid W\mid\mathrm{E}\mid Y_{34}^*\mid\cdot\left(\frac{t^2}{n}\mathrm{E}\mid Y_{56}^*\zeta_5\zeta_6\mid\right)\cdot\mid\varphi_n(t)\mid^{\lambda_n-6}\\
&\quad+\mathrm{E}\mid W(Y_{13}^*+Y_{23}^*)\mid\left(\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\mid Y_{45}^*Y_{56}^*\zeta_6\mid\right)\mid\varphi_n(t)\mid^{\lambda_n-6}\\
&\quad+\mid\mathrm{E}W\exp[\mathrm{i}t(\zeta_1+\zeta_2)/\sqrt{n}]\mid\cdot C\cdot\mid\varphi_n(t)\mid^{\lambda_n-6}\Big\}\\
&\leqslant Cn^{-1}(t^4+\mid t\mid^5)\mathrm{e}^{-\mu t^2}.
\end{aligned}$$

类似地,$\#(A)=7$ 或 8 的那些项的和的绝对值不超过 $Cn^{-1}(t^6+t^8)\mathrm{e}^{-\mu t^2}$,当 $\mid t\mid\leqslant\sqrt{n}\eta$. 由此即得(30)式,引理 8 得证.

引理 9 $\exists\,\eta,\ \mu,$ s. t.

$$\mid\mathrm{E}(S_n')^j\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid\leqslant Cn^{-1/2}(\mid t\mid+\mid t\mid^{j+3})\mathrm{e}^{-\mu t^2}+Cn^{-1}t^2,\ j=0,\ 1,\ 2, \tag{36}$$

$$\mid\mathrm{E}(S_n')^3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid\leqslant\frac{C}{\sqrt{n}}(\mid t\mid+\mid t\mid^{11})\mathrm{e}^{-\mu t^2}+Cn^{-3/2}(t^2+t^4)+Cn^{-2}(\mid t\mid^5+\mid t\mid^7). \tag{37}$$

证 (36)式由赵林城工作可得到. 今证(37)式,有

$$\begin{aligned}
\mid\mathrm{E}(S_n')^3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid&\leqslant Cn^{-1/2}\mid\mathrm{E}\zeta_1^3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid\\
&\quad+Cn^{1/2}\mid\mathrm{E}\zeta_1^2\zeta_2\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid\\
&\quad+Cn^{3/2}\mid\mathrm{E}\zeta_1\zeta_2\zeta_3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid.
\end{aligned} \tag{38}$$

容易验证,$\exists\,\eta,\ \mu,$ s. t.

$$Cn^{-1/2}\mid\mathrm{E}\zeta_1^3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)\mid\leqslant Cn^{-1/2}(\mid t\mid+t^2)\mathrm{e}^{-\mu t^2}+Cn^{-3/2}t^2. \tag{39}$$

由引理 8, $\exists\, \eta,\ \mu$, s. t.

$$Cn^{1/2}\mid E\zeta_1^2\zeta_2 e^{itS_n'}(e^{it\Delta_{n1}^*}-1)\mid \leqslant Cn^{-1/2}(\mid t\mid+t^8)e^{-\mu t^2}+Cn^{-3/2}t^4. \tag{40}$$

今证, $\exists\, \eta,\ \mu$, s. t.

$$\begin{aligned}&Cn^{3/2}\mid E\zeta_1\zeta_2\zeta_3 e^{itS_n'}(e^{it\Delta_{n1}^*}-1)\mid\\&\leqslant Cn^{-1/2}(t^2+\mid t\mid^{11})e^{-\mu t^2}+Cn^{-3/2}(t^2+t^4)+Cn^{-2}(\mid t\mid^5+\mid t\mid^7).\end{aligned} \tag{41}$$

为此,记

$$\begin{aligned}\Delta_{n1}^* &= b_n(Y_{12}^*+Y_{13}^*+Y_{23}^*)+b_n\sum_{k\geqslant 4}{}'(Y_{1k}^*+Y_{2k}^*+Y_{3k}^*)+b_n\sum_{j\geqslant 4}{}'Y_{jk}^*\\&=V_0+V_1+V',\end{aligned}$$

则

$$\begin{aligned}(41)\text{式左方}\leqslant\ & Cn^{3/2}\mid E\zeta_1\zeta_2\zeta_3 e^{it(S_n'+V')}(e^{itV_0}-1)(e^{itV_1}-1)\mid\\&+Cn^{3/2}\mid E\zeta_1\zeta_2\zeta_3 e^{it(S_n'+V')}(e^{itV_0}-1)\mid\\&+Cn^{3/2}\mid E\zeta_1\zeta_2\zeta_3 e^{it(S_n'+V')}(e^{itV_1}-1)\mid\\&+Cn^{3/2}\mid E\zeta_1\zeta_2\zeta_3 e^{itS_n'}(e^{itV'}-1)\mid\\&\triangleq \sum_{j=1}^{4}J_j(t).\end{aligned} \tag{42}$$

由引理 7,当 $\mid t\mid\leqslant\sqrt{n}\,\eta$ 时,有

$$\begin{aligned}J_4(t)&\leqslant Cn^{3/2}\mid E\zeta_1(e^{it\zeta_1/\sqrt{n}}-1)\mid^3\cdot\left|E(e^{itV'}-1)\exp\left(it\sum_{j\geqslant 4}{}'\zeta_j/\sqrt{n}\right)\right|\\&\leqslant C\mid t\mid^3\left|E(e^{itV'}-1)\exp\left(it\sum_{j\geqslant 4}{}'\zeta_j/\sqrt{n}\right)\right|\\&\leqslant Cn^{-1/2}(\mid t\mid^5+t^{10})e^{-\mu t^2}+Cn^{-2}\mid t\mid^7.\end{aligned} \tag{43}$$

由对称性,有

$$\begin{aligned}J_1(t)&\leqslant Cn^{-3/2}t^2\mid E\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*+Y_{23}^*)\sum{}'Y_{3k}^*e^{it(S_n'+V')}\mid\\&\quad+Cn^{-9/2}t^4E\mid\zeta_1\zeta_2\zeta_3\mid(Y_{12}^*+Y_{13}^*+Y_{23}^*)^2\left(\sum{}'Y_{3k}^*\right)^2\\&\triangleq J_{11}(t)+J_{12}(t).\end{aligned} \tag{44}$$

而

$$\begin{aligned}J_{12}(t)&\leqslant Cn^{-9/2}t^4E\mid\zeta_1\zeta_2\zeta_3\mid(Y_{12}^{*2}+Y_{13}^{*2}+Y_{23}^{*2})\sum{}'Y_{3k}^{*2}\\&\leqslant Cn^{-9/2}t^4\cdot Cn^2E\mid\zeta_1\zeta_2\zeta_3\mid Y_{12}^{*2}\\&\leqslant Cn^{-5/2}t^4,\end{aligned} \tag{45}$$

$$\begin{aligned}J_{11}(t)&\leqslant Cn^{-3/2}t^2\mid E\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)\sum{}'Y_{3k}^*e^{itS_n'}\mid\\&\quad+Cn^{-3/2}t^2\mid E(\zeta_1\zeta_2\zeta_3)(Y_{12}^*+Y_{13}^*)\sum{}'Y_{3k}^*e^{itS_n'}(e^{itV'}-1)\mid\\&\triangleq\sum_{j=1}^{2}J_{11j}(t).\end{aligned} \tag{46}$$

当 $|t|\leqslant\sqrt{n}\eta$ 时,我们有

$$J_{111}(t)\leqslant Cn^{-1/2}t^2\{|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{12}^*Y_{34}^*\mathrm{e}^{itS_n'}|+|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{13}^*Y_{34}^*\mathrm{e}^{itS_n'}|\}$$
$$\leqslant Cn^{-1/2}t^2\{C\mathrm{e}^{-\mu t^2}+\mathrm{E}|\zeta_2|(\mathrm{E}|\zeta_1\zeta_3|^3)^{1/3}(\mathrm{E}|Y_{13}^*|^3)^{1/3}(\mathrm{E}|Y_{34}^*|^3)^{1/3}\mathrm{e}^{-\mu t^2}\}$$
$$\leqslant\frac{Ct^2}{\sqrt{n}}\mathrm{e}^{-\mu t^2}.\tag{47}$$

$$J_{112}(t)\leqslant Cn^{-1/2}t^2|\mathrm{E}\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)Y_{34}^*\mathrm{e}^{itS_n'}(\mathrm{e}^{itV'}-1)|$$
$$\leqslant Cn^{-2}|t|^3\cdot\left|\mathrm{E}\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)Y_{34}^*\sum_{j\geqslant4}{}'Y_{jk}^*\mathrm{e}^{itS_n'}\right|$$
$$+Cn^{-7/2}t^4\mathrm{E}\left|\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)Y_{34}^*\Big(\sum_{j\geqslant4}{}'Y_{jk}^*\Big)^2\right|$$
$$\triangleq M_1(t)+M_2(t),\tag{48}$$

$$M_2(t)\leqslant Cn^{-7/2}t^4\mathrm{E}|\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)Y_{34}^*|\Big(\sum{}'Y_{4k}^{*2}+\sum_{j\geqslant5}{}'Y_{jk}^{*2}\Big)$$
$$\leqslant Cn^{-7/2}t^4\{C\sqrt{n}\mathrm{E}|\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*)|\cdot\mathrm{E}\sum{}'Y_{4k}^{*2}+[\mathrm{E}|\zeta_1\zeta_2Y_{12}^*|\mathrm{E}|\zeta_3Y_{34}^*|$$
$$+\mathrm{E}|\zeta_2|(\mathrm{E}|\zeta_1\zeta_3|^3)^{1/3}(\mathrm{E}|Y_{13}^*|^3)^{1/3}(\mathrm{E}|Y_{34}^*|^3)^{1/3}]\mathrm{E}\sum_{j\geqslant5}{}'Y_{jk}^{*2}\}$$
$$\leqslant Cn^{-7/2}t^4\{C\sqrt{n}\cdot C\cdot Cn+C\cdot Cn^2\}$$
$$\leqslant Cn^{-3/2}t^4.\tag{49}$$

由(*)及(24)式,当 $|t|\leqslant\sqrt{n}\eta$ 时, 有

$$M_1(t)\leqslant\frac{C}{n}|t|^3(|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{12}^*Y_{34}^*Y_{45}^*\mathrm{e}^{itS_n'}|+|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{13}^*Y_{34}^*Y_{45}^*\mathrm{e}^{itS_n'}|)$$
$$+C|t|^3(|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{12}^*Y_{34}^*Y_{56}^*\mathrm{e}^{itS_n'}|+|\mathrm{E}\zeta_1\zeta_2\zeta_3Y_{13}^*Y_{34}^*Y_{56}^*\mathrm{e}^{itS_n'}|)$$
$$\leqslant\frac{C}{n}|t|^3(\mathrm{E}|\zeta_1\zeta_2Y_{12}^*|\cdot\mathrm{E}|\zeta_3Y_{34}^*Y_{45}^*|\mathrm{e}^{-\mu t^2}$$
$$+C\sqrt{n}\mathrm{E}|\zeta_2|\mathrm{E}|\zeta_1\zeta_3Y_{13}^*|\mathrm{E}|Y_{45}^*|\mathrm{e}^{-\mu t^2})$$
$$+C|t|^3\Big(\mathrm{E}|\zeta_1\zeta_2Y_{12}^*|\mathrm{E}|\zeta_3Y_{34}^*|\cdot\frac{|t|}{\sqrt{n}}\mathrm{E}|Y_{56}^*\zeta_6|\mathrm{e}^{-\mu t^2}$$
$$+\frac{|t|}{\sqrt{n}}\mathrm{E}\zeta_2^2(\mathrm{E}|\zeta_1\zeta_3|^3)^{1/3}(\mathrm{E}|Y_{13}^*|^3)^{1/3}(\mathrm{E}|Y_{34}^*|^3)^{1/3}\mathrm{E}|Y_{56}^*|\mathrm{e}^{-\mu t^2}\Big)$$
$$\leqslant\frac{C}{\sqrt{n}}(|t|^3+t^4)\mathrm{e}^{-\mu t^2}.\tag{50}$$

由(44)~(50)式,即得

$$J_1(t)\leqslant Cn^{-1/2}(t^2+t^4)\mathrm{e}^{-\mu t^2}+Cn^{-3/2}t^4,\text{当 }|t|\leqslant\sqrt{n}\eta\text{ 时}.\tag{51}$$

又由对称性及引理 7,当 $|t|\leqslant\sqrt{n}\eta$ 时,有

$$J_2(t)\leqslant Cn^{3/2}|\mathrm{E}\zeta_1\zeta_2\zeta_3\mathrm{e}^{it(S_n'+V')}(\mathrm{e}^{itV_0}-1)|$$
$$\leqslant C|t|\cdot|\mathrm{E}\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*+Y_{23}^*)\mathrm{e}^{it(S_n'+V')}|$$
$$+Cn^{-3/2}t^2\mathrm{E}|\zeta_1\zeta_2\zeta_3(Y_{12}^*+Y_{13}^*+Y_{23}^*)^2|$$

$$
\begin{aligned}
&\leqslant C\mid t\mid\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3 Y_{12}^{*}\mathrm{e}^{\mathrm{i}t(S_n'+V')}\mid + Cn^{-3/2}t^2\mathrm{E}\mid \zeta_1\zeta_2\zeta_3 Y_{12}^{*2}\mid \\
&\leqslant C\mid t\mid\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3 Y_{12}^{*}\mathrm{e}^{\mathrm{i}tS_n'}\mid + C\mid t\mid\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3 Y_{12}^{*}\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}tV'}-1)\mid \\
&\quad + Cn^{-3/2}t^2(\mathrm{E}\mid\zeta_1\zeta_2\mid^3)^{1/3}(\mathrm{E}\mid Y_{12}^{*}\mid^3)^{2/3}\mathrm{E}\mid\zeta_3\mid \\
&\leqslant C\mid t\mid\cdot \mathrm{E}\mid \zeta_1\zeta_2 Y_{12}^{*}\mid\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_3^2\cdot\mathrm{e}^{-\mu t^2} \\
&\quad + C\mid t\mid \mathrm{E}\mid\zeta_1\zeta_2 Y_{12}^{*}\mid \mathrm{E}\mid\zeta_3\mid\cdot\left|\mathrm{E}\mathrm{e}^{\mathrm{i}t\sum_{j\geqslant 4}'\zeta_j/\sqrt{n}}(\mathrm{e}^{\mathrm{i}tV'}-1)\right| + Cn^{-3/2}t^2 \\
&\leqslant \frac{Ct^2}{\sqrt{n}}\mathrm{e}^{-\mu t^2} + \frac{C}{\sqrt{n}}(\mid t\mid^3 + t^8)\mathrm{e}^{-\mu t^2} + Cn^{-2}\mid t\mid^5 + Cn^{-3/2}t^2. \qquad (52)
\end{aligned}
$$

最后，我们估计 $J_3(t)$. 由对称性及 $V_1 = b_n\sum_{k\geqslant 4}'(Y_{1k}^{*}+Y_{2k}^{*}+Y_{3k}^{*})$，有

$$
\begin{aligned}
J_3(t) &= Cn^{3/2}\mid \mathrm{E}\zeta_1\zeta_2\zeta_3\mathrm{e}^{\mathrm{i}t(S_n'+V')}(\mathrm{e}^{\mathrm{i}tV_1}-1)\mid \\
&\leqslant \sum_{j=1}^{3}Cn^{-\frac{3}{2}(j-1)}\mid t\mid^j\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3(\textstyle\sum' Y_{3k}^{*})^j\mathrm{e}^{\mathrm{i}t(S_n'+V')}\mid + Cn^{-9/2}t^4\mathrm{E}\mid\zeta_1\zeta_2\zeta_3(\textstyle\sum' Y_{3k}^{*})^4\mid \\
&\triangleq \sum_{j=1}^{4}J_{3j}(t). \qquad (53)
\end{aligned}
$$

由 Marcinkiewicz 不等式，容易验证

$$
J_{3j}(t)\leqslant Cn^{-3/2}t^4,\ j=2,\ 3,\ 4. \qquad (54)
$$

事实上，以 $J_{33}(t)$为例，有

$$
\begin{aligned}
J_{33}(t) &\leqslant Cn^{-3}\mid t\mid^3\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_1^2\cdot\mathrm{E}\mid\zeta_2\mid\cdot\mathrm{E}\mid\zeta_3(\textstyle\sum' Y_{3k}^{*})^3\mid \\
&\leqslant Cn^{-7/2}t^4\mathrm{E}\mid\zeta_3(\textstyle\sum' Y_{3k}^{*2})^{3/2}\mid \\
&\leqslant Cn^{-2}t^4\mathrm{E}\textstyle\sum'\mid\zeta_3\mid\mid Y_{3k}^{*}\mid^3/(\lambda_n-3) \\
&\leqslant Cn^{-2}t^4\mathrm{E}(\mid\zeta_3\mid\cdot\mid Y_{34}^{*}\mid^3) \\
&\leqslant Cn^{-3/2}t^4, \qquad (55)
\end{aligned}
$$

这就是所要证明的. 最后，若记 $\overline{S}_n = \sum_{j\geqslant 3}'\zeta_j/\sqrt{n}$, $\overline{U}_n = V' = b_n\sum_{j\geqslant 4}'Y_{jk}^{*}$，则由(*)，(24) 式及引理 8，当 $\mid t\mid\leqslant\sqrt{n}\eta$ 时，我们有

$$
\begin{aligned}
J_{31}(t) &\leqslant Cn\mid t\mid\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3 Y_{34}^{*}\mathrm{e}^{\mathrm{i}tS_n'}\mid + Cn\mid t\mid\cdot\mid \mathrm{E}\zeta_1\zeta_2\zeta_3 Y_{34}^{*}\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}tV'}-1)\mid \\
&\leqslant Cn\mid t\mid\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_1^2\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_2^2\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\mid\zeta_3 Y_{34}^{*}\zeta_4\mid\cdot\mid\varphi_n(t)\mid^{\lambda_n-4} \\
&\quad + Cn\mid t\mid\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_1^2\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\zeta_2^2\cdot\mid \mathrm{E}\zeta_3 Y_{34}^{*}\mathrm{e}^{\mathrm{i}t\overline{S}_n}(\mathrm{e}^{\mathrm{i}t\overline{U}_n}-1)\mid \\
&\leqslant Cn^{-1/2}t^4\mathrm{e}^{-\mu t^2} + C\mid t\mid^3[Cn^{-1}(\mid t\mid + t^8)\mathrm{e}^{-\mu t^2} + Cn^{-2}t^4] \\
&\leqslant Cn^{-1/2}(t^4 + \mid t\mid^{11})\mathrm{e}^{-\mu t^2} + Cn^{-2}\mid t\mid^7. \qquad (56)
\end{aligned}
$$

由(53)～(56)式，得

$$
J_3(t)\leqslant Cn^{-1/2}(t^4+\mid t\mid^{11})\mathrm{e}^{-\mu t^2} + Cn^{-3/2}t^4 + Cn^{-2}\mid t\mid^7,\text{当}\mid t\mid\leqslant\sqrt{n}\eta. \qquad (57)
$$

由(42),(43),(51),(52)及(57)式,即得(41)式.再由(38)～(41)式,即得(37)式.这就完成了引理 9 的证明.

引理 10 $\mathrm{i}^3\mathrm{E}(S_n+\Delta_{n1}^*)^3\mathrm{e}^{\mathrm{i}t(S_n+\Delta_{n1}^*)}\sim \mathrm{i}^3\mathrm{E}(S_n+\Delta_{n1}^*)^3\mathrm{e}^{\mathrm{i}tS_n}$.

证 只需证明:

$$\mathrm{E}S_n^m\Delta_{n1}^{*\,3-m}\mathrm{e}^{\mathrm{i}t(S_n+\Delta_{n1}^*)}\sim \mathrm{E}S_n^m\Delta_{n1}^{*\,3-m}\mathrm{e}^{\mathrm{i}tS_n},\ m=0,\ 1,\ 2,\ 3. \tag{58}$$

$m=0,1,2$ 的情形由引理 1 和 6 得出,以 $m=1$ 为例,当 $|t|\leqslant\sqrt{n}\eta$ 时,

$$\begin{aligned}
|\mathrm{E}S_n'\Delta_{n1}^{*2}\mathrm{e}^{\mathrm{i}tS_n}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)| &\leqslant |\mathrm{E}S_n'\Delta_{n1}^{*2}\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)|\cdot|\mathrm{E}\mathrm{e}^{\mathrm{i}tS_n''}|\\
&\leqslant |t|\,\mathrm{E}|S_n'\Delta_{n1}^{*3}|\exp(-\mu n^{-1/3}t^2)\\
&\leqslant |t|[\mathrm{E}(S_n'\Delta_{n1}^*)^2\mathrm{E}\Delta_{n1}^{*4}]^{1/2}\exp(-\mu n^{-1/3}t^2)\\
&\leqslant Cn^{-3/2}|t|\exp(-\mu n^{-1/3}t^2),\\
|\mathrm{E}S_n''\Delta_{n1}^{*2}\mathrm{e}^{\mathrm{i}tS_n}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)| &\leqslant |\mathrm{E}\Delta_{n1}^{*2}\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)|\cdot|\mathrm{E}S_n''\mathrm{e}^{\mathrm{i}tS_n''}|\\
&\leqslant |t|\,\mathrm{E}|\Delta_{n1}^{*3}|\cdot Cn^{-1/3}|t|\exp(-\mu n^{-1/3}t^2)\\
&\leqslant Cn^{-3/2}|t|\cdot Cn^{-1/3}|t|\exp(-\mu n^{-1/3}t^2)\\
&\leqslant Cn^{-11/6}t^2\exp(-\mu n^{-1/3}t^2).
\end{aligned}$$

故由“～”的定义,(58)式 ($m=1$) 成立.为证 $m=3$ 的情形,只需证

$$|\mathrm{E}(S_n')^j(S_n'')^{3-j}\mathrm{e}^{\mathrm{i}tS_n}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)|\sim 0,\ j=0,\ 1,\ 2,\ 3. \tag{59}$$

这四种情形可同法处理,以 $j=3$ 为例.由引理 6 和 9,当 $|t|\leqslant\sqrt{n}\eta$ 时,我们有

$$\begin{aligned}
(59)\text{式左方}&\leqslant|\mathrm{E}(S_n')^3\mathrm{e}^{\mathrm{i}tS_n'}(\mathrm{e}^{\mathrm{i}t\Delta_{n1}^*}-1)|\cdot|\mathrm{E}\mathrm{e}^{\mathrm{i}tS_n''}|\\
&\leqslant Cn^{-1/2}(|t|+|t|^{11})\mathrm{e}^{-\mu t^2}\\
&\quad +C[n^{-3/2}(t^2+t^4)+n^{-2}(|t|^5+|t|^7)]\exp(-\mu n^{-1/3}t^2).
\end{aligned}$$

由“～”的定义,即得(59)式.引理 10 证毕.

引理 11 记 $\psi_n(t)=\mathrm{i}^3\mathrm{E}S_n^3\mathrm{e}^{\mathrm{i}tS_n}+\mathrm{i}^3[3\mathrm{E}S_n^2\Delta_{n1}^*+3\mathrm{E}S_n\Delta_{n1}^{*2}+\mathrm{E}\Delta_{n1}^{*3}]\mathrm{e}^{\frac{1}{2}t^2}$,则

$$\mathrm{i}^3\mathrm{E}(S_n+\Delta_{n1}^*)^3\mathrm{e}^{\mathrm{i}tS_n}\sim\psi_n(t).$$

证 只需证明:

$$\mathrm{E}S_n^m\Delta_{n1}^{*\,3-m}\mathrm{e}^{\mathrm{i}tS_n}\sim \mathrm{E}S_n^m\Delta_{n1}^{*\,3-m}\mathrm{e}^{-\frac{1}{2}t^2},\ m=0,\ 1,\ 2. \tag{60}$$

以 $m=2$ 为例,其他情形可类似地处理.只需证明:

$$n^{-1}b_n\sum_{j=1}^n\mathrm{E}(\zeta_j^2{\sum}'Y_{lm}^*\mathrm{e}^{\mathrm{i}tS_n})\sim 0, \tag{61}$$

$$n^{-1}b_n\sum_{j<k}\mathrm{E}(\zeta_j\zeta_k{\sum}'Y_{lm}^*\mathrm{e}^{\mathrm{i}tS_n})\sim n^{-1}b_n\sum_{l<m}{}'\mathrm{E}\zeta_l\zeta_mY_{lm}^*\mathrm{e}^{-\frac{1}{2}t^2}. \tag{62}$$

由(24)式,当 $|t|\leqslant\sqrt{n}\eta$ 时,

$$\begin{aligned}\left|n^{-1}b_n\sum_{j=1}^{n}\mathrm{E}(\zeta_j^2\sum{}'Y_{lm}^{*}\mathrm{e}^{\mathrm{i}tS_n})\right| &\leqslant Cn^{-1/2}\sum_{j=1}^{n}\mid \mathrm{E}\zeta_j^2Y_{12}^{*}\mathrm{e}^{\mathrm{i}tS_n}\mid\\ &\leqslant 2Cn^{-1/2}\mid \mathrm{E}\zeta_1^2Y_{12}^{*}\mathrm{e}^{\mathrm{i}tS_n}\mid + Cn^{1/2}\mid \mathrm{E}\zeta_3^2Y_{12}^{*}\mathrm{e}^{\mathrm{i}tS_n}\mid\\ &\leqslant Cn^{-1/2}\cdot\frac{\mid t\mid}{\sqrt{n}}\mathrm{E}\mid\zeta_1^2Y_{12}^{*}\zeta_2\mid\cdot\mid\varphi_n(t)\mid^{n-2}\\ &\quad + Cn^{1/2}\mathrm{E}\zeta_3^2\cdot\frac{t^2}{n}\mathrm{E}\mid Y_{12}^{*}\zeta_1\zeta_2\mid\cdot\mid\varphi_n(t)\mid^{n-3}\\ &\leqslant Cn^{-1/2}(\mid t\mid + t^2)\mathrm{e}^{-\mu t^2}\sim 0.\end{aligned}$$

此即(61)式.为证(62)式,只需证明

$$Cn^{1/2}\mathrm{E}Y_{12}^{*}\zeta_1\zeta_3\mathrm{e}^{\mathrm{i}tS_n}\sim 0, \tag{63}$$

$$Cn^{3/2}\mathrm{E}Y_{12}^{*}\zeta_3\zeta_4\mathrm{e}^{\mathrm{i}tS_n}\sim 0, \tag{64}$$

$$\left|\frac{1}{\sqrt{n}\sigma_g}\binom{n}{2}^{-1}\binom{\lambda_n}{2}[\mathrm{E}Y_{12}^{*}\zeta_1\zeta_2\mathrm{e}^{\mathrm{i}tS_n}-\mathrm{E}Y_{12}^{*}\zeta_1\zeta_2\mathrm{e}^{-\frac{1}{2}t^2}]\right|\sim 0. \tag{65}$$

(63), (64)式的证明与前面类似. 今证(65). 由文献[4], p. 109, 引理 1, $\exists\,\eta$, s. t.

$$\mid \mathrm{E}\mathrm{e}^{\mathrm{i}tS_n}-\mathrm{e}^{-\frac{1}{2}t^2}\mid\leqslant Cn^{-1/2}\mid t\mid^3\mathrm{e}^{-\frac{1}{3}t^2}.$$

由此及(24)式,当 $\mid t\mid\leqslant\sqrt{n}\eta$ 时, 有

$$\begin{aligned}(65)式左方 &\leqslant Cn^{-1/2}\mid \mathrm{E}(Y_{12}^{*}\zeta_1\zeta_2-\mathrm{E}Y_{12}^{*}\zeta_1\zeta_2)\mathrm{e}^{\mathrm{i}tS_n}\mid\\ &\quad + Cn^{-1/2}\mid \mathrm{E}Y_{12}^{*}\zeta_1\zeta_2\mid\cdot\mid \mathrm{E}\mathrm{e}^{\mathrm{i}tS_n}-\mathrm{e}^{-\frac{1}{2}t^2}\mid\\ &\leqslant Cn^{-1/2}\mathrm{E}\mid(Y_{12}^{*}\zeta_1\zeta_2-\mathrm{E}Y_{12}^{*}\zeta_1\zeta_2)(\mathrm{e}^{\mathrm{i}t(\zeta_1+\zeta_2)/\sqrt{n}}-1)\mid\cdot\mid\varphi_n(t)\mid^{n-2}\\ &\quad + Cn^{-1}\mid t\mid^3\mathrm{e}^{-\frac{1}{3}t^2}\\ &\leqslant Cn^{-1}\mid t\mid \mathrm{E}(\mid Y_{12}^{*}\zeta_1\zeta_2-\mathrm{E}Y_{12}^{*}\zeta_1\zeta_2\mid\cdot\mid\zeta_1+\zeta_2\mid)\mathrm{e}^{-\mu t^2}\\ &\quad + Cn^{-1}\mid t\mid^3\mathrm{e}^{-t^2/3}\\ &\leqslant Cn^{-1}\mid t\mid\mathrm{e}^{-\mu t^2}+Cn^{-1}\mid t\mid^3\mathrm{e}^{-t^2/3}\sim 0.\end{aligned}$$

此即(65)式,故(60)式 ($m=2$) 得证. 引理 11 证毕.

3 定理 1 的证明

令 $a_2=\frac{1}{2}\mathrm{E}\Delta_{n1}^{*2}$, $a_3=\frac{1}{6}(3\mathrm{E}S_n^2\Delta_{n1}^{*}+3\mathrm{E}S_n\Delta_{n1}^{*2}+\mathrm{E}\Delta_{n1}^{*3})$. 易见

$$\mid a_j\mid\leqslant C/\sqrt{n},\ j=2,\ 3. \tag{66}$$

定义

$$h_n(t)=\mathrm{E}\mathrm{e}^{\mathrm{i}tS_n}+a_2(\mathrm{i}t)^2\mathrm{e}^{-\frac{1}{2}t^2}+a_3(\mathrm{i}t)^3\mathrm{e}^{-\frac{1}{2}t^2}, \tag{67}$$

$$g_n(t)=\mathrm{E}\mathrm{e}^{\mathrm{i}t(S_n+\Delta_{n1}^{*})}. \tag{68}$$

由(66)式易见

$$| h_n^{(3)}(t) - \psi_n(t) | \leqslant Cn^{-1/2}(| t | + t^6) e^{\frac{1}{2}t^2},$$

其中,$\psi_n(t)$的定义见引理 11. 由引理 10 和引理 11,有

$$g_n^{(3)}(t) \sim \psi_n(t) \sim h_n^{(3)}(t),$$

亦即,存在与 n 无关的常数 $\eta > 0$, 使

$$\int_{|t| \leqslant \sqrt{n}\eta} | t |^{-1} \cdot | g_n^{(3)}(t) - h_n^{(3)}(t) | \, dt \leqslant C / \sqrt{n}. \tag{69}$$

但 $g_n^{(j)}(0) = h_n^{(j)}(0)$, $j = 0, 1, 2$. 故由引理 5,有

$$\int_{|t| \leqslant \sqrt{n}\eta} | t |^{j-4} \cdot | g_n^{(j)}(t) - h_n^{(j)}(t) | \, dt \leqslant C / \sqrt{n}, \ j = 0, 1, 2, 3. \tag{70}$$

由引理 6 和 7,当 $| t | \leqslant \sqrt{n}\eta$ 时, 有

$$\begin{aligned} | \mathrm{E}e^{itS_n}(e^{it\Delta_{n1}^*} - 1) | &\leqslant | \mathrm{E}e^{itS_n'}(e^{it\Delta_{n1}^*} - 1) | \cdot | \mathrm{E}e^{itS_n''} | \\ &\leqslant Cn^{-1/2}(t^2 + | t |^7) e^{-\mu t^2} + Cn^{-2} t^4 \exp(-\mu n^{-1/3} t^2), \end{aligned}$$

因此,$g_n(t) \sim \mathrm{E}e^{itS_n}$. 由(66), (67)式,$h_n(t) \sim \mathrm{E}e^{itS_n}$,故 $g_n(t) \sim h_n(t)$,即

$$\int_{|t| \leqslant \sqrt{n}\eta} | t |^{-1} \cdot | g_n(t) - h_n(t) | \, dt \leqslant C / \sqrt{n}. \tag{71}$$

令

$$H_n(x) = P(S_n \leqslant x) + (-1)^2 a_2 \Phi''(x) + (-1)^3 a_3 \Phi'''(x), \tag{72}$$

$$G_n(x) = P(S_n + \Delta_{n1}^* \leqslant x). \tag{73}$$

由文献[4],p. 125,定理 14,我们有

$$| P(S_n \leqslant x) - \Phi(x) | \leqslant Cn^{-1/2}(1 + | x |)^{-3}.$$

由(72)及(66)式,有

$$| H_n(x) - \Phi(x) | \leqslant Cn^{-1/2}(1 + | x |)^{-3}. \tag{74}$$

易见,$G_n(x)$非降,$H_n(x)$在 R^1 上有有界变差, $G_n(\pm\infty) = H_n(\pm\infty)$, $\int | x |^3 \, | \mathrm{d}(G_n(x) - H_n(x)) | < \infty$, 且(74)式成立. 故由引理 4,

$$\begin{aligned} | G_n(x) - H_n(x) | \leqslant C(1 + | x |)^{-3} \Big\{ &\int_{|t| \leqslant \sqrt{n}\eta} | t |^{-1} \cdot | g_n(t) - h_n(t) | \, dt \\ &+ \int_{|t| \leqslant \sqrt{n}\eta} | t |^{-1} \cdot | \delta_3^{(n)}(t) | \, dt + C / \sqrt{n} \Big\}, \end{aligned} \tag{75}$$

此处 $\delta_3^{(n)}(t) = \int_{-\infty}^{\infty} e^{itx} \mathrm{d}\{x^3(G_n(x) - H_n(x))\}$. 由文献[4], p. 153,引理 7,并注意到(70), (71)和(75)式,我们有

$$| G_n(x) - H_n(x) | \leqslant C(1 + | x |)^{-3} \Big\{ \int_{|t| \leqslant \sqrt{n}\eta} | t |^{-1} \cdot | g_n(t) - h_n(t) | \, dt$$

$$+\sum_{j=0}^{3}\int_{|t|\leqslant\sqrt{n}\eta} |t|^{j-4}\cdot|g_n^{(j)}(t)-h_n^{(j)}(t)|\,\mathrm{d}t+C/\sqrt{n}\Big\}$$
$$\leqslant Cn^{-1/2}(1+|x|)^{-3}. \tag{76}$$

由(74)式,我们有

$$|G_n(x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}. \tag{77}$$

由引理 1,对 $|x|\geqslant 1$, 有

$$\begin{aligned}P(|\Delta_{n2}^{*}|\geqslant|x|/\sqrt{n}) &\leqslant n^{3/2}|x|^{-3}\mathrm{E}|\Delta_{n2}^{*}|^{3}\\ &\leqslant n^{3/2}|x|^{-3}\cdot Cn^{-2}\\ &\leqslant Cn^{-1/2}(1+|x|)^{-3}.\end{aligned} \tag{78}$$

由引理 3,有

$$|P(S_n+\Delta_n^{*}\leqslant x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}. \tag{79}$$

又,由引理 2,对 $|x|\geqslant 1$, 我们有

$$\begin{aligned}&P\left\{\left|\frac{\sqrt{n}}{2\sigma_g}U_n-(S_n+\Delta_n^{*})\right|\geqslant|x|/\sqrt{n}\right\}\\ &\leqslant P\left(|\Delta_n-\widetilde{\Delta}_n^{*}|\geqslant\frac{|x|}{2\sqrt{n}}\right)+P\left(|\widetilde{\Delta}_n^{*}-\Delta_n^{*}|\geqslant\frac{|x|}{2\sqrt{n}}\right)\\ &\leqslant 4nx^{-2}\mathrm{E}(\Delta_n-\widetilde{\Delta}_n^{*})^2+16n^2x^{-4}\mathrm{E}(\widetilde{\Delta}_n^{*}-\Delta_n^{*})^4\\ &\leqslant 4nx^{-2}\cdot Cn^{-3/2}(1+|x|)^{-1}+16n^2x^{-4}\cdot Cn^{-5/2}(1+|x|)\\ &\leqslant Cn^{-1/2}(1+|x|)^{-3}.\end{aligned} \tag{80}$$

由引理 3,

$$\left|P\left(\frac{\sqrt{n}}{2\sigma_g}U_n\leqslant x\right)-\Phi(x)\right|\leqslant Cn^{-1/2}(1+|x|)^{-3}. \tag{81}$$

由(6)式易见, $\left|\frac{2}{\sqrt{n}}\sigma_g/\sigma_n-1\right|\leqslant C/n$, 再一次用引理 3,有

$|P(U_n/\sigma_n\leqslant x)-\Phi(x)|\leqslant Cn^{-1/2}(1+|x|)^{-3}$, 对充分大的 n 和所有 x.
因而(4)式对所有 n 和 x 都成立. 这就完成了定理 1 的证明.

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8

On One Conjecture of R. S. Singh

Abstract In 1979 R. S. Singh (*Ann. Statist.*, 1979, p. 890) made a conjecture concerning the convergence rate of EB estimates of the parameter θ in an one-dimensional continuous exponential distribution family, under the square loss function, the prior distribution family being confined to a bounded interval. The conjecture asserts that the rate cannot reach $o(1/n)$ or even $O(1/n)$. In this article, the weaker part of this conjecture (i. e. the $o(1/n)$ part) is shown to be correct.

1 Introduction

Suppose that there are given a sample space $\mathscr{X}$, a family of probability distributions depending on an one-dimensional parameter θ, and loss function $(\theta - a)^2$. Suppose also that there are given a family $\mathscr{F}$ of prior distributions, and the "historical" samples $X_1, \cdots, X_n$ and "contemporary" sample X, in a standard empirical Bayes structure. Denote by $\delta_G(X)$ the Bayes estimate of θ when the prior is G. In the case we know only $G \in \mathscr{F}$ and $\mathscr{F}$ contains more than one member. δ_G can no longer be determined, and one may resort to the empirical Bayes (EB) estimation based on $X_1, \cdots, X_n$ and X. The Bayes risk of δ_G and the "over-all" Bayes risk of an EB estimate $\delta_n \triangleq \delta_n(X) \triangleq \delta_n(X_1, \cdots, X_n, X)$ will be denoted by $R(G)$ and $R(\delta_n, G)$, respectively. If

$$\lim_{n \to \infty} R(\delta_n, G) = R(G), \text{ for each } G \in \mathscr{F}, \tag{1}$$

then δ_n is said to be asymptotically optimal (a. o.) with respect to the prior family $\mathscr{F}$. Recently, there has been much interest in the rate of convergence in (1), in the case of exponential family. Under various (rather complicate) conditions, Lin, Singh and Zhao, among others, obtained rates of convergence in the form

$$R(\delta_n, G) - R(G) = O(n^{-t}) \tag{2}$$

(see [1]~[3]).

In 1979, Singh[2] considered the Lebesgue-exponential family

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$$f(x, \theta)\mathrm{d}x = C(\theta)h(x)\mathrm{e}^{\theta x}\,\mathrm{d}x. \tag{3}$$

He showed that under certain conditions, EB estimate δ_n can be constructed such that in (2), t may take values arbitrarily close to, but less than, one (Zhao established in [3] a similar result for discrete exponential family). In view of this result Singh made the conjecture that however the concrete form of (3) may be, and even if we assume a "well-behaved" prior family such as

$$\mathscr{F}_{a,b} = \{G: G \text{ has a support within } (a, b)\} \tag{4}$$

(where $-\infty < a < b < \infty$), we cannot find an EB estimate δ_n such that the left hand side of (2) is of the order $o\left(\frac{1}{n}\right)$ or even $O\left(\frac{1}{n}\right)$ for any G belonging to the prior family.

The purpose of this article is to show that the weaker part of Singh's conjecture, the $o\left(\frac{1}{n}\right)$ part, is correct. It remains unknown whether the $O\left(\frac{1}{n}\right)$ part of the conjecture is correct or not.

2 The Main Theorem

For simplicity we shall consider the case of discrete exponential family. The result, together with its proof, carries over to the continuous case (3) without any difficulty.

Consider the exponential family

$$P(X = x \mid \theta) = h(x)\beta(\theta)\theta^x,\ x = 0, 1, 2, \cdots, \tag{5}$$

where

$$h(x) > 0,\ x = 0, 1, 2, \cdots \tag{6}$$

and the parameter space is

$$\Theta = \left\{\theta: \theta > 0,\ \sum_{x=0}^{\infty} h(x)\theta^x < \infty\right\}.$$

We may assume $2 \in \Theta$, for otherwise we can choose suitable $b > 0$ and use $b\theta$ as the new parameter. Thus $\sum_{x=0}^{\infty} h(x) < \infty$, and

$$0 < \left(\sum_{x=0}^{\infty} h(x)\right)^{-1} \leqslant \beta(\theta) \leqslant \frac{1}{h(0)} < \infty,\ 0 < \theta < 1. \tag{7}$$

Theorem 1 Under the above assumptions and suppose the prior distribution family is $\mathscr{F}_{0,1}$ (see (4)), for any EB estimate δ_n there exists $G \in \mathscr{F}_{0,1}$ such that

$$\liminf_{n\to\infty} n(R(\delta_n, G) - R(G)) > 0. \tag{8}$$

Hence, at least for this G, $R(\delta_n, G) - R(G)$ is not of the form $o\left(\frac{1}{n}\right)$.

Proof Take a subfamily $\mathscr{F}^*$ of $\mathscr{F}_{0,1}$ as follows

$$\mathscr{F}^* = \left\{G_\lambda : \mathrm{d}G_\lambda(\theta) = \frac{M(\lambda)}{\beta(\theta)}\theta^\lambda I_{(0,1)}(\theta)\mathrm{d}\theta,\ 0 < \lambda < \infty\right\}, \tag{9}$$

where

$$M(\lambda) = \left(\int_0^1 \frac{1}{\beta(\theta)}\theta^\lambda \mathrm{d}\theta\right)^{-1}$$

which is continuously differentiable in $\lambda > 0$.

Under the prior G_λ, X has a marginal distribution

$$p_\lambda(x) = P_\lambda^*(X = x) = M(\lambda)h(x)(1+\lambda+x)^{-1},\ x = 0, 1, 2, \cdots \tag{10}$$

and the Bayes estimate of θ is

$$\delta_\lambda(x) = \frac{h(x)}{h(x+1)}\,\frac{p_\lambda(x+1)}{p_\lambda(x)} = \frac{1+\lambda+x}{2+\lambda+x}. \tag{11}$$

Now suppose in the contrary that there is an EB estimate δ_n such that

$$R(\delta_n, G_\lambda) - R(G_\lambda) = o\left(\frac{1}{n}\right), \text{ for each } \lambda > 0. \tag{12}$$

By the well-known formula

$$R(\delta_n, G_\lambda) - R(G_\lambda) = \sum_{x=0}^{\infty} p_\lambda(x)\mathrm{E}_\lambda[\delta_n(X_1, \cdots, X_n, x) - \delta_\lambda(x)]^2 \tag{13}$$

and the fact that $p_\lambda(x) > 0$ (which is an easy consequence of (6)), one finds that

$$\lim_{n\to\infty} \mathrm{E}_\lambda[\delta_n(X_1, \cdots, X_n, x) - \delta_\lambda(x)]^2 = 0 \tag{14}$$

for each $\lambda > 0$ and $x = 0, 1, 2, \cdots$, and this in turn contains that as $n \to \infty$

$$g_n(\lambda, x) \triangleq \mathrm{E}_\lambda[\delta_n(X_1, \cdots, X_n, x)] \to \delta_\lambda(x) \tag{15}$$

for each $\lambda > 0$ and $x = 0, 1, 2, \cdots$. Take arbitrarily positive integer x_0, and for simplicity write $\delta(\lambda)$ for $\delta_\lambda(x_0)$. We can view $\delta_n(X_1, \cdots, X_n, x_0)$ as an estimate of $\delta(\lambda)$, with the iid. samples $X_1, X_2, \cdots$ drawn from a population possessing probability distribution (10).

It is easy to verify the following facts:

(a) The parameter space of distribution family (10) is an interval (i.e., $(0, \infty)$).

(b) $p_\lambda(x) > 0$, $\partial p_\lambda(x)/\partial\lambda$ exists and is a continuous function of λ, for $\lambda > 0$ and $x = 0, 1, 2, \cdots$.

(c) For each $\lambda > 0$

$$\sum_{x=0}^{\infty} \partial p_\lambda(x)/\partial\lambda = 0. \tag{16}$$

Since, as pointed out earlier, $\sum_{x=0}^{\infty} h(x) < \infty$ and $M(\lambda)$ is continuously differentiable, it follows easily that the series $\sum_{x=0}^{\infty}\partial p_\lambda(x)/\partial\lambda$ converges uniformly in $a \leqslant \lambda \leqslant b$ for any $0 < a < b < \infty$. This, together with $\sum_{x=0}^{\infty} p_\lambda(x) = 1$, establishes (16).

(d) $d\delta(\lambda)/d\lambda$ is continuous for $\lambda > 0$, and

$$I(\lambda) = \mathrm{E}_\lambda[\partial \log p_\lambda(X)/\partial\lambda]^2 = \sum_{x=0}^{\infty} p_\lambda(x)\left(\frac{M'(\lambda)}{M(\lambda)} - \frac{1}{1+\lambda+x}\right)^2.$$

Hence $I(\lambda)$ is continuous and positive for $\lambda > 0$. So

$$0 < \frac{1}{A} \triangleq \sup_{1\leqslant\lambda\leqslant 2} I(\lambda) < \infty. \tag{17}$$

Now we observe that, under any prior distribution $G_\lambda \in \mathscr{F}^*$, the Bayes estimate $\delta_\lambda(\cdot)$ can take values only in $(0, 1)$. Hence, if we introduce a new EB estimate $\bar{\delta}_n = \bar{\delta}_n(x_1, \cdots, x_n, x)$ as follows

$$\bar{\delta}_n = \begin{cases} 0, & \text{for } \delta_n \leqslant 0, \\ \delta_n, & \text{for } 0 < \delta_n < 1, \\ 1, & \text{for } \delta_n \geqslant 1, \end{cases}$$

then we have

$$0 \leqslant R(G_\lambda) \leqslant R(\bar{\delta}_n, G_\lambda) \leqslant 1,\ R(\bar{\delta}_n, G_\lambda) \leqslant R(\delta_n, G_\lambda) \tag{18}$$

for any $\lambda > 0$. From (12), (18), it follows that

$$R(\bar{\delta}_n, G_\lambda) - R(G_\lambda) = o\left(\frac{1}{n}\right), \text{ for any } \lambda > 0. \tag{19}$$

The argument leading to (15) applies to $\bar{\delta}_n$, rendering

$$g_n(\lambda) \triangleq \mathrm{E}_\lambda[\bar{\delta}_n(X_1, \cdots, X_n, x_0)] \to \delta(\lambda) \tag{20}$$

for any $\lambda > 0$ as $n \to \infty$. Since $0 \leqslant \bar{\delta}_n \leqslant 1$, it follows easily that the series

$$g_n(\lambda) = \sum_{x_1=0}^{\infty} \cdots \sum_{x_n=0}^{\infty} \bar{\delta}_n(x_1, \cdots, x_n, x_0) M^n(\lambda) \prod_{i=1}^{n} \frac{h(x_i)}{1+\lambda+x_i}$$

and be differentiated under the summation sign, and hence $g_n(\lambda)$ is continuously differentiable in $\lambda > 0$. From this and the facts (a)—(d) proved above, it follows that (see [4]) the conditions for applying the Cramer-Rao inequality are fulfilled. Hence, noticing (17), we get

$$\mathrm{var}_\lambda(\bar{\delta}_n(X_1, \cdots, X_n, x_0)) \geqslant \frac{(g_n'(\lambda))^2}{nI(\lambda)} \geqslant \frac{A}{n}(g_n'(\lambda))^2,\ 1 \leqslant \lambda \leqslant 2. \tag{21}$$

From (18) and (19), using Fatou's lemma, we have

$$\lim_{n\to\infty} \sup \int_1^2 n(R(\bar{\delta}_n, G_\lambda) - R(G_\lambda))d\lambda$$

$$\leqslant \int_1^2 \lim_{n\to\infty} \sup[n(R(\bar{\delta}_n, G_\lambda) - R(G_\lambda))]\mathrm{d}\lambda = 0. \tag{22}$$

On the other hand, denote

$$d \triangleq \inf_{1\leqslant\lambda\leqslant 2} p_\lambda(x_0) \geqslant \frac{h(x_0)}{3+x_0} \inf_{1\leqslant\lambda\leqslant 2} M(\lambda) > 0,$$

by (13) (replacing δ_n by $\bar{\delta}_n$), (20) and Cauchy-Schwarz inequality, we also have as $n \to \infty$

$$\begin{aligned}
&\int_1^2 n(R(\bar{\delta}_n, G_\lambda) - R(G_\lambda))\mathrm{d}\lambda \\
&\geqslant \int_1^2 np_\lambda(x_0)\mathrm{E}_\lambda[\bar{\delta}_n(X_1, \cdots, X_n, x_0) - \delta(\lambda)]^2\mathrm{d}\lambda \\
&\geqslant \int_1^2 nd\,\mathrm{var}_\lambda[\bar{\delta}_n(X_1, \cdots, X_n, x_0)]\mathrm{d}\lambda \\
&\geqslant Ad\int_1^2 [g_n'(\lambda)]^2\mathrm{d}\lambda \\
&\geqslant Ad\left(\int_1^2 g_n'(\lambda)\mathrm{d}\lambda\right)^2 = Ad(g_n(2) - g_n(1))^2 \\
&\to Ad(\delta(2) - \delta(1))^2 = Ad/[(4+x_0)^2(3+x_0)^2] > 0
\end{aligned} \tag{23}$$

which contradicts evidently (22), and the theorem is proved.

3 Some Further Remarks

(a) It is easily seen that the conclusion and the proof of the theorem remain valid when the prior family is changed from $\mathscr{F}_{0,1}$ to $\mathscr{F}_{a,b}$ for any $a \in \Theta$, $b \in \Theta$, and $a < b$.

(b) In case of (6), not being satisfied, it has been shown in [5] that there exists no a.o. EB estimate of θ, unless the prior distribution family satisfies a condition of very peculiar nature. With some necessary modifications to look after this point, the whole argument in section 2 goes through without difficulty.

(c) The method of proof employed in this paper applies to a wide class of distributions, not only the exponential. Supposing that the distribution family of X, $\{P_\theta, \theta \in \Theta\}$, is rather "well-behaved", one can usually find a prior family $\{G_\lambda, \lambda \in I\}$, where I is an interval of $(-\infty, \infty)$, such that

(i) The family of marginal distribution of X, i. e.

$$P_\lambda^*(X \in A) = \int_I P_\theta(A)\mathrm{d}G_\lambda(\theta),\ \lambda \in I$$

satisfies the conditions for employing the Cramer-Rao inequality.

(ii) The Bayes estimate of θ, $\delta_\lambda(x)$, under the prior distribution G_λ, is a sufficiently smooth function of λ. for each $x \in \mathscr{X}$ with the possible exception of a $\mathscr{P}^*$-null set.

If this is the case, then, starting from the basic equality

$$R(\delta_n, G_\lambda) - R(G_\lambda) = \int_{\mathscr{X}} \mathrm{E}_\lambda[\delta_n(X_1, \cdots, X_n, x) - \delta_\lambda(x)]^2 \mathrm{d}P_\lambda^*(x),$$

one has no much difficulty in carring through the arguments of section 2 to reach the conclusion that it is impossible to have

$$R(\delta_n, G_\lambda) - R(G_\lambda) = o\left(\frac{1}{n}\right), \text{ for all } \lambda \in I,$$

with any possible choice of δ_n. Particularily, in case of exponential family this can be easily done.

However, the $O\left(\frac{1}{n}\right)$ part of Singh's conjecture looks difficult. The author does not know whether an easy and satisfactory answer can be reached.

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9

Exponential Bounds of Posterior Risk for k-NN Prediction

1 Introduction and Result

Let (X, θ), (X_1, θ_1), $\cdots$, (X_n, θ_n) be $R^d \times R^1$ -valued random vectors, it is desired to predict the value of θ, based on the observed value of X and with the help of the training sample $Z^n = \{(X_i, \theta_i), i = 1, \cdots, n\}$. Cover[1] used the k-Nearest Neighbor method to this problem. The method is as follows: Introduce a metric $\| X - Y \|$ in R_d. Rearrange X_1, $\cdots$, X_n into X_{n1}, $\cdots$, X_{nn}, such that $\| X_{n1} - X \| \leqslant \| X_{n2} - X \| \leqslant \cdots \leqslant \| X_{nn} - X \|$, and break ties by comparing indices. Choose positive integer k. Denote by θ_{ni} the θ-value associated with X_{ni}, i. e., $\theta_{ni} = \theta_j$ when $X_{ni} = X_j$. Under the square loss $L(\theta, a) = (\theta - a)^2$, Cover proposed the predictor $\theta_n^* = \frac{1}{k} \sum_{i=1}^{k} \theta_{ni}$, with prediction risk $R_n = \mathrm{E}(\theta_n - \theta)^2$. If the joint distribution of (X, θ) is known, one can calculate the Bayes predictor $\delta(x) = \mathrm{E}(\theta \mid X = x)$ of θ, whose risk $R = \mathrm{E}(\theta - \delta(X))^2$ attains the minimum value. Under certain conditions Cover proved that

$$R \triangleq \lim_{n \to \infty} R_n = \left(1 + \frac{1}{k}\right) R = \left(1 + \frac{1}{k}\right) (\mathrm{E}\theta^2 - \mathrm{E}\delta^2(X)). \tag{1}$$

In case where θ assumes only a finite number of values, usually each value represents one of the classes to which the sample item may belong, and in this case the problem is known as the discrimination problem. One can use the k-Nearest Neighbor method in this case, which means to base the discrimination of θ on θ_{n1}, $\cdots$, θ_{nk}. Denote by $\hat{\theta}_n$ the result of discrimination. Then the error probability is $P(\theta \neq \hat{\theta}_n)$. Wagner pointed out in [2] that it is of more practical interest to consider the posterior (or conditional) error probability $L_n = P(\theta \neq \hat{\theta}_n \mid Z^n)$, for in the process of discrimination one can only use the available training samples. This argument is also valid in the problem of prediction. Write

$$L_n = \mathrm{E}\{(\theta - \theta_n)^2 \mid Z_n\}. \tag{2}$$

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Wagner studied in [2] the asymptotic properties of the posterior error probability L_n under the condition that θ assumes only two values and $k=1$. His result was improved by Fritz[3], who showed that as $n\to\infty$, L_n converges to $\lim\limits_{n\to\infty} P(\theta\neq\hat{\theta}_n)$ with an exponential rate. The present author consider the case of $k>1$ (see [4]). The purpose of this paper is to consider the convergence rate of L_n to R, utilzing the idea of the method of [4]. The result can be formulated in the following theorem:

Theorem Suppose that

1° There is a constant H such that $P(|\theta|\leqslant H)=1$;

2° $\delta(x)=\mathrm{E}(\theta\mid X=x)$ and $\mu(x)=\mathrm{E}(\theta^2\mid X=x)$ are a. s. continuous;

3° The probability distribution of X is non-atomic. Then for any $\varepsilon>0$ we have

$$P(|L_n-R|\geqslant\varepsilon)=O\{\exp(-cn^{1/3})\},\tag{3}$$

for any c such that

$$c<c^*=128(4^{1/3}+2^{1/3})\mathrm{e}^{1/3}H^{4/3}\varepsilon^{2/3}.\tag{4}$$

2 Proof of the Theorem

(A) Introduce some preliminary facts.

1. Put

$$V_{ni}^{(j)}=\{x\colon x_{nj}=X_i\};\ U_{ni}^{(j)}=Q(V_{ni}^{(j)}),\ \widetilde{U}_n^{(j)}=\max_{1\leqslant i\leqslant n}U_{ni}^{(j)}.$$

If Q, the distribution of X, is atomic, then for any given $\lambda>0$ we have

$$\mathrm{E}\{\exp(-\lambda^2/2U_{ni}^{(j)})\}=O\{n^{\frac{(2j-1)M+1}{2}}\exp(-[(M-2)^2+2\lambda^2(n-(j-1)M-2)]^{1/2})\},\tag{5}$$

$$\mathrm{E}\{\exp(-\lambda^2/2\sqrt{U_{ni}^{(j)}})\}=O\{-\alpha(1+4^{-1/6})\lambda^{2/3}n^{1/3}\},\ \text{for any }\alpha<1,\tag{6}$$

where M is a constant depending solely upon d.

2. The Inequality of Tau-bo and Chen Ping. Suppose that $Y_1,\cdots,Y_n$ are independent and with mean zero, $a_1,\cdots,a_n$ are constants such that $\sum\limits_{i=1}^n a_i^2=1$. Then

$$\mathrm{E}\Big(\sum_{i=1}^n a_iY_i\Big)^{2r}\leqslant\Big(\frac{3}{2}\Big)^r(2r-1)!!\max_{1\leqslant i\leqslant n}\mathrm{E}\,|Y_i|^{2r},\ r=1,2,\cdots.\tag{7}$$

3. The Inequality of Marcinkiewicz-Zygmund-Burkholder. Let $Y_1,\cdots,Y_n$ be a martingale-difference sequence, then

$$\mathrm{E}\Big|\sum_{i=1}^n Y_i\Big|^p\leqslant(64p2^q)^p\mathrm{E}\Big(\sum_{i=2}^n Y_i^2\Big)^{p/2},\ q=p/(p-1).\tag{8}$$

The proof of these assertions can be found in [4], [5] and [6], respectively.

(B) We have

$$L_n^* = E\theta^2 - 2\sum_{j=1}^{k} E(\theta\theta_{nj} \mid z^n)/k + \sum_{j=1}^{k} E(\theta_{nj}^2 \mid z^n)/k^2 + 2\sum_{1\leqslant i<j\leqslant k} E(\theta_{ni}\theta_{nj} \mid z^n)/k^2. \quad (9)$$

Write

$$V_{nuv}^{(ij)} = \{x: x_{ni} = X_u,\ x_{nj} = X_v\},\ 1 \leqslant i < j \leqslant k,$$

$$U_{nuv}^{(ij)} = Q(V_{nuv}^{(ij)}).$$

Then

$$\begin{aligned} E(\theta\theta_{nj} \mid Z^n) &= \sum_{i=1}^{n} U_{ni}^{(j)} E(\theta\theta_{nj} \mid Z^n,\ X \in V_{ni}^{(j)}) \\ &= \sum_{i=1}^{n} U_{ni}^{(j)} \theta_i \int_{V_{ni}^{(j)}} \delta(x) dQ(x)/U_{ni}^{(j)} \\ &= \sum_{i=1}^{n} \theta_i \int_{V_{ni}^{(j)}} \delta(x) dQ(x) \\ &= \sum_{i=1}^{n} (\theta_i - \delta(X_i)) \int_{V_{ni}^{(j)}} \delta(x) dQ(x) \\ &\quad - \sum_{i=1}^{n} \int_{V_{ni}^{(j)}} \delta(x)[\delta(x) - \delta(X_i)] dQ(x) + E\delta^2(X) \\ &\triangleq J_{1n} - J_{2n} + E\delta^2(X). \end{aligned} \quad (10)$$

Now we proceed to show that for any $\varepsilon_1 > 0$ there exists constant $b > 0$, such that

$$P(|J_{2n}| \geqslant \varepsilon_1) = O(e^{-bn}). \quad (11)$$

To establish this fact, set $\eta = \varepsilon_1/(8H^2 + 2H)$. Find $\xi > 0$ and set B_ξ such that $Q(B_\xi) > 1 - \eta$, also, for $x \in B_\xi$ and $\| x - y \| \leqslant \xi$, we have $|\delta(x) - \delta(y)| < \eta$. Further, find $B \subset B_\xi$, such that

$$Q(B) > 1 - 2\eta,\ \inf_{x\in B} Q(S_{x\xi}) = a > 0,$$

where $S_{x\xi} = \{y: \| y - x \| \leqslant \xi\}$. The existence of such ξ and B follows from the a. s. continuity of δ. Put $\widetilde{V}_{ni}^{(j)} = V_{ni}^{(j)} \cap B \cap S_{X_{i\xi}}^c$, we have

$$\begin{aligned} |J_{2n}| &\leqslant \sum_{i=1}^{n} \int_{V_{ni}^{(j)}} |\delta(x)|\,|\delta(x) - \delta(X_i)|\, dQ(x) \\ &\leqslant 2H^2 2\eta + \sum_{i=1}^{n} \int_{V_{ni}^{(j)} \cap B} |\delta(x)|\,|\delta(x) - \delta(X_i)|\, dQ(x) \\ &\leqslant 4H^2\eta + H\eta + 2H^2 \sum_{i=1}^{n} Q(\widetilde{V}_{ni}^{(j)}) \\ &= \varepsilon_1/2 + 2H^2 \sum_{i=1}^{n} Q(\widetilde{V}_{ni}^{(j)}) \\ &\leqslant \varepsilon_1/2 + 2H^2 P\{X \in B,\ \| X - X_{nj} \| > \xi \mid X^n\}, \end{aligned}$$

here $X^n = (X_1, \cdots, X_n)$. Hence

$$P(|J_{2n}| \geqslant \varepsilon_1) \leqslant P\{P(X \in B,\ \| X - X_{nj} \| > \xi \mid X^n) > \varepsilon_1/4H^2\}$$

$$\leqslant 4M^2\int_B P\{X\in B,\ \|X-X_{nj}\|>\xi\}\mathrm{d}Q(x)$$
$$=4H^2\int_B P(\|x-X_{nj}\|>\xi)\mathrm{d}Q(x), \tag{12}$$

Since $Q(S_{x\xi})\geqslant a$ for $x\in B$, we have

$$P(\|x-X_{nj}\|>\xi)\leqslant\sum_{r=0}^{j-1}\binom{n}{r}a^r(1-a)^{n-r}\leqslant jn^{j-1}(1-a)^{n-j+1}$$
$$\leqslant kn^{k-1}\mathrm{e}^{-b'(n-k+1)}\leqslant\mathrm{e}^{-b'n/2}$$

uniformly for $x\in B$ when n large, where $b'=-\log(1-a)>0$. From this and (12) it follows that (11) is true for $b=b'/3$.

Now consider J_{1n} in (10). Put

$$T_n=\sum_{r=1}^{n}\left(\int_{V_{nr}^{(j)}}\delta(x)\mathrm{d}Q(x)\right)^2,\ J'_{1n}=J_{1n}/\sqrt{T_n}.$$

Since $\theta_i-\delta(X_i)$, $i=1,\cdots,n$, are conditionally independent when X^n is given, and $\mathrm{E}\{\theta_i-\delta(X_i)\mid X_i\}=0$, By (7) we have

$$P(|J_{1n}|\geqslant\varepsilon_2\mid X^n)=P(|J'_{1n}|\geqslant\varepsilon_2/\sqrt{T_n}\mid X^n)$$
$$\leqslant\exp(-\varepsilon_2^2/T_n)\mathrm{E}\{\exp(J'^2_{1n})\mid X^n\}$$
$$=\exp(-\varepsilon_2^2/T_n)\sum_{r=0}^{\infty}\frac{1}{r!}\mathrm{E}(J'^{2r}_{1n}\mid X^n)$$
$$\leqslant\exp(-\varepsilon_2^2/T_n)\left\{1+\sum_{r=1}^{\infty}\frac{1}{r!}\left(\frac{3}{2}\right)^r(2r-1)!!\right.$$
$$\left.\cdot\max_{1\leqslant i\leqslant n}\mathrm{E}((\theta_i-\delta(X_i))^{2r}\mid X^n)\right\},$$

Since $|\theta_i-\delta(X_i)|\leqslant 2H$, $(2r-1)!!<2^r r!$, hence when $H\leqslant 1/5$ we have

$$P(|J_{1n}|\geqslant\varepsilon_2\mid X^n)\leqslant\exp(-\varepsilon_2^2/T_n)\left\{1+\sum_{r=1}^{\infty}(24H^2)^r\right\}$$
$$\leqslant 25\exp(-\varepsilon_2^2/T_n). \tag{13}$$

Since

$$T_n\leqslant H^2\sum_{i=1}^{n}(U_{ni}^{(j)})^2\leqslant H^2\,\widetilde{U}_n^{(j)},$$

from (13) we have

$$P(|J_{1n}|\geqslant\varepsilon_2\mid X^n)\leqslant 25\exp(-\varepsilon_2^2/H^2\,\widetilde{U}_n^{(j)})$$
$$\leqslant 25\sum_{i=1}^{n}\exp(-\varepsilon_2^2/H^2U_{ni}^{(j)}). \tag{14}$$

Hence, utilizing (5), we see that there exists $b_1>0$ such that

$$P(|J_{1n}|\geqslant\varepsilon_2)=O\{\exp(-b\sqrt{n})\}. \tag{15}$$

When $H > 1/5$, we can define $J_{1n}^{*} = J_{1n}/5H$, and (15) holds true when J_{1n} is replaced by J_{1n}^{*}. As $P(|J_{1n}| \geqslant \varepsilon_2) = P(|J_{1n}^{*}| \geqslant \varepsilon_2/5H)$, (15) still holds true even when $H > 1/5$.

(C) Now consider

$$
\begin{aligned}
\mathrm{E}(\theta_{nj}^2 \mid Z^n) &= \sum_{i=1}^{n} \theta_i^2 U_{ni}^{(j)} \\
&= \sum_{i=1}^{n} (\theta_i^2 - \mu(X_i)) U_{ni}^{(j)} + \sum_{i=1}^{n} \int_{V_{ni}^{(j)}} (\mu(X_i) - \mu(x)) \mathrm{d}Q(x) + \mathrm{E}\theta^2 \\
&\triangleq J_{1n}' + J_{2n}' + \mathrm{E}\theta^2.
\end{aligned} \tag{16}
$$

Since $|\theta| \leqslant H$, we have $\mu(x) \leqslant H^2$. Also, $\mu(x)$ is continuous a. s. , copying the argument employed in (B) for J_{2n}, we see that for any given $\varepsilon_3 > 0$, there exists $b'' > 0$, such that

$$
P(|J_{2n}'| \geqslant \varepsilon_3) = O(\mathrm{e}^{-b''n}). \tag{17}
$$

Similarily, copying the argument used in (B) for J_{1n}, we see that for any given $\varepsilon_4 > 0$ there exists $b_2 > 0$, such that

$$
P(|J_{1n}'| \geqslant \varepsilon_4) = O\{\exp(-b_2\sqrt{n})\}. \tag{18}
$$

(D) Finally we have to treat the term $\mathrm{E}(\theta_{ni}\theta_{nj} \mid Z^n)$. Put $f_{nuv} = U_{nuv}^{(ij)} + U_{nvu}^{(ij)}$, we have

$$
\begin{aligned}
\mathrm{E}(\theta_{ni}\theta_{nj} \mid Z^n) &= \sum_{1\leqslant u<v\leqslant n} \theta_n\theta_v f_{nuv} \\
&= \sum_{1\leqslant u<v\leqslant n} (\theta_u - \delta(X_u))(\theta_v - \delta(X_v)) f_{nuv} \\
&\quad + \sum_{1\leqslant n<v\leqslant n} \delta(X_u)(\theta_v - \delta(X_v)) f_{nuv} \\
&\quad + \sum_{1\leqslant u<v\leqslant n} \delta(X_v)(\theta_u - \delta(X_u)) f_{nuv} \\
&\quad + \sum_{1\leqslant u<v\leqslant n} \int_{V_{nuv}^{(ij)} \cup V_{nvu}^{(ij)}} [\delta(X_u)\delta(X_v) - \delta^2(x)] \mathrm{d}Q(x) + \mathrm{E}\delta^2(X) \\
&\triangleq J_{1n}'' + J_{2n}'' + J_{3n}'' + J_{4n}'' + \mathrm{E}\delta^2(X).
\end{aligned} \tag{19}
$$

Copying the argument previously used in treating J_{2n}, we easily see that for any given $\varepsilon_5 > 0$, there exists $b''' > 0$ such that

$$
P(|J_{4n}''| \geqslant \varepsilon_5) = O(\mathrm{e}^{-b'''n}). \tag{20}
$$

Now consider J_{2n}''. Put

$$
C_{nv}(X^n) = \sum_{u=1}^{v-1} \delta(X_u) f_{nuv}.
$$

Then

$$
J_{2n}'' = \sum_{v=2}^{n} C_{nv}(X^n)(\theta_v - \delta(X_v)).
$$

Since $|\delta(x)| \leqslant H$, and $\sum_{u=1}^{n} f_{nuv} = U_{nv}^{(i)} + U_{nv}^{(j)}$, we have

$$| C_{mv}(X^n) | \leqslant H(U_{mv}^{(i)} + U_{mv}^{(j)}).$$

Therefore

$$T_n \triangleq \sum_{v=2}^{n} C_{mv}^2(X^n) \leqslant 2H^2 \left\{ \sum_{v=1}^{n} (U_{mv}^{(i)})^2 + \sum_{v=1}^{n} (U_{mv}^{(j)})^2 \right\} \leqslant 2H^2 (\widetilde{U}_n^{(i)} + \widetilde{U}_n^{(j)}).$$

Copying the method previously used in treating J_{1n}, we obtain the following inequality similar to (14):

$$P(| J''_{2n} | \geqslant \varepsilon_6 \mid X^n) \leqslant 25 \sum_{r=1}^{n} \exp(-\varepsilon_6^2/4H^2 U_{nr}^{(i)}) + 25 \sum_{r=1}^{n} \exp(-\varepsilon_6^2/4H^2 U_{nr}^{(j)}).$$

Therefore it follows from (5) that there exists $b_2 > 0$ such that

$$P(| J''_{2n} | \geqslant \varepsilon_6) = O(\mathrm{e}^{-b_2\sqrt{n}}). \tag{21}$$

Likewisely,

$$P(| J''_{3n} | \geqslant \varepsilon_6) = O(\mathrm{e}^{-b_2\sqrt{n}}). \tag{22}$$

Lastly we turn to J''_{1n}. Define

$$H_r = \sum_{t=n-r+1}^{n} (\theta_{n-r} - \delta(X_{n-r}))(\theta_t - \delta(X_t)) f_{n,\, n-r,\, t}, \quad r = 1, \cdots, n-1.$$

Then $J''_{1n} = H_1 + \cdots + H_{n-1}$, and $H_1, \cdots, H_{n-1}$ form a martingale-difference sequence under the condition of giving X^n. In fact,

$$\mathrm{E}(H_1 \mid X^n) = f_{n,\, n-1,\, n} \mathrm{E}\{\theta_{n-1} - \delta(X_{n-1}) \mid X^n\} \mathrm{E}\{\theta_n - \delta(X_n) \mid X^n\} = 0,$$

$$\mathrm{E}(H_r \mid H_1, \cdots, H_{r-1}, X^n) = \mathrm{E}\{\mathrm{E}(H_r \mid \theta_{n-r+1}, \cdots, \theta_n, X^n) \mid H_1, \cdots, H_{r-1}, X^n\}.$$

It is easily seen that $\mathrm{E}(H_r \mid \theta_{n-r+1}, \cdots, \theta_n, X^n) = 0$. Hence

$$\mathrm{E}(H_r \mid H_1, \cdots, H_{r-1}, X^n) = 0, \quad r = 2, \cdots, n-1.$$

This proves the above-mentioned assertion. Further

$$| H_r | \leqslant 4H^2 \sum_{t=n-r+1}^{n} f_{n,\, n-r,\, t} \leqslant 4H^2 (U_{n,\, n-r}^{(i)} + U_{n,\, n-r}^{(j)}).$$

Therefore

$$\sum_{r=1}^{n-1} H_r^2 \leqslant 16H^4 \sum_{r=1}^{n} (U_{nr}^{(i)} + U_{nr}^{(j)})^2 \leqslant 32H^4 \left\{ \sum_{r=1}^{n} (U_{nr}^{(i)})^2 + \sum_{r=1}^{n} (U_{nr}^{(j)})^2 \right\} \leqslant 32H^4 (\widetilde{U}_n^{(i)} + \widetilde{U}_n^{(j)}) \triangleq D_n.$$

Choose arbitrarily $g > (128\mathrm{e})^2$. From (8), for any given $\varepsilon_7 > 0$ we have

$$
\begin{aligned}
P(|J''_{1n}| \geqslant \varepsilon_7 \mid X^n) &= P(|J''_{1n}/\sqrt{gD_n}| \geqslant \varepsilon_7/\sqrt{gD_n} \mid X^n) \\
&\leqslant \exp(-\varepsilon_7/\sqrt{gD_n})\mathrm{E}\{\exp(|J''_{1n}/\sqrt{gD_n}|) \mid X^n\} \\
&= \exp(-\varepsilon_7/\sqrt{gD_n}) \sum_{p=0}^{\infty} (gD_n)^{-p/2}\mathrm{E}(|J''_{1n}|^p \mid X^n) \\
&\leqslant \exp(-\varepsilon_7/\sqrt{gD_n}) \sum_{p=0}^{\infty} \frac{1}{p!}(gD_n)^{-p/2}(64p2^q)^p D_n^{p/2} \\
&= \exp(-\varepsilon_7/\sqrt{gD_n}) \sum_{p=0}^{\infty} \frac{1}{p!} g^{-p/2}(64p2^q)^p. \qquad (23)
\end{aligned}
$$

Since $p! > p^p e^{-p}$, we have

$$
\frac{1}{p!} g^{-p/2}(64p2^q)^p < (64\mathrm{e}2^q/\sqrt{g})^p.
$$

Noticing that $q \to 1$ as $p \to \infty$, and $g > (128\mathrm{e})^2$, we see that the series appearing in the last expression of (23) converges. Denote the sum of this series by C_g, we have

$$
\begin{aligned}
P(|J''_{1n}| \geqslant \varepsilon_7) &\leqslant C_g \mathrm{E}\{\exp(-\varepsilon_7/\sqrt{gD_n})\} \\
&\leqslant C_g \mathrm{E}\{\exp(-\varepsilon_7/(16\sqrt{2}g^{1/2}H^2(\sqrt{\widetilde{U}_n^{(i)}} + \sqrt{\widetilde{U}_n^{(j)}})))\} \\
&\leqslant C_g \mathrm{E}\{\exp(-\varepsilon_7/(32\sqrt{2}g^{1/2}H^2\sqrt{\widetilde{U}_n^{(i)}}))\} \\
&\quad + C_g \mathrm{E}\{\exp(-\varepsilon_7/(32\sqrt{2}g^{1/2}H^2\sqrt{\widetilde{U}_n^{(j)}}))\} \\
&\leqslant C_g \sum_{r=1}^{n} \mathrm{E}\{\exp(-\varepsilon_7/(32\sqrt{2}g^{1/2}H^2\sqrt{U_{nr}^{(i)}}))\} \\
&\quad + C_g \sum_{r=1}^{n} \mathrm{E}\{\exp(-\varepsilon_7/(32\sqrt{2}g^{1/2}H^2\sqrt{U_{nr}^{(j)}}))\}. \qquad (24)
\end{aligned}
$$

By (6) and (24), we get for any $\alpha < 1$

$$
P(|J''_{1n}| \geqslant \varepsilon_7) = O\{-\alpha(1 + 4^{-1/6})8g^{1/3}H^{4/3}\varepsilon_7^{2/3}n^{1/3}\}. \qquad (25)
$$

Since α can be chosen arbitrarily near 1, (25) can be rewritten as

$$
P(|J''_{1n}| \geqslant \varepsilon_7) = O\{\exp(-\alpha \cdot 128(4^{1/3} + 2^{1/3})\mathrm{e}^{2/3}H^{4/3}\varepsilon_7^{2/3}n^{1/3})\} \qquad (26)
$$

for any $\alpha < 1$.

Now let $\varepsilon > 0$ be arbitrarily given. Choose $\varepsilon_i > 0$, $i = 1, \cdots, 7$, such that $\varepsilon = 2(\varepsilon_1 + \varepsilon_2) + (\varepsilon_3 + \varepsilon_4)/k + \varepsilon_5 + 2\varepsilon_6 + \varepsilon_7$. Combining (1), (9)—(11), (15)—(22) and (26), we see that

$$
P(|L_n^* - R| \geqslant \varepsilon) = \text{The right hand side of (26)}.
$$

Since α and ε_7 can be chosen arbitrarily near 1 and ε respectively, we finally get (3), where c satisfies (4), and the proof of the theorem is concluded.

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10

Uniform Convergence Rates of Kernel Density Estimators

1 Introduction and Main Results

Let $X_1, \cdots, X_n$ be iid. samples drawn from a population with density f in R^m. The problem of estimating f from these samples have attracted much attention for twenty years. Various methods were suggested. In particular, substantial results about the large sample theory of kernel estimates, which was originally proposed in [1], [2], were obtained; see, for example, [3]. On the other hand, Farrell first considered in [4], [5] the problem: What is the best attainable asymptotic rate for estimating the value of f at a given point (e. g., $f(0)$)? He proved in [5] the following theorem:

Theorem(Farrell) Let k be a positive integer and $\alpha > 0$ be a constant. Denote by $C_{k\alpha}$ the class of densities f in R^m satisfying the following condition:

$$| f^{(k)}(k_1, \cdots, k_m; x) | \triangleq | \partial^k f / \partial x_1^{k_1} \cdots \partial x_m^{k_m} | \leqslant \alpha \tag{1}$$

for all $x = (x_1, \cdots, x_m) \in R^m$ and $k_1 + \cdots + k_m = k$. Let $r_n(X_1, \cdots, X_n)$ be any estimator of $f(0)$. If for some sequence of constants $\{a_n\}$ we have

$$\lim_{n\to\infty} \{ \inf_{f\in C_{k\alpha}} P_f(| r_n(X_1, \cdots, X_n) - f(0) | \leqslant a_n) \} = 1,$$

then $a_n n^{k/(2k+m)} \to \infty$ as $n \to \infty$.

This result can be interpreted roughly as: the accuracy of estimating $f(0)$ in the whole class $C_{k\alpha}$ will never exceed $O(n^{-k/(2k+m)})$. But, as pointed out by Farrell in [5], in the case $m = 1$ it follows from Parzen[2] that this accuracy can be reached by a suitably chosen kernel estimator. Farrell conjectured that this would be true also in the case $m > 1$. (This can be easily verified; see the Remark at the end of this paper.) This establishes an optimal property of the kernel estimator in the large sample setting.

Farrell, however, considered only the estimation of densities at a fixed point, while the more natural and more difficult problem is to determine the best asymptotic uniform conver-

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gence rate obtainable in estimating the whole density function. The study of this problem is the purpose of this paper. Interestingly, our result shows that even in this stronger sense of convergence, essentially the accuracy level remains. To be precise, we have

Theorem 1 There exists a kernel estimate $f_n(x) = f_n(X_1, \cdots, X_n; x)$ such that

$$\lim_{a\to\infty}\lim_{n\to\infty}\{\inf_{f\in C_{k\alpha}} P_f(|\sup_x |f_n(x)-f(x)| \leqslant a(\log n/n)^{k/(2k+m)})\} = 1. \tag{2}$$

Farrell's result guarantees that the uniform asymptotic rate of any estimator cannot exceed $O(n^{-k/(2k+m)})$, while Theorm 1 indicates that the rate $O(n^{-k/(2k+m)}(\log n)^{k/(2k+m)})$ can be reached for some kernel estimator. Thus, for the main part of the rate, the kernel estimator attains the best possible order. Probably the rate in (2) is exactly the best possible, and this deserves further investigation.

A by-product in proving Theorem 1 is the following

Theorem 2 There exists a kernel estimator f_n such that

$$\lim_{n\to\infty}(\log n)^{-k/(2k+2m)} n^{k/(2k+m)} \sup_x |f_n(x)-f(x)| = 0, \text{ a. s.} \tag{3}$$

for any $f\in C_{k\alpha}$ ($\alpha<0$ arbitrarily chosen).

The problem of determining the rate of convergence of $\sup_x |f_n(x)-f(x)|$ for the kernel estimator f_n has been explored under various conditions. Under the conditions that the kernel K is continuous and of bounded variation on R^1, and

$$\int_{-\infty}^{\infty} |xK(x)|\,dx < \infty,$$

Schuster showed in 1969[6] for $m=1$ that for a suitably chosen h_n,

$$n^{1/4-\epsilon}\sup_x |f_n(x)-f(x)| \to 0, \text{ a. s.}$$

for any $\epsilon>0$. In 1977, Singh[7] proved, also for $m=1$, that

$$\sup_x |f_n(x)-f(x)| = O(n^{-k/(2k+2)}\sqrt{\log\log n}), \text{ a. s.}$$

under the conditions that the kernel K is continuous and of bounded variation on R^1, and $\sup_x |Ef_n(x)-f(x)| = O(h_n^k)$. These are far weaker than (3). Silverman's work[8] is comparable to (3). He proved that if the kernel K satisfies a pile of conditions (C1) and (C2), then for the suitably chosen h_n one has $\sup_x |Ef_n(x)-f_n(x)| = O((nh_n)^{-1/2}\sqrt{-\log|h_n|})$ a. s. To obtain from this a result concerning the rate of $\sup_x |f_n(x)-f(x)|$ comparable to (3), one still needs to impose the condition $\int_{-\infty}^{\infty} x^i K(x)dx = 0$, $i=1,\cdots,k-1$. But it is not clear whether this condition is consistent with (C1) and (C2) or not.

2 Lemmas

First we establish a purely analytical result, namely, f in $C_{k\alpha}$ is uniformly bounded on

R^m. In fact we can show:

Lemma 1 There is a constant B depending only on α, k and m such that

$$| f^{(l)}(l_1, \cdots, l_m; x) | = | \partial^l f / \partial x_1^{l_1} \cdots x_m^{l_m} | \leqslant B \tag{4}$$

for each $f \in C_{k\alpha}$, $x = (x_1, \cdots, x_m) \in R^m$, and $l_1 + \cdots + l_m = l$. $l \leqslant k-1$.

Proof Suppose in contrary that such B does not exist. Then one can find sequences $\{x_r\} \subset R^m$, $\{f_r\} \subset C_{k\alpha}$, nonnegative integers $\tilde{l}_1, \cdots, \tilde{l}_m$ with the sum $\tilde{l} \leqslant k-1$, such that

(1) $| f^{(\tilde{l})}(l_1, \cdots, l_m; x_r) | \geqslant r$; (5)

(2) for any nonnegative integers $s, s_1, \cdots, s_m, u, u_1, \cdots, u_m$ with $\sum_{i=1}^m s_i = s \leqslant k-1$ and $\sum_{i=1}^m u_i = u \leqslant k-1$, the limit

$$A(s_1, \cdots, s_m; u_1, \cdots, u_m) = \lim_{r \to \infty} | f_r^{(s)}(s_1, \cdots, s_m; x_r) / f_r^{(u)}(u_1, \cdots, u_m; x_r) |$$

exists (which may be ∞). It is easily seen that there exists $(s_1, \cdots, s_m)$ such that $A(s_1, \cdots, s_m; u_1, \cdots, u_m) > 0$ for any $(u_1, \cdots, u_m)$ satisfying the above-mentioned conditions. Denote by S the set of all such $(s_1, \cdots, s_m)$, and choose an element $(s_1^0, \cdots, s_m^0) \in S$ by the following procedure: Let $d_1 = \min\{s_1 : (s_1, \cdots, s_m) \in S\}$, $S_1 = \{(s_1, \cdots, s_m) : s_1 = d_1\}$. If S_1 contains only a single element (i. e. $\#(S_1) = 1$), then it will be chosen as $(s_1^0, \cdots, s_m^0)$. In general, if $(s_1^0, \cdots, s_m^0)$ is still not determined after defining d_i and S_i, $i = 1, \cdots, j-1$, then let $d_j = \min\{s_j : (s_1, \cdots, s_m) \in S_{j-1}\}$ and $S_j = \{(s_1, \cdots, s_m) : (s_1, \cdots, s_m) \in S_{j-1}; s_j = d_j\}$. If $\#(S_j) = 1$, its single element will be chosen as $(s_1^0, \cdots, s_m^0)$; otherwise proceed to $j+1$. In this way eventually a unique $(s_1^0, \cdots, s_m^0)$ will be obtained. In the following we assume that this element is obtained just after the definition of S_j. Let

$$U = s_1^0 + \cdots + s_m^0. \tag{6}$$

Obviously,

(i) $A_r(s_1^0, \cdots, s_m^0; u_1, \cdots, u_m) \triangleq | f_r^{(v)}(s_1^0, \cdots, s_m^0; x_r) / f_r^{(u)}(u_1, \cdots, u_m; x_r) | \to A(s_1^0, \cdots, s_m^0; u_1, \cdots, u_m) > 0$, for each $(u_1, \cdots, u_m)$ satisfying the above-mentioned conditions;

(ii) write $U = \{(u_1, \cdots, u_m) : A(s_1^0, \cdots, s_m^0; u_1, \cdots, u_m) = \infty\}$; then $u_1 \geqslant s_1^0$ for each $(u_1, \cdots, u_m) \overline{\in} U$, and if $u_1 = s_1^0$, an integer $i \leqslant j$ can be found such that $u_t = s_t^0$ for $t \leqslant i-1$, but $u_i > s_i^0$.

Now choose $\{h_{r1}, \cdots, h_{rj}, r = 1, 2, \cdots\}$ that satisfies the following conditions:

(a) $0 < h_{ri} \to 0$ as $r \to \infty$, $i = 1, \cdots, j$.

(b) $\lim_{r \to \infty} h_{ri} h_{r,i+1}^{-v} = 0$, $i = 1, \cdots, j-1$.

(c) $\lim_{r \to \infty} r h_{r1}^{2v} = \infty$.

(d) $\lim_{r \to \infty} h_{r1}^{v} A_r(s_1^0, \cdots, s_m^0; u_1, \cdots, u_m) = \infty$, for any $(u_1, \cdots, u_m) \in U$.

Write $H_{rt} = (t_1 h_{r1}, \cdots, t_j h_{rj}, t_{j+1}, \cdots, t_m)$, $\frac{1}{2} \leqslant t_i \leqslant 1$, $i = 1, \cdots, m$. By Taylor's formula

$$f_r(x_r + H_{rt}) = \sum_{l=0}^{k-1} \sum_{l}^{(*)} \frac{f_r^{(l)}(l_1, \cdots, l_m; x_r)}{l_1! \cdots l_m!} (t_1 h_{r1})^{l_1} \cdots (t_j h_{rj})^{l_j} t_{j+1}^{l_{j+1}} \cdots t_m^{l_m}$$
$$+ \sum_{k}^{(*)} \frac{f_r^{(k)}(k_1, \cdots, k_m; \tilde{x}_r)}{k_1! \cdots k_m!} (t_1 h_{r1})^{k_1} \cdots (t_j h_{rj})^{k_j} t_{j+1}^{k_{j+1}} \cdots t_m^{k_m} \tag{7}$$

where $\tilde{x}_r$ lies somewhere in the interval with vertices x_r and $x_r + H_{rt}$, and the summation $\sum\limits_{w}^{(*)}$ extends to all $w_1, \cdots, w_m$ with $w_i \geqslant 0$, $\sum\limits_{i=1}^{m} w_i = w$. By the choice of $(s_1^0, \cdots, s_m^0)$ and $\{h_{r1}, \cdots, h_{rj}\}$, it is easy to see that among all the terms on the right side of (7), the term with $l_i = s_i^0$, $i = 1, \cdots, m$, possesses the highest order uniformly on

$$T = \left\{(t_1, \cdots, t_m): \frac{1}{2} \leqslant t_i \leqslant 1,\ i = 1, \cdots, m\right\}.$$

In fact, denote this term by G, and compare it with an arbitrarily chosen term

$$Q = \frac{f_r^{(l)}(l_1, \cdots, l_m; x_r)}{l_1! \cdots l_m!} (t_1 h_{r1})^{l_1} \cdots (t_j h_{rj})^{l_j} t_{j+1}^{l_{j+1}} \cdots t_m^{l_m}.$$

Selecting a subsequence if necessary, we can assume $f_r^{(v)}(s_1^0, \cdots, s_m^0; x_r) \geqslant 0$, $r = 1, 2, \cdots$, without loss of generality. If $(l_1, \cdots, l_m) \in U$, then by condition (d) we see that $G/|Q| \to \infty$, as $r \to \infty$, uniformly on T. For $(l_1, \cdots, l_m) \overline{\in} U$, there exists $i \leqslant j$ such that $l_w = s_w^0$ when $w \leqslant i - 1$, and $l_i > s_i^0$. Hence on T(take $h_{r,j+1} = 1$)

$$G/|Q| \geqslant (k!2^k)^{-1} A_r(s_1^0, \cdots, s_m^0; l_1, \cdots, l_m) h_{ri} h_{r,i+1}^{-v}.$$

By condition (b) and $\lim\limits_{r\to\infty} A_r(s_1^0, \cdots, s_m^0; l_1, \cdots, l_m) > 0$, it again follows that $G/|Q| \to \infty$ uniformly on T. Therefore, from (7) it follows that there exists a constant $C_1 > 0$ such that

$$f_r(x_r + H_{rt}) \geqslant C_1 f_r^{(v)}(s_1^0, \cdots, s_m^0; x_r) h_{r1}^v \tag{8}$$

holds on T. Further, by (5) and the choice of $(s_1^0, \cdots, s_m^0)$, there exists a constant $C_2 > 0$ such that for r large enough

$$f_r^{(v)}(s_1^0, \cdots, s_m^0; x_r) \geqslant C_2 |f_r^{(\tilde{l})}(\tilde{l}_1, \cdots, \tilde{l}_m; x_r)| \geqslant C^2 r. \tag{9}$$

By (8) and (9), $f_r(x_r + H_{rt}) \geqslant C_1 C_2 r h_{r1}$ uniformly on T for r large enough. Hence, denoting $D_r = \{x: x = x_r + H_{rt},\ t \in T\}$ and noticing that $h_{r1} < h_{ri}$ for r sufficiently large and $i = 2, \cdots, j$, we get

$$\int_{D_r}\cdots\int f_r(x)\mathrm{d}x \geqslant C_1 C_2 r h_{r1}^{2v} 2^{-k}. \tag{10}$$

By condition (c) we have $C_1 C_2 r h_{r1}^{2v} 2^{-k} > 1$ for r sufficiently large. This contradicts the fact that f_r is a density function. The lemma is proved.

Choose a polynomial P such that

$$\int_{-1/2}^{1/2} P(x)x^i \mathrm{d}x = \begin{cases} 1, & \text{for } i = 0, \\ 0, & \text{for } i = 1, \cdots, k \end{cases}$$

and define the kernel as $K(u) = K(u_1, \cdots, u_m) = \prod_{i=1}^{m} \{P(u_i) I_{(-\frac{1}{2} \leqslant u_i < \frac{1}{2})}\}$ satisfying the condition

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \cdots \int_{-\frac{1}{2}}^{\frac{1}{2}} K(u) \prod_{i=1}^{m} u_i^{n_i} \mathrm{d}u_1 \cdots \mathrm{d}u_m = \begin{cases} 1, & \text{for } n_1 = \cdots = n_m = 0, \\ 0, & \text{for } n_1 \leqslant k, \cdots, n_m \leqslant k, \ \sum_{i=1}^{m} n_i > 0. \end{cases} \tag{11}$$

The kernel K generates a kernel estimator

$$f_n(x) = f_n(X_1, \cdots, X_n; x) = (nh_n^m)^{-1} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h_n}\right) \tag{12}$$

where

$$h_n = n^{-1/(2k+m)} (\log n)^{1/(2k+2m)}. \tag{13}$$

Also, let

$$\hat{Y}_{n1}(x) = h_n^m f_n(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h_n}\right), \tag{14}$$

$$Y_n(x) = \mathrm{E}\, \hat{Y}_{n1}(x) = h_n^m \mathrm{E} f_n(x) = h_n^m \int_{R^m} K(u) f(x - h_n u) \mathrm{d}u. \tag{15}$$

From Lemma 1 and the choice of K, it follows that a constant M (depending upon k, m, α but not f) exists such that

$$f(x) \leqslant M, \text{ for any } x \in R^m, \tag{16}$$

$$|K(u)| \leqslant M, \ |\partial K / \partial u_i| \leqslant M, \ i = 1, \cdots, m, \text{ for } |u_i| \leqslant \frac{1}{2}, i = 1, \cdots, m. \tag{17}$$

By (16), (17), we have

$$|Y_n(x)| \leqslant M^2 h_n^m, \ x \in R^m. \tag{18}$$

Further, let $X_1, \cdots, X_{2n}$ be iid. samples drawn from a population with density $f \in C_{k\alpha}$, and define

$$\hat{Y}_{n2}(x) = \frac{1}{n} \sum_{i=n+1}^{2n} K\left(\frac{x - X_i}{h_n}\right), \ Y_n(x) = \frac{1}{2n} \sum_{i=1}^{2n} K\left(\frac{x - X_i}{h_n}\right).$$

Lemma 2 For any given $\eta > 0$, when n is sufficiently large, the event that the inequality

$$|\hat{Y}_{n1}(x) - \hat{Y}_{n1}(y)| \leqslant 9mMn^{-1+\eta} \tag{19}$$

holds whenever $\|x - y\| = \max_{1 \leqslant i \leqslant m} |x_i - y_i| \leqslant h_n / n$ has a probability no less than 1 −

$\exp(-n^{\eta/3})$.

Proof For any given $\alpha_1 > 0, \cdots, \alpha_m > 0$, introduce a norm in $u \in R^m$ as $\| u \|_{\alpha_1 \cdots \alpha_m} = \max_{1 \leqslant i \leqslant m} | u_i/\alpha_i |$, and for simplicity write $\| u \|$ for $\| u \|_{1 \cdots 1}$. Under this norm an interval $\{(x_1, \cdots, x_m) : a_i \leqslant x_i \leqslant b_i, i = 1, \cdots, m\}$ with $(b_1 - a_1)/\alpha_1 = \cdots = (b_m - a_m)/\alpha_m$ is a closed sphere. Denote by $S(\alpha_1, \cdots, \alpha_m)$ the set of such spheres with volumes less than or equal to h_n^m/n. By (13), (16), we have for n large enough

$$P(X_1 \in B) \leqslant Mh_n^m/n, \text{ for } B \in S(\alpha_1, \cdots, \alpha_m). \tag{20}$$

Taking $b = n^{-1+\eta}$, $\varepsilon = n^{-1+\eta}$, $0 < \eta < 1$, in formula (2.5) of [3], we obtain for n sufficiently large:

$$\begin{aligned} & P\Big(\sup_{B \in S(\alpha_1, \cdots, \alpha_m)} \Big| \frac{\#(B \cap \{X_1, \cdots, X_n\})}{n} - P(X_1 \in B) \Big| \geqslant n^{-1+\eta}\Big) \\ & \leqslant 4(2n)^{2m} \exp(-n^{\eta}/68) + 8n \exp(-n^{\eta}/10) \\ & \leqslant \exp(-n^{\eta/2}). \end{aligned} \tag{21}$$

By (20), (21), for large n we have

$$P(\sup_{B \in S(\alpha_1, \cdots, \alpha_m)} \#(B \cap \{X_1, \cdots, X_n\}) \geqslant 2n^{\eta}) \leqslant \exp(-n^{\eta/2}). \tag{22}$$

Define

$$S = \bigcup_{\alpha_1, \cdots, \alpha_m = 1}^{n} S(\alpha_1, \cdots, \alpha_m). \tag{23}$$

From (21) for n sufficiently large,

$$P(\sup_{B \in S} \#(B \cap \{X_1, \cdots, X_n\}) \geqslant 2n^{\eta}) \leqslant n^m \exp(-n^{\eta/2}) \leqslant \exp(-n^{\eta/3}). \tag{24}$$

For any $z \in R^m$, denote by ρ_{zn} the set $\{x : x \in R^m, \| x - z \| \leqslant h_n\}$. Assume $\| x - y \| \leqslant h_n/n$. It is easily seen that the symmetric difference $\rho_{xn} \Delta \rho_{yn}$ can be contained in at most $2m$ sets in S. According to (24), the event that the number of sample points $X_1, \cdots, X_n$ contained in these $2m$ sets do not exceed $4mn^{\eta}$ has a probability no less than $1 - \exp(-n^{\eta/3})$. Again, since $\| x - y \| \leqslant h_n/n$, by (17) we see that for $X_i \in \rho_{xn} \Delta \rho_{yn}$

$$\Big| K\Big(\frac{x - X_i}{h_n}\Big) - K\Big(\frac{y - X_i}{h_n}\Big) \Big| \leqslant mM/n.$$

Hence we obtain, with probability no less than $1 - \exp(-n^{\eta/3})$, that for n sufficiently large and $\| x - y \| \leqslant h_n/n$,

$$| \hat{Y}_{n1}(x) - \hat{Y}_{n2}(y) | \leqslant \frac{1}{n}(8mMn^{\eta} + mM) \leqslant 9mMn^{-1+\eta}.$$

This is just (19), and the lemma is proved.

Lemma 3 If $\lim_{n \to \infty} \varepsilon_n n^{(k+m)/(2k+m)} (\log n)^{-m/(4k+4M)} = \infty$, then for n sufficiently large

$$P(\sup_x |\hat{Y}_{n1}(x) - Y_n(x)| \geqslant \varepsilon_n) \leqslant 2P(\sup_x |\hat{Y}_{n1}(x) - \hat{Y}_{n2}(x)| \geqslant \varepsilon_n/2). \tag{25}$$

Proof Since $\hat{Y}_{n1}$ and Y_n are right continuous, if we arrange all the rational points in R^m as $r_1, r_2, \cdots$, we have

$$\sup_x |\hat{Y}_{n1}(x) - Y_n(x)| = \sup_i |\hat{Y}_{n1}(r_i) - Y_n(r_i)|,$$

$$\sup_x |\hat{Y}_{n1}(x) - \hat{Y}_{n2}(x)| = \sup_i |\hat{Y}_{n1}(r_i) - \hat{Y}_{n2}(r_i)|.$$

Write $Z_i = \hat{Y}_{n1}(r_i) - Y_n(r_i)$, $Z_i' = \hat{Y}_{n2}(r_i) - Y_n(r_i)$, $i = 1, 2, \cdots$. Then $Z_i^* = Z_i - Z_i'$ is the symmetrization of Z_i. Since $\mathrm{E}\hat{Y}_{n1}(x) = Y_n(x)$ and

$$\mathrm{Var}(\hat{Y}_{n1}(x)) \leqslant M^3 h_n^m / n = M^3 n^{-2(k+m)/(2k+m)} (\log n)^{m/(2k+2m)}, \tag{26}$$

it follows that if $\varepsilon_n n^{(k+m)/(2k+m)} (\log n)^{-m/(4k+4m)} \to \infty$ and n is sufficiently large, we have $P(|Z_i| \leqslant \varepsilon_n/2) < 1/2$. Hence, denoting by μ the medium, we have $|\mu(Z_i)| \leqslant \varepsilon_n/2$ uniformly for i when n is large. Thus

$$P(\sup_i |Z_i| \geqslant \varepsilon_n) \leqslant P(\sup_i |Z_i - \mu(Z_i)| \geqslant \varepsilon_n/2).$$

Using p. 247 (ii) of [9], we finally get

$$P(\sup_i |Z_i| \geqslant \varepsilon_n) \leqslant 2P(\sup_i |Z_i^*| \geqslant \varepsilon_n/2).$$

This is just (25), and the lemma is proved.

The following lemma is essentially a known result:

Lemma 4 A population contains N balls bearing numbers $a_1, \cdots, a_N$, respectively. Suppose that $|a_i| \leqslant R$, $i = 1, \cdots, N$, $a = \frac{1}{N}\sum_{i=1}^{N} a_i$, $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(a_i - \bar{a})^2$. Let Z be the arithmetic mean of the n numbers on the n balls drawn randomly and without replacement from the population. Then for $0 < t < 2R$ we have

$$P(|Z - a| \geqslant t) \leqslant 2\exp(-nt^2/(2\sigma^2 + 2Rt)).$$

The proof follows from Theorems 3 and 4 of [10]. Also, use is made of the elementary inequality $\log(1+x) \geqslant \frac{2x}{2+x}$ $(x \geqslant 0)$.

Lemma 5 Take $\eta \in (0, 2k/(2k+m))$ and

$$\varepsilon_n = n^{-(k+m)/(2k+m)} \sqrt{\log n a} \quad (a > 0). \tag{27}$$

Then we have for n sufficiently large

$$\begin{aligned} J &\triangleq P(\sup_x |\hat{Y}_{n1}(x) - \hat{Y}_{n2}(x)| \geqslant \varepsilon_n/2) \\ &\leqslant 2n^{4m - a^2/(432M^3 + aM)} + \exp(-n^{\eta/3}) + 4n^{m+1}\exp\left(-\frac{M}{4}n^{2k/(2k+m)}\right). \end{aligned} \tag{28}$$

Proof Denote by T_{ni}, $i = 1, \cdots, d_n$, the different nonempty intersections of the set $\{X_1, \cdots, X_n\}$ with all possible intervals in R^m. Evidently $d_n \leqslant (2n)^{2m}$. Write

$$U_{ni} = \{x : \rho_{xn} \cap \{X_1, \cdots, X_{2n}\} = T_{ni}\}, \ i = 1, \cdots, d_n.$$

The set U_{ni} has a form $\{(x_1, \cdots, x_m) : a_j \leqslant x_j \leqslant b_j, \ j = 1, \cdots, m\}$ with $\max\limits_{1 \leqslant j \leqslant m}(b_j - a_j) \leqslant 2h_n$. Divide each interval $[a_j, b_j]$, $j = 1, \cdots, m$, into n sub-intervals, thus dividing U_{ni} into n^m m-dimensional intervals $\{U_{nij}\}$. Also, use u_{nij} to denote the center of U_{nij}. Note that d_n, T_{ni}, U_{ni}, U_{nij} and u_{nij} all depend upon $X_1, \cdots, X_{2n}$. Now we have

$$\begin{aligned} J &\leqslant P(\sup_x |\hat{Y}_{n1}(x) - \hat{Y}_{n2}(x)| \geqslant \varepsilon_n/2, \ \sup_x \widetilde{Y}_n(x) \leqslant b) + P(\sup_x \widetilde{Y}_n(x) > b) \\ &\triangleq J_1 + J_2 \end{aligned} \tag{29}$$

where $\widetilde{Y}_n(x) = \dfrac{1}{2n}\sum\limits_{i=1}^{2n}\left|K\left(\dfrac{x - X_i}{h_n}\right)\right|$, and b will be taken as

$$b = 6M^2 n^{-m/(2k+m)}. \tag{30}$$

We have

$$J_1 = \int_{R^{2mn}} \frac{1}{(2n)!}\sum{}' I(\sup_x |\hat{Y}_{n1}(x) - \hat{Y}_{n2}(x)| \geqslant \varepsilon_n/2) I(\sup_x \widetilde{Y}_n(x) \leqslant b)\mathrm{d}Q,$$

where Q is the probability measure of $(X_1, \cdots, X_{2n})$, and $\sum'$ denotes summation over all permutations of $X_1, \cdots, X_{2n}$. By Lemma 2 and the definition of U_{nij}, one sees that for n large enough the event "if $x \in U_{nij}$, then $|\hat{Y}_{nt}(x) - \hat{Y}_{nt}(u_{nij})| \leqslant 9mMn^{-1+\eta} = o(\varepsilon_n)$, for all i, j and $t = 1, 2$" has a probability not less than $1 - \exp(n^{-\eta/3})$. So for n large enough

$$\begin{aligned} J_1 &\leqslant \int_{R^{2mn}} \frac{1}{(2n)!}\sum{}' I(\sup_{i,j} |\hat{Y}_{n1}(u_{nij}) - \hat{Y}_{n2}(u_{nij})| \geqslant \varepsilon_n/3) I(\sup_x \widetilde{Y}_n(x) \leqslant b)\mathrm{d}Q \\ &\quad + \exp(-n^{\eta/3}) \\ &\triangleq J_1' + \exp(-n^{\eta/3}). \end{aligned} \tag{31}$$

Noticing that $|\hat{Y}_{n1}(u_{nij}) - \hat{Y}_{n2}(u_{nij})| + 2|\hat{Y}_{n1}(u_{nij}) - \hat{Y}_n(u_{nij})|$, we have

$$\begin{aligned} J_1' &\leqslant \int_{R^{2mn}} \frac{1}{(2n)!} I(\sup_x \widetilde{Y}_n(x) \leqslant b)\sum{}' I(\sup_{i,j} |\hat{Y}_{n1}(u_{nij}) - \hat{Y}_n(u_{nij}) \geqslant \varepsilon_n/6)\mathrm{d}Q \\ &\leqslant \sum_{i,j}\int_{R^{2mn}} I(\sup_x \widetilde{Y}_n(x) \leqslant b)\ \frac{1}{(2n)!} I(|\hat{Y}_{n1}(u_{nij}) - \hat{Y}_n(u_{nij})| \geqslant \varepsilon_n/6)\mathrm{d}Q. \end{aligned}$$

Applying Lemma 4 to $\dfrac{1}{(2n)!}\sum' I(|\hat{Y}_{n1}(u_{nij}) - \hat{Y}_n(u_{nij})| \geqslant \varepsilon_n/6)$ and noticing that $\sigma^2 \leqslant Mb$ since $\sup\limits_x \widetilde{Y}_n(x) \leqslant b$, we have

$$\frac{1}{(2n)!}\sum{}' I(|\hat{Y}_{n1}(u_{nij}) - \hat{Y}_n(u_{nij})| \geqslant \varepsilon_n/6) \leqslant 2\exp(-n\varepsilon_n^2/72M(b + \varepsilon_n)).$$

Since for fixed n the number of u_{nij}'s does not exceed n^{4m}, we get

$$J_1' \leqslant n^{4m} 2\exp(-n\varepsilon_n^2/72M(b + \varepsilon_n)) \leqslant 2n^{4m - a^2/(432m^3 + aM)}. \tag{32}$$

Now consider the term J_2 in (29). Since the x maximizing $\widetilde{Y}_n$ necessarily satisfies $\rho_{xn} \cap$

$\{X_1, \cdots, X_n\} \neq \varnothing$, we assume for definiteness $X_1 \in \rho_{xn}$. Thus x falls into the cube with vertices $X_1 \pm (h_n, \cdots, h_n)$. Divide this cube into n^m equal parts V_{xi}, $i = 1, \cdots, m$, by dividing equally the intervals $[X_{1i} - h_n, X_{1i} + h_n]$, $i = 1, \cdots, m$ (here $X_1 = (X_{11}, \cdots, X_{1m})$). Denote the centers of these parts as v_{xi}, $i = 1, \cdots, m$. From the choice of η, an argument similar to that employed in the proof of Lemma 2 gives $P(H_n) \geqslant 1 - \exp(-n^{\eta/3})$ for n large enough, where

$$H_n = \{x \in V_{xj} \Rightarrow |\widetilde{Y}_n(x) - \widetilde{Y}_n(v_{xj})| \leqslant 9mMn^{-1-\eta} \leqslant b/2\}.$$

For such n we have

$$\widetilde{Y}_n(x) \geqslant b \Rightarrow \widetilde{Y}_n(v_{xj}) \geqslant b/2. \tag{33}$$

Define $\widetilde{Y}_{n1}(x) = \dfrac{1}{2n-1}\sum_{j=2}^{2n}\left|K\left(\dfrac{x-X_j}{h_n}\right)\right|$. Then by (17) we have $\widetilde{Y}_{n1}(x) \geqslant \widetilde{Y}_n(x) - M/2n$. From this and (33) we have

$$\widetilde{Y}_n(x) \geqslant b \Rightarrow \widetilde{Y}_{n1}(v_{xj}) \leqslant b/3.$$

As there are only n^m v_{xj}'s and $X_1, \cdots, X_{2n}$ are independently and identically distributed, we get

$$P(\sup_x \widetilde{Y}_n(x) \geqslant b) \leqslant 2n^{m+1} \sup_x P(\widetilde{Y}_{n1}(x) \geqslant b/3). \tag{34}$$

Noticing (30) and that $\mathrm{E}\,\widetilde{Y}_{n1}(x) \leqslant M^2 h_n^m$ following from (16), (17), we have

$$P(\widetilde{Y}_{n1}(x) \leqslant b/3) \leqslant P(\widetilde{Y}_{n1}(x) - \mathrm{E}\,\widetilde{Y}_{n1}(x) \geqslant M^2 h_n^m).$$

Since $\mathrm{Var}\left(\left|K\left(\dfrac{x-X_1}{h_n}\right)\right|\right) \leqslant M^3 h_n^m$, employing Lemma 4 (take $R = M$) and (13), we have

$$P(\widetilde{Y}_{n1}(x) \geqslant b/3) \leqslant 2\exp\left(-\frac{M}{4}n^{2k/(2k+m)}\right). \tag{35}$$

It follows from (34) and (35) that

$$J_2 \leqslant 4n^{m+1}\exp\left(-\frac{M}{4}n^{2k/(2k+m)}\right). \tag{36}$$

Finally, (28) follows from (29), (31), (32) and (36), and Lemma 5 is proved.

3 Proof of the Theorem

Employing (25), (28) and noticing (12), (14), (15), we see that for any $f \in C_{k\alpha}$, $\eta > 0$ sufficiently small and n sufficiently large,

$$\begin{aligned}
&P_f(\sup_x |f_n(x) - \mathrm{E}f_n(x)| \geqslant an^{-k/(2k+m)}(\log n)^{k/(2k+2m)}) \\
&= P_f(\sup_x |Y_{n1}(x) - \mathrm{E}Y_{n1}(x)| \geqslant \varepsilon_n) \\
&\leqslant 4n^{4m-a^2/(432M^3+aM)} + 2\exp(n^{-\eta/3}) + 4n^{m+1}\exp\left(-\frac{M}{4}n^{2k/(2k+m)}\right) \\
&\leqslant 5n^{4m-a^2/(432M^3+aM)}.
\end{aligned} \tag{37}$$

On the other hand, it is easily seen that

$$\mathrm{E}f_n(x) - f(x) = \int_{R^m} K(u)[f(x - h_n u) - f(x)]\mathrm{d}u. \tag{38}$$

Taking Taylor's expansion with the remainder term of f at point x:

$$\begin{aligned} f(x - h_n u) = & \sum_{l=0}^{k-1} \sum_{l}^{(*)} \frac{f^{(l)}(l_1, \cdots, l_m; x)}{l_1! \cdots l_m!}(-h_n)^l u_1^{l_1} \cdots u_m^{l_m} \\ & + \sum_{k}^{(*)} \frac{f^{(k)}(k_1, \cdots, k_m; \tilde{x})}{k_1! \cdots k_m!}(-h_n)^k u_1^{k_1} \cdots u_m^{k_m}, \end{aligned} \tag{39}$$

where $\tilde{x}$ lies in a interval with vertices x and $x - h_n u$. Substituting (39) into (38) and making use of (1), (11), we get

$$| \mathrm{E}f_n(x) - f(x) | \leqslant m^k \alpha h_n^k = \alpha m^k n^{-k/(2k+m)} (\log n)^{k/(2k+2m)}, \tag{40}$$

which holds uniformly for $x \in R^m$. From this and (37) we see that for large n, a,

$$\begin{aligned} & P_f(\sup_x | f_n(x) - f(x) | \geqslant an^{-k/(2k+m)} (\log n)^{-k/(2k+2m)}) \\ & \leqslant P_f\left(\sup_x | f_n(x) - \mathrm{E}f_n(x) | \geqslant \frac{a}{2} n^{-k/(2k+m)} (\log n)^{-k/(2k+2m)}\right) \\ & \leqslant 5n^{4m - a^2/(1728M^3 + 2aM)} \end{aligned} \tag{41}$$

holds uniformly for $f \in C_{k\alpha}$. This proves (2) and terminates the proof of Theorem 1.

For a proof of Theorem 2, notice that, by (41), for any given $\varepsilon > 0$ and constant sequence $a_n \to \infty$ we have

$$\begin{aligned} & P_f(a_n^{-1} n^{k/(2k+m)} (\log n)^{-k/(2k+2m)} \sup_x | f_n(x) - f(x) | \geqslant \varepsilon) \\ & = P_f(\sup_x | f_n(x) - f(x) | \geqslant a_n \varepsilon_n^{-k/(2k+m)} (\log n)^{k/(2k+2m)}) \\ & \leqslant 5n^{4m - a_n^2 \varepsilon^3/(1728M^3 + 2a_n M)} \\ & = O(n^{-2}). \end{aligned}$$

Hence

$$\sum_n P_f(a_n^{-1} n^{-k/(2k+m)} (\log n)^{k/(2k+2m)} \sup_x | f_n(x) - f(x) | \geqslant \varepsilon) < \infty$$

for any $\varepsilon > 0$. This proves that

$$\lim_{n \to \infty} W_n / a_n = 0, \text{ a. s.} \tag{42}$$

where $W_n = n^{-k/(2k+2m)} (\log n)^{k/(2k+2m)} \sup_x | f_n(x) - f(x) |$. Since $a_n \to \infty$ is arbitrarily chosen, from (42) one evidently gets

$$\limsup_{n \to \infty} | W_n | < \infty, \text{ a. s.}$$

and this proves Theorem 2.

Remark By changing h_n, defined by (13), to $n^{-1/(2k+m)}$ the log term in (26) and (40) disappears, and we have for $f \in C_{k\alpha}$ uniformly that

$$\mathrm{E}_f(f_n(x)-f(x))^2 = \mathrm{Var}_f(f_n(x)) + (\mathrm{E}f_n(x)-f(x))^2$$
$$\leqslant (M^3+\alpha m^k)n^{-2k/(2k+m)}.$$

From this we get

$$P_f(|f_n(x)-f(x)| \geqslant an^{-k/(2k+m)}) \leqslant (M^3+\alpha m^k)/a^2 \to 0,$$

when $a\to\infty$ uniformly for $f\in C_{ka}$. Hence it follows that

$$\lim_{a\to\infty}\lim_{n\to\infty}\left\{\inf_{f\in C_{ka}} P_f(|f_n(x)-f(x)| \leqslant an^{-k/(2k+m)})\right\} = 1.$$

This verifies the correctness of Farrell's conjecture mentioned in Section 1.

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密度核估计的一致收敛速度

陈希孺

摘　　要

以 C_{ka} 记 R^m 中一概率密度族,其任意 k 阶混合偏导数绝对值都不超过 α. Farrell 在[5]中证明:为估计 $f(0)(f\in C_{ka})$,任何估计量的收敛速度不会超过 $O(n^{-k/(2k+m)})$,且对适当选择的核估计这个速度可以达到.在本文中,我们证明了对适当选择的核估计 f_n, $\sup_x|f_n(x)-f(x)|$ 收敛于零的速度可达到 $O(n^{-k/(2k+m)}\cdot(\log n)^{k/(2k+2m)})$. 因此,对 $\sup_x|f_n(x)-f(x)|$ 收敛的阶的主要部分而言,本文的结果已无可改善,这个结果显著改进了文献中的已有结果.

11

The Optimal Rate of Convergence of Error for k-NN Median Regression Estimates

Abstract Let (X, Y), (X_1, Y_1), $\cdots$, (X_n, Y_n) be iid. random vectors, where Y is one-dimensional. It is desired to estimate the conditional median $\xi(X)$ of Y, by use of $Z_n = \{(X_i, Y_i), i = 1, \cdots, n\}$ and X. Denote by $\xi_{nk}(X, Z_n)$ the k-NN estimate of $\xi(X)$, and put $H_{nk}(Z_n) = \mathrm{E}\{| \xi_{nk}(X, Z_n) - \xi(X) | | Z_n\}$, the conditional mean absolute error. This artical establishes the optimal convergence rate of $P(H_{nk_n}(Z_n) \geqslant \varepsilon)$, under fairly general assumptions on (X, Y) and k_n, which tends to ∞ in some suitable way.

1 Introduction and Main Result

Let (X, Y), (X_1, Y_1), $\cdots$, (X_n, Y_n) be $R^d \times R^1$-valued iid. random vectors. Denote

$$Z_n = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}, \quad X^n = \{X_1, \cdots, X_n\}. \tag{1}$$

Z_n is the sample of (X, Y). The conditional distribution function of Y given $X = x$ is denoted by $F(y \mid x) = P(Y < y \mid X = x)$.

Suppose that $R(x)$ is a quantity determined by $F(y|x)$, and we wish to estimate it, basing on the sample Z_n. The case that $R(x) = \mathrm{E}(Y \mid X = x)$, the mean-value regression function, has been much studied in the literature. On considering the robustness, Zheng Zhongguo proposed in [1] the conditional median $\xi(x)$—the median of $F(y|x)$, as the object of estimation. He introduced the k-Nearest Neighbor (k-NN) estimate $\xi_{nk}(x, Z_n)$ of $\xi(x)$ as follows: Introduce a suitable distance $\| x-x' \|$ in R^d. Rearrange $X_1, \cdots, X_n$ according to their distances from x:

$$\| X_{n1}(x) - x \| \leqslant \| x_{n2}(x) - x \| \leqslant \cdots \leqslant \| X_{nn}(x) - x \| .$$

Call $\{X_{n1}(x), \cdots, X_{nk}(x)\}$ the k-NN of x. Denote by $Y_{ni}(x)$ the Y-value corresponding to $X_{ni}(x)$, that is, $Y_{ni}(x) = Y_j$ if $X_{ni}(x) = X_j$. $\xi_{nk}(x, Z_n)$ is defined as the sample-median of $\{Y_{n1}(x), \cdots, Y_{nk}(x)\}$. Zheng studied in [1] the strong consistency and asymptotic normality

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of this estimate.

The mean absolute error, and the conditional mean absolute error for given Z_n, of this estimate, are

$$H_{nk} = \mathrm{E} \mid \xi_{nk}(X, Z_n) - \xi(X) \mid, \text{ and } H_{nk}(Z_n) = \mathrm{E}\{ \mid \xi_{nk}(X, Z_n) - \xi(X) \mid \mid Z_n \}$$

respectively. From a practical point of view the latter is more sensible than the former. As pointed out by Wagner in [2], in practice Z_n is not always easily available. On the contrary, Z_n is gathered during a period of time, and will be used repeatedly in problems with the same nature.

The purpose of this paper is to study the behavior of $H_{nk_n}(X, Z_n)$, as $n \to \infty$ and $k = k_n$ varies with n. A remarkable work of this kind of study is due to Beck [3]. In his work Beck used $\sum_{i=1}^{k} Y_{ni}(X)/k$ to estimate $\mathrm{E}(Y|x)$. Denote by $G_{nk_n}(Z_n)$ the conditional mean absolute error $\mathrm{E}\left\{ \left| \sum_{i=1}^{k} Y_{ni}(x)/k - \mathrm{E}(Y \mid X) \right| \mid Z_n \right\}$. Beck established the following exponential rate of convergence

$$P(G_{nk_n}(Z_n) \geqslant \varepsilon) = O(\mathrm{e}^{-cn})$$

under a number of conditions, among which the crucial one is the boundedness of Y. From a theoratical point of view the boundedness condition on Y is too restrictive. We shall tackle the problem under more reasonable condition imposed on Y.

Define $m(x, p)$ as the largest possible absolute value of the p-percentile of $F(y \mid x)$, $0 < p < 1$, and

$$G(x, \delta) = \sup\left\{ m\left(x', \frac{1}{2} \pm \delta\right) : \| x' \| \leqslant 2 \| x \| \right\}, \ 0 < \delta < \frac{1}{2}.$$

The main result of this paper is as follows:

Theorem 1 Suppose that

1° $F(y|x)$ has a unique median $\xi(x)$ for any x.

2° $F(y \mid x') \xrightarrow{\mathscr{L}} F(y \mid x)$ for $x' \to x$.

3° $\mathrm{E}[G(X, \delta_0)] < \infty$ for some $\delta_0 \in \left(0, \frac{1}{2}\right)$.

4° The distribution Q of X possesses a density f, and for $a>0$ small enough the set $\{x: f(x) > a\}$ differs from some open set by only a Lebesgue null-set.

Also, suppose that

$$k_n/n \to 0, \ \log n/k_n \to 0, \text{ for } n \to \infty. \tag{2}$$

Then for any $\varepsilon > 0$ there exists $C > 0$ (depending on ε but not n) such that

$$P(H_{nk_n}(Z_n) \geqslant \varepsilon) = O(\mathrm{e}^{-ck_n}). \tag{3}$$

This rate cannot be improved further: For any given $\{k_n\}$ with $k_n \to \infty$ and $k_n/n \to 0$, one can find (X, Y) whose distribution satisfies the four conditions of Theorem 1, but for any

$\varepsilon > 0$ and $C > \log 2$ it is true that

$$\lim_{n\to\infty} \sup\{ e^{ck_n} P(H_{nk_n}(Z_n) \geqslant \varepsilon)\} \geqslant 1. \tag{4}$$

Condition 3° is a mild restriction on Y. In order that $H_{nk_n}(Z_n)$ is meaningful, we must have $\mathrm{E} \mid \xi(X) \mid < \infty$. Condition 3° is stronger than this. Hence there is a question: Can condition 3° be replaced by the weaker one $\mathrm{E} \mid \xi(X) \mid < \infty$? We shall show by an example that this is impossible: The condition $\mathrm{E} \mid \xi(X) \mid < \infty$ is not enough even for the far-weaker assertion that

$$\lim_{n\to\infty} P(H_{nk_n}(Z_n) \geqslant \varepsilon) = 0.$$

Since in (3) k_n may tend to ∞ with any rate slower than n, the rate $O(e^{-ck_n})$ given by Theorem 1 can be made arbitrarily near $O(e^{-cn})$ but cannot reach it. One naturally asks: Whether or not there exists (X, Y) with Y unbounded, such that $P(H_{nk_n}(Z_n) \geqslant \varepsilon) = O(e^{-cn})$ for some $\{k_n\}$, with $k_n \to \infty$ and $k_n/n \to 0$? The question is interesting but the probable answer is not apparent to guess.

We shall give the proof of Theorem 1 in Section 2, and then make some remarks related to this theorem.

2 Proof of Theorem 1

(I) Some preliminary facts.

1. Hoeffding inequality ([4]). Suppose that the random variable ξ obeys the Binomial Law $B(n, p)$, then for any $\varepsilon > 0$ we have

$$P\left(\left| \frac{\xi}{n} - p \right| \geqslant \varepsilon \right) \leqslant 2\exp(-n\varepsilon^2/(2p+\varepsilon)).$$

From this the following corollary follows easily: Let $A_1, \cdots, A_n$ be n independent events,

$$\min_{1\leqslant i\leqslant n} P(A_i) = p^* > p > 0.$$

Put

$$\xi = \sum_{i=1}^{n} I_{A_i},$$

then

$$P(\xi \leqslant np) \leqslant 2\exp\{-n(p^* - p)^2/(3p^* - p)\}. \tag{5}$$

2. Under conditions 1°, 2° of Theorem 1, the median $\xi(x)$ of $F(y \mid x)$ is a continuous function of x.

The proof is easy and therefore omitted.

3. For $\varepsilon > 0$, define

$$\begin{aligned} f_{1\varepsilon}(x) &= P(Y \leqslant \xi(x) - \varepsilon \mid x), \\ f_{2\varepsilon}(x) &= P(Y \geqslant \xi(x) + \varepsilon \mid x). \end{aligned} \tag{6}$$

Suppose that conditions 1°, 2° of Theorem 1 hold. Then for any constant b, the sets

$$\{x: f_{1\varepsilon}(x) < b\}, \{x: f_{2\varepsilon}(x) < b\}$$

are both open.

Proof Evidently one only has to show that

$$x_n \to x \quad \Rightarrow \quad \limsup_{n\to\infty} f_{i\varepsilon}(x_n) \leqslant f_{i\varepsilon}(x), \ i = 1, 2. \tag{7}$$

Given $\varepsilon > 0$, find $\delta > 0$ sufficiently small, such that $\xi(x) \mp \varepsilon \pm \delta$ are continuity points of the distribution $F(y|x)$. Since $\xi(x)$ is continuous, we have

$$| \xi(x_n) - \xi(x) | < \delta$$

for large n. Hence for large n we have

$$\xi(x_n) - \varepsilon < \xi(x) - \varepsilon + \delta, \ \xi(x_n) + \varepsilon > \xi(x) + \varepsilon - \delta.$$

Therefore we have for large n,

$$f_{1\varepsilon}(x_n) \leqslant F(\xi(x) - \varepsilon + \delta \mid x_n), \ f_{2\varepsilon}(x_n) \leqslant 1 - F(\xi(x) + \varepsilon - \delta \mid x_n). \tag{8}$$

Since $\xi(x) \mp \varepsilon \pm \delta$ are continuity points, from condition 2° and (8), we have

$$\limsup_{n\to\infty} f_{1\varepsilon}(x_n) \leqslant \lim_{n\to\infty} F(\xi(x) - \varepsilon + \delta \mid x_n) = F(\xi(x) - \varepsilon + \delta \mid x).$$

Setting $\delta \downarrow 0$, we get $\limsup_{n\to\infty} f_{1\varepsilon}(x_n) \leqslant f_{1\varepsilon}(x)$. The case $i = 2$ in (7) can be handled in a similar fashion.

4. Under condition 2° of Theorem 1, we have

$$\sup\{| m(x, p) | : \| x \| \leqslant r\} < \infty$$

for any $p \in (0, 1)$ and $r < \infty$.

5. There exist $\eta > 0$ and r sufficiently large such that $Q(S_{x, \|x\|}) \geqslant \eta$ for any x belonging to the support of Q and $\| x \| \geqslant r$, where $S_{a,\rho}$ is the sphere with radius ρ and centered at a.

These two facts are easily proved and details omitted.

(Ⅱ) Since $\xi(x)$ is unique, we have

$$f_{i, \varepsilon/6}(x) < \frac{1}{2}$$

for all x and $i = 1, 2$. Choose $b < \frac{1}{2}$ so that $Q(B_1) > 1 - \varepsilon_1\varepsilon$, where $B_1 = \{x: f_{i, \varepsilon/6}(x) < b, i = 1, 2\}$, ε_1 is a positive number to be specified. It follows from (Ⅰ) 3 that B_1 is open. By condition 4° of Theorem 1 one can find $a > 0$ such that $\{x: f(x) > a\}$ differs from some open set B_2 by only a Lebesgue null-set, and $Q(B_2) > 1 - \varepsilon_1\varepsilon$. So there is an open set $B_3 \subset B_1 \cap B_2$ such that $Q(B_3) > 1 - 3\varepsilon_1\varepsilon$, and $\xi(x)$ is continuous uniformly on B_3.

For simplicity we shall call the set

$$\{x = (x_1, \cdots, x_d): a_i \leqslant x_i \leqslant a_i + h, \ i = 1, \cdots, d\}$$

a regular supercube with size h. Find regular supercubes $V_1, \cdots, V_N$ with the same size such that $V_i \subset B_3$, $V_i \cap V_j = \varnothing$ for $i, j = 1, \cdots, N$, $i \neq j$, and

$$\sum_{i=1}^{N} Q(V_i) > 1 - 4\varepsilon_1\varepsilon,$$
$$\sup\{|\xi(x) - \xi(x')| : x \in V_i, x' \in V_i\} < \varepsilon/6, \quad i = 1, \cdots, N.$$

Denote by $\rho(S_1, S_2)$ the distance of two sets S_1 and S_2 in R^d, and by $D(S)$ the diameter of set S. Find regular supercubes $V_1^*, \cdots, V_N^*$ with the same size q^* such that $V_i^* \subset V_i^0$ (the inner of V_i), $i = 1, \cdots, N$, and

$$\sum_{i=1}^{N} Q(V_i^*) > 1 - 5\varepsilon_1\varepsilon. \tag{9}$$

We have $t \triangleq \min_{1 \leqslant i \leqslant N} \rho(V_i^*, V_i - V_i^0) > 0$. Denote by $[a]$ the largest integer not exceeding a, and

$$c_n = [q^*(an/(2k_n))^{1/d}], \quad b_n = q^*/c_n.$$

From $k_n/n \to 0$ it follows that $b_n \to 0$.

For each i ($i = 1, \cdots, N$) there exists a family of regular supercubes $H_i = \{H_{i1}, H_{i2}, \cdots\}$ such that

1° Each H_{iu} has a size b_n.

2° $H_{iu}^0 \cap H_{iv}^0 = \varnothing$ for $u \neq v$.

3° $R^d = \bigcup_{j=1}^{\infty} H_{ij}$.

4° There is a subset of H_i, to be denoted by $\mathscr{G}_{ni}^* = \{G_{i1}^*, \cdots, G_{il_n}^*\}$ ($l_n = c_n^d$), such that

$$V_i^* = \bigcup_{j=1}^{l_n} G_{ij}^*.$$

Write $\mathscr{G}_n^* = \bigcup_{i=1}^{N} \mathscr{G}_{ni}^*$. Similarly, the set $\{H_{iu} : H_{iu} \subset V_i, u = 1, 2, \cdots\}$ will be denoted by $\mathscr{G}_{ni} = \{G_{i1}, G_{i2}, \cdots\}$, and $\mathscr{G}_n = \bigcup_{i=1}^{N} \mathscr{G}_{ni}$. Evidently one has $\mathscr{G}_{ni} \supset \mathscr{G}_{ni}^*$ and $\mathscr{G}_n \supset \mathscr{G}_{nj}^*$. Also, if the size of V_i is q, then the number of elements in $\mathscr{G}_{ni}$ does not exceed

$$(q/b_n)^d = (q/q^*)^d c_n^d \leqslant \frac{1}{2} a q^d n/k_n.$$

Hence, the number of elements in $\mathscr{G}_n$ does not exceed $\frac{a}{2} N q^d n/k_n$. Note that $\frac{1}{2} a N q^d$ does not depend on n.

Let $\mathscr{F}$ be a family of point sets in R^d. For simplicity, we shall denote by $(\mathscr{F})$ the union of all sets contained in $\mathscr{F}$. Also, for a set $B \subset R^d$, the number of X_i's ($i = 1, \cdots, n$) contained in B will be denoted by $\#(B)$. Now define

$$\widehat{\mathscr{G}}_{ni}^* = \{G : G \in \mathscr{G}_{ni}^*, \#(G) < k_n\}, \quad i = 1, \cdots, N;$$
$$\widehat{\mathscr{G}}_n^* = \bigcup_{i=1}^{N} \widehat{\mathscr{G}}_{ni}^*.$$

For $G \in \mathscr{G}_{ni}^* - \widehat{\mathscr{G}}_{ni}^*$, define

$$W(G) = \bigcup \{H: H \in H_i, \rho(G, H) \leqslant D(G)\}.$$

From the definition of $\hat{\mathscr{G}}_m^*$ it follows that if $x \in G$ $(G \in \mathscr{G}_m^* - \hat{\mathscr{G}}_m^*)$, then $X_{ni}(x) \in W(G)$ for $i = 1, \cdots, k_n$. Also, when n is sufficiently large such that $b_n < t/d$, then $W(G) \subset (\mathscr{G}_{ni})$. It is obvious that there exists constant m_d depending only upon d, such that for $G \in \mathscr{G}_m^* - \hat{\mathscr{G}}_m^*$, $W(G)$ can be expressed as the union of less than m_d regular supercubes in H_i.

Since the volume of G_{ij} is $b_n^d \geqslant 2k_n/(an)$ and $f(x) > a$ for $x \in B_3$, a. e., we have

$$Q(G_{ij}) \geqslant 2k_n/n. \tag{10}$$

By Hoeffding inequality one finds easily that

$$P(\#(G_{ij}) < k_n) \leqslant 2\exp(-k_n/5). \tag{11}$$

Put

$$S_n = \{\#(G_{ij}) \geqslant k_n, \text{ for all } G_{ij} \in \mathscr{G}_n\}.$$

By (11), and noticing that the number of elements in $\mathscr{G}_n$ does not exceed $\frac{1}{2}aNq^d n/k_n$, one finds that

$$P(S_n) \geqslant 1 - aNq^d nk_n^{-1}\exp(-k_n/5). \tag{12}$$

Since $\log n/k_n \to 0$, we have $n = 0(\exp(k_n/15))$. Therefore it follows from (12) that there exists constant $C > 0$ not depending on n, such that

$$P(S_n) \geqslant 1 - \exp(-Ck_n). \tag{13}$$

For simplicity the symbol C will be employed to denote any positive constant not depending on n. C can assume different values in each of its appearance, even within the same expression.

(Ⅲ) Now we proceed to prove the first half of Theorem 1. We have

$$\begin{aligned} & P(H_{nk_n}(Z_n) \geqslant \varepsilon) \\ & \quad \leqslant P\left\{\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n) - \xi(X)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)}(X) \mid Z_n] \geqslant \frac{1}{2}\varepsilon\right\} \\ & \qquad + P\left\{\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n) - \xi(X)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \geqslant \frac{1}{2}\varepsilon\right\} \\ & \quad \triangleq J_{1n} + J_{2n}. \end{aligned} \tag{14}$$

Take J_{2n} first. We have

$$\begin{aligned} J_{2n} & \leqslant P\left\{\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \geqslant \frac{1}{4}\varepsilon\right\} \\ & \quad + P\left\{\mathrm{E}[\,|\,\xi(X)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \geqslant \frac{1}{4}\varepsilon\right\} \\ & \triangleq J'_{2n} + J''_{2n}. \end{aligned} \tag{15}$$

It follows easily from condition 3° of Theorem 1 that $\mathrm{E}\,|\,\xi(X)\,| < \infty$. Find M sufficiently large such that

$$\int_{\{x:|\xi(x)|>M\}} |\xi(x)|\, dQ(x) < \frac{\varepsilon}{8}.$$

Then we have

$$J''_{2n} \leqslant P\left\{Q((\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c) \geqslant \frac{\varepsilon}{8M}\right\}. \tag{16}$$

Choose $\varepsilon_1 \in \left(0, \frac{1}{40M}\right)$ (see (9)). Since from (9) we have

$$Q((\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c) = Q((\hat{\mathscr{G}}_n^*)) + 1 - Q((\mathscr{G}_n^*)) \leqslant Q((\hat{\mathscr{G}}_n^*)) + 5\varepsilon_1\varepsilon,$$

it follows from (16) and the choice of ε_1 that

$$J''_{2n} \leqslant P(Q((\hat{\mathscr{G}}_n^*)) > 0) \leqslant P(\hat{\mathscr{G}}_n^* \neq \varnothing). \tag{17}$$

Since $\{\hat{\mathscr{G}}_n^* \neq \varnothing\} \subset S_n^c$, by (13), (17), we have

$$J''_{2n} \leqslant P(R_n^c) \leqslant \exp(-Ck_n). \tag{18}$$

To deal with J'_{2n}, define a set T_n containing all points Z_n satisfying the following condition: "For each sphere $S \subset R^d$ such that $\#(S) \geqslant k_n$, if

$$S \cap \{X_1, \cdots, X_n\} = \{X_{i_1}, \cdots, X_{i_l}\},$$

then the number of elements in $\{Y_{i_1}, \cdots, Y_{i_l}\}$ satisfying the inequality

$$Y_{i_j} \leqslant m\left(X_{i_j}, \frac{1}{2} + \delta_0\right)$$

is not smaller than $\frac{1}{2}l(1+\delta_0)$, the number of elements in $\{Y_{i_1}, \cdots, Y_{i_l}\}$ satisfying the inequality $Y_{i_j} \geqslant -m\left(X_{i_j}, \frac{1}{2} - \delta_0\right)$ is also not smaller than $\frac{1}{2}l(1+\delta_0)$", where δ_0 is the number appearing in condition 3° of Theorem 1. Note that the number of different sets formed by $S \cap \{X_1, \cdots, X_n\}$ with all possible sphere S, does not exceed n^{2d}.

By the definition of $m(x, p)$ and use (5), it is not difficult to show that there exists constant $C > 0$ not depending on $X^n = \{X_1, \cdots, X_n\}$, such that under the condition of given X_n, if a sphere S satisfies $\#(S) \geqslant k_n$, then the conditional probability that S fulfils the requirement specified in the definition of T_n, is not smaller than $1 - \exp(-Ck_n)$. Considering this and the remark made in the end of the last paragragh, we get

$$P(T_n \mid X^n) \geqslant (1 - e^{-Ck_n})^{n^{2d}} \geqslant 1 - n^{2d}e^{-Ck_n}.$$

Hence $P(T_n) \geqslant 1 - n^{2d}e^{-Ck_n}$. Since $\log n/k_n \to 0$, we get

$$P(T_n) \geqslant 1 - \exp(-Ck_n). \tag{19}$$

(Notice the meaning of the symbol C explained earlier.)

According to (I) 5 and $k_n/n \to 0$, it is easily seen from the Hoeffding inequality that

$$P(T_{nr}^c) < \exp(-Cn), \text{ for } n \text{ sufficiently large,} \tag{20}$$

where T_{nr} denotes the event "$\#(S_{x,\|x\|}) \geqslant k_n$, for any x belonging to the support of Q with $\|x\| \geqslant r$".

Now we have

$$J'_{2n} \leqslant P(T_n^c) + P(T_{nr}^c) + P\Big(T_n \cap T_{nr} \cap \Big\{\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \geqslant \frac{\varepsilon}{4}\Big\}\Big). \tag{21}$$

By the definition of T_n, T_{nr}, $m(x, p)$ and $G(x, \delta)$, one has for $Z_n \in T_n \cap T_{nr}$ that

$$|\,\xi_{nk_n}(x, Z_n)\,| \leqslant G(x, \delta_0), \quad \|x\| > r \text{ and } n \text{ large.} \tag{22}$$

Further, from (Ⅰ) 4, it follows that there exists constant M_1 such that

$$|\,\xi_{nk_n}(x, Z_n)\,| \leqslant M_1, \quad \|x\| \leqslant r,\ Z_n \in T_n \cap T_{nr} \text{ and } n \text{ large.} \tag{23}$$

From (22), (23) and noticing that

$$P((\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c) \leqslant 5\varepsilon_1\varepsilon + P((\hat{\mathscr{G}}_n^*)),$$

we get

$$\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \leqslant \int_{\|x\|>r} G(x, \delta_0)\, \mathrm{d}Q(x) + 5M_{1\varepsilon_1\varepsilon}$$

when $\hat{\mathscr{G}}_n^* = \varnothing$, $Z_n \in T_n \cap T_{nr}$ and n large. By condition 3° of Theorem 1, choose r sufficiently large, we can make the integral on the right hand side not exceed $\varepsilon/10$. Fix this r and hence M_1 is also fixed, take $\varepsilon_1 \in \left(0, \frac{1}{50M_1}\right)$, we get under the above conditions ($\hat{\mathscr{G}}_n^* = \varnothing$, $Z_n \in T_n \cap T_{nr}$ and n large) that

$$\mathrm{E}[\,|\,\xi_{nk_n}(X, Z_n)\,|\, I_{(\mathscr{G}_n^* - \hat{\mathscr{G}}_n^*)^c}(X) \mid Z_n] \leqslant \varepsilon/5 < \varepsilon/4.$$

Therefore, on noticing that $\{\hat{\mathscr{G}}_n^* \neq \varnothing\} \subset S_n^c$, and (13), (19)—(21), we obtain

$$J'_{2n} \leqslant P(T_n^c) + P(T_{nr}^c) + P(S_n^c) \leqslant \exp(-Ck_n). \tag{24}$$

Now turn to J_{1n}. Define a set $T_n^{(1)}$ formed by all Z_n satisfying the following condition: "For each sphere $S \subset R^d$ such that $\#(S) \geqslant k_n$, if

$$S \cap \{X_1, \cdots, X_n\} = \{X_{i_1}, \cdots, X_{i_l}\},$$

then the number of elements in $\{Y_{i_1}, \cdots, Y_{i_l}\}$ satisfying the inequality $Y_{i_j} \geqslant \xi(X_{i_j}) - \varepsilon/6$ is larger than $l/2$, and the number of elements in $\{Y_{i_1}, \cdots, Y_{i_l}\}$ satisfying the inequality $Y_{i_j} \leqslant \xi(X_{i_j}) + \varepsilon/6$ is also large than $l/2$." Considering the choice of B_1 in (Ⅱ), the definition of $f_{i,\varepsilon/6}(x)$, the fact that in the definition of B_1 we have $b < \frac{1}{2}$, the definition of V_i, V_i^*, $W(G)$, $\mathscr{G}_n^*$, and the fact that

$$\{X_{n1}(x), \cdots, X_{nk}(x)\} \subset W(G_{ij}^*) \subset V,$$

for $x \in G_{ij}^* \in \mathscr{G}_n^* - \widehat{\mathscr{G}}_n^*$, we can show that

$$P(T_n^{(1)}) \geqslant 1 - \exp(-Ck_n)$$

by an argument similar to that employed in dealing with T_n. Hence, if we put

$$T_n^{(2)} = \left\{ Z_n : \text{for each } x \in \mathscr{G}_n^* - \widehat{\mathscr{G}}_n^*,\ x \in V_i \text{ for some } i = 1, \cdots, n \right\},$$

we have

$$\inf_{x' \in V_i} \xi(x') - \frac{\varepsilon}{6} \leqslant \xi_{nk_n}(x, Z_n) \leqslant \sup_{x' \in V_i} \xi(x') + \frac{\varepsilon}{6},$$

we shall have $T_n^{(2)} \supset T_n^{(1)}$, and therefore

$$P(T_n^{(2)}) \geqslant P(T_n^{(1)}) \geqslant 1 - \exp(-Ck_n). \tag{25}$$

Since $\sup\{ | \xi(x) - \xi(x') | : x \in V_i,\ x' \in V_i \} < \varepsilon/6$, for $Z_n \in T_n^{(2)}$ and $x \in \mathscr{G}_n^* - \widehat{\mathscr{G}}_n^*$, we have

$$| \xi(x) - \xi_{nk_n}(x, Z_n) | \leqslant \varepsilon/3.$$

Hence for $Z_n \in T_n^{(2)}$ we have

$$\mathrm{E}[\, | \xi_{nk_n}(X, Z_n) - \xi(X) | \, I_{(\mathscr{G}_n^* - \widehat{\mathscr{G}}_n^*)}(X) \mid Z_n] \leqslant \varepsilon/3.$$

From this and (25) we get

$$J_{1n} \leqslant 1 - P(T_n^{(2)}) \leqslant \exp(-Ck_n). \tag{26}$$

Finally, summing up (14), (18), (24) and (26), we get (3). This concludes the proof of the first half of the theorem.

(Ⅳ) To prove the remaining part of Theorem 1, suppose that a sequence of positive integers $\{k_n\}$ is given to satisfy $k_n \to \infty$, $k_n/n \to 0$. Take a sequence of positive integers $n_1 < n_2 < \cdots$ such that $\sum_{i=1}^{\infty} k_{n_i}/n_i < 1$. Choose a such that

$$\sum_{i=1}^{\infty} k_{n_i}/n_i < \frac{1}{a} < 1,$$

$a < \sqrt{2}$. Define a one-dimensional density function f satisfying the following conditions:

1. $\int_{2i}^{2i+1} f(x)\mathrm{d}x = ak_{n_i}/n_i,\ i = 1, 2, \cdots;$ (27)

2. $\int_{-\infty}^{0} f(x)\mathrm{d}x = 1 - a\sum_{i=1}^{\infty} k_{n_i}/n_i;$

3. f is everywhere continuous on $(-\infty, \infty)$.

Now take $d = 1$, and define a two-dimensional random vector (X, Y) such that X has a marginal density f, and the conditional distribution $F(y|x)$ of Y given $X = x$ satisfies the following four conditions:

(a) $F(y|x)$, as a function of (x, y), is continuous everywhere on R^2.

(b) $F(y|x)$, as a function of y for fixed x, is strictlyincreasing.

(c) For all x we have

$$F(1 \mid x)=\frac{a}{2},\ F(0 \mid x)=\frac{1}{2},\ F(-1 \mid x)=1-\frac{a}{2}.$$

(d) $F\left(\frac{in_i}{k_{n_i}}\middle| x\right)=\frac{a^2}{2}$ for $x \in (2i, 2i+1)$, $i=1, 2, \cdots$.

It is easily seen that such a (X, Y) satisfies all conditions $1^\circ \sim 4^\circ$ of Theorem 1.

Denote again by $Z_n=\{(X_i, Y_i), i=1, \cdots, n\}$ the random sample drawn from (X, Y), $X^n=\{X_1, \cdots, X_n\}$. Define the event

$$B_i=\{a^{\frac{1}{2}} k_{n_i} \leqslant \#\{(2i, 2i+1)\} \leqslant a^2 k_{n_i}\},\ i=1, 2, \cdots$$

(Remember that $\#(G)$ is the number of elements of the set $G \cap X^n$.) Noticing (27) and employing the Hoeffding inequality, one sees easily that

$$\lim_{i \to \infty} P(B_i)=1. \tag{28}$$

Denote the elements of the set $X^{n_i} \cap (2i, 2i+1)$ by $X_{i1}, \cdots, X_{iN_i}$, and their corresponding Y-values are denoted by $Y_{i1}, \cdots, Y_{iN_i}$. Put

$$K_i=\{Y_{i_j} \geqslant in_i/k_{n_i},\ j=1, \cdots, N_i\},$$

$$G_i=B_i \cap K_i,\ i=1, 2, \cdots.$$

Then by condition d of $F(y|x)$, we see that for $X^{n_i} \in B_i$:

$$P(K_i \mid X^{n_i})=\left(\frac{1}{2}a^2\right)^{n_i} \geqslant \left(\frac{1}{2}a^2\right)^{a^2 k_{n_i}} \triangleq C_a^{-k_{n_i}}, \tag{29}$$

where $C_a=\left(\frac{1}{2}a^2\right)^{-a^2}$.

On the other hand, by the definition of k-NN and the three conditions satisfied by the density f, it is easily seen that $\{X_{n_i1}(x), \cdots, X_{n_ik_{n_i}}(x)\} \in (2i, 2i+1)$ and $\xi_{n_ik_{n_i}}(x, Z_{n_i}) \geqslant in_i/k_{n_i}$ for $x \in (2i, 2i+1)$ and $Z_{n_i} \in G_i$. Hence

$$H_{n_ik_{n_i}}(Z_{n_i}) \geqslant \int_{2i}^{2i+1} \mid \xi_{n_ik_{n_i}}(x, Z_{n_i})-0 \mid f(x)\mathrm{d}x \geqslant ia > i$$

for $Z_{n_i} \in G_i$. From this and (28), (29), also noticing that $P(G_i) \geqslant C_a^{-k_{n_i}} P(B_i)$, we get

$$\limsup_{i \to \infty}\{C_a^{k_{n_i}} P(H_{n_ik_{n_i}}(Z_{n_i}) \geqslant \varepsilon)\} \geqslant \limsup_{i \to \infty}\{C_a^{k_{n_i}} P(G_i)\} \geqslant \lim_{i \to \infty} P(B_i)=1. \tag{30}$$

Since a can be chosen arbitrarily near $\sqrt{2}$, (4) follows from (30). This proves the latter conclusion of Theorem 1. The proof is completed.

3 Some Remarks

1. If the condition 3° of Theorem 1 is replaced by the weaker one

$$\mathrm{E} \mid \xi(X) \mid < \infty \tag{31}$$

and maintain the other conditions, then the probability $P(H_{nk_n}(Z_n) \geqslant \varepsilon)$ may not tend to zero.

Choose the (X, Y) as in Section 2 (Ⅳ), but modify the conditions c and d imposed on $F(y|x)$ to: " $F(0 \mid x) = \frac{1}{2}$, $F(in_i/k_{n_i} \mid x) - F(in_i/(2k_{n_i}) \mid x) + F(-in_i/(4k_{n_i}) \mid x) - F(-in_i/(8k_{n_i}) \mid x) = 1 - k_{n_i}^{-2}$ for $x \in (2i, 2i+1)$, $i = 1, 2, \cdots$." Then the condition (31) and the conditions 1°, 2°, 4° of Theorem 1 are satisfied. But an argument similar to those employed in Section 2 (Ⅳ) yields

$$\lim_{i\to\infty} P(H_{n_i k_{n_i}}(Z_{n_i}) \geqslant \varepsilon) = 1, \text{ for any } \varepsilon > 0.$$

2. In case Y is bounded, we have the following

Theorem 2 Maintain the conditions 1°, 2° of Theorem 1, and modify the conditions 3°, 4° as follows: 3′. Y is bounded. 4′. For sufficiently small $a > 0$ and sufficiently large A, the set $\{x: a < f(x) < A\}$ differs from an open set by only a Lebesgue nullset. Suppose that $k_n \to \infty$, $k_n/n \to 0$. Then for any given $\varepsilon > 0$ there exists $C > 0$ (depending on ε) such that

$$P(H_{nk_n}(Z_n) \geqslant \varepsilon) = 0(\mathrm{e}^{-Cn}).$$

This can be proved by combining the methods of Beck [3] and the present paper. Details are not presented here.

3. Even if Y is bounded, one cannot establish in general a rate faster than the type of $O(\mathrm{e}^{-Cn})$.

Example Take $d = 1$. Define (X, Y), whose distribution $F(x, y)$ is as follows: X has a marginal density $(1 - |x|)I_{(-1,1)}(x)$, and the conditional distribution $F(y|x)$ is $R(x-1, x+1)$. Here we have $|Y| \leqslant 2$. Noticing that

$$P(Y > 3/2) = \int_{\frac{1}{2}}^{2} \left(x - \frac{1}{2}\right)(1-x)\mathrm{d}x = \frac{1}{48}.$$

One gets $P(Y_i > 3/2, i = 1, \cdots, n) = 48^{-n}$. But $|\xi(x)| \leqslant 1$. Hence

$$P\left(H_{nk_n}(Z_n) \geqslant \frac{1}{2}\right) \geqslant 48^{-n} = \mathrm{e}^{-Cn}, \ C = \log 48.$$

References

[1] Zheng Zhongguo. Asymptotic properties of k-NN estimates of conditional median (in Chinese, to appear).

[2] Wagner T J. *IEEE Trans. Inform. Theory*, 1971, 17: 566

[3] Beck J. *Problems of Control and Information Theory*, 1979, 8: 303

[4] Hoeffding W. *JASA*, 1963, 13

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只有一个转变点的模型的假设检验和区间估计

摘要 对最多只含一个转变点 t_0 的模型 $X(i/n)=f(i/n)+e(i/n)$，其中 $f(t)=a+\theta I_{(t_0,1)}(t)$，$0\leqslant t\leqslant 1$，$e(1/n)$，…，$e(n/n)$ 独立同分布. 本文讨论了关于转变点 t_0，跃度 θ 以及 $e(t)$ 的方差 σ^2 的假设检验和区间估计问题.

1 引言

考虑模型

$$X(t)=f(t)+e(t),\ 0\leqslant t\leqslant 1, \tag{1}$$

其中 f 为形如

$$f(t)=\begin{cases}a,\ 0\leqslant t\leqslant t_0\\ a+\theta,\ t_0<t\leqslant 1\end{cases} \tag{2}$$

的非随机函数，a，θ，t_0 为未知常数，$0<t_0<1$，t_0 称为"转变点"，而 θ 则是在转变点 t_0 处的跃度，$e(t)$ 为随机变量，其分布 F 与 t 无关. 又假定 $e(t)$ 的均值为 0，其方差 σ^2 存在有限.

我们的目的是要利用对 $X(t)$ 的观察值去对 t_0 和 θ 及 σ^2 作统计推断. 在本文中，假定对等距的 t 值处作观察. 确切地说，我们观察 $X(i/n)$，$i=1$，…，n. 为简化符号，将 $X(i/n)$ 记为 X_i，并设 X_1，…，X_n 独立，以上记号及假定通用本文，将不再一一提及.

只有一个转变点的模型(1)和(2)，是关于转变点的最简单模型. 有大量的文献讨论这个在应用上很重要的模型，详细的综合性介绍可参阅文献[1，4].

因为本文方法不用到似然函数及正态性假定，所以是非参数性的. 我们利用了文献[3]中的一个想法，即通过局部性的比较去发现转变点. 我们发现，这方法有如下的优点：为检验原假设

$$H_0:\theta=0(\text{即不存在转变点}) \tag{3}$$

而导出的统计量，有简单的渐近分布，这不仅便于对 H_0 作检验，而且对估计检验的功效及 t_0 的区间估计，也提供了方便的做法. 这些将在下面第 2、3、5 节中，就关于 F 的不同假定依次讨论之. 第 4 节和第 6 节则分别与 σ^2 和 θ 的估计有关.

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2 F 为正态分布而方差已知

在本节中设

$$F \sim N(0, \sigma^2) \tag{4}$$

且 σ^2 已知. 我们的方法基于下述定理:

定理 1 设 $X_1, \cdots, X_n$ 独立同分布(iid.), X_1 有分布 $N(a, \sigma^2)$. 设 $l = l_n$ 为正整数,适合条件

$$\lim_{n\to\infty} l/n = 0, \ \lim_{n\to\infty} (\log n)^2/l = 0. \tag{5}$$

定义

$$Y_m = \Big(\sum_{m-l+1}^{m} X_i - \sum_{m-2l+1}^{m-l} X_i\Big)\Big/\sqrt{2l}, \ m = 2l, \cdots, n, \tag{6}$$

$$\xi_n = \max\{|Y_m|: m = 2l, \cdots, n\}, \tag{7}$$

$$A_n(x) = [2\log(3n/2l-3)]^{-1/2}\Big(x + 2\log(3n/2l-3) + \frac{1}{2}\log\log(3n/2l-3) - \frac{1}{2}\log \pi\Big). \tag{8}$$

则

$$\lim_{n\to\infty} P(\xi_n/\sigma \leqslant A_n(x)) = \exp(-2e^{-x}), \ -\infty < x < \infty. \tag{9}$$

证 作一个标准布朗运动 $\{W(t), t \geqslant 0\}$, 使

$$W(3m/2l) = \sqrt{3/2l}(X_1 + \cdots + X_m - ma)/\sigma, \ m = 1, 2, \cdots,$$

令

$$Z(t) = 3^{-1/2}[W(t) - 2W(t+3/2) + W(t+3)], \ t \geqslant 0,$$

则有 $Y_m = \sigma Z(3m/2l-3)$, $m = 2l, 2l+1, \cdots$. 令

$$\zeta_n = \sup\{|Z(t)|: 0 \leqslant t \leqslant 3n/2l-3\}, \ \eta_n = \zeta_n - \xi_n/\sigma, \tag{10}$$

往证

$$\lim_{n\to\infty} \eta_n \sqrt{\log n} = 0, \text{ a. s.} \tag{11}$$

事实上,有

$$0 \leqslant \eta_n \leqslant (4/\sqrt{3})\sup\{|W(t+s) - W(t)|: 0 \leqslant s \leqslant 3/2l, \ 0 \leqslant t \leqslant (3n-3)/2l-3\}.$$

在文献[4]的引理 1.2.1 中取 $T = 3n/2l-3$, $h = 3/2l$, $\varepsilon = 1$ 以及 $v = (\sqrt{2}/4)\sqrt{l/\log n}\,\delta$, 得

$$P(\eta_n \sqrt{\log n} \geqslant \delta) \leqslant Cn\exp(-\delta^2 l/24\log n). \tag{12}$$

这里 $\delta > 0$ 给定, C 为与 n 无关的常数. 由于 $l/(\log n)^2 \to \infty$, (11)式由(12)式及 Borel-Cantelli 引理得出.

$Z(t)$为平稳 Gauss 过程，$EZ(t) = 0$，且

$$\rho(\tau) = \operatorname{cov}(Z(t),\ Z(t+\tau)) = \begin{cases} 1-|\tau|, & 0 \leqslant |\tau| \leqslant 3/2, \\ -1+|\tau|/3, & 2/3 < |\tau| < 3, \\ 0, & 3 \leqslant |\tau| < \infty, \end{cases}$$

故定理 5.2① 的条件全满足. 得

$$\lim_{n\to\infty} P(\xi_n \leqslant A_n(x)) = \exp(-2e^{-x}). \tag{13}$$

又当 n 充分大时，有

$$A_n(x+\Delta x) - A_n(x) \begin{matrix} \geqslant \\ \leqslant \end{matrix} \Delta x/\sqrt{2\log n},\quad \begin{matrix} \Delta x > 0, \\ \Delta x < 0. \end{matrix}$$

故当 n 充分大时，有

$$\begin{aligned} & P(\xi_n \leqslant A_n(x-|\Delta x|)) - P(\eta_n \geqslant |\Delta x|/\sqrt{2\log n}) \\ & \leqslant P(\xi_n/\sigma \leqslant A_n(x)) \\ & \leqslant P(\xi_n \leqslant A_n(x+|\Delta x|)) + P(\eta_n \geqslant |\Delta x|/\sqrt{2\log n}). \end{aligned} \tag{14}$$

由(12)～(14)式，令 $n\to\infty$，然后 $\Delta x \to 0$，得(9)式，定理证毕.

定理 1 提出了一个检验假设(3)式的方法：取水平 $\alpha \in (0,\ 1)$. 解方程 $\exp(-2e^{-x}) = 1-\alpha$，其解为

$$x(\alpha) = -\log\left(-\frac{1}{2}\log(1-\alpha)\right).$$

计算 $d = 2l/n$，$C_n(\alpha,\ d) = A_n(x(\alpha))$，然后当且仅当

$$\xi_n > \sigma C_n(\alpha,\ d) \tag{15}$$

时否定假设(3)式. 当 $n\to\infty$时，此检验有渐近水平 α.

为估计此检验的功效 $\beta(\theta) = \beta_n(\theta,\ \sigma)$，以 r 记适合条件

$$r/n \leqslant t_0 < (r+1)/n \tag{16}$$

的整数，则 $Y_{r+l} \sim N(\sqrt{l/2}\,\theta,\ \sigma^2)$. 故

$$\begin{aligned} \beta(\theta) & \geqslant P(|Y_{r+l}| > \sigma C_n(\alpha,\ d)) \\ & = \Phi\left(\frac{|\theta|\sqrt{l}}{\sqrt{2}\sigma} - C_n(\alpha,\ d)\right) + \Phi\left(-\frac{|\theta|\sqrt{l}}{\sqrt{2}\sigma} - C_n(\alpha,\ d)\right) \\ & > \Phi\left(\frac{|\theta|\sqrt{l}}{\sqrt{2}\sigma} - C_n(\alpha,\ d)\right), \end{aligned} \tag{17}$$

此处 Φ 为标准正态分布函数. 从此式，且从直观上也可明显看出：为使功效 $\beta(\theta)$ 增大，l 应取较大之值. 还可看出：当 $|\theta|/\sigma$ 增加时，$\beta(\theta)$ 应增加. 从实用的观点看，只有在 $|\theta|/\sigma$ 较显著地大于

① Yao Y C, David R A. The Asymptotic Behavior of the Likelihood Ratio Statistic for Testing a Shift in Mean in Sequence of Independent Normal Variates. Technical Report, 1984. 80

0 时,转变点才有意义.

现考虑转变点 t_0 的区间估计问题. t_0 的存在可以是事先的假定,也可以是检验假设(3)式的结果((3)式被否定). 不论是何种情况,我们都采用如下的

法则 找 k,使 $|Y_k| = \xi_n$. 以 $[(k-2l)/n, k/n]$ 作为 t_0 的区间估计.

此区间之长为 $d = 2l/n$. 故为提高估计精度,应取较小的 l. 但这带来两个问题:一是若 t_0 的存在要由上述检验来决定时,l 取得过小使当(3)式不成立时其被接受的危险增加. 二是 l 太小时,这区间估计的置信系数会很低. 为把这一点搞得更清楚,我们来对此区间的置信系数 γ 作估计:

$$\begin{aligned}\gamma &= P((k-2l)/n \leqslant t_0 \leqslant k/n) \\ &\geqslant P(\{\sup_{m \overline{\in} (r, r+2l)} |Y_m| \leqslant \sigma C_n(\alpha, d)\} \cap \{|Y_{r+l}| > \sigma C_n(\alpha, d)\}),\end{aligned}$$

因为 $X_m, X_{m-1}, \cdots, X_{m-2l-1}$ 为 iid. 当 $m \overline{\in} (r, r+2l)$, 由定理 1 得

$$P(\sup_{m \overline{\in} (r, r+2l)} |Y_m| \leqslant \sigma C_n(\alpha, d)) \leqslant P(\xi_n \leqslant \sigma C_n(\alpha, d)) \approx 1-\alpha, \tag{18}$$

这里 $P(\xi_n \leqslant \sigma C_n(\alpha, d))$ 是在 $X_1, \cdots, X_n$ 为 iid. 的条件下计算的. 因此,若把(18)式最后一式中涉及的误差忽略不计,则有

$$\begin{aligned}\gamma &\geqslant (1-\alpha) + P(|Y_{r+l}| \geqslant \sigma C_n(\alpha, d)) - 1 \\ &\geqslant \Phi(|\theta|\sqrt{l}/\sqrt{2}\sigma - C_n(\alpha, d)) - \alpha.\end{aligned} \tag{19}$$

此式显示,γ 应随 $|\theta|/\sigma$ 的增加而增加.

用(19)式,可给予下述重要问题以一近似解:为求 t_0 的区间估计,使具有给定的长度 d_0 与置信系数 $1-\alpha_0$, 问 n 和 l 应取为多少? 为此,在(19)式中令 $\alpha = \alpha_0/2$ 而解方程

$$\Phi\left(|\theta|\sqrt{l}/\sqrt{2}\sigma - C_n\left(\frac{\alpha_0}{2}, d_0\right)\right) = 1-\alpha_0/2$$

得到

$$l \approx 2(\sigma/|\theta|)^2[C_n(\alpha_0/2, d_0) + u_{\alpha_0/2}]^2, \quad n \approx 2l/d_0, \tag{20}$$

这里 $u_{\alpha_0/2}$ 是 $N(0, 1)$ 的上 $\alpha_0/2$ 分位点.

注意 l 表达式中的 $C_n(\alpha_0/2, d_0)$ 其实与 n 无关,但 l 依赖于 θ 与 σ,前者必然未知而后者常为未知. 在有些情况下,可设想作些初步观察以大致估计它们,但我们推荐如下的做法:根据实际应用上的考虑定出常数 M,使只有在 $|\theta|/\sigma \geqslant M$ 时,转变点 t_0 才有实际意义. 在(20)式中以 M 代替 $|\theta|/\sigma$ 去计算 l.

例如,取 $\alpha_0 = 0.05$, $d_0 = 0.1$, $M = 2$, 由(20)式给出

$$l = 19.46, \quad n = 389.2.$$

此结果看来相当不错,因为它与在估计二项分布概率 p 时所需的样本大小相当:为了找二项分布 p 的一长度不超过 0.1,而置信系数不小于 0.95 的区间估计,所需样本大小约为 384,而 $|\theta|/\sigma \geqslant 2$ 的要求在应用上看来是适中的.

3 F 为正态分布而方差未知

当方差 σ^2 未知时，要用样本 $X_1, \cdots, X_n$ 去估计它，以 $\hat{\sigma}^2$ 记 σ^2 的估计，用 $\hat{\sigma}$ 代(15)式中的 σ 以进行检验. 为使这样得到的检验仍具有渐近水平 α，$\hat{\sigma}^2$ 必须适合下面定理表述的性质.

定理 2 设定理 1 的条件满足，而 $\hat{\sigma}^2$ 适合条件

$$\lim_{n\to\infty} |\hat{\sigma}^2 - \sigma^2| \log n \overset{P}{=} 0, \tag{21}$$

$\overset{P}{=}$表依概率收敛. 则有

$$\lim_{n\to\infty} P(\xi_n/\hat{\sigma} \leqslant A_n(x)) = \exp(-2e^{-x}). \tag{22}$$

证明易由定理 1 得出. 因由(21)式知 $|\hat{\sigma} - \sigma| \log n \overset{P}{=} 0$，故对任给 $\varepsilon > 0$，有

$$\begin{aligned} & P(\xi_n/\sigma \leqslant A_n(x)(1-\varepsilon/\log n)) - P(|\hat{\sigma}-\sigma|\log n \geqslant \varepsilon) \\ & \quad \leqslant P(\xi_n/\hat{\sigma} \leqslant A_n(x)) \\ & \quad \leqslant P(\xi_n/\sigma \leqslant A_n(x)(1+\varepsilon/\log n)) + P(|\hat{\sigma}-\sigma|\log n \geqslant \varepsilon). \end{aligned}$$

给定 $\delta > 0$，当 n 充分大时，有

$$A_n(x-\delta) \leqslant A_n(x)(1-\varepsilon/\log n) \leqslant A_n(x)(1+\varepsilon/\log n) \leqslant A_n(x+\delta).$$

故当 n 充分大时，有

$$\begin{aligned} & P(\xi_n/\sigma \leqslant A_n(x-\delta)) - P(|\hat{\sigma}-\sigma|\log n \geqslant \varepsilon) \\ & \quad \leqslant P(\xi_n/\hat{\sigma} \leqslant A_n(x)) \\ & \quad \leqslant P(\xi_n/\sigma \leqslant A_n(x+\delta)) + P(|\hat{\sigma}-\sigma|\log n \geqslant \varepsilon). \end{aligned}$$

令 $n\to\infty$，然后 $\delta\to 0$，得(21)式. 定理证毕.

方差 σ^2 的估计 在正态假定下，σ^2 的极大似然估计有下文(26)式的形式，下一定理证明，在很宽的条件下此估计适合(21)式的要求.

定理 3 设 $N_1, N_2, \cdots$ 为一串趋于无穷的正整数. 对每个 n，给定了 N_n 个独立随机变量 X_{n_i}，$i = 1, \cdots, N_n$，且

$$X_{n_i} \sim F(x),\ i = 1, \cdots, m_n;\ X_{n_i} \sim F(x-\theta),\ i = m_n+1, \cdots, N_n,$$

这里分布 F 有 δ 阶矩，对某个 $\delta > 2$，而 θ 为常数. 记

$$\overline{X}_{n1c} = \sum_{i=1}^{c} X_{n_i}/c,\ \overline{X}_{x2c} = \sum_{i=c+1}^{N_n} X_{n_i}/(N_n - c),\ \overline{X}_n = \sum_{i=1}^{N_n} X_{n_i}/N_n, \tag{23}$$

$$S_{nc}^2 = \sum_{i=1}^{c} (X_{n_i} - \overline{X}_{n1c})^2 + \sum_{i=c+1}^{N_n} (X_{n_i} - \overline{X}_{n2c})^2,\ c = 1, \cdots, N_n, \tag{24}$$

$$S^2(n) = \min(S_{nc}^2 : c = 1, \cdots, N_n), \tag{25}$$

$$\hat{\sigma}_n^2 = S^2(n)/N_n. \tag{26}$$

并以 σ^2 记 F 的方差,则有

$$\lim_{n\to\infty} |\hat{\sigma}_n^2 - \sigma^2| \log N_n \overset{P}{=\!=} 0. \tag{27}$$

证 为简化记号,在以下我们将略去各记号中的 n,故 X_{n_i}, m_n, N_n, $\overline{X}_{n1c}$, $\overline{X}_n$, S_{nc}^2, $S^2(n)$, $\hat{\sigma}_n^2$ 等将简记为 X_i, m, N, $\overline{X}_{1c}$, $\overline{X}$, S_c^2, S^2, σ^2. 令 $Y_i = X_i - \mathrm{E}X_i$, $T_i = Y_1 + \cdots + Y_i$, $\varepsilon > 0$. 由 Kolmogorov 不等式有

$$P(|T_c/c| \leqslant \varepsilon/\log N : N\varepsilon/\log N \leqslant c \leqslant N) \geqslant 1 - D(\log N)^4/N, \tag{28}$$

$$P(|(T_N - T_c)/(N-c)| \leqslant \varepsilon/\log N : 1 \leqslant c \leqslant N(1-\varepsilon/\log N)) \geqslant 1 - D(\log N)^4/N. \tag{29}$$

此处及以下用 D 记一与 n 无关的常数,其值每次出现时可不同. 由(28)和(29)式,得

$$P(|T_c/c| \leqslant \varepsilon/\log N,\ |(T_N - T_c)/(N-c)| \leqslant \varepsilon/\log N : N\varepsilon/\log N \leqslant c \leqslant N(1-\varepsilon/\log N)) \geqslant 1 - D(\log N)^4/N. \tag{30}$$

首先考虑 $\theta = 0$ 的情况. 因为

$$S_c^2 - S_N^2 = -(c(N-c)/N)(\overline{X}_{1c} - \overline{X}_{2c})^2, \tag{31}$$

且当 $\theta = 0$ 时,有 $\overline{X}_{1c} - \overline{X}_{2c} = T_c/c - (T_N - T_c)/(N-c)$, 由(30)和(31)式知,

$$\begin{aligned} &P(S_c^2 - S_N^2 \geqslant -N\varepsilon^2/(\log N)^2 : N\varepsilon/\log N \leqslant c \leqslant N(1-\varepsilon/\log N)) \\ &\quad \geqslant 1 - D(\log N)^4/N. \end{aligned} \tag{32}$$

对 $c \leqslant N\varepsilon/\log N$, 有

$$\begin{aligned} S_c^2 - S_N^2 &\geqslant -\sum_{i=1}^{N\varepsilon/\log N} (X_i - \overline{X}_{2c})^2 \\ &\geqslant -2\Big(\sum_{i=1}^{N\varepsilon/\log N} X_i^2 + (\varepsilon/\log N)(1-\varepsilon/\log N)^{-1}\sum_{i=1}^{N} X_i^2\Big). \end{aligned} \tag{33}$$

对 $c \geqslant N(1-\varepsilon/\log N)$ 类似得到

$$S_c^2 - S_N^2 \geqslant -2\sum_{i=N(1-\varepsilon/\log N)}^{N} X_i^2 + (\varepsilon/\log N)(1-\varepsilon/\log N)^{-1}\sum_{i=1}^{N} X_i^2). \tag{34}$$

由(32)~(34)式,并令

$$Q = \varepsilon^2/\log N + \frac{2\varepsilon}{1-\varepsilon}\frac{1}{N}\sum_{i=1}^{N} X_i^2 + 2\varepsilon\,\frac{\log N}{N\varepsilon}\Big(\sum_{i=1}^{N\varepsilon/\log N} X_i^2 + \sum_{i=N(1-\varepsilon/\log N)} X_i^2\Big), \tag{35}$$

我们得到

$$P(|\hat{\sigma}^2 - S_N^2/N| \log N \geqslant Q) \leqslant D(\log N)^4/N. \tag{36}$$

记 $a = \mathrm{E}(X_1)$, 得

$$\lim_{n\to\infty} P(Q \leqslant 6\varepsilon(1-\varepsilon)^{-1}(\sigma^2 + a^2 + 1)) = 1. \tag{37}$$

其次,在本定理条件下(利用 Marcinkiewicz 大数律),易证存在 $\delta' > 0$, 使

$$\lim_{n\to\infty} N^{\delta'}(\sigma^2 - S_N^2/N) \stackrel{P}{=} 0. \tag{38}$$

由(36)～(38)式,即得(27)式.

现设 $\theta \neq 0$. 记

$$\theta_i = \begin{cases} 0, & i = 1, \cdots, m, \\ \theta, & i = m+1, \cdots, N, \end{cases} \tag{39}$$

$$Z_i = X_i - \theta_i, \ i = 1, \cdots, N. \tag{40}$$

定义 $\bar{\theta}_{1c}, \bar{\theta}_{2c}, \bar{Z}_{1c}, \bar{Z}_{2c}, \bar{Z}$,其意义与 $\bar{X}_{1c}, \bar{X}_{2c}, \bar{X}$ 类似. 则

$$\begin{aligned} &\sum_{i=1}^{c}(X_i - \bar{X}_{1c})^2 + \sum_{i=c+1}^{N}(X_i - \bar{X}_{2c})^2 \\ &= \sum_{i=1}^{c}(Z_i - \bar{Z}_{1c} + \theta_i - \bar{\theta}_{1c})^2 + \sum_{i=c+1}^{N}(Z_i - \bar{Z}_{2c} + \theta_i - \bar{\theta}_{2c})^2 \qquad (41) \\ &= \sum_{i=1}^{c}(Z_i - \bar{Z}_{1c})^2 + \sum_{i=c+1}^{N}(Z_i - \bar{Z}_{2c})^2 + R_{1n}(c) + R_{2n}(c). \qquad (42) \end{aligned}$$

其中

$$R_{1n}(c) = \sum_{i=1}^{c}(\theta_i - \bar{\theta}_{1c})^2 + \sum_{i=c+1}^{N}(\theta_i - \bar{\theta}_{2c})^2, \tag{43}$$

$$R_{2n}(c) = 2\sum_{i=1}^{c}(Z_i - \bar{Z}_{1c})(\theta_i - \bar{\theta}_{1c}) + 2\sum_{i=c+1}^{N}(Z_i - \bar{Z}_{2c})(\theta_i - \bar{\theta}_{2c}). \tag{44}$$

由于 $|\theta_i - \bar{\theta}_{1c}| \leqslant 2\theta$, $|\theta_i - \bar{\theta}_{2c}| \leqslant 2\theta$, 且它们是非随机的,于是由 Chebyshev 不等式易见,对于 $1 \leqslant c \leqslant N$ 一致地有

$$R_{2n}(c) = O_p(N^{1/2}). \tag{45}$$

注意

$$\min_{1\leqslant c\leqslant N} R_{1n}(c) = R_{1n}(m) = 0, \tag{46}$$

由此从(42)和(45)式,我们得到

$$S^2(n) \geqslant \min_{1\leqslant c\leqslant N}\left\{\sum_{i=1}^{c}(Z_i - \bar{Z}_{1c})^2 + \sum_{i=c+1}^{N}|(Z_i - \bar{Z}_{2c})^2\right\} + O_p(N^{1/2}). \tag{47}$$

又因为

$$R_{2n}(m) = 0, \tag{48}$$

再根据 $S^2(n)$ 之定义,又得

$$S^2(n) \leqslant \sum_{i=1}^{m}(Z_i - \bar{Z}_{1m})^2 + \sum_{i=m+1}^{N}(Z_i - \bar{Z}_{2m})^2 \leqslant \sum_{i=1}^{N}(Z_i - \bar{Z})^2. \tag{49}$$

由定义(39)和(40)式知,$Z_1, \cdots, Z_N$ 为 iid. $\sim F(x)$. 于是由(47)和(49)式,以及上面关于 $\theta = 0$ 的情况的证明知,当 $\theta \neq 0$ 时,(27)式亦真,从而定理 3 得证①.

① 此处“$\theta = 0$”的证明是据审查人的建议而改写的,它简化了本文原给的证明,特此说明,顺致谢意.

4 方差估计的偏差

仍用第 3 节的记号. 因为 $S^2 = \min(S_c^2) \leqslant S_m^2$ 及 $ES_m^2/(N-2) = \sigma^2$, 知即使用 $\tilde{\sigma}^2 = S^2/(N-2)$ 代替 $\hat{\sigma}^2$, $\tilde{\sigma}^2$ 作为 σ^2 的估计仍系统偏低. 直观上明显:当 $|\theta|/\sigma$ 愈小时,偏差应愈严重. 故以下我们较仔细地考察一下 $\theta = 0$ 时的偏差.

由(31)式得

$$
\begin{aligned}
(S_N^2 - S^2)/\sigma^2 &= \max_{1\leqslant c\leqslant N} \frac{N}{c(N-c)}\left(T_c - \frac{c}{N}T_N\right)^2 \\
&= \left\{\sup_{0<t<1} |Z_N(t)| / \sqrt{t(1-t)}\right\}^2 \triangleq \eta_n.
\end{aligned}
$$

此处 $T_c = \sum_{i=1}^{c}(X_i - a)/\sigma$, 且

$$
Z_N(t) = \begin{cases} (T_{[(N+1)t]} - [(N+1)t]T_N/N)/\sqrt{N}, & 0 \leqslant t < 1, \\ 0, & t = 1. \end{cases}
$$

若 X_1 为正态,或更一般地, $E|X_1|^\delta < \infty$ 对某个 $\delta > 2$, 则有下述结果[1,5]:

$$
\lim_{n\to\infty} P\left(\sqrt{2\log_2 n}\,\sqrt{\eta_n} - \left(2\log_2 n + \frac{1}{2}\log_3 n - \frac{1}{2}\log\pi\right) < x\right) = \exp(-2e^{-x}).
$$

此处 $\log_{k+1}(x) = \log\log_k(x)$, $\log_1 x = \log x$. 由此推出偏差的渐近分布:

$$
\lim_{n\to\infty} P((S_N^2 - S^2)/\sigma^2 < 2\log_2 N + \log_3 N - \log\pi + x) = \exp(-2e^{-x/2}). \tag{50}
$$

这个结果提示,偏差有如下的形状:

$$
S_N^2 - S^2 = 2\log_2 N + \log_3 N + O_p(1).
$$

其次,因分布函数 $\exp(-2e^{-x/2})$ 有期望值 $2\gamma + 2\log 2 = 2.54$ (γ 为 Euler 常数),(50)式显示:当 N 甚大时,以

$$
\begin{aligned}
&S^2/[N - 1 - (2\log_2 N + \log_3 N - \log\pi) - 2(\gamma + \log 2)] \\
&= S^2/(N - 2\log_2 N - \log_3 N - 2.4)
\end{aligned}
$$

作为 σ^2 的估计,应能降低偏差.

在某些应用中,我们可能有根据在事先肯定 t_0 介于 λ_1 和 λ_2 之间 $(0 < \lambda_1 < \lambda_2 < 1)$. 这时可用

$$
S^2(\lambda_1, \lambda_2) = \min(S_c^2 : \lambda_1 N \leqslant c \leqslant \lambda_2 N)
$$

来代替 S^2. 我们有

$$
(S_N^2 - S^2(\lambda_1, \lambda_2))/\sigma^2 = \sup_{\lambda_1\leqslant t\leqslant\lambda_2} |Z_N(t)| / \sqrt{t(1-t)} \triangleq v_N.
$$

在文献[5]中证明:当 $N\to\infty$ 时, v_N 依分布收敛于

$$v=\sup\left\{|V(S)|:0\leqslant S\leqslant\frac{1}{2}\log[\lambda_2(1-\lambda_1)/\lambda_1(1-\lambda_2)]\right\}.$$

这里$\{V(S):S\geqslant 0\}$为 Ornstein-Uhlenbeck 过程，即一个具有均值为 0，协方差函数为$\exp(-|t-s|)$的 Gauss 过程. v的分布曾由 Delong 在文献[5]中造表. 若记

$$\mathrm{E}(v)=C(\lambda_1,\lambda_2),$$

则用$S^2(\lambda_1,\lambda_2)/(N-1-C(\lambda_1,\lambda_2))$估计$\sigma^2$是合适的.

我们可以用下面的方法估出$C(\lambda_1,\lambda_2)$的一个上界. 记$\lambda=\min(\lambda_1,1-\lambda_2)$，有

$$\begin{aligned}(S_N^2-S^2(\lambda_1,\lambda_2))/\sigma^2&\leqslant\max\left\{\frac{N}{c(N-c)}\left(T_c-\frac{c}{N}T_N\right)^2:\lambda N\leqslant c\leqslant(1-\lambda)N\right\}\\&\leqslant\lambda^{-1}(1-\lambda)^{-1}\max_{\lambda N\leqslant c\leqslant(1-\lambda)N}\left|\frac{1}{\sqrt{N}}T_c-\frac{1}{\sqrt{N}}\frac{c}{N}T_N\right|^2\\&\leqslant\lambda^{-1}(1-\lambda)^{-1}\max_{\lambda\leqslant t\leqslant 1-\lambda}B^2(t)\leqslant\lambda^{-1}(1-\lambda)^{-1}\max_{0\leqslant t\leqslant 1}B^2(t),\end{aligned}$$

这里$\{B(t):0\leqslant t\leqslant 1\}$为标准 Brown 桥.

$\max\limits_{0\leqslant t\leqslant 1}|B(t)|$的分布为$1-\sum\limits_{i=1}^{\infty}2(-1)^{i+1}\exp(-2i^2x^2)$. 由此易得

$$\mathrm{E}(\max_{0\leqslant t\leqslant 1}B^2(t))=1-1/2^2+1/3^2-1/4^2+\cdots\approx 0.85.$$

因此

$$C(\lambda_1,\lambda_2)\leqslant 0.85\lambda^{-1}(1-\lambda)^{-1}.$$

故若用

$$\hat{\sigma}^2(\lambda_1,\lambda_2)=S^2(\lambda_1,\lambda_2)/(N-2)$$

去估计σ^2，则相对偏差不超过

$$(\sigma^2-\mathrm{E}\hat{\sigma}^2(\lambda_1,\lambda_2))/\sigma^2=(0.85\lambda^{-1}(1-\lambda)^{-1}-1)/(N-2).\tag{51}$$

例如，取$\lambda_1=1-\lambda_2=0.1$. 在应用上，这个值可认为是比较小的. 令$N=100$. 依(51)式，相对偏差不超过 8.6%. 如果考虑到在这样一个较复杂的模型中去估计方差，而样本大小不过 100，这结果可以认为是好的.

5 *F* 为非正态

当随机误差$e(t)$的分布为非正态时，可以用 iid. 变量的部分和通过 Brown 运动的强逼近，以给出定理 1 的推广. 这样，前几节的方法仍可以使用.

定理 4 将定理 1 中的条件$X_1\sim N(a,\sigma^2)$代之以

$$\mathrm{E}(\exp(tX_1))<\infty,\ \text{当}\ |t|>0\ \text{充分小},\tag{52}$$

则定理 1 的结论仍有效.

证 记

$$S_k=(X_1+\cdots+X_k-ka)/\sigma,\ k=1,2,\cdots,$$

依文献[6],存在 Brown 运动过程 $\{W(t):t\geqslant 0\}$,使

$$\lim_{n\to\infty}\sup\{\sup_{k\leqslant n}|S_k-W(k)|/\log n\}<\infty,\ \text{a. s.}\tag{53}$$

因为

$$Y_m/\sigma=[(S_m-S_{m-l})-(S_{m-l}-S_{m-2l})]/\sqrt{2l},$$

对 $m\leqslant n$ 我们有

$$|Y_m/\sigma-[W(m)-2W(m-l)+W(m-2l)]/\sqrt{2l}|\leqslant 4\sup_{1\leqslant k\leqslant n}|S_k-W(k)|/\sqrt{2l}.\tag{54}$$

依(53)式,并注意到 $\log n/\sqrt{l}\to 0$ (当 $n\to\infty$)得

$$\lim_{n\to\infty}\{\sup_{m\leqslant n}|Y_m/\sigma-[W(m)-2W(m-l)+W(m-2l)]|/\sqrt{2l}\}=0,\ \text{a. s.}\tag{55}$$

由定理 1,我们有

$$\begin{aligned}&\lim_{n\to\infty}P\{\sup([W(m)-2W(m-l)+W(m-2l)]/\sqrt{2l}:m=2l,\ \cdots,\ n)\leqslant A_n(x)\}\\&=\exp(-2\mathrm{e}^{-x}),\ -\infty<x<\infty.\end{aligned}\tag{56}$$

最后,由(55)和(56)式得(7)式. 定理 4 证毕.

因为在假定(52)式之下,定理 3 的结论仍成立,故由此可知,以上两节的方法在条件(52)式之下适用.

第 2 节和第 3 节中的渐近定理成立所需之条件还可以显著减弱.

定理 5 将定理 1 的条件 $X_1\sim N(a,\ \sigma^2)$和$(\log n)^2/l\to 0$ 改为

$$\mathrm{E}|X_1|^\delta<\infty,\tag{57}$$

$$\lim_{n\to\infty}n^{2/\delta}/l=0.\tag{58}$$

对某个 $\delta>2$ 成立,则定理 1 的结论仍成立.

本定理的证法与定理 4 平行,只须用到文献[6]中的另一结果:在条件(57)式之下有

$$\lim_{n\to\infty}\{\sup_{k\leqslant n}|S_k-W(k)|n^{-1/\delta}\}=0,\ \text{a. s.}$$

6 跃度 θ 的估计

我们用下面的步骤去构造 θ 的一个点估计:

1. 找 k,使 $|Y_k|=\xi_n$.
2. 计算

$$\theta=\hat{\theta}_n=\frac{1}{n-k}\sum_{i=k+1}^{n}X_i-\frac{1}{k-2l}\sum_{i=1}^{k-2l}X_i,\tag{59}$$

$\hat{\theta}$ 即取为 θ 之点估计.

当 $k=2l$ 或 $k=n$ 时,$\hat{\theta}$ 无定义. 一般地,若 k 过于接近 $2l$ 或 n,则可能表示转变点 t_0 太接

近 0 或 1，而我们现有的样本数太小，不足以对 θ 作出合理的估计.

为作出 θ 的区间估计，先证明关于 $\hat{\theta}$ 的下述极限定理.

定理 6 设对某个 $\delta>2$，F 的 δ 阶矩存在. 又 $\theta\neq 0$，l 适合(58)式且 $l/n\to 0$. 则当 $n\to\infty$ 时，有

$$(\sqrt{t_0(1-t_0)}/\sigma)(\hat{\theta}-\theta)\xrightarrow{\mathscr{L}} N(0,1), \tag{60}$$

这里 $\xrightarrow{\mathscr{L}}$ 表示依分布收敛.

证 不失普遍性，设 $a=0$，$\sigma=1$ (a，σ 的意义见第 1 节). 用定理 5，并将第 2 节中的论证略加修改，不难证明

$$\lim_{n\to\infty} P(nt_0\leqslant k\leqslant nt_0+2l)=1. \tag{61}$$

定义 $n_1=[nt_0-2l]$，$n_2=n-[nt_0+2l]-1$，以及

$$T_n=\sqrt{nt_0(1-t_0)}\Big(\sum_{i=n-n_2+1}^{n}(X_i-\theta)/n_2-\sum_{i=1}^{n_1}X_i/n_1\Big). \tag{62}$$

因为 $l/n\to 0$，$0<t_0<1$，当 $n\to\infty$ 时，有

$$T_n\xrightarrow{\mathscr{L}} N(0,1). \tag{63}$$

现有

$$\begin{aligned}
&T_n-\sqrt{nt_0(1-t_0)}(\hat{\theta}-\theta)\\
&=\sqrt{nt_0(1-t_0)}\Big\{\frac{n-k-n_2}{n_2(n-k)}\sum_{i=n-n_2+1}^{n}(X_i-\theta)-\frac{1}{n-k}\sum_{i=k+1}^{n-n_2}(X_i-\theta)\\
&\quad-\frac{k-2l-n_1}{n_1(k-2l)}\sum_{i=1}^{n_1}X_i+\frac{1}{k-2l}\sum_{i=1}^{k-2l}X_i\Big\}.
\end{aligned}$$

故有

$$\begin{aligned}
&\sup\{|T_n-\sqrt{nt_0(1-t_0)}(\hat{\theta}-\theta)|:nt_0\leqslant k\leqslant nt_0+2l\}\\
&\leqslant C\{n^{-3/2}l\cdot\sup(|\sum_{j=1}^{n}(X_i-\theta)|:j\geqslant n-n_2+1)\\
&\quad+n^{-1/2}\sup(|\sum_{i=j}^{n-n_2}(X_i-\theta)|:[nt_0]+1\leqslant j\leqslant n-n_2)\\
&\quad+n^{-3/2}l\cdot\sup(|\sum_{i=1}^{j}X_i|:1\leqslant j\leqslant n_1)\\
&\quad+n^{-1/2}\sup(|\sum_{i=n_1+1}^{j}X_i|:n_1+1\leqslant j\leqslant[nt_0])\}\\
&\triangleq\sum_{i=1}^{4}I_{n_i}.
\end{aligned} \tag{64}$$

这里 C 为与 n 无关的常数. 因 $l/n\to 0$，X_1，…，X_n 独立且有方差 1，$EX_i=\theta$(当 $i\geqslant n-n_2+1$)，$EX_i=0$(当 $i\leqslant[nt_0]$)，由 Donsker 定理，有

$$I_{n1} \xrightarrow{P} 0,\ \text{当}\ n \to \infty, \tag{65}$$

类似地

$$I_{n3} \xrightarrow{P} 0,\ \text{当}\ n \to \infty, \tag{66}$$

现定义 $\tilde{X}_i = X_i + \theta$, 当 $1 \leqslant i \leqslant [nt_0]$, $\tilde{X}_i = X_i$, 当 $[nt_0]+1 \leqslant i \leqslant n$. 给定 $\varepsilon > 0$, 由 Donsker 定理得

$$\begin{aligned} I_{n2} &\leqslant n^{-1/2} \sup\left\{\left|\sum_{i=j}^{n-n_2} (\tilde{X}_i - \theta)\right| : [n(t_0 - \varepsilon)] \leqslant j \leqslant n - n_2\right\} \\ &\xrightarrow{\mathscr{L}} \sup\{|W(t)| : t_0 - \varepsilon \leqslant t \leqslant t_0\},\ \text{当}\ n \to \infty, \\ &\to 0,\ \text{a.s.},\ \text{当}\ \varepsilon \to 0, \end{aligned}$$

故有

$$I_{n2} \xrightarrow{P} 0,\ \text{当}\ n \to \infty. \tag{67}$$

类似地

$$I_{n4} \xrightarrow{P} 0,\ \text{当}\ n \to \infty. \tag{68}$$

总结(65)～(68)式,得知,当 $n\to\infty$时,有

$$\sup\{|T_n - \sqrt{nt_0(1-t_0)}(\hat{\theta} - \theta)| : nt_0 \leqslant k \leqslant nt_0 + 2l\} \xrightarrow{P} 0.$$

由此从(61)和(63)式,得到(60)式. 定理证毕.

易见 $\hat{t}_0 = (k-l)/n$ 为 t_0 之一相合估计(当然,要求 $\theta \neq 0$ 使 t_0 有定义). 前面我们引进了 σ 的一个相合估计 $\hat{\sigma}$. 以 $\hat{t}_0$ 和 $\hat{\sigma}$ 分别代替 t_0 和 σ,得下述定理:

定理 7 若定理 6 的条件全成立,则当 $n\to\infty$时,有

$$(\sqrt{n\hat{t}_0(1-\hat{t}_0)}/\hat{\sigma})(\hat{\theta} - \theta) \xrightarrow{\mathscr{L}} N(0,1). \tag{69}$$

当 $\theta = 0$ 因而 t_0 无定义时,统计量 $\hat{t}_0$ 仍有意义. 现还不知道(69)式当 $\theta = 0$ 时是否正确,因而定理 7 不能作为检验假设"$\theta = 0$"(即由(3)式定义的原假设 H_0)之用. 但如事先已知 $\theta \neq 0$, 或用前几节方法检验假设(3)式否定了 $\theta = 0$, 则定理 7 可用来给出 θ 之一大样本置信区间.

当然,若 X_1 为正态分布,或更一般地,适合条件(52)式,则定理 6 和定理 7 中的条件(58)式可减弱为 $(\log n)^2/l \to 0$.

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13

Strong Consistency of *M*-Estimates in Linear Models

Abstract This article studies the strong consistency of M-estimates in linear regression models directly from the minimization problem

$$\sum_{i=1}^{n}\rho(Y_i-\alpha-\boldsymbol{X}_i'\boldsymbol{\beta}):=\min,$$

where $\boldsymbol{X}_1,\boldsymbol{X}_2,\cdots$ can be random observations of a p-dimensional random vector $\boldsymbol{X}$, or that they are simply known nonrandom p-vectors. It is shown that the solution $(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n')$ of this minimization problem converges with probability one to the true parameter $(\alpha_0, \boldsymbol{\beta}_0')$ under very general conditions on the function ρ and the sequence $\{(\boldsymbol{X}_i', Y_i)\}$.

1 Introduction

Consider the linear regression model

$$Y_i=\alpha_0+\boldsymbol{X}_i'\boldsymbol{\beta}_0+e_i,\ i=1,\ 2,\ \cdots, \tag{1.1}$$

where $(\alpha_0, \boldsymbol{\beta}_0')$ is the unknown parameter, $e_1, e_2, \cdots$ are random errors. As for $\{\boldsymbol{X}_i\}$, two cases will be considered: (1) $\{\boldsymbol{X}_i\}$ is a sequence of known p-dimensional vectors. (2) $(\boldsymbol{X}_1', Y_1)$, $(\boldsymbol{X}_2', Y_2)$, $\cdots$ are iid. observations of a $(p+1)$-dimensional random vector $(\boldsymbol{X}', Y)$.

The M-estimate, introduced by Huber [5], takes the solution $(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n')$ of the minimization problem

$$\sum_{i=1}^{n}\rho(Y_i-\alpha-\boldsymbol{X}_i'\boldsymbol{\beta}):=\min \tag{1.2}$$

as the estimate of $(\alpha_0, \boldsymbol{\beta}_0')$. This paper seeks the conditions under which $(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n')$ is strongly consistent:

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$$\hat{\alpha}_n \to \alpha_0,\ \hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0, \quad \text{a. s.} \quad \text{as} \quad n \to \infty. \tag{1.3}$$

In (1.2), ρ is a suitably chosen function on R and $(\alpha, \boldsymbol{\beta}')$ varies over some set $\Theta \subset R^{p+1}$, Θ is the parameter space. Two cases are often considered in the literature: (i) $\Theta = R^{p+1}$, (ii) Θ is a closed subset of R^{p+1} containing the true parameter $(\alpha_0, \boldsymbol{\beta}'_0)$ as an interior point. In the following, unless stated otherwise, we shall only consider the (more general) first case.

An often-made assumption in the literature, for example, [6, 7, 10], is that $\rho'(u) = \mathrm{d}\rho(u)/\mathrm{d}u$ exists everywhere on R. In this case the solution of (1.2) must satisfy the equations

$$\sum_{i=1}^{n} \rho'(Y_i - \alpha - \boldsymbol{X}'_i\boldsymbol{\beta}) = 0, \quad \sum_{i=1}^{n} \boldsymbol{X}_i\rho'(Y_i - \alpha - \boldsymbol{X}'_i\boldsymbol{\beta}) = 0. \tag{1.4}$$

If, in addition, ρ is convex, then (1.2) and (1.4) are equivalent. However, in many important examples of *M*-estimates, $\rho'(u)$ does not exist for some u. In such cases, although one may formally write down Eq. (1.4), it may have no solution, or none of its solutions is a solution of (1.2). A well-known example is furnished by $\rho(u) = |u|$ (minimum L_1-norm estimate). Consistency results of the *M*-estimate in this case were given by [3, 8, 11]. A more sophisticated example, considered in [4], is that $\rho(u) = (1-\delta)u^2 + \delta|u|$. In the standard form of linear regression $Y_i = \boldsymbol{X}'_i\boldsymbol{\beta} + e_i$, for this choice of ρ, formally (1.4) reduces to

$$2(1-\delta)\sum_{i=1}^{n} \boldsymbol{X}_i(Y_i - \boldsymbol{X}'_i\boldsymbol{\beta}) + \delta\sum_{i=1}^{n} \boldsymbol{X}_i \mathrm{sgn}(Y_i - \boldsymbol{X}'_i\boldsymbol{\beta}) = 0, \tag{1.5}$$

where $\mathrm{sgn}\,(0) = 0$, $\mathrm{sgn}(u) = u/|u|$ for $u \neq 0$. Although [4] asserts that (1.5) is equivalent to

$$(1-\delta)\sum_{i=1}^{n} (Y_i - \boldsymbol{X}'_i\boldsymbol{\beta})^2 + \delta\sum_{i=1}^{n} |Y_i - \boldsymbol{X}'_i\boldsymbol{\beta}| := \min,$$

this is not true, as has been shown in [1].

We also note that the convexity assumption excludes many functions with practical significance, such as $\rho(u) = \min(|u|, k)$ for some constant $k > 0$. Another example is

$$\rho(u) = \begin{cases} |u|, & |u| \leqslant k \\ k/2 + |u|/2, & |u| > k. \end{cases} \tag{1.6}$$

So it makes good sense to tackle this estimation problem starting directly from the original formulation (1.2). This we shall do in the following sections.

2 Formulation of Results

First consider the case where $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots$ are iid. random vectors.

Theorem 1 Suppose that $(\boldsymbol{X}'_1, Y_1), (\boldsymbol{X}'_2, Y_2), \cdots$ are iid. observations of a random vector $(\boldsymbol{X}', Y)$, and the following conditions are satisfied:

(a) The function ρ is continuous everywhere on R, nondecreasing on $[0, \infty)$, nonincreasing on $(-\infty, 0]$, and $\rho(0)=0$.

(b) Either $\rho(\infty)=\rho(-\infty)=\infty$ and

$$P(\alpha+\boldsymbol{X}'\boldsymbol{\beta}=0)<1 \quad \text{when} \quad (\alpha, \boldsymbol{\beta}') \neq (0, \boldsymbol{0}') \tag{2.1}$$

or $\rho(\infty)=\rho(-\infty)\in(0, \infty)$ and

$$P(\alpha+\boldsymbol{X}'\boldsymbol{\beta}=0)=0 \quad \text{when} \quad (\alpha, \boldsymbol{\beta}') \neq (0, \boldsymbol{0}'). \tag{2.2}$$

(c) For every $(\alpha, \boldsymbol{\beta}')\in R^{p+1}$ we have

$$Q(\alpha, \boldsymbol{\beta}') \equiv \mathrm{E}\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})<\infty \tag{2.3}$$

and Q attains its minimum uniquely at $(\alpha_0, \boldsymbol{\beta}_0')$.

Then (1.3) is true.

When ρ is a convex function, condition (2.3) can be somewhat weakened.

Theorem 2 If ρ is a convex function, then (1.3) is still true when condition (a) of Theorem 1 is satisfied, condition (b) is deleted, and condition (c) is replaced by condition (c′):

(c′) For every $(\alpha, \boldsymbol{\beta}')\in R^{p+1}$ we have

$$Q^*(\alpha, \boldsymbol{\beta}') \equiv \mathrm{E}\{\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})-\rho(Y-\alpha_0-\boldsymbol{X}'\boldsymbol{\beta}_0)\} \tag{2.4}$$

exists and is finite, and that

$$Q^*(\alpha, \boldsymbol{\beta}')>0, \quad \text{for any } (\alpha, \boldsymbol{\beta}') \neq (\alpha_0, \boldsymbol{\beta}_0'). \tag{2.5}$$

The following theorem gives an exponential convergence rate of the estimate $(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n')$.

Theorem 3 Suppose that the conditions of Theorem 1 are met, and in addition that the moment generating function of $\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})$ exists in some neighbourhood of 0, then for arbitrarily given $\varepsilon>0$ there exists a constant $c>0$ independent of n such that

$$P(|\hat{\alpha}_n-\alpha_0|\geqslant\varepsilon)=O(\mathrm{e}^{-cn}), \quad P(\|\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0\|\geqslant\varepsilon)=O(\mathrm{e}^{-cn}). \tag{2.6}$$

This conclusion remains valid if the conditions of Theorem 2 are met, and the moment generating function of $\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})-\rho(Y-\alpha_0-\boldsymbol{X}'\boldsymbol{\beta}_0)$ exists in some neighbourhood of 0.

We next consider the case where $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots$ are nonrandom p-vectors.

Theorem 4 Suppose that in model (1.1) $\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots$ are nonrandom p-vectors and the following conditions are satisfied:

(a) Condition (a) of Theorem 1 is true, $\rho(\infty)=\rho(-\infty)=\infty$.

(b) $\{\boldsymbol{X}_i\}$ is bounded, and if λ_n denotes the smallest eigenvalue of the matrix $\sum_{i=1}^n(\boldsymbol{X}_i-\overline{\boldsymbol{X}}_n)(\boldsymbol{X}_i-\overline{\boldsymbol{X}}_n)'$ $(\overline{X}_n=\sum_{i=1}^n\boldsymbol{X}_i/n)$, then

$$\liminf_{n\to\infty}\lambda_n/n>0. \tag{2.7}$$

(c) $\{e_i\}$ is a sequence of iid. random errors.

(d) For any $t\in R$, $\mathrm{E}\rho(e_1+t)<\infty$, $\mathrm{E}\{\rho(e_1+t)-\rho(e_1)\}>0$ for any $t\neq0$, and there exists a constant $c_1 > 0$ such that

$$\mathrm{E}\{\rho(e_1 + t) - \rho(e_1)\} \geqslant c_1 t^2 \tag{2.8}$$

for $|t|$ sufficiently small.

Then (1.3) is true. This conclusion remains valid if (a), (b) are replaced by

(a′) Condition (a) of Theorem 1 is true,

$$0 < \rho(\infty) = \rho(-\infty) < \infty. \tag{2.9}$$

(b′) $\lim_{\varepsilon\to 0}\limsup_{n\to\infty} \#\{i: 1 \leqslant i \leqslant n,\ |\alpha + \boldsymbol{X}_i'\boldsymbol{\beta}| \leqslant \varepsilon\}/n = 0,$

$$(\alpha, \boldsymbol{\beta}') \neq (0, \boldsymbol{0}'), \tag{2.10}$$

where $\#(B)$ denotes the number of elements in set B. Note that condition (2.10) corresponds to condition (2.2) of Theorem 1.

Also, when ρ is convex, the condition $\mathrm{E}\rho(e_1 + t) < \infty$ can be weakened to $\mathrm{E}\,|\rho(e_1 + t) - \rho(e_1)| < \infty$.

Before proving the theorems, we shall make some comments concerning the conditions assumed:

1. Condition (c) of Theorem 1, which stipulates that Q attains its minimum uniquely at the point $(\alpha_0, \boldsymbol{\beta}_0')$, is closely related to the meaning of the regression. The essence is that the selection of ρ must be compatible with the type of regression considered. For example, when $\alpha_0 + \boldsymbol{x}'\boldsymbol{\beta}_0$ is the conditional median of Y given $\boldsymbol{X} = \boldsymbol{x}$ (median regression), we may choose $\rho(u) = |u|$. Likewisely, when $\alpha_0 + \boldsymbol{x}'\boldsymbol{\beta}_0 = \mathrm{E}(Y \mid \boldsymbol{X} = \boldsymbol{x})$ (the usual mean regression), we may choose $\rho(u) = |u|^2$. An important case is that the conditional distribution of Y given $\boldsymbol{X} = \boldsymbol{x}$ is symmetric and unimodal with center $\alpha_0 + \boldsymbol{x}'\boldsymbol{\beta}_0$. In this case, ρ can be chosen as any even function satisfying condition (a), and such that $\rho(t) > 0$ when $u \neq 0$. This gives us some freedom in the choice of ρ with the aim of obtaining more robust estimates.

2. Condition (2.8) of Theorem 4 reveals a difference between the two cases of $\{\boldsymbol{X}_i\}$ mentioned earlier. In the case that $\{\boldsymbol{X}_i\}$ is a sequence of nonrandom vectors we can no longer assume only that 0 is the unique minimization point of $\mathrm{E}\rho(e_1 + u)$, as shown in the counterexample given in [2] for $\rho(u) = |u|$.

Condition (2.8) holds automatically when $\rho(u) = u^2$ and $\mathrm{E}e_1 = 0$. When $\rho(u) = |u|$, it holds when e_1 has median 0 and a density which is bounded away from 0 in some neighborhood of 0. When ρ is even and e_1 is symmetric and unimodal with center 0, (2.8) holds if one of the following two conditions is satisfied: (i) $\inf\{(\rho(u_2) - \rho(u_1))/(u_2 - u_1): \varepsilon \leqslant u_1 < u_2 < \infty\} > 0$ for any $\varepsilon > 0$, (ii) there exist positive constants $a < b$ and c, such that

$$(\rho(u_2) - \rho(u_1))/(u_2 - u_1) \geqslant c,\quad |f(u_2) - f(u_1)|/(u_2 - u_1) \geqslant c$$

for any $a \leqslant u_1 < u_2 \leqslant b$, where f is the density of e_1.

3 Proof of Theorems 1—3

Our main task is to prove Theorem 1. The proof of Theorem 1 can be easily modified to prove Theorems 2 and 3. For any constant $l > 0$, define the sets

$$A_l = [-l, l]^{p+1}, \tilde{A}_l = [-l, l]^p. \tag{3.1}$$

Without loss of generality, we shall assume in the sequel that

$$\alpha_0 = 0, \boldsymbol{\beta}_0 = \boldsymbol{0}. \tag{3.2}$$

Lemma 1 Suppose that the conditions of Theorem 1 are satisfied. Denote by $(\tilde{\alpha}_n, \tilde{\boldsymbol{\beta}}'_n)$ a Borel measurable solution of the constrained minimization problem

$$\sum_{i=1}^{n} \rho(Y_i - \alpha - \boldsymbol{X}'_i\boldsymbol{\beta}) := \text{min over } (\alpha, \boldsymbol{\beta}') \in A_l, \tag{3.3}$$

where $l > 0$ is a given constant. Then as $n \to \infty$,

$$\tilde{\alpha}_n \to 0, \tilde{\boldsymbol{\beta}}_n \to \boldsymbol{0} \quad \text{a. s.} \tag{3.4}$$

Proof Denote the $T = 2^{p+1}$ vertices $(\pm l, \pm l, \cdots, \pm l)$ of A_l by $(a_1, \boldsymbol{b}'_1), \cdots, (a_T, \boldsymbol{b}'_T)$. From condition (a) it can be easily shown that

$$0 \leqslant \rho(Y - \alpha - \boldsymbol{X}'\boldsymbol{\beta}) \leqslant \sum_{j=1}^{T} \rho(Y - a_j - \boldsymbol{X}'\boldsymbol{b}_j) \tag{3.5}$$

for any $(X', Y) \in R^{p+1}$ and $(\alpha, \boldsymbol{\beta}') \in A_l$. From this, the continuity of ρ, and the dominated convergence theorem, one sees that the function $\boldsymbol{Q}$, defined by (2.3), is continuous. Since $(0, \boldsymbol{0}')$ is the unique minimum point of $\boldsymbol{Q}$, for any $\varepsilon > 0$ we have

$$q \equiv \inf\{Q(\alpha, \boldsymbol{\beta}') - Q(0, \boldsymbol{0}') : (\alpha, \boldsymbol{\beta}') \in A_l - A_\varepsilon\} > 0. \tag{3.6}$$

Choose $\varepsilon_1 \in (0, q/6)$ and m sufficiently large such that

$$E\{I((\boldsymbol{X}', Y) \notin A_m)\rho(Y - \alpha - \boldsymbol{X}'\boldsymbol{\beta})\} < \varepsilon_1, \text{ when } (\alpha, \boldsymbol{\beta}') \in A_l. \tag{3.7}$$

The existence of such m follows from (2.3) and (3.5). Write

$$\{(\boldsymbol{X}_1^{*\prime}, Y_1^*), \cdots, (\boldsymbol{X}_{n^*}^{*\prime}, Y_{n^*}^*)\} = \{(\boldsymbol{X}'_1, Y_1), \cdots, (\boldsymbol{X}'_n, Y_n)\} \cap A_m. \tag{3.8}$$

Put $g = \sup\{|Y - a - \boldsymbol{X}'\boldsymbol{b}| : (\boldsymbol{X}', Y) \in A_m, (a, \boldsymbol{b}') \in A_l\}$. Choose $\varepsilon_2 > 0$ such that

$$\sup\{|\rho(u_2) - \rho(u_1)| : |u_1| \leqslant g, |u_2| \leqslant g, |u_2 - u_1| \leqslant \varepsilon_2\} < \varepsilon_1. \tag{3.9}$$

Choose $\varepsilon_3 > 0$ such that

$$\sup\{|a + \boldsymbol{X}'\boldsymbol{b} - (\tilde{a} + \boldsymbol{X}'\tilde{\boldsymbol{b}})| : (a, \boldsymbol{b}') \in A_l, (\tilde{a}, \tilde{\boldsymbol{b}}') \in A_l,$$
$$|a - \tilde{a}| \leqslant \varepsilon_3, \|\boldsymbol{b} - \tilde{\boldsymbol{b}}\| \leqslant \varepsilon_3, \|\boldsymbol{X}\| \leqslant pm\} < \varepsilon_2. \tag{3.10}$$

Choose a finite set $G = \{(\alpha_1, \boldsymbol{\beta}'_1), \cdots, (\alpha_k, \boldsymbol{\beta}'_k)\} \subset A_l - A_\varepsilon$, such that for any $(\alpha, \boldsymbol{\beta}') \in$

$A_l - A_\varepsilon$ there exists j satisfying $|\alpha - \alpha_j| \leqslant \varepsilon_3$, $\|\boldsymbol{\beta} - \boldsymbol{\beta}_j\| \leqslant \varepsilon_3$.

In the following we shall repeatedly use the phrase "with probability one for n sufficiently large". For simplicity we shall abbreviate it by "wpln". Also, "strong law of large numbers" will be simply written as "SLLN".

Now by SLLN, (3.6) and (3.7), we have wpln:

$$n^{-1}\sum_{i=1}^{n'}\rho(Y_i^* - \alpha_j - \boldsymbol{X}_i^{*\prime}\boldsymbol{\beta}_j) > \mathrm{E}\{I((\boldsymbol{X}', Y) \in A_m)\rho(Y - \alpha_j - \boldsymbol{X}'\boldsymbol{\beta}_j)\} - \varepsilon_1$$

$$> \mathrm{E}\rho(Y - \alpha_j - \boldsymbol{X}'\boldsymbol{\beta}_j) - 2\varepsilon_1$$

$$> Q(0, \boldsymbol{0}') + 4\varepsilon_1, \ j = 1, \cdots, k. \tag{3.11}$$

Fix $(\alpha, \boldsymbol{\beta}') \in A_l - A_\varepsilon$. Find j such that $|\alpha - \alpha_j| \leqslant \varepsilon_3$, $\|\boldsymbol{\beta} - \boldsymbol{\beta}_j\| \leqslant \varepsilon_3$. According to (3.9)—(3.11), we have

$$\sum_{i=1}^{n}\rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}) \geqslant \sum_{i=1}^{n'}\rho(Y_i' - \alpha_j - \boldsymbol{X}_i^{*\prime}\boldsymbol{\beta}_j)$$

$$- \sum_{i=1}^{n'} |\rho(Y_i' - \alpha_j - \boldsymbol{X}_i^{*\prime}\boldsymbol{\beta}_j) - \rho(Y_i - \alpha - \boldsymbol{X}_i^{*\prime}\boldsymbol{\beta})|$$

$$\geqslant n[Q(0, \boldsymbol{0}') + 4\varepsilon_1] - n'\varepsilon_1$$

$$\geqslant n[Q(0, \boldsymbol{0}') + 3\varepsilon_1]. \tag{3.12}$$

This holds simultaneously for all $(\alpha, \boldsymbol{\beta}') \in A_l - A_\varepsilon$ wpln. On the other hand, by SLLN, we have wpln:

$$\sum_{i=1}^{n}\rho(Y_i) < n[Q(0, \boldsymbol{0}') + \varepsilon_1]. \tag{3.13}$$

From (3.12) and (3.13), it follows that $|\tilde{\alpha}_n| \leqslant \varepsilon$ and $\|\tilde{\boldsymbol{\beta}}_n\| \leqslant \varepsilon$ wpln, so (3.4) is proved.

Lemma 2 Suppose that the conditions of Theorem 1 are satisfied. Then there exists a constant $l > 0$ such that $(\tilde{\alpha}_n, \tilde{\boldsymbol{\beta}}'_n) \in A_l$ wpln, where $(\tilde{\alpha}_n, \tilde{\boldsymbol{\beta}}'_n)$ is defined as a solution of (3.3).

Proof Write $S = \{(\alpha, \boldsymbol{\beta}') : (\alpha, \boldsymbol{\beta}') \in R^{p+1}, \alpha^2 + \|\boldsymbol{\beta}\|^2 = 1\}$. By (2.1) we can find $\varepsilon > 0$ such that

$$v \equiv \inf\{P(|\alpha + \boldsymbol{X}'\boldsymbol{\beta}| > \varepsilon) : (\alpha, \boldsymbol{\beta}') \in S\} > 0. \tag{3.14}$$

Choose $m > 0$ such that $P(\boldsymbol{X} \in \tilde{A}_m) > 1 - v/4$, and put $u = 3^{-1}(1 + pm)^{-1}\varepsilon$. Choose a finite set $S_1 \subset S$ such that for each $\boldsymbol{\theta} \in S$ there exists $\boldsymbol{\theta}_1 \in S_1$ for which $\|\boldsymbol{\theta} - \boldsymbol{\theta}_1\| < u$. By (3.14), we have wpln:

$$\#\{i : 1 \leqslant i \leqslant n, \ |\alpha + \boldsymbol{X}_i'\boldsymbol{\beta}| > \varepsilon\} \geqslant nv/2, \quad \text{for every } (\alpha, \boldsymbol{\beta}') \in S_1. \tag{3.15}$$

First consider the case $\rho(\infty) = \rho(-\infty) = \infty$. By SLLN, we have wpln:

$$\#\{i : 1 \leqslant i \leqslant n, \ \boldsymbol{X}_i \in A_m\} \geqslant n(1 - v/4). \tag{3.16}$$

Choose a constant $K > 8[Q(0, \mathbf{0}') + 1]/v$. Since ρ is continuous and $\rho(\pm\infty) = \infty$, we can find $h > 0$ such that $\rho(x) \geqslant K$ when $|x| \geqslant h$. Choose $l > 0$ large enough such that

$$\varepsilon l \geqslant 4h, \qquad P(|Y| \leqslant \varepsilon l/4) > 1 - v/8. \tag{3.17}$$

By SLLN, we have wpln:

$$\#\{i: 1 \leqslant i \leqslant n, |Y_i| \leqslant \varepsilon l/4\} \geqslant n(1 - v/8). \tag{3.18}$$

Now choose arbitrarily $(\tilde{\alpha}, \tilde{\boldsymbol{\beta}}') \notin A_l$. Then $(\tilde{\alpha}, \tilde{\boldsymbol{\beta}}') = r(\alpha, \boldsymbol{\beta}')$ for some $r > l$ and $(\alpha, \boldsymbol{\beta}') \in S$. If $(\alpha, \boldsymbol{\beta}') \in S_1$, then from (3.15) and (3.18), we have wpln:

$$\#\{i: 1 \leqslant i \leqslant n, |Y_i - \tilde{\alpha} - \boldsymbol{X}_i' \tilde{\boldsymbol{\beta}}| \geqslant 3l\varepsilon/4\} \geqslant 3nv/8. \tag{3.19}$$

If $(\alpha, \boldsymbol{\beta}') \notin S_1$, then choose $(\alpha^*, \boldsymbol{\beta}^{*\prime}) \in S_1$ such that $|\alpha - \alpha^*| < u$, $\|\boldsymbol{\beta} - \boldsymbol{\beta}^*\| < u$. When $|\alpha^* + \boldsymbol{X}_i'\boldsymbol{\beta}^*| > \varepsilon$ and $\boldsymbol{X}_i \in \tilde{A}_m$, we have

$$\begin{aligned} |\alpha + \boldsymbol{X}_i'\boldsymbol{\beta}| &\geqslant \varepsilon - |\alpha^* - \alpha + \boldsymbol{X}_i'(\boldsymbol{\beta}^* - \boldsymbol{\beta})| \\ &\geqslant \varepsilon - |\alpha^* - \alpha| - \|\boldsymbol{X}_i\| \, \|\boldsymbol{\beta}^* - \boldsymbol{\beta}\| \\ &\geqslant \varepsilon - u - pmu \\ &\geqslant \varepsilon - (1 + pm)u \\ &> \varepsilon/2. \end{aligned}$$

(Recall that $u = 3^{-1}(1 + pm)^{-1}\varepsilon$.) Hence $|\tilde{\alpha} + \boldsymbol{X}_i' \tilde{\boldsymbol{\beta}}| > l\varepsilon/2$. From this, (3.15), (3.16), and (3.18), we have wpln:

$$\#\{i: 1 \leqslant i \leqslant n, |Y_i - \tilde{\alpha} - \boldsymbol{X}_i' \tilde{\boldsymbol{\beta}}| > l\varepsilon/4\} \geqslant nv/8. \tag{3.20}$$

By (3.19), (3.20), (3.17), and the choice of h, we have wpln:

$$\sum_{i=1}^{n} \rho(Y_i - \tilde{\alpha} - \boldsymbol{X}_i'\tilde{\boldsymbol{\beta}}) \geqslant vKn/8 \geqslant [Q(0, \mathbf{0}') + 1]n, \tag{3.21}$$

simultaneous for all $(\tilde{\alpha}, \tilde{\boldsymbol{\beta}}') \notin A_l$. Taking $\varepsilon_1 < \frac{1}{2}$ in (3.13), we see that $(\tilde{\alpha}_n, \tilde{\boldsymbol{\beta}}_n) \in A_l$ wpln.

We now consider the case $0 < \rho(\pm\infty) = c < \infty$. First note that $Q(0, \mathbf{0}') < c$ by condition (a). Further, condition (2.2) ensures the existence of $\varepsilon > 0$ for given $t < 1$ such that

$$\inf\{P(|\alpha + \boldsymbol{X}'\boldsymbol{\beta}| > \varepsilon): (\alpha, \boldsymbol{\beta}') \in S\} > t. \tag{3.22}$$

Based on (3.22) and modifying the previous argument appropriately, we can choose $l > 0$ such that for given $\varepsilon_1 > 0$, we have wpln:

$$\sum_{i=1}^{n} \rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}) > n(c - \varepsilon_1), \quad \text{for all } (\alpha, \boldsymbol{\beta}') \notin A_l. \tag{3.23}$$

Choose $\varepsilon_1 = [c - Q(0, \mathbf{0}')]/3$. From (3.13) and (3.23), it follows that $|\tilde{\alpha}_n| \leqslant \varepsilon$, and $\|\tilde{\boldsymbol{\beta}}_n\| < \varepsilon$ wpln, as before. This concludes the proof of Lemma 2.

Proof of Theorem 1 Apply Lemmas 1 and 2.

Proof of Theorem 2 Since ρ is a convex function, we need only prove that the conclusion of Lemma 1 holds under the assumptions of Theorem 2. For this purpose put $\rho^*(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta}) = \rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})-\rho(Y)$, and define q^* as

$$q^* = \inf\{Q^*(\alpha, \boldsymbol{\beta}'): (\alpha, \boldsymbol{\beta}') \in A_l - A_\varepsilon\}, \tag{3.24}$$

where Q^* is defined in (2.4).

Now denote by $(a_1, \boldsymbol{b}_1'), \cdots, (a_T, \boldsymbol{b}_T')$ the $T=2^{p-1}$ vertices $(\pm l, \cdots, \pm l)$ of A_l, $(a_{T+1}, \boldsymbol{b}_{T+1}'), \cdots, (a_{2T}, \boldsymbol{b}_{2T}')$ the vertices of A_{2l}. We proceed to show that

$$\begin{aligned}&\sup\{|\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})-\rho(Y)|:(\alpha, \boldsymbol{\beta}') \in A_l\}\\ &\leqslant 2\max_{1\leqslant j\leqslant 2T}|\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j)-\rho(Y)|\\ &\equiv K(\boldsymbol{X}', Y).\end{aligned} \tag{3.25}$$

Indeed, if $\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta}) \geqslant \rho(Y)$, then by condition (a) we have $|\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})-\rho(Y)| \leqslant \max_{1\leqslant j\leqslant T}|\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j)-\rho(Y)|$. If $\rho(Y)>\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})$, two cases are possible: $\alpha+\boldsymbol{X}'\boldsymbol{\beta}>0$ and $\alpha+\boldsymbol{X}'\boldsymbol{\beta}<0$. The handling of these cases being similar, we shall consider only the former case. By convexity of ρ, we have

$$\rho(Y+c)-\rho(Y+c-\alpha-\boldsymbol{X}'\boldsymbol{\beta}) \geqslant \rho(Y)-\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta}), \text{ for any } c \geqslant 0. \tag{3.26}$$

Since $(\alpha, \boldsymbol{\beta}') \in A_l$, there exists $j \leqslant T$ such that $\alpha+\boldsymbol{X}'\boldsymbol{\beta} \geqslant a_j+\boldsymbol{X}'\boldsymbol{b}_j$. Write $\tilde{a} = \alpha - a_j$, $\tilde{\boldsymbol{b}} = \boldsymbol{\beta}-\boldsymbol{b}_j$, and set $c = \tilde{a}+\boldsymbol{X}'\tilde{\boldsymbol{b}}$ in (3.26). We obtain

$$\rho(Y)-\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta}) \leqslant \rho(Y+\tilde{a}+\boldsymbol{X}'\tilde{\boldsymbol{b}})-\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j).$$

Obviously, $(\tilde{a}, \tilde{\boldsymbol{b}}') \in A_{2l}$. Hence by condition (a) there exists $k \leqslant 2T$ such that $\rho(Y+\tilde{a}+\boldsymbol{X}'\tilde{\boldsymbol{b}}) \leqslant \rho(Y-a_k-\boldsymbol{X}'\boldsymbol{b}_k)$, and we get

$$\begin{aligned}&|\rho(Y)-\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})|\\ &=\rho(Y)-\rho(Y-\alpha-\boldsymbol{X}'\boldsymbol{\beta})\\ &\leqslant \rho(Y-a_k-\boldsymbol{X}'\boldsymbol{b}_k)-\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j)\\ &\leqslant |\rho(Y-a_k-\boldsymbol{X}'\boldsymbol{b}_k)-\rho(Y)|+|\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j)-\rho(Y)|\\ &\leqslant 2\max_{1\leqslant j\leqslant 2T}|\rho(Y-a_j-\boldsymbol{X}'\boldsymbol{b}_j)-\rho(Y)|\end{aligned}$$

and (3.25) is proved. (3.25) and condition (c′) together ensure that Q^* is continuous, and therefore $q^*>0$. The rest of the proof is similar to that of Lemma 1.

Applying Theorem 2 to the case $\rho(u)=|u|$, we obtain the following corollary, which was proved in [3] with the additional conditions that $\mathrm{E}|Y|<\infty$, $Y-\alpha_0-\boldsymbol{X}'\boldsymbol{\beta}_0$ and $\boldsymbol{X}$ are independent, and $P(\alpha+\boldsymbol{X}'\boldsymbol{\beta}=0)=0$ when $(\alpha, \boldsymbol{\beta}') \neq (0, \boldsymbol{0}')$.

Corollary Suppose that $(\boldsymbol{X}_1', Y_1)$, $(\boldsymbol{X}_2', Y_2)$, $\cdots$ are iid. samples of the random vector $(\boldsymbol{X}', Y)$, which satisfies the conditions:

1. $\mathrm{E}\|\boldsymbol{X}\| < \infty$.

2. The conditional distribution of Y given $\boldsymbol{X} = \boldsymbol{x}$ has a unique median $\alpha_0 + \boldsymbol{x}'\boldsymbol{\beta}_0$.

Denote by $(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n)$ a solution of (1.2). Then (1.3) holds.

Proof of Theorem 3 The proof follows from the following two lemmas.

Lemma 1′ Suppose that the conditions of Theorem 3 are satisfied, and $l > 0$ is a given constant. Then for any $\varepsilon > 0$ there exists a constant $c > 0$ independent of n, such that

$$P(|\tilde{\alpha}_n - \alpha_0| \geqslant \varepsilon) = O(\mathrm{e}^{-cn}),\ P(\|\boldsymbol{\beta}_n - \boldsymbol{\beta}_0\| \geqslant \varepsilon) = O(\mathrm{e}^{-cn}),$$

where $(\tilde{\alpha}_n, \tilde{\boldsymbol{\beta}}'_n)$ is defined as a solution of (3.3).

Lemma 2′ Suppose that the conditions of Theorem 3 are satisfied. Then there exist constants $l > 0$ and $c > 0$ such that

$$P\{(\hat{\alpha}_n - \alpha_0, \hat{\boldsymbol{\beta}}'_n - \boldsymbol{\beta}'_0) \notin A_l\} = O(\mathrm{e}^{-cn}).$$

These lemmas can be proved by the same method used in proving Lemma 1 and Lemma 2, together with the following fact (see [9], p. 288):

Suppose that $\xi_1, \xi_2, \cdots$ are iid. random variables, $\mathrm{E}\xi_1 = 0$ and there exists $\delta > 0$ such that $\mathrm{E}\exp(t\xi_1) < \infty$ when $|t| < \delta$. Then for any given $\varepsilon > 0$ we can find a constant $c > 0$ such that

$$P(|\sum_{i=1}^{n}\xi_i/n| \geqslant \varepsilon) = O(\mathrm{e}^{-cn}).$$

4 Proof of Theorem 4

We give only the proof of Theorem 4 under conditions (a)—(d). It is easy to modify the proof when (a) and (b) are replaced by (a′) and (b′).

Lemma 3 Suppose that function ρ is defined on R, $\rho(0) = 0$, is nondecreasing on $[0, \infty)$ and nonincreasing on $(-\infty, 0]$. Let $\{Y_i, i = 1, 2, \cdots\}$ be a sequence of iid. variables such that

$$\mathrm{E}\rho(Y_1 + c) < \infty, \quad \text{for any } c \in R, \tag{4.1}$$

and $\{c_i, i = 1, 2, \cdots\}$ be a sequence of bounded real constants. Then

$$\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}[\rho(Y_i - c_i) - \mathrm{E}\rho(Y_i - c_i)] = 0, \quad \text{a. s.} \tag{4.2}$$

Proof Apply a standard truncation argument.

Lemma 4 Suppose that the conditions of Lemma 3 are satisfied, and that ρ is continuous everywhere on R, $\{\boldsymbol{X}_i\}$ is a bounded sequence, and B is a bounded set in R^{p+1}. Then, with probability one, the sequence $\{(1/n)(\sum_{i=1}^{n}\rho(Y_i - \alpha - \boldsymbol{X}'_i\boldsymbol{\beta}) - \sum_{i=1}^{n}\mathrm{E}\rho(Y_i - \alpha - \boldsymbol{X}'_i\boldsymbol{\beta})) : n = 1, 2, \cdots\}$ of functions of $(\alpha, \boldsymbol{\beta}')$ is equicontinuous and uniformly bounded on B.

Proof Denote by F the probability distribution of Y_1. Construct the probability space $(R^{\infty}, \mathscr{R}^{\infty}, F^{\infty})$. Fix integer $m > 0$, find $h > 0$ such that

$$\mathrm{E}\{\rho(Y_1 + T)I(\rho(Y_1 + T) \geqslant h) + \rho(Y_1 - T)I(\rho(Y_1 - T) \geqslant h)\} < 1/(3m), \quad (4.3)$$

where $T = \sup\{|\alpha + \boldsymbol{X}_i'\boldsymbol{\beta}| : i = 1, 2, \cdots, (\alpha, \boldsymbol{\beta}') \in B\}$ ($<\infty$ by the boundedness of $\{\boldsymbol{X}_i\}$ and B). From the assumptions on ρ, it follows that there exists $\varepsilon_m' > 0$ such that we have $|\rho(u_1) - \rho(u_2)| \leqslant 1/(3m)$ when $|u_1 - u_2| \leqslant \varepsilon_m'$ and $\min(\rho(u_1), \rho(u_2)) < h$. Find $\varepsilon_{m1} > 0$ such that $|(\alpha + \boldsymbol{X}_i'\boldsymbol{\beta}) - (\alpha^* + \boldsymbol{X}_i'\boldsymbol{\beta}^*)| \leqslant \varepsilon_m'$, $i = 1, 2, \cdots$, whenever $(\alpha, \boldsymbol{\beta}') \in B$, $(\alpha^*, \boldsymbol{\beta}^{*\prime}) \in B$ and $\|(\alpha, \boldsymbol{\beta}') - (\alpha^*, \boldsymbol{\beta}^{*\prime})\| \leqslant \varepsilon_{m1}$.

Now by (4.3) and SLLN, we can find a positive integer N_m and a set $D_m \in \mathscr{R}^{\infty}$ with $F^{\infty}(D_m) < 2^{-m}$, such that

$$n^{-1}\sum_{i=1}^{n}\rho(Y_i + T)I(\rho(Y_i + T) \geqslant h) + n^{-1}\sum_{i=1}^{n}\rho(Y_i - T)I(\rho(Y_i - T) \geqslant h) < 1/(3m), \quad (4.4)$$

whenever $n \geqslant N_m$ and $Y^* \equiv (Y_1, Y_2, \cdots) \notin D_m$. Since $\{\boldsymbol{X}_i\}$ and B are bounded, for any $Y^* \in R^{\infty}$ we can find $\varepsilon_{m2}(Y^*)$ such that

$$\left|n^{-1}\sum_{i=1}^{n}(\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*) - \rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}))\right| < 1/(3m), \quad (4.5)$$

whenever $1 \leqslant n \leqslant N_m$, $(\alpha, \boldsymbol{\beta}') \in B$, $(\alpha^*, \boldsymbol{\beta}^{*\prime}) \in B$, and $\|(\alpha^*, \boldsymbol{\beta}^{*\prime}) - (\alpha, \boldsymbol{\beta}')\| \leqslant \varepsilon_{m2}(Y^*)$. Take $\varepsilon_m(Y^*) = \min(\varepsilon_{m1}, \varepsilon_{m2}(Y^*))$.

Now suppose that $(\alpha, \boldsymbol{\beta}') \in B$, $(\alpha^*, \boldsymbol{\beta}^{*\prime}) \in B$, $\|(\alpha^*, \boldsymbol{\beta}^{*\prime}) - (\alpha, \boldsymbol{\beta}')\| \leqslant \varepsilon_m(Y^*)$, and $Y^* \notin D_m$. Then for $n \leqslant N_m$ we have (4.5). If $n > N_m$, then

$$\begin{aligned}
&\left|n^{-1}\sum_{i=1}^{n}(\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*) - \rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}))\right| \\
&\leqslant n^{-1}\sum_{i=1}^{n}\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*)I(\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*) \geqslant h) \\
&\quad + n^{-1}\sum_{i=1}^{n}\rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta})I(\rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}) \geqslant h) \\
&\quad + n^{-1}\left|\sum{}'[\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*) - \rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta})]\right| \\
&\equiv J_1 + J_2 + J_3,
\end{aligned} \quad (4.6)$$

where the summation $\sum'$ is over all i such that $1 \leqslant i \leqslant n$ and $\min(\rho(Y_i - \alpha^* - \boldsymbol{X}_i'\boldsymbol{\beta}^*), \rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta})) < h$. From (4.4), the definition of T and the conditions imposed on ρ, we have

$$J_1 \leqslant \text{the left-hand side of (4.4)} < 1/(3m).$$

Likewise, $J_2 \leqslant 1/(3m)$. Finally, by the definition of ε_m', ε_{m1}, and $\varepsilon_m(Y^*)$, for each i belonging to the range of summation $\sum'$, we have $|\rho(Y_i-\alpha^*-X_i'\beta^*)-\rho(Y_i-\alpha-X_i'\beta)|<1/(3m)$. Hence $J_3<1/(3m)$. Summing up, we find that (4.5) is still true when $1/(3m)$ on the right-hand side of (4.5) is replaced by $1/m$.

Now write $D=\bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty}D_m$. Since $F^{\infty}(D_m)<2^{-m}$, we have $F^{\infty}(D)=0$. From the above discussion we see that for any $Y^*=(Y_1, Y_2, \cdots)\notin D$ and any positive integer m, we can find $\varepsilon_m(Y^*)>0$ such that

$$\left|\sum_{i=1}^{n}(\rho(Y_i-\alpha^*-X_i'\beta^*)-\rho(Y_i-\alpha-X_i'\beta))\right|/n\leqslant 1/m, \quad \text{for} \quad n=1,2,\cdots$$

whenever $(\alpha, \beta')\in B$, $(\alpha^*, \beta^{*\prime})\in B$, and $\|(\alpha^*, \beta^{*\prime})-(\alpha, \beta')\|\leqslant\varepsilon_m(Y^*)$. This proves the equicontinuity of $\{\sum_{i=1}^{n}\rho(Y_i-\alpha-X_i'\beta)/n:n=1,2,\cdots\}$ over B, with probability one. The uniform boundedness of this sequence of functions follows from the fact that when $Y^*\notin D$, we have $Y^*\notin D_m$ for some m. Repeating the above argument, we find that

$$\sum_{i=1}^{n}\rho(Y_i-\alpha-X_i'\beta)/n\leqslant h+1/(3m)$$

for $n\geqslant N_m$ and $(\alpha, \beta')\in B$, while for $n<N_m$ we have

$$\sum_{i=1}^{n}\rho(Y_i-\alpha-X_i'\beta)/n\leqslant\sum_{i=1}^{N_m}(\rho(Y_i+T)+\rho(Y_i-T))$$

for any $(\alpha, \beta')\in B$.

Therefore, in order to prove Lemma 4, we have only to establish that $\{\sum_{i=1}^{n}\mathrm{E}\rho(Y_i-\alpha-X_i'\beta)/n:n=1,2,\cdots\}$ is uniformly bounded and equicontinuous on B. This is simple, since $\mathrm{E}\rho(Y_1+c)$ is continuous for each c, $\sup\{|\alpha+X_i'\beta|:i=1,2,\cdots,(\alpha,\beta')\in B\}=T<\infty$, and

$$\mathrm{E}\rho(Y_i-\alpha-X_i'\beta)\leqslant\mathrm{E}\rho(Y_1+T)+\mathrm{E}\rho(Y_1-T),\ (\alpha,\beta')\in B,\ i=1,2,\cdots$$

Combining Lemma 3 and Lemma 4, we obtain

Lemma 5 If the conditions of Lemma 3 and Lemma 4 are satisfied, then there exists a set $D\in\mathscr{R}^{\infty}$ such that $F^{\infty}(D)=0$, and when $(Y_1, Y_2, \cdots)\notin D$ we have $\lim_{n\to\infty}\sum_{i=1}^{n}[\rho(Y_i-\alpha-X_i'\beta)-\mathrm{E}\rho(Y_i-\alpha-X_i'\beta)]/n=0$ uniformly for $(\alpha,\beta')\in B$, B is a given bounded set in R^{p+1}.

In the following we adhere to (3.2), and put

$$S_M=\{(\alpha,\beta'):\alpha^2+\|\beta\|^2\leqslant M\},$$

$$\overline{S}_M=\{(\alpha,\beta'):\alpha^2+\|\beta\|^2=M\}$$

for any $M>0$.

Lemma 6 Suppose that the conditions (a)—(d) of Theorem 4 are satisfied.

(i) There exist $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$ such that

$$\inf\{\#(i:1\leqslant i\leqslant n,\ |\alpha+\boldsymbol{X}_i'\boldsymbol{\beta}|\geqslant\varepsilon_1)/n:(\alpha,\boldsymbol{\beta}')\in\overline{S}_1\}\geqslant\varepsilon_2 \tag{4.7}$$

for n sufficiently large.

(ii) For each $M>0$ there exists a constant $\varepsilon_M>0$ such that

$$\sum_{i=1}^{n}\mathrm{E}[\rho(Y_i-\alpha-\boldsymbol{X}_i'\boldsymbol{\beta})-\rho(Y_i)]/n\geqslant\|(\alpha,\boldsymbol{\beta}')\|^2\varepsilon_M \tag{4.8}$$

for $(\alpha,\boldsymbol{\beta}')\in S_M$ and n sufficiently large.

Proof Consider

$$\sum_{i=1}^{n}(\alpha+\boldsymbol{X}_i'\boldsymbol{\beta})^2=n(\alpha-\overline{\boldsymbol{X}}_i'\boldsymbol{\beta})^2+\boldsymbol{\beta}'H_n\boldsymbol{\beta},$$

where $\overline{\boldsymbol{X}}_n=(\boldsymbol{X}_1+\cdots+\boldsymbol{X}_n)/n$, $H_n=\sum_{i=1}^{n}(\boldsymbol{X}_i-\overline{\boldsymbol{X}}_n)(\boldsymbol{X}_i-\overline{\boldsymbol{X}}_n)'$. Suppose that $(\alpha,\boldsymbol{\beta}')\in\overline{S}_1$. Write $M_0=\sup\{1,\ \|\boldsymbol{X}_i\|:i=1,2,\cdots\}$. If $\|\boldsymbol{\beta}\|\geqslant(2pM_0)^{-1}$, we have, according to (2.7), $\boldsymbol{\beta}'H_n\boldsymbol{\beta}\geqslant(2pM_0)^{-2}\delta_1 n$ for some constant $\delta_1>0$ and n large. If $\|\boldsymbol{\beta}\|<(2pM_0)^{-1}$, then $\|\overline{\boldsymbol{X}}_n'\boldsymbol{\beta}\|\leqslant(2p)^{-1}$, and $|\alpha|\geqslant\sqrt{1-(2p)^{-2}}\geqslant\sqrt{3}/2$. Hence $|\alpha-\overline{\boldsymbol{X}}_n'\boldsymbol{\beta}|\geqslant\sqrt{3}/2-\frac{1}{2}>\frac{1}{3}$, and so $n(\alpha-\overline{\boldsymbol{X}}_n'\boldsymbol{\beta})^2\geqslant n/9$. Summing up the above gives

$$\sum_{i=1}^{n}(\alpha+\boldsymbol{X}_i'\boldsymbol{\beta})^2\geqslant\delta n,\quad\text{for all }(\alpha,\boldsymbol{\beta}')\in\overline{S}_1\text{ and }n\text{ large}, \tag{4.9}$$

for some $\delta>0$.

Now suppose that (4.7) is false. Then we can find $n_j\to\infty$, $0<\varepsilon_{1j}\to0$, $0<\varepsilon_{2j}\to0$, $(\alpha_j,\boldsymbol{\beta}_j')\in\overline{S}_1$, such that

$$\#\{i:1\leqslant i\leqslant n_j,\ |\alpha_j+\boldsymbol{X}_i'\boldsymbol{\beta}_j|\geqslant\varepsilon_{1j}\}\leqslant\varepsilon_{2j}n_j,\ j=1,2,\cdots,$$

which entails that

$$\sum_{i=1}^{n_j}(\alpha_j+\boldsymbol{X}_i'\boldsymbol{\beta}_j)^2/n_j\leqslant\varepsilon_{1j}^2+\varepsilon_{2j}T^2,\ j=1,2,\cdots,$$

where $T=M_0+1$. This contradicts (4.9), and (4.7) is proved.

For a proof of (4.8), we notice that since $\mathrm{E}\rho(Y_1+t)>\mathrm{E}\rho(Y_1)$ when $t\neq0$, $\mathrm{E}\rho(Y_1+t)$ is continuous in t and $\alpha+\boldsymbol{X}_i'\boldsymbol{\beta}$ is uniformly bounded for $i=1,2,\cdots$ and $(\alpha,\boldsymbol{\beta}')\in S_M$. Hence it follows from (2.8) that there exists a constant $\delta_M>0$, depending only on M, such that $\mathrm{E}[\rho(Y_i-\alpha-\boldsymbol{X}_i'\boldsymbol{\beta})-\rho(Y_i)]\geqslant\delta_M|\alpha+\boldsymbol{X}_i'\boldsymbol{\beta}|^2$ for $(\alpha,\boldsymbol{\beta}')\in S_M$. From this and (4.9), (4.8) follows.

Lemma 7 Suppose that the conditions (a)—(d) of Theorem 4 are satisfied. Given $l>0$, denote by $(\tilde{\alpha}_n,\tilde{\boldsymbol{\beta}}_n')$ the solution of the constrained minimization problem (3.3). Then (3.4) holds. Moreover, the conclusion of Lemma 2 holds.

Proof Fix $\varepsilon \in (0, R)$. Let D be the set mentioned in Lemma 5. Since $\|(\alpha, \boldsymbol{\beta}')\| \geqslant \varepsilon$ when $(\alpha, \boldsymbol{\beta}') \notin A_\varepsilon$ (see (3.1)), it follows from Lemma 5 and Lemma 6(b) that

$$\inf\left\{\sum_{i=1}^{n}[\rho(Y_i - \alpha - \boldsymbol{X}_i'\boldsymbol{\beta}) - \rho(Y_i)]/n : (\alpha, \boldsymbol{\beta}') \in A_l - A_\varepsilon\right\} \leqslant 2^{-1}\varepsilon^2\varepsilon_{(p-1)l},$$

for all $(Y_1, Y_2, \cdots) \notin D$.

By Lemma 6(i), we still have (3.15) in a slightly different notation. Moreover, (3.16) remains true by the boundedness assumption of $\{\boldsymbol{X}_i\}$. Hence the proof of Lemma 2 remains valid in the present setting.

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Estimation of Multivariate Binary Density Using Orthogonal Functions

Abstract In this paper, the authors studied certain properties of the estimate of Liang and Krishnaiah (1985, *J. Multivariate Anal.*, 16: 162～172) for multivariate binary density. An alternative shrinkage estimate is also obtained. The above results are generalized to general orthonormal systems.

1 Introduction

In a number of situations, the experimenter is confronted with the statistical analysis of the data which is binary in nature. For example, one may be interested in diagnosis of the disease on the basis of symptoms. The reliability of complicated systems can be studied by examining whether the components are functioning or not. In image processing, a picture is classified on the basis of two grey levels like white and black using some threshold value. We may assign a score of 1 or 0 according as the grey level is white or black, respectively. So, it is important to study the problems of estimation of multivariate binary density.

Cencov [3] expressed continuous multivariate density as a series of orthonormal functions. Bahadur [1] expressed the multivariate binary density as a series. Ott and Kronmal [5] expressed the density as a series involving Walsh functions. Liang and Krishnaiah [4] also expressed the density in terms of Walsh functions but the coefficients in their series are different from those used by Ott and Kronmal. The present paper is a continuation of the work done by Liang and Krishnaiah.

In Section 2, we established the necessary and sufficient condition for the difference between the mean integrated square error (MISE) of the estimate of Liang and Krishnaiah and the MISE of a "natural" shrinkage estimator to be of order $O(n^{-2})$, where n denotes the sample size. It is also shown that $O(n^{-2})$ is the best possible rate than can be obtained. In

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Section 3, a new shrinkage estimate of the multivariate binary density estimate is proposed. It is shown that the difference between the MISE of this estimate and the "natural" shrinkage estimator is of order $O(n^{-2})$. The results of Sections 2 and 3 are generalized to general orthonormal systems in Section 4.

2 Shrinkage Estimates Based on Walsh Functions

Denote by R the sets

$$R = \{(x_1, \cdots, x_d): x_i = 0 \text{ or } 1, i = 1, \cdots, d\} \tag{1}$$

and by $\widetilde{P} = \{p(x)\}$ the family of probability functions on R: $\widetilde{P}$ includes all functions on R satisfying

$$p(x) \geqslant 0 \quad \text{for} \quad x \in R; \quad \sum_{x \in R} p(x) = 1. \tag{2}$$

The system of Walsh functions $\{\phi_r\}$ is defined by

$$\phi_r(x) = (-1)^{r'x}/2^{d/2}, \quad x \in R; r \in R. \tag{3}$$

Each $p(x) \in \widetilde{P}$ can be expressed as

$$p(x) = \sum_{r \in R} a_r \phi_r(X). \tag{4}$$

So the estimation of $p(x)$ involves that of $\{a_r\}$.

Suppose that X is a random vector with range R whose distribution $p(x)$ is unknown, and $X_1, \cdots, X_n$ are iid. observations of X. Then, since

$$a_r = \mathrm{E}\phi_r(X), \quad r \in R, \tag{5}$$

we obtain the moment estimate of a_r as

$$\hat{a}_r = \sum_{i=1}^{n} \phi_r(X_i)/n. \tag{6}$$

As argued in [4], there are reasons to consider the "shrinkage estimator"

$$\hat{p}_\lambda(x) = \sum_{r \in R} \lambda_r \hat{a}_r \phi_r(x) \tag{7}$$

in replacing the "natural" estimator $\hat{p}(x) = \sum_{r \in R} \hat{a}_r \phi_r(x)$ derived directly from (4) and (6). As was shown in [4], the mean integrated square error of $\hat{p}_\lambda(x)$ to be denoted by MISE($\hat{p}_\lambda$), attains its minimum upon choosing

$$\lambda_r = n a_r^2/(2^{-d} + (n-1)a_r^2), \quad r \in R. \tag{8}$$

Since in practice we do not know $\{a_r\}$, the constants $\{\lambda_r\}$ need to be estimated from the sample. Liang and Krishnaiah proposed such an estimate in [4] via some iteration considerations,

$$\lambda_r^* = \begin{cases} 0, & \text{if } \hat{a}_r^2 < 4(n-1)2^{-d}/n^2, \\ [n + (n^2 - 4(n-1)2^{-d}\hat{a}_r^{-2})^{1/2}]/2(n-1), & \text{otherwise.} \end{cases} \tag{9}$$

Denote

$$\hat{p}_{\lambda^*}(x) = \sum_{r\in R} \lambda_r^* \hat{a}_r \phi_r(x). \tag{10}$$

Liang and Krishnaiah proved the following theorem.

Theorem [[4], Theorem 2] If $0 < |a_r| < 2^{-d/2}$ for all r, then ①

$$\mathrm{MISE}(\hat{p}_{\lambda^*}) - \mathrm{MISE}(\hat{p}_\lambda) = O(n^{-2}). \tag{11}$$

Here we improve and complete this result as follows:

Theorem 1 (1) The necessary and sufficient condition for (11) to be true is that

$$a_r \neq 0, \quad \text{for all } r \in R. \tag{12}$$

(2) $O(n^{-2})$ is the best rate obtainable. Suppose that a_r^* is any estimate of a_r (based on $X_1, \cdots, X_n$) and form the estimator $p^*(x) = \sum_{r\in R} a_r^* \phi_r(x)$. Then for at least one $p(x) \in \widetilde{P}$, we have

$$\limsup_{n\to\infty} n^2 \,|\, \mathrm{MISE}(p^*) - \mathrm{MISE}(\hat{p}_\lambda) \,| > 0. \tag{13}$$

Proof (1) Since ϕ_r is bounded by 1 on R, and $\phi_r(X_1), \cdots, \phi_r(X_n)$ are iid. random variables with mean a_r, by Bennett inequality (see [2]), we have

$$P(|\hat{a}_r - a_r| \geqslant \varepsilon) \leqslant 2\exp\left(-\frac{n\varepsilon^2}{2(1+\varepsilon)}\right) \tag{14}$$

for arbitrarily $\varepsilon > 0$. The validity of (14) does not depend on the value assumed by a_r. Use (14) to replace (4-5) of [4], and the rest of the proof for sufficiency runs exactly along the same way as in [4].

To prove the necessity of (12), suppose that $a_{r_0} = 0$ for some $r_0 \in R$. Denote by $\widetilde{R}$ the set $\{r: r \in R;\ a_r \neq 0\}$. Then we have

$$\begin{aligned} &|\, \mathrm{MISE}(\hat{p}_\lambda^*) - \mathrm{MISE}(\hat{p}_\lambda) - \mathrm{E}(\lambda_{r_0}^* \hat{a}_{r_0})^2 \,| \\ &\leqslant \sum_{r\in \widetilde{R}} |\, \mathrm{E}(a_r - \lambda_r \hat{a}_r)^2 - \mathrm{E}(a_r - \lambda_r^* \hat{a}_r)^2 \,|. \end{aligned} \tag{15}$$

In the proof of sufficiency, we actually proved that

$$|\, \mathrm{E}(a_r - \lambda_r \hat{a}_r)^2 - \mathrm{E}(a_r - \lambda_r^* \hat{a}_r)^2 \,| = O(n^{-2}), \quad r \in \widetilde{R}. \tag{16}$$

On the other hand, since $a_{r_0} = 0$, we have

$$\mathrm{Var}(\hat{a}_{r_0}) = 2^{-d/n}.$$

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① We point out here that "$0 < |a_r| < 2^{-d/2}$ for all r" should be corrected as "$0 < |a_r| < 2^{-d/2}$ for all $r \neq (0, 0, \cdots, 0)$", since $a_r = 2^{-d/2}$ for $r = (0, 0, \cdots, 0)$.

Hence, by the Central Limit Theorem, we obtain

$$\sqrt{n}\hat{a}_{r_0} \xrightarrow{\mathscr{L}} N(0,\ 2^{-d}) \quad (n \to \infty). \tag{17}$$

From (9) we see that $\lambda_r^* \geqslant \frac{1}{2}$ when $\lambda_r^* \neq 0$. Therefore,

$$\liminf_{n\to\infty} P\left(\lambda_{r_0}^* \geqslant \frac{1}{2}\right) \geqslant \liminf_{n\to\infty} P(\hat{a}_{r_0}^2 \geqslant 4(n-1)2^{-d}/n^2)$$
$$= 1 - \frac{1}{\sqrt{2\pi}}\int_{-2}^{2} e^{-t^2/2}\,dt \triangleq g > 0. \tag{18}$$

So, when n is large, we have

$$P(\lambda_{r_0}^* \geqslant 1/2) \geqslant g/2. \tag{19}$$

Use again (17). We get for arbitrarily given $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|\ \hat{a}_{r_0}\ | \geqslant \varepsilon/\sqrt{n}) = 1 - \frac{1}{\sqrt{2\pi}}\int_{-\varepsilon 2^{d/2}}^{\varepsilon 2^{d/2}} e^{-t^2/2}\,dt \triangleq h_\varepsilon. \tag{20}$$

Choose $\varepsilon > 0$ small enough. We have $h_\varepsilon > 1 - g/3$. Therefore, for n sufficiently large, we have

$$\begin{aligned} E(\lambda_{r_0}^* \hat{a}_{r_0})^2 &\leqslant \left(\frac{1}{2}\frac{\varepsilon}{\sqrt{n}}\right)^2 P\left(\lambda_{r_0}^* > \frac{1}{2},\ |\ \hat{a}_{r_0}\ | \geqslant \frac{\varepsilon}{\sqrt{n}}\right) \\ &\geqslant \varepsilon^2/4n\left(P\left(\lambda_{r_0}^* \geqslant \frac{1}{2}\right) + P\left(|\ \hat{a}_{r_0}\ | \geqslant \frac{\varepsilon}{\sqrt{n}}\right) - 1\right) \\ &\geqslant \varepsilon^2/4n(g/2 + 1 - g/3 - 1) \\ &= g\varepsilon^2/24n. \end{aligned} \tag{21}$$

From (15), (16), and (21), we see that (11) cannot be true.

(2) To prove the second part of this theorem, we proceed to show that, even in the case of $d = 1$, (13) holds.

In the case of $d = 1$, x can only take two values, 0 and 1, and we shall write $p(0) = p$, $p(1) = 1 - p(0) = q$. In this case, we have

$$a_0 = 1/\sqrt{2}, \quad a_1 = (2p - 1)/\sqrt{2}. \tag{22}$$

Suppose that (a_0^* , a_1^*) is an estimate of (a_0, a_1). Then, since it is well known in the theory of point estimation that the estimate

$$\hat{a}_1 = \sqrt{2}\,\overline{X}_n - 1/\sqrt{2}$$

is admissible under quadratic loss $L(d,\ a_1) = (d - a_1)^2$. It follows that there exists $p \neq \frac{1}{2}$, such that

$$E_p(a_1^* - a_1)^2 \geqslant E(\hat{a}_1 - a_1)^2 = 2pq/n. \tag{23}$$

Here $\lambda_1 = (2p-1)^2/[(2p-1)^2+4pq/n]$, and simple manipulations shows that

$$\mathrm{E}_p(\lambda_1\hat{a}_1 - a_1)^2 = \frac{2pq(2p-1)^2}{n((2p-1)^2+4q/n)}$$
$$= \frac{2pq}{n} - \frac{8p^2q^2}{n^2(2p-1)^2} + O(n^{-3}). \tag{24}$$

Here $a_0 = 1/\sqrt{2}$, $\lambda_0 = 1$, $\phi_0(0) = \phi_0(1) = 1/\sqrt{2}$. Hence, $\mathrm{E}_p(\lambda_0\hat{a}_0 - a_0)^2 = 0$. From this and (23), (24), we see that, denoting $p^*(x) = \sum_{r=0}^{1} a_r^*\phi_r(\lambda)$,

$$\mathrm{MISE}_p(p^*) - \mathrm{MISE}_p(\hat{p}_\lambda) \geqslant \frac{8p^2q^2}{n^2(2p-1)^2} + O(n^{-3}),$$

for some $p \neq \frac{1}{2}$, $0 < p < 1$, and we have

$$\limsup_{n\to\infty} n^2 \mid \mathrm{MISE}_p(p^*) - \mathrm{MISE}_p(\hat{p}_\lambda) \mid > 0.$$

This proves (13).

The above method of proof can be extended to the general case of d without difficulty. We need only consider a subset of $\widetilde{P}$ with the form

$$\widetilde{P}_1 = \{p(x):\ p(x)\in\widetilde{P},\ p(x^{(1)}) = p,\ p(x^{(2)}) = 1-p,\ 0<p<1\},$$

where $x^{(1)}$, $x^{(2)}$ are two chosen points in R and $x^{(1)} \neq x^{(2)}$.

3 A New Shrinkage Estimate

We see from Theorem 1 that in order to have (11), condition (12) is necessary. In this section we shall introduce a new shrinkage estimator, for which (11) holds without any restriction on $\{a_r\}$. For this purpose, define

$$\hat{\lambda}_r = \begin{cases} n\hat{a}_r^2/(2^{-d}+(n-1)\hat{a}_r^2), & \text{if } |\hat{a}_r| \geqslant n^{-1/3} \\ 0, & \text{otherwise} \end{cases} \tag{25}$$

and the shrinkage estimator

$$\hat{p}_{\hat{\lambda}}(x) = \sum_{r\in R}\hat{\lambda}_r\hat{a}_r\phi_r(x). \tag{26}$$

Theorem 2 For $\hat{p}_{\hat{\lambda}}$ defined in (26), we have

$$\mathrm{MISE}(\hat{p}_{\hat{\lambda}}) - \mathrm{MISE}(\hat{p}_\lambda) = O(n^{-2}). \tag{27}$$

Proof The proof is a slight modification of the proof of Theorem 2 in [4]. First, according to (4-3) of [4], we only need to show that

$$\mathrm{E}(a_r - \lambda_r'\hat{a}_r)^2\mathrm{E}(\lambda_r - \tilde{\lambda}_r)^2 = O(n^{-4}) \tag{28}$$

for each $r \in R$, where λ_i' lies between λ_r and $\tilde{\lambda}_r$. Consider separately two cases:

(1) $a_r = 0$. In this case, using Bennett inequality (14), we find that

$$P(\tilde{\lambda}_r \neq 0) = P(|\hat{a}_r| \geqslant n^{-1/3}) \leqslant 2\exp(-n^{1/3}/4) = O(n^{-4}).$$

Since $\tilde{\lambda}_r$ is bounded by 1, we get

$$E(\lambda_r - \tilde{\lambda}_r)^2 = E\tilde{\lambda}_r^2 = O(n^{-4}).$$

As a_r, λ_r', $\hat{a}_r$ are bounded, we get (28).

(2) $a_r \neq 0$. Again, using Bennett inequality, we have

$$P(|a_r - \hat{a}_r| \geqslant |a_r|/2) \leqslant c^{-cn}, \quad \text{for some } c > 0. \tag{29}$$

Put $f_n(t) = nt/(2^{-d} + (n-1)t)$. There exists $c_1 > 0$ such that $f_n'(t) \leqslant c_1 n^{-1}$ for $t \geqslant |a_r|^2/4$, and c_1 does not depend on n. Hence

$$|\lambda_r - \tilde{\lambda}_r| \leqslant c_1 n^{-1} |\hat{a}_r^2 - a_r^2|.$$

Since $E|\hat{a}_r^2 - a_r^2|^2 \leqslant 2E|\hat{a}_r - a_r|^2 = 2\mathrm{Var}(\hat{a}_r) = O(n^{-1})$, we have

$$E(\lambda_r - \tilde{\lambda}_r) = O(n^{-3}). \tag{30}$$

On the other hand, we have

$$E(a_r - \lambda_r'\hat{a}_r)^2 \leqslant 2\mathrm{Var}(\hat{a}_r) + E(1 - \lambda_r')^2. \tag{31}$$

We have

$$1 - \lambda_r = O(n^{-1}) \tag{32}$$

and

$$1 - \tilde{\lambda}_r = O(n^{-1}), \quad \text{when } |\hat{a}_r| \geqslant |a_r|/2. \tag{33}$$

From (32)—(33), and observing that λ_r' lies between λ_r and $\tilde{\lambda}_r$, we get in view of (29)

$$\begin{aligned} E(1 - \lambda_r')^2 &\leqslant O(n^{-2}) + P(|\hat{a}_r| < |a_r|/2) \\ &\leqslant O(n^{-2}) + P(|\hat{a}_r - a_r| \geqslant |a_r|/2) \\ &= O(n^{-2}). \end{aligned} \tag{34}$$

Since $\mathrm{Var}(\hat{a}_r) = O(n^{-1})$, from (31) and (34) we get

$$E(a_r - \lambda_r'\hat{a}_r)^2 = O(n^{-1}). \tag{35}$$

From (30) and (35) we get (28), and the theorem is proved.

4 Estimates Based on General Orthonormal System

In the above discussions we chose Walsh functions as our orthonormal system. There are infinitely many orthonormal systems which can be chosen for estimation purposes. The particular choice depends upon convenience and the application needs. For instance, the following system

$$\phi_r(x_r) = 1, \quad \phi_r(x) = 0 \quad \text{for} \quad x \neq x_r, \ r \in R$$

leads to the usual frequency estimate and its shrinkage. We now indicate that the results of Section 2 and 3 are still valid for any choice of orthonormal system $\{\phi_r\}$. The only modification is that in the definitions of λ_r, λ_r^*, and $\tilde{\lambda}_r$ in (8), (9), and (25), 2^{-d} should be replaced by

$$b_r = \sum_{r \in R} p(x)\phi_r^2(x). \tag{36}$$

Theorem 3 Suppose that $\{\phi_r\}$ is any orthonormal function system on R, then the conclusions of Theorem 2 and Theorem 3 remain valid if the definitions of λ_r, λ_r^*, and $\tilde{\lambda}_r$ are modified as stated earlier.

The proof is obvious, since in the proofs of Theorem 2 and Theorem 3, the special form of Walsh functions plays no special role, except one point which we now discuss.

In the proof of Theorem 1 (2), we made use of the fact that $a_0 = 1/\sqrt{2}$ is a constant, so that $\lambda_0 \hat{a}_0$ gives an exact estimation of a_0. In the general case this situation no longer holds. In general, we have

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} \phi_0(0) & \phi_1(0) \\ \phi_0(1) & \phi_1(1) \end{pmatrix}^{-1} \begin{pmatrix} p \\ 1-p \end{pmatrix}$$

or

$$a_0 = ap + b, \quad a_1 = cp + d$$

for some known constants a, b, c, d. An inspection of the proof of Theorem 1(2) convinces us that we have to prove that $(a\overline{X}_n + b, \ c\overline{X}_n + d)$ is an admissible estimate of (a_0, a_1) under quadratic loss. That is to say, if there exists an estimate (a_0^*, a_1^*) such that

$$\mathrm{E}_p(a_0' - a_0)^2 + \mathrm{E}_p(a_1' - a_1)^2 \leqslant (a^2 + c^2)p(1-p)/n \tag{37}$$

for all $p \in (0, 1)$, then in (37) we have equality for each $p \in (0, 1)$. For a proof of this, we put

$$\xi_0 = (a_0^* - b)/a, \quad \xi_1 = (a_1^* - d)/c \tag{38}$$

and $t_0 = a^2/(a^2 + c^2)$, $t_1 = c^2/(a^2 + c^2)$. Note that $t_0 > 0$, $t_1 > 0$, and $t_0 + t_1 = 1$. Owing to the convexity of the function $f(x) = x^2$, we have

$$(t_0\xi_0 + t_1\xi_1 - p)^2 \leqslant t_0(\xi_0 - p)^2 + t_1(\xi_1 - p)^2. \tag{39}$$

Noticing that

$$\mathrm{E}_p(\xi_0 - p)^2 = \mathrm{E}_p(a_0^* - a_0)^2/a^2,$$
$$\mathrm{E}_p(\xi_1 - p)^2 = \mathrm{E}_p(a_1^* - a_1)^2/c^2$$

and (37), we have

$$\mathrm{E}(t_0\xi_0 + t_1\xi_1 - p)^2 \leqslant (a^2 + c^2)[\mathrm{E}_p(a_0^* - a_0)^2 + \mathrm{E}_p(a_1^* - a_1)^2] = pq/n. \tag{40}$$

But $\overline{X}_n$ is an admissible estimate for p. Hence, we must have equality in (40), and $t_0\xi_0+t_1\xi_1=\overline{X}_n$. Also, since x^2 is strictly convex, in order to have equality in (39), we must have $\xi_0=\xi_1$. Therefore $\xi_0=\xi_1=\overline{X}_n$, and from (38) we see that (a_0^*, a_1^*) is no other than $(a\overline{X}_n+b, c\overline{X}_n+d)$, and we have equality in (37) for each p. This completes the proof of Theorem 3.

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15

线性模型中最小一乘估计的渐近正态性

摘要 考虑线性回归模型(1.1),其中 $\boldsymbol{\beta}_0$ 为未知的回归系数向量,$e_1, e_2, \cdots$ 独立且各有中位数零.定义 $\boldsymbol{\beta}_0$ 的最小一乘估计 $\hat{\boldsymbol{\beta}}_n$ 为极值问题(1.2)的解.本文首先论述了文献中关于 $\hat{\boldsymbol{\beta}}_n$ 渐近正态性的工作,并指出其中错误所在,然后在最一般的条件下证明了 $\hat{\boldsymbol{\beta}}_n$ 的渐近正态性(定理 1).特别,对 $\{\boldsymbol{x}_i\}$ 施加的条件与最小二乘估计的渐近正态性条件相同(见(1.6)式),说明这一条件是无可改进的.

1 引言

考虑线性回归模型

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta}_0 + e_i,\ i = 1, \cdots, n, \cdots \tag{1.1}$$

这里假定 $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots$ 是已知的 p 维向量,$\boldsymbol{\beta}_0$ 是未知的 p 维回归系数向量,$e_1, e_2, \cdots$ 是iid.的随机误差序列,其中位数为零,公共密度为 f, f 在零点连续且 $f(0) > 0$. $\boldsymbol{\beta}_0$ 的最小一乘(简记为 ML_1N)估计 $\hat{\boldsymbol{\beta}}_n$ 定义为下述极值问题之一解:

$$\sum_{i=1}^{n} | Y_i - \boldsymbol{x}_i'\hat{\boldsymbol{\beta}}_n | = \inf\Big\{ \sum_{i=1}^{n} | Y_i - \boldsymbol{x}_i'\boldsymbol{\beta} | : \boldsymbol{\beta} \in \boldsymbol{R}^p \Big\}, \tag{1.2}$$

此处设参数空间为 p 维欧氏空间 $\boldsymbol{R}^p$.以下(见系 3)将指出:若将参数空间改为 $\boldsymbol{R}^p$ 之任一子集,以 $\boldsymbol{\beta}_0$ 为内点,则一切均无变化.

最小一乘原则可追溯到 Laplace,但在长时期中未能吸引很多的注意力.理由之一在于其计算困难.此问题现因高速计算机的出现以及 Charnes 等的文章[1]的发表而解决了,该文将 ML_1N 估计的计算与线性规划的求解联系起来.另一个理由是缺乏适当的渐近理论:如所周知,在估计总体中位数这个简单场合,样本中位数这个估计,在一定条件下是渐近正态的.但对一般线性模型(1.1)而言,问题大为复杂.受上述简单情况的启发,令

$$\boldsymbol{S}_n = \sum_{i=1}^{n} \boldsymbol{x}_i\boldsymbol{x}_i'. \tag{1.3}$$

可以猜测,在一定条件下应有

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$$2f(0)\boldsymbol{S}_n^{1/2}(\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0)\xrightarrow{\mathscr{L}}N(\boldsymbol{0},\ \boldsymbol{I}_p),\ \text{当}\ n\to\infty,\tag{1.4}$$

此处 $\boldsymbol{I}_p$ 为 p 阶单位阵，$\xrightarrow{\mathscr{L}}$ 表依分布收敛.

证明(1.4)式的首次努力是由 Bassett 等在文献[2]中作出. 他们假定 $\{e_i\}$ 满足前述条件，又假定极值问题(1.2)的解唯一(这不总成立且难验证)，以及

$$\boldsymbol{S}_n/n\to\boldsymbol{Q},\ \boldsymbol{Q}>\boldsymbol{0}.\tag{1.5}$$

不幸的是，他们的论证包含了不易解决的严重错误. 仅举一事，他们忽略了如下事实：在文献[2]的(3.10)式上面那个方程右端的 $o(1)$ 应为 $o_h(1)$，因而无法肯定当 $T\to\infty$ 时，对 H 中的 h 一致地有 $o_h(1)\to 0$. 另外，文献[2]中的(3.9)式不对. 一反例如下(用文献[2]中之记号)：

$$Y_t=x_t\beta+u_t,\ t=1,\ 2,\ \cdots(\beta\ \text{为一维}),$$
$$x_1=1/\sqrt{2},\ x_2=1+\sqrt{2}/10,\ x_i=1\ \text{当}\ i\geqslant 3,$$
$$u_1,\ u_2,\ \cdots\ \text{为 iid.},\ u_1\sim N(0,\ 1).$$

容易验证，包括唯一性和非格子点条件在内的一切条件都满足. 但可以证明当 $T=2,\ 4,\ 6,\ \cdots$ 时，对 $h=1$ (它属于集 $H=\{1,\ 2,\ \cdots,\ T\}$)有

$$P_r(Z_T(\delta,\ h)\in C[0,\ 1])=0,$$

因而(3.9)式失效(见附录 1).

Bloomfield 等[3]试图提出一个证明，他们假定：$\boldsymbol{x}_1,\ \boldsymbol{x}_2,\ \cdots$ 为某一随机向量 $\boldsymbol{X}$ 的观察值，$\boldsymbol{X}$ 的协方差阵为正定，且 $(\boldsymbol{x}_1,\ Y_1),\ (\boldsymbol{x}_2,\ Y_2),\ \cdots$ 为平稳遍历的. 可惜的是，他们未能注意到：为了使文献[3]中的 p. 45，(6)式所定义的 $\{g_n(\underset{\sim}{c})\}$ 为等度连续，$g_n(\underset{\sim}{c})$ 必须定义为 $\sum_{i=1}^{n}h_n(r_i(\underset{\sim}{c}))/n$，而不能像他们在书中那样定义为 $\sum_{i=1}^{n}h_n(r_i(\underset{\sim}{c}))$. 但这样一来，文献[3]p. 47 的断言 $n^{-1}[\boldsymbol{D}_n(\underset{\sim}{c}_n)]\to 2f(0)\boldsymbol{C}$ 应改为 $[\boldsymbol{D}_n(\underset{\sim}{c}_n)]\to 2f(0)\boldsymbol{C}$，由此只能得到 $\underset{\sim}{c}_n-\underset{\sim}{a}_n\xrightarrow{P}\underset{\sim}{0}$，而非文献[3]中 p. 46 上的关键断言(8)，因而其证明失效. 除此之外，他们还忽视了文献[3]中p. 45 所定义的 $h_n(t)$ 在 $t=\pm n^{-p}$ 处无二阶导数，这使其 p. 47 的(12)式失效.

另外，Amemiya 在文献[4]中曾试图给出(1.4)式的一个证明，他的想法是用二阶可导函数去逼近绝对值函数. 除了(1.5)式外，还假定 $\{\boldsymbol{x}_i\}$ 有界，且 $\boldsymbol{\beta}_0$ 局限在一紧集内. 可惜的是，他的证明也属错误. 问题之一在于(以下至本段完用文献[4]的记号)：(3.12)式 $A_1\to 0$ 及(3.22)式 $B_1\to 0$ 都不对. 正相反，我们证明了其实有 $A_1\xrightarrow{P}\infty,\ B_1\xrightarrow{P}\infty$ (见附录 2). 另一个关键之点在于：为证当 $T\to\infty$ 时 $T(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}^*)'(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}^*)\xrightarrow{P}0$，(3.11)式应理解为 $\sup\{|S^*(\boldsymbol{\beta})-S(\boldsymbol{\beta})|:\boldsymbol{\beta}\in D\}\xrightarrow{P}0$($D$ 为参数空间). 而文献[4]中之论证，即使不计上述错误，也是不充分的.

当回归模型包含一常数项，即形如 $Y_i=\alpha_0+\boldsymbol{x}_i'\boldsymbol{\beta}_0+e_i,\ i=1,\ 2,\ \cdots$ 时，文献[3](p. 62，引理 1)注意到：$\boldsymbol{\beta}_0$ 的 ML_1N 估计 $\hat{\boldsymbol{\beta}}_n$ 是 Jaeckel 在文献[5]中引进的一族秩估计的特例. Jaeckel 注意到，他的估计与 Jurečková 在文献[6]中引进的一个估计等价. 由此出发，可利用文献[6]中的一个结果去证明 $\hat{\boldsymbol{\beta}}_n$ 的渐近正态性. 但由于下述理由，未能给出问题的一个满意的解决. 首先，文献[6]对 $\{\boldsymbol{x}_i\}$ 提出了很难验证的苛刻条件. 其次，其中假定密度 f 有 Fisher 信息量，故 f

必须处处为正且绝对连续,甚至最简单的均匀分布也不满足这个条件. 还有,这个方法不能对付模型中无常数项 α_0 的情况,若 α_0 出现,也只能处理 $\boldsymbol{\beta}_0$ 的问题而无法处理 α_0 的问题.

Dupačová[7] 证明了一个关于 $\hat{\boldsymbol{\beta}}_n$ 的渐近正态性的定理,其中假定 $\{\boldsymbol{x}_i\}$ 为随机观察值,她的结果用于参数无约束的情况时,大致与文献[3]相当,但包含了一个在数学上不理想的假定,即要求 $\|\boldsymbol{x}_i\|$ 有三阶矩. 且其方法不能处理 $\boldsymbol{x}_i$ 非随机这个更重要的情况.

在回顾了本问题的历史后,我们指出本文的目的是:在最一般的条件下给出(1.4)式的一个正确证明. 主要结果如下:

定理 1 设在(1.1)式中 $e_1, e_2, \cdots$ 为 iid. ,且以下两条件满足:

1. 以 F 记 e_1 的分布. 存在 $\Delta>0$, 使当 $|u|\leqslant\Delta$ 时, $f(u)=F'(u)$ 存在. f 在零点连续, $f(0)>0$ 且 $F(0)=1/2$(这表示零为 e_1 的唯一中位数).

2. 对某自然数 m, $\boldsymbol{S}_m^{-1}$ 存在,且

$$\lim_{n\to\infty}\max_{1\leqslant i\leqslant n}\boldsymbol{x}_i'\boldsymbol{S}_n^{-1}\boldsymbol{x}_i=0, \tag{1.6}$$

则(1.4)式成立.

注 条件(1.6)正是保证 $\boldsymbol{\beta}_0$ 的最小二乘估计为渐近正态的条件(当 $\mathrm{E}e_1=0$, $0<\mathrm{E}e_1^2<\infty$ 且 $e_1, e_2, \cdots$ 为 iid.). 我们曾预料: ML_1N 估计渐近正态条件可能会比最小二乘估计的渐近正态条件为强,因前者为非线性估计而后者为线性的. 这一点使我们相信,(1.6)式是无可改进了.

由这个定理可得出一些推论,它包含了前面提到的诸位学者企图证明的结果.

系 1 设在(1.1)式中 $e_1, e_2, \cdots$ 为 iid.,定理 1 的条件 1 满足,且存在常数列 $\{g_n\}$, $g_n\to\infty$, $g_n/g_{n+1}\to 1$,使

$$\boldsymbol{S}_n/g_n\to\boldsymbol{A}>\boldsymbol{0}, \tag{1.7}$$

则(1.4)式成立.

吴健福[8]在讨论最小二乘估计的相合性时,提到了条件(1.7). 这个系包含了 Bassett 等在文献[2]中提到的结果为其特例. 他还立即推出下述结果:

系 2 设 $e_1, e_2, \cdots$ 为 iid. , 定理 1 的条件 1 满足. 又 $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots$ 为某随机向量 $\boldsymbol{X}$ 的 iid. 观察值而 $\mathrm{E}(\boldsymbol{XX}')>\boldsymbol{0}$, $\{\boldsymbol{x}_i\}$ 与 $\{e_i\}$ 独立,则对 a. s. $\{\boldsymbol{x}_i\}$, (1.4)式成立. 因而(1.4)式也无条件成立. 这就是文献[3]的结果.

系 3 在定理 1 条件下, ML_1N 估计 $\hat{\boldsymbol{\beta}}_n$ 为 $\boldsymbol{\beta}_0$ 的弱相合估计(见引理 4 后的注).

据(1.4)式,为证此,只需证当(1.6)式成立时有

$$\boldsymbol{S}_n^{-1}\to\boldsymbol{0},\ \text{当}\ n\to\infty. \tag{1.8}$$

(1.8)式证明如下:固定 m,使 $\boldsymbol{S}_m>\boldsymbol{0}$. 对 $N>m$,以 $\rho_{N_1}\leqslant\cdots\leqslant\rho_{N_p}$ 记 $\boldsymbol{S}_N$ 的特征根. 据 Neumann[9] 的一个结果,有

$$\mathrm{tr}(\boldsymbol{S}_m\boldsymbol{S}_n^{-1})\geqslant\sum_{i=1}^{p}\rho_{mi}/\rho_{ni},\ m\leqslant n. \tag{1.9}$$

因而,由(1.6)式得

$$\operatorname{tr}(\boldsymbol{S}_m\boldsymbol{S}_n^{-1}) = \sum_{j=1}^{m}\operatorname{tr}(\boldsymbol{x}_j\boldsymbol{x}_j'\boldsymbol{S}_n^{-1}) = \sum_{j=1}^{m}\boldsymbol{x}_j'\boldsymbol{S}_n^{-1}\boldsymbol{x}_j$$
$$\leqslant m\max_{1\leqslant i\leqslant m}\boldsymbol{x}_i'\boldsymbol{S}_n^{-1}\boldsymbol{x}_i \to 0,\ 当\ n\to\infty. \tag{1.10}$$

因为 $\rho_{m1}>0$,由(1.9)及(1.10)式,有 $\lim\limits_{n\to\infty}\rho_{n1}=\infty$,(1.8)式得证.

由本引理可知,若在(1.2)式中以 G 代 R^p,其中 G 是一含 $\boldsymbol{\beta}_0$ 为内点的集,而将(1.2)式的解记为 $\hat{\boldsymbol{\beta}}_n(G)$,则将有 $P(\hat{\boldsymbol{\beta}}_n(G)\neq\hat{\boldsymbol{\beta}}_n)\to 0$ 当 $n\to\infty$. 因此,当以 $\hat{\boldsymbol{\beta}}_n(G)$ 代 $\hat{\boldsymbol{\beta}}_n$ 时,(1.4)式仍有效.

在此顺便提到,吴月华[10]在比定理 1 的条件稍强的条件下,证明了 $\hat{\boldsymbol{\beta}}_n$ 为 $\boldsymbol{\beta}_0$ 的强相合估计. 看来在定理 1 的条件下无法证明这一点.

在实际应用中,回归模型有常数项. 代替(1.1)式有

$$Y_i=\alpha_0+\boldsymbol{x}_i'\boldsymbol{\beta}_0+e_i,\ i=1,\ \cdots,\ n,\ \cdots. \tag{1.11}$$

虽则(1.11)为(1.1)式的特例,可用定理 1 处理. 但为统计推断之目的,以下面的形式为方便:

定理 2 以 $(\hat{\alpha}_n,\ \hat{\boldsymbol{\beta}}_n')$ 记 $(\alpha_0,\ \boldsymbol{\beta}_0')$ 的 ML_1N 估计,及

$$\overline{\boldsymbol{x}}_n=(\boldsymbol{x}_1+\cdots+\boldsymbol{x}_n)/n,\ \boldsymbol{T}_n=\sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}}_n)(\boldsymbol{x}_i-\overline{\boldsymbol{x}}_n)'. \tag{1.12}$$

设 $\{e_i\}$ 为 iid.,定理 1 的条件 1 满足,对某自然数 m 有 $\boldsymbol{T}_m>0$,且

$$\lim_{n\to\infty}\max_{1\leqslant i\leqslant n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}}_n)'\boldsymbol{T}_n^{-1}(\boldsymbol{x}_i-\overline{\boldsymbol{x}}_n)=0. \tag{1.13}$$

则当 $n\to\infty$ 时有

$$2f(0)\boldsymbol{T}_n^{1/2}(\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0)\xrightarrow{\mathscr{L}}N(\boldsymbol{0},\ \boldsymbol{I}_p), \tag{1.14}$$

$$\frac{2f(0)\sqrt{n}}{\sqrt{1+n\overline{\boldsymbol{x}}_n'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n}}(\hat{\alpha}_n-\alpha_0)\xrightarrow{\mathscr{L}}N(0,\ 1). \tag{1.15}$$

而且,当 $n\to\infty$ 时,$2f(0)\boldsymbol{T}_n^{1/2}(\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0)$ 与 $2f(0)\sqrt{n}\{(\hat{\alpha}_n-\alpha_0)+\overline{\boldsymbol{x}}_n'(\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0)\}$ 为渐近独立.

我们注意,$\hat{\alpha}_n$ 和 $\hat{\boldsymbol{\beta}}_n$ 仍有弱相合性. 对 $\hat{\boldsymbol{\beta}}_n$,结论由(1.14)式及 $\boldsymbol{T}_n^{-1}\to\boldsymbol{0}$(它是(1.13)式的推论)得出. 对 $\hat{\alpha}_n$ 则可由(1.15)式及 $\overline{\boldsymbol{x}}_n'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n\to 0$ 得出,后者是(1.13)式及 $\boldsymbol{T}_n^{-1}\to\boldsymbol{0}$ 的推论.

定理 1 的系 2 亦可稍加修改而用于此处,只需将条件 $\mathrm{E}(\boldsymbol{X}\boldsymbol{X}')>\boldsymbol{0}$ 改为 $\mathrm{E}\{(\boldsymbol{X}-\mathrm{E}\boldsymbol{X})(\boldsymbol{X}-\mathrm{E}\boldsymbol{X})'\}>\boldsymbol{0}$.

上述两定理中 $e_1,\ e_2,\ \cdots$ 为 iid. 的条件可以减轻. 我们有如下的结果:

定理 3 设在(1.1)式中,$e_1,\ e_2,\ \cdots$ 独立,且存在 $\Delta>0$,使 e_i 的分布 F_i 在 $[-\Delta,\ \Delta]$ 内可导,$F_i(0)=1/2$,$i=1,\ 2,\ \cdots$,记 $f_i(u)=F_i'(u)$. 假定 $\{f_i\}$ 在 $u=0$ 处等度连续,且 $0<\inf\limits_i f_i(0)\leqslant\sup\limits_i f_i(0)<\infty$. 又设(1.6)式成立,则有

$$2\boldsymbol{S}_n^{-1/2}\sum_{i=1}^{n}f_i(0)\boldsymbol{x}_i\boldsymbol{x}_i'(\hat{\boldsymbol{\beta}}_n-\boldsymbol{\beta}_0)\xrightarrow{\mathscr{L}}N(\boldsymbol{0},\ \boldsymbol{I}_p),\ 当\ n\to\infty. \tag{1.16}$$

我们的主要工作为证明定理 1. 定理 2, 3 的证明只需做少许修改. 为证定理 1,将原模型改写为下述形式是方便的:令

$$S_n^{-1/2}\boldsymbol{x}_i = \boldsymbol{x}_{ni},\ i = 1,\ \cdots,\ n,$$
$$S_n^{1/2}\boldsymbol{\beta}_0 = \boldsymbol{\beta}_{n0},\ Y_{ni} = Y_i,\ e_{ni} = e_i.$$

则(1.1)式有形式

$$Y_{ni} = \boldsymbol{x}'_{ni}\boldsymbol{\beta}_{n0} + e_{ni},\ i = 1,\ \cdots,\ n,\ n = 1,\ 2,\cdots \tag{1.17}$$

且

$$\sum_{i=1}^{n}\boldsymbol{x}_{ni}\boldsymbol{x}'_{ni} = \boldsymbol{I}_p,\ n = m,\ m+1,\ \cdots. \tag{1.18}$$

仍以 $\hat{\boldsymbol{\beta}}_n$ 记(1.17)式中的 $\boldsymbol{\beta}_{n0}$ 的 ML_1N 估计,则我们有下述一般定理,它包含了定理 1 作为其特例:

定理 1′ 设在(1.17)式中, $\boldsymbol{x}_{n1},\ \cdots,\ \boldsymbol{x}_{nn}$ 是适合(1.18)式的已知 p 维向量, $e_{n1},\ \cdots,\ e_{nn}$ 为 iid. ,其公共分布 F 不依赖 n 且满足定理 1 的条件 1. 又设

$$d_n \equiv \max_{1\leqslant i\leqslant n}\|\boldsymbol{x}_{ni}\| \to 0,\ 当\ n\to\infty, \tag{1.19}$$

此处 $\|\cdot\|$ 表欧氏长度. 则当 $n\to\infty$ 时有

$$2f(0)(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_{n0}) \xrightarrow{\mathscr{L}} N(\boldsymbol{0},\ \boldsymbol{I}_p). \tag{1.20}$$

本定理的证明将在第 3 节给出,证明的关键部分将以引理形式在第 2 节给出.

2 若干引理

引理 1(Bennett) 设随机变量 $\xi_1,\ \cdots,\ \xi_n$ 独立, $E\xi_i = 0,\ |\xi_i| \leqslant b < \infty,\ i = 1,\ \cdots,\ n,\ b$ 为常数. 记 $V = \sum_{i=1}^{n}\mathrm{Var}(\xi_i)/n$,则对任何 $\varepsilon > 0$ 有

$$P\left(\left|\sum_{i=1}^{n}\xi_i/n\right| \geqslant \varepsilon\right) \leqslant 2\exp(-n\varepsilon^2/(2V + 2b\varepsilon)). \tag{2.1}$$

证明见文献[11].

在模型(1.17)中,不失普遍性可设

$$\boldsymbol{\beta}_{n0} = \boldsymbol{0}, \tag{2.2}$$

以下总设(2.2)式成立,因而 $Y_{ni} = e_{ni}$. 对 R^p 中任何向量 $\boldsymbol{a} = (a_1,\ \cdots,\ a_p)'$, 记 $|a| = \max_{1\leqslant i\leqslant p}|a_i|$, 又以 $I(A)$ 记集 A 的指示函数.

引理 2 设在(1.17)式中,除(1.19)式可能例外,定理 1′的其他条件全部成立,则有

$$I\left(\left|\sum_{i=1}^{n}\mathrm{sgn}(Y_{ni} - \boldsymbol{x}'_{ni}\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni}\right| \geqslant (p+1)d_n\right) \leqslant I(|\hat{\boldsymbol{\beta}}_n| \geqslant \Delta/(\sqrt{p}d_n)). \tag{2.3}$$

以概率 1 成立,此处 $\mathrm{sgn}(a) = a/|a|$ 当 $a \neq 0$, $\mathrm{sgn}(0) = 0$.

证 按 $\hat{\boldsymbol{\beta}}_n$ 的定义,通过取方向导数易知:对 R^p 的任一单位向量 $\boldsymbol{\theta}$ 有

$$-\sum_{i=1}^{n}\operatorname{sgn}(Y_{ni}-\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni}'\boldsymbol{\theta}I(Y_{ni}\neq\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)+\sum_{i=1}^{n}|\boldsymbol{x}_{ni}'\boldsymbol{\theta}|I(Y_{ni}=\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\geqslant 0.$$

由于 $\boldsymbol{\theta}$ 的任意性,由此得

$$\left|\sum_{i=1}^{n}\operatorname{sgn}(Y_{ni}-\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni}\right|\leqslant\sum_{i=1}^{n}|\boldsymbol{x}_{ni}|I(Y_{ni}=\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n).\tag{2.4}$$

现设 $|\hat{\boldsymbol{\beta}}_n|<\Delta/(\sqrt{p}d_n)$. 若 $1\leqslant i_1\leqslant\cdots\leqslant i_{p+1}\leqslant n$ 且 $Y_{nk}=\boldsymbol{x}_{nk}'\hat{\boldsymbol{\beta}}_n$, $k=i,\cdots,i_{p+1}$,则找不同时为零的常数 $c_1,\cdots,c_{p+1}$,使 $c_1\boldsymbol{x}_{ni_1}+\cdots+c_{p+1}\boldsymbol{x}_{ni_{p+1}}=\boldsymbol{0}$,将有

$$c_1Y_{ni_1}+\cdots+c_{p+1}Y_{ni_{p+1}}=0.\tag{2.5}$$

由此,并注意到 $|\hat{\boldsymbol{\beta}}_n|<\Delta/(\sqrt{p}d_n)$ 蕴含 $|\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n|\leqslant\Delta$,可得:

$$\begin{aligned}&\text{事件}\left\{\sum_{i=1}^{n}|\boldsymbol{x}_{ni}|I(Y_{ni}=\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\geqslant(p+1)d_n\text{ 且 }|\hat{\boldsymbol{\beta}}_n|<\Delta/(\sqrt{p}d_n)\right\}\\&\quad\subset\text{事件}\Big\{\bigcup_{1\leqslant i_1<\cdots<i_{p+1}\leqslant n}\Big(\text{存在只依赖于 }\boldsymbol{x}_{ni_1},\cdots,\boldsymbol{x}_{ni_{p+1}}\text{ 的不全为零的常数}\\&\qquad c_1,\cdots,c_{p+1},\text{使}\sum_{j=1}^{p+1}c_jY_{ni_j}=0\text{ 且 }|Y_{ni_j}|<\Delta,\ j=1,\cdots,p+1\Big)\Big\}.\end{aligned}\tag{2.6}$$

由于 $Y_{ni}=e_{ni}$ 且 $e_{n1},\cdots,e_{nn}$ 为 iid.,其公共分布在 $(-\Delta,\Delta)$ 内连续,知(2.6)式右端事件之概率为零. 这事实与(2.4)式一起蕴含(2.3)式,而引理获证.

为引入关键的引理 3,先定义几个记号. 由(1.19)式知,存在正的常数序列 $\{\mu_n\}$,使

$$\mu_n\to\infty,\ \mu_n^2d_n\to0,\ \sqrt{p}\mu_n^2d_n<1/2,\text{对充分大的 }n.\tag{2.7}$$

以下设定理 1′条件满足,记

$$M\equiv M_n\equiv[\log n/\log\mu_n],\ r_m\equiv\mu_n^m,\ m=1,\cdots,M,\ r_{M+1}=n.\tag{2.8}$$

$$D\equiv D_n\equiv\{\boldsymbol{\beta}=(\beta_1,\cdots,\beta_p)':-\mu_n\leqslant\beta_i<\mu_n,\ i=1,\cdots,p\}.\tag{2.9}$$

注意当 n 大时 $M\geqslant1$. 此因(1.18)式有 $nd_n^2\geqslant1$, 故 $\mu_n<n^{1/4}$. 将 D 剖分为一些超长方体 $\widetilde{D}_1,\cdots,\widetilde{D}_{J_1}$,有形如 $\widetilde{D}_i=\{\boldsymbol{u}=(u^{(1)},\cdots,u^{(p)})':a_i\leqslant u^{(i)}<b_i,\ i=1,\cdots,p\}$,使

$$J_1\leqslant\mu_n^{2p},\ L(\widetilde{D}_i)\leqslant1,\ \widetilde{D}_i\cap\widetilde{D}_j=\varnothing,\ i,j=1,\cdots,J_1,\ i\neq j,$$

这里 $L(A)$ 定义为 $\sup\{|\boldsymbol{u}-\boldsymbol{v}|:\boldsymbol{u}\in A,\ \boldsymbol{v}\in A\}$. 然后,每个 $\widetilde{D}_k$ 又剖分为一些超长方体 $\widetilde{D}_{k1},\cdots,\widetilde{D}_{kJ_2}$,而这些超长方体又被进一步如此剖分. 这一步骤可以归纳地定义如下:设在经过 $m-1$ 轮剖分后,D 被剖分为一些超长方体 $\{\widetilde{D}_{j_1\cdots j_{m-1}}\}$. 则在第 m 轮剖分时,每个 $\widetilde{D}_{j_1\cdots j_{m-1}}$ 被剖分为一些超长方体 $\{\widetilde{D}_{j_1\cdots j_{m-1}l}:l=1,\cdots,J_m\}$,使得

$$J_m\leqslant\mu_n^{2p},\ L(\widetilde{D}_{j_1\cdots j_{m-1}l})\leqslant\mu_n^{-2(m-1)},\ l=1,\cdots,J_m.\tag{2.10}$$

此过程进行到 $M+1$ 轮剖分结束时为止. 记

$$G_m=\{\widetilde{D}_{j_1\cdots j_m}\}=D\text{ 在第 }m\text{ 轮剖分后的结果},m=1,\cdots,M+1.\tag{2.11}$$

G_m 中的一超长方体,即 $\widetilde{D}_{j_1\cdots j_m}$,有时也记为 B. B 中任选一点记为 $\boldsymbol{b}_{j_1\cdots j_m}$ 或 $\boldsymbol{b}$. 令

$$\psi_{ni}(\boldsymbol{\beta}) = \mathrm{sgn}(e_{ni} - \boldsymbol{x}'_{ni}\boldsymbol{\beta}) - \mathrm{sgn}(e_{ni}), \tag{2.12}$$

$$\lambda_{ni}(\boldsymbol{\beta}) = 2(F(\boldsymbol{x}'_{ni}\boldsymbol{\beta}) - 1/2) = -\mathrm{E}\psi_{ni}(\boldsymbol{\beta}), \tag{2.13}$$

$$t_{ni}(\boldsymbol{\beta}) = \psi_{ni}(\boldsymbol{\beta}) + \lambda_{ni}(\boldsymbol{\beta}). \tag{2.14}$$

由(2.7)式，当 n 大时，对 $\boldsymbol{\beta} \in D$ 及 $i = 1, \cdots, n$ 有

$$|\boldsymbol{x}'_{ni}\boldsymbol{\beta}| \leqslant \sqrt{p}d_n |\boldsymbol{\beta}| < (2\mu_n)^{-1}, \tag{2.15}$$

又当 n 大时，对 $\boldsymbol{b} \in B$, $\boldsymbol{b}^* \in B$ $(B \in G_m)$ 有

$$|\boldsymbol{x}'_{ni}(\boldsymbol{b} - \boldsymbol{b}^*)| \leqslant \sqrt{p}d_n |\boldsymbol{b} - \boldsymbol{b}^*| \leqslant (2\mu_n^{2m})^{-1}, \ 1 \leqslant i \leqslant n. \tag{2.16}$$

由(2.15)式及加在 e_{ni} 的分布 F 上的条件可知：可找到常数 $C > 1$，使对 $\boldsymbol{\beta} \in D$ 及 $i = 1, \cdots, n$ 有

$$2\{F(\boldsymbol{x}'_{ni}\boldsymbol{\beta} + \mu_n^{-2m}) - F(\boldsymbol{x}'_{ni}\boldsymbol{\beta} - \mu_n^{-2m})\} \leqslant C\mu_n^{-2m}, \tag{2.17}$$

$$|\lambda_{ni}(\boldsymbol{\beta})| \leqslant C\mu_n^{-1}, \tag{2.18}$$

$$P(|e_{ni}| \leqslant \sqrt{p}d_n\mu_n) \leqslant Cd_n\mu_n. \tag{2.19}$$

设 $\{a_{ni}, 1 \leqslant i \leqslant n, n \geqslant 1\}$ 为满足如下条件的实数三角阵列：

$$\sum_{i=1}^{n} a_{ni}^2 = 1, \ |a_{ni}| \leqslant \mu_n^{-m/2} \text{ for } i \geqslant r_m + 1,$$
$$n = 1, 2, \cdots, \quad m = 1, 2, \cdots, M+1. \tag{2.20}$$

定义以下诸量：

$$Q_m = \sum_{B \in G_m} P\left\{\sup_{\boldsymbol{\beta} \in B} \left|\sum_{i=r_m+1}^{n} a_{ni}(t_{ni}(\boldsymbol{\beta}) - t_{ni}(\boldsymbol{b}))\right| \geqslant \varepsilon 3^{-m}\right\}, \ 1 \leqslant m \leqslant M+1, \tag{2.21}$$

$$U_m = \sum_{B \in G_m} U(m, B), \ 1 \leqslant m \leqslant M \text{，其中}$$

$$U(m, B) = P\left\{\sup_{\boldsymbol{\beta} \in B} \left|\sum_{i=r_m+1}^{r_{m+1}} a_{ni}(t_{ni}(\boldsymbol{\beta}) - t_{ni}(\boldsymbol{b}))\right| \geqslant \varepsilon 3^{-(m+1)}\right\}, \tag{2.22}$$

$$V_m = \sum_{j_1, \cdots, j_{m+1}} P\left\{\left|\sum_{i=r_{m+1}+1}^{n} a_{ni}(t_{ni}(\boldsymbol{b}_{j_1\cdots j_m}) - t_{ni}(\boldsymbol{b}_{j_1\cdots j_m j_{m+1}}))\right| \geqslant \varepsilon 3^{-(m+1)}\right\}$$
$$\equiv \sum_{j_1, \cdots, j_{m+1}} V(m, j_1, \cdots, j_{m+1}), \ 1 \leqslant m \leqslant M, \ V_M = 0. \tag{2.23}$$

在 V_m 的定义中，$j_1, \cdots, j_{m+1}$ 的求和范围为 $1 \leqslant j_i \leqslant J_i$, $i = 1, \cdots, m+1$．由这些定义易见

$$Q_m \leqslant U_m + V_m + Q_{m+1}, \ m = 1, \cdots, M. \tag{2.24}$$

所以，注意到 $Q_{M+1} = 0$，得

$$Q_1 \leqslant \sum_{m=1}^{M}(U_m + V_m). \tag{2.25}$$

有了这些准备，现在可以证明下面的

引理 3 设定理 1′的条件都满足，$\{a_{ni}\}$ 满足(2.20)式，$0 < \varepsilon < 1$．则当 n 大时有

$$U_m + V_m \leqslant 4\mu_n^{2pm} \exp\{-48^{-1}\varepsilon(9^{-1}\mu_n)^{m/2}\},\ m = 1,\ \cdots,\ M, \tag{2.26}$$

这里 C 是在(2.17)～(2.19)式中出现的常数.

证 定义

$$\eta_{ni} \equiv \eta_{ni}(B) = \begin{cases} 2,\ 当\ |e_{ni} - \boldsymbol{x}'_{ni}\boldsymbol{b}| < \mu_n^{-2m}, \\ 0,\ 否则, \end{cases} \tag{2.27}$$

这里 $B \in G_m$ 而 $\boldsymbol{b}$ 为 B 中选定之一点. 由(2.14),(2.16)和(2.17)式有

$$\sup_{\boldsymbol{\beta}\in B}\left|\sum_{i=r_m+1}^{r_{m+1}} a_{ni}(t_{ni}(\boldsymbol{\beta}) - t_{ni}(\boldsymbol{b}))\right| \leqslant \sum_{i=r_m+1}^{r_{m+1}} |a_{ni}|(\eta_{ni} - \mathrm{E}\eta_{ni}) + 2\sum_{i=r_m+1}^{r_{m+1}} |a_{ni}|\mathrm{E}\eta_{ni}. \tag{2.28}$$

由(2.16),(2.17)和(2.27)式,有

$$\mathrm{E}\eta_{ni} \leqslant C\mu_n^{-2m},\ i = r_m + 1,\ \cdots,\ r_{m+1}. \tag{2.29}$$

由(2.20)式知,当 n 大时有

$$\begin{aligned} 2\sum_{i=r_m+1}^{r_{m+1}} |a_{ni}|\mathrm{E}\eta_{ni} &\leqslant \left(2\sum_{i=r_m+1}^{r_{m+1}} (\mathrm{E}\eta_{ni})^2\right)^{1/2} \leqslant 2(\mu_n^{m+1}C^2\mu_n^{-4m})^{1/2} \\ &\leqslant 2C\mu_n^{-m} < \varepsilon 2^{-1}3^{-(m+1)}. \end{aligned} \tag{2.30}$$

由(2.1),(2.28)和(2.30)式以及引理1,得到

$$\begin{aligned} U(m,\ B) &\leqslant P\left(\left|\sum_{i=r_m+1}^{r_{m+1}} |a_{ni}|(\eta_{ni} - \mathrm{E}\eta_{ni})\right| \geqslant \varepsilon 2^{-1}3^{-(m+1)}\right) \\ &\leqslant 2\exp\{-\varepsilon^2 2^{-2}3^{-(2m+2)}/[2(\max_{r_m<i\leqslant r_{m+1}} \mathrm{E}\eta_{ni}^2 + \varepsilon 3^{-(m+1)}\max_{i>r_m}|a_{ni}|)]\}. \end{aligned} \tag{2.31}$$

$U(m,\ B)$ 的定义见(2.22)式. 由(2.17),(2.20)和(2.29)式知,当 n 大时有

$$\max_{i>r_m}|a_{ni}| \leqslant \mu_n^{-m/2},\quad \max_{r_m<i\leqslant r_{m+1}} \mathrm{E}\eta_{ni}^2 \leqslant 2C\mu_n^{-2m} \leqslant \mu_n^{-m/2}\varepsilon 3^{-(m+1)},$$

故有

$$U(m,\ B) \leqslant 2\exp(-\varepsilon 3^{-(m+1)}\mu_n^{m/2}/16), \tag{2.32}$$

此式蕴含

$$U_m \leqslant 2\mu_n^{2pm}\exp(-\varepsilon 3^{-(m+1)}\mu_n^{m/2}/16). \tag{2.33}$$

由(2.20)式有

$$4\max_{i>r_{m+1}}|a_{ni}|\varepsilon 3^{-(m+1)} \leqslant 4\varepsilon 3^{-(m+1)}\mu_n^{-(m+1)/2}. \tag{2.34}$$

记 $g_{ni} = t_{ni}(\boldsymbol{b}_{j_1\cdots j_m}) - t_{ni}(\boldsymbol{b}_{j_1\cdots j_{m+1}})$,则由(2.16)和(2.17)式得

$$\begin{aligned} \mathrm{E}g_{ni}^2 &\leqslant 4|F(\boldsymbol{x}'_{ni}\boldsymbol{b}_{j_1\cdots j_m}) - F(\boldsymbol{x}'_{ni}\boldsymbol{b}_{j_1\cdots j_{m+1}})| \\ &\leqslant 2C\mu_n^{-2m} \leqslant 4\varepsilon 3^{-(m+1)}\mu_n^{-(m+1)/2}. \end{aligned} \tag{2.35}$$

由(2.23),(2.34)和(2.35)式及 Bennett 不等式(2.1),有

$$V(m, j_1, \cdots, j_m) \leqslant 2\exp\{-\varepsilon^2 3^{-(2m+2)}/[2(\max_{i>r_{m+1}} \mathrm{E}g_{ni}^2 + 4\max_{i>r_{m+1}} |a_{ni}| \varepsilon 3^{-(m+1)})]\}$$
$$\leqslant 2\exp\{-\varepsilon(\mu_n/9)^{(m+1)/2}/16\},$$

此式蕴含

$$V_m \leqslant 2\mu_n^{2p(m+1)}\exp(-\varepsilon(\mu_n/9)^{(m+1)/2}/16) \leqslant 2\mu_n^{2pm}\exp(-\varepsilon(\mu_n/9)^{m/2}/16). \qquad (2.36)$$

最后,当 n 大时,(2.26)式由(2.33)和(2.36)式得出.引理 3 证毕.

引理 4 在定理 1′的条件下并假定(2.2)式,则对任何满足 $v_n \to \infty$ 的常数列 $\{v_n\}$ 有

$$\lim_{n\to\infty} P(|\hat{\boldsymbol{\beta}}_n| > v_n) = 0. \qquad (2.37)$$

换句话说,$\hat{\boldsymbol{\beta}}_n = O_p(1)$.

证 不失普遍性,可设 $d_n v_n^2 \leqslant 1$.令

$$\widetilde{D} \equiv \widetilde{D}_n \equiv \{\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p)': |\beta_i| \leqslant v_n, i = 1, \cdots, p\},$$

$$\boldsymbol{\Phi}_{ni}(\boldsymbol{\beta}) = |e_{ni}| - |e_{ni} - \boldsymbol{x}'_{ni}\boldsymbol{\beta}|, \ \Lambda_{ni}(\boldsymbol{\beta}) = \mathrm{E}(\Phi_{ni}(\boldsymbol{\beta})),$$

$$\boldsymbol{R}_{ni}(\beta) = \boldsymbol{\Phi}_{ni}(\boldsymbol{\beta}) - \Lambda_{ni}(\boldsymbol{\beta}), \ i = 1, \cdots, n; \ n = 1, 2, \cdots.$$

首先要验证

$$v_n^{-2} \sup\left\{\left|\sum_{i=1}^n R_{ni}(\boldsymbol{\beta})\right| : \boldsymbol{\beta} \in \widetilde{D}\right\} \xrightarrow{P} 0. \qquad (2.38)$$

为此,我们对 $\widetilde{D}$ 进行多次剖分,其方式与前面对由(2.9)式定义的 D 所作的剖分相同,只是那里的 μ_n 现改为 v_n,而且那里的 M 也改为满足下式

$$\sqrt{np}\, v_n^{-2M} < 2^{-1}\varepsilon 3^{-(M+1)} v_n^2 \qquad (2.39)$$

的一个整数 M, $\varepsilon > 0$ 为给定常数.这种 M 的存在由 $v_n \to \infty$ 看出.把 $\widetilde{D}$ 经第 m 轮剖分后所得的全部超长方体之集记为 $\widetilde{G}_m$,$\widetilde{G}_m$ 中任一元将记为 $B_{j_1\cdots j_m}$ 或 B,而从 B 中选定一点记为 $\boldsymbol{b}_{j_1\cdots j_m}$ 或 $\boldsymbol{b}$.令

$$\widetilde{V}_0 = \sum_{j_1} P\left(\left|\sum_{i=1}^n R_{ni}(\boldsymbol{b}_{j_1})\right| \geqslant \varepsilon v_n^2/3\right),$$

$$\widetilde{V}_m = \sum_{j_1\cdots j_{m+1}} P\left(\left|\sum_{i=1}^n (R_{ni}(\boldsymbol{b}_{j_1\cdots j_m}) - R_{ni}(\boldsymbol{b}_{j_1\cdots j_{m+1}}))\right| \geqslant \varepsilon v_n^2/3^{m+1}\right), \ m = 1, \cdots, M,$$

$$\widetilde{Q}_m = \sum_{j_1\cdots j_m} P\left(\sup_{\boldsymbol{\beta}\in B_{j_1\cdots j_m}} \left|\sum_{i=1}^n (R_{ni}(\boldsymbol{\beta}) - R_{ni}(\boldsymbol{b}_{j_1\cdots j_m}))\right| \geqslant \varepsilon v_n^2/3^m\right), \ m = 1, \cdots, M+1.$$

由(2.39)式知,对任何 $B \in \widetilde{G}_{M+1}$ 及 $\boldsymbol{\beta} \in B$,有

$$\left|\sum_{i=1}^n (\boldsymbol{\Phi}_{ni}(\boldsymbol{\beta}) - \boldsymbol{\Phi}_{ni}(\boldsymbol{b}))\right| \leqslant \sum_{i=1}^n |\boldsymbol{x}'_{ni}(\boldsymbol{\beta} - \boldsymbol{b})| \leqslant \left(n\sum_{i=1}^n (\boldsymbol{x}'_{ni}(\boldsymbol{\beta} - \boldsymbol{b}))^2\right)^{1/2}$$
$$\leqslant \sqrt{n}\,\|\boldsymbol{\beta} - \boldsymbol{b}\| \leqslant \sqrt{np}\,|\boldsymbol{\beta} - \boldsymbol{b}|$$
$$\leqslant \sqrt{np}\, v_n^{-2M} < 2^{-1}\varepsilon 3^{-(M+1)} v_n^2,$$

故

$$\sup_{\boldsymbol{\beta}\in \tilde{B}}\left|\sum_{i=1}^{n}(R_{ni}(\boldsymbol{\beta})-R_{ni}(\boldsymbol{b}))\right|<\varepsilon 3^{-(M+1)}v_n^2,$$

这蕴含

$$\widetilde{Q}_{M+1}=0. \tag{2.40}$$

易见

$$\widetilde{Q}_m\leqslant \widetilde{V}_m+\widetilde{Q}_{m+1},\ m=1,\cdots,M, \tag{2.41}$$

$$P\left(v_n^{-2}\sup_{\boldsymbol{\beta}\in \tilde{D}}\left|\sum_{i=1}^{n}R_{ni}(\boldsymbol{\beta})\right|\geqslant\varepsilon\right)\leqslant \widetilde{V}_0+\widetilde{Q}_1. \tag{2.42}$$

由(2.40)～(2.42)式,得

$$P\left(v_n^{-2}\sup_{\boldsymbol{\beta}\in \tilde{D}}\left|\sum_{i=1}^{n}R_{ni}(\boldsymbol{\beta})\right|\geqslant\varepsilon\right)\leqslant \sum_{m=0}^{M}\widetilde{V}_m. \tag{2.43}$$

因为

$$\begin{aligned}
&|\Phi_{ni}(\boldsymbol{b}_{j_1})|\leqslant|\boldsymbol{x}'_{ni}\boldsymbol{b}_{j_1}|\leqslant d_nv_n\leqslant v_n^{-1}(\text{用到 } d_nv_n^2\leqslant 1),\\
&\sum_{i=1}^{n}\mathrm{Var}(\Phi_{ni}(\boldsymbol{b}_{j_1}))\leqslant\sum_{i=1}^{n}(\boldsymbol{x}'_{ni}\boldsymbol{b}_{j_1})^2=\|\boldsymbol{b}_{j_1}\|^2\leqslant pv_n^2,\\
&|\Phi_{ni}(\boldsymbol{b}_{j_1\cdots j_m})-\Phi_{ni}(\boldsymbol{b}_{j_1\cdots j_{m+1}})|\leqslant|\boldsymbol{x}'_{ni}(\boldsymbol{b}_{j_1\cdots j_m}-\boldsymbol{b}_{j_1\cdots j_{m+1}})|\\
&\qquad\leqslant\|\boldsymbol{x}_{ni}\|\cdot\|\boldsymbol{b}_{j_1\cdots j_m}-\boldsymbol{b}_{j_1\cdots j_{m+1}}\|\\
&\qquad\leqslant d_npv_n^{-2m+2}\leqslant pv_n^{-2m},
\end{aligned}$$

以及

$$\begin{aligned}
\sum_{i=1}^{n}\mathrm{Var}(\Phi_{ni}(\boldsymbol{b}_{j_1\cdots j_m})-\Phi_{ni}(\boldsymbol{b}_{j_1\cdots j_{m+1}}))&\leqslant\sum_{i=1}^{n}(\boldsymbol{x}'_{ni}(\boldsymbol{b}_{j_1\cdots j_m}-\boldsymbol{b}_{j_1\cdots j_{m+1}}))^2\\
&=\|\boldsymbol{b}_{j_1\cdots j_m}-\boldsymbol{b}_{j_1\cdots j_{m+1}}\|^2\\
&\leqslant p|\boldsymbol{b}_{j_1\cdots j_m}-\boldsymbol{b}_{j_1\cdots j_{m+1}}|^2\\
&\leqslant pv_n^{-4m+4}.
\end{aligned}$$

利用引理 1,得

$$\widetilde{V}_0\leqslant 2v_n^{2p}\exp(-3^{-2}\varepsilon^2v_n^4/(2pv_n^2+2\varepsilon v_n))\leqslant 2v_n^{2p}\exp(-cv_n^2), \tag{2.44}$$

$$\begin{aligned}
\widetilde{V}_m&\leqslant 2v_n^{2p(m+1)}\exp\{-(\varepsilon 3^{-m-1}v_n^2)^2/(2pv_n^{-4m+4}+4\varepsilon 3^{-m-1}pv_n^{-2m+2})\}\\
&\leqslant 2v_n^{2p(m+1)}\exp(-c3^{-(m+1)}v_n^{2m+2}),\ m\geqslant 1.
\end{aligned} \tag{2.45}$$

此处 $c>0$ 为一与 n 和 m 无关的常数.由(2.43)～(2.45)式得

$$P\left(v_n^{-2}\sup_{\boldsymbol{\beta}\in \tilde{D}}\left|\sum_{i=1}^{n}R_{ni}(\boldsymbol{\beta})\right|\geqslant\varepsilon\right)\leqslant 2\sum_{m=0}^{\infty}v_n^{2p(m+1)}\exp(-c3^{-m-1}v_n^{2m+2}). \tag{2.46}$$

因为(2.46)左右边当 $n\to\infty$ 时收敛于零,得到(2.38)式.

现注意

$$\sup_{\beta \in \widetilde{D}} \max_{1 \leqslant i \leqslant n} | x_{ni}'\beta | \leqslant d_n v_n \leqslant v_n^{-1} \to 0, \text{ 当 } n \to \infty. \tag{2.47}$$

因此，考虑到定理 1 的条件 1，得

$$\begin{aligned} \mathrm{E}\Phi_{ni}(\beta) &= \begin{cases} \int_0^{x_{ni}'\beta} (2u - 2x_{ni}'\beta) f(u) \mathrm{d}u, & \text{当 } x_{ni}'\beta \geqslant 0 \\ \int_{x_{ni}'\beta}^0 (2x_{ni}'\beta - 2u) f(u) \mathrm{d}u, & \text{当 } x_{ni}'\beta < 0 \end{cases} \\ &= - f(0)(x_{ni}'\beta)^2 (1 + o(1)). \end{aligned} \tag{2.48}$$

这里 $\lim\limits_{n\to\infty} o(1) = 0$ 一致地对 $\beta \in \widetilde{D}$ 和 $1 \leqslant i \leqslant n$ 成立，这一点由(2.47)式得出. 由(2.38)和(2.48)式，得知当 $n \to \infty$ 时

$$u_n^{-2} \sup_{\beta \in \widetilde{D}} \left| \sum_{i=1}^n (| e_{ni} | - | e_{ni} - x_{ni}\beta |) + f(0) \| \beta \|^2 \right| \xrightarrow{P} 0,$$

此式蕴含

$$v_n^{-2} \left(\sum_{i=1}^n | e_{ni} | - \inf_{|\beta| = v_n} \sum_{i=1}^n | e_{ni} - x_{ni}'\beta | \right) \leqslant - f(0)(1 + o_p(1)), \tag{2.49}$$

此处 $o_p(1) \xrightarrow{P} 0$ 当 $n \to \infty$. 因为 $\sum\limits_{i=1}^n | e_{ni} - x_{ni}'\beta |$ 为 β 的凸函数，由(2.49)式可知

$$v_n^{-2} \left(\sum_{i=1}^n | e_{ni} | - \inf_{|\beta| \geqslant v_n} \sum_{i=1}^n | e_{ni} - x_{ni}'\beta | \right) \leqslant - f(0)(1 + o_p(1)). \tag{2.50}$$

因为 $f(0) > 0$，(2.50)式蕴含(2.37)式. 引理 4 证毕.

注 让我们暂回到模型(1.1)而考虑由(1.2)式定义的 ML_1N 估计，暂记为 $\bar{\beta}_n$，$\hat{\beta}_n$ 与 $\bar{\beta}_n$ 有关系 $S_n^{1/2}\bar{\beta}_n = \hat{\beta}_n$，这里 $\hat{\beta}_n$ 就是(2.37)式中那个 $\hat{\beta}_n$. 由(2.37)及(1.8)式即知 $\bar{\beta}_n \xrightarrow{P} \mathbf{0}$，即 $\bar{\beta}_n$ 为 β_0 的弱相合估计(注意此处已假定了(2.2)式，因而 $\beta_0 = \mathbf{0}$). 前面我们曾把这一结果作为定理 1 的推论(系 3). 现在看出，事实上这是为证明定理 1 必经的一个步骤.

3 定理的证明

主要是证明定理 1，略加修改可证明定理 2. 定理 3 的证明也可在定理 1 方法的基础上略加修改得到，此处将从略.

定理 1 的证明 前面已指出，只需证定理 1′. 为此先验证下面的事实：

$$\sup\left\{ \left\| \sum_{i=1}^n t_{ni}(\beta) x_{ni} \right\| : \beta \in D \right\} \xrightarrow{P} 0, \text{ 当 } n \to \infty, \tag{3.1}$$

D 和 $t_{ni}(\beta)$ 曾在(2.9)和(2.12) ~ (2.14)式中定义. 由(1.18)式只需证当 $n \to \infty$ 时

$$Q_0 \equiv P\left(\sup\left\{ \left| \sum_{i=1}^n a_{ni} t_{ni}(\beta) \right| : \beta \in D \right\} \geqslant \varepsilon \right) \to 0, \tag{3.2}$$

这里 $\{a_{ni}\}$ 为满足 $\sum\limits_{i=1}^n a_{ni}^2 = 1$ 的实数三角阵列，$\varepsilon \in (0, 1)$ 给定.

不失普遍性,可设 $|a_{n1}|\geqslant\cdots\geqslant|a_{nn}|$. 按(2.7)和(2.8)式选 μ_n 和 r_m,则(2.20)式显然成立. 对 $1\leqslant m\leqslant M$,按(2.21)～(2.23)式定义 U_m, V_m 和 Q_m,以及

$$U_0=P\Big\{\sup_{\boldsymbol{\beta}\in D}\Big|\sum_{i=1}^{r_1}a_{ni}t_{ni}(\boldsymbol{\beta})\Big|\geqslant\varepsilon/3\Big\},\tag{3.3}$$

$$V_0=\sum_{B\in G_1}P\Big\{\Big|\sum_{i=r_1+1}^{n}a_{ni}t_{ni}(\boldsymbol{b})\Big|\geqslant\varepsilon/3\Big\},\tag{3.4}$$

其中 $\boldsymbol{b}\in B$. 易见当 Q_0, U_0 和 V_0 如此定义时,(2.24)式对 $m=0$ 也成立,故

$$Q_0\leqslant U_0+V_0+Q_1\leqslant\sum_{m=0}^{M}(U_m+V_m).\tag{3.5}$$

此处 M 按(2.8)式定义,现往证

$$U_0+V_0\to 0,\text{当 } n\to\infty.\tag{3.6}$$

由(2.18)式知,对 $\boldsymbol{\beta}\in D$ 当 $n\to\infty$ 时有

$$\Big|\sum_{i=1}^{n}a_{ni}\lambda_{ni}(\boldsymbol{\beta})\Big|\leqslant\Big(\sum_{i=1}^{n}\lambda_{ni}^2(\boldsymbol{\beta})\Big)^{1/2}\leqslant\{\mu_n(c/\mu_n)^2\}^{1/2}\to 0.\tag{3.7}$$

其次,当 $\boldsymbol{\beta}\in D$ 且 $|e_{ni}|>\sqrt{p}d_n\mu_n$ 时,有 $|e_{ni}|>|\boldsymbol{x}_{ni}'\boldsymbol{\beta}|$,因而按(2.12)式有 $\psi_{ni}(\boldsymbol{\beta})=0$. 由(2.7),(2.19)和(3.7)式,得

$$U_0\leqslant P\Big\{\sup_{\boldsymbol{\beta}\in D}\Big|\sum_{i=1}^{r_1}a_{ni}\psi_{ni}(\boldsymbol{\beta})\Big|\neq 0\Big\}\leqslant\sum_{i=1}^{r_1}P(|e_{ni}|\leqslant\sqrt{p}d_n\mu_n)\leqslant Cd_n\mu_n^2\to 0.\tag{3.8}$$

利用在证明(2.36)式时使用的方法,及(2.17)～(2.20)式,易证存在常数 $C_1>0$,使

$$V_0\leqslant 2\mu_n^{2p}\exp(-C_1\mu_n^{1/2})\to 0,\text{当 } n\to\infty.\tag{3.9}$$

现由(3.8)和(3.9)式得出(3.6)式. 其次,由引理3知当 n 大时有

$$\sum_{m=1}^{M}(U_m+V_m)\leqslant 4\sum_{m=1}^{\infty}\mu_n^{2pm}\exp(-48^{-1}\varepsilon(\mu_n/9)^{m/2}).\tag{3.10}$$

因为函数 $x^{3pm}\exp(-a(x/b)^{m/2})$, $x>0$,在 $x=b(6p/a)^{2/m}$ 处达到最大值,我们有

$$\begin{aligned}\mu_n^{2pm}\exp\{-48^{-1}\varepsilon(\mu_n/9)^{m/2}\}&=\mu_n^{-pm}\cdot\mu_n^{3pm}\exp\{-48^{-1}\varepsilon(\mu_n/9)^{m/2}\}\\&\leqslant\mu_n^{-pm}9^{3pm}(288p/\varepsilon)^{6p}\mathrm{e}^{-6p}\\&\leqslant(\mu_n/729)^{pm}(288p/\varepsilon)^{6p}.\end{aligned}\tag{3.11}$$

因此,注意到 $\mu_n\to\infty$,得知当 $n\to\infty$ 时,

$$\sum_{m=1}^{M}(U_m+V_m)\leqslant(288p\varepsilon^{-1})^{6p}\sum_{m=1}^{\infty}(\mu_n/729)^{-pm}\to 0.\tag{3.12}$$

由(3.5),(3.6)和(3.12)式,即推出(3.2)式,这证明了(3.1)式.

由(3.1)式知当 $n\to\infty$ 时有

$$\sum_{i=1}^{n} t_{ni}(\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni} I(\hat{\boldsymbol{\beta}}_n \in D) \xrightarrow{P} \boldsymbol{0}, \tag{3.13}$$

当 n 大时，由(2.7)式得出 $\Delta/(\sqrt{p}dn) > \mu_n^2$（$\Delta$ 是在定理 1 条件 1 中出现的常数）. 故由引理 2 得

$$I\left(\left|\sum_{i=1}^{n}\operatorname{sgn}(e_{ni} - \boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni}\right| \geqslant (p+1)d_n\right) I(|\hat{\boldsymbol{\beta}}_n| < \mu_n) = 0, \text{ a. s.}$$

由此由(1.19)式得

$$\left|\sum_{i=1}^{n}\operatorname{sgn}(e_{ni} - \boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni}\right| I(|\hat{\boldsymbol{\beta}}_n| < \mu_n) \leqslant (p+1)d_n \to 0, \text{ a. s.} \tag{3.14}$$

由(2.12)～(2.14)式，(3.13)和(3.14)式，得知当 $n \to \infty$ 时，

$$\left(\sum_{i=1}^{n}\lambda_{ni}(\hat{\boldsymbol{\beta}}_n)\boldsymbol{x}_{ni} - \sum_{i=1}^{n}\operatorname{sgn}(e_{ni})\boldsymbol{x}_{ni}\right) I(|\hat{\boldsymbol{\beta}}_n| < \mu_n) \xrightarrow{P} \boldsymbol{0}. \tag{3.15}$$

因为 $\sup\{|\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n| : |\hat{\boldsymbol{\beta}}_n| < \mu_n,\ 1 \leqslant i \leqslant n\} \leqslant \mu_n d_n \leqslant \mu_n^2 d_n \to 0$，按定理 1 的条件 1 知 $\lambda_{ni}(\hat{\boldsymbol{\beta}}_n) = 2f(0)\boldsymbol{x}_{ni}'\hat{\boldsymbol{\beta}}_n(1 + o_p(1))$，此处 $o_p(1) \xrightarrow{P} 0$ 当 $n \to \infty$，一致地对 $i = 1, \cdots, n$. 由此和(1.18)及(3.15)式，得知当 $n \to \infty$ 时，

$$\left(2f(0)\hat{\boldsymbol{\beta}}_n - \sum_{i=1}^{n}\operatorname{sgn}(e_{ni})\boldsymbol{x}_{ni}\right) I(|\hat{\boldsymbol{\beta}}_n| < \mu_n) \xrightarrow{P} \boldsymbol{0}. \tag{3.16}$$

考虑到(1.18)和(1.19)式及加在 $\{e_{ni}\}$ 上的条件，按 Lindeberg 中心极限定理，得知当 $n \to \infty$ 时，

$$\sum_{i=1}^{n}\operatorname{sgn}(e_{ni})\boldsymbol{x}_{ni} \xrightarrow{\mathscr{L}} N(\boldsymbol{0}, \boldsymbol{I}_p). \tag{3.17}$$

由(3.16)和(3.17)式并用引理 4，且注意到(2.2)式，我们得到(1.20)式. 这完成了定理 1′的证明，因而也完成了定理 1 的证明.

定理 2 的证明 定义 $\boldsymbol{T}_n$ 如(1.12)式，以及

$$\boldsymbol{\beta}_{n0} = \boldsymbol{T}_n^{1/2}\boldsymbol{\beta}_0,\ \alpha_{n0} = \sqrt{n}(\alpha_0 + \overline{\boldsymbol{x}}_n'\boldsymbol{\beta}_0),\ \boldsymbol{\gamma}_{n0} = (\alpha_{n0}, \boldsymbol{\beta}_{n0}')',$$
$$\boldsymbol{x}_{ni} = \boldsymbol{T}_n^{-1/2}(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n),\ \boldsymbol{z}_{ni} = (1/\sqrt{n}, \boldsymbol{x}_{ni}')',\ i = 1, \cdots, n.$$

将模型(1.11)转化为

$$Y_i = \boldsymbol{z}_{ni}'\boldsymbol{\gamma}_{n0} + e_i,\ i = 1, \cdots, n.$$

由于 $\sum_{i=1}^{n}\boldsymbol{z}_{ni}\boldsymbol{z}_{ni}' = \boldsymbol{I}_{p+1}$，且(1.13)式保证了 $\lim_{n\to\infty}\max_{1\leqslant i\leqslant n}|\boldsymbol{z}_{ni}| = 0$，定理 1′可用. 得到

$$2f(0)(\hat{\boldsymbol{\gamma}}_{n0} - \boldsymbol{\gamma}_{n0}) \xrightarrow{\mathscr{L}} N(\boldsymbol{0}, \boldsymbol{I}_{p+1}),\ \text{当 } n \to \infty. \tag{3.18}$$

这里 $\hat{\boldsymbol{\gamma}}_{n0} = (\hat{\alpha}_{n0}, \hat{\boldsymbol{\beta}}_{n0}')'$，而 $\hat{\alpha}_{n0}$ 和 $\hat{\boldsymbol{\beta}}_{n0}$ 分别是 α_{n0} 和 $\boldsymbol{\beta}_{n0}$ 的 ML_1N 估计. 由(3.18)式得出(1.14)式，以及 $\hat{\alpha}_{n0}$ 和 $\hat{\boldsymbol{\beta}}_{n0}$ 渐近独立. 最后，(1.15)式易由已证部分得出. 定理证毕.

附录 I

在本附录中,除有明确定义者外,一切符号及公式编号均同文献[2].

考虑模型 $y_t = x_t\beta + u_t$, $t = 1, \cdots, T, \cdots$. 这里 β 为一维未知参数,x_t 为已知实数:$x_1 = 1/\sqrt{2}$, $x_2 = 1+\sqrt{2}/10$, $x_i = 1$ 当 $i \geqslant 3$, $u_1, u_2, \cdots$ 为 iid., $u_1 \sim N(0, 1)$. 首先验证:函数 $J(\beta) = \sum_{t=1}^{T} | y_t - x_t\beta |$ 有唯一最小值点. 用反证法. 若不然,则由 $J(\beta)$ 之凸性,知存在区间 $[a, b]$, $a < b$,使 J 在 $[a, b]$ 上处处为最小. 取 $r \in (a, b)$, $r \neq y_t/x_t$, $1 \leqslant t \leqslant T$. 则因 $J'(r) = 0$,有

$$-\frac{1}{\sqrt{2}}\operatorname{sgn}(y_1 - r/\sqrt{2}) - \left(1+\frac{\sqrt{2}}{10}\right)\operatorname{sgn}\left(y_2 - \left(1+\frac{\sqrt{2}}{10}\right)r\right) - \sum_{t=3}^{T}\operatorname{sgn}(y_t - r) = 0,$$

但这不可能,因上式左边前两项之和为无理数,而第三项为整数.

在此模型中 $H = \{1, 2, \cdots, T\}$. 由 $\{x_t\}$ 之选择,$Z_T(\delta, 1)$ 之分布不是格子点分布. 故按文献[2]的(3.9)式,应有

$$\lim_{T\to\infty} T^{1/2} P_r(Z_T(\delta, 1) \in C[0, 1]) \text{ 存在非零}, \tag{$*$}$$

但是

$$Z_T(\delta, 1) = \left(\sqrt{2}+\frac{1}{5}\right)\operatorname{sgn}\left(u_2 - T^{-1/2}\left(1+\frac{\sqrt{2}}{10}\right)\delta\right) + \sum_{t=3}^{T}\sqrt{2}\operatorname{sgn}(u_t - T^{-1/2}\delta),$$

所以,以概率 1 成立下述事实:当 T 为偶数时上式等于$\sqrt{2}$的奇数倍$\pm 1/\sqrt{5}$,而这种数总在区间 $[-1, 1]$ 之外,因此有 $P_r(Z_T(\delta, 1) \in C[0, 1]) = 0$,当 $T = 2, 4, 6, \cdots$,而($*$)式失效.

附录 II

在本附录中,除有明确定义者外,一切符号及公式编号均同文献[4].

1. 以 $A_1(\beta_0)$ 记(3.12) 式(是文献[4] 的(3.12) 式!) 中的 A_1 在 β_0 点所取之值. 往证当 $T \to \infty$ 时,$A_1(\beta_0) \xrightarrow{P} \infty$. 定义 $\xi_t = I(| y_t - \boldsymbol{x}_t'\beta_0 | \leqslant C_T^{-1})$. 由于 $C_T = \boldsymbol{T}^d$ 而 $1/3 < d < 1/2$,知 $C_T^{-1} < T^{-1/3}$,故有 $Z_t \geqslant \xi_t$, 以及

$$\begin{aligned} A_1(\beta_0) &= 2C_T^{-1}\sum_{i=1}^{T} Z_t \log(1+\exp(-C_T | y_t - \boldsymbol{x}_t'\beta_0 |)) \\ &\geqslant 2C_T^{-1}\sum_{t=1}^{T}\xi_t \log(1+\exp(-C_T | y_t - \boldsymbol{x}_t'\beta_0 |)) \\ &\geqslant 2C_T^{-1}\log(1+\mathrm{e}^{-1})\sum_{t=1}^{T}\xi_t \equiv \widetilde{A}. \end{aligned}$$

由于 $y_t - \boldsymbol{x}_t'\beta_0 = u_t$, $t = 1, 2, \cdots$ 为 iid.,其公共密度 f 满足 $f(0) > 0$ 且 f 在零处连续,知存在两个不依赖 t 的常数 h_1 和 h_2,使 $h_1 C_T^{-1} \leqslant P(\xi_t = 1) \leqslant h_2 C_T^{-1}$, $t = 1, 2, \cdots$. 由此得出

$$E(\widetilde{A}) \geqslant 2\log(1+e^{-1})h_1 TC_T^{-2} \to \infty,$$

又按 C_T 的定义,得

$$\begin{aligned}\mathrm{Var}(\widetilde{A}) &\leqslant (2C_T^{-1}\log(1+e^{-1}))^2\sum_{t=1}^{T}E\xi_t^2\\ &\leqslant (2C_T^{-1}\log(1+e^{-1}))^2 Th_2 D_T^{-1}\\ &= 4h_2\log^2(1+e^{-1})T/C_T^3 \to 0.\end{aligned}$$

由 $E(\widetilde{A}) \to \infty$ 及 $\mathrm{Var}(\widetilde{A}) \to 0$,知当 $T\to\infty$ 时 $\widetilde{A} \xrightarrow{P} \infty$. 因为 $A_1(\beta_0) \geqslant \widetilde{A}$,知当 $T\to\infty$ 时有 $A_1(\beta_0) \xrightarrow{P} \infty$.

2. 以 $B_1(\beta_0)$ 记(3.22)式定义的 B_1 在 β_0 点所取的值,往证当 $T\to\infty$ 时,有 $B_1(\beta_0) \xrightarrow{P} \infty$. 定义 ξ_t 如前. 因 $y_t - \boldsymbol{x}_t'\beta_0 = u_t$,有

$$\begin{aligned}B_1(\beta_0) &= T^{-1/2}\sum_{t=1}^{T}I(|u_t| < T^{1/3}C_T^{-1})\,|W_t - G_0(u_t)|\,|x_{it}|\\ &\geqslant (M(e+1))^{-1}T^{-1/2}\sum_{t=1}^{T}\xi_t |x_{it}|^2 \equiv \widetilde{B},\end{aligned}$$

此处按假定,有 $M \equiv \sup\{|x_t|: t\geqslant 1\} < \infty$. 我们得到

$$\begin{aligned}E(B) &\geqslant (M(e+1))^{-1}T^{-1/2}h_1 C_T^{-1}\sum_{t=1}^{T}x_{it}^2\\ &= (M(e+1))^{-1}h_1 T^{1/2}C_T^{-1}T^{-1}\sum_{t=1}^{T}\boldsymbol{x}_{it}^2 \to \infty, \text{当 } T\to\infty.\end{aligned}$$

此系依假定,当 $T\to\infty$ 时,$\sum_{t=1}^{T}x_{it}^2/T$ 趋于一个正极限,及 $C_T = T^d$, $1/3 < d < 1/2$. 又当 $T\to\infty$ 时

$$\begin{aligned}\mathrm{Var}(\widetilde{B}) &\leqslant M^{-2}(e+1)^{-2}T^{-1}\sum_{t=1}^{T}|x_{it}|^4 E\xi_t^2\\ &\leqslant M^2(e+1)^{-2}T^{-1}\sum_{t=1}^{T}E\xi_t^2\\ &\leqslant M^2(e+1)^{-2}T^{-1}Th_2C_T^{-1} \to 0.\end{aligned}$$

与前面一样,由此得出 $\widetilde{B} \xrightarrow{P} \infty$ 当 $T\to\infty$. 因为 $B_1(\beta_0) \geqslant \widetilde{B}$,知 $B_1(\beta_0) \xrightarrow{P} \infty$.

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16

Asymptotic Theory of Least Distances Estimate in Multivariate Linear Models

Abstract We consider the multivariate linear model

$$Y_i = X_i'\beta_0 + \varepsilon_i,\ i = 1, \cdots, n$$

where Y_i is a p-vector random variable, X_i is a $q \times p$ matrix, β_0 is an unknown q-vector parameter and $\{\varepsilon_i\}$ is a sequence of iid. p-vector random variable with median vector zero. The estimate $\hat{\beta}_n$ of β_0 such that

$$\min_{\beta} \sum_{i=1}^{n} \| Y_i - X_i'\beta \| = \sum_{i=1}^{n} \| Y_i - X_i'\hat{\beta}_n \|$$

is called the least distances (LD) estimator. It may be recalled that the least squares (LS) estimator is obtained by minimizing the sum of norm squares.

In this paper, it is shown that the LD estimator is unique, consistent and has an asymptotic q-variate normal distribution with mean β_0 and covariance matrix V which depends on the distribution of the error vectors $\{\varepsilon_i\}$. A consistent estimator of V is proposed which together with $\hat{\beta}_n$ enables an asymptotic inference on β_0. In particular, tests of linear hypotheses on β_0 analogous to those of analysis of variance in the Gauss-Markov linear model are developed. Explicit expressions are obtained in some cases for the asymptotic relative efficiency of the LD compared to the LS estimator.

1 Introduction and Summary

Consider the multivariate linear model

$$Y_i = X_i'\beta_0 + \varepsilon_i,\ i = 1, \cdots, n, \cdots \tag{1.1}$$

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where Y_i is a p-vector random variable, X_1, X_2, $\cdots$ are $q\times p$ known matrices, β_0 is an unknown q-vector parameter and $\{\varepsilon_i\}$ is a sequence of iid. p-vector random variables with median vector zero. [The median vector is here defined as a vector a_0 which minimizes $\mathrm{E}(\|\varepsilon-a\|-\|\varepsilon\|)$ where $\|\cdot\|$ denotes the Euclidean norm. Such a_0 always exists and is unique when ε is not concentrated on any straight line. See the Appendix and also Milasevic *et al.* (1987).] We note that the model (1.1) is more general than the usual MANOVA model.

Define the function

$$Q(\beta)\equiv Q(Y_1,\cdots,Y_n;\beta)\equiv\sum_{i=1}^{n}\|Y_i-X_i'\beta\|.$$

We propose to estimate β_0 by the statistic $\hat{\beta}_n=\hat{\beta}_n(Y_1,\cdots,Y_n)$ which minimizes $Q(\beta)$, i.e., $Q(\hat{\beta}_n)=\min\{Q(\beta):\beta\in\mathrm{R}^q\}$. $\hat{\beta}_n$ will be called the Least Distances (LD) estimate of β_0. In the special case $p=1$, $\hat{\beta}_n$ is the much studied LAD (least absolute deviations) estimate of β_0, and when $X_i=I_p$ for all i, $\hat{\beta}_n$ is the spatial median defined by Haldane (1948) and studied by Brown (1983) and Gower (1974) among others. For a review of LAD estimation and other definitions of the median in the multivariate case the reader is referred to Roussecuw and Leroy (1987), Rao (1988) and Liu (1989).

It is trivially true that there always exists Borel-measurable $\hat{\beta}_n$ minimizing $Q(\beta)$. Under certain conditions on $\{X_i\}$ it can be shown that $\hat{\beta}_n$ is unique with probability one (see the Appendix), but the asymptotic theory developed in this paper is irrelevant to this uniqueness. Some elementary properties of $\hat{\beta}_n$ can be easily verified:

1° If $\mathrm{E}\|\varepsilon_1^{\delta}\|<\infty$ for some $\delta>0$, then $\mathrm{E}\|\hat{\beta}_n\|<\infty$, for n large.

2° If the distribution of ε_1 is symmetric about 0, then the same is true for $\hat{\beta}_n$. Hence in this case $\hat{\beta}_n$ will be an unbiased estimate of β_0, if $\mathrm{E}\|\hat{\beta}_n\|<\infty$. Of course, $\hat{\beta}_n$ is median-unbiased regardless of whether $\mathrm{E}\|\hat{\beta}_n\|<\infty$ or not.

The paper is organized as follows: Section 2 proves the asymptotic normality of the LD estimate $\hat{\beta}_n$. The covariance matrix V of the limiting distribution depends on the error distribution F, which is usually unknown. In Section 3 we propose a method for estimating V and prove that the estimate is consistent. In this way we obtain a form of asymptotic distribution of $\hat{\beta}_n$, which is directly applicable to large-sample inferences on β_0. Section 4 considers the testing of a general linear hypothesis in the context of LD framework. A general limit theorem for the test-statistic under the null hypothesis is derived. Section 5 gives some remarks concerning the Asymptotic Relative Efficiency (ARE) of $\hat{\beta}_n$ compared to $\hat{\beta}_n$, the least squares (LS) estimate of β_0, in the special case when $X_i=I$ for all i.

With the asymptotic theory of the LDE's developed in this paper, questions of robustness compared to other estimators of β_0 can be studied in greater detail than what is done here. We hope to do this in a future communication.

2 Asymptotic Normality of $\hat{\boldsymbol{\beta}}_n$

We shall make the following assumptions on the model (1.1).

(a) The random errors, ε_1, ε_2, $\cdots$ are iid. with a common distribution F. F has a bounded density on the set $\{u: \| u \| < \delta\}$ for some $\delta > 0$ and $P(c'\varepsilon_1 = 0) < 1$, for every $c \neq 0$.

(b) $\int \frac{u}{\| u \|} \mathrm{d}F(u) = 0$.

(c) There exists an integer n_0 such that the matrix $(X_1 : \cdots : X_{n_0})$ has rank q. Define the matrices

$$A = \int \frac{1}{\| u \|}\Big(I_p - \frac{uu'}{\| u \|^2}\Big)\mathrm{d}F(u), \tag{2.1}$$

$$B = \int \frac{uu'}{\| u \|^2}\mathrm{d}F(u). \tag{2.2}$$

From Condition (a) it is easy to see that A and B exist and are positive definite when $p \geqslant 2$. Hence, by Condition (c), the matrices

$$S_n = \sum_{i=1}^{n} X_i B X_i', \qquad T_n = \sum_{i=1}^{n} X_i A X_i', \tag{2.3}$$

are positive definite when $n \geqslant n_0$. In the sequel, we shall always assume that $n \geqslant n_0$. Define

$$d_n = \max_{1 \leqslant i \leqslant n} \| S_n^{-1/2} X_i \|. \tag{2.4}$$

Now we can introduce the following condition:

(d) $\lim\limits_{n\to\infty} d_n = 0$.

Theorem 1 If $p \geqslant 2$ and the conditions (a)—(d) are met, then we have as $n \to \infty$

$$S_n^{-1/2} T_n (\hat{\beta}_n - \beta_0) \xrightarrow{\mathcal{L}} N(0, I_q). \tag{2.5}$$

We make some remarks before proving the theorem.

1. Since $B > 0$, Condition (d) is equivalent to

$$\max_{1 \leqslant i \leqslant n} \Big\| \Big(\sum_{j=1}^{n} X_j X_j'\Big)^{-1/2} X_i \Big\| \to 0.$$

Hence, this condition involves only $\{X_i\}$. It is exactly the same as that required in proving the asymptotic normality of the LS estimate $\tilde{\beta}_n$.

2. It is easily seen that if ε is a random vector with distribution F, and c is a constant vector such that $P(\varepsilon = c) = 0$, then

$$c \text{ is a median-vector of } \varepsilon \Leftrightarrow \int \frac{u - c}{\| u - c \|} \mathrm{d}F(u) = 0.$$

Therefore, under Condition (a), Condition (b) is the same as that 0 is the only median vector of ε_1.

3. The assumption of F (in (a)) is made to ensure the existence and positive definiteness of A and B, and further that 0 is the unique median-vector of ε_1. This assumption can be replaced by a slightly weaker one: $P(c'\varepsilon_1 = 0) < 1$ when $c \neq 0$ and

$$\sup\left\{\int \frac{1}{\| u-c \|} \mathrm{d}F(u): \| c \| < r\right\} < \infty \quad \text{for some} \quad r > 0.$$

4. Note the restriction $p \geqslant 2$ in the theorem. In the univariate case ($p=1$), the matrices A and B are not defined and the asymptotic variance of $\hat{\beta}_n$ depends on the distribution F only through its derivative at 0, viz., $f(0)$. In the multivariate case ($p \geqslant 2$), the asymptotic covariance matrix of $\hat{\beta}_n$ depends on F in a far more complicated way. The difference stems from the fact that while the univariate median is a local property, it is not so in the multivariate case.

It may be noted that the limiting distribution (2.5) does not change much even if there is a large change in the original distribution in the vicinity of a point far away from the median center. In this sense LD estimates are more robust than LS estimates.

5. In the one-dimensional case, $f(0)>0$ is a necessary condition for the asymptotic normality of a LAD estimate. In the multidimensional case, the asymptotic normality of the LD estimate is still true even if the distribution F has no measure in a neighborhood of the origin. This shows an important difference between the one and multidimensional L_1-norm estimation.

Now turn to the proof of Theorem 1. The proof follows closely on the lines of the univariate case (Bai *et al.*, 1987) and the details are much the same. So we content ourselves with a brief sketch.

To begin with, by introducing the notations

$$X_{ni} = (S_n)^{-1/2} X_i, \quad Y_{ni} = Y_i, \quad \varepsilon_{ni} = \varepsilon_i, \quad i=1, \cdots, n, \quad \beta_{n0} = (S_n)^{1/2}\beta_0,$$

we transform the model (1.1) into a more general form

$$Y_{ni} = X'_{ni}\beta_{n0} + \varepsilon_{ni}, \quad i=1, \cdots, n, \quad n=1, 2, \cdots \tag{2.6}$$

with

$$\sum_{i=1}^{n} X_{ni} B X'_{ni} = I_q, \quad n=1, 2, \cdots \tag{2.7}$$

Denote by β_n^* the LD estimate of β_{n0} under this model. It is easily seen that Theorem 1 is a consequence of the following theorem:

Theorem 1′ Suppose that in (2.6), $\varepsilon_{n1}, \cdots, \varepsilon_{nn}$ are iid. with a common distribution F not depending on n, and F satisfies Conditions (a) and (b) of Theorem 1. Further, suppose that (2.7) is true and

$$d_n^* = \max_{1 \leqslant i \leqslant n} \| X_{ni} \| \to 0, \quad \text{as} \quad n \to \infty. \tag{2.8}$$

Then as $n \to \infty$ we have

$$T_n^* (\beta_n^* - \beta_{n0}) \xrightarrow{\mathfrak{L}} N(0, I_q), \quad T_n^* = \sum_{i=1}^{n} X_{ni} A X'_{ni}. \tag{2.9}$$

Note an important feature of Theorem 1′. The dimension of β_{n0} may depend on n and tend to ∞ as $n \to \infty$. According to Huber (1973), this may be important in some practical applications. Also, the assumptions of Theorem 1 can be somewhat weakend. In particular, the distribution of ε_{n1} can be allowed to depend on n.

To prove Theorem 1$'$, we set

$$\beta_{n0} = 0, \tag{2.10}$$

without loss of generality. Since $d_n^* \to 0$, we can choose a sequence of positive constants $\{v_n\}$ such that

$$\lim_{n\to\infty} v_n = \infty, \quad \lim_{n\to\infty} v_n^4 d_n^* = 0. \tag{2.11}$$

Define

$$D_n = \{\beta: \|\beta\| \leqslant v_n\}, \quad \varphi_{ni}(\beta) = [(\varepsilon_{ni} - X'_{ni}\beta)/\|\varepsilon_{ni} - X'_{ni}\beta\| - \varepsilon_{ni}/\|\varepsilon_{ni}\|,$$
$$\lambda_{ni}(\beta) = \mathrm{E}\varphi_{ni}(\beta), \quad i = 1, \cdots, n.$$

Then it is easily shown

Lemma 1

$$\beta_n^* = O_p(1). \tag{2.12}$$

Now we turn to prove Theorem 1$'$. We can show that, as in the univariate case,

$$\limsup_{n\to\infty}\left\{ \left\| \sum_{i=1}^n X_{ni}(\varphi_{ni}(\beta) - \lambda_{ni}(\beta)) \right\| : \beta \in D_n \right\} = 0, \text{ in pr.} \tag{2.13}$$

and, with probability one,

$$\left\{ \|\beta_n^*\| \leqslant \frac{1}{d_n^*} \right\} \Rightarrow \left\{ \left| \sum_{i=1}^n X_{ni} \frac{\varepsilon_{ni} - X'_{ni}\beta_n^*}{\|\varepsilon_{ni} - X'_{ni}\beta_n^*\|} \right| \leqslant (q+1)d_n^* \right\}. \tag{2.14}$$

Now we show that

$$\mathrm{E}\frac{\varepsilon_{ni} - a}{\|\varepsilon_{ni} - a\|} = -(1+o(1))Aa, \quad \text{as} \quad \|a\| \to 0, \tag{2.15}$$

where A is as defined in (2.1). Indeed, elementary arguments give

$$\left\| \frac{\varepsilon_{n1} - a}{\|\varepsilon_{n1} - a\|} - \frac{\varepsilon_{n1}}{\|\varepsilon_{n1}\|} \right\| \leqslant 2\frac{\|a\|}{\|\varepsilon_{n1}\|} \quad (\varepsilon_{n1} \neq 0,\ a). \tag{2.16}$$

Hence

$$\left\| \frac{1}{\|a\|}\left\{ \frac{\varepsilon_{n1} - a}{\|\varepsilon_{n1} - a\|} - \frac{\varepsilon_{n1}}{\|\varepsilon_{n1}\|} + \frac{1}{\|\varepsilon_{n1}\|}\left(I_p - \frac{\varepsilon_{n1}\varepsilon'_{n1}}{\|\varepsilon_{n1}\|^2}\right)a \right\} \right\| \leqslant \frac{p+1}{\|\varepsilon_{n1}\|}. \tag{2.17}$$

Also, the left-hand side of (2.17) tends to zero as $a \to 0$. Hence, observing that $\mathrm{E}\|\varepsilon_{n1}\|^{-1} < \infty$ and $\mathrm{E}\varepsilon_{n1}/\|\varepsilon_{n1}\| = 0$, (2.15) follows from (2.17) and the Lebesgue convergence theorem. From (2.15), Condition (b) and the definition of T_n^*, we get

$$\sum_{i=1}^n X_{ni}\lambda_{ni}(\beta) = -T_n^*\beta + r_n(\beta), \tag{2.18}$$

where $\limsup_{n\to\infty}\{\|r_n(\beta)\|:\beta\in D_n\}=0$ in pr. From (2.12), (2.13) and (2.18), we have

$$\sum_{i=1}^{n}X_{ni}\varphi_{ni}(\beta_n^*)=-T_n^*\beta_n^*+o_p(1). \tag{2.19}$$

Since $d_n^*\to 0$, from (2.12), (2.14) and (2.19), we obtain

$$T_n^*\beta_n^*=\sum_{i=1}^{n}X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|+o_p(1). \tag{2.20}$$

Now (2.9) follows from (2.20), on considering Condition (b) and (2.7), by applying the multivariate Lindeberg's theorem.

3 Estimation of the Matrices A and B

The matrices S_n, T_n in (2.5) depend on A and B, which in turn depends on the error distribution F. Since F is usually unknown, (2.5) cannot be directly used in asymptotic inference on β. However, if we succeed in finding some consistent estimates $(\hat{A}_n, \hat{B}_n)$, of (A, B) and define $\hat{S}_n=\sum_{i=1}^{n}X_i\hat{B}_nX_i'$, $\hat{T}_n=\sum_{i=1}^{n}X_i\hat{A}_nX_i'$, then it is easy to verify that

$$(\hat{S}_n)^{-1/2}\hat{T}_n(\hat{\beta}_n-\beta_0)\xrightarrow{\mathcal{L}}N(0, I_q) \quad \text{as} \quad n\to\infty. \tag{3.1}$$

Here $\hat{S}_n$ and $\hat{T}_n$ depend solely on the observations, and an asymptotic inference on β_0 can be carried out by using (3.1).

Denote the residuals by $\hat{\varepsilon}_{ni}=Y_i-X_i'\hat{\beta}_n$, $i=1, \cdots, n$. An apparent candidate for $(\hat{A}_n, \hat{B}_n)$ is:

$$\begin{aligned}\hat{A}_n&=\frac{1}{n}\sum_{i=1}^{n}\|\hat{\varepsilon}_{ni}\|^{-1}(I_p-\hat{\varepsilon}_{ni}\hat{\varepsilon}_{ni}'/\|\hat{\varepsilon}_{ni}\|^2),\\ \hat{B}_n&=\frac{1}{n}\sum_{i=1}^{n}\hat{\varepsilon}_{ni}\hat{\varepsilon}_{ni}'/\|\hat{\varepsilon}_{ni}\|^2.\end{aligned} \tag{3.2}$$

Theorem 2 Under conditions of Theorem 1, $\hat{A}_n$ and $\hat{B}_n$ are weakly consistent estimates of A and B, respectively.

To prove the theorem, we again transform the model (1.1) to (2.6). Note that the residuals $\hat{\varepsilon}_{n1}, \cdots, \hat{\varepsilon}_{nn}$ are not changed under this transformation, but now $\hat{\varepsilon}_{ni}$ should be understood as $\varepsilon_{ni}-X_{ni}'\beta_n^*$ (recall (2.10)).

Some preliminary results are needed in the proof. Consider the model M_i obtained from (2.6) by deleting the i-th equation. Denote by β_{ni}^* the LD estimate of β_{n0} under model M_i. Then it is easily shown:

Lemma 2 $\max_{1\leq i\leq n}\|\beta_{ni}^*\|=O_p(1)$.

Lemma 3 $\max_{1\leq i\leq n}\|\beta_n^*-\beta_{ni}^*\|=o_p(1)$.

Let

$$A_n^* = \frac{1}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1}(I_p - \varepsilon_{ni}\varepsilon'_{ni}/ \| \varepsilon_{ni} \|^2), \qquad B_n^* = \frac{1}{n}\sum_{i=1}^{n}\varepsilon_{ni}\varepsilon'_{ni}/ \| \varepsilon_{ni} \|^2.$$

Then by the law of large numbers (LLN), it suffices to show that

$$\hat{A}_n - A_n^* = o_p(1), \tag{3.3}$$

$$\hat{B}_n - B_n^* = o_p(1). \tag{3.4}$$

Choose v_n according to (2.11). From (2.16) we easily deduce

$$| \hat{B}_n - B_n^* | \leqslant (4d_n^* v_n/n)\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} + 2I(\| \beta_n^* \| \geqslant v_n) = J_1 + J_2,$$

where I (.) denotes the indicator function. Since $\mathrm{E} \| \varepsilon_{ni} \|^{-1} < \infty$ and $d_n^* v_n \to 0$, from LLN we have $J_1 = o_p(1)$. The same is true for J_2 in view of (2.12). Hence (3.4) follows.

The proof of (3.3) is a bit involved. To begin with, choose a positive constant sequence $\{r_n\}$ such that

$$r_n \to 0, \; d_n^* v_n/r_n^2 \to 0, \quad \text{as} \quad n \to \infty. \tag{3.5}$$

From (2.12) and (2.16) we easily deduce that

$$\begin{aligned} &\left| \frac{1}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1}(\hat{\varepsilon}_{ni} \hat{\varepsilon}'_{ni}/ \| \hat{\varepsilon}_{ni} \|^2 - \varepsilon_{ni}\varepsilon'_{ni}/ \| \varepsilon_{ni} \|^2) \right| \\ &\leqslant \frac{2}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} I(\| \varepsilon_{ni} \| \leqslant r_n) + \frac{4d_n^* v_n}{nr_n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} \\ &\quad + \frac{4}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} I(\| \beta_n^* \| \geqslant v_n) \\ &= H_1 + H_2 + o_p(1). \end{aligned} \tag{3.6}$$

Here $H_j = o_p(1)$, $j = 1, 2$, following from LLN and (3.5). Therefore, since $| \hat{\varepsilon}_{ni} \hat{\varepsilon}'_{ni}|/ \| \hat{\varepsilon}_{ni} \|^2$ and $|\varepsilon_{ni}\varepsilon'_{ni}|/ \| \varepsilon_{ni} \|^2$ are bounded by 1, to prove (3.3), we have only to verify that

$$\frac{1}{n}\sum_{i=1}^{n} | \; \| \hat{\varepsilon}_{ni} \|^{-1} - \| \varepsilon_{ni} \|^{-1} | = o_p(1). \tag{3.7}$$

By the mean value theorem and (2.16) we have, for any two vectors $e \neq 0$ and a,

$$\begin{aligned} | \; \| \varepsilon - a \| - \| \varepsilon \| + \varepsilon' a \| \varepsilon \|^{-1} | &= | ((\varepsilon - a^*) \| \varepsilon - a^* \|^{-1} - \varepsilon \| \varepsilon \|^{-1})' a | \\ &\leqslant 2 \| a^* \| \| a \| \| \varepsilon \|^{-1} \\ &\leqslant 2 \| a \|^2 \| \varepsilon \|^{-1}, \end{aligned} \tag{3.8}$$

where $a^* = \vartheta a$ for some $\vartheta \in [0, 1]$. From (3.5) and (3.8) we have

$$\begin{aligned} &\frac{1}{n}\sum_{i=1}^{n} | \; \| \hat{\varepsilon}_{ni} \|^{-1} - \| \varepsilon_{ni} \|^{-1} | \, I(\| \varepsilon_{ni} \| \geqslant r_n) \\ &\leqslant 6d_n^* v_n r_n^{-2} \frac{1}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} + \frac{1}{n}\sum_{i=1}^{n} | \; \| \varepsilon_{ni} \|^{-1} - \| \hat{\varepsilon}_{ni} \|^{-1} | \, I(\| \beta_n^* \| \geqslant v_n) \\ &= o_p(1). \end{aligned} \tag{3.9}$$

Now we notice the fact that ε_{ni} and β_{ni}^* are independent. This, together with (3.5), gives

$$\mathrm{E}\left\{\frac{1}{n}\sum_{i=1}^{n} \mid \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \|^{-1} \mid I(\| \varepsilon_{ni} \| \leqslant r_n, \max_{1\leqslant i\leqslant n} \| \beta_{ni}^* \| \leqslant v_n)\right\}$$
$$\leqslant \sup_{\|a\|\leqslant d_n^* v_n} \mathrm{E}(\| \varepsilon - a \|^{-1} I(\| \varepsilon \| \leqslant r_n)) \to 0, \quad \text{as} \quad n\to\infty. \tag{3.10}$$

By Lemma 2, (3.9) and (3.10), and the easy fact that

$$\frac{1}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} \|^{-1} I(\| \varepsilon_{ni} \| < r_n) \to 0,$$

to prove (3.7) we need only to verify

$$\frac{1}{n}\sum_{i=1}^{n} \mid \| \hat{\varepsilon}_{ni} \|^{-1} - \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \|^{-1} \mid I(\| \varepsilon_{ni} \| \leqslant r_n) = o_p(1). \tag{3.11}$$

By Lemma 3, we can find a sequence of positive constants $\{\delta_n\}$ such that $\delta_n \to 0$ and

$$\lim_{n\to\infty} P(\max_{1\leqslant i\leqslant n} \| \beta_n^* - \beta_{ni}^* \| \geqslant \delta_n) = 0. \tag{3.12}$$

By Lemma 1, Condition (a) of Theorem 1 and (2.8), we can find a constant M such that for n sufficiently large and $1\leqslant i\leqslant n$, we have

$$P(\| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| \leqslant 2\| X_{ni} \| \delta_n, \ \| \beta_{ni}^* \| \leqslant v_n)$$
$$\leqslant \mathrm{E}\{P(\| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| \leqslant 2\| X_{ni} \| \delta_n \mid, \varepsilon_{n1}, \cdots, \varepsilon_{n,i-1}, \varepsilon_{n,i+1}, \cdots, \varepsilon_{nn}) I(\| \beta_{ni}^* \| \leqslant v_n)\}$$
$$\leqslant M \| X_{ni} \|^2 \delta_n^2.$$

By (2.7) and $B > 0$, we can find a constant M_1, such that $\sum_{i=1}^{n} \| X_{ni} \|^2 \leqslant M_1$.

Hence,

$$\sum_{i=1}^{n} P(\| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| \leqslant 2\| X_{ni} \| \delta_n, \ \| \beta_{ni}^* \| \leqslant v_n)$$
$$\leqslant M\delta_n^2 \sum_{i=1}^{n} \| X_{ni} \|^2 \leqslant MM_1\delta_n^2 \to 0. \tag{3.13}$$

Further, by (3.10)

$$\frac{1}{n}\sum_{i=1}^{n} \mid \| \hat{\varepsilon}_{ni} \|^{-1} - \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \|^{-1} \mid \tag{3.14}$$
$$\times I(\| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| \geqslant 2\| X_{ni} \| \delta_n, \ \| \varepsilon_{ni} \| \leqslant r_n, \| \beta_{ni}^* \| < v_n, \ \| \beta_n^* - \beta_{ni}^* \| < \delta_n)$$
$$\leqslant \frac{1}{n}\sum_{i=1}^{n} \frac{2\| \hat{\varepsilon}_{ni} \| \ \| X'_{ni}(\beta_n^* - \beta_{ni}^*) \| + 3\| X'_{ni}(\beta_n^* - \beta_{ni}^*) \|^2}{\| \hat{\varepsilon}_{ni} \| \ \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| (\| \hat{\varepsilon}_{ni} \| + \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \|)}$$
$$\times I(\| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \| \geqslant 2\| X_{ni} \| \delta_n, \ \| \varepsilon_n \| \leqslant r_n, \| \beta_{ni}^* \| < v_n, \ \| \beta_n^* - \beta_{ni}^* \| < \delta_n)$$
$$\leqslant \frac{20}{n}\sum_{i=1}^{n} \| \varepsilon_{ni} - X'_{ni}\beta_{ni}^* \|^{-1} I(\| \varepsilon_{ni} \| \leqslant r_n)$$
$$= o_p(1).$$

Now (3. 11) follows from (3. 12) — (3. 14), and the proof is concluded.

4 Testing Linear Hypotheses

Consider the model (1. 1). Let H be an $m \times q$ matrix of rank m, and H_0 be the null hypothesis

$$H_0: H(\beta - \beta_0) = 0. \tag{4.1}$$

In LS theory, the testing of H_0 is based on the Residual Sum of Squares (RSS). Similarly, in the median-regression model (1. 1) we may construct a test of H_0 based on the Residual Sum of Distances (RSD). Define

$$\mathrm{RSD}_n = \sum_{i=1}^{n} \| Y_i - X_i' \hat{\beta}_n \| ,$$

where, as before, $\hat{\beta}_n$ is the LD estimate of β_0 in the original (unconstrained) model (1. 1). Let $\hat{\beta}_{nH}$ be the LD estimate of β_0 under the constraint (4. 1): $\sum_{i=1}^{n} \| Y_i - X_i' \hat{\beta}_{nH} \| = \min\{ \sum_{i=1}^{n} \| Y_i - X_i'\beta \| : H(\beta - \beta_0) = 0 \}$, and denote the constrained RSD as

$$\mathrm{RSD}_{nH} = \sum_{i=1}^{n} \| Y_i - X_i' \hat{\beta}_{nH} \| .$$

Surely $\mathrm{RSD}_{nH} \geqslant \mathrm{RSD}_n$; the difference $\mathrm{RSD}_{nH} - \mathrm{RSD}_n$ which measures the discrepancy between the true model and the model constrained by (4. 1) can be an appropriate test statistic. We reject H_0 if

$$\mathrm{RSD}_{nH} - \mathrm{RSD}_n > c, \tag{4.2}$$

where c is the critical value determined by the level of the test. To obtain c, we must have sufficient knowledge about the distribution of $\mathrm{RSD}_{nH} - \mathrm{RSD}_n$. Following is a result along this direction.

Theorem 3 Suppose that $p \geqslant 2$ in the model (1. 1), Conditions (a)—(d) of Theorem 1 are met, and H_0 is true. Then

$$\mathrm{RSD}_{nH} - \mathrm{RSD}_n = \frac{1}{2} \Big(\sum_{i=1}^{n} \frac{X_i \varepsilon_i}{\| \varepsilon_i \|} \Big)' H_n' H_n \Big(\sum_{i=1}^{n} \frac{X_i \varepsilon_i}{\| \varepsilon_i \|} \Big) + o_p(1), \tag{4.3}$$

where H_n is defined in the following way: First, find a $(q - m) \times q$ matrix Z_n such that $HZ_n' = 0$ and $Z_n S_n Z_n' = I_{q-m}$. Using this Z_n, find an $m \times q$ matrix H_n such that

$$H_n T_n Z_n' = 0, \qquad H_n T_n H_n' = I_n,$$

where T_n is defined by (2. 3). When $m = q$, Z_n is not defined, and in this case $H_n' H_n$ will be replaced by T_n^{-1}. One verifies easily that $H_n H_n'$ is uniquely determined by the above description, notwithstanding the fact that H_n may not be uniquely defined.

The first term on the right-hand side of (4.3), when properly standardized, has an asymptotic distribution of the weighted χ^2 type, which involves the matrices of A and B defined by (2.3). So it is not an easy matter to use this theorem in determining the critical value c. However, in one important special case this difficulty disappears.

Corollary If the conditions of Theorem 3 are met, and in addition the error distribution F is spherically symmetric, then we have as $n \to \infty$

$$2((p-1)\mathrm{E}\|\varepsilon_1\|^{-1})^{-1}(\mathrm{RSD}_{nH} - \mathrm{RSD}_n) \to \chi_m^2. \tag{4.4}$$

Since, as shown in the proof of the consistency of $\hat{A}$ in last section, $\frac{1}{n}\sum_{i=1}^n \|Y_i - X_i'\hat{\beta}_n\|^{-1}$ gives a consistent estimate of $\mathrm{E}\|\varepsilon_1\|^{-1}$, an asymptotic threshold c can easily be determined by using (4.4).

Note that Theorem 3, as Theorem 1, is valid only in the case of $p \geqslant 2$. In the univariate case $p = 1$ the asymptotic distribution of $\mathrm{RSD}_{nH} - \mathrm{RSD}_n$ has been much studied (see Bai, Rao and Yin (1987), Basset and Koenker (1978), Koenker and Basset (1982), and Mckean and Schrader (1987) among others). The theory for the univariate case is simpler in one important respect: The asymptotic distribution involves only the simple χ^2.

Proof of Theorem 3 As before, first we transform the model (1.1) to (2.6). Since the residuals are unchanged by this transformation, so RSD_n and RSD_{nH} remain unchanged. Further, note that in the new model (2.6) the hypothesis (4.1) should be written as $H(S_n)^{-1/2}(\beta - \beta_{n0}) = 0$. It is easy to verify that if we denote by H_n^* the matrix determined by the way described in the formulation of Theorem 3, with H and T_n being changed to $H(S_n)^{-1/2}$ and $(S_n)^{-1/2}T_n(S_n)^{-1/2}$ respectively, we shall have $H_n^{*\prime}H_n^* = (S_n)^{1/2}H_n'H_n(S_n)^{1/2}$. Therefore, the first term on the right-hand side of (4.3) is also unchanged under this transformation. In the remainder of this section all discussions are within the framework of (2.6).

Now we proceed to show that

$$\mathrm{E}(\|\varepsilon_{ni} - a\| - \|\varepsilon_{ni}\|) = \frac{1}{2}a'Aa + o(\|a\|^2), \quad \text{as} \quad \|a\| \to 0, \tag{4.5}$$

where A is defined by (2.1). Indeed, by (2.16) and (3.8) we have

$$\left|\,\|\varepsilon_{ni} - a\| - \|\varepsilon_{ni}\| - \varepsilon_{ni}'a/\|\varepsilon_{ni}\| - \frac{a'}{2\|\varepsilon_{ni}\|}(I_p - \varepsilon_{ni}\varepsilon_{ni}'/\|\varepsilon_{ni}\|^2)a\,\right| / \|a\|^2$$
$$\leqslant (p+3)/(2\|\varepsilon_{ni}\|). \tag{4.6}$$

Further, the left-hand side of (4.6) tends to 0 as $a \to 0$. Therefore, (4.5) follows from the Lebesgue convergence theorem and the fact that $\mathrm{E}\|\varepsilon_{ni}\|^{-1} < \infty$.

Define

$$\Phi_{ni}(\beta) = \|\varepsilon_{ni} - X_{ni}'\beta\| - \|\varepsilon_{ni}\|, \; \Lambda_{ni}(\beta) = \mathrm{E}\Phi_{ni}(\beta), \quad i = 1, \cdots, n. \tag{4.7}$$

Then we can show that

$$\sup\left\{\left|\sum_{i=1}^{n}\Phi_{ni}(\beta)-\Lambda_{ni}(\beta)+\beta' X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|\right|: \beta\in D_n\right\}=o_p(1). \tag{4.8}$$

Here recall that $D_n=\{\beta:\|\beta\|\leqslant v_n\}$ and $\{v_n\}$ is chosen according to (2.11). The proof of (4.8) is essentially the same as the proof of Lemma 1 in Bai, Rao and Yin (1987).

From (4.5), (4.8) and (2.20), we obtain

$$\begin{aligned}\mathrm{RSD}_n &= -\beta_n^{*\prime}\sum_{i=1}^{n}X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|+\frac{1}{2}\beta_n^{*\prime}T_n^*\beta_n+\sum_{i=1}^{n}\|\varepsilon_{ni}\|+o_p(1)\\ &= -\frac{1}{2}\Big(\sum_{i=1}^{n}X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|\Big)'T_n^{*-1}\Big(\sum_{i=1}^{n}X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|\Big)+\sum_{i=1}^{n}\|\varepsilon_{ni}\|+o_p(1).\end{aligned} \tag{4.9}$$

Recall that β_n^* is the LD estimate of β_{n0} under the model (2.6), and T_n^* is defined by (2.9).

Suppose first that $m<q$. The hypothesis (4.1) under the model (1.1) becomes $K_n(\beta-\beta_{n0})=0$ under the model (2.6), where $K_n=HS_n^{-1/2}$. Let G_n be a matrix such that $K_nG_n'=0$, $G_nG_n'=I_q$. Then

$$\{\beta:K_n(\beta-\beta_{n0})=0\}=\{\beta_{n0}+G_n'\gamma:\gamma\in\mathrm{R}^{q-m}\}.$$

From this and (4.9), it follows that

$$\mathrm{RSD}_{nH}=-\frac{1}{2}\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)'G_n'(G_nT_n^*G_n')^{-1}G_n\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)+\sum_{i=1}^{n}\|\varepsilon_{ni}\|+o_p(1). \tag{4.10}$$

Therefore,

$$\mathrm{RSD}_{nH}-\mathrm{RSD}_n=\frac{1}{2}\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)'(T_n^{*-1}-G_n'(G_nT_n^*G_n')^{-1}G_n)\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)+o_p(1). \tag{4.11}$$

Noting the fact that $T_n^{*1/2}G_n'(G_nT_n^*G_n)^{-1}G_nT_n^{*1/2}$ is a projection matrix (projecting each vector to the linear subspace generated by the column vectors of $T_n^{*1/2}G_n'$), we have $I_q-T_n^{*1/2}G_n'(G_nT_n^*G_n')^{-1}G_nT_n^{*1/2}=Q_n'Q_n$, where the matrix Q_n satisfies the conditions $G_nT_n^{*1/2}Q_n'=0$, and $Q_nQ_n'=I_m$. Writing $H_n^*=Q_nT_n^{*1/2}$, we have, by (4.11),

$$\mathrm{RSD}_{nH}-\mathrm{RSD}_n=\frac{1}{2}\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)'H_n^{*\prime}H_n^*\Big(\sum_{i=1}^{n}\frac{X_{ni}\varepsilon_{ni}}{\|\varepsilon_{ni}\|}\Big)+o_p(1), \tag{4.12}$$

where H_n^* satisfies

$$G_nT_n^*H_n^{*}=0,\qquad H_nT_n^*H_n^*=I_m. \tag{4.13}$$

Returning to the original model (1.1), we note that

$$\begin{aligned}&X_{ni}\varepsilon_{ni}/\|\varepsilon_{ni}\|=(S_n)^{-1/2}X_i\varepsilon_i/\|\varepsilon_i\|,\ i=1,\cdots,n,\\ &T_n^*=(S_n)^{-1/2}T_n(S_n)^{-1/2}.\end{aligned}$$

So the constraints in (4.13) now read, after setting $Z_n = G_n(S_n)^{-1/2}$ and $H_n = H_n^*(S_n)^{1/2}$,

$$Z_n T_n H_n' = 0, \qquad H_n T_n H_n' = I_m. \tag{4.14}$$

Further G_n satisfies $K_n G_n' = 0$, $G_n G_n' = I_{q-m}$, where $K_n = H(S_n)^{1/2}$. Replacing G_n by $Z_n(S_n)^{1/2}$. We get

$$HZ_n' = 0, \qquad Z_n S_n Z_n' = I_{q-m}. \tag{4.15}$$

Therefore, from (4.12) we obtain (4.3), with H_n constrained by (4.14) and (4.15). This completes the discussion of the case when $m < q$.

If $m = q$, then obviously we have $\mathrm{RSD}_{nH} = \sum_{i=1}^{n} \|\varepsilon_i\|$, and (4.3) follows directly from (4.9). Theorem 3 is proved.

5 Case of a Location Vector

When $X_1 = X_2 = \cdots = I_p$, the model (1.1) reduces to

$$Y_i = \beta_0 + \varepsilon_i, \quad i = 1, \cdots, n, \cdots \tag{5.1}$$

where β_0 is a location vector in the model. As mentioned earlier, the LD estimate of β_0 in this case is called the spatial median center. According to Theorem 1, if $\varepsilon_1, \varepsilon_2, \cdots$ are iid. and its common distribution F satisfies the Conditions (a) and (b) of Theorem 1, then we have

$$\begin{aligned} &\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{\mathfrak{L}} N(0, V), \\ &V = A^{-1}BA^{-1}, \end{aligned} \tag{5.2}$$

where A and B are as defined in (2.1) and (2.2). On the other hand, under the conditions that $\mathrm{E}\varepsilon_1 = 0$ and $\mathrm{Cov}(\varepsilon_1) = W$ exists, we have

$$\sqrt{n}(\tilde{\beta}_n - \beta_0) \xrightarrow{\mathfrak{L}} N(0, W), \tag{5.3}$$

where $\tilde{\beta}_n$ is the LS estimate of β_0; i. e., $\tilde{\beta}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$.

In some simple cases V and W are readily obtained. Using (5.2) and (5.3) we obtain the relative efficiencies of the two estimators, $\hat{\beta}_n$ and $\tilde{\beta}_n$. Two cases will be considered.

a. Spherically symmetric error distribution

When the error distribution F possesses a density $g(u)$ which depends only on $\|u\|$: $g(u) = f(\|u\|)$, then simple calculations give

$$V = c(f)I_p, \qquad c(f) = p(p-1)^{-2}\left(\int_0^{\infty} r^{p-1}f(r)\mathrm{d}r \Big/ \int_0^{\infty} r^{p-2}f(r)\mathrm{d}r\right), \tag{5.4}$$

$$W = d(f)I_p, \qquad d(f) = p^{-1}\left(\int_0^{\infty} r^{p+1}f(r)\mathrm{d}r \Big/ \int_0^{\infty} r^{p-1}f(r)\mathrm{d}r\right). \tag{5.5}$$

Defining as usual the Asymptotic Relative Efficiency $\mathrm{ARE}(\hat{\beta}, \tilde{\beta})$ as the inverse ratio of the determinants of covariance matrices, we obtain from (5.4) and (5.5) that

$$\mathrm{ARE}(\hat{\beta}, \tilde{\beta}) = (d(f)/c(f))^p. \tag{5.6}$$

From (5.4) and (5.5) we see that in the spherically symmetric case the p components of $\hat{\beta}_n$ (or $\tilde{\beta}_n$) become asymptotically independent with the same variance $c(f)$ (or $d(f)$). So in this case it also makes sense to define the ARE in the following way:

$$\mathrm{ARE}^*(\hat{\beta}, \tilde{\beta}) = d(f)/c(f). \tag{5.7}$$

Example 1 $F \sim N(0, \sigma^2 I_p)$ for some $\sigma^2 > 0$. In this case we have, for $p \geqslant 2$,

$$\mathrm{ARE}^*(\hat{\beta}, \tilde{\beta}) = (2p)^{-1}(p-1)^2\Gamma^2\left(\frac{p-1}{2}\right)\Big/\Gamma^2\left(\frac{p}{2}\right). \tag{5.8}$$

which is less than one and increases steadily to one as $p \to \infty$. In fact it is easy to show that

$$\mathrm{ARE}(\hat{\beta}, \tilde{\beta}) \to 1, \quad \text{as } p \to \infty. \tag{5.9}$$

Example 2 F has a multivariate Laplace distribution with density function const. $\exp(-c\,\|u\|)$. In this case we have

$$\begin{aligned} &\mathrm{ARE}^*(\hat{\beta}_n, \tilde{\beta}_n) = 1 + 1/p, \\ &\mathrm{ARE}(\hat{\beta}_n, \tilde{\beta}_n) = (1+1/p)^p \to \mathrm{e}, \quad \text{as } p \to \infty. \end{aligned} \tag{5.10}$$

Laplace distribution is more heavy-tailed than the normal, and in accordance $\hat{\beta}_n$ is more efficient (than $\tilde{\beta}_n$) in this case.

Let us now take a closer look at Example 1. As is well-known, in the univariate case we have $\mathrm{ARE}(\hat{\beta}_n, \tilde{\beta}_n) = \mathrm{ARE}*(\hat{\beta}_n, \tilde{\beta}_n) = 2/\pi$, which is considerably smaller than one. Example 1 shows that the advantage of $\tilde{\beta}_n$ over $\hat{\beta}_n$ diminishes as the dimension p gets larger, and for very large p the LS estimate $\tilde{\beta}_n$ practically loses its merit over $\hat{\beta}_n$ (from the point of view of efficiency). Of course, $\tilde{\beta}_n$ is much easier to compute than $\hat{\beta}_n$. Example 2 provides a case where $\hat{\beta}_n$ is superior to $\tilde{\beta}_n$.

There is yet another interesting phenomenon in Example 1. Suppose that we have in hand p completely unrelated estimation problems:

$$Y_{j1}, \cdots, Y_{jn} \quad \text{all iid.}, \quad \sim N(\beta_j, \sigma^2), \quad j = 1, 2, \cdots, p,$$

where the pn variables $\{Y_{jl'}\}$ are mutually independent. Since these p problems are not related in any sense, it seems a reasonable guess that the best we can do is to treat these p problems separately, that is to say, use only $Y_{j1}, \cdots, Y_{jn}$ in estimating β_j. But when $p \geqslant 3$ and we use the LS estimate, then a classical paper of Stein reveals the surprising fact that this is not true, and we can indeed gain something by lumping these p problems together. Now since the right-hand side of (5.8) increases with increasing p, we see that the use of LD estimation after lumping together the p problems gives us greater efficiency than that obtained by treating these p problems separately with the LD method. This fact is, to some extent, similar to the now famous Stein phenomenon.

b. $p=2$ and F has a normal distribution $N(0,0;\sigma^2,\sigma^2,\rho)$ with $\sigma^2>0$ and $0\leqslant\rho<1$

By applying Theorem 1, simple calculations give

$$\sqrt{n}(\hat{\beta}_n-\beta_0)\xrightarrow{\mathfrak{L}} N(0,\sigma^2A^{-1}BA^{-1}),\tag{5.11}$$

where

$$A=(2\sqrt{2\pi\rho})^{-1}\begin{bmatrix}\rho K_1 & (1-\rho^2)K_2-K_1\\ (1-\rho^2)K_2-K_1 & \rho K_1\end{bmatrix}\tag{5.12}$$

$$B=(2\rho\sqrt{1-p^2})^{-1}\begin{pmatrix}\sin\xi & \cos\xi\\ \cos\xi & \sin\xi\end{pmatrix}\begin{bmatrix}\rho & \sqrt{1-\rho^2}-1\\ \sqrt{1-\rho^2}-1 & \rho\end{bmatrix}\begin{pmatrix}\sin\xi & \cos\xi\\ \cos\xi & \sin\xi\end{pmatrix}\tag{5.13}$$

$$K_1=\int_0^\pi(1+\cos\vartheta)^{-1/2}\mathrm{d}\vartheta,\quad K_2=\int_0^\pi(1+\cos\vartheta)^{-3/2}\mathrm{d}\vartheta,\quad \xi=\frac{1}{2}\cos^{-1}\rho+\frac{\pi}{4}.\tag{5.14}$$

On the other hand, we have

$$\sqrt{n}(\hat{\beta}_n,\beta_0)\xrightarrow{\mathfrak{L}} N(0,W),\quad W=\sigma^2\begin{pmatrix}1 & \rho^2\\ \rho^2 & 1\end{pmatrix}.\tag{5.15}$$

(5.11)—(5.15) give the following expression for the ARE:

$$\mathrm{ARE}(\hat{\beta},\tilde{\beta})=(1-\rho^2)^{5/2}(\rho^2K_2^2)-(K_2-K_1)^2/(32\pi^2\rho^2(1-\sqrt{1-\rho^2})).\tag{5.16}$$

Some numerical results are given in the following table:

ρ	0.0	0.1	0.2	0.3	0.4	0.5
$\mathrm{ARE}(\hat{\beta},\tilde{\beta})$	0.6168	0.6161	0.6137	0.6095	0.6033	0.5943
ρ	0.6	0.7	0.8	0.9	0.96	0.97
$\mathrm{ARE}(\hat{\beta},\tilde{\beta})$	0.5816	0.5630	0.5341	0.4802	0.4064	0.3835

From the table we see that the ARE of $\hat{\beta}$ to $\tilde{\beta}$ decreases slowly as $|\rho|$ increases. It can be shown that it tends to 0 as $|\rho|\to 1$. This example can be compared with the case when the error distribution is $N(0,0,1,r^2,0)$, $0<r^2<1$, considered by Brown (1983) in that the two components of $\hat{\beta}_n$ have the same ARE, while in Brown's case the component of $\hat{\beta}_n$ corresponding to the smaller variance r^2 has a far more smaller ARE. On the other hand, the "over-all" AREs of $\hat{\beta}$ to $\tilde{\beta}$ of these cases are closely related: If we denote by $\varepsilon_1(\rho)$ the $\mathrm{ARE}(\hat{\beta},\tilde{\beta})$ in our case and $\varepsilon_2(\rho)$ the $\mathrm{ARE}(\hat{\beta},\tilde{\beta})$ in Brown's case, then $\varepsilon_1(\rho)=\varepsilon_2(\sqrt{(1-|\rho|)/(1+|\rho|)})$.

Appendix Uniqueness of LD Estimate

Consider the problem of minimizing

$$Q(\beta) = \sum_{i=1}^{n} \| Y_i - X_i'\beta \| , \tag{A. 1}$$

where $Y_i = (Y_{i1}, \cdots, Y_{ip})'$, $i \geqslant 1$, are p-dimensional random vectors, $p \geqslant 2$, and β runs over R^q.

Theorem A1 Suppose that the following two conditions are true:

1° There exists a subset of $m \geqslant 2$ integers $j_1 < j_2 < \cdots < j_m$ of the set $(1, \cdots, p)$ such that

$$(Y_{1j_1}, \cdots, Y_{1j_m}, \cdots, Y_{nj_1}, \cdots, Y_{nj_m})$$

possesses a probability density with respect to the Lebesgue measure $L(\cdot)$.

2° For any subset $S \subset \{1, \cdots, n\}$ containing not less than $n - [(2q-2)/(m-1)]$ elements, $\sum X_i X_i' > 0$ where the summation is taken over the integers in S.

Then with probability one, $Q(\beta)$ has one and only one minimum point.

Proof As the existence part is trivial, only the uniqueness part is of interest. We need the following simple lemma which is a consequence of Sard's theorem given in Sternberg (1964, p. 45).

Lemma A1 Let f be a continuously-differentiable function defined on R^a and taking values on R^b. If $a < b$, then

$$L(\{f(x): x \in \mathrm{R}^a\}) = 0, \tag{A. 2}$$

where L is the Lebesgue measure in R^b.

Although (A. 2) is known, here we still give a simple proof. For any given integer $c > 0$, write $E_c = \{X = (x_1, \cdots, x_a)' : | x_i | \leqslant c, i = 1, \cdots, a\}$. Partition E_c into $(2nc)^a \equiv N$ cubes $F_1, F_2, \cdots, F_N$ each having edge-length $1/n$. Since f has bounded partial derivatives on E_c, we see that for each i, the set $G_i \equiv \{f(x): x \in F_i\}$ is contained in a cube in R^b with edge-length not exceeding M/n for some constant M. Therefore,

$$L(\{f(x): x \in E_c\}) \leqslant \sum_{i=1}^{N} L(G_i) \leqslant N\left(\frac{M}{n}\right)^b \leqslant (2c)^a M^b n^{-(b-a)} \to 0, \quad \text{as} \quad n \to \infty.$$

The lemma is proved.

Now turn to the proof of the theorem. For simplicity we assume that $m = p$. The arguments apply to general m without essential change.

Put $c = [(2q-2)/(p-1)] + 1$, $1 \leqslant i_1 < \cdots < i_c \leqslant n$. Consider the following subset of R^{pc}.

$$S_{i_1 \cdots i_c} = \left\{ \begin{pmatrix} a_{i1} \\ \vdots \\ a_{i_c} \end{pmatrix} : a_{i_j} = X_{i_j}'(\beta + t_j l), \ t_j \in \mathrm{R}, \quad i = 1, \cdots, c, \quad \beta \in \mathrm{R}^q \right\}$$

$$= \left\{ \begin{pmatrix} a_{i_1} \\ \vdots \\ a_{i_c} \end{pmatrix} : \begin{array}{l} a_{i_1} = X_{i_1}'\beta, \quad a_{i_2} = X_{i_2}'(\beta + l), \\ a_{i_j} = X_{i_j}'(\beta + t_j l), \quad t_j \in \mathrm{R}, \quad j = 3, \cdots, c, \quad \beta \in \mathrm{R}^q \end{array} \right\}.$$

Since $pc > 2q + c - 2$, according to Lemma A1, we have $L(S_{i_1 \cdots i_c}) = 0$. Hence

$$L(\bigcup_{1 \leqslant i_1 < \cdots < i_c \leqslant n} S_{i_1 \cdots i_c}) = 0. \tag{A.3}$$

Suppose now that for some $Y_1, \cdots, Y_n$, $Q(\beta)$ has two minimizing points β^* and $\beta^* + l$, $l \neq 0$. By convexity of $Q(\beta)$, each $\beta = \beta^* + tl$ with $0 \leqslant t \leqslant 1$ is also a minimizing point of $Q(\beta)$. Hence, if we define $g(t) = Q(\beta^* + tl)$, then $g''(t) = 0$ for $t \in [0, 1)$. Put

$$B_1 = \{i: 1 \leqslant i \leqslant n, Y_i - X_i'\beta^* = 0\},$$
$$B_2 = \{i: i \in B_1, X_i'l = 0\}.$$

For $i \in B_1 - B_2$, we have $Y_i - X_i'\tilde{\beta} \neq 0$, $\tilde{\beta} = \beta^* + tl$, for any $t \in (0, 1)$. So we can choose some $t_0 \in (0, 1)$ such that on replacing β^* by $\tilde{\beta} = \beta^* + t_0 l$ and l by $(1 - t_0)l$, we have

$$Y_i - X_i'\tilde{\beta} \neq 0, \quad \text{for} \quad 1 \leqslant i \leqslant n, \quad i \notin B_2.$$

Accordingly we redefine $g(t) = Q(\tilde{\beta} + t\tilde{l})$, $\tilde{l} = (1 - t_0)l$. Since $g''(t) = 0$ for $t \in [0, 1)$ and $\| Y_i - X'_i(\tilde{\beta} + t\tilde{l}) \| \equiv 0$ as a function of t for $i \in B_2$, we have, after some simple calculations,

$$g''(0) = \sum_{i \notin B_2} \frac{l'X_i}{\| Y_i - X_i'\tilde{\beta} \|} \left[I - \frac{(Y_i - X_i'\tilde{\beta})(Y_i - X_i'\tilde{\beta})'}{\| Y_i - X_i'\tilde{\beta} \|^2} \right] X_i'\tilde{l} = 0.$$

From this we see that for each $i \in \{1, \cdots, n\} - B_2$, one of the following two assertions must be true: $X_i'\tilde{l} = 0$, or there exists t_i such that $Y_i - X_i'\tilde{\beta} = t_i X_i'\tilde{l}$ since when $i \in B_2$ we have $Y_i - X_i'\tilde{\beta} = t_i X_i'\tilde{l}$. Therefore we have proved that for each $i = 1, \cdots, n$, one of the following two cases must be true:

1° $Y_i - X_i'\tilde{\beta} = t_i X_i'\tilde{l}$, for some t_i;

2° $X_i'\tilde{l} = 0$.

Denote by N the number of i's belonging to the first case. Since $\tilde{l} \neq 0$ we must have $N \geqslant c$. For, if $N < c$, then $n - N \geqslant n - c + 1 = n - [(2q - 2)/(p - 1)]$. Write $S = \{i: 1 \leqslant i \leqslant n, i \text{ belongs to case } 2°\}$. Then $\#(S) \geqslant n - [(2q - 2)/(p - 1)]$, and by assumption 2° of the theorem we have $\sum_{i \in S} X_i X_i' > 0$. From $\sum_{i \in S} \tilde{l}' X_i X_i' \tilde{l} = 0$, it would follow that $\tilde{l} = 0$. Therefore, we have proved $N \geqslant c$. Now the proof of the theorem follows by using (A.3).

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On a Problem of Existence of Consistent Estimate

1 Introduction

Suppose that we have a linear regression model

$$Y_i = \alpha + \beta' x_i + e_i, \quad i = 1, \cdots, n, \cdots$$

where $\{x_i\}$ is a sequence of known p-vectors, $\{e_i\}$ is a sequence of random errors, and α, β are parameters to be estimated. Let $c_0\alpha + c'\beta$ be an estimable function. A basic problem is under what conditions a consistent estimate of $c_0\alpha + c'\beta$ exists or does not exist.

When $\mathrm{E}e_i = 0$ and $\{e_i\}$ satisfies the so-called Gauss-Markov condition, Drygas [2] obtained the necessary and sufficient (n. & s.) condition for the consistency of the least squares (LS) estimate. Assuming the independence of the variables e_1, e_2, $\cdots$, Chen [1] obtained the sufficient condition imposed on $\{e_i\}$ (this condition is also necessary in a certain sense) that when the LS estimate is not consistent, there exists no consistent linear estimates (i. e., estimates of the form $c_{n0} + \sum_{i=1}^{n} c_{ni} Y_i$). Li [3] further showed that when e_1, e_2, $\cdots$ are iid. with a finite variance and finite Fisher information, the non-consistency of the LS estimate implies the non-existence of any consistent estimate for $c_0\alpha + c'\beta$.

An important condition in Li's result is that the density function $f(x)$ of e_i must be continuous everywhere on $(-\infty, \infty)$. It is easy to construct examples showing that when this condition is not met, the conclusion of Li's result may not be true. In particular, when the distribution of the random errors is bounded, a consistent estimate may exist in the case that the LS estimate is not consistent.

In the present paper we shall study this problem under the assumption that the distribution of the random errors is bounded. We have only succeeded in giving a complete solution for the model

$$Y_i = \alpha + \beta x_i + e_i, \quad i = 1, \cdots, n, \cdots \tag{1}$$

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where x_1, x_2, $\cdots$ are known constants.

We mention an interesting fact concerning the LS estimate $(\hat{\alpha}_n, \hat{\beta}_n)$ of (α, β) in (1). As shown by Drygas [2], the necessary and sufficient condition for $\hat{\alpha}_n$ is

$$\lim_{n\to\infty} \bar{x}_n^2 / S_n^2 = 0 \quad \left(\bar{x}_n = \sum_{i=1}^n x_i/n, \quad S_n^2 = \sum_{i=1}^n (x_i - \bar{x}_n)^2\right), \tag{2}$$

and that for $\hat{\beta}_n$ is

$$\lim_{n\to\infty} S_n^2 = \infty. \tag{3}$$

One is easily convinced that (2) follows from (3), so that when $\hat{\beta}_n$ is consistent, $\hat{\alpha}_n$ must be so.

2 Necessary and Sufficient Condition for the Consistent Estimate of β

Suppose that in model (1), e_1, e_2, $\cdots$ are independent and identically distributed, and e_1 possesses a density function $f(x)$ with the form

$$f(x) = h(x) I_{(\sigma_1 \leqslant x \leqslant \sigma_2)}, \tag{4}$$

where $-\infty < \sigma_1 < \sigma_2 < \infty$, σ_1, σ_2 and $h(x)$ are all unknown. We assume that $h(x)$ satisfies the Lipschitz condition on $[\sigma_1, \sigma_2]$:

$$|h(t_2) - h(t_1)| \leqslant c\, |t_2 - t_1|, \quad \text{for all } t_i \in [\sigma_1, \sigma_2], \quad i = 1, 2 \tag{5}$$

and

$$r = \inf_{\sigma_1 \leqslant x \leqslant \sigma_2} h(x) > 0. \tag{6}$$

Of course, we also suppose that

$$\int_{\sigma_1}^{\sigma_2} h(x)\mathrm{d}x = 1, \quad \int_{\sigma_1}^{\sigma_2} x h(x)\mathrm{d}x = 0, \tag{7}$$

which makes $f(x)$ a probability density with mean value zero. Notice that by (7) we have $\sigma_1 < 0 < \sigma_2$.

Denote by $\bar{x}_n$ the arithmetic mean of x_1, $\cdots$, x_n, and

$$H_n = \sum_{i=1}^n |x_i - \bar{x}_n|, \quad n = 1, 2, \cdots. \tag{8}$$

We have the following theorem.

Theorem 1 The n. & s. condition for the existence of a consistent estimate of β is

$$\lim_{n\to\infty} H_n = \infty. \tag{9}$$

Remark It is easy to construct a sequence $\{x_i\}$ (for example, $x_i = i^{-1}$) for which (9) is true but (3) is not. For such sequences $\{x_i\}$, there will be no consistent linear estimate of β, but a consistent nonlinear estimate of β does exist.

Proof of Theorem 1 Denote by m_n the median of $x_1, \cdots, x_n$, and

$$M_n = \sum_{i=1}^{n} | x_i - m_n |. \tag{10}$$

It can be easily seen that (9) is equivalent to

$$\lim_{n\to\infty} M_n = \infty. \tag{11}$$

We only need to verify that (9)⇒(11), and this follows from the fact that

$$| H_n - M_n | \leqslant n | m_n - \bar{x}_n | \leqslant \sum_{i=1}^{n} | x_i - m_n | = M_n \tag{12}$$

which gives $M_n \geqslant H_n/2$. We also note that $M_1 \leqslant M_2 \leqslant M_3 \leqslant \cdots$, so that if (11) fails, $\{M_n\}$ will be bounded. In view of $H_n \leqslant 2M_n$, when (9) fails, $\{H_n\}$ will be bounded.

Now we divide the main body of the proof into three parts.

(I) Construct a suitable estimate for σ_1 and σ_2.

We assume from the start that $\{x_i\}$ is bounded. If not, then (3) is true and the LSE of β, $\hat{\beta}_n$, is consistent. Put

$$x_{ni} = x_i - \bar{x}_n, \quad Y_{ni} = Y_i - \bar{Y}_n, \quad i = 1, \cdots, n \quad \left(\bar{Y}_n = \sum_{i=1}^{n} Y_i/n\right)$$

and introduce the sets:

$$\begin{aligned}
B_n &= \{i: 1 \leqslant i \leqslant n,\ x_{ni} > 0\},\\
C_n &= \{i: 1 \leqslant i \leqslant n,\ x_{ni} < 0\},\\
A_n &= B_n \cup C_n,\\
b_n &= \#(B_n) = \text{the number of elements in } B_n,\\
c_n &= \#(C_n).
\end{aligned}$$

By (9) and the boundedness of $\{X_i\}$ we have

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} c_n = \infty. \tag{13}$$

Find two sequences of numbers $\{b_n^*\}$, $\{c_n^*\}$, with the conditions

1° $b_n^* \geqslant b_n n^{-\frac{1}{4}}$, $c_n^* \geqslant c_n n^{-\frac{1}{4}}$, $n = 1, 2, \cdots$,

2° $b_n^* \to \infty$, $c_n^* \to \infty$ (as $n \to \infty$),

3° $b_n^*/b_n = c_n^*/c_n \to 0$ (as $n \to \infty$).

Such sequences do exist. An example is: for $b_n \leqslant c_n$, choose

$$b_n^* = \begin{cases} b_n n^{-\frac{1}{4}}, & \text{for } b_n \geqslant n^{\frac{1}{3}},\\ b_n^{\frac{1}{4}}, & \text{for } b_n < n^{\frac{1}{3}}, \end{cases} \qquad b_n^* = b_n \frac{c_n^*}{c_n},$$

and for $b_n > c_n$, we choose

$$c_n^* = \begin{cases} c_n n^{-\frac{1}{4}}, & \text{for } c_n \geqslant n^{\frac{1}{3}}, \\ c_n^{\frac{1}{4}}, & \text{for } c_n < n^{\frac{1}{3}}, \end{cases} \qquad b_n^* = b_n \frac{c_n^*}{c_n}.$$

Put $\bar{e}_n = \sum_{i=1}^{n} e_i / n$, and

$$S_{n1}^* = \max_{\tilde{\theta}} \min_{i \in A_n} (Y_{ni} - x_{ni}\tilde{\theta}), \qquad S_{n1} = S_{n1}^* - 2b_n^*/b_n,$$

$$S_{n2}^* = \min_{\tilde{\theta}} \max_{i \in A_n} (Y_{ni} - x_{ni}\tilde{\theta}), \qquad S_{n2} = S_{n2}^* + 2b_n^*/b_n.$$

We proceed to prove:

(a) $S_{n1} \geqslant \sigma_1 - 2b_n^*/b_n - |\bar{e}_n|$, $\lim_{n\to\infty} P(S_{n1} \leqslant \sigma_1 - |\bar{e}_n|) = 1$,

(b) $S_{n2} \leqslant \sigma_2 + 2b_n^*/b_n + |\bar{e}_n|$, $\lim_{n\to\infty} P(S_{n2} \geqslant \sigma_2 + |\bar{e}_n|) = 1$.

Remark Since $b_n^*/b_n \to 0$, $\bar{e}_n \xrightarrow{P} 0$ under our assumption on $\{e_i\}$, it is seen from (a), (b) that S_{n1} and S_{n2} are consistent estimates of σ_1 and σ_2, respectively. Of course, the more simple S_{n1}^* and S_{n2}^* are also consistent, but S_{n1} and S_{n2} meet our special needs here.

To prove (b), notice that

$$Y_{ni} - x_{ni}\tilde{\theta} = x_{ni}(\beta - \tilde{\theta}) + e_i - \bar{e}_n. \tag{14}$$

Hence $S_{n2}^* \leqslant \max_{i \in A_n} (e_i - \bar{e}_n) \leqslant \sigma_2 + |\bar{e}_n|$, which proves the first part of (b). For the latter part, define the event

$$G_n = \{\max_{i \in B_n} e_i > \sigma_2 - b_n^*/b_n + |\bar{e}_n|, \ \max_{i \in C_n} e_i > \sigma_2 - b_n^*/b_n + |\bar{e}_n|\}.$$

Then we have

$$\begin{aligned} P(G_n) &\geqslant P(\max_{i \in B_n} e_i > \sigma_2 - b_n^*/b_n + |\bar{e}_n|) + P(\max_{i \in C_n} e_i > \sigma_2 - b_n^*/b_n + |\bar{e}_n|) - 1 \\ &\geqslant P(\max_{i \in B_n} e_i > \sigma_2 - b_n^*/b_n + n^{-\frac{1}{3}}) + P(\max_{i \in C_n} e_i > \sigma_2 - b_n^*/b_n + n^{-\frac{1}{3}}) \\ &\quad + 2P(|\bar{e}_n| \leqslant n^{-\frac{1}{3}}) - 3 \\ &\geqslant 1 - [1 - \gamma(b_n^*/b_n - n^{-\frac{1}{3}})]^{b_n} + 1 - [1 - \gamma(b_n^*/b_n - n^{-\frac{1}{3}})]^{c_n} \\ &\quad + 2P(|\bar{e}_n| \leqslant n^{-\frac{1}{3}}) - 3. \end{aligned} \tag{15}$$

We have as $n \to \infty$

$$b_n(b_n^*/b_n - n^{-\frac{1}{3}}) = b_n^* - b_n n^{-\frac{1}{3}} = b_n^* - b_n n^{-\frac{1}{4}} n^{-\frac{1}{12}} \geqslant b_n^* (1 - n^{-\frac{1}{12}}) \to \infty,$$

$$c_n(b_n^*/b_n - n^{-\frac{1}{3}}) = c_n(c_n^*/c_n - n^{-\frac{1}{3}}) \to \infty,$$

and there exists constant K not depending on n such that

$$P(|\bar{e}_n| \leqslant n^{-\frac{1}{3}}) \geqslant 1 - Kn^{-\frac{1}{3}} \to 1, \quad \text{as } n \to \infty. \tag{16}$$

From these facts and (15), it follows that

$$\lim_{n\to\infty} P(G_n) = 1. \tag{17}$$

Define

$$E_n = \{i : i \in B_n,\ e_i \geqslant 0\}, \qquad F_n = \{i : i \in C_n,\ e_i \geqslant 0\}. \tag{18}$$

The occurrence of G_n implies that

$$\max_{i\in E_n} e_i > \sigma_2 - b_n^* / b_n + |\bar{e}_n|, \qquad \max_{i\in F_n} e_i > \sigma_2 - b_n^* / b_n + |\bar{e}_n|. \tag{19}$$

Write $\Delta = \beta - \tilde{\theta}$. If $\Delta \geqslant 0$, then, noticing that $x_{ni} > 0$ for $i \in B_n$, from (14) and (19) we have

$$\max_{i\in A_n}(Y_{ni} - x_{ni}\tilde{\theta}) \geqslant \max_{i\in E_n}(Y_{ni} - x_{ni}\tilde{\theta}) \geqslant \max_{i\in E_n} e_i - |\bar{e}_n| \geqslant \sigma_2 - b_n^* / b_n,$$

while for $\Delta < 0$ we have

$$\max_{i\in A_n}(Y_{ni} - x_{ni}\tilde{\theta}) \geqslant \max_{i\in F_n}(Y_{ni} - x_{ni}\tilde{\theta}) \geqslant \max_{i\in F_n} e_i - |\bar{e}_n| \geqslant \sigma_2 - b_n^* / b_n.$$

Therefore, when the event G_n occurs, we always have $S_{n2}^* \geqslant \sigma_2 - b_n^* / b_n$, which in turn implies $S_{n2}^* \geqslant \sigma_2 + b_n^* / b_n$. Since $b_n^* / b_n \geqslant n^{-\frac{1}{4}}$, we have

$$\begin{aligned} P(S_{n2} \geqslant \sigma_2 + |\bar{e}_n|) &\geqslant P(\{S_{n2} \geqslant \sigma + |\bar{e}_n|\} \cap G_n) \\ &\geqslant P(b_n^* / b_n \geqslant |\bar{e}_n|) + P(G_n) - 1 \\ &\geqslant P(|\bar{e}_n| \leqslant n^{-\frac{1}{4}}) + P(G_n) - 1. \end{aligned}$$

By (17) and the fact $\lim\limits_{n\to\infty} P(|\bar{e}_n| \leqslant n^{-\frac{1}{4}}) = 1$, the latter part of (b) follows. The proof of (a) is similar.

(Ⅱ) Sufficiency of condition (9).

For each positive integer n, we choose n' (also a positive integer) satisfying the following conditions:

$$b_{n'}/b_n \to 0,\ n' b_n^* / b_n \to 0,\ 2^{n'} n^{-\frac{1}{3}} \to 0,\ n' \to \infty \quad (\text{as } n \to \infty). \tag{20}$$

Introduce an estimate of β:

$$T_n = \max\left\{\max_{i\in B_{n'}^*}\left(\frac{Y_{ni} - S_{n2}}{x_{ni}}\right),\ \max_{i\in C_{n'}^*}\left(\frac{Y_{ni} - S_{n1}}{x_{ni}}\right)\right\} \tag{21}$$

where

$$\begin{aligned} B_{n'}^* &= \{i = 1 \leqslant i \leqslant n',\ x_{ni} > 0\}, \\ C_{n'}^* &= \{i = 1 \leqslant i \leqslant n',\ x_{ni} < 0\}. \end{aligned}$$

Since $\beta = \dfrac{Y_{ni}}{x_{ni}} - \dfrac{e_i - \bar{e}_n}{x_{ni}}$ for $i \in A_n$, it follows that

$$\begin{aligned} i \in B_{n'}^* &\Rightarrow \beta \geqslant \frac{Y_{ni}}{x_{ni}} - \frac{\sigma_2 + |\bar{e}_n|}{x_{ni}}, \\ i \in C_{n'}^* &\Rightarrow \beta \geqslant \frac{Y_{ni}}{x_{ni}} - \frac{|\sigma_1| + |\bar{e}_n|}{|x_{ni}|}. \end{aligned} \tag{22}$$

From (21), (22), we obtain (noticing that $\sigma_1 < 0$)

$$\{S_{n1} \leqslant \sigma_1 - |\bar{e}_n|, \ S_{n2} \geqslant \sigma_2 + |\bar{e}_n|\} \Rightarrow \{\beta > T_n\}.$$

Hence, by the assertions (a), (b) proved above, we have

$$\lim_{n\to\infty} P(T_n \leqslant \beta) = 1. \tag{23}$$

On the other hand, for arbitrarily given $\varepsilon > 0$, we have

$$\begin{aligned} P(T_n \leqslant \beta - \varepsilon) &\leqslant P\Big(\frac{Y_{ni}}{x_{ni}} - \frac{\sigma_2 + 2b_n^*/b_n + |\bar{e}_n|}{x_{ni}} \leqslant \beta - \varepsilon, \quad i \in B_{n'}^*, \\ &\qquad \frac{Y_{ni}}{x_{ni}} - \frac{|\sigma_1| + 2b_n^*/b_n + |\bar{e}_n|}{x_{ni}} \leqslant \beta - \varepsilon, \quad i \in C_{n'}^* \Big) \\ &\leqslant P(e_i \leqslant \sigma_2 + 2b_n^*/b_n + 2|\bar{e}_n| - \varepsilon x_{ni}, \quad i \in B_{n'}^*, \\ &\qquad e_i \geqslant \sigma_1 - 2b_n^*/b_n - 2|\bar{e}_n| + \varepsilon |x_{ni}|, \quad i \in C_{n'}^*). \end{aligned} \tag{24}$$

In view of (16), we have

$$\begin{aligned} &P(e_i \leqslant \sigma_2 + 2b_n^*/b_n + 2|\bar{e}_n| - \varepsilon x_{ni}) \\ &\quad \leqslant P(e_i \leqslant \sigma_2 + 2b_n^*/b_n + 2n^{-\frac{1}{3}} - \varepsilon x_{ni}) + P(|\bar{e}_n| \geqslant n^{-\frac{1}{3}}) \\ &\quad \leqslant 1 - r(\varepsilon x_{ni} - 2b_n^*/b_n - 2n^{-\frac{1}{3}}) + Kn^{-\frac{1}{3}}, \quad i \in B_{n'}^*. \end{aligned}$$

Likewise,

$$\begin{aligned} &P(e_i \geqslant \sigma_1 - 2b_n^*/b_n - 2|\bar{e}_n| + \varepsilon |x_{ni}|) \\ &\quad \leqslant 1 - r(\varepsilon |x_{ni}| - 2b_n^*/b_n - 2n^{-\frac{1}{3}}) + Kn^{-\frac{1}{3}}, \quad i \in C_{n'}^*. \end{aligned}$$

Therefore, putting $A_{n'}^* = B_{n'}^* \cup C_{n'}^*$, we have

$$\begin{aligned} P(T_n \leqslant \beta - \varepsilon) \leqslant Q_n &\triangleq \prod_{i \in A_{n'}^*} \{[1 - r(\varepsilon |x_{ni}| - 2b_n^*/b_n - 2n^{-\frac{1}{3}})] + Kn^{-\frac{1}{3}}\} \\ &\triangleq Q_{n1} + Q_{n2}, \end{aligned} \tag{25}$$

where

$$\begin{aligned} Q_{ni} &= \prod_{i \in A_{n'}^*} [1 - r(\varepsilon |x_{ni}| - 2b_n^*/b_n - 2n^{-\frac{1}{3}})] \\ &\leqslant \exp\Big[- r\Big(\varepsilon \sum_{i=1}^{n'} |x_{ni}| - 2b_n^*/b_n - 2n'n^{-\frac{1}{3}}\Big)\Big]. \end{aligned}$$

Since $n' \to \infty$, by (11) (which is a consequence of (9)), we have

$$\sum_{i=1}^{n'} |x_{ni}| \geqslant \sum_{i=1}^{n'} |x_i - m_{n'}| = M_{n'} \to \infty, \quad \text{as } n \to \infty.$$

Also, $\lim\limits_{n\to\infty} n' b_n^*/b_n = 0$, $\lim\limits_{n\to\infty} 2^{n'} n^{-\frac{1}{3}} = 0$, by the choice of n'. Hence

$$\lim_{n\to\infty} Q_{ni} = 0. \tag{26}$$

As for Q_{n2}, it is a sum of $2^{n'} - 1$ terms; each does not exceed $Kn^{-\frac{1}{3}}$. So from $2^{n'} n^{-\frac{1}{3}} \to 0$, we have

$$\lim_{n\to\infty} Q_{n2} = 0. \tag{27}$$

Combining (26) and (27), we get

$$\lim_{n\to\infty} P(T_n \leqslant \beta - \varepsilon) = 0. \tag{28}$$

Now it follows from (23) and (28) that T_n is a consistent estimate of β. This proves the sufficiency of (9).

(Ⅲ) Necessity of condition (9).

Now suppose that (9) fails. As mentioned earlier, there is a constant $H < \infty$ such that $H_n \leqslant H$ for all n. We now show that there exists a constant h such that

$$\sum_{i=1}^{\infty} | x_i - h | < \infty. \tag{29}$$

In fact, since $\{\bar{x}_n\}$ is bounded, a subsequence $\{\bar{x}_{n_i}\}$ can be found such that x_{n_i} tends to some constant h as $i \to \infty$. For fixed n and i sufficiently large such that $n_i > n$, we have

$$\sum_{i=1}^{n} | x_i - \bar{x}_{n_i} | \leqslant H_{n_i} \leqslant H.$$

Letting $i \to \infty$, we get (29).

We need only to prove the non-existence of the consistent estimate of β in the case that σ_1, σ_2 and the function $h(x)$ are known. Take θ_0 such that

$$0 < \theta_0 < \frac{\min(r/(2c),\ Q^{-1})}{\sup_{i\geqslant 1} | x_i - h |} \quad (Q = \max_{\sigma_1 \leqslant x \leqslant \sigma_2} h(x)).$$

Choose a prior distribution of (α, β) in the following way:

$$G(\alpha = \beta = 0) = G(\alpha = -\theta_0 h,\ \beta = \theta_0) = 1/2$$

and the square loss function $(d-\beta)^2$. Put

$$J_{n1} = \{(Y_1, \cdots, Y_n): \sigma_1 \leqslant Y_i \leqslant \sigma_2, \quad i = 1, \cdots, n\},$$
$$J_{n2} = \{(Y_1, \cdots, Y_n): \sigma_1 \leqslant Y_i - (x_i - h)\theta_0 \leqslant \sigma_2, \quad i = 1, \cdots, n\},$$
$$L_n = \prod_{i=1}^{n} \frac{h(Y_i - (x_i - h)\theta_0)}{h(Y_i)}, \qquad l_n = \frac{L_n}{1 + L_n}.$$

Then it is easy to see that, under the above prior and loss, the Bayesian estimate of β is

$$\tilde{\beta}_n = \begin{cases} 0, & \text{when } (Y_1, \cdots, Y_n) \in J_{n1} \cap J_{n2}^c, \\ \theta_0, & \text{when } (Y_1, \cdots, Y_n) \in J_{n1}^c \cap J_{n2}, \\ l_n\theta_0, & \text{when } (Y_1, \cdots, Y_n) \in J_{n1} \cap J_{n2}. \end{cases}$$

By (5), we have $| h(Y_i) - h(Y_i - (x_i - h)\theta_0) | \leqslant c | x_i - h | \theta_0$, for $(Y_1, \cdots, Y_n) \in J_{n1} \cap J_{n2}$. By the choice of θ_0, it is true that

$$c \mid x_i - h \mid \frac{\theta_0}{h(Y_i)} \leqslant \frac{c \mid x_i - h \mid \theta_0}{r} \leqslant \frac{1}{2}.$$

Also, $\log(1-x) \geqslant -2x$, for $0 \leqslant x \leqslant 1/2$. Hence we have

$$\log L_n \geqslant \sum_{i=1}^{n} \log\left(1 - \frac{c \mid x_i - h \mid \theta_0}{h(Y_i)}\right) \geqslant -\frac{2c\theta_0}{r} \sum_{i=1}^{n} \mid x_i - h \mid > -\frac{2c\theta_0 H}{r} \stackrel{\triangle}{=} -f.$$

So we have

$$l_n \geqslant (1+e^f)^{-1} \stackrel{\triangle}{=} q.$$

Now observe that

$$P(J_{n1} \cap J_{n2} \mid \alpha = \beta = 0) = \sum_{i=1}^{n} P(\sigma_1 \leqslant Y_i \leqslant \sigma_2,\ \sigma_1 \leqslant Y_i - (x_i - h)\theta_0 \leqslant \sigma_2 \mid \alpha = \beta = 0)$$

since

$$P(\sigma_1 \leqslant Y_i \leqslant \sigma_2,\ \sigma_1 \leqslant Y_i - (x_i - h)\theta_0 \leqslant \sigma_2 \mid \alpha = \beta = 0) \geqslant 1 - Q\theta_0 \mid x_i - h \mid > 0.$$

Hence, considering (29), we obtain

$$\liminf_{n\to\infty} P(J_{n1} \cap J_{n2} \mid \alpha = \beta = 0) \geqslant \liminf_{n\to\infty} \prod_{i=1}^{n} (1 - Q\theta_0 \mid x_i - h \mid)$$

$$\stackrel{\triangle}{=} u > 0. \tag{30}$$

Denoting by $R_G(\tilde{\beta}_N)$ the Bayesian risk of $\tilde{\beta}_n$ under the prior G, we get in view of (30):

$$\liminf_{n\to\infty} R_G(\tilde{\beta}_n) \geqslant uq^2\theta_0^2/2 > 0. \tag{31}$$

But if there exists a consistent estimate β_n^* of β, then we define

$$\beta_n^{**} = \begin{cases} 0, & \text{when } \mid \beta_n^* \mid \leqslant \mid \beta_n^* - \beta_0 \mid, \\ \theta_0, & \text{otherwise,} \end{cases}$$

which is bounded and converges in probability to β, when $\beta = 0$ or θ_0, and we shall have

$$\lim_{n\to\infty} R_G(\beta_n^{**}) = 0. \tag{32}$$

This contradicts with (31), in view of the fact that $\tilde{\beta}_n$ is the Bayesian estimate. Therefore, a consistent estimate of β cannot exist, and the proof of Theorem 1 is complete.

3 Necessary and Sufficient Condition for the Existence of Consistent Estimate of α

The purpose of this section is to prove the following theorem:

Theorem 2 Under the assumptions imposed earlier on $\{e_i\}$, the n. & s. condition for the existence of a consistent estimate of α is

$$\lim_{n\to\infty}\frac{\overline{X}_n}{H_n}=0, \tag{33}$$

where H_n is defined by (8).

Proof (Ⅰ) Sufficiency. We separate two cases.

1° $\lim\limits_{n\to\infty} H_n=\infty$.

In this case, as shown in Theorem 1, the estimate T_n defined by (21) is a consistent estimate of β. Since $\{x_i\}$ is bounded (see the beginning of the proof of Theorem 1), $\{\overline{X}_n\}$ is also bounded. Define

$$\tilde{\alpha}_n=\overline{Y}_n-T_n\overline{X}_n.$$

Then $\tilde{\alpha}_n-\alpha=(\beta-T_n)\overline{X}_n+\bar{e}_n\xrightarrow{P}0$, as $n\to\infty$. Hence $\tilde{\alpha}_n$ is a consistent estimate of α.

2° H_n does not tend to ∞.

In this case, as stated earlier, $\{H_n\}$ is bounded. So from (33) we have $\lim\limits_{n\to\infty}\overline{X}_n=0$. It is easy to verify that $\overline{Y}_n$ is a consistent estimate of α.

(Ⅱ) Necessity. Suppose that (33) is not true. Then it is easily seen that H_n cannot tend to ∞ as $n\to\infty$, for otherwise (33) would be true. As noticed earlier, it follows that $\{H_n\}$ is bounded; hence we can choose a subsequence $\{n_i\}$ such that $\overline{X}_{n_i}\to a\neq 0$, and $\sum\limits_{n=1}^{\infty}|x_n-a|<\infty$.

Exactly as in Section 2, we only have to prove the non-existence of a consistent estimate for α in the case that σ_1, σ_2 and $h(x)$ are completely known. Take θ_0, with

$$0<\theta_0<\frac{\min(r/(2c),\ Q^{-1})}{\sup\limits_{n\geqslant 1}|x_n-a|}$$

and introduce the following prior distribution G:

$$G(\alpha=\beta=0)=G(\alpha=-\theta_0 a,\ \beta=\theta_0)=1/2$$

and the square loss $(d-a)^2$. Denote by $\tilde{\alpha}_n$ the Bayesian estimate of α under this prior and loss function. Then, by exactly the same argument employed in proving (31), it can readily be shown that

$$\liminf_{n\to\infty} R_G(\tilde{\alpha}_n)>0$$

and this will lead to a contradiction, as at the end of Section 2. So the proof is complete.

Remark As mentioned in Section 1, in LS theory we have the interesting fact that when the LS estimate of β is consistent, the same must be true for α. This, together with Li's result, shows that under the conditions proposed by Li, if a consistent estimate of β ever exists, the same must be true for α. From Theorems 1 and 2 of the present paper, it is seen that this phenomenon persists also in the case where the distribution of e_i is bounded, as we have pointed out that (33) is a consequence of (9). Whether or not this is true in general remains an open problem.

4 Median Regression

Sometimes it will be more practical or more convenient to consider the median regression instead of the mean regression. In this section we shall show that, even in this case, the conclusion of Theorems 1 and 2 holds true without change.

Theorem 3 Suppose that in model (1) the random errors e_1, e_2, $\cdots$ are iid. and e_1 fulfills all the conditions as in Theorem 1 except that we now assume that the median of e_1 is zero. Then the n. & s. condition for the existence of a consistent estimate of α or β is (33) or (9), respectively.

Proof First consider the case for β. The proof of the necessity part in Theorem 1 remains valid, for in the proof we have not made use of the assumption that $\mathrm{E}e_1 = 0$. For sufficiency, we write $\alpha_0 = \mathrm{E}e_1$, $\alpha^* = \alpha + \alpha_0$, $e_i^* = e_i - \alpha_0$, and rewrite the model (1) as

$$Y_i = \alpha^* + \beta x_i + e_i^*, \quad i = 1, \cdots, n. \tag{34}$$

Now we write $\mathrm{E}e_i^* = 0$, and the sequence $\{e_i^*\}$ satisfies all the conditions of $\{e_i\}$ in Theorem 1. Hence, by Theorem 1, condition (9) guarantees the existence of a consistent estimate of β.

Now turn to the case for α. The necessity part of the proof of Theorem 2 remains in force in the present case, and it suffices to prove the sufficiency of (33). Three cases are in order:

1° $\{x_i\}$ is not bounded.

For simplicity of writing and without loss of generality, we may assume that $\lim\limits_{i\to\infty} |x_i| = \infty$. Put

$$\beta_n^* = \frac{Y_n - Y_1}{x_n - x_1}.$$

Choose $m = m_n$ such that $\lim\limits_{n\to\infty} m_n = \infty$, and

$$\max_{1\leqslant i\leqslant m} \frac{|x_i|}{|x_n - x_1|} \to 0, \qquad \text{as } n \to \infty. \tag{35}$$

Finally we define

$$\alpha_n^* = \text{median of } \{Y_i - x_i\beta_n^*, \quad i = 1, \cdots, m\}.$$

Then

$$\alpha_n^* - \alpha = \text{median of } \{x_i(\beta - \beta_n^*) + e_i, \quad i = 1, \cdots, m\}.$$

By (35), we have

$$\begin{aligned} |x_i(\beta - \beta_n^*)| &\leqslant |x_i| \frac{|e_n - e_1|}{|x_n - x_1|} \\ &\leqslant \left(\max_{1\leqslant i\leqslant m} \frac{|x_i|}{|x_n - x_1|}\right)(|\sigma_1| + \sigma_2) \\ &\to 0, \quad \text{as } n \to \infty. \end{aligned}$$

So we have

$$| \alpha_n^* - \alpha - \text{median of } \{e_1, \cdots, e_m\} | \to 0, \quad \text{as } n \to \infty.$$

Since $e_1, e_2, \cdots$ are iid. and e_1 has a unique median 0, it follows that the median of $\{e_1, \cdots, e_m\}$ tends to zero in probability, which proves the consistency of α_n^*.

2° $\{X_i\}$ is bounded, but (9) holds.

In this case we know that there exists a consistent estimate of β which we shall denote by β_n^*. Divide $(-\infty, \infty)$ into half-open intervals $I_j = [j/\sqrt{n}, (j+1)/\sqrt{n})$, $j = 0, \pm 1, \pm 2, \cdots$. For fixed n, we choose $j = j_n$ such that $l_j \geqslant l_k$ for all k, where

$$l_{jn} = \# \{i: 1 \leqslant i \leqslant n, x_i \in I_j\}.$$

(If there are more than one such j, choose arbitrarily one.) Since $\{x_i\}$ is bounded, we have

$$\lim_{n\to\infty} l_{jn} = \infty. \tag{36}$$

Denote all the members in the set $\{i: 1 \leqslant i \leqslant n, x_i \in I_j\}$ by $n_1, n_2, \cdots, n_{n'}$, and $x_n^* = (x_{n_1} + x_{n_2} + \cdots + x_{n_{n'}})/n'$. Now define

$$\alpha_n^* = \text{median of } \{Y_{n_i} - \beta_n^* x_n^*, \quad i = 1, 2, \cdots, n'\}.$$

Since

$$\begin{aligned} Y_{n_i} - \beta_n^* x_n^* &= \alpha + (\beta - \beta_n^*) x_n^* + \beta(x_{n_i} - x_n^*) + e_{n_i}, \\ | x_{n_i} - x_n^* | &\leqslant \frac{1}{\sqrt{n}}, \quad i = 1, \cdots, n' \end{aligned} \tag{37}$$

and $\{x_n^*\}$ is bounded, it follows that

$$(\beta - \beta_n^*) x_n^* + \beta(x_{n_i} - x_n^*) \xrightarrow{P} 0, \quad \text{as } n \to \infty.$$

Also, in view of (36) and the assumptions imposed upon $\{e_i\}$, we have

$$\text{median of } \{e_{n_1}, \cdots, e_{n_{n'}}\} \xrightarrow{P} 0, \quad \text{as } n \to \infty.$$

Combining these facts, from (37), we get $\alpha_n^* \xrightarrow{P} \alpha$, as $n \to \infty$.

Remark A seemingly simpler proof runs as follows. Since $\{x_i\}$ is bounded, there is a subsequence $x_{n_i} \to a$, as $i \to \infty$. Denote by $i(n)$ the largest i such that $n_i \leqslant n$, and define

$$\alpha_n^{**} = \text{median of} \{Y_j - a\beta_n^*, \quad j = n_{[i(n)/2]}, n_{[i(n)/2]+1}, \cdots, n_{i(n)}\}.$$

Then it is readily shown that $\alpha_n^{**} \xrightarrow{P} \alpha$, as $n \to \infty$. This proof has the unpleasant feature of using the values of x_i not only for $i \leqslant n$.

3° $H_n = \sum_{i=1}^{n} | x_i - \bar{x}_n |$ is bounded.

In this case we have $\lim_{n\to\infty} \bar{x}_n = 0$. From this, and the boundedness of H_n, we obtain further that $\sum_{n=1}^{\infty} | x_n | < \infty$, hence $\lim_{n\to\infty} x_n = 0$. Define

$$\alpha_n^* = \text{median of } \{Y_i,\ i = [n/2],\ [n/2]+1,\ \cdots,\ n\}.$$

It is readily shown that $\alpha_n^* \xrightarrow{P} \alpha$, as $n \to \infty$. This concludes the proof of the existence of a consistent estimate for α. Theorem 3 is proved.

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Consistency of Minimum L_1-norm Estimates in Linear Models

1 Introduction

Consider a standard linear regression model

$$Y_i = \boldsymbol{x}_i'\boldsymbol{\beta}_0 + e_i,\ i = 1, 2, \cdots, n, \cdots, \tag{1.1}$$

where $\{\boldsymbol{x}_i\}$ is a sequence of known p-vectors, $\boldsymbol{\beta}_0$ is the unknown parameter-vector, and e_1, e_2, $\cdots$ are non-observable random errors. Denote by $\hat{\boldsymbol{\beta}}_n$ a Borel measurable solution of the minimization problem:

$$\sum_{i=1}^{n} | Y_i - \boldsymbol{x}_i' \hat{\boldsymbol{\beta}}_n | = \min_{\boldsymbol{\beta}\in R^p} \sum_{i=1}^{n} | Y_i - \boldsymbol{x}_i'\boldsymbol{\beta} |. \tag{1.2}$$

Different names have been given to $\hat{\boldsymbol{\beta}}_n$. Here we call it the ML_1NE (minimum L_1-norm estimate) of $\boldsymbol{\beta}_0$.

This method of estimation enjoys a long history (for an account of historical developments the reader is referred to [1]), and in recent years it has aroused much attention among statisticians and econometricians (see [2]). Since no explicit form of $\hat{\boldsymbol{\beta}}_n$ is available, a workable small-sample theory is out of the question. On the large-sample side, take for example the basic problem of consistency; a number of results have appeared but are far from conclusive (see [1]—[5]). In this paper we develop a new method to tackle this problem, enabling us to give a basically complete solution in an elegant form.

The paper will be organized as follows. Section 2 gives the basic result concerning consistency of $\hat{\boldsymbol{\beta}}_n$ in the homogeneous model (1.1). Section 3 extends the result to the non-homogeneous case. Finally, in Section 4 we present some counter examples which give further insight into the main results of the paper.

2 Consistency and Exponential Rate of $\hat{\boldsymbol{\beta}}_n$ (Homogeneous Case)

First we introduce some notations and preliminary lemmas. For $\boldsymbol{a} = (a_1, \cdots, a_p)' \in R^p$,

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$$\| \boldsymbol{a} \| = \left(\sum_{i=1}^{n} a_i^2 \right)^{\frac{1}{2}}$$

and

$$| \boldsymbol{a} | = \max(| a_1 |, \cdots, | a_p |).$$

For a square matrix $\boldsymbol{A}$, its largest and smallest eigenvalues will be denoted by $\bar{\lambda}(\boldsymbol{A})$ and $\underline{\lambda}(\boldsymbol{A})$, and $\operatorname{tr}(\boldsymbol{A})$ denotes the trace of $\boldsymbol{A}$. Also, write

$$\boldsymbol{S}_n = \sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i', \quad d_n = \max_{1 \leqslant i \leqslant n} \boldsymbol{x}_i' \boldsymbol{S}_n^{-1} \boldsymbol{x}_i \quad (\text{when } \boldsymbol{S}_n^{-1} \text{ exists}).$$

Lemma 2.1 (Hoeffding's inequality) Suppose that $Z_1, \cdots, Z_n$ are independent r. v. s with $\mathrm{E}Z_i = 0$, $a_i \leqslant Z_i \leqslant b_i$, $i = 1, \cdots, n$. Then for any $\varepsilon > 0$ we have

$$P\left(\left| \sum_{i=1}^{n} Z_i \right| \geqslant \varepsilon \right) \leqslant 2\exp\left(-2\varepsilon^2 \Big/ \sum_{i=1}^{n} (b_i - a_i)^2 \right). \tag{2.1}$$

For a proof, see [6].

Lemma 2.2 Suppose that $\boldsymbol{S}_m^{-1}$ exists for some m. Then there exists a constant $C_0 > 0$ such that

$$d_n \geqslant C_0 / \underline{\lambda}(\boldsymbol{S}_n), \quad n \geqslant m. \tag{2.2}$$

Proof Denote by $\rho_{n_1} \leqslant \cdots \leqslant \rho_{n_p}$ the eigenvalues of $\boldsymbol{S}_n$ and $\rho_{m_1} \leqslant \cdots \leqslant \rho_{m_p}$ the eigenvalues of $\boldsymbol{S}_m$. By a result of von Neumann [7], we have

$$\operatorname{tr}(\boldsymbol{S}_m \boldsymbol{S}_n^{-1}) \geqslant \sum_{i=1}^{p} \rho_{m_i} / \rho_{n_i}.$$

Hence

$$\rho_{m_1} / \underline{\lambda}(\boldsymbol{S}_n) \leqslant \sum_{i=1}^{p} \rho_{m_i} / \rho_{n_i} \leqslant m d_n,$$

for $n \geqslant m$. Since $\rho_{m_1} > 0$, (2.2) is true for $C_0 = \rho_{m_1} / m > 0$. The lemma is proved.

Now follows the main result of this section:

Theorem 2.1 Suppose that in (1.1) the following conditions are met:

(a) $e_1, e_2, \cdots$ are independent with median 0.

(b) There exist constants $l_1 > 0$ and $l_2 > 0$, such that $(i = 1, 2, \cdots)$

$$P(0 < e_i < h) \geqslant l_2 h \leqslant P(-h < e_i < 0), \text{ for } h \in (0, l_1). \tag{2.3}$$

Then the following assertions are true:

(1) $d_n = o(1) \Rightarrow \hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0$ in pr.;

(2) $d_n = o(1/\log n) \Rightarrow \hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0$ a. s.;

(3) $d_n = O(1/n) \Rightarrow \hat{\boldsymbol{\beta}}_n$ tends to $\boldsymbol{\beta}_0$ exponentially in the sense that $P(\| \hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0 \| \geqslant \varepsilon) = O(e^{-Cn})$ for any $\varepsilon > 0$, with C independent of n but possibly depend on ε.

Remark 2.1 It is noticeable that the consistency of $\hat{\boldsymbol{\beta}}_n$ is related here solely to the convergence rate of d_n. This elegant form substantially improves Oberhofer [3], and also a result of one of the authors [5].

Proof of Theorem 2.1 Put $Y_{ni} = Y_i$, $e_{ni} = e_i$, $\boldsymbol{x}_{ni} = \boldsymbol{S}_n^{-1/2}\boldsymbol{x}_i$, $i = 1, 2, \cdots, n$, and $\boldsymbol{\beta}_{n_0} = \boldsymbol{S}_n^{1/2}\boldsymbol{\beta}_0$. Model (1.1) is transformed to

$$Y_{ni} = \boldsymbol{x}'_{ni}\boldsymbol{\beta}_{n_0} + e_{ni}, \ 1 \leqslant i \leqslant n, \ n = 1, 2, \cdots. \tag{2.4}$$

Denote by $\boldsymbol{\beta}_n^*$ the ML$_1$NE of $\boldsymbol{\beta}_{n_0}$ under (2.4). We have

$$\hat{\boldsymbol{\beta}}_n = \boldsymbol{S}_n^{-1/2}\boldsymbol{\beta}_n^*, \ \sum_{i=1}^n \boldsymbol{x}_{ni}\boldsymbol{x}'_{ni} = \boldsymbol{I}_p, \ d_n = \max_{1\leqslant i\leqslant n} \| \boldsymbol{x}_{ni} \|^2. \tag{2.5}$$

Without loss of generality, assume $\boldsymbol{\beta}_0 = \boldsymbol{0} = \boldsymbol{\beta}_{n_0}$. Choose $\eta \in (0, l_1/2)$, and define

$$V_n = 2[\eta/\sqrt{d_n p}], \tag{2.6}$$

where $[a]$ denotes the integer part of a. In any case ((1), (2), or (3)) we have $V_n \to \infty$, so $V_n > 2$ for n large. Fix $\varepsilon' \in (0, l_2/2)$. For such n, a sufficiently large integer M_n can be found such that

$$\sqrt{np}V_n^{-2t} < \varepsilon' 2^{-1} 3^{-t-1} V_n^2, \ t \geqslant M_n. \tag{2.7}$$

Write

$$D_n = \{\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p): -V_n \leqslant \beta_i < V_n, \ i = 1, \cdots, p\}$$

and

$$\begin{aligned} \Phi_{ni}(\boldsymbol{\beta}) &= | e_{ni} | - | e_{ni} - \boldsymbol{x}'_{ni}\boldsymbol{\beta} |, \\ \Lambda_{ni}(\boldsymbol{\beta}) &= \mathrm{E}\Phi_{ni}(\boldsymbol{\beta}), \\ R_{ni}(\boldsymbol{\beta}) &= \Phi_{ni}(\boldsymbol{\beta}) - \Lambda_{ni}(\boldsymbol{\beta}). \end{aligned} \tag{2.8}$$

For an interval

$$B = \{\boldsymbol{u} = (u_1, \cdots, u_p): a_i \leqslant u_i \leqslant b_i, \ 1 \leqslant i \leqslant p\},$$

the length of its longest edge, i.e.

$$\max_{1\leqslant i\leqslant p}(b_i - a_i)$$

will be denoted by $L(B)$. Partition D_n into V_n^{2p} disjoint intervals

$$\{B_{j_1}: 1 \leqslant j_1 \leqslant V_n^{2p}\},$$

such that

$$L(B_{j_1}) = 1, \ 1 \leqslant j_1 \leqslant V_n^{2p}.$$

Next, each B_{j_1} is partitioned into V_n^{2p} disjoint intervals

$$\{B_{j_1 j_2}: 1 \leqslant j_2 \leqslant V_n^{2p}\}$$

with

$$L(B_{j_1 j_2}) = V_n^{-2}, \ 1 \leqslant j_2 \leqslant V_n^{2p}.$$

Inductively, after partitioning D into

$$\{B_{j_1 \cdots j_m} : 1 \leqslant j_i \leqslant V_n^{2p}, \ 1 \leqslant i \leqslant m\}$$

with

$$L(B_{j_1 \cdots j_m}) = V_n^{-2(m-1)},$$

we go a step further in partitioning each $B_{j_1, \cdots, j_m}$ into disjoint intervals

$$\{B_{j_1, \cdots, j_m, j_{m+1}}, \ 1 \leqslant j_{m+1} \leqslant V_n^{2p}\}$$

with

$$L(B_{j_1, \cdots, j_{m+1}}) = V_n^{-2m}.$$

Denote by $\boldsymbol{b}_{j_1, \cdots, j_m}$ the centre of $B_{j_1, \cdots, j_m}$ and put

$$U_0 = \sum_{j_1} P\Big\{ \Big| \sum_{i=1}^n R_{n_i}(\boldsymbol{b}_{j_1}) \Big| \geqslant \varepsilon' V_n^2/3 \Big\},$$

$$U_m = \sum_{j_1, \cdots, j_{m+1}} P\Big\{ \Big| \sum_{i=1}^n (R_{n_i}(\boldsymbol{b}_{j_1 \cdots j_m}) - R_{n_i}(\boldsymbol{b}_{j_1 \cdots j_{m+1}})) \Big| \geqslant \varepsilon' V_n^2/3^{m+1} \Big\}, \ m \geqslant 1,$$

$$Q_m = \sum_{j_1, \cdots, j_m} P\Big\{ \sup_{\boldsymbol{\beta} \in B_{j_1, \cdots, j_m}} \Big| \sum_{i=1}^n (R_{n_i}(\boldsymbol{\beta}) - R_{n_i}(\boldsymbol{b}_{j_1 \cdots j_{m+1}})) \Big| \geqslant \varepsilon' V_n^2/3^m \Big\}, \ m \geqslant 1,$$

where the summations are taken over $1 \leqslant j_i \leqslant V_n^{2p}$, $i = 1, 2, \cdots$. If $m \geqslant M_n + 1$ and $\boldsymbol{\beta} \in B_{j_1, \cdots, j_m}$, we have, in view of (2.5), (2.7) and (2.8),

$$\begin{aligned}
\Big| \sum_{i=1}^n [\Phi_{n_i}(\boldsymbol{\beta}) - \Phi(\boldsymbol{b}_{j_1, \cdots, j_m})] \Big| &\leqslant \sum_{i=1}^n | \boldsymbol{x}_{n_i}' (\boldsymbol{\beta} - \boldsymbol{b}_{j_1, \cdots, j_m}) | \\
&\leqslant \Big(n \sum_{i=1}^n [\boldsymbol{x}_{n_i}' (\boldsymbol{\beta} - \boldsymbol{b}_{j_1, \cdots, j_m})]^2 \Big)^{1/2} \\
&= \sqrt{n} \, \| \boldsymbol{\beta} - \boldsymbol{b}_{j_1, \cdots, j_m} \| \\
&\leqslant \sqrt{np} V_n^{-2(m-1)} \\
&< \varepsilon' 2^{-1} 3^{-m} V_n^2.
\end{aligned}$$

Hence $Q_m = 0$, for $m \geqslant M_n + 1$. Since $Q_m \leqslant U_m + Q_{m+1}$, $m \geqslant 1$ and

$$P\Big\{ V_n^{-2} \sup\Big(\Big| \sum_{i=1}^n R_{n_i}(\boldsymbol{\beta}) \Big| : \boldsymbol{\beta} \in D_n \Big) \geqslant \varepsilon' \Big\} \leqslant U_0 + Q_1,$$

it follows that

$$P\Big\{ V_n^{-2} \sup\Big(\Big| \sum_{i=1} R_{n_i}(\boldsymbol{\beta}) \Big| : \boldsymbol{\beta} \in D_n \Big) \geqslant \varepsilon' \Big\} \leqslant \sum_{m=1}^{M_n} U_m. \tag{2.9}$$

Since

$$\sum_{i=1}^{n} |\Phi_{n_i}(\boldsymbol{b}_{j_1})|^2 \leqslant \sum_{i=1}^{n} |\boldsymbol{x}'_{n_i}\boldsymbol{b}_{j_1}|^2 = \|\boldsymbol{b}_{j_1}\|^2 \leqslant V_n^2$$

and

$$\begin{aligned}\sum_{i=1}^{n} |\Phi_{n_i}(\boldsymbol{b}_{j_1,\cdots,j_m}) - \Phi_{n_i}(\boldsymbol{b}_{j_1,\cdots,j_m,j_{m+1}})|^2 &\leqslant \sum_{i=1}^{n} |\boldsymbol{x}'_{n_i}(\boldsymbol{b}_{j_1,\cdots,j_m} - \boldsymbol{b}_{j_1,\cdots,j_m,j_{m+1}})|^2 \\ &= \|\boldsymbol{b}_{j_1,\cdots,j_m} - \boldsymbol{b}_{j_1\cdots,j_m,j_{m+1}}\|^2 \\ &\leqslant pV_n^{-4(m-1)},\end{aligned}$$

by Lemma 2.1 we have

$$U_0 \leqslant 2V_n^{2p}\exp\left(-\frac{2(\varepsilon' V_n^2/3)^2}{4pV_n^2}\right) \leqslant V_n^{2p}\exp(-4CV_n^2), \tag{2.10}$$

$$\begin{aligned}U_m &\leqslant 2V_n^{2p(m+1)}\exp\left\{-\frac{2(\varepsilon' V_n^2/3^{m+1})^2}{4pV_n^{-4(m-1)}}\right\} \\ &\leqslant V_n^{2p(m+1)}\exp\left(-\frac{2CV_n^{4m}}{9^{m+1}}\right),\ m \geqslant 1,\end{aligned} \tag{2.11}$$

where $C > 0$ is a constant independent of n. From (2.9)−(2.11), we have

$$\begin{aligned}&P\left\{V_n^{-2}\sup\left\{\left|\sum_{i=1}^{n} R_{n_i}(\boldsymbol{\beta})\right| : \boldsymbol{\beta} \in D_n\right\} \geqslant \varepsilon'\right\} \\ &\leqslant V_n^{2p}\exp(-4CV_n^2) + \sum_{m=1}^{\infty} V_n^{2p(m+1)}\exp\left(-\frac{2CV_n^{4m}}{9^{m+1}}\right).\end{aligned} \tag{2.12}$$

Since $V_n \to \infty$, for n large we have

$$V_n^{2p}\exp(-4CV_n^2) < \exp(-2CV_n^2)$$

and

$$V_n^{2p(m+1)}\exp\left(-\frac{2CV_n^{4m}}{9^{m+1}}\right) < \exp\left(-\frac{CV_n^{4m}}{81^m}\right),$$

we have, for n large,

$$\begin{aligned}&P\left\{V_n^{-2}\sup\left(\left|\sum_{i=1}^{n} R_{n_i}(\boldsymbol{\beta})\right| : \boldsymbol{\beta} \in D_n\right) \geqslant \varepsilon'\right\} \\ &\leqslant \exp(-2CV_n^2) + \exp\left(-\frac{CV_n^4}{81}\right)\sum_{m=0}^{\infty}\exp\left(-\frac{CV_n^{4m}}{81^m}\right) \\ &\leqslant \exp(-2CV_n^2) + 2\exp\left(-\frac{CV_n^4}{81}\right) \\ &\leqslant \exp(-CV_n^2).\end{aligned} \tag{2.13}$$

According to (2.6), when $\boldsymbol{\beta} \in D_n$ we have

$$| x_{ni}'\boldsymbol{\beta} | \leqslant \| x_{ni} \| \, \| \boldsymbol{\beta} \| \leqslant \sqrt{d_n p} V_n \leqslant 2\eta < l_1.$$

Hence, denoting by F_i the d. f. of e_i and noticing (2. 3), we have

$$\Lambda_{ni}(\boldsymbol{\beta}) = \int_0^{x_{ni}'\boldsymbol{\beta}} (2u - 2x_{ni}'\boldsymbol{\beta}) \mathrm{d}F_i(u) \leqslant - x_{ni}'\boldsymbol{\beta} \int_0^{x_{ni}'\boldsymbol{\beta}} \mathrm{d}F_i(u) \leqslant - l_2 (x_{ni}'\boldsymbol{\beta})^2. \qquad (2.14)$$

In deriving (2. 14) we assume tacitly that $x_{ni}'\boldsymbol{\beta} \leqslant 0$. It is not difficult to see that it is also true when $x_{ni}'\boldsymbol{\beta} < 0$. Hence

$$\sum_{i=1}^n \Lambda_{ni}(\boldsymbol{\beta}) \leqslant - l_2 \sum_{i=1}^n (x_{ni}'\boldsymbol{\beta})^2 = - l_2 \| \boldsymbol{\beta} \|^2. \qquad (2.15)$$

From (2. 13), (2. 15), and recalling (2. 8), we have, for n large,

$$P\{ \sup_{|\boldsymbol{\beta}| = V_n} V_n^{-2} (\sum_{i=1}^n (| e_{ni} | - | e_{ni} - x_{ni}'\boldsymbol{\beta} |)) > \varepsilon' - l_2/2 \} \leqslant \exp(- CV_n^2).$$

This implies, in view of $\varepsilon' < l_2/2$, that

$$P\{ \sum_{i=1}^n | e_{ni} | - \inf_{|\boldsymbol{\beta}| = V_n} \sum_{i=1}^n | e_{ni} - x_{ni}'\boldsymbol{\beta} | > 0 \} \leqslant \exp(- CV_n^2). \qquad (2.16)$$

Since $\sum_{i=1}^n | e_{ni} - x_{ni}'\boldsymbol{\beta} |$ is a convex function of $\boldsymbol{\beta}$, (2. 16) implies that

$$P\{ (\sum_{i=1}^n | e_{ni} | - \inf_{|\boldsymbol{\beta}| \geqslant V_n} \sum_{i=1}^n | e_{ni} - x_{ni}'\boldsymbol{\beta} |) > 0 \} \leqslant \exp(- CV_n^2). \qquad (2.17)$$

From this, also recalling the definition of $\mathrm{ML_1NE}$, we get

$$P(| \boldsymbol{\beta}_n^* | > V_n) \leqslant \exp(- CV_n^2), \text{ for } n \text{ large}. \qquad (2.18)$$

An inspection of the argument leading to (2. 18) convinces us that it is still true that

$$P(| \boldsymbol{\beta}_n^* | > qV_n) \leqslant \exp(- CV_n^2), \text{ for } n \text{ large}, \qquad (2.19)$$

for any fixed constant $q > 0$, where, of course, C may depend on q. Also, for any positive constant sequence $\{\mu_n\}$ with $\mu_n \to \infty$ and $\mu_n \sqrt{d_n} \to 0$, we have

$$P(| \boldsymbol{\beta}_n^* | > q\mu_n) \leqslant \exp(- CV_n^2), \text{ for } n \text{ large}, \qquad (2.20)$$

where C depends on q and $\{\mu_n\}$.

Now by Lemma 2. 2, and (2. 5), (2. 6), we have

$$\{ \| \hat{\boldsymbol{\beta}}_n \| \geqslant \varepsilon \} \Rightarrow \{ \| S_n^{1/2} \hat{\boldsymbol{\beta}}_n \| \geqslant \sqrt{\lambda(S_n)} \varepsilon \}$$

$$\Rightarrow \{ \| \boldsymbol{\beta}_n^* \| \geqslant \varepsilon \sqrt{C_0/d_n} \}$$

$$\Rightarrow \{ \| \boldsymbol{\beta}_n^* \| \geqslant V_n \varepsilon \sqrt{pC_0}/(2\eta) \}.$$

From this and (2. 19), taking $q = \varepsilon \sqrt{pC_0}/(2\eta)$, we have

$$P(\|\hat{\boldsymbol{\beta}}_n\| \geqslant \varepsilon) \leqslant \exp(-CV_n^2), \text{ for } n \text{ large.} \tag{2.21}$$

When $d_n \to 0$ we have $V_n \to \infty$. From (2.21) we have $P(\|\hat{\boldsymbol{\beta}}_n\| \geqslant \varepsilon) \to 0$ as $n \to \infty$. Since we have assumed that $\boldsymbol{\beta}_0 = \mathbf{0}$, this amounts to saying that $\hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0$ in pr., and (1) is proved. If $d_n = o(1/\log n)$, from (2.6) it follows that $V_n \geqslant \sqrt{2\log n/c}$ for n large, and by (2.21) we obtain $P(\|\hat{\boldsymbol{\beta}}_n\| \geqslant \varepsilon) \leqslant n^{-2}$, for n large. This implies $\hat{\boldsymbol{\beta}} \to \boldsymbol{\beta}_0$, a. s., in view of the Borel-Cantelli lemma and $\boldsymbol{\beta}_0 = \mathbf{0}$. Finally, when $d_n = O(1/n)$ we have $V_n^2 \geqslant bn$ for some $b > 0$, so (3) also follows from (2.21). Theorem 2.1 is proved.

Remark 2.2 In cases (1) and (2) the following slightly general results are true:

(1)′ $d_n \to 0 \Rightarrow \boldsymbol{S}_n^{1/2}\hat{\boldsymbol{\beta}}_n = O_p(1)$;

(2)′ $d_n = o(1/\log n) \Rightarrow \boldsymbol{S}_n^{1/2}\hat{\boldsymbol{\beta}}_n/\sqrt{\log n} = O(1)$, a. s.

The proof can be obtained by slightly modifying the previous argument, and details are omitted. That (1)′ and (2)′ are indeed stronger than (1) and (2) respectively can be seen from Lemma 2.2.

Remark 2.3 In (1.1) we tacitly assumed that $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots$ and $\boldsymbol{\beta}_0$ do not depend on n. An inspection of the proof reveals that even when $\boldsymbol{x}_i = \boldsymbol{Z}_{ni}$, $i = 1, \cdots, n$ and $\boldsymbol{\beta}_0 = \boldsymbol{\gamma}_{n_0}$ depend on n, the conclusion of Theorem 2.1 remains valid if $\boldsymbol{S}_n$ is defined as $\sum_{i=1}^{n}\boldsymbol{Z}_{ni}\boldsymbol{Z}_{ni}'$. This fact is of importance in the following section.

3 Model with a Constant Term

In practical applications the linear regression function usually contains an additive constant term, and (1.1) is written in the form

$$Y_i = \alpha_0 + \boldsymbol{x}_i'\boldsymbol{\beta}_0 + e_i,\ i = 1, \cdots, n, \cdots, \tag{3.1}$$

where $\boldsymbol{\beta}_0$ is p-dimensional. In principle (3.1) can be regarded as a special case of (1.1), so Theorem 2.1 is also applicable to this case. But, by first centering the model (3.1) at the sample mean $\bar{\boldsymbol{x}}_n = \sum_{i=1}^{n}\boldsymbol{x}_i/n$, we get a more elegant and easily-understandable form of the result.

Denote by $\hat{\alpha}_n$ and $\hat{\boldsymbol{\beta}}_n$ the $\mathrm{ML_1NE}$ of α_0 and $\boldsymbol{\beta}_0$ in (3.1), and define

$$\tilde{\alpha}_0 = \alpha_0 + \bar{\boldsymbol{x}}_n'\boldsymbol{\beta}_0,\ \boldsymbol{\gamma}_0 = (\tilde{\alpha}_0, \boldsymbol{\beta}_0')',\ \boldsymbol{Z}_i = (1, \boldsymbol{x}_i' - \bar{\boldsymbol{x}}_n')',\ 1 \leqslant i \leqslant n.$$

We transform (3.1) to

$$Y_i = \boldsymbol{Z}_i'\boldsymbol{\gamma}_0 + e_i,\ 1 \leqslant i \leqslant n. \tag{3.2}$$

Note that here the unknown parameter-vector γ_0 depends on n. But, as noted in Remark 2.3, Theorem 2.1 still applies in this case. Define

$$T_n = \sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)', \ \delta_n = \max_{1\leqslant i\leqslant n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)'T_n^{-1}(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n). \tag{3.3}$$

It is easy to see that $T_n > 0 \Rightarrow T_N > 0$ for $N > n$.

Theorem 3.1 Suppose that in model (3.1) the error sequence $\{e_i\}$ satisfies conditions (a), (b) of Theorem 2.1. Then

(1) $\delta_n = o(1) \Rightarrow \hat{\alpha}_n \to \alpha_0$ in pr., $\hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0$ in pr.;

(2) $\delta_n = o(1/\log n) \Rightarrow \hat{\alpha}_n \to \alpha_0$, $\hat{\boldsymbol{\beta}}_n \to \boldsymbol{\beta}_0$ a. s.;

(3) $\delta_n = o(1/n) \Rightarrow (\hat{\alpha}_n, \hat{\boldsymbol{\beta}}'_n)$ tends to $(\alpha_0, \boldsymbol{\beta}'_0)$ exponentially in the sense of Theorem 2.1, (3).

Proof First show that

$$\| \boldsymbol{T}_n^{-1/2}\overline{\boldsymbol{x}}_n \| = O(\sqrt{\delta_n}). \tag{3.4}$$

Indeed, by Lemma 2.2, which is applicable also in the present case, we have $\boldsymbol{x}_1'\boldsymbol{T}_n^{-1}\boldsymbol{x}_1 \leqslant \| \boldsymbol{x}_1 \|^2/\lambda(\boldsymbol{T}_n) \leqslant C\delta_n$, for some constant C independent of n. Hence

$$| \boldsymbol{x}_1'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n | \leqslant \| \boldsymbol{T}_n^{-1/2}\boldsymbol{x}_1 \| \cdot \| \boldsymbol{T}_n^{-1/2}\overline{\boldsymbol{x}}_n \| \leqslant \sqrt{C\delta_n}\, \| \boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n \|.$$

Thus

$$\overline{\boldsymbol{x}}_n'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n \leqslant \delta_n + 2\,| \boldsymbol{x}_1'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n | \leqslant \delta_n + \sqrt{C\delta_n}\, \| \boldsymbol{T}_n^{-1}\boldsymbol{x}_n \|.$$

We would reach a contradiction if

$$\limsup_{n\to\infty} \overline{\boldsymbol{x}}_n'\boldsymbol{T}_n^{-1}\overline{\boldsymbol{x}}_n/\delta_n = \infty$$

and (3.4) is proved.

Write

$$\hat{\boldsymbol{\gamma}}_n = (\hat{\alpha}_n + \overline{\boldsymbol{x}}_n'\hat{\boldsymbol{\beta}}, \ \hat{\boldsymbol{\beta}}')', \ \boldsymbol{S}_n = \begin{pmatrix} n & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{T}_n \end{pmatrix}$$

and

$$d_n = \max_{1\leqslant i\leqslant n} \boldsymbol{Z}_i'\boldsymbol{S}_n^{-1}\boldsymbol{Z}_i.$$

Since $d_n = 1/n + \delta_n$, on assuming $\boldsymbol{\gamma}_0 = \boldsymbol{0}$ without loss of generality, we have, according to Theorem 2.1:

$$\delta_n = o(1) \Rightarrow \hat{\boldsymbol{\gamma}}_n \to \boldsymbol{0} \text{ in pr.} \tag{3.5}$$

$$\delta_n = o(1/\log n) \Rightarrow \hat{\boldsymbol{\gamma}}_n \to \boldsymbol{0} \text{ a. s.} \tag{3.6}$$

$$\delta_n = O(1/n) \Rightarrow \hat{\boldsymbol{\gamma}}_n \to \boldsymbol{0} \text{ exponentially.} \tag{3.7}$$

This proves the part of Theorem 3.1 concerning $\hat{\boldsymbol{\beta}}_n$. For a proof concerning $\hat{\alpha}_n$, let us transform the model (3.2) in just the same way as (1.1) was transformed to (2.4), and denote by $(a_0, \boldsymbol{b}_0')$ the parameter in the resulting model. The ML_1NE of $(a_0, \boldsymbol{b}_0')'$ will be denoted by $(\hat{a}_n, \hat{\boldsymbol{b}}_n)'$. Then we have, as can be seen from the proof of Theorem 2. 1:

$$\hat{\boldsymbol{b}}_n = T_n^{1/2}\hat{\boldsymbol{\beta}}_n, \tag{3.8}$$

$$P(\|\hat{\boldsymbol{b}}_n\| \geqslant q/\sqrt{\delta_n}) \leqslant e^{-C/\delta_n}, \tag{3.9}$$

where $q > 0$, and $C > 0$ may depend on q but not n. From (3. 4) we have $\|\boldsymbol{T}_n^{-1/2}\overline{\boldsymbol{x}}_n\| \leqslant k\sqrt{\delta_n}$ for some constant k. Combining (2. 29), (2. 30), we see that for any $\varepsilon > 0$,

$$\begin{aligned} P(|\overline{\boldsymbol{x}}_n'\hat{\boldsymbol{\beta}}_n| \geqslant \varepsilon) &= P(|\overline{\boldsymbol{x}}_n'\boldsymbol{T}_n^{-1/2}\boldsymbol{T}_n^{1/2}\hat{\boldsymbol{\beta}}_n| \geqslant \varepsilon) \\ &\leqslant P(k\sqrt{\delta_n}\,\|\hat{\boldsymbol{b}}_n\| \geqslant \varepsilon) \\ &= P(\|\hat{\boldsymbol{b}}_n\| \geqslant q/\sqrt{\delta_n}) \\ &\leqslant e^{-C/\delta_n}, \end{aligned} \tag{3.10}$$

where $q = \varepsilon/k$. Now the assertion on $\hat{\boldsymbol{a}}_n$ follows readily from (3. 5)—(3. 7) and (3. 10), Theorem 3. 1 is proved.

4 Two Examples

Although the conditions in Theorem 2. 1 are quite sharp, one still wonders what can be said about the necessary conditions of consistency. Apparently there is little hope of obtaining elegant necessary and sufficient conditions. The following examples, which give insight into this problem, are of interest.

Example 4. 1 This example shows that the convergence rate specified in Theorem 2. 1 is not necessary in guaranteeing the conclusions of the theorem.

Consider the model

$$Y_i = 10^i\boldsymbol{\beta}_0 + e_i,\ i = 1, \cdots, n, \cdots,$$

where $e_1, e_2, \cdots$ are iid. with a common distribution $N(0, 1)$. Note that $\boldsymbol{\beta}_0$ is one-dimensional. Without loss of generality assume $\boldsymbol{\beta}_0 = \boldsymbol{0}$. Given $\varepsilon > 0$, then for $|\boldsymbol{\beta}| \geqslant \varepsilon$,

$$\sum_{i=1}^n |e_i - 10^i\boldsymbol{\beta}| \geqslant \left(10^n - \sum_{i=1}^{n-1}10^i\right)\varepsilon - \sum_{i=1}^n |e_i| \geqslant \frac{\varepsilon}{2}10^n - \sum_{i=1}^n |e_i|.$$

Since $e_1, e_2, \cdots$ are iid. and $e_1 \sim N(0, 1)$, we have

$$P\left(\left|\sum_{i=1}^n e_i\right| \geqslant n\right) = O(e^{-Cn}),$$

for some constant $C > 0$. As

$$\frac{\varepsilon}{2}10^n > 2n,$$

for n large, we have for n large,

$$P(|\hat{\boldsymbol{\beta}}_n| \geqslant \varepsilon) \leqslant P\left(\left|\sum_{i=1}^{n} e_i\right| \geqslant n\right) = O(e^{-Cn}).$$

So $\hat{\boldsymbol{\beta}}_n$ converges to $\boldsymbol{\beta}_0$ exponentially; yet it is easy to verify that in this example even the weakest condition $d_n = o(1)$ is not true.

Conditions (a), (b) of Theorem 2.1 guarantee that each e_i has a unique median 0, and moreover, there should be "enough" mass of e_i in the vicinity of the median 0. An interesting question is whether or not the mere uniqueness of median is sufficient. If this is the case, condition (b) can be dispensed with. A simple example shows that this is not true when e_1, e_2, $\cdots$ are not identically distributed. Of more interest, and somewhat surprisingly, is the fact that this is still not true even when e_1, e_2, $\cdots$ are iid., as shown by the following example.

Example 4.2 Choose a sequence of positive constants $\{x_i\}$ satisfying the following two conditions:

(1) $x_1^2 + x_2^2 + \cdots = \infty$, $T \equiv x_1^3 + x_2^3 + \cdots < \infty$;

(2) Write $D_n = x_1 + \cdots + x_n$. For any $i_1, \cdots, i_t$ such that $1 \leqslant i_1 < \cdots < i_t \leqslant n$, we have $x_{i_1} + \cdots + x_{i_t} \neq D_n/2$.

The existence of such a sequence $\{x_i\}$ can be seen as follows. Put $x_1 = 1$. Suppose that we have determined $x_1, \cdots, x_m$ such that $x_i \in (1/\sqrt{i}, 1/\sqrt{i-1})$ for $i = 2, \cdots, m$ and condition (2) holds for $n \leqslant m$. Define

$$H_m = \{D_m\} \cup \{\pm(2(x_{i_1} +, \cdots x_{i_t}) - D_m) : 1 \leqslant i_1 < \cdots < i_t \leqslant m, t \leqslant 1\}$$

and choose

$$x_{m+1} \in \left(\frac{1}{\sqrt{m-1}}, \frac{1}{\sqrt{m}}\right) - H_m.$$

The sequence $\{x_i\}$ constructed inductively in this way satisfies the two conditions specified above.

Now consider the model

$$Y_i = \boldsymbol{x}_i\boldsymbol{\beta}_0 + e_i, \ i = 1, \cdots, n, \cdots,$$

where e_1, e_2, $\cdots$ are iid. With a common density

$$f(u) = 2u^2 I(|u| < 2^{-1/3}).$$

This density has a unique median 0. Also, since $\{x_i\}$ is bounded and

$$S_n \equiv \sum_{i=1}^{n} x_i^2 \to \infty,$$

as $n \to \infty$, we have

$$d_n = \max_{1 \leqslant i \leqslant n} \frac{x_i^2}{S_n} \to 0$$

as $n\to\infty$. If the conclusion of Theorem 2.1 is true for this example, we should have $\hat{\boldsymbol{\beta}}_n\to\boldsymbol{\beta}_0$ in pr. However, this is not the case.

Assume $\boldsymbol{\beta}_0=\mathbf{0}$ as earlier. For given values of $e_1, e_2, \cdots, e_n$ define a random variable ξ_n with distribution

$$P(\xi_n=e_i/x_i)=x_i/D_n,\ i=1,\cdots,n.$$

Since $Y_i=e_i$,

$$\sum_{i=1}^n |Y_i-x_i\boldsymbol{\beta}|=\sum_{i=1}^n x_i|e_i/x_i-\boldsymbol{\beta}|,$$

we see that the ML_1NE $\hat{\boldsymbol{\beta}}_n$ of $\boldsymbol{\beta}_0$ equals $\mathrm{med}(\xi_n)$, the median of ξ_n. Therefore, if $\hat{\boldsymbol{\beta}}_n\to 0$ in pr., we should have

$$P(\mathrm{med}(\xi_n)\geqslant\varepsilon)\to 0,\ \text{as } n\to\infty, \tag{4.1}$$

for any given $\varepsilon>0$. Define

$$Z_n=\sum_{i=1}^n x_iI(e_i/x_i\geqslant\varepsilon)/D_n.$$

Then we should have

$$P(Z_n>1/2)\to 0,\ \text{as } n\to\infty, \tag{4.2}$$

when (4.1) is true.

Choose $\omega\in(0,1)$ such that $1-u>\mathrm{e}^{-2u}$ when $0<u<\omega$. Choose $\varepsilon\in(0,(\omega/2)^{1/3})$. Since $\sup\limits_{i\geqslant 1}2(x_i\varepsilon)^3\leqslant 2\varepsilon^3<\omega$, for any $i\geqslant 1$ we have

$$P\left(\frac{e_i}{x_i}\geqslant\varepsilon\right)=\frac{1}{2}-\int_0^{x_i\varepsilon}3u^2\mathrm{d}u=\frac{1}{2}\{1-2(x_i\varepsilon)^3\}<\frac{1}{2}\exp(-4(x_i\varepsilon)^3).$$

Therefore, for any $1\leqslant i_1<\cdots<i_t\leqslant n$ we have

$$\begin{aligned}
&P(e_j/x_j\geqslant\varepsilon,\ j=i_1,\cdots,i_t;\ e_j/x_j<\varepsilon,\ j\neq i_1,\cdots,i_t,\ 1\leqslant j\leqslant n)\\
&>\left(\frac{1}{2}\right)^t\exp\Big(-4\sum_{j=1}^t(x_{i_j}\varepsilon)^3\Big)\left(\frac{1}{2}\right)^{n-t}\\
&=\frac{1}{2^n}\exp\Big(-4\sum_{j=1}^t(x_{i_j}\varepsilon)^3\Big)\\
&\geqslant\frac{1}{2^n}\exp(-4\varepsilon^3T).
\end{aligned}\tag{4.3}$$

Define

$$A_n=\{(i_1,\cdots,i_t):1\leqslant t\leqslant n,\ 1\leqslant i_1<\cdots<i_t\leqslant n,\ x_{i_1}+\cdots+x_{i_t}>D_n/2\},$$

then the number of elements in A_n is $2^n/2$. This follows from condition (2). Therefore

$$\begin{aligned}
&P(Z_n>1/2)\\
&=\sum_{(i_1,\cdots,i_t)\in A_n}P(e_j/x_j\geqslant\varepsilon,\ j=i_1,\cdots,i_t;\ e_j/x_j<\varepsilon,\ j\neq i_1,\cdots,i_t,\ 1\leqslant j\leqslant n)
\end{aligned}$$

$$> \frac{1}{2}2^n \frac{1}{2^n}\exp(-4\varepsilon^3 T)$$

$$= \frac{1}{2}\exp(-4\varepsilon T).$$

This show that (4. 2) is not true; hence (4. 1) is also not true. As indicated earlier, this proves that $\hat{\boldsymbol{\beta}}_n$ does not converge in probability to $\boldsymbol{\beta}_0$.

So we see that although it does not seem possible to prove the necessity of (b) in Theorem 2. 1, there is not much room left for improving this condition.

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线性模型参数 M 估计的线性表示

摘要 本文在非凸的情况下,研究了线性回归参数的 M 估计的强、弱线性表示. 所得余项估计的阶或阶的主部是确切的. 作为应用,导出了 M 估计的收敛速度、重对数律及 Berry-Esseen型界限.

1 引言与结果

考虑线性回归模型

$$y_i = x_i'\beta_0 + e_i,\ i = 1,\ \cdots,\ n, \tag{1.1}$$

其中 $x_1,\ \cdots,\ x_n$ 为已知的 p 维向量,$e_1,\ \cdots,\ e_n$ 为随机误差. 未知参数向量 β_0 属于$\mathbb{R}^p$中的一个给定的有界闭域 B.

设 ρ 为定义于$\mathbb{R}^1$的函数. β_0 的 M 估计定义为满足下式的 $\hat{\beta}_n$:

$$\sum_{i=1}^{n}\rho(y_i - x_i'\hat{\beta}_n) = \min_{\beta\in B}\sum_{i=1}^{n}\rho(y_i - x_i'\beta). \tag{1.2}$$

因 B 有界闭,只要 ρ 连续,这样的极值点 $\hat{\beta}_n$ 必存在,但不必唯一. 样本中位数是一个熟悉的例子.

除去 $\rho(u) = u^2$ 这一特例,求不出 $\hat{\beta}_n$ 的显式解,这给其研究带来困难. M 估计的线性表示,是把 $\hat{\beta}_n$ 表为一个独立和加上一个剩余项,它以一定速度 in pr. 或者 a. s. 收敛于 0. 前者称为弱表示而后者称为强表示,这种表示是研究 M 估计的更深层次的渐近性质的基础.

这方面的研究始于 Bahadur[1] 关于样本分位数强表示的工作. 1989 年 Babu[2] 研究了(1.1)式中参数 β_0 的最小一乘估计的强表示. 1992 年 Rao 和 Zhao[3] 将其推广到 ρ 为一般凸函数的情形. ρ 非凸的情况下至今没有任何结果,但这种情况有很大的重要性,尤其是在 M 估计稳健性和崩溃点的研究中. 例如,1984 年 Huber[4] 讨论了位置参数估计中 ρ 有界、或无界但 $\lim\limits_{|x|\to\infty}\rho(x)/x = 0$ 的情况,这种 ρ 都不是凸的(也可参见文献[5]). 从方法上说,ρ 为凸时的方法不能移用于非凸情况. 由此看来,ρ 非凸时 M 估计强表示的研究有较大的意义.

本文的目的就是研究这个问题及其若干重要应用,包括 M 估计强弱相合的速度,重对数

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律和 Berry-Esseen 型界限等. 首先我们引入一些记号,以后将不再一一说明. 记 $S_n = x_1x_1' + \cdots + x_nx_n'$.

设 n 甚大时 S_n^{-1} 存在. S_n 的最大和最小特征根分别记为 μ_n 和 λ_n. 记 $x_{ni} = S_n^{-1/2}x_i$, $1 \leqslant i \leqslant n$. 有

$$\sum_{i=1}^{n} x_{ni}x_{ni}' = I_p, \quad \sum_{i=1}^{n} \| x_{ni} \|^2 = p. \tag{1.3}$$

令

$$d_n^2 = \max_{1\leqslant i\leqslant n} \| x_{ni} \|^2 = \max_{1\leqslant i\leqslant n} x_i'S_n^{-1}x_i. \tag{1.4}$$

易见若 $\{x_i\}$ 有界,则存在常数 $c_1 > 0, c_2 > 0$, 使

$$c_1\mu_n^{-1} \leqslant d_n^2 \leqslant c_2\lambda_n^{-1}. \tag{1.5}$$

若 $a_n = O(b_n)$, $b_n = O(a_n)$, 则记为 $a_n \approx b_n$. 由(1.5)式知,若 $\{x_i\}$ 有界且 $\lambda_n \approx \mu_n$, 则 $d_n^2 \approx \lambda_n^{-1}$.

对矩阵 $A = (a_{ij})$, 以 $|A|$ 记 $\max|a_{ij}|$. 向量附以足标 j 记其第 j 分量. 例如, $\hat{\beta}_n$ 和 x_{ni} 的第 j 分量分别记为 $\hat{\beta}_{nj}$ 和 x_{nij}. 但为避免与 x_{ni} 混淆, x_i 的第 j 分量将记为 $x_{i(j)}$. 取极限将略去 $n\to\infty$. 例如 $\lim a_n$ 和 $\limsup a_n$ 分别指 $\lim\limits_{n\to\infty} a_n$ 和 $\limsup\limits_{n\to\infty} a_n$. Φ 记 $N(0, 1)$的分布函数. 现列述本文的假定.

(1) $e_1, e_2, \cdots$ 为 iid.

(2) ρ 非负, ρ 的二阶导数在 $\mathbb{R}^1$ 处处存在有界,且 ρ'' 满足 Lipschitz 条件, $\rho(e_1)$ 的各阶矩有限.

(3) 记 $h(t) = \mathrm{E}\rho(e_1 + t)$, 则 $h(t)$ 以 0 为唯一的最小值点. 记 $\psi = \rho'$, 则 $\sigma^2 \equiv \mathrm{E}\psi^2(e_1) > 0$, $g \equiv \mathrm{E}\psi'(e_1) > 0$.

(4) $\{x_i\}$ 有界,且存在 $\delta > 0$, 使

$$\liminf \lambda_n n^{-\delta} > 0. \tag{1.6}$$

令 $r_n = S_n^{1/2}(\hat{\beta}_n - \beta_0) - g^{-1}\sum_{i=1}^{n}\psi(e_i)x_{ni}$. 本文主要结果为

定理 1 在假定(1)~(4)满足时有

$$r_n = O(d_n \log n), \ \text{a. s.}; \ r_n = O_p(d_n), \tag{1.7}$$

后一式中的 $O_p(d_n)$, 除了 $\rho(u) = u^2$ 这一平凡情况及某种极特殊的情况(见例 1)外,不能改进为 $o_p(d_n)$.

又若还满足条件 $\lambda_n \approx \mu_n$, 则

$$r_n = O(d_n \log_2 n), \ \text{a. s.} \ (\log_2 n = \log\log n). \tag{1.8}$$

作为这个主要定理的应用,有

定理 2 在假定(1)~(4)满足时有

$$\hat{\beta}_n - \beta_0 = O(d_n \log^{1/2} n), \ \text{a. s.} \tag{1.9}$$

下一定理表明,(1.9)式在某种情况下尚可改进,但改进余地有限.

定理 3($\hat{\beta}_n$ 的重对数律) 以 ξ_n 记 $S_n^{1/2}(\hat{\beta}_n-\beta_0)$ 的第 j 分量，$\sigma_n^2=\mathrm{Var}(\xi_n)$. 设假定(1)～(4)满足，并有常数列 $\{c_n\}$ 和正定方阵 V，使

$$\lim S_n/c_n = V. \tag{1.10}$$

则

$$\lim \sup_{\inf} \xi_n/(2\sigma_n^2\log_2 n)^{1/2} = \pm 1, \text{ a. s.} \tag{1.11}$$

(在此处，inf 部分并不能从其 sup 部分直接推出，这与独立和不同.)

又若将 ξ_n 中心化，即以 $\xi_n-\mathrm{E}\xi_n$ 代 ξ_n，(1.11)式仍成立. (1.11)式显示，在特殊情况(1.10)式之下，(1.9)式中的 $\log^{1/2}n$ 可以改进为 $\log_2^{1/2}n$. 是否当(1.10)式不成立时仍有此改进？有此可能，但目前尚不得而知.

定理 4 ξ_n 的意义同定理 3. 以 F_n 记 $g\xi_n/\sigma$ 的分布函数，g，σ 的意义见假定 3. 若假定(1)～(4)成立，则有

$$\sup_{|x|<\infty} |F_n(x)-\Phi(x)| = O(d_n\log n). \tag{1.12}$$

假定 3 中关于 $h(t)$ 有唯一最小值点的要求是本质的，否则 $\hat{\beta}_n$ 甚至可以不为弱相合. 但通常，回归函数 $x'\beta_0$ 中总含有常数项 α_0（即 $x_{i(1)}=1$，$i\geqslant 1$）. 这时上述唯一最小值点 c 不必为 0，而不致影响对其他回归系数的估计. 但 α_0 应调整为 α_0-c.

ρ 的光滑性在保证余项 r_n 达到较高的数量级上是不能少的. 如在文献[3]中，在假定 ρ 虽然是凸函数但 ρ' 不处处存在时，只能达到 $O(d_n^{1/2}\log^{3/4}n)$ 的数量级. 且不难证明，这里 $d_n^{1/2}$ 中指数 1/2 已不能再升高. 一个重要的例子是最小一乘估计—— $\rho(u)=|u|$. 此问题将在另文中仔细讨论.

2 一些引理

引理 1 设假定(1)～(3)满足，则

1° $h(t)$ 处处有限，连续.

2° $\psi(e_1)$ 的各阶矩有限，$\mathrm{E}\,\psi(e_1)=0$.

3° 对给定的常数 $L>0$，存在只与 L 有关的常数 $c_1>0$，$c_2>0$，使当 $|t|\leqslant L$ 时有

$$h(t)-h(0)\geqslant c_1t^2;\ \mathrm{Var}(\rho(e_1+t)-\rho(e_1))\leqslant c_2t^2. \tag{2.1}$$

4° 给定常数 $L>0$，$m>0$. 存在只与 L,m 有关的常数 $c>0$，使当 $|t|\leqslant L$ 时有

$$\mathrm{E}\,|\rho(e_1+t)-\rho(e_1)|^m\leqslant c. \tag{2.2}$$

证 作 Taylor 展开

$$\rho(e_1\pm t)=\rho(e_1)\pm t\psi(e_1)+2^{-1}t^2\psi'(e_1+\theta_\pm t),\ |\theta_\pm|\leqslant 1, \tag{2.3}$$

相加，并注意 ρ 非负及 $M=\sup|\psi'|<\infty$，有 $\rho(e_1+t)\leqslant 2\rho(e_1)+Mt^2$. 因 $\rho(e_1)$ 的各阶矩有限，证明了 1°. 在(2.3)式中取 $t=1$ 得 $|\psi(e_1)|\leqslant\rho(e_1)+\rho(e_1+1)+M$. 这证明了 $\psi(e_1)$ 各阶矩有限. 又 $|\rho(e_1+t)-\rho(e_1)|\leqslant|t\psi(e_1)|+Mt^2$，可知(2.1)式后一式成立. 前一式由 $\mathrm{E}\psi(e_1)=0$（见下），$h(t)$ 连续，$h(t)>h(0)$ 当 $t\neq 0$ 及 $\lim\limits_{t\to 0}t^{-2}(h(t)-h(0))=g/2>0$ 推出. $\mathrm{E}\psi(e_1)=0$

则由 $h'(0)=\mathrm{E}\psi(e_1)$ 及 0 为 h 的极值点推出. 因 $\rho(e_1+t)$ 各阶矩有限,(2.2)式也显然. 引理证毕.

引理 2 若假定(1)～(4)满足,则

$$\sum_{i=1}^{n}\psi(e_i)x_{ni}=O(\log^{1/2}n),\ \text{a. s.}, \tag{2.4}$$

$$A_n\equiv\sum_{i=1}^{n}(\psi'(e_i)-g)x_{ni}x'_{ni}=O(d_n\log^{1/2}n),\ \text{a. s.} \tag{2.5}$$

若还满足条件 $\lambda_n\approx\mu_n$, 则可以改进为

$$\sum_{i=1}^{n}\psi(e_i)x_{ni}=O(\log_2^{1/2}n),\ \text{a. s.}, \tag{2.6}$$

$$A_n=O(d_n\log_2^{1/2}n),\ \text{a. s.}. \tag{2.7}$$

证 取 $\sum_{i=1}^{n}\psi(e_i)x_{ni}$ 的第 j 分量 $T_n\equiv\sum_{i=1}^{n}\psi(e_i)x_{nij}$, 任取常数 $c_n\to\infty$. 注意 $\mathrm{E}\psi(e_1)=0$ 且 $\psi(e_1)$各阶矩有限. 记

$$T_n/(c_n\log^{1/2}n)\equiv\sum_{i=1}^{n}a_{ni}\psi(e_i),\ a_{ni}=x_{nij}/(c_n\log^{1/2}n).$$

有 $\sum_{i=1}^{n}a_{ni}^2=\sum_{i=1}^{n}x_{nij}^2/(c_n^2\log n)\leqslant\sum_{i=1}^{n}\|x_{ni}\|^2/(c_n^2\log n)=p/(c_n^2\log n)=o(1/\log n)$. 又 $\max\limits_{1\leqslant i\leqslant n}|a_{ni}|\leqslant d_n/(c_n\log^{1/2}n)=o(d_n)=o(\lambda_n^{-1/2})=o(n^{-\delta/2})$ (见(1.5)和(1.6)式). 故按文献[7]的定理 4.1.3 有 $T_n/(c_n\log^{1/2}n)\to 0$, a. s.

由于 $c_n\to\infty$的任意性,证明了(2.4)式. (2.5)式的证明类似.

现设 $\lambda_n\approx\mu_n$ 成立. 先证(2.6)式. 取 $\sum_{i=1}^{n}\psi(e_i)x_i$ 的第 j 分量 $T_n\equiv\sum_{i=1}^{n}\psi(e_i)x_{i(j)}$. $\mathrm{E}T_n=0$, $\mathrm{Var}(T_n)\equiv\sigma_n^2\equiv\sigma^2\sum_{i=1}^{n}x_{i(j)}^2$. 以 F_n 记 T_n/σ_n 之分布函数,因为 $\mathrm{E}|\psi(e_i)|^3<\infty$,有

$$\begin{aligned}c_n&\equiv\sum_{i=1}^{n}\mathrm{E}|\psi(e_i)x_{i(j)}|^3=O\Big(\sum_{i=1}^{n}|x_{i(j)}^3|\Big)\\&=O(\max_{1\leqslant i\leqslant n}\|x_i\|\sigma_n^2).\end{aligned}$$

故 $c_n/\sigma_n^3=O(\max\limits_{1\leqslant i\leqslant n}\|x_i\|/\sigma_n)=O(\max\limits_{1\leqslant i\leqslant n}\|S_n^{1/2}x_{ni}\|/\sigma_n)$. 但 σ_n^2 为 S_n 的(j,j)元. 由 $\lambda_n\approx\mu_n$ 知存在 $c>0$ 使 $\sigma_n^2\geqslant c\mu_n$. 另一方面, $\max\limits_{1\leqslant i\leqslant n}\|S_n^{1/2}x_{ni}\|\leqslant p|S_n^{1/2}|d_n\leqslant p\mu_n^{1/2}d_n$. 总结以上可知 $c_n/\sigma_n^3=O(d_n)$. 因此由文献[6] p. 111 的定理 3,得

$$\sup_{|x|<\infty}|F_n(x)-\Phi(x)|=O(d_n). \tag{2.8}$$

据(1.5)与(1.6)式,因 $\lambda_n\approx\mu_n$, 有 $O(d_n)=O(n^{-\delta/2})$. 另一方面,因$\{x_i\}$有界,有 $\sigma_n^2=O(n)$. 故由(2.8)式有

$$\sup_{|x|<\infty} |F_n(x) - \Phi(x)| = O((\log \sigma_n^2)^{-2}).$$

因此据文献[6] p. 305 的定理 3,知对 T_n 适用重对数律:

$$\lim{}_{\inf}^{\sup} T_n/(2\sigma_n^2 \log_2 \sigma_n^2)^{1/2} = \pm 1, \text{ a. s.}. \tag{2.9}$$

这对 $\sum_{i=1}^n \psi(e_i)x_i$ 的每一分量都对. 由于 $\sum_{i=1}^n \psi(e_i)x_{ni} = S_n^{-1/2}\sum_{i=1}^n \psi(e_i)x_i$, 及由 $\lambda_n \approx \mu_n$ 得出的 $|S_n^{-1/2}| = O(\mu_n^{-1/2})$ 和 $\sigma_n = O(\mu_n^{1/2})$, 以及由 $n^{\delta/2} = O(\lambda_n) = O(\sigma_n^2)$ 和 $\sigma_n^2 = O(n)$ 而推出的 $\log_2\sigma_n^2 \approx \log_2 n$, 由(2.9)式即得(2.4)式.

现证(2.7)式. 记 $Z_i = \psi'(e_i) - g$. 注意因 ψ' 有界,知$\{Z_i\}$一致有界. $EZ_i = 0$. 以 σ_0^2 记 Z_i 的方差. 令 $D_n = \sum_{i=1}^n Z_i x_i x_i'$. 取 D_n 的(j, k)元 $T_n \equiv \sum_{i=1}^n Z_i x_{i(j)} x_{i(k)}$,记 $b_n^2 = \sum_{i=1}^n x_{i(j)}^2 x_{i(k)}^2$. 分两种情况讨论:

(1) $b_n \to \infty$. 这时 Kolmogorov 的重对数律可用,这是因为$\{Z_i\}$及$\{x_i\}$都有界因而$\{Z_i x_{i(j)} x_{i(k)}, i \geqslant 1\}$也有界. 因此

$$\lim{}_{\inf}^{\sup} T_n/(2b_n^2\sigma_0^2 \log_2(b_n^2\sigma_0^2))^{1/2} = \pm 1, \text{ a. s.}. \tag{2.10}$$

因为 $\lambda_n \approx \mu_n$, 有 $\sum_{i=1}^n x_{i(j)}^2 \leqslant \sum_{i=1}^n \|x_i\|^2 = \mathrm{tr}(S_n) = O(\lambda_n)$, 又在前面已证

$$\max_{1\leqslant i\leqslant n} |x_{i(k)}| = O(\mu_n^{1/2} d_n) = O(\lambda_n^{1/2} d_n),$$

因而

$$b_n^2 \leqslant \max_{1\leqslant i\leqslant n} x_{i(k)}^2 \sum_{i=1}^n x_{i(j)}^2 = O(\lambda_n^2 d_n^2).$$

由此式以及 $\lambda_n = O(n)$, 从(2.10)式得

$$|D_n| = O(\lambda_n d_n \log_2^{1/2} n), \text{ a. s.}. \tag{2.11}$$

因为 $A_n = S_n^{-1/2} D_n S_n^{-1/2}$, 且前已指出$|S_n^{-1/2}| = O(\lambda_n^{-1/2})$,由(2.11)式即得(2.7)式.

(2) b_n 有界. 我们需要由文献[8]推出的一个结果:设 $\xi_1, \xi_2, \cdots$为 iid., $E\xi_1 = 0$, $E\xi_1^{2r} < \infty$, r 为自然数,则有

$$\sup\left\{E\left|\sum_{i=1}^n a_i\xi_i\right|^{2r}: \sum_{i=1}^n a_i^2 = 1, n \geqslant 1\right\} < \infty. \tag{2.12}$$

取 D_n 的(j, k)元 $T_n \equiv \sum_{i=1}^n Z_i x_{i(j)} x_{i(k)}$. 注意到 $EZ_i = 0$, Z_i 有界,因而各阶矩有限,以及 b_n 有界,由(2.12)式易知 $E|d_n T_n|^{2r} = O(d_n^r) = O(n^{-r\delta/2})$(这后一步用到 $\lambda_n \approx \mu_n$). 取 $r > 4/\delta$ 得 $E|d_n T_n|^{2r} = O(n^{-2})$, 因而 $d_n T_n \to 0$, a. s., 即 $T_n = O(d_n^{-1})$, 故 $|D_n| = O(d_n^{-1})$. 再利用 $A_n = S_n^{-1/2} D_n S_n^{-1/2}$ 及 $|S_n^{-1/2}| = O(\lambda_n^{-1/2}) = O(d_n)$, 即得 $A_n = O(d_n)$, 从而证明了(2.7)式. 引理证毕.

引理 3 设假定(1)～(4)满足,仍设 $\beta_0 = 0$. 则对任给 $\alpha>0$,有

$$P(\|S_n^{1/2}\hat{\beta}_n\| \geqslant n^\alpha, \text{ i. o.}) = 0. \tag{2.13}$$

证 我们需要 Fuk 和 Nagaev 的一个不等式[9]：设 $\xi_1, \cdots, \xi_n$ 独立，$\mathrm{E}\xi_i = 0$，$1 \leqslant i \leqslant n$. 则对任何 $m \geqslant 2$ 存在只与 m 有关的常数 $c_1 > 0$，$c_2 > 0$，使

$$P\left(\left|\sum_{i=1}^{n} \xi_i\right| \geqslant u\right) \leqslant c_1 u^{-m} \sum_{i=1}^{n} \mathrm{E} \mid \xi_i \mid^m + 2\exp\left(-c_2 u^2 \Big/ \sum_{i=1}^{n} \mathrm{Var}(\xi_i)\right), \quad u > 0. \tag{2.14}$$

现为方便计且不失普遍性，令 $\beta_0 = 0$. 这时有 $y_i = e_i$. 记 $B_n = B \cap \{\beta: \| S_n^{1/2}\beta \| \geqslant n^a\}$. 把 $\mathbb{R}^p$ 剖分为一些立方体 $J(l_1, \cdots, l_p) = \{(u_1, \cdots, u_p): l_i n^{-3} \leqslant u_i < (l_i + 1)n^{-3}, 1 \leqslant i \leqslant p, l_i = 0, \pm 1, \cdots, 1 \leqslant i \leqslant p\}$. 若 $J(l_1, \cdots, l_p)$ 与 B_n 有非空交，则从交中取出一点. 所有取出之点构成一集 G_n. 因 B 有界，当 n 充分大时 G_n 所含点数 $\#(G_n) \leqslant n^{3p+1}$. 现取定 $\beta \in G_n$. 记

$$\xi_i = \rho(e_i - x_i'\beta) - \rho(e_i) - (h(-x_i'\beta) - h(0)), \ i = 1, 2, \cdots.$$

$$\Delta_n(\beta) = \sum_{i=1}^{n} [\rho(e_i - x_i'\beta) - \rho(e_i)]$$

$$= \sum_{i=1}^{n} [h(-x_i'\beta) - h(0)] + \sum_{i=1}^{n} \xi_i,$$

$$b_n(\beta) = \sum_{i=1}^{n} [h(-x_i'\beta) - h(0)],$$

$$V_n^2(\beta) = \sum_{i=1}^{n} \mathrm{Var}(\rho(e_i - x_i'\beta) - \rho(e_i)).$$

因 $\{x_i\}$ 和 B 都有界，按(2.1)式，存在常数 $h_1 > 0$，$h_2 > 0$，使

$$b_n(\beta) \geqslant h_1 \| S_n^{1/2}\beta \|^2 \geqslant h_1 n^{2a}, \ V_n^2(\beta) \leqslant h_2 \| S_n^{1/2}\beta \|^2. \tag{2.15}$$

取 $m \geqslant 2$ 待定. 按(2.2)式，存在常数 $a_m > 0$，使 $\sup\limits_n \mathrm{E}|\xi_n|^m \leqslant a_m < \infty$. 于是由(2.14)，(2.15)式，得

$$P(\Delta_n(\beta) \leqslant 2^{-1}h_1 n^{2a}) \leqslant P\left(\sum_{i=1}^{n} \xi_i \leqslant -2^{-1}h_1 \| S_n^{1/2}\beta \|\right)$$

$$\leqslant c_1(2^{-1}h_1 n^{2a})^{-m} n a_m + 2\exp(-c_2 h_1^2 \| S_n^{1/2}\beta \|^4 / (4h_2 \| S_n^{1/2}\beta \|^2))$$

$$\leqslant c_1(2^{-1}h_1 n^{2a})^{-m} n a_m + 2\exp(-c_2 h_1^3 n^{2a}/4k_2).$$

由此式及 $\#(G_n) \leqslant n^{3p+1}$，得

$$P(\min_{\beta \in G_n} \Delta_n(\beta) \leqslant 2^{-1}h_1 n^{2a}) \leqslant c_1' n^{-2am+3p+2} + 2n^{3p+1}\exp(-c_2' n^{2a}), \tag{2.16}$$

其中 $c_1' > 0$，$c_2' > 0$，都与 n 无关.

现任取 $\beta \in B_n$. 找 $(l_1, \cdots, l_p)$，使 $\beta \in J(l_1, \cdots, l_p)$. 按 G_n 的定义，存在 $\beta^* \in G_n \cap J(l_1, \cdots, l_p)$. 有

$$\| \beta - \beta^* \| \leqslant \sqrt{p} n^{-3}.$$

因 $\{x_i\}$，ψ' 和 B 都有界，把 $\rho(e_i - x_i'\beta) - \rho(e_i - x_i'\beta^*)$ 在点 $e_i - x_i'\beta$ 处作 Taylor 展开，易见存在与 n 无关的常数 M，使

$$A_n(\beta) \equiv \sum_{i=1}^{n}[\rho(e_i - x_i'\beta) - \rho(e_i - x_i'\beta^*)]$$

$$\leqslant M\Big[\sum_{i=1}^{n} |\psi(e_i)| n^{-3} + n^{-3} + \mu_n \|\beta - \beta^*\|^2\Big].$$

因 $\mu_n = O(n)$, 有 $\mu_n \|\beta - \beta^*\|^2 = O(n^{-5})$. 又

$$P\Big(\sum_{i=1}^{n} |\psi(e_i)| n^{-3} \geqslant 1\Big) \leqslant n^{-6}\mathrm{E}\Big(\sum_{i=1}^{n} |\psi(e_i)|\Big)^2 \leqslant \sigma^2 n^{-4}.$$

由以上可知,存在 M_1 与 n 无关,使

$$P(\sup_{\beta\in B_n} A_n(\beta) \geqslant M_1) = O(n^{-4}). \tag{2.17}$$

现定义事件 $E_n = \{\min_{\beta\in B_n}\Delta_n(\beta) \leqslant 0\}$. 因 $\Delta_n(\beta) = -A_n(\beta) + \Delta_n(\beta^*)$,有

$$W_n \equiv \min_{\beta\in B_n}\Delta_n(\beta) \geqslant -\sup_{\beta\in B_n} A_n(\beta) + \min_{\beta\in G_n}\Delta_n(\beta).$$

由此式及(2.17)式,知

$$P(W_n < -M_1 + \min_{\beta\in G_n}\Delta_n(\beta)) = O(n^{-4}). \tag{2.18}$$

但易见

$$\{\|S_n^{1/2}\hat{\beta}_n\| \geqslant n^a\} \subset E_n,$$

$$E_n \subset \{W_n < -M_1 + \min_{\beta\in G_n}\hat{\beta}\,\Delta_n(\beta)\} \cup \{\min_{\beta\in G_n}\Delta_n(\beta) \leqslant 2^{-1}h_1 n^{2a}\}.$$

由此及(2.16),(2.18)式,得

$$P(\|S_n^{1/2}\hat{\beta}_n \geqslant n^a) \leqslant O(n^{-4}) + c_1' n^{-2am+3p+2} + 2n^{3p+1}\exp(-c_2' n^{2a}).$$

取 $m \geqslant 2$ 使 $2am - 3p - 2 \geqslant 4$. 由上式得

$$P(\|S_n^{1/2}\hat{\beta}_n\| \geqslant n^a) = O(n^{-4}). \tag{2.19}$$

由(2.19)式,据 Borel-Cantelli 引理,即得(2.13)式. 引理证毕.

3 定理 1 的证明

以下仍设 $\beta_0 = 0$. 令

$$\begin{aligned} \beta_n &= S_n^{1/2}\hat{\beta}_n, \quad \bar{\beta}_n = g^{-1}\sum_{i=1}^{n}\psi(e_i)x_{ni}, \\ H_n &= \sum_{i=1}^{n}[\rho(e_i - x_{ni}'\beta_n) - \rho(e_i - x_{ni}'\bar{\beta}_n)]. \end{aligned} \tag{3.1}$$

按 M 估计的定义,应有 $H_n \leqslant 0$. 另一方面,把 $\rho(e_i - x_{ni}'\beta_n)$在点 $e_i - x_{ni}'\bar{\beta}_n$ 作 Taylor 展开,有

$$
\begin{aligned}
H_n = & \sum_{i=1}^{n}\psi(e_i - x'_{ni}\bar{\beta}_n)x'_{ni}(\bar{\beta}_n - \beta_n) \\
& + (\beta_n - \bar{\beta}_n)'\sum_{i=1}^{n}\psi'(e_i - x'_{ni}\bar{\beta}_n + \theta_{1i}x'_{ni}(\bar{\beta}_n - \beta_n))x_{ni}x'_{ni}(\beta_n - \bar{\beta}_n)/2 \\
= & \sum_{i=1}^{n}\psi(e_i)x'_{ni}(\bar{\beta}_n - \beta_n) - \bar{\beta}'_n\sum_{i=1}^{n}\psi'(e_i)x_{ni}x'_{ni}(\bar{\beta}_n - \beta_n) \\
& - \bar{\beta}'_n\sum_{i=1}^{n}[\psi'(e_i - \theta_{2i}x'_{ni}\bar{\beta}_n) - \psi'(e_i)]x_{ni}x'_{ni}(\bar{\beta}_n - \beta_n) \\
& + 2^{-1}g\|\beta_n - \bar{\beta}_n\|^2 + 2^{-1}(\beta_n - \bar{\beta}_n)'\sum_{i=1}^{n}(\psi'(e_i) - g)x_{ni}x'_{ni}(\beta_n - \bar{\beta}_n) \\
& + 2^{-1}(\beta_n - \bar{\beta}_n)'\sum_{i=1}^{n}[\psi'(e_i - x'_{ni}\bar{\beta}_n + \theta_{3i}x'_{ni}(\bar{\beta}_n - \beta_n)) - \psi'(e_i)]x_{ni}x'_{ni}(\beta_n - \bar{\beta}_n)'.
\end{aligned}
$$

把此式右边第 3，5，6 项依次记为 J_2，J_3，J_4. 把第 2 项中的 $\sum_{i=1}^{n}\psi'(e_i)x_{ni}x_{ni}$ 表为

$$
gI_p + \sum_{i=1}^{n}(\psi'(e_i) - g)x_{ni}x'_{ni} \equiv gI_p + A_n,
$$

而将$-\bar{\beta}'_nA_n(\bar{\beta}_n - \beta_n)$记为 J_1. 再注意到$\bar{\beta}_n$ 的定义(3.1),得

$$
H_n = 2^{-1}g\|\beta_n - \bar{\beta}_n\|^2 + \sum_{i=1}^{n}J_i. \tag{3.2}
$$

考虑到 $|\theta_{ji}| \leqslant 1$, $j = 1, 2, 3$, $\sum_{i=1}^{n}\|x_{ni}\|^2 = p$, 以及 ψ'满足 Lipschitz 条件,得

$$
|J_1| \leqslant C|A_n| \|\bar{\beta}_n\| \|\beta_n - \bar{\beta}_n\|, \quad |J_2| \leqslant Cd_n\|\bar{\beta}_n\|^2\|\beta_n - \bar{\beta}_n\|,
$$

$$
|J_3| \leqslant C|A_n| \|\beta_n - \bar{\beta}_n\|^2, \quad |J_4| \leqslant C(d_n\|\beta_n\| + d_n\|\bar{\beta}_n\|)\|\beta_n - \bar{\beta}_n\|^2,
$$

此处 C 为常数,与 n 无关. 由这些估计及(3.2)式,得

$$
\begin{aligned}
0 \geqslant H_n \geqslant & [(2^{-1}g - C|A_n| - Cd_n\|\beta_n\| - Cd_n\|\bar{\beta}_n\|)\|\beta_n - \bar{\beta}_n\| \\
& - C|A_n|\|\bar{\beta}_n\| - Cd_n\|\bar{\beta}_n\|^2]\|\beta_n - \bar{\beta}_n\|.
\end{aligned} \tag{3.3}
$$

据引理 2 及 $d_n = O(n^{-\delta/2})$, 有 $|A_n| = o(1)$, a. s. , $d_n\|\bar{\beta}_n\| = o(1)$, a. s. . 在引理 3 中取 $\alpha < \delta/2$, 得到 $d_n\|\beta_n\| = o(1)$, a. s. . 又由引理 2 有 $|A_n|\|\bar{\beta}_n\| = O(d_n\log n)$, a. s. , $d_n\|\bar{\beta}_n\|^2 = O(d_n\log n)$, a. s. . 综合这些及(3.3)式,得

$$
0 \geqslant H_n \geqslant [(g/2 - o(1))\|\beta_n - \bar{\beta}_n\| - O(d_n\log n)]\|\beta_n - \bar{\beta}_n\|, \text{ a. s. .} \tag{3.4}
$$

因 $g > 0$, 这显然只在 $\|\beta_n - \bar{\beta}_n\| = O(d_n\log n)$ 才可能. 由此证明了(1.7)式的第一式.

若 $\lambda_n \approx \mu_n$, 则可用(2.6)和(2.7)式代替(3.4)式,将有

$$
0 \geqslant H_n \geqslant [(g/2 - o(1))\|\beta_n - \bar{\beta}_n\| - O(d_n\log_2 n)]\|\beta_n - \bar{\beta}_n\|, \text{ a. s. .} \tag{3.5}
$$

由此推出(1.8)式.

(1.7)式的第二式可根据(3.3)式以及

$$|A_n| = O_p(d_n),\ \|\bar{\beta}_n\| = O(1)$$

推出,这是因为 $\mathrm{E}\|\bar{\beta}_n\|^2 = g^{-2}\sigma^2$,且对 A_n 之任一分量 T_n 有 $\mathrm{E}|T_n^2| = O(d_n^2)$.

为证(1.7)式第二式的数量级 $O_p(d_n)$一般不能改善,考虑位置参数模型

$$y_i = \beta_0 + e_i,\ i = 1,\ \cdots,\ n. \tag{3.6}$$

定理 1 的假定(4)自动满足且 $d_n = n^{-1/2}$. 选 ρ 和 $\{e_i\}$ 使假定(1)~(3)满足,且 $0 < \sigma_0^2 \equiv \mathrm{Var}(\psi'(e_1)) < \infty$, ψ''在$\mathbb{R}^1$有界一致连续,且 $\mathrm{E}\psi''(e_1) = 0$, $\psi'''(0)$存在有限. 例如,$\rho(u) \equiv u^4/(1+u^2)$,而 e_i 为 iid., $e_1 \sim N(0,\ 1)$. 且上述条件皆满足(见注 1).

在上述条件下,β_0 的 M 估计 $\hat{\beta}_n$ 为方程 $\sum_{i=1}^n \psi(\gamma_i - \beta) = 0$ 的解. 为方便计,设 $\beta_0 = 0$, 而在点 $y_i = e_i$ 作 Taylor 展开,有

$$\sum_{i=1}^n \psi(e_i) - \sum_{i=1}^n \psi'(e_i)\hat{\beta}_n + 2^{-1}\sum_{i=1}^n \psi''(e_i - \theta_i\hat{\beta}_n)\hat{\beta}_n = 0,\ 0 \leqslant \theta_i \leqslant 1. \tag{3.7}$$

前已证 $\hat{\beta}_n = O_p(n^{-1/2}) = o_p(1)$. 故由 $\psi'''(0)$存在有限,知

$$\psi''(e_i - \theta_i\hat{\beta}_n) = \psi''(e_i) + O_p(1)\hat{\beta}_n.$$

以此代入(3.7)式,得

$$\begin{aligned}&\Big(\sum_{i=1}^n \psi(e_i) - ng\hat{\beta}_n\Big) - \sum_{i=1}^n (\psi'(e_i) - g)\hat{\beta}_n + 2^{-1}\sum_{i=1}^n \psi''(e_i)\hat{\beta}_n^2 + O_p(1)n\hat{\beta}_n^3 \\ &\equiv L_1 - L_2 + L_3 + L_4 = 0.\end{aligned} \tag{3.8}$$

由 $\mathrm{E}\psi''(e_1) = 0$ 及 ψ''有界,知

$$\sum_{i=1}^n \psi''(e_i) = O_p(n^{1/2}).$$

又 $\hat{\beta}_n = O_p(n^{-1/2})$,故

$$L_3 = O_p(n^{-1/2}),\ L_4 = O_p(n^{-1/2}). \tag{3.9}$$

注意对模型(3.6)有 $\bar{\beta}_n = g^{-1}\sum_{i=1}^n \psi(e_i)/\sqrt{n}$, $S_n^{1/2}\hat{\beta}_n = \sqrt{n}\hat{\beta}_n$, 故 $L_1 = -\sqrt{n}gr_n$. 现若有 $r_n = o_p(d_n) = o_p(n^{-1/2})$, 则将有 $L_1 = o_p(1)$, 因而由(3.8)及(3.9)式将得 $L_2 = o_p(1)$. 但这显然不可能,因为

$$\sum_{i=1}^n (\psi'(e_i) - g)/\sqrt{n} \xrightarrow{\mathscr{L}} N(0,\ \sigma_0^2),\ \sqrt{n}\hat{\beta}_n \xrightarrow{\mathscr{L}} N(0,\ \sigma^2/g^2).$$

故 r_n 不能是 $o_p(n^{-1/2}) = o_p(d_n)$, 从而完成了定理 1 的证明.

注 1　为验证本例满足上述一切条件,只需验证一条:即函数

$$J(a) = \int_{-\infty}^{\infty} \exp(-(t-a)^2/2)t^4(1+t^2)^{-1}\mathrm{d}t$$

有惟一最小值点 $a=0$. 其他条件皆显然. 有

$$J'(a)=\int_{-\infty}^{\infty}(t-a)\exp(-(t-a)^2/2)t^4(1+t^2)^{-1}\mathrm{d}t$$

$$=\int_0^{\infty}\mathrm{e}^{-u^2/2}u\left(\frac{(u+a)^2}{1+(u+a)^2}-\frac{(u-a)^2}{1+(u-a)^2}\right)\mathrm{d}u.$$

第一个等式给出 $J'(0)=0$, $J'(-a)=J'(a)$. 因函数 $u^4(1+u^2)^{-1}$ 在 $u\geqslant 0$ 处严增,第二等式给出 $J'(a)>0$ 当 $a>0$. 这证明了 $a=0$ 为惟一最小值点.

注 2 回到(3.8)式,若以概率 1 有 $\psi'(e_1)=g$, 则 $L_2=0$. 而

$$r_n=O_p(n^{-1})=O_p(d_n^2),$$

且仅在 $\mathrm{Var}(\psi''(e_1))=0$, $\mathrm{E}\psi''(e_1)=0$ 和 $\psi'''(0)=0$ 时,r_n 才能有更高的数量级,一个平凡例子是 $\rho(u)=u^2$, 即 LS 估计. 也可以造出非平凡的例子. 为简单计,以下给出一个满足 $r_n=O_p(d_n^2)$ 的例子.

例 1 仍考虑位置参数模型(3.6). 取

$$\rho(u)=\frac{u^2}{1+u^2},$$

$e_1, e_2, \cdots$ 为 iid. 且 e_1 有分布

$$P(e_1=a)=P(e_1=-a)=1/2,$$

此处 $0<a<1/(2\sqrt{3})$.

易验证假定(1)~(4)都成立. 因 ψ'为偶函数而 $P(|e_1|=a)=1$, 有 $\mathrm{Var}(\psi'(e_1))=0$. 据前面的论证,有 $r_n=O_p(n^{-1})=O_p(d_n^2)$.

4 定理 2~4 的证明

仍设 $\beta_0=0$.

定理 2 的证 有

$$\hat{\beta}_n=S_n^{-1/2}\bar{\beta}_n+S_n^{-1/2}r_n. \tag{4.1}$$

据(1.7)式及 $|S_n^{-1/2}|=O(\lambda_n^{-1/2})=O(n^{-\delta/2})$ 知 $S_n^{-1/2}r_n=O(d_n\log^{1/2}n)$. 其次,由(2.4)式有 $\|\bar{\beta}_n\|=O(\log^{1/2}n)$. 故由(4.1)式得

$$\hat{\beta}_n=O(\lambda_n^{-1/2}\log^{1/2}n)+O(d_n\log^{1/2}n). \tag{4.2}$$

现往证 $\lambda_n^{-1/2}=O(d_n)$. 若此得证,则由(4.2)式立即证得(1.9)式,为此要利用 Neumann 的一个如果:若 $m\leqslant n$ 且 S_n^{-1} 存在,以 $\theta_{k1}\leqslant\cdots\leqslant\theta_{kp}$ 记 S_k 的特征根,则

$$\sum_{j=1}^m x_j'S_n^{-1}x_j\geqslant\sum_{i=1}^p\theta_{mi}/\theta_{ni}$$

(见文献[10], p. 205). 由此式得

$$md_n^2 \geqslant \sum_{j=1}^{m} x_j' S_n^{-1} x_j \geqslant \theta_{m1}/\theta_{n1} = \lambda_m/\lambda_n.$$

因为 $\lambda_m > 0$，由上式即得 $\lambda_n^{-1/2} = O(d_n)$. 于是证明了(1.9)式的第一式. 第二式由弱表示定理(1.7)式，$|S_n^{-1/2}| = O(d_n)$ 及 $\|\bar{\beta}_n\| = O_p(1)$，由(4.1)式立即得出. 定理 2 证毕.

定理 3 的证 首先注意：当(1.6)和(1.10)式成立，且$\{x_i\}$有界时，必有

$$n^{\delta} = O(c_n),\ c_n = O(n),\ \lim(\log_2 c_n/\log_2 n) = 1. \tag{4.3}$$

令 $T_n = \sum_{i=1}^{n} V^{-1/2} x_i \psi(e_i)$. 以 a_{ij} 记 $V^{-1/2}x_i$ 的 j 分量，则 T_n 的 j 分量 T_{nj} 为

$$\sum_{i=1}^{n} a_{ij}\psi(e_i).$$

引理 2 中证明(2.9)式的论据完全适用于此处的 T_{nj}. 因为

$$\sum_{i=1}^{n} a_{ij}^2 = \text{方阵 } V^{-1/2} S_n V^{-1/2} \text{ 的}(j,\ j)\text{ 元},$$

而由(1.10)式可得

$$V^{-1/2} S_n V^{-1/2} = c_n I_p + o(c_n).$$

于是有

$$\lim\Big(\sum_{i=1}^{n} a_{ij}^2 / c_n\Big) = 1. \tag{4.4}$$

把(2.9)式用于 T_{nj}，并注意(4.3)和(4.4)式，得

$$\lim\nolimits_{\inf}^{\sup} T_{nj}/(c_n \sigma^2 2\log_2 n)^{1/2} = \pm 1,\ \text{a. s.} \tag{4.5}$$

因为

$$\bar{\beta}_n = g^{-1} S_n^{-1/2} V^{/12} T_n = g^{-1} c_n^{-1} T_n + o(c_n^{-1/2} T_n),$$

由(4.5)式得

$$\lim\nolimits_{\inf}^{\sup} \bar{\beta}_{nj}/(g^{-2}\sigma^2 2\log_2 n)^{1/2} = \pm 1,\ \text{a. s.} \tag{4.6}$$

由(1.7)及(4.6)式，得

$$\lim\nolimits_{\inf}^{\sup} \beta_{nj}/(g^{-2}\sigma^2 2\log_2 n)^{1/2} = \pm 1,\ \text{a. s.} \tag{4.7}$$

由(4.7)式看出，(1.11)式的证明归结为

$$\lim(\mathrm{Var}(\beta_{nj})/g^{-2}\sigma^2) = 1. \tag{4.8}$$

往证(4.8)式. 为简化表述，以下用 c_0 表示一个与 n 无关的常数，每次出现时可取不同之正值. 定义事件

$$E_n = \{\|\beta_n\| \leqslant n^{\alpha}\} \cap \{|A_n| < \varepsilon\} \cap \{d_n \|\bar{\beta}_n\| < \varepsilon\}. \tag{4.9}$$

$\varepsilon \in (0,\ g/8C)$，$C$ 见(3.3)式，A_n 见(2.5)式. 取 A_n 之$(j,\ k)$元并记为 T_n. 注意到

$$\sum_{i=1}^{n} x_{nij}^2 x_{nik}^2 \leqslant d_n^2 = O(n^{-\delta}),$$

由(2.12)式可得 $\mathrm{E}T_n^{2r} = O(n^{-r\delta/2}) = O(n^{-3})$，只要 $r \geqslant 6\delta^{-1}$. 于是

$$P(|T_n| \geqslant \varepsilon) = O(n^{-3}).$$

类似地证明 $P(d_n \|\bar{\beta}_n\| > \varepsilon) = O(n^{-3})$. 由此及(2.19)式，得

$$P(E_n^c) = O(n^{-3}). \tag{4.10}$$

取引理 3 中的 $\alpha < \delta/4$. 因 $d_n = O(n^{-\delta/2})$，当事件 E_n 发生时有 $d_n \|\beta_n\| = O(n^{-\delta/4})$. 于是由(3.3)式知当事件 E_n 发生时，有

$$0 \geqslant H_n \geqslant [(g/2 - g/4 - O(n^{-\delta/4}))\|\beta_n - \bar{\beta}\| - 2C\varepsilon\|\bar{\beta}_n\|]\|\beta_n - \bar{\beta}_n\|. \tag{4.11}$$

由(4.11)式知存在常数 $c_0 > 0$ 使

$$|\beta_n - \bar{\beta}_n| \leqslant c_0\varepsilon\|\bar{\beta}_n\| \tag{4.12}$$

在 E_n 上. 有

$$\begin{aligned}
|\mathrm{E}\beta_{nj}^2 - g^{-2}\sigma^2| &= |\mathrm{E}\beta_{nj}^2 - \mathrm{E}\bar{\beta}_{nj}^2| \\
&\leqslant |\mathrm{E}(I(E_n)\beta_{nj}^2) - \mathrm{E}(I(E_n)\bar{\beta}_{nj})| + \mathrm{E}(I(E_n^c)\beta_{nj}^2) + \mathrm{E}(I(E_n^c)\bar{\beta}_{nj}^2) \\
&\equiv J_1 + J_2 + J_3.
\end{aligned}$$

此处 I 为指示函数. 在 E_n 上有

$$\begin{aligned}
|\beta_{nj}^2 - \bar{\beta}_{nj}^2| &\leqslant |\beta_{nj} - \bar{\beta}_{nj}|(\|\beta_n\| + \|\bar{\beta}_n\|) \\
&\leqslant c_0\varepsilon\|\beta_n\|(2 + c_0\varepsilon)\|\beta_n\| \\
&= c_0\varepsilon(2 + c_0\varepsilon)\|\beta_n\|^2.
\end{aligned}$$

因此 $J_1 \leqslant c_0\varepsilon(2+c_0\varepsilon)\mathrm{E}\|\bar{\beta}_n\|^2 = c_0\varepsilon(2+c_0\varepsilon)p\sigma^2$，当取 $\varepsilon > 0$ 充分小时，它可以任意小. 因 $\{x_i\}$ 和 B 都有界，有 $|\beta_n| \leqslant c_1 n^{1/2}$，对某常数 c_1. 故由(4.10)式有 $J_2 \leqslant c_1^2 nO(n^{-3}) \to 0$. 又因 $\{\mathrm{E}\bar{\beta}_{nj}^3 : n \geqslant 1\}$ 有界，知 $\{\bar{\beta}_{nj}^2 : n \geqslant 1\}$ 一致可积，故 $J_3 \to 0$. 这证明了(4.8)式因而得(4.11)式. 最后，从(4.12)式出发. 用类似于上述的论证，易证 $\mathrm{E}\beta_{nj} \to 0$，故(1.11)式中的 $\xi_n - \mathrm{E}\xi_n$ 可用 ξ_n 取代. 定理 3 证毕.

定理 4 的证 需要以下的不等式：设 $\xi_1, \cdots, \xi_n$ 独立，各有期望 0，又 $|\xi_i| \leqslant b < \infty$, $i = 1, \cdots, n$. 记 $q = \sum_{i=1}^{n} \mathrm{E}\xi_i^2$，则对任何 $\varepsilon > 0$ 有[11]

$$P\left(\left|\sum_{i=1}^{n}\xi_i\right| \geqslant \varepsilon\right) \leqslant 2\exp\left(-\frac{\varepsilon^2}{2q + 2b\varepsilon}\right). \tag{4.13}$$

现回到(3.3)式，在推导(4.11)式的过程中我们曾证明

$$g/2 - C|A_n| - Cd_n\|\beta_n\| - Cd_n\|\bar{\beta}_n\| \geqslant g/4 - O(n^{-\delta/4}).$$

此式当 E_n 发生时成立. 由此及(3.3)式知存在常数 C_0,使

$$\| \beta_n - \bar{\beta}_n \| \leqslant c_0 \mid A_n \mid \| \bar{\beta}_n \| + c_0 d_n \| \bar{\beta}_n \|^2. \tag{4.14}$$

以$\bar{F}_n$ 记 $g\bar{\beta}_{nj}/\sigma$ 的分布函数. 用文献[6] p.111 的定理 3,易得

$$\sup_{|x|<\infty} \mid \bar{F}_n(x) - \Phi(x) \mid = O(d_n). \tag{4.15}$$

定义事件 $E_{n1} = \{ \mid \bar{\beta}_{nj} \mid \leqslant g^{-1}\sigma \log^{1/2} n \}$. 则由(3.18)式得

$$P(E_{n1}^c) \leqslant O(d_n) + 2[1 - \Phi(\log^{1/2} n)] = O(d_n) + O(n^{-1/2}) = O(d_n). \tag{4.16}$$

取 A_n 的 $(j,\ k)$ 元 $T_n \equiv \sum_{i=1}^n (\psi'(e_i) - g) x_{nij} x_{nik}$. 记 $\mathrm{Var}(\psi'(e_1)) = \sigma_0^2$. $\mid \psi'(e_i) - g \mid \leqslant \sup \mid \psi' \mid + g \equiv b < \infty$, 又 $\sum_{i=1}^n x_{nij}^2 x_{nik}^2 \leqslant d_n^2$. 于是 $\mathrm{Var}(T_n) \leqslant \sigma_0^2 d_n^2$. 定义事件

$$E_{n2} = \{ \mid T_n \mid \leqslant k d_n \log^{1/2} n \},$$

其中常数 $k>0$ 选择之使 $k^2 = 2\sigma_0^2 d_n + 2bkd_n \log^{1/2}$. 则由不等式(4.13),有

$$P(E_{n2}^c) \leqslant 2\exp(-\log n) = 2/n = O(d_n). \tag{4.17}$$

因此由(4.10), (4.16)和(4.17)式,得

$$P(E_n \cap E_{n1} \cap E_{n2}) \geqslant 1 - O(d_n). \tag{4.18}$$

但由(4.14)式及 E_{n1}, E_{n2}的定义,知 $E_n \cap E_{n1} \cap E_{n2}$ 发生时有 $g\sigma^{-1} \mid \beta_{nj} - \bar{\beta}_{nj} \mid \leqslant c_1 d_n \log n$, 对某个与 n 无关的常数 c_1. 于是由(4.15)式及 Φ' 在$\mathbb{R}^1$ 上有界,得

$$\begin{aligned}
F_n(x) &= P(g\sigma^{-1}\beta_{nj} \leqslant x) \\
&\leqslant P(g\sigma^{-1}\bar{\beta}_{nj} \leqslant x + c_1 d_n \log n) + O(d_n) \\
&= \bar{F}_n(x + c_1 d_n \log n) + O(d_n) \\
&= (\bar{F}_n(x + c_1 d_n \log n) - \Phi(x + c_1 d_n \log n)) \\
&\quad + [\Phi(x + c_1 d_n \log n) - \Phi(x)] + \Phi(x) + O(d_n) \\
&= \Phi(x) + O(d_n \log n) + O(d_n) \\
&= \Phi(x) + O(d_n \log n).
\end{aligned}$$

同理,由 $F_n(x) \geqslant \bar{F}_n(x - c_1 d_n \log n) - O(d_n)$ 出发,也可得

$$F_n(x) \geqslant \Phi(x) + O(d_n \log n).$$

结合以上两式,即得(1.12)式,定理 4 证毕.

自然地可以问,定理中的数量级 $O(d_n \log n)$可否改进? 在 $\rho(u) = u^2$ 即 LS 估计的特例,熟知本定理中的 $O(d_n \log n)$可用 $O(d_n)$取代,而且对任何 $\delta \in (0,\ 1/2]$, $d_n \approx n^{-\delta}$, 可作出这样的例子,使(1.12)式中的 $O(d_n \log n)$不能被 $o(d_n)$取代. 这显示:对满足定理 4 条件的一般的 ρ, $O(d_n)$是所能期望的最佳数量级. 它是否真能达到尚待研究.

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20

低阶矩条件下线性回归最小二乘估计弱相合的充要条件

摘要 考虑线性回归模型 $Y_i = x_i'\beta + e_i,\ 1 \leqslant i \leqslant n,\ n \geqslant 1$. 假定误差 e_i 独立,同分布或不同分布,其 $r(1 \leqslant r < 2)$ 阶矩有限. 得出了为使 β 或其一线性函数 $c'\beta$ 的最小二乘估计为弱相合,$\{x_i\}$所应满足的充要条件.

考虑线性回归模型

$$Y_i = x_i'\beta + e_i,\ 1 \leqslant i \leqslant n,\ n \geqslant 1, \tag{0.1}$$

这里 $x_1,\ x_2,\ \cdots$是已知的 p 维向量,$e_1,\ e_2,\ \cdots$是随机误差,$\beta = (\beta_1,\ \cdots,\ \beta_p)'$ 是未知的参数向量. 记 $S_n = \sum_{i=1}^n x_i x_i'$. 设 n 足够大时 S_n^{-1} 存在. 令 $Y_{(n)} = (Y_1,\ \cdots,\ Y_n)'$, $e_{(n)} = (e_1,\ \cdots,\ e_n)'$, 及

$$U_n = S_n^{-1}(x_1 \vdots x_2 \vdots \cdots \vdots x_n) = \begin{pmatrix} u_{n11} & u_{n12} & \cdots & u_{n1n} \\ \vdots & \vdots & \cdots & \vdots \\ u_{np1} & u_{np2} & \cdots & u_{npn} \end{pmatrix}, \tag{0.2}$$

则 β 的最小二乘(LS)估计为

$$\hat{\beta}_n = (\hat{\beta}_{n1},\ \cdots,\ \hat{\beta}_{np})' = U_n Y_{(n)} = \beta + U_n e_{(n)}. \tag{0.3}$$

故 $\hat{\beta}_n$ 在指定意义下的相合性,等价于 $U_n e_{(n)}$ 在该意义下收敛于 0. 例如,$\hat{\beta}_{nj}$ 的弱相合性等价于

$$\sum_{i=1}^n u_{nji} e_i \xrightarrow{P} 0. \tag{0.4}$$

自 20 世纪 60 年代以来,LS 估计的相合性问题引起了许多学者的兴趣,到 70 年代末完满地解决了误差 e_i 有二阶矩的情况[1~3]. 此后注意力转向于误差 e_i 只有较低阶矩时的情况,这种研究有重要意义. 因为均值回归的定义只要求误差有一阶矩,且 LS 估计作为线性估计,从其形式和统计背景看,其相合性问题应视为大数律的延伸,不应只局限在二阶矩有限的假定下去处理. 同时,通过研究 LS 估计在误差只有低阶矩时的表现,可以揭示这一极重要方法的一些前所未知的性质.

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部分地由于二阶矩有限时的结果的影响，早先关于低阶矩条件下 LS 估计相合性的工作，多把注意力集中在相合性与 S_n^{-1} 收敛于 0 的速度的关系上[4~6]. 结果表明，循着这一途径只能获得一些充分条件，它们表面上看改进余地不大，其实与必要条件相去甚远(参见本文定理 1 和定理 2). 到 1989 年，朱力行[7] 在这方面做出了本质的推进，他考虑了在误差 $e = (e_1, e_2, \cdots)$ 的分布属于类

$$\mathscr{F}_r' = \left\{(F_1 \times F_2 \times \cdots): \int_{|x|<\infty} x \mathrm{d}F_i = 0,\ \lim_{a\to\infty}\left[\sup_i \int_{|x|>a} |x|^r \mathrm{d}F_i\right] = 0\right\} \tag{0.5}$$

$(1 \leqslant r < 2)$ 的情况下 $\hat{\beta}_n$ 的一个分量 $\hat{\beta}_{nj}$ 的相合问题，得到了 $\hat{\beta}_{nj}$ r 阶平均相合的充要条件. 这是在低阶矩条件下首次在一种相合意义下得到充要条件，是值得一提的.

在往下讨论之前要明确一下此处及本文以下所指的“充要条件”的含义. 记 $x = (x_1, x_2, \cdots)$, $\mathscr{X} = (\mathbb{R}^p)^\infty$. 设给定了误差 $e = (e_1, e_2, \cdots)$ 的一个分布类 $\mathscr{F}$, 例如(0.5), 及某种相合性意义 C(例如弱相合 W, r 阶平均相合 M_r, 强相合 S), 则有两种可以考虑的“充要条件”含义：

1° 称条件 A 为 $\hat{\beta}_{nj}$ C 相合的充要条件，若当 x 满足条件 A 时，只要 e 的分布属于 $\mathscr{F}$, 就有 $\hat{\beta}_{nj}$ 按意义 C 收敛于 β_j. 反之，若 x 不满足条件 A, 则这一断言不真：即至少存在一个分布 $F \in \mathscr{F}$, 使当 e 有分布 F 时，$\hat{\beta}_{nj}$ 不依意义 C 收敛于 β_j. 暂把这种充要条件称为 Ⅰ 型的.

2° 称条件 A 为 $\hat{\beta}_{nj} C$ 相合的充要条件，若 x 满足条件 A, 则与 1°同. 反之，若 x 不满足条件 A, 则不论 e 的分布是 $\mathscr{F}$ 中的那一个，$\hat{\beta}_{nj}$ 都不按意义 C 收敛于 $\hat{\beta}_j$. 把这种充要条件称为 Ⅱ 型的.

显然，Ⅱ 型充要条件优于 Ⅰ 型. 可惜的是，Ⅱ 型条件往往不存在，而 Ⅰ 型条件则总是存在的，这容易从以下的推理中看出：固定 $F \in \mathscr{F}$, 它相应于 $\mathscr{X}$ 的一子集 $A_j(F, C)$, 若误差 e 有分布 F, 则当且仅当 $x \in A_j(F, C)$ 时 $\hat{\beta}_{nj}$ 依意义 C 收敛于 β_j. 记 $\underline{A}_j(\mathscr{F}, C) = \bigcap_{F\in\mathscr{F}} A_j(F, C)$, $\overline{A}_j(\mathscr{F}, C) = \bigcup_{F\in\mathscr{F}} A_j(F, C)$. 显然，$\{x \in \underline{A}_j(\mathscr{F}, C)\}$ 是 $\hat{\beta}_{nj}$ C 相合的 Ⅰ 型充要条件. 但当且仅当 $\underline{A}_j(\mathscr{F}, C) = \overline{A}_j(\mathscr{F}, C)$ 时，Ⅱ 型充要条件才存在，与 Ⅰ 型条件同. 这等于要求 $A_j(F, C)$ 与 F 无关，因而更多地出于例外而非常有. 以下将指出，对误差只有低于二阶矩的情形，Ⅱ 型充要条件不存在. 这实在是 LS 估计表现的一个特异性质. 因众所周知，当误差有二阶矩时，Ⅱ 型充要条件是存在的.

文献[7]也致力于解决在误差分布类 $\mathscr{F}_r'$ 之下 LS 估计弱相合的条件问题，得到了比前人工作有改进的充分条件和必要条件，但未能得出充要条件，还有一点：它没有研究从统计的观点看最有兴趣的 iid. 情形，即误差 e 有分布类

$$\mathscr{F}_r = \left\{(F \times F \times \cdots): \int_{|x|<\infty} x \mathrm{d}F = 0,\ \int_{|x|<\infty} |x|^r \mathrm{d}F < \infty\right\} \tag{0.6}$$

的情形. 固然，由于 $\mathscr{F}_r \subset \mathscr{F}_r'$, 对 $\mathscr{F}_r'$ 定出的充要条件对 $\mathscr{F}_r$ 仍为充分，但不一定必要. 事实的确如此.

1 主要结果

先引进一些记号. 以下总把 u_{nji} 简化为 u_{ni}. 将 $|u_{n1}|, \cdots, |u_{nn}|$ 排序为 $u_{n(1)} \geqslant \cdots \geqslant u_{n(n)}$.

记 $W_r(n)=\max\limits_{1\leqslant i\leqslant n} i u_{n(i)}^r$, $V(n,j)=\sum\limits_{i=1}^{n} u_{ni} I(|u_{ni}|\geqslant u_{n(j)})$, I 为指示函数, $V(n)=\max\limits_{1\leqslant j\leqslant n}|V(n,j)|$.

定理 1 设模型(0.1)中误差分布类为 $\mathscr{F}_r$, 若 $1<r<2$, 则 $\hat{\beta}_{nj}$ 弱相合的充要条件(Ⅰ型,下同)为

$$\lim_{n\to\infty} S_n^{-1}(j)=0 \tag{1.1}$$

和

$$W_r(n)=O(1) \tag{1.2}$$

同时成立;若 $r=1$, 则充要条件为(1.1)式及

$$W_1(n)=O(1) \tag{1.3}$$

和

$$V(n)=O(1) \tag{1.4}$$

同时成立,Ⅱ型充要条件不存在.

定理 2 设模型(0.1)中误差分布类为 $\mathscr{F}'_r$, $1\leqslant r<2$, 则 $\hat{\beta}_{nj}$ 弱相合的充要条件为(1.1)式及

$$\sum_{i=1}^{n}|u_{ni}|^r=O(1) \tag{1.5}$$

同时成立,Ⅱ型充要条件不存在.

Ⅱ型充要条件的不存在容易证明,取分布类

$$\mathscr{F}_2=\left\{(F\times F\times\cdots):\int_{|x|<\infty} x\mathrm{d}F=0,\ 0<\int_{|x|<\infty} x^2\mathrm{d}F<\infty\right\}. \tag{1.6}$$

则 $\mathscr{F}_2\subset\mathscr{F}_r$, $1\leqslant r<2$, 据文献[2],有 $\underline{A}_j(\mathscr{F}_2,W)=\overline{A}_j(\mathscr{F}_2,W)=\{(1.1)\text{式}\}$ (W 表弱相合). 由此并结合定理 1, 2,即知

$$\overline{A}_j(\mathscr{F}_r,W)=\overline{A}_j(\mathscr{F}'_r,W)=\{(1.1)\text{式}\},\ 1\leqslant r<2.$$

但由(1.1)式推不出(1.2)或(1.3)~(1.5)式(见注 1),故 $\underline{A}_j(\mathscr{F}_r,W)$ 和 $\underline{A}_j(\mathscr{F}'_r,W)$ 分别是 $\overline{A}_j(\mathscr{F}_r,W)$ 和 $\overline{A}_j(\mathscr{F}'_r,W)$ 的真子集,因而Ⅱ型充要条件不存在.

在证明定理之前先作一些注解.

注 1 定理中的条件没有多余的,即任一部分都不能推出其余部分,这一点有必要验明,因为 $\{u_{ni},1\leqslant i\leqslant n\}$ 并非可任意取值,而是受到(0.2)式结构的制约,为了验证需构造反例:在模型(0.1)中取 $p=1$, 这时有 $u_{ni}=x_i\Big/\sum\limits_{i=1}^{n}x_i^2$, $1\leqslant i\leqslant n$, $S_n^{-1}=\left(\sum\limits_{i=1}^{n}x_i^2\right)^{-1}$.

1° (1.1)式推不出(1.2)式:取 $x_i=1/\sqrt{i}$, $i\geqslant 1$.

2° (1.2)式推不出(1.1)式:取 $x_i=1/i$, $i\geqslant 1$.

3° (1.3)式+(1.4)式推不出(1.1)式:取 $x_i=1/i^2$, $i\geqslant 1$.

4° (1.1)式+(1.4)式推不出(1.3)式:取

$x_1=1,\ x_2=-1,\ x_3=1/\sqrt{2},\ x_4=-1/\sqrt{2},\ \cdots,\ x_{2n-1}=1/\sqrt{n},\ x_{2n}=-1/\sqrt{n},\ \cdots.$

5° (1.1)式＋(1.3)式推不出(1.4)式:取

$x_1=1,\ x_2=1,\ x_3=1/2,\ \cdots,\ x_{k(k-1)/2+1}=1,\ x_{k(k-1)/2+2}=1/2,\ \cdots,\ x_{k(k+1)/2}=1/k,\ \cdots.$

6° (1.1)式推不出(1.5)式:取 x_i 与 1°同.

7° (1.5)式推不出(1.1)式:取 x_i 与 3°同.

注 2 易见当 $1<r<2$ 时,(1.5)式⇒(1.2)式,但其逆不真;当 $r=1$ 时,(1.5)式⇒(1.3)式＋(1.4)式,但其逆不真.反例如下:取模型(0.1)中 $p=1$. 找一串自然数 $n_1<n_2<\cdots$ 满足条件 $\lim\limits_{k\to\infty}k^r/\log n_k=0$, 而后令 $x_1,\ x_2,\ \cdots$ 依次等于下述序列中的相应项:

$1,\ -2^{-1/r},\ 3^{-1/r},\ \cdots,\ (-1)^{n_1-1}n_1^{-1/r},\ \cdots,\ 1,\ -2^{-1/r},\ 3^{-1/r},\ \cdots,\ (-1)^{n_k-1}n_k^{-1/r},\ \cdots,$

由此看出:在 e_i 只有 r 阶矩而 $1\leqslant r<2$ 时,同分布下 LS 估计弱相合的充要条件,比不同分布之下的充要条件要宽一些,这也与二阶矩有限时的结果不同.

注 3 定理 2 确定的条件与文献[7]中定理 2.2.1 所给出的在分布类 $\mathscr{F}_r'$之下 $\hat{\beta}_{nj}$ r 阶平均相合的充要条件相同.由此推出一个有趣的结果:尽管对 $\mathscr{F}_r'$中一个具体分布而言,$\hat{\beta}_{nj}$ 的弱相合与 r 阶平均相合不一定等价(反例见文献[3]),但从分布类 $\mathscr{F}_r'$的整体看二者却是等价的.

注 4 注意到文献[7]定理 2.2.1 的必要部分证明中,所构造的反例内 e_i 为 iid.,从而得出结论:在误差分布类为 $\mathscr{F}_r(1\leqslant r<2)$ 时,$\hat{\beta}_{nj}$ r 阶平均相合的充要条件也是(1.1)式及(1.5)式,与 $\mathscr{F}_r'$同,这样,文献[7]和本文结果结合,彻底解决了在误差只有低于 2 阶的矩时,LS 估计 $\hat{\beta}_{nj}$ 的弱相合与 r 阶平均相合的条件问题,有趣的是:对 iid. 情况 $\mathscr{F}_r$, r 阶平均相合的条件严于弱相合的条件,而在非 iid. 情况 $\mathscr{F}_r'$,二者却一致.

注 5 定理的结论也可用于任一线性函数 $c'\beta$ 的 LS 估计 $c'\hat{\beta}_n$,只需将(1.1)式改为 $c'S_n^{-1}c\to 0$, 而在(1.2)~(1.5)式中,用 $c'U_n$ 代替$(u_{n1},\ \cdots,\ u_{nn})$.

2 定理的证明

先证明一个预备事实.记

$$V_r(n,\ j)=\sum_{i=1}^{n}u_{ni}I(|\ u_{ni}\ |\geqslant u_{n(j)})u_{n(j)}^{r-1},\ V_r(n)=\max_{1\leqslant j\leqslant n}|\ V_r(n,\ j)\ |.$$

注意当 $r=1$ 时,$V_r(n,\ j)$和 $V_r(n)$就是前面定义过的 $V(n,\ j)$和 $V(n)$. 令 $B(n,\ j)=\{i:1\leqslant i\leqslant n,\ 2^{-j}<|\ u_{ni}\ |^r\leqslant 2^{-j+1}\}$, 及

$$Z_r(n,\ j)=\sum_{i\in B(n,\ j)}|\ u_{ni}\ |^r,\ Z_r(n)=\sup_{j\geqslant 1}Z_r(n,\ j).$$

引理 1 设 $1\leqslant r<2$, 则

$$\{W_r(n)=O(1)\}\Rightarrow\{Z_r(n)=O(1)\};\tag{2.1}$$

若 $1<r<2$, 则

$$\{W_r(n)=O(1)\}\Rightarrow\{V_r(n)=O(1)\}.\tag{2.2}$$

证 先证(2.1)式.用反证法,设 $Z_r(n)=O(1)$ 不对,则必要时取子列,可设存在自然数 j_n,致 $Z_r(n, j_n)\to\infty$. 以 t_n 记集合 $B(n, j_n)$中所含元素个数,则有 $Z_r(n, j_n)\leqslant t_n 2^{-j_n+1}$,故 $t_n 2^{-j_n}\to\infty$. 对每个 $i\in B(n, j_n)$ 定出自然数 m_i,使 $|u_{ni}|=u_{n(m_i)}$. 记 $m=\max\limits_{i\in B(n, j_n)} m_i$,则 $m\geqslant t_n$,所以

$$W_r(n)\geqslant m u_{n(m)}^r\geqslant m2^{-j_n}\geqslant t_n 2^{-j_n}\to\infty$$

与 $W_r(n)=O(1)$ 不合. 现证(2.2)式. 设 $1<r<2$. 由 $W_r(n)=O(1)$ 知存在常数 M,使 $iu_{n(i)}^r\leqslant M^r$,故 $u_{n(i)}\leqslant Mi^{-1/r}$, $1\leqslant i\leqslant n$, 现有

$$V_r(n, j)=\sum_{i=1}^n u_{ni}I(|u_{ni}|>u_{n(j)})u_{n(j)}^{r-1}+\sum_{i=1}^n u_{ni}I(|u_{ni}|=u_{n(j)})u_{n(j)}^{r-1}\equiv J_1+J_2. \quad (2.3)$$

因为 $\left|\sum\limits_{i=1}^n u_{ni}I(|u_{ni}|>u_{n(j)})\right|\leqslant\sum\limits_{i=1}^{j-1}u_{n(i)}$, 且 $r>1$, 故当 $j\to\infty$ 时,

$$|J_1|\leqslant M\sum_{i=1}^{j} i^{-1/r}M^{r-1}j^{-(r-1)/r}=O(j^{1-1/r})O(j^{-(r-1)/r})=O(1). \quad (2.4)$$

由此知 $J_1=O(1)$. 其次,$|J_2|\leqslant Z_r(n)$. 由(2.1)式知 $J_2=O(1)$. 最后,由(2.3)式得 $V_r(n)=O(1)$. 引理证毕.

从这个引理看出为何定理 1 中 $r=1$ 的情况要单列. 因 $r=1$ 时,(2.4)式已失效. 确实,当 $r=1$ 时(2.2)式已不成立,此在注 1 的 5°已指出过了.

定理 1 的证 充分性. 不失普遍性,以下总设 $|u_{n1}|\geqslant\cdots\geqslant|u_{nn}|>0$, 故 $|u_{ni}|=u_{n(i)}$. 以 F 记 e_i 的公共分布,有 $\int_{|x|<\infty}x\mathrm{d}F=0$, $\int_{|x|<\infty}|x|^r\mathrm{d}F<\infty$.

设(1.1)式+(1.2)式,或(1.1)式+(1.3)式+(1.4) 式成立,则据引理 1,不论 $1<r<2$ 或 $r=1$, 总有

$$W_r(n)=O(1),\ V_r(n)=O(1),\ Z_r(n)=O(1),$$

因此 $r>1$ 和 $r=1$ 两种情况可合并处理. 因为

$$S_n^{-1}(j)=\sum_{i=1}^n u_{ni}^2, \quad (2.5)$$

由(1.1)式有 $u_{n1}\to 0$, 再加上 e_i 为 iid.,即知 $\{u_{ni}e_i: 1\leqslant i\leqslant n,\ n\leqslant 1\}$ 满足"一致渐近无穷小"条件(uan 条件[8]),于是据中心极限定理,为证 $\hat{\beta}_{nj}$ 弱相合,即(0.4)式,需验证以下三条同时成立:

$$\sigma_n^2(1)\equiv\sum_{i=1}^n u_{ni}^2\int_{|x|<|u_{ni}|^{-1}}x^2\mathrm{d}F\to 0; \quad (2.6)$$

$$a_n(1)\equiv\sum_{i=1}^n u_{ni}\int_{|x|<|u_{ni}|^{-1}}x\mathrm{d}F\to 0; \quad (2.7)$$

$$p_n(1)\equiv\sum_{i=1}^n\int_{|x|>|u_{ni}|^{-1}}\mathrm{d}F\to 0. \quad (2.8)$$

考虑(2.6)式.由 $Z_r(n, j)$ 的定义及 $u_{n1}\to 0$，知存在自然数 $m_n\to\infty$，使

$$Z_r(n, j)=0,\ 1\leqslant j\leqslant m_n,\ n\text{ 充分大}. \tag{2.9}$$

定义

$$b(t)=\int_{|x|<t}x^2\mathrm{d}F/t^{2-r},\ t>0, \tag{2.10}$$

$$p_i=\int_{2^{(i-1)/r}\leqslant|x|<2^{i/r}}|x|^r\mathrm{d}F,\ i\geqslant 1. \tag{2.11}$$

则易见

$$b(t_2)\leqslant 2b(t_1),\text{当 }0<t_1/2\leqslant t_2\leqslant t_1, \tag{2.12}$$

$$M\equiv\sum_{i=1}^{\infty}p_i\leqslant\int_{|x|<\infty}|x|^r\mathrm{d}F<\infty. \tag{2.13}$$

现有

$$\sigma_n^2(1)=\sum_{i=1}^{n}|u_{ni}|^r b(|u_{ni}|^{-1})=\sum_{j=1}^{\infty}\sum_{i\in B(n,\ j)}|u_{ni}|^r b(|u_{ni}|^{-1}), \tag{2.14}$$

$$b(2^{j/r})=2^{-j(2-r)/r}\int_{|x|<2^{j/r}}x^2\mathrm{d}F\leqslant\sum_{t=1}^{j}2^{-(j-t)(2-r)/r}p_t. \tag{2.15}$$

由(2.12)，(2.14)式，并注意 $Z_r(n, j)$ 的定义，有

$$\sum_{i\in B(n,\ j)}|u_{ni}|^r b(|u_{ni}|^{-1})\leqslant 2b(2^{j/r})\sum_{i\in B(n,\ j)}|u_{ni}|^r=2b(2^{j/r})Z_r(n,\ j).$$

由此式及(2.9)，(2.14)，(2.15)式，并利用(2.1)式，知存在常数 M_1，使当 n 充分大时有

$$\begin{aligned}\sigma_n^2(1)&\leqslant M_1\sum_{j=m_n}^{\infty}\sum_{t=1}^{j}2^{-(j-t)(2-r)/r}p_t\\&\leqslant M_1\sum_{j=m_n}^{\infty}\sum_{t=1}^{c}2^{-(j-t)(2-r)/r}p_t+M_1\sum_{j=m_n}^{\infty}\sum_{t=c+1}^{j}2^{-(j-t)(2-r)/r}p_t\\&\equiv J_1+J_2.\end{aligned} \tag{2.16}$$

据(2.13)式，有

$$J_1\leqslant M_1M\sum_{j=m_n}^{\infty}2^{-(j-c)(2-r)/r}\to 0\ (n\to\infty). \tag{2.17}$$

又

$$\begin{aligned}J_2&=M_1\sum_{t=c+1}^{\infty}\sum_{j=\max(m_n,\ t)}2^{-(j-t)(2-r)/r}p_t\\&\leqslant M_1(1-2^{-(2-r)/r})^{-1}\sum_{t=c+1}^{\infty}p_t.\end{aligned}$$

按(2.13)式,有 $\lim\limits_{c\to\infty} J_2 = 0$. 此与(2.16), (2.17)式结合,即得(2.6)式.

为证(2.7)式,令

$$q_i = \int_{|u_{ni}|^{-1}\leqslant|x|<|u_{ni+1}|^{-1}} x\mathrm{d}F,\ r_i = \int_{|u_{ni}|^{-1}\leqslant|x|<|u_{ni+1}|^{-1}} |x|^r\mathrm{d}F,\ 1\leqslant i\leqslant n-1,$$

$$q_n = \int_{|x|\geqslant|u_{nn}|^{-1}} x\mathrm{d}F,\qquad r_n = \int_{|x|\geqslant|u_{nn}|^{-1}} |x|^r\mathrm{d}F.$$

因 $\int_{|x|<\infty} x\mathrm{d}F = 0$, 有

$$\begin{aligned}-a_n(1) &= u_{n1}(q_1+\cdots+q_n)+u_{n2}(q_2+\cdots+q_n)+\cdots+u_{nn}q_n\\ &= \sum_{i=1}^n (u_{n1}+\cdots+u_{ni})q_i;\end{aligned}$$

因 $q_i\leqslant|u_{ni}|^{r-1}r_i = u_{n(i)}^{r-1}r_i$, 有

$$|a_n(1)|\leqslant\sum_{i=1}^n |(u_{n1}+\cdots+u_{ni})u_{n(i)}^{(r-1)}|r_i. \tag{2.18}$$

若 $|u_{ni}|>|u_{ni+1}|$, 则 $(u_{n1}+\cdots+u_{ni})u_{n(i)}^{r-1} = V_r(n,\ i)$. 若 $|u_{ni}|=|u_{ni+1}|$, 则 $r_i=0$. 因此由(2.18)式得

$$|a_n(1)|\leqslant V_r(n)(r_1+\cdots+r_n) = V_r(n)\int_{|x|\geqslant|u_{n1}|^{-1}} |x|^r\mathrm{d}F. \tag{2.19}$$

由(2.2)式及假定(1.4)有 $V_r(n)=O(1)$, 又因 $u_{n1}\to 0$ 以及 $\int_{|x|<\infty}|x|^r\mathrm{d}F<\infty$, 由(2.19)式知(2.7)式成立.

最后证明(2.8)式,令

$$l_i = \int_{|u_{ni}|^{-1}\leqslant|x|<|u_{ni+1}|^{-1}}\mathrm{d}F,\ 1\leqslant i\leqslant n-1,\ l_n = \int_{|x|\geqslant|u_{nn}|^{-1}}\mathrm{d}F. \tag{2.20}$$

则有

$$p_n(1) = (l_1+\cdots+l_n)+(l_2+\cdots+l_n)+\cdots+l_n = l_1+2l_2+\cdots+nl_n. \tag{2.21}$$

由(1.2), (1.3)式知存在常数 M,使 $i\leqslant M|u_{ni}|^{-r}$, 故由(2.21)式,

$$p_n(1)\leqslant M\sum_{i=1}^n l_i|u_{ni}|^{-r}\leqslant M\int_{|x|\geqslant|u_{n1}|^{-1}}|x|^r\mathrm{d}F\to 0,$$

即(2.8)式,充分性证毕.

必要性. (1.1)式的必要性由 $\mathscr{F}_2\subset\mathscr{F}_r$ ($\mathscr{F}_2$ 见(1.6)式)及(1.1)式对 $\mathscr{F}_2$ 为必要而推出,事实上,(1.1)式是Ⅱ型必要条件,即如在分布类 $\mathscr{F}_r$ 中去掉那些不足道的退化于0的情况,则当(1.1)式不成立时,不论误差 e 取 $\mathscr{F}_r$ 中哪一个分布, $\hat{\beta}_{nj}$ 必不弱收敛于 β_j. 这可由文献[9]中引理2推出.

为证(1.2)~(1.4)式的必要性,可设(1.1)式成立,因若不然,则由上述已知 $\hat{\beta}_{nj}$ 不为弱相合,毋庸另证. 因有(1.1)式,据(2.5)式知对 $\{u_{ni}e_i:1\leqslant i\leqslant n,\ n\geqslant 1\}$ uan 条件成立,故为证 $\hat{\beta}_{nj}$

不相合,只需证明存在分布 $F \in \mathscr{F}_r$, 使当误差 $e=(e_1, e_2, \cdots)$ 有分布 F 时,(2.6)～(2.8)式中至少有一条不成立.

先设(1.2)或(1.3)式不成立,为避免多重足标,以 $u(n, i)$和 $u(n, (i))$分别记 u_{ni} 和 $u_{n(i)}$,又约定

$$u(n, n+1)=|u(n, n)|/(1+|u(n, n)|).$$

必要时取子列,可设存在自然数 $i(n) \leqslant n$, 使

$$i(n)|u(n, i(n))|^r \to \infty, \ |u(n, i(n))|>|u(n, i(n)+1)|.$$

由 $u(n, 1) \to 0$ 可知 $i(n) \to \infty$. 故可找到 $\varepsilon(n) \downarrow 0$,使

$$\varepsilon(n) i(n) \to \infty, \ \varepsilon(n)|u(n, i(n))|^{-r} \to 0.$$

据此可找到自然数子列 $n_1<n_2<\cdots$, 使

$$\sum_{k=1}^{\infty} \varepsilon(n_k)|u(n_k, i(n_k))|^{-r}<\infty, \tag{2.22}$$

$$|u(n_{k+1}, 1)|<|u(n_k, n_k)|, \ k \geqslant 1. \tag{2.23}$$

因 $u(n, 1) \to 0$,由(2.22)式有

$$0<c \equiv \sum_{k=1}^{\infty} \varepsilon(n_k)<\infty. \tag{2.24}$$

作概率分布 F_0 如下:

$$F_0(\{|u(n_k, i(n_k))|^{-1}\})=F_0(\{-|u(n_k, i(n_k))|^{-1}\})=(2c)^{-1} \varepsilon(n_k), \ k \geqslant 1,$$

F_0 是关于 0 对称的分布,且据(2.22)式有 $\int_{|x|<\infty}|x|^r \mathrm{d} F_0<\infty$, 故 $\int_{|x|<\infty} x \mathrm{d} F_0$ 存在且等于 0, 因此无穷维分布 $F=(F_0)^{\infty}$ 属于分布类 $\mathscr{F}_r$. 取模型(0.1)中误差 $e=(e_1, e_2, \cdots)$ 的分布为 F, 则据(2.21)和(2.20)式,有

$$p_{n_k}(1) \geqslant i(n_k) l_{i(n_k)} \geqslant i(n_k) \varepsilon(n_k)/c \to \infty,$$

即(2.8)式不成立,而 $\hat{\beta}_{nj}$ 不依概率收敛于 β_j.

剩下要证明当 $r=1$ 时条件(1.4)的必要性,先证明下面的简单引理:

引理 2 令

$$\widetilde{V}(n, j)=\sum_{i=1}^n u(n, i) I(|u(n, i)| \leqslant|u(n, j)|), \ 1 \leqslant j \leqslant n,$$

而 $\widetilde{V}(n)=\max\limits_{1 \leqslant j \leqslant n}|\widetilde{V}(n, j)|$, 则当(1.4)式不成立时,必有

$$\limsup_{n \to \infty} \widetilde{V}(n)=\infty. \tag{2.25}$$

事实上,若(2.25)式不对,则存在常数 M, 使对一切 n 有 $\widetilde{V}(n) \leqslant M$. 记 $h(n)=\sum_{i=1}^n u(n, i)$, 则 $|h(n)|=|\widetilde{V}(n, 1)| \leqslant \widetilde{V}(n) \leqslant M$. 现任取 j, $1 \leqslant j \leqslant n$. 若 $|u(n, j)|=$

$|u(n,n)|$，则 $V(n,j)=h(n)$，因而 $|V(n,j)|\leqslant M$；若 $|u(n,j)|>|u(n,n)|$，则找最小的 j'，使 $|u(n,j)|>|u(n,j')|$，有 $V(n,j)=h(n)-\widetilde{V}(n,j')$，因而 $|V(n,j)|\leqslant 2M$. 这将得出对一切 n，$V(n)\leqslant 2M$，与(1.4)式不成立之假定不合，引理证毕.

现设(1.4)式不成立. 据(2.25)式，必要时取子列，可设存在自然数 $j(n)\leqslant n$，使 $|\widetilde{V}(n,j(n))|\to\infty$，因此可找到 $\varepsilon(n)\downarrow 0$，使

$$\varepsilon(n)\,|\widetilde{V}(n,j(n))|\to\infty. \tag{2.26}$$

约定 $u(n,0)=2u(n,1)$，定义

$$r(n)=\min\{|u(n,i)|: 0\leqslant i\leqslant n,\ |u(n,i)|>|u(n,j(n))|\}.$$

找自然数子列 $n_1<n_2<\cdots$，满足条件

$$\varepsilon(n_k)\leqslant 2^{-k},\ 2|u(n_k,1)|<|u(n_{k-1},n_{k-1})|,\ k\geqslant 2. \tag{2.27}$$

由 $\sum\varepsilon(n_k)<\infty$ 及 $u(n,1)\to 0$ 因而 $r(n)\to 0$，知

$$0<c\equiv\sum_{k=1}^{\infty}\varepsilon(n_k)(r(n_k)+|u(n_k,n_k)|)<\infty. \tag{2.28}$$

定义分布 F_0 如下：

$$F_0(\{1/r(n_k)\})=c^{-1}r(n_k)\varepsilon(n_k),\ F_0(\{-1/|u(n_k,n_k)|\})=c^{-1}\varepsilon(n_k)|u(n_k,n_k)|. \tag{2.29}$$

由 $\sum\varepsilon(n_k)<\infty$ 知 F_0 的期望存在，此期望为 0，因按(2.27)式和 $r(n)$ 的定义以及 F_0 的定义(2.29)式，易见

$$\int_{|x|\leqslant|u(n_k,n_k)|^{-1}}x\mathrm{d}F_0=0, \tag{2.30}$$

此可由 F_0 在区间 $[-|u(n_k,n_k)|^{-1},|u(n_k,n_k)|^{-1}]$ 上的支撑点集为 $\{1/r(n_i),-|u(n_i,n_i)|^{-1};1\leqslant i\leqslant k\}$ 及(2.29)式看出，因为 $u(n_k,n_k)\to 0$，在(2.30)式左边令 $k\to\infty$ 即得 $\int_{|x|<\infty}x\mathrm{d}F_0=0$，因此，若记 $F=(F_0)^{\infty}$，则 $F_0\in\mathscr{F}_1$. 现计算

$$a_{n_k}(1)=\sum_{i=1}^{n_k}u(n_k,i)\int_{|x|<|u(n_k,i)|^{-1}}x\mathrm{d}F_0. \tag{2.31}$$

按得出(2.30)式的类似推理，易得

$$\int_{|x|<|u(n_k,i)|^{-1}}x\mathrm{d}F_0=\begin{cases}c^{-1}\varepsilon(n_k), & 当\ |u(n_k,i)|\leqslant|u(n_k,j(n_k))|,\\ 0, & 当\ |u(n_k,i)|>|u(n_k,j(n_k))|.\end{cases} \tag{2.32}$$

由(2.31)，(2.32)式，得

$$\begin{aligned}a_{n_k}(1)&=c^{-1}\varepsilon(n_k)\sum_{i=1}^{n_k}u(n_k,i)I(|u(n_k,i)|\leqslant|u(n_k,j(n_k))|)\\&=c^{-1}\varepsilon(n_k)\,\widetilde{V}(n_k,j(n_k)).\end{aligned}$$

由此及(2.26)式知 $a_{n_k}(1)\to\infty$，即(2.7)式不成立，而 $\hat{\beta}_{nj}$ 不依概率收敛于 β_j，定理 1 证毕.

定理 2 的证 充分性部分是文献[7]中定理 2.2.1 的推论，往证必要性. 可设(1.1)式成立，理由同前，因此有 $u_{n1}\to 0$，再加上 $\{|e_i|^r, i\geqslant 1\}$ 一致可积，不难知道对 $\{u(n, i)e_i: 1\leqslant i\leqslant n, n\geqslant 1\}$ uan 条件成立，故为证 $\hat{\beta}_{nj}$ 不为弱相合，只需证明存在分布 $F\in\mathscr{F}'_r$，使当$(e_1, e_2, \cdots)$有分布 F 时，(2.6)～(2.8)式中至少有一条不成立，记 $a^+=\max(a, 0)$，$a^-=\max(-a, 0)$.

现设(1.5)式不成立，必要时取子列，可设

$$\lim_{n\to\infty}\sum_{i=1}^{n}(u^+(n, i))^r=\infty,$$

或

$$\lim_{n\to\infty}\sum_{i=1}^{n}(u^{-1}(n, u))^r=\infty.$$

为确定计，设为前者，找一串常数 $\{\varepsilon(n)\}$，$1>\varepsilon(1)>\varepsilon(2)>\cdots\downarrow 0$，使

$$\lim_{n\to\infty}\varepsilon(n)\sum_{i=1}^{n}(u^+(n, i))^r=\infty. \tag{2.33}$$

记

$$P(n)=\{i: 1\leqslant i\leqslant n, u(n, i)>0\}.$$

用归纳法找一串自然数 $n_1<n_2<\cdots$ 如下：首先找 n_1 充分大，使

$$\varepsilon(n_1)\sum_{i\in P(n_1)}u^r(n_1, i)/2\geqslant 1,\ |u(n, 1)|<1,\ 当\ n\geqslant n_1.$$

现设 $n_1, \cdots, n_{k-1}$ 已定出，找 $n_k>n_{k-1}$，使

$$2^{-1}\varepsilon(n_k)\sum_{i\in P(n_k),\ i>n_{k-1}}u^r(n_k, i)-\sum_{i=1}^{n_{k-1}}|u(n_k, i)|\geqslant 1, \tag{2.34}$$

这种 n_k 的存在由(2.33)式保证.

现在定一串一维分布$\{F_i\}$如下：设 $k\geqslant 2$，$n_{k-1}<i\leqslant n_k$，$i\in P(n_k)$，则

$$\begin{aligned}
&F_i(\{2^{-1}|u(n_k, i)|^{-1}\})=u^r(n_k, i)\varepsilon(n_k),\\
&F_i(\{-2|u(n_k, i)|^{-1}\})=4^{-1}u^r(n_k, i)\varepsilon(n_k),\\
&F_i(\{0\})=1-5u^r(n_k, i)\varepsilon(n_k)/4.
\end{aligned} \tag{2.35}$$

对其他的 $i\geqslant 1$ 令 $F_i\sim N(0, 1)$，有 $\int_{|x|<\infty}x\mathrm{d}F_i=0$，$i\geqslant 1$. 又若 F_i 按(2.34)式定义，则 $\int_{|x|<\infty}|x|^r\mathrm{d}F_i<2\varepsilon(n_k)\to 0$，由此得到 $\lim\limits_{a\to\infty}\left(\sup\limits_i\int_{|x|\geqslant a}|x|^r\mathrm{d}F_i\right)=0$. 故若取模型(0.1)中误差$(e_1, e_2, \cdots)$的分布为 $F=F_1\times F_2\times\cdots$，则 $F\in\mathscr{F}'_r$. 但当 $n_{k-1}<i\leqslant n_k$ 而 $i\in P(n_k)$ 时，有

$$\int_{|x|<|u(n_k, i)|^{-1}}x\mathrm{d}F_i=2^{-1}u^{r-1}(n_k, i)\varepsilon(n_k);$$

而对其他的 $i\leqslant n_k$，则有

$$\int_{|x|<|u(n_k,\ i)|^{-1}} |x| \, \mathrm{d}F_i \leqslant \max\left(\sqrt{2\pi}\int_{|x|<\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x,\ \varepsilon(m_t)\ |\ u(m_t,\ 1)\ |^{r-1},\ t \leqslant n_k\right) < 1,$$

由此得出,对一切 $k \geqslant 2$,

$$a_{n_k}(1) \geqslant 2^{-1}\varepsilon(n_k) \sum_{i \in P(n_k),\ i > n_{k-1}} u^r(n_k,\ i) - \sum_{i=1}^{n_{k-1}} |\ u(n_k,\ i)\ | \geqslant 1,$$

因此(2.7)式不成立,定理 2 证毕.

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21

Gauss-Markov 条件下最小二乘估计的强相合性

摘要 设 $Y_i = x_i'\beta + e_i$, $1 \leqslant i \leqslant n$ 为线性模型, $\hat{\beta}_n = (\hat{\beta}_{n1}, \cdots, \hat{\beta}_{np})'$ 为 $\beta = (\beta_1, \cdots, \beta_p)'$ 的最小二乘估计,以 u_n 记 $\left(\sum_{i=1}^{n} x_i x_i'\right)^{-1}$ 的(1, 1)元, $v_n = u_n^{-1}$. 证明了在 $\mathrm{E}e_i = 0$ 且 $\{e_i\}$ 满足 Gauss-Markov 条件时, $v_i \to \infty$ 及 $\sum_{i=2}^{\infty} v_i^{-2}(v_i - v_{i-1})\log^2 i < \infty$ 为 $\hat{\beta}_{n1}$ 强相合的充分条件,且对任何 $\varepsilon_n \to 0$, $v_i \to \infty$ 及 $\sum_{i=2}^{\infty} \varepsilon_i v_i^{-2}(v_i - v_{i-1})\log^2 i < \infty$ 已不再充分. 提出了 $\hat{\beta}_{n1}$ 强相合的一个充要条件,它把 $\hat{\beta}_{n1}$ 强相合归结为正交随机变量级数的收敛问题.

考虑线性回归模型

$$Y_i = x_i'\beta + e_i, \ 1 \leqslant i \leqslant n, \ n \geqslant 1, \tag{0.1}$$

这里 $x_1, x_2, \cdots$ 是已知的 p 维向量, $\beta = (\beta_1, \cdots, \beta_p)'$ 是未知的回归系数向量, $e_1, e_2, \cdots$ 是随机误差. 本文统一使用以下的记号:

$S_n = x_1 x_1' + \cdots + x_n x_n'$;

$u_n = S_n^{-1}$ 的(1, 1)元, $v_n = u_n^{-1}$, 当 S_n^{-1} 存在时;

$\hat{\beta}_n = (\hat{\beta}_{n1}, \cdots, \hat{\beta}_{np})' = \beta$ 的最小二乘(LS)估计;

$\delta_{ij} = 1$ 或 0, 视 $i = j$ 或否而定;

$\mathscr{D} = \{(e_1, e_2, \cdots): \mathrm{E}e_i = 0, \ \mathrm{E}e_i e_j = \delta_{ij}, \ i, j = 1, 2, \cdots\}$.

本文的目的是研究 $\hat{\beta}_n$ 的强相合性(SC). 关于这个 $\hat{\beta}_n$ 的相合问题,有两种情况研究较多,一是随机误差 $e_1, e_2, \cdots$ 独立[1~4];另一是 $e_1, e_2, \cdots$ 满足 Gauss-Markov(GM)条件 $\mathrm{E}e_i e_j = \sigma^2 \delta_{ij}$. 因为 σ^2 之值无关紧要,可以说 GM 条件相当于说 $(e_1, e_2, \cdots) \in \mathscr{D}$. 如果 $\mathrm{E}\xi\eta = 0$, 称随机变量 ξ 和 η 正交.

在 GM 条件下只知道 $(e_1, e_2, \cdots) \in \mathscr{D}$ 而不知其确切的分布. 命题"某条件 C(施加于$(x_1$,

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$x_2,\cdots)$)是 $\hat{\beta}_n$ 强相合的充分条件"意义明确:若条件 C 满足,则对任何 $(e_1,e_2,\cdots)\in\mathscr{D}$ 有 $\hat{\beta}_n\to\beta$, a. s. 而命题"条件 C 是 $\hat{\beta}_n$ 强相合的必要条件". 则可以有两种理解:1° 若$(x_1,x_2,\cdots)$ 不满足条件 C,则对任何 $(e_1,e_2,\cdots)\in\mathscr{D}$, $\hat{\beta}_n$ 不能 a. s. 收敛于 β(A 型);2° 若$(x_1,x_2,\cdots)$不满足条件 C,则存在 $(e_1,e_2,\cdots)\in\mathscr{D}$, 使 $\hat{\beta}_n$ 不 a. s. 收敛于 β(B 型). 以上定义也可用于 $\hat{\beta}_n$ 的一分量 $\hat{\beta}_{n1}$.

本文第 1 节证明:所建立的一个 $\hat{\beta}_{n1}$ 强相合充分条件在该方向上已毫无改进余地,在第 2 节中我们讨论 $\hat{\beta}_{n1}$ 的必要及充分条件问题,其中建立了在 GM 条件下 $\hat{\beta}_{n1}$ 为强相合的一个充要条件,但这个条件有赖于一个有关正交级数收敛性问题的彻底解决.

1 一个充分条件

本节的主要结果是

定理 1 在 GM 条件下, $\hat{\beta}_{n1}$ 强相合的一个充分条件是

$$v_i\to\infty,\ \sum_{i=2}^{\infty}v_i^{-2}(v_i-v_{i-1})\log^2 i<\infty, \tag{1.1}$$

此处若 S_i^{-1} 不存在,则 $v_i^{-2}(v_i-v_{i-1})$ 被认为是 0. 条件(1.1)在下述意义上是确切的:对任何非负数列$\{\varepsilon_n\}$,若 $\lim\limits_{n\to\infty}\varepsilon_n=0$, 则条件

$$v_i\to\infty,\ \sum_{i=2}^{\infty}\varepsilon_i v_i^{-2}(v_i-v_{i-1})\log^2 i<\infty \tag{1.2}$$

对 $\hat{\beta}_{n1}$ 的强相合已不再充分.

我们将证明的主要部分列成几条引理.

引理 1 设 $\xi_1',\xi_2',\cdots$ 为定义在某概率空间$(\Omega_1,\mathscr{F}_1,P_1)$上的一串正交随机变量, $\sum\limits_{i=1}^{\infty}\xi_i'$ 不 a. s. 收敛. 则存在概率空间$(\Omega,\mathscr{F},P)$及定义在其上的两串随机变量 $\{\xi_i,i\geqslant 1\}$ 及 $\{\eta_i,i\geqslant 1\}$, 满足以下条件:

1° $\{\xi_1,\xi_2,\cdots,\eta_1,\eta_2,\cdots\}$ 全体正交;

2° $\mathrm{E}\xi_i=\mathrm{E}\eta_i=0,\ i\geqslant 1$;

3° $\mathrm{E}\xi_i^2=\mathrm{E}\xi_i'^2,\ \mathrm{E}\eta_i^2=1$, 当 $\mathrm{E}\xi_i'^2\neq 0,\ i\geqslant 1$;

4° $\sum\limits_{i=1}^{\infty}\xi_i$ 不 a. s. 收敛.

证 设$(\Omega_2,\mathscr{F}_2,P_2)$是$(\Omega_1,\mathscr{F}_1,P_1)$的一个复制品(copy),且 $\Omega_1\cap\Omega_2=\varnothing$. 定义新概率空间$(\Omega_3,\mathscr{F}_3,P_3)$如下: $\Omega_3=\Omega_1\cup\Omega_2$. $\mathscr{F}_3$ 由一切形如 $A=A_1\cup A_2$ 之集构成,其中 $A_1\in\mathscr{F}_1$, $A_2\in\mathscr{F}_2$, 而 $P_3(A)=(P_1(A_1)+P_2(A_2))/2$. 在$(\Omega_3,\mathscr{F}_3,P_3)$上定义一串随机变量 $\{\tilde{\xi}_i,i\geqslant 1\}$ 如下:

$$\xi_i(\omega)=\begin{cases}\xi_i'(\omega), & \text{当 } \omega\in\Omega_1,\\ -\xi_i'(\omega'), & \text{当 } \omega\in\Omega_2 \text{ 且 } \omega \text{ 是由 } \omega'\in\Omega_1 \text{ 复制而来},\end{cases}\quad i\geqslant 1.$$

则易见 $\tilde{\xi}_1,\tilde{\xi}_2,\cdots$ 正交, $\mathrm{E}\tilde{\xi}_i=0$, $\mathrm{E}\tilde{\xi}_i^2=\mathrm{E}\xi_i'^2$, $i\geqslant 1$,且 $\sum\limits_{i=1}^{\infty}\tilde{\xi}_i$ 不 a. s. 收敛.

现在取$(\Omega_3, \mathscr{F}_3, P_3)$的一个复制品$(\Omega_4, \mathscr{F}_4, P_4)$且$\Omega_3 \cap \Omega_4 = \varnothing$. 将此两者结合成概率空间$(\Omega, \mathscr{F}, P)$,其方法正如将$(\Omega_1, \mathscr{F}_1, P_1)$和$(\Omega_2, \mathscr{F}_2, P_2)$结合而得到$(\Omega_3, \mathscr{F}_3, P_3)$. 在$(\Omega, \mathscr{F}, P)$上定义两串随机变量如下:

$$\xi_i(\omega) = \begin{cases} \sqrt{2}\,\tilde{\xi}_i(\omega), & \omega \in \Omega_3, \\ 0, & \omega \in \Omega_4, \end{cases} \quad i \geqslant 1;$$

$$\eta_i(\omega) = \begin{cases} 0, & \omega \in \Omega_3, \\ \sqrt{2}\,\tilde{\xi}_i(\omega'), & \omega \in \Omega_4 \text{ 且系由 } \omega' \in \Omega_3 \text{ 复制而来}, \end{cases} \quad i \geqslant 1.$$

则易见除了$E\eta_i^2 = 1$之外,引理中的一切要求都满足. 若$E\xi_i'^2 \neq 0$, 则以$\eta_i/(E\xi_i'^2)^{1/2}$取代η_i,就可以满足$E\eta_i^2 = 1$. 引理证毕.

引理 2 对任给非负常数串$\{\varepsilon_i, i \geqslant 1\}$, $\lim\limits_{i\to\infty}\varepsilon_i = 0$, 可找到一串正交随机变量$\{\xi_i, i \geqslant 1\}$及一串常数$0 = \lambda_0 < 1/2 = \lambda_1 < \lambda_2 < \cdots \uparrow \infty$, 满足以下条件:

1° $\sum\limits_{i=1}^{\infty}\xi_i$ a. s. 发散;

2° $\sum\limits_{i=1}^{\infty}\varepsilon_i E\xi_i^2 \log^2 i < \infty$;

3° $E\xi_i = 0, \ i \geqslant 1$;

4° $E\xi_i^2 = \lambda_i^{-2}(\lambda_i - \lambda_{i-1}), \ i \geqslant 1$.

证 记$\tilde{\varepsilon}_n = \sup\limits_{i \geqslant n}\varepsilon_i$, 有$\tilde{\varepsilon}_n \to 0$. 故可找到一列自然数$n_1 < n_2 < \cdots$, 使

$$\sum_{i=1}^{\infty}\tilde{\varepsilon}_{n_i} < \infty. \tag{1.3}$$

找一串自然数$N_1 < N_2 < \cdots$, 满足:

(a) $N_i \geqslant n_i, \ i \geqslant 1$;

(b) $N_i - N_{i-1}$ 非降, $i \geqslant 2$;

(c) $(N_2 - N_1)\log^2 N_2 > 4$;

(d) $\log^2 N_i / \log^2 N_{i+1} < 1/8$;

(e) $(N_1 - 1)/(N_2 - 1) + \sum\limits_{i=1}^{\infty}\log^{-2} N_i < 1/2$;

(f) $(N_{k+1} - N_k)^{-1}\log^{-2} N_{k+1} > 4\sum\limits_{i=k+1}^{\infty}(N_{i+1} - N_i)^{-1}\sum\limits_{j=N_i+1}^{N_{i+1}}\log^{-2} j$.

只要取N_i足够大,就不难满足条件(a)~(e). (f)能够满足是因为当$m \to \infty$时, $(m - N_i)^{-1}\sum\limits_{j=N_i+1}^{m}\log^{-2} j \to 0$. 现定义

$$c_1 = 2, \ c_j = (N_2 - N_1)^{-1}, \ 2 < j \leqslant N_1,$$

$$c_j = (N_{i+1} - N_i)^{-1}\log^{-2} j, \ N_i < j \leqslant N_{i+1}, \ i \geqslant 1.$$

由(b)易见$\{c_j\}$非增. 由(e)有

$$c_2 + c_3 + \cdots < 1/2. \tag{1.4}$$

显然有

$$\sum_{j=1}^{\infty} c_j \log^2 j = \infty. \tag{1.5}$$

又利用条件(a)及 $\tilde{\varepsilon}_n$ 的定义,并注意到(1.3)式,易见

$$\sum_{j=1}^{\infty} \varepsilon_j c_j \log^2 j < \infty. \tag{1.6}$$

由$\{c_j\}$的非增及(1.5)式,据 Tandori[5]的一个结果,知存在正交变量序列 $\{\xi_i',\ i \geqslant 1\}$,使

$$\sum_{i=1}^{\infty} \xi_i' \quad \text{a. s. 发散}, \mathrm{E}\xi_i'^2 = c_i,\ i \geqslant 1.$$

现证可找到一串常数 $0 = \mu_0 < 1/2 = \mu_1 < \mu_2 < \cdots < 1$, 使

$$q_i \equiv \mu_i^{-2}(\mu_i - \mu_{i-1}) = c_i,\ i \geqslant 1. \tag{1.7}$$

由 $\mu_0 = 0,\ \mu_1 = 1/2$ 有 $q_1 = 2 = c_1$. 用归纳法,设已定了 $\mu_0 < \mu_1 < \cdots < \mu_m$, 满足

$$q_i = c_i,\ 1 \leqslant i \leqslant m,\ \mu_m \leqslant 1/2 + c_2 + \cdots + c_m < 1, \tag{1.8}$$

这里用了(1.4)式. 令 $b(u) = u/(\mu_m + u)^2$, 有 $b(0) = 0 < c_{m+1}$. 又按归纳假设(1.8)及(1.4)式,有 $b(c_{m+1}) > c_{m+1}$. 故存在 $u' \in (0,\ c_{m+1})$ 使 $b(u') = c_{m+1}$. 取 $\mu_{m+1} = \mu_m + u'$, 则见(1.8)式可以由 m 拓展至 $m+1$. 这完成了(1.7)式的归纳证明.

定义

$$\lambda_i = \mu_i,\ 0 \leqslant i \leqslant N_2 - 1.$$

记 $b(u) = u/(\lambda_{N_2-1} + u)^2$, 有 $b(1/c_{N_2} - \lambda_{N_2-1}) < c_{N_2}$. 又因 $\lambda_{N_2-1} < 1$ 及 $c_{N_2} < 1/4$ (据条件(c)), 有 $b((2c_{N_2})^{-1} - \lambda_{N_2-1}) = 2c_{N_2}(1 - 2c_{N_2}\lambda_{N_2-1}) > c_{N_2}$. 故存在 $u' \in ((2c_{N_2})^{-1} - \lambda_{N_2-1},\ c_{N_2}^{-1} - \lambda_{N_2-1})$ 使 $b(u') = c_{N_2}$. 令

$$\lambda_{N_2} = \lambda_{N_2-1} + u',\ \text{有}\ \lambda_{N_2}^{-2}(\lambda_{N_2} - \lambda_{N_2-1}) = c_{N_2}\ \text{且}\ (2c_{N_2})^{-1} < \lambda_{N_2} < c_{N_2}^{-1}. \tag{1.9}$$

这样定义了 $\lambda_i,\ 0 \leqslant i \leqslant N_2$. 现在要把上述做法拓展到 $N_2 < i \leqslant N_3$, 方法类似:找一串 $\lambda_{N_2} = \theta_{N_2} < \theta_{N_2+1} < \cdots < 2\lambda_{N_2}$, 使 $q_i' \equiv \theta_i^{-2}(\theta_i - \theta_{i-1}) = c_i,\ i > N_2$. 为此令 $b(u) = u/(\lambda_{N_2} + u)^2$, 则 $b(0) < c_{N_2+1}$. 又据(f)及(1.9)式,有 $c_{N_2+1} < 4^{-1}c_{N_2} < 4^{-1}/\lambda_{N_2}$. 故 $b(4\lambda_{N_2}^2 c_{N_2+1}) > 4\lambda_{N_2}^2 c_{N_2+1}/(2\lambda_{N_2})^2 = c_{N_2+1}$, 则存在 $u' \in (0,\ 4\lambda_{N_2}^2 c_{N_2+1})$ 使 $b(u') = c_{N_2+1}$. 令 $\theta_{N_2+1} = \lambda_{N_2} + u'$, 有 $q'_{N_2+1} = c_{N_2+1}$. 以下与前面一样,用归纳法,并注意到由(f)得出的事实

$$4\lambda_{N_2}^2(c_{N_2+1} + c_{N_2+2} + \cdots) < 4\lambda_{N_2}^2 4^{-1}/\lambda_{N_2} = \lambda_{N_2},$$

即可证明具有上述性质的 $\{\theta_i,\ i \geqslant N_2\}$ 存在. 现令

$$\lambda_i = \theta_i,\ N_2 + 1 \leqslant i \leqslant N_3 - 1.$$

仿前面证法,可找到 λ_{N_3}, 使

$$\lambda_{N_3}^{-2}(\lambda_{N_3} - \lambda_{N_3-1}) = c_{N_3}\ \text{且}\ (2c_{N_3})^{-1} < \lambda_{N_3} < c_{N_3}^{-1},$$

这最后一式用到了 $\lambda_{N_3-1} < 2\lambda_{N_2} < 2/c_{N_2}$，因而

$$2c_{N_3}(1-2c_{N_3}\lambda_{N_3-1}) > 2c_{N_3}(1-4c_{N_3}/c_{N_2}) > 2c_{N_3}(1-4/8) = c_{N_3},$$

其中 $c_{N_3}/c_{N_2} < 1/8$ 是根据条件(d).

用这个办法我们又可把 λ_i 的定义拓展到 $N_3+1 \leqslant i \leqslant N_4$. 如此无限地进行下去,我们就定义了 $\{\lambda_i, i\geqslant 0\}$ 满足引理中的一切要求,其中 $\lambda_i \to \infty$ 是因为 $\lambda_{N_k} > (2c_{N_k})^{-1}$，而 $c_i \to 0$ 当 $i\to\infty$.

所造出的 $\{\xi'_i\}$ 也满足引理中除了 $E\xi'_i = 0$ 之外的一切要求. 因此我们再应用引理 1,得以造出一串正交变量 $\{\xi_i, i\geqslant 1\}$，满足本引理的一切要求,引理证毕.

引理 3 设 $\xi_1, \xi_2, \cdots$ 为一串正交变量, $B_n = E\xi_1^2 + \cdots + E\xi_n^2 \to \infty$. 则

$$(\mathrm{I})\ B_n^{-1}\sum_{i=1}^n \xi_i \to 0 \text{ a. s.} \Leftrightarrow (\mathrm{II})\ \sum_{i=1}^\infty \xi_i/B_i \text{ a. s. 收敛.}$$

证 记 $\eta_n = \xi_1 + \cdots + \xi_n$. (Ⅱ) ⇒ (Ⅰ)由 Kronecker 引理推出(正交性用不着). 为证(Ⅰ) ⇒ (Ⅱ),注意

$$\sum_{i=m}^n \xi_i/B_i = \sum_{i=m}^n (\eta_i - \eta_{i-1})/B_i = \sum_{i=m}^{n-1}(B_i^{-1} - B_{i+1}^{-1})\eta_i + \eta_n/B_n - \eta_{m-1}/B_m, \tag{1.10}$$

此处 m 固定之,使 $B_m > 0$, $n > m$. 往证

$$\sum_{i=m}^\infty (B_i^{-1} - B_{i+1}^{-1})\eta_i \text{ a. s. 收敛.} \tag{1.11}$$

因 $B_i^{-1} - B_{i+1}^{-1} \geqslant 0$，又 $E|\eta_i| \leqslant (E\eta_i^2)^{1/2} = B_i^{1/2}$，故知为证(1.11)式,只需证

$$\sum_{i=m}^\infty B_i^{-1/2}B_{i+1}^{-1}(B_{i+1} - B_i) < \infty. \tag{1.12}$$

令 $Q = \{i_1, i_2, \cdots\} = \{i: i\geqslant m, B_{i+1}\geqslant 2B_i\}$. 若 $i \notin Q$，则有

$$B_i^{-1/2}B_{i+1}^{-1}(B_{i+1} - B_i) \leqslant \sqrt{2}B_{i+1}^{-3/2}(B_{i+1} - B_i) \leqslant \sqrt{2}\int_{B_i}^{B_{i+1}} x^{-3/2}\,dx.$$

因此

$$\sum_{i\notin Q} B_i^{-1/2}B_{i+1}^{-1}(B_{i+1} - B_i) < \infty;$$

另一方面,因 $B_i \uparrow$，有 $B_{i_k} \geqslant B_{i_{k-1}+1} \geqslant 2B_{i_{k-1}} \geqslant \cdots \geqslant 2^{k-1}B_{i_1}$. 故

$$\sum_{i\in Q} B_i^{-1/2}B_{i+1}^{-1}(B_{i+1} - B_i) \leqslant \sum_{j=1}^\infty B_{i_j}^{-1/2} \leqslant B_{i_1}^{-1/2}\sum_{j=1}^\infty 2^{-(j-1)/2} < \infty.$$

合以上两式证明了(1.12)式,因而得(1.11)式. 由(1.10)与(1.11)式立即得出(Ⅰ)⇒(Ⅱ). 引理证毕.

现转向模型(0.1). 若 $p>1$，则将 x_i 记为 $(x_{i1}, T'_i)'$，T_i 为 $(p-1)$ 维,记 $K_j = \sum_{i=1}^j x_{i1}T_i$，$H_j = \sum_{i=1}^j T_iT'_i$. 若当 $j\geqslant N$ 时 S_j^{-1} 存在,则 H_j^{-1} 亦然. 令

$$h_{ji}=\begin{cases}x_{i1}, & p=1,\\ x_{i1}-K_j'H_j^{-1}T_j, & p>1,\end{cases}\quad 1\leqslant i\leqslant j, \tag{1.13}$$

$$w_N=\sum_{i=1}^{N}h_{Ni}e_i,\ w_n=\sum_{i=1}^{n}h_{ni}e_i-\sum_{i=1}^{n-1}h_{n-1,i}e_i,\ n>N. \tag{1.14}$$

引理 4(Anderson-Taylor) 有

$$w_j=h_{jj}\big(e_j-T_j'H_{j-1}^{-1}\sum_{i=1}^{j-1}T_ie_i\big),\ j>N, \tag{1.15}$$

$$\hat{\beta}_{n1}=\beta_1+(w_N+w_{N+1}+\cdots+w_n)/v_n,\ n>N. \tag{1.16}$$

又若 $(e_1,e_2,\cdots)\in\mathscr{D}$，则 $w_N,w_{N+1},\cdots$ 是一个正交序列且

$$\mathrm{E}w_j=0,\ j\geqslant N;\ \mathrm{Var}(w_N)=v_N,\ \mathrm{Var}(w_j)=v_j-v_{j-1},\ j>N. \tag{1.17}$$

证明见文献[1]或文献[6]，p. 37.

定理 1 的证 (1.1)式的充分性由引理 3、引理 4 及 Rademacher-Menchoff 定理[7,8]得出. 为证定理的后一部分，设 $0\leqslant\varepsilon_n\to 0$. 据引理 2，可找到 $0=\lambda_0<\lambda_1<\cdots\uparrow\infty$ 及一串正交随机变量 $\xi_1,\xi_2,\cdots$，使 $P\big(\sum_{i=1}^{\infty}\xi_i$ 发散$\big)>0$. $\mathrm{E}\xi_i=0$，$q_i\equiv\lambda_i^{-2}(\lambda_i-\lambda_{i-1})=\mathrm{E}\xi_i^2>0$，$i\geqslant 1$，且

$$\sum_{i=1}^{\infty}\varepsilon_iq_i\log^2 i<\infty. \tag{1.18}$$

考虑线性回归模型

$$Y_i=x_i\beta_1+e_i,\ 1\leqslant i\leqslant n,\ n\geqslant 1, \tag{1.19}$$

其中 $x_i=(\lambda_i-\lambda_{i-1})^{1/2}$，$e_i=\xi_i/(\mathrm{E}\xi_i^2)^{1/2}$，$i\geqslant 1$. 此处有 $S_n=x_1^2+\cdots+x_n^2=\lambda_n$，故当 $\lambda_n>0$ 时有 $v_n=\lambda_n$. 此模型中 β_1 的最小二乘估计为

$$\hat{\beta}_{n1}=\beta_1+\sum_{i=1}^{n}(\lambda_i-\lambda_{i-1})^{1/2}e_i/\lambda_n.$$

因 $\lambda_n\uparrow\infty$，按引理 3，为使 $\hat{\beta}_{n1}\to\beta_1$ a. s.，必须有

$$\sum_{i=1}^{\infty}(\lambda_i-\lambda_{i-1})^{1/2}\lambda_i^{-1}e_i\ \text{a. s. 收敛}. \tag{1.20}$$

但是，据随机变量 e_i 的定义，以及 $\mathrm{E}\xi_i^2=\lambda_i^{-2}(\lambda_i-\lambda_{i-1})$，知(1.20)式正好就是 $\sum_{i=1}^{\infty}\xi_i$ a. s. 收敛. 因为这不成立，故知 $\hat{\beta}_{n1}$ 不能 a. s. 收敛于 β_1，对上述特定的 $(e_1,e_2,\cdots)\in\mathscr{D}$. 这证明了条件(1.2)不充分. 定理 1 证毕.

注 可能认为，本定理后一结论暗示，(1.1)式也是必要条件. 这不正确，反例如下：考虑模型(1.19)，其中

$$x_{2^n}=\sqrt{2n-1},\ n\geqslant 1;\ x_i=0,\ \text{其他}\ i.$$

对此模型有 $v_i=n^2$，当 $2^n\leqslant i<2^{n+1}$，$n\geqslant 1$. 故

$$\sum_{i=2}^{\infty} v_i^{-2}(v_i - v_{i-1})\log^2 i = \sum_{n=1}^{\infty} n^{-4}(n^2 - (n-1)^2)(\log 2^n)^2 = \infty,$$

故(1.1)式不满足.但

$$\hat{\beta}_{n1} = \beta_1 + N^{-2}\sum_{i=1}^{N}\sqrt{2i-1}e_{2^i},\ 2^N \leqslant n < 2^{N+1},$$

据引理 2 及 Rademacher-Menchoff 定理,只要 $(e_1, e_2, \cdots) \in \mathscr{D}$, 就有 $\hat{\beta}_{n1} \to \beta_1$ a. s. ,即 $\hat{\beta}_{n1}$ 为强相合的,这证明了条件(1.1)并非必要.

就简单模型(1.19)而言,若附加条件"当 i 充分大时 $|x_i|\downarrow$",则据文献[5]中一个结果可知,在 Gauss-Markov 假定下,(1.1)式就是 $\hat{\beta}_{n1}$ 强相合的充要条件.

2 关于充分和必要条件

本节仍在 $(e_1, e_2, \cdots) \in \mathscr{D}$ 的假定下,来讨论一些有关 $\hat{\beta}_{n1}$ 强相合的条件问题.

(a) $u_n \to 0$ 是最强的 A 型必要条件. 事实上,若 $u_n \not\to 0$, 则按文献[9],对任何 $(e_1, e_2, \cdots) \in \mathscr{D}$, $\hat{\beta}_{n1}$ 不能 a. s. 收敛于 β_1,故 $u_n \to 0$ 为 A 型必要条件. 其次,若某条件 C 比 $u_n \to 0$ 强,则存在 $(x_1, x_2, \cdots)$ 满足 $u_n \to 0$ 但不满足条件 C. 由 $u_n \to 0$ 可知(据文献[4]),若 $e_1, e_2, \cdots$ 为 iid. , $Ee_1 = 0$, $Ee_1^2 = 1$ (这样的 $(e_1, e_2, \cdots)$ 属于 $\mathscr{D}$),将有 $\hat{\beta}_{n1} \to \beta_1$ a. s. . 这证明了条件 C 不是 A 型必要条件.

(b) B 型充要条件必存在,且这个条件严格强于 $u_n \to 0$. 事实上,固定 $e \equiv (e_1, e_2, \cdots) \in \mathscr{D}$. 记 $\mathscr{B}_e = \{(x_1, x_2, \cdots): \hat{\beta}_{n1} \to \beta_1$ a. s. . 对这个 $e\}$, $B = \bigcap_{e \in \mathscr{D}} B_e$. 则"$(x_1, x_2, \cdots) \in B$"就是 B 型充要条件.

此条件必严格地强于 $u_n \to 0$. 因由此条件为充分,结合 Drygas 的结果[9],知此条件要强于 $u_n \to 0$. 两者不能一致. 因若不然,则 $u_n \to 0$ 将是充分条件,而据定理 1 的后一部分这是不对的. 事实上,在(1.2)式中取 $\varepsilon_1 = 2$, $\varepsilon_i = 1/\log i$, $i \geqslant 2$. 据定理 1 的后半,知条件 $\sum_{i=2}^{\infty} v_i^{-2}(v_i - v_{i-1})\log i < \infty$ 尚不充分,而这个条件已经比 $v_i \to \infty$ (即 $u_n \to 0$)强.

(c) 以上的论证说明了:A 型充要条件不存在(因 $u_n \to 0$ 为 A 型必要但非充分),但 B 型充要条件存在. 因此, $\hat{\beta}_{n1}$ 相合问题的最好解决,只能期望找出上述集合 B. 关于这个问题有以下的结果. 先定义集合

$$C = \{(c_1, c_2, \cdots): \text{对任何满足条件 } E\xi_i^2 = c_i,\ i \geqslant 1 \text{ 的正交变量序列} \{\xi_i,\ i \geqslant 1\},$$
$$\text{有} \sum_{i=1}^{\infty} \xi_i \text{ a. s. 收敛}\}.$$

目前只知道若 $c_i \geqslant 0$, $\sum_{i=1}^{\infty} c_i \log^2 i < \infty$, 则 $(c_1, c_2, \cdots) \in C$. 但 C 中不止这些点. 又据文献[5],若 $0 \leqslant c_1 \geqslant c_2 \geqslant \cdots$ 而 $\sum_{i=1}^{\infty} c_i \log^2 i = \infty$, 则 $(c_1, c_2, \cdots) \notin C$.

定理 2 在 GM 条件下, $\hat{\beta}_{n1}$ 强相合的 B 型充要条件是:对某个自然数 N, S_{N-1}^{-1} 存在,且

$$\lim_{i\to\infty} v_i = \infty, \ (q_1, q_2, \cdots) \in C, \tag{2.1}$$

其中 $q_i = v_{i+N-1}^{-2}(v_{i+N-1} - v_{i+N-2})$.

证 充分性由引理 3 和引理 4 得出. 为证必要性,设(2.1)式不成立,若 $v_i \not\to \infty$,则 $\hat{\beta}_{n1}$ 不为强相合已如前述,故设 $(q_1, q_2, \cdots) \notin C$. 按 C 的定义,存在一串正交随机变量 $\{\tilde{\rho}_i, i \geqslant 1\}$,使 $\mathrm{E}\tilde{\rho}_i^2 = q_i$, $i \geqslant 1$,且 $\sum_{i=1}^{\infty} \tilde{\rho}_i$ 不 a. s. 收敛. 据引理 1,不失普遍性可设 $\mathrm{E}\tilde{\rho}_i = 0$. 且因 $\sum_{i=1}^{\infty} \tilde{\rho}_i$ 不 a. s. 收敛,必有无穷个 i 使 $\mathrm{E}\tilde{\rho}_i^2 > 0$. 因此据引理 1,存在一串随机变量 $\{\theta_i, i \geqslant 1\}$,使 $\mathrm{E}\theta_i = 0$, $\mathrm{E}\theta_i^2 = 1$ 且 $\{\tilde{\rho}_i, \theta_i, i \geqslant 1\}$ 全体正交.

记 $c_i^2 = v_{i+N-1} - v_{i+N-2}$. 据(1.15)和(1.17)式,有

$$c_i^2 = h_{i+N-1,\, i+N-1}^2 \Big(1 + \sum_{r=1}^{i+N-2} (T'_{i+N-1} H_{i+N-2}^{-1} T_r)^2\Big), \ i \geqslant 1. \tag{2.2}$$

记 $\rho_i = v_{i+N-1}\tilde{\rho}_i$, $i \geqslant 1$. 往证存在 $(e_1, e_2, \cdots) \in \mathscr{D}$,使

$$\rho_i = h_{i+N-1,\, i+N-1}\Big(e_{i+N-1} - T'_{i+N-1} H_{i+N-2}^{-1} \sum_{r=1}^{i+N-2} T_r e_r\Big), \ r \geqslant 1. \tag{2.3}$$

利用(2.2)式,可将(2.3)式变为

$$\rho_i = c_i \sum_{r=1}^{\infty} a_{ir} e_r, \ r \geqslant 1, \tag{2.4}$$

其中

$$\sum_{r=1}^{\infty} a_{ir}^2 = 1, \ i \geqslant 1. \tag{2.5}$$

为证(2.4)式,考虑 Hilbert 空间 $l_2 = \left\{(a_1, a_2, \cdots): \sum_{i=1}^{\infty} a_i^2 < \infty\right\}$. l_2 中任两元 a 和 b 的内积记为 $a \cdot b$. 由(2.5)式及引理 4,知 $a_i \equiv (a_{i1}, a_{i2}, \cdots)$, $i \geqslant 1$ 是 l_2 中一组法正交向量. 在 l_2 中选一组向量 $M = \{b_t = (b_{t1}, b_{t2}, \cdots): t \geqslant 1\}$,使 $\{a_i: i \geqslant 1\} \cup M$ 构成 l_2 之一法正交基(M 中可以只含有限个向量或为空集,这时证明无实质不同). 将这个基中之全部向量按 $a_1, b_1, a_2, b_2, \cdots$ 的次序排列,并记 $d_r = (a_{1r}, b_{1r}, a_{2r}, b_{2r}, \cdots)$, $r \geqslant 1$. 则因 $\{a_i, i \geqslant 1\} \cup M$ 为法正交基,有

$$d_i \cdot d_j = \delta_{ij}, \ i, j = 1, 2, \cdots. \tag{2.6}$$

定义

$$e_r = \sum_{i=1}^{\infty} a_{ir}\rho_i / c_i + \sum_{t=1}^{\infty} b_{tr}\theta_t, \ r \geqslant 1. \tag{2.7}$$

则因 $\mathrm{E}\rho_i^2 = v_{i+N-1} - v_{i+N-2} = c_i^2$,知 $\{\rho_i/c_i: i \geqslant 1\} \cup \{\theta_t: t \geqslant 1\}$ 为一组正交随机变量,每一个的二阶矩为 1. 这一事实连同(2.6)式,证明了 $(e_1, e_2, \cdots) \in \mathscr{D}$.

以 τ 记在 L_2 意义下由 $\{\rho_i/c_i: i \geqslant 1\} \cup \{\theta_t: t \geqslant 1\}$ 所生成的闭线性空间. 往证 $(e_1, e_2, \cdots)$ 为 τ 之一法正交基. 事实上,任取 $\eta = f_1\rho_1/c_1 + f_2\theta_1 + f_3\rho_2/c_2 + f_4\theta_2 + \cdots \in \tau$,满足 $\mathrm{E}(\eta e_r) = 0$, $r \geqslant 1$,则按(2.7)式,有

$$0 = \mathrm{E}(\eta e_r) = f \cdot d_r,\ r \geqslant 1,\ f = (f_1, f_2, \cdots) \in l_2.$$

由于 $\{d_r, r \geqslant 1\}$ 为 l_2 的基,知 $f = 0$. 这证明了所说的事实.

现取 $\rho_i/c_i \in \tau$. 由于 $\{e_i, i \geqslant 1\}$ 为 τ 的基底,ρ_i/c_i 可表为

$$\rho_i c_i = \sum_{r=1}^{\infty} g_r e_r, \tag{2.8}$$

其中 $g \equiv (g_1, g_2, \cdots) \in l_2$. 借助于(2.7)式去计算 $\mathrm{E}(e_r\rho_i/c_i)$, 易见 $g_r = a_{ir}$. 故(2.8)式变成

$$\rho_i/c_i = \sum_{r=1}^{\infty} a_{ir} e_r,$$

这证明了(2.4)式,因而得(2.3)式.

现于模型(0.1)中取刚才决定的 $\{e_i, i \geqslant 1\}$ 作为随机误差,则按引理 4,在此模型下有

$$\hat{\beta}_{n1} = \beta_1 + (w + \rho_1 + \cdots + \rho_{n-N+1})/v_n = \beta_1 + (w + v_N\tilde{\rho}_1 + \cdots + v_n\tilde{\rho}_{n-N+1})/v_n. \tag{2.9}$$

由(2.9)式,用引理 3,知为要 $\hat{\beta}_{n1} \to \beta_1$ a. s. 必须成立 $\sum_{i=1}^{\infty} \tilde{\rho}_i$ a. s. 收敛. 由于这不是事实,故 $\hat{\beta}_{n1}$ 不能 a. s. 收敛于 β_1. 这就完成了定理的证明.

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22

多元线性回归最小二乘估计的一个非正常表现

摘要 通过一个实例揭示线性回归最小二乘估计的一个非正常性质，证明了在误差方差无限时，在模型中人为地引入赘余参数有可能在相合性意义下改善原来的估计，并对这个现象的含义作了讨论.

一般地说，在随机误差的方差有限时，不论在小样本或大样本的意义下，线性回归的最小二乘估计(LSE)都有良好的表现. 当方差有限的假定不真时情形如何？目前对此所知不多.

有几个理由使此问题有重要性. 首先，通常均值回归的定义只需要随机误差有有限的一阶矩，而且，由 LSE 的线性形式及其统计背景，知其相合性问题可视为通常大数律的自然推广，因而不应仅限于在误差二阶矩有限的假定下去研究. 其次，通过在较弱的矩限制下对 LSE 的性质进行研究，可以揭示其一些迥异于二阶矩存在时的性质，从而有助于对这十分重要的估计获得新的了解.

本文通过揭示在误差方差无限时 LSE 的一个非正常性质，对以上所述提供一点实际证据. 为此考虑线性回归模型

$$y_i = x_i^{\mathrm{T}}\beta + z_i^{\mathrm{T}}\gamma + e_i,\ 1 \leqslant i \leqslant n, \tag{0.1}$$

其中 x_i, z_i 都是已知常向量，β, γ 是未知参数向量，y_i 为目标变量观察值，e_i 为随机误差. 若要估计 β，则 γ 是赘余参数向量. 设基于某种理由事先已知 γ 之值，且决定用 LSE 去估计 β，则有两种方法可用：

A 令 $\tilde{y}_i = y_i - z_i^{\mathrm{T}}\gamma$ 而将(0.1)式转化为

$$\tilde{y}_i = x_i^{\mathrm{T}}\beta + e_i,\ 1 \leqslant i \leqslant n, \tag{0.2}$$

然后在模型(0.2)中用 LSE $\tilde{\beta}_n$ 估计 β.

B 直接在原模型(0.1)中用 LSE 估计(β, γ)，得($\hat{\beta}_n$, $\hat{\gamma}_n$). 取 $\hat{\beta}_n$作为 β 的估计.

按常理，看不出使用方法 B 有何好处. 在随机误差为 iid.，有均值 0 和非 0 有限方差时，这一点易由理论证明. 例如，当$\tilde{\beta}_n$ 为相合时 $\hat{\beta}_n$也必相合，但其逆不真.

直观上看，此一论断(即方法 A 优于 B)不应依赖于误差方差的有限性. 因为在以 B 代 A 时，人为地将一赘余参数向量 γ 引入模型. 然而这一直观看法并不正确：若把条件 $Ee_1^2 < \infty$ 削

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弱为 $E|e_1|^r<\infty$ 对某一 $r\in[1,2)$，则可构造出反例，其中 $\hat{\beta}_n$ 为相合而 $\tilde{\beta}_n$ 则否. 为简单计本文设 $r=1$.

1 例子

本文最初源于作者之一的一项工作，对误差有 r 阶矩而 $r\in[1,2)$ 的情形，得到了 LSE 弱相合的充要条件，基于它可造出所要的反例，但细节极复杂. 以后另一作者成功地简化了例子，并使有关论证不依赖上引结果.

取最简单的模型

$$y_i=\alpha+\beta x_i+e_i,\ 1\leqslant i\leqslant n, \tag{1.1}$$

其中 $x_1,\ x_2,\ \cdots$ 是已知实数. 回归系数 β 的 LSE 为

$$\hat{\beta}_n=\sum_{i=1}^n(x_i-\bar{x}_n)y_i\Big/\sum_{i=1}^n(x_i-\bar{x}_n)^2\quad\left(\bar{x}_n=\sum_{i=1}^n x_i/n\right).$$

若事先已知 $\alpha=0$，则有 $y_i=\beta x_i+e_i$，而 β 的 LSE 为

$$\tilde{\beta}_n=\sum_{i=1}^n x_iy_i\Big/\sum_{i=1}^n x_i^2.$$

要找一串实数 $\{x_i\}$ 及一串 iid. 随机误差 $\{e_i\}$，满足 $Ee_i=0$，$E|e_i|<\infty$，使 $\hat{\beta}_n$ 弱收敛于 β 而 $\tilde{\beta}_n$ 则否，这等于要求

$$\sum_{i=1}^n(x_i-\bar{x}_n)e_i\Big/\sum_{i=1}^n(x_i-\bar{x}_n)^2\rightarrow 0,\ \text{in pr.} \tag{1.2}$$

而

$$\sum_{i=1}^n x_ie_i\Big/\sum_{i=1}^n x_i^2\nrightarrow 0,\ \text{in pr.} \tag{1.3}$$

以 $[a]$ 记不超过 a 的最大整数. 令 $S_1=10$，而

$$S_k=[\log^6 S_{k-1}](S_1+\cdots+S_{k-1}),\ k\geqslant 2, \tag{1.4}$$

$$n_k=S_1+\cdots+S_k,\ k\geqslant 1,\ n_0=0, \tag{1.5}$$

显然 $S_k\uparrow\infty$，$n_k\uparrow\infty$. 定义

$$a_i=(\log S_k)^{-2},\ n_{k-1}<i\leqslant n_k,\ k=1,2,\cdots, \tag{1.6}$$

$$b_i=\begin{cases}(S_k\log S_k)^{1/2}, & i=n_k,\ k=1,2,\cdots,\\ 0, & \text{其他 } i\geqslant 1,\end{cases} \tag{1.7}$$

$$x_i=a_i+b_i,\ i\geqslant 1, \tag{1.8}$$

定理 1 若在模型(1.1)中 x_i 按(1.8)式定义，$e_1,\ e_2,\ \cdots$ 为 iid.，其公共分布 F 关于 0 对称，且满足

$$1 - F(x) \propto x^{-1}\log x, \text{当 } x \to \infty, \tag{1.9}$$

则 $\hat{\beta}_n$ 为 β 的弱相合估计. 但即使 $\alpha = 0$, $\tilde{\beta}_n$ 也非 β 的弱相合估计.

此处 $f(x) \propto g(x)$ 当 $x \to \infty$ 是指 $f(x) = O(g(x))$, $g(x) = O(f(x))$ 当 $x \to \infty$. 注意由定理假设推出 $\mathrm{E}e_1 = 0$, $\mathrm{E}|e_1| < \infty$.

2 定理 1 的证明

关键在于证明(1.2)和(1.3)式,分 6 步进行.

(i) $\{S_k\}$ 的若干性质.

按 S_k 的定义有

$$\begin{aligned} S_k &= [\log^6 S_{k-1}]\{S_{k-1} + (S_1 + \cdots + S_{k-2})[\log^6 S_{k-2}]/[\log^6 S_{k-2}]\} \\ &= [\log^6 S_{k-1}]\{S_{k-1} + S_{k-1}/[\log^6 S_{k-2}]\} \\ &= S_{k-1}[\log^6 S_{k-1}](1 + 1/[\log^6 S_{k-1}]). \end{aligned} \tag{2.1}$$

因此,考虑到 $S_k \to \infty$, 有

$$\log S_k = \log S_{k-1} + 6\log\log S_{k-1} + o(1), \tag{2.2}$$

$$\log S_k/\log S_{k-1} = 1 + 6\log\log S_{k-1}/\log S_{k-1} + o(1/\log S_k), \tag{2.3}$$

$$\log\log S_k = \log\log S_{k-1} + o(1), \tag{2.4}$$

此处 $o(1) \to 0$ 当 $k \to \infty$. 由(2.3)式有

$$\begin{aligned} (\log S_{k-1})^{-2} - (\log S_k)^{-2} &= (\log S_k/\log S_{k-1})^{-2}\{(\log S_k/\log S_{k-1})^2 - 1\} \\ &= 12\log\log S_{k-1}(\log S_k)^{-2}(\log S_{k-1})^{-1}(1 + o(1)) \\ &= 12\log\log S_{k-1}(\log S_k)^{-3}(1 + o(1)). \end{aligned} \tag{2.5}$$

任一自然数 n 可表为

$$n = n_k + m,\ 0 \leqslant m < S_{k+1},\ k = 0, 1, \cdots, \tag{2.6}$$

另由(1.4)式有

$$n_{k-1} = \sum_{i=1}^{k-1} S_i = S_k/[\log^6 S_{k-1}]. \tag{2.7}$$

由(2.6)和(2.7)式有

$$(S_k + m)/n = \{n - (S_1 + \cdots + S_{k-1})\}/n = 1 - O(\log^{-6} S_k). \tag{2.8}$$

(ii) 估计 $\bar{a}_n$ 和 $\bar{b}_n$.

首先,有

$$\begin{aligned} \sum_{i=1}^{n} a_i &= \sum_{i=n_k+1}^{n_k+m} a_i + \sum_{i=n_{k-1}+1}^{n_k} a_i + \sum_{i=1}^{n_{k-1}} a_i \\ &= m(\log S_{k+1})^{-2} + S_k(\log S_k)^{-2} + \sum_{j=1}^{k-1} S_j(\log S_j)^{-2} \\ &= (S_k + m)(\log S_k)^{-2} - m((\log S_k)^{-2} - (\log S_{k+1})^{-2}) + O(S_k\log^{-6} S_k), \end{aligned}$$

由此式及(2.5)和(2.8)式,得

$$\bar{a}_n = \sum_{i=1}^{n} a_i = (\log S_k)^{-2} - 12m(\log S_k)^{-3}(S_k+m)^{-1}\log\log S_k(1+o(1)) + O(\log^{-6} S_k), \tag{2.9}$$

利用(2.5)式(改其中的 k 为 $k+1$),可将(2.9)式改写为

$$\bar{a}_n = (\log S_{k+1})^{-2} + 12S_k(\log S_k)^{-3}(S_k+m)^{-1}\log\log S_k(1+o(1)) + O(\log^{-6} S_k). \tag{2.10}$$

其次,有

$$\bar{b}_n = \sum_{i=1}^{n} b_i/n = n^{-1}\left\{(S_k\log S_k)^{1/2} + \sum_{j=1}^{k-1}(S_j\log S_j)^{1/2}\right\}.$$

因 $n \geqslant S_k$ 且当 j 充分大时 $(S_j\log S_j)^{1/2} \leqslant S_j/2$,有

$$\begin{aligned}\bar{b}_n &\leqslant S_k^{-1/2}(\log S_k)^{1/2} + \sum_{j=1}^{k-1} S_j/S_k \\ &\leqslant S_k^{-1/2}(\log S_k)^{1/2} + (\log S_{k-1})^{-6} \\ &= O(\log^{-6} S_k).\end{aligned} \tag{2.11}$$

(iii) 估计 $\sum_{i=1}^{n} | x_i - \bar{x}_n |$.

因 $\bar{x}_n = \sum_{i=1}^{n} x_i/n = \bar{a}_n + \bar{b}_n$ 且 $b_i \geqslant 0$,有

$$\sum_{i=1}^{n} | x_i - \bar{x}_n | \leqslant \sum_{i=1}^{n} | a_i + b_i - \bar{a}_n - \bar{b}_n | \leqslant \sum_{i=1}^{n} | a_i - \bar{a}_n | + 2n\bar{b}_n. \tag{2.12}$$

由 a_i 的定义及(2.8)~(2.10)式,得

$$\begin{aligned}\sum_{i=1}^{n} | a_i - \bar{a}_n | &= \sum_{i=n_k+1}^{n_k+m} | a_i - \bar{a}_n | + \sum_{i=n_{k-1}+1}^{n_k} | a_i - \bar{a}_n | + \sum_{i=1}^{n_{k-1}} | a_i - \bar{a}_n | \\ &= 12mS_k(\log S_k)^{-3}(S_k+m)^{-1}\log\log S_k(1+o(1)) \\ &\quad + 12mS_k(\log S_k)^{-3}(S_k+m)^{-1}\log\log S_k(1+o(1)) \\ &\quad + \sum_{j=1}^{k-1} | \bar{a}_n - (\log S_j)^{-2} | S_j + O(n\log^{-6} S_k).\end{aligned}$$

因 $|\bar{a}_n|$ 和 $|(\log S_j)^{-2}|$ 都有界,上式第三项不超过 $C\sum_{j=1}^{k-1} S_j$,对某常数 C,故它有数量级 $O(S_k \cdot \log^{-6} S_k)$,因此

$$\begin{aligned}\sum_{i=1}^{n} | a_i - \bar{a}_n | &= O(S_k(\log S_k)^{-3}\log\log S_k) + O(S_k\log^{-6} S_k) \\ &= O(S_k(\log S_k)^{-3}\log\log S_k) + O(n\log^{-6} S_k),\end{aligned} \tag{2.13}$$

由此式及(2.11)和(2.12)式得

$$\sum_{i=1}^{n} | x_i - \overline{x}_n | = O(S_k (\log S_k)^{-3} \log\log S_k) + O(n \log^{-6} S_k). \tag{2.14}$$

(iv) 估计 $\sum_{i=1}^{n}(x_i - \overline{x}_n)^2$.

因 $\{a_i - \overline{a}_n - \overline{b}_n\}$ 有界,有

$$\left| \sum_{i=1}^{n} b_i (a_i - \overline{a}_n - \overline{b}_n) \right| = O(\sum_{i=1}^{n} b_i) = O(\sum_{j=1}^{k} (S_j \log S_j)^{1/2})$$
$$= o(\sum_{j=1}^{k} S_j \log S_j) = o(S_k \log S_k). \tag{2.15}$$

又 $\sum_{i=1}^{n} b_i^2 \geqslant b_{n_k}^2 = S_k \log S_k$, 故

$$\begin{aligned}\sum_{i=1}^{n}(x_i - \overline{x}_n)^2 &= \sum_{i=1}^{n}(a_i - \overline{a}_n - \overline{b}_n + b_i)^2 \\ &\geqslant S_k \log S_k - o(S_k \log S_k) \\ &= S_k \log S_k (1 + o(1)).\end{aligned} \tag{2.16}$$

(v) 证明(1.2)式.

因 $e_1, e_2, \cdots$ 为 iid. 且 $\mathrm{E}\,|e_i| < \infty$, 只需证

$$\sum_{i=1}^{n} | x_i - \overline{x}_n | \Big/ \sum_{i=1}^{n}(x_i - \overline{x}_n)^2 \to 0. \tag{2.17}$$

有

$$\begin{aligned} n &\leqslant \sum_{j=1}^{k} S_j + S_{k+1} = \sum_{j=1}^{k} S_j + \sum_{j=1}^{k} S_j [\log S_k]^6 \\ &= O(\sum_{j=1}^{k} S_j [\log S_k]^6) = O(S_k \log^6 S_k).\end{aligned} \tag{2.18}$$

结合(2.14)、(2.16)和(2.18)式,并注意当 $n \to \infty$ 时 $S_k \to \infty$, 知当 $n \to \infty$ 时有

$$\begin{aligned}&\sum_{i=1}^{n} | x_i - \overline{x}_n | \Big/ \sum_{i=1}^{n}(x_i - \overline{x}_n)^2 \\ &\leqslant \{O((S_k \log S_k)^{-3} \log\log S_k) + O(n \log^{-6} S_k)\} / S_k \log S_k (1 + o(1)) \\ &= O((\log S_k)^{-4} \log\log S_k) + O(\log^{-1} S_k) \\ &\to 0.\end{aligned}$$

(vi) 证明(1.3)式.

应用下述结果(见文献[1], p. 74, Exe. 17):设随机变量 $e_1, \cdots, e_n$ 为 iid.,其公共分布 F 关于 0 对称,则对任何 a 有

$$P(\sum_{i=1}^{n} e_i \geqslant a/2) \geqslant 2^{-1} n P(e_1 > a)(P(e_1 \leqslant a))^{n-1}. \tag{2.19}$$

令 $m_k = n_{k+1} - 1, k = 1, 2, \cdots$, 则

$$\begin{aligned}\sum_{i=1}^{m_k} x_i^2 &= \sum_{i=n_k+1}^{m_k} a_i^2 + \sum_{i=1}^{n_k} a_i^2 + 2\sum_{i=1}^{m_k} a_i b_i + \sum_{i=1}^{m_k} b_i^2 \\ &= (S_{k+1}-1)(\log S_{k+1})^{-4} + \sum_{j=1}^{k} S_j(\log S_j)^{-4} + 2\sum_{j=1}^{k} S_j^{1/2}(\log S_j)^{-3/2} + \sum_{j=1}^{k} S_j \log S_j \\ &= (S_{k+1}-1)(\log S_{k+1})^{-4} + O(S_{k+1}(\log S_k)^{-6}) + O(S_k^{1/2}(\log S_k)^{-3/2}) + O(S_k \log S_k) \\ &= S_{k+1}(\log S_{k+1})^{-4}(1+o(1)) + O(S_k \log S_k) \\ &= S_{k+1}(\log S_{k+1})^{-4}(1+o(1)) + O(S_{k+1}[\log S_k]^{-6}\log S_k) \\ &= S_{k+1}(\log S_{k+1})^{-4}(1+o(1)). \end{aligned} \tag{2.20}$$

再利用 $\mathrm{E}\,|\,e_i\,| < \infty$，得

$$\begin{aligned}\sum_{i=1}^{m_k} x_i e_i &= \sum_{i=1}^{m_k} a_i e_i + \sum_{i=1}^{m_k} b_i e_i = \sum_{i=n_k+1}^{n_{k+1}-1} a_i e_i + \sum_{i=1}^{n_k} a_i e_i + \sum_{j=1}^{k} (S_j \log S_j)^{1/2} e_{n_j} \\ &= (\log S_{k+1})^{-2} \sum_{i=n_k+1}^{n_{k+1}-1} e_i + O_p\Big(\sum_{i=1}^{n_k} a_i\Big) + O_p\Big(\sum_{i=1}^{k} (S_i \log S_i)^{1/2}\Big) \\ &= (\log S_{k+1})^{-2} \sum_{i=n_k+1}^{n_{k+1}-1} e_i + O_p\Big(\sum_{i=1}^{k} S_i(\log S_i)^{-2}\Big) + O_p(S_k(\log S_k)^{-6}) \\ &= (\log S_{k+1})^{-2} \sum_{i=n_k+1}^{n_{k+1}-1} e_i + O_p(S_{k+1}(\log S_{k+1})^{-6}). \end{aligned} \tag{2.21}$$

由(2.20)和(2.21)式得

$$\sum_{i=1}^{m_k} x_i e_i \Big/ \sum_{i=1}^{m_k} x_i^2 = S_{k+1}^{-1}(\log S_{k+1})^2(1+o(1)) \sum_{i=n_k+1}^{n_{k+1}-1} e_i + O_p((\log S_{k+1})^{-2}). \tag{2.22}$$

下证当 $k\to\infty$ 时，(2.22)式左边不依概率收敛于 0. 据(2.22)式，只须证当 $k\to\infty$ 时

$$\xi_k := S_{k+1}^{-1}(\log S_{k+1})^2 \sum_{i=n_k+1}^{n_{k+1}} e_i$$

不依概率收敛于 0. 在(2.19)式中取 $a = S_{k+1}(\log S_{k+1})^{-2}$，并注意到 $n_{k+1} - n_k = S_{k+1}$，得

$$P(\xi_k \geqslant 1/2) = P\Big(\sum_{i=n_k+1}^{n_{k+1}} e_i \geqslant a/2\Big) \geqslant 2^{-1} S_{k+1} P(e_1 > a)(P(e_1 \leqslant a))^{S_{k+1}-1},$$

由此及(1.9)式，知存在常数 $C_1, C_2 > 0$，使

$$\begin{aligned}P(\xi_k \geqslant 1/2) &\geqslant 2^{-1}\frac{S_{k+1}C_1\{S_{k+1}(\log S_{k+1})^{-2}\}^{-1}}{\log^2\{S_{k+1}(\log S_{k+1})^{-2}\}} \cdot \left\{1 - \frac{C_2(S_{k+1}(\log S_{k+1})^{-2})^{-1}}{\log^2(S_{k+1}(\log S_{k+1})^{-2})}\right\}^{S_{k+1}-1} \\ &\to 2^{-1}C_1\exp(-C_2) > 0, \text{当 } k\to\infty,\end{aligned}$$

这证明了(1.3)式，因而完成了定理 1 的证明.

顺便指出：模型(1.1)中的 α 的 LSE $\hat{\alpha}_n$ 也相合，这是因为

$$\hat{\alpha}_n = \bar{y}_n - \hat{\beta}_n \bar{x}_n = \alpha + (\beta - \hat{\beta}_n)\bar{x}_n + e_n,$$

由于 $\hat{\beta}_n \to \beta$ in pr.，$\bar{e}_n \to 0$ in pr.，知 $\hat{\alpha}_n \to \alpha$ in pr.

3 讨论

一般认为,当误差方差无限,或更普遍一些,当误差有厚尾分布时,LSE 的表现不佳. 本文在一个基本的渐近性质上,以确切的意义,加强了这一信念. 另外,LSE 的表现很依赖于设计矩阵 $X=(x_1, \cdots, x_n)$. 当矩阵 XX^{T} 呈现病态时,LSE 表现变坏. 这些提醒我们要对这一应用最广的估计采取一种更谨慎的态度,而给予稳健估计的重要性以更多的理由. 在这方面,文献[2]提出了线性回归的一种估计,它在对模型的最小假定之下具有相合性.

能否对本文的例子给予某种直观的解释? 我们试着提出以下的看法,如果模型误差比较显著,则允许模型中包含较多参数也许有益,因为这可能把误差的影响多少分散一些,而其总合效应则是增加了回归估计的精度. 如果这一看法有几分真实性,则对通常接受的变量选择原则,应采取一种较灵活的看法:如果两个备选模型在一些通常采用的判据下看起来差不多,则并不一定变元少者一定就好一些.

可以怀疑,LSE 的所述非正常表现与其非稳健性有关联. 因此可以合理地提出如下问题:在 M-估计类中,特别对最小绝对偏差估计(LADE),这种非正常性质是否能出现? 此问题的回答与 LADE 的相合问题有密切关系,为说明这一点,考虑线性模型

$$y_i = x_i^{\mathrm{T}}\beta + e_i,\ 1 \leqslant i \leqslant n, \tag{3.1}$$

此处 $x_1, x_2, \cdots$ 是已知的 p 维向量,$e_1, e_2, \cdots$ 为 iid,其公共分布 F 在 0 的某邻域 J 内有导数 F',且 F' 在 J 上有界并有非 0 下界,又 $\operatorname{med}(e_1)=0$. 以 β_n^* 记(3.1)式中 β 的 LADE,文献[3]中证明了:对 β_n^* 的弱相合,条件

$$S_n^{-1} \to 0\ (S_n = x_1x_1^{\mathrm{T}} + \cdots + x_nx_n^{\mathrm{T}}) \tag{3.2}$$

是充分的. 我们猜测,条件(3.2)式也属必要,这一点尚未得到证明,但在文献[4]中已证得:条件

$$\sum_{i=1}^{n} \| x_i \|^2 \to \infty \tag{3.3}$$

是必要的. 以这些事实为基础,可以证明下面断言.

定理 2 下述两个命题 A 和 B 等价:

A 若在(3.1)式中 β 的 LADE 非相合,则引入赘余的线性参数不能使其成为相合的.

B 为使(3.1)式中 β 的 LADE 为相合,条件(3.3)是必要的.

证 设命题 A 为真,而 LADE β_n^* 为相合. 任意固定一个 p 维向量 $c \neq 0$,再选 $p-1$ 个向量 $c_2, \cdots, c_p$,使矩阵

$$C = \begin{pmatrix} c^{\mathrm{T}} \\ c_2^{\mathrm{T}} \\ \vdots \\ c_p^{\mathrm{T}} \end{pmatrix}$$

非异,令 $D=(C^{\mathrm{T}})^{-1}$, $\gamma=(\gamma_1, \cdots, \gamma_p)^{\mathrm{T}} = D\beta$, $\tilde{x}_i = Cx_i$, 而将模型(3.1)改写为

$$y_i = \tilde{x}_i^{\mathrm{T}}\gamma + e_i, \ 1 \leqslant i \leqslant n. \tag{3.4}$$

因 γ 的 LADE γ_n^* 等于 $D\beta_n^*$，由 β_n^* 相合知 γ_n^* 为相合. 现设 $\gamma_2, \cdots, \gamma_p$ 已知为 0，则由命题 A，知在模型

$$y_i = \tilde{x}_{i1}^{\mathrm{T}}\gamma_1 + e_i, \ 1 \leqslant i \leqslant n$$

($\tilde{x}_{i1}$ 为 $\tilde{x}_i$ 的第一分量)下，γ_1 的 LADE 应为相合，由此，应用(3.3)式知 $\sum_{i=1}^{n} \tilde{x}_{i1}^2 \to \infty$. 因 $\tilde{x}_{i1} = c^{\mathrm{T}}x_i$，此可写为

$$\sum_{i=1}^{n}(c^{\mathrm{T}}x_i)^2 = c^{\mathrm{T}}S_n c \to \infty,$$

此式对任何 $c \neq 0$ 成立，故必有(3.2)式. 这证明了命题 B 的正确性.

反过来，若命题 B 正确，则 A 必正确. 这由条件(3.2)式的充分性及以下事实推出：记

$$x_i^{\mathrm{T}} = (x_i^{\mathrm{T}}(1), \ x_i^{\mathrm{T}}(2)), \ S_{nuv} = \sum_{i=1}^{n} x_i(u)x_i^{\mathrm{T}}(v), \ u, \ v = 1, \ 2,$$

$$S_n = \begin{pmatrix} S_{n11} & S_{n12} \\ S_{n21} & S_{n22} \end{pmatrix}, \ S_n^{-1} = \begin{pmatrix} \tilde{S}_{n11} & \tilde{S}_{n12} \\ \tilde{S}_{n21} & \tilde{S}_{n22} \end{pmatrix},$$

则有 $S_{n11}^{-1} \leqslant \tilde{S}_{n11}$.

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On a Problem of Strong Consistency of Least Absolute Deviation Estimates

Abstract Under some assumptions on the random error e_i of the linear model $Y_i = x'_i\beta_0 + e_i$, $1 \leqslant i \leqslant n$, it was shown in Chen *et al*. (1992) that the condition $d_n \equiv \max_{1\leqslant i\leqslant n} x'_i\left(\sum_{j=1}^{n} x_j x'_j\right)^{-1} x_i = o(1/\log n)$ is sufficient for the strong consistency of the LAD estimate $\hat{\beta}_n$ of β_0. In this paper we prove that for any constant sequence $D_n \uparrow \infty$, the condition $d_n = o(D_n/\log n)$ is no longer sufficient.

1 Introduction

Consider the linear regression model

$$Y_i = x'_i\beta_0 + e_i,\ 1 \leqslant i \leqslant n,\ n \geqslant 1, \tag{1.1}$$

where $x_1, \cdots, x_n$ are known p-vectors, $e_1, \cdots, e_n$ are random errors, β_0 is an unknown p-vector of regression coefficients, and $Y_1, \cdots, Y_n$ are known observations of the target variable. The Least Absolute Deviation (LAD) estimate of β_0, to be denoted by $\hat{\beta}_n$, is defined as a minimum point of the function

$$A(\beta) = \sum_{i=1}^{n} | Y_i - x'_i\beta |. \tag{1.2}$$

Usually we do not know the exact distribution of the random error sequence $e \equiv (e_1, e_2, \cdots)$ and only know that it belongs to a certain class $\mathscr{F}$ of distributions in R^∞. Under this framework the consistency of $\hat{\beta}_n$ is understood as $\hat{\beta}_n \to \beta_0$ in some sense as $n \to \infty$ for any $\beta_0 \in R^p$ and $e \overset{d}{=} F$ for any $F \in \mathscr{F}$. An adequate selection for $\mathscr{F}$, which is adopted in most studies on consistency of $\hat{\beta}_n$, is as follows:

$$\begin{aligned}\mathscr{F} = \{F:\ (e \overset{d}{=} F) \Rightarrow\ & e_1, e_2, \cdots \text{ are independent, } \operatorname{med}(e_i) = 0,\ i \geqslant 1,\\ & \text{and there exist constants } l_e > 0 \text{ and } b_e > 0 \text{ such that for}\\ & \text{any } i \geqslant 1 \text{ and } u \in (0, b_e), \text{ we have}\\ & P(0 < e_i < u) \geqslant l_e u \leqslant P(-u < e_i < 0)\}.\end{aligned}$$

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In the following we shall simply write $e(\mathscr{F})$ to express the fact or assumption that the distribution of e belongs to $\mathscr{F}$. It guarantees that each e_i has a unique median 0. This is a must in dealing with the consistency problem of LAD estimates. But $e(\mathscr{F})$ requires more: For each e_i, there must be at least linear accumulation of probability in the vicinity of zero. Other assumptions about the degree of accumulation of probability may be considered — different assumptions result in different sufficient conditions on $\{x_i\}$ for the consistency of $\hat{\beta}_n$. Among these the linear assumption (2) is the most interesting as it contains the important special case that $e_1, e_2, \cdots$ are iid., e_1 possesses a density f in some vicinity of 0, $f(0) > 0$ and f is continuous at 0.

Under $e(\mathscr{F})$ the weak consistency problem of $\hat{\beta}_n$ has been satisfactorily solved by Zhao *et al.* (1993). They proved that

$$S_n^{-1} \to 0 \ (S_n = x_1x'_1 + \cdots + x_nx'_n) \tag{1.3}$$

is sufficient for $\hat{\beta}_n$ to be weakly consistent. There are strong indications that this condition is also necessary (see Chen *et al.* (1993)). This result, together with the many results concerning the consistency of LS estimates, gives one the hope that (1.3), or some stronger condition (for example, $S_n^{-1} \to 0$ with some rate) may be sufficient for $\hat{\beta}_n$ to be strongly consistent. But a simple example shows that this is not the case: For any $\varepsilon_n \downarrow 0$, the condition $S_n^{-1} = O(\varepsilon_n)$ remains insufficient. This is a remarkable difference between LAD and LS estimates, and indicates the inherent complications in the strong consistency problem of the LAD estimate.

The first result in this respect, with a certain degree of generality, is due to Wu (1988): Define

$$t_n = \max_{1 \leq i \leq n} \| x_i \|,$$

$$\lambda_n = \text{the smallest eigenvalue of } S_n,$$

$$d_n = \max_{1 \leq i \leq n} x'_i S_n^{-1} x_i.$$

Wu showed that if

$$t_n^2/\lambda_n = o(1/\log n) \text{ and } t_n = O(n^a) \text{ for some } a > 0, \tag{1.4}$$

then $\hat{\beta}_n$ is strongly consistent. Later Chen *et al.* (1992) improved this result by getting rid of the second condition in (1.4), and replacing the first by a weaker one:

$$d_n = o(1/\log n). \tag{1.5}$$

A forward logical step is to ask whether or not the rate $o(1/\log n)$ can further be improved and if so, what is the best rate. At first, some indications led us to believe that this is possible, but somewhat unexpectedly, it turns out that the following assertion is true:

Theorem 1 Under the assumption $e(\mathscr{F})$ for any $D_n \uparrow \infty$ the condition

$$d_n = o(D_n/\log n) \tag{1.6}$$

is no longer sufficient for $\hat{\beta}_n$ to be strongly consistent.

2 Some Lemmas

To prove Theorem 1, we have to construct a random error sequence $e \equiv (e_1, e_2, \cdots) (\mathscr{F})$, a sequence of p-vectors $\{x_i\}$ satisfying (1.6), such that the LAD estimate $\hat{\beta}_n$ of β_0 in model (1.1) is not strongly consistent. In this section, we give the technical details of the construction in the form of several lemmas.

Lemma 1 Suppose that in model (1.1) $p = 1$ and $x_i \neq 0$, $i \geqslant 1$. Write $M(n) = \sum_{i=1}^{n} |x_i|$. If there exist integers $i_1, \cdots, i_m$, $1 \leqslant i_1 < \cdots < i_m \leqslant n$, and a constant a such that

$$Y_{i_j}/x_{i_j} \geqslant a,\ 1 \leqslant j \leqslant m,\ \sum_{j=1}^{m} |x_{i_j}| > M(n)/2,$$

then $\hat{\beta}_n \geqslant a$.

Proof Define a random variable with the conditional distribution (given the x_i)

$$P(\xi = Y_i/x_i) = |x_i|/M(n),\ 1 \leqslant i \leqslant n.$$

It is easily seen, by rewriting $A(\beta)$ in the form

$$A(\beta) = M(n) \sum_{i=1}^{n} \frac{|x_i|}{M(n)} \left| \frac{Y_i}{x_i} - \beta \right|,$$

that $\hat{\beta}_n$ must be a median of ξ. From this the assertion of the lemma follows readily.

Without loss of generality take $D_1 = 1$. Define a sequence $\{C_n, n \geqslant 1\}$ in the following way:

$$C_1 = \sqrt{D_1},\ C_n = \min(C_{n-1} + (n \log n)^{-1}, \sqrt{D_n}),\ n \geqslant 2. \tag{2.1}$$

Lemma 2 The sequence $\{C_n\}$ has the following properties:

$$C_n = o(D_n). \tag{2.2}$$

$$1 = C_1 \leqslant C_2 \leqslant \cdots \uparrow \infty. \tag{2.3}$$

$$nC_n/\log n \uparrow \infty,\ n \geqslant 3. \tag{2.4}$$

$$C_n/\log n \downarrow 0. \tag{2.5}$$

$$nC_n/\log n - (n-1)C_{n-1}/\log(n-1)$$
$$\geqslant C_n(\log(n-1) - 1)/(\log n \cdot \log(n-1)),\ n \geqslant 3. \tag{2.6}$$

$$nC_n/\log n - (n-1)C_{n-1}/\log(n-1) \leqslant C_n/\log n,\ n \geqslant 3. \tag{2.7}$$

$$(\log n)^{-1} \exp(C_n/\log n) \text{ increases with } n, \text{ for } n \text{ large}. \tag{2.8}$$

Proof By (2.1) we have $C_n \leqslant \sqrt{D_n} = o(D_n)$. To prove (2.3), observe that if $C_{n-1} +$

$(n\log n)^{-1} \leqslant \sqrt{D_n}$, then $C_n = C_{n-1} + (n\log n)^{-1} > C_{n-1}$. Otherwise $C_n = \sqrt{D_n} \geqslant \sqrt{D_{n-1}} \geqslant C_{n-1}$. So we always have $C_{n-1} \leqslant C_n$, and $C \equiv \lim_{n\to\infty} C_n \leqslant \infty$ exists. If $C < \infty$, then since $D_n \to \infty$ we have $\sqrt{D_n} > C_{n-1} + (n\log n)^{-1}$ for n large, hence $C_n = C_{n-1} + (n\log n)^{-1}$ for n large. This entails $C_n = C_m + \sum_{i=m+1}^{n} (i\log i)^{-1} \to \infty$ as $n \to \infty$, a contradiction. This proves (2.3). (2.4) is obvious since $C_n \uparrow \infty$ and $n/\log n \uparrow \infty$, $n \geqslant 3$. Since $C_n \leqslant C_{n-1} + (n\log n)^{-1}$, we have

$$C_n \leqslant 1 + \sum_{i=2}^{n} (i\log i)^{-1} = O(\log\log n).$$

Hence, $C_n/\log n \to 0$. Further, noting that $\log n - \log(n-1) \geqslant n^{-1}$, we have

$$\begin{aligned} C_{n-1}/\log(n-1) - C_n/\log n &\geqslant C_{n-1}/\log(n-1) - (C_{n-1} + (n\log n)^{-1})/\log n \\ &= C_{n-1}(\log n - \log(n-1))/(\log n \cdot \log(n-1)) - (n\log^2 n)^{-1} \\ &\geqslant (n\log n \cdot \log(n-1))^{-1} - (n\log^2 n)^{-1} \\ &> 0,\ n \geqslant 3. \end{aligned}$$

This proves (2.5). (2.7) follows by using (2.5):

$$\begin{aligned} &nC_n/\log n - (n-1)C_{n-1}/\log(n-1) \\ &\quad = C_n/\log n - (n-1)(C_{n-1}/\log(n-1) - C_n/\log n) \\ &\quad \leqslant C_n/\log n. \end{aligned} \tag{2.9}$$

To prove (2.6), start from (2.9), we have

$$\begin{aligned} &nC_n/\log n - (n-1)C_{n-1}/\log(n-1) \\ &\quad \geqslant C_n/\log n - (n-1)(C_{n-1}/\log(n-1) - C_{n-1}/\log n) \\ &\quad = C_n/\log n - (n-1)C_{n-1}(\log n - \log(n-1))/(\log n \cdot \log(n-1)) \\ &\quad \geqslant C_n/\log n - C_n/(\log n \cdot \log(n-1)) \\ &\quad = C_n(\log(n-1) - 1)/(\log n \cdot \log(n-1)). \end{aligned}$$

Finally, (2.8) follows by calculating the derivative of the function $(\log x)^{-1} \cdot \exp(C_x/\log x)$.

Lemma 3 Define

$$x_1 = 1,\ x_i = \sqrt{C_i/\log i}\ \exp(2^{-1} iC_i/\log i),\ i \geqslant 2,\ S_n = \sum_{i=1}^{n} x_i^2. \tag{2.10}$$

Then, for i sufficiently large, x_i increases with i, $x_i \to \infty$, and

$$\max_{1\leqslant i\leqslant n} x_i^2/S_n = o(D_n/\log n). \tag{2.11}$$

Proof Since $\sqrt{\log(n-1)/\log n} > 1 - 2(n\log n)^{-1}$, we have

$$x_n/x_{n-1} \geqslant (1 - 2(n\log n)^{-1})\exp(2^{-1}(nC_n/\log n - (n-1)C_{n-1}/\log(n-1))).$$

From (2.6), we have

$$nC_n/\log n-(n-1)C_{n-1}/\log(n-1)\geqslant 2^{-1}C_n/\log n\geqslant(2\log n)^{-1}$$

for n sufficiently large. Hence

$$\begin{aligned}x_n/x_{n-1}&\geqslant(1-2(n\log n)^{-1})\exp((4\log n)^{-1})\\&\geqslant(1-2(n\log n)^{-1})(1+(4\log n)^{-1})\\&>1\end{aligned}$$

for n sufficiently large. $x_i\to\infty$ is obvious.

To prove (2.11), we first derive an approximate expression of S_n:

$$S_n=(1+o(1))\exp(nC_n/\log n). \tag{2.12}$$

By (2.5), we have

$$\begin{aligned}S_n&\geqslant\frac{C_n}{\log n}\sum_{i=2}^{n}\exp(iC_n/\log n)\\&=\frac{C_n}{\log n}\frac{\exp((n+1)C_n/\log n)-\exp(2C_n/\log n)}{\exp(C_n/\log n)-1}\\&=(1+o(1))\exp((n+1)C_n/\log n)\\&=(1+o(1))\exp(nC_n/\log n).\end{aligned}\tag{2.13}$$

On the other hand,

$$\begin{aligned}S_n&\leqslant 1+\sum_{i=2}^{n}\frac{C_n}{\log i}\exp\left(\frac{iC_n}{\log i}\right)\\&=1+\sum_{i=2}^{[n/2]-1}+\sum_{i=[n/2]}^{n}\\&\equiv 1+J_{1n}+J_{2n}.\end{aligned}\tag{2.14}$$

We have

$$J_{1n}\leqslant nC_n\exp\left(\frac{nC_n}{2\log(n/2-2)}\right)\leqslant nC_n\exp\left(\frac{2}{3}\frac{nC_n}{\log n}\right).$$

Hence by (2.5), for n sufficiently large,

$$J_{1n}/\exp(nC_n/\log n)\leqslant n\log n\ \exp(-n/(3\log n))\to 0,\ \text{as } n\to\infty. \tag{2.15}$$

For J_{2n}, define the function $H(x)=\int_2^x\exp(C_nt/\log t)\mathrm{d}t$. By (2.8), for n large we have

$$\begin{aligned}J_{2n}&\leqslant\int_{[n/2]}^{n+1}C_n(\log x)^{-1}\mathrm{d}H(x)\\&\leqslant C_nH(n+1)/\log(n+1)+C_n\int_{[n/2]}^{n+1}H(x)\mathrm{d}(-(\log x)^{-1})\\&\leqslant C_nH(n+1)/\log(n+1)+C_nH(n+1)((\log^{-1}[n/2]-\log^{-1}(n+1))\\&=(1+o(1))C_nH(n+1)/\log(n+1).\end{aligned}\tag{2.16}$$

Since $\mathrm{d}(C_nx/\log x)=C_n(\log x-1)\log^{-2}x\mathrm{d}x=C_n(1+o(1))\log^{-1}x\mathrm{d}x$ as $x\to\infty$, we have

$$H(n+1) = (1+o(1))C_n^{-1}\int_2^{n+1} \log x d(\exp(C_n x/\log x))$$
$$= (1+o(1))C_n^{-1}\log(n+1)\exp(C_n(n+1)/\log(n+1)). \tag{2.17}$$

Summing up (2. 14)—(2. 17), we obtain

$$S_n \leqslant (1+o(1))\exp(C_n(n+1)/\log(n+1)) = (1+o(1))\exp(C_n n/\log n). \tag{2.18}$$

The last step follows from (2. 4), (2. 5), (2. 7), and the fact that

$$C_n n/\log n \leqslant C_n(n+1)/\log(n+1) \leqslant C_{n+1}(n+1)/\log(n+1).$$

Now (2. 12) follows from (2. 13) and (2. 18). From (2. 12), and the fact that x_i increases with i and $x_i \to \infty$, we have, for n sufficiently large,

$$\max_{1\leqslant i\leqslant n} x_i^2/S_n = x_n^2/S_n \leqslant (1+o(1))C_n\log^{-1} n\exp(nC_n/\log n)\exp(-nC_n/\log n)$$
$$= (1+o(1))C_n/\log n = o(D_n/\log n).$$

The last step follows by (2. 2). Lemma 3 is proved.

Lemma 4 Let $M(n) = x_1 + \cdots + x_n$, where x_i is given by (2. 10). Then

$$M(n-[2\log n/C_n]) < M(n)/2$$

for n sufficiently large.

Proof We have the following approximate expression for $M(n)$:

$$M(n) = (2+o(1))\sqrt{\log n/C_n}\exp(2^{-1}nC_n/\log n). \tag{2.19}$$

The method of proving (2. 19) is exactly the same as that used in deriving (2. 12), and the details are omitted. By (2. 19) and (2. 5), we have

$$M(n-[2\log n/C_n])$$
$$\leqslant (2+o(1))\sqrt{\log n/C_n}\exp(2^{-1}(n-[2\log n/C_n])C_n/\log(n-[2\log n/C_n])).$$

Since

$$\lim_{n\to\infty}[2\log n/C_n]C_n/\log(n-[2\log n/C_n]) = 2,$$

$$nC_n/\log(n-[2\log n/C_n]) = nC_n/\log n + O(1/\log n),$$

we have

$$M(n-[2\log n/C_n]) \leqslant (2+o(1))\sqrt{\log n/C_n}\exp(2^{-1}nC_n/\log n)e^{-1} < M(n)/2$$

for n sufficiently large. The lemma is proved.

3 Proof of Theorem 1

First consider the case $p=1$ in model (1. 1). Define x_i by (2. 10), and a sequence of iid. random variables $e_1, e_2, \cdots$ with the common distribution specified as follows: Take a positive integer N large enough such that $l \equiv \exp(-C_N/4) < 1/2$. Let

$$P(e_1 = x_n) = P(e_1 = -x_n) = \exp(-C_n/4) - \exp(-C_{n+1}/4),\ n \geqslant N,$$

and within the interval $(-x_N, x_N)$ e_1 has a density $(1-2l)/(2x_N)$. Evidently $e \equiv (e_1, e_2, \cdots)$ $(\mathscr{F})$, and according to Lemma 3 the condition (1. 6) is fulfilled. Yet, as we shall show below, the LAD estimate $\hat{\beta}_n$ of β_0 is not strongly consistent.

Without loss of generality assume $\beta_0 = 0$, so $Y_i = e_i$. For each integer $k \geqslant 2$, define $n_k = [k \log k]$, $m_k = n_k - [2\log n_k/C_{n_k}]$, and the events

$$E_k = \{e_i \geqslant x_i,\ m_k \leqslant i \leqslant n_k\},\ k \geqslant 1,$$

$$\widetilde{E}_k = \{e_i \geqslant x_{n_k},\ m_k \leqslant i \leqslant n_k\},\ k \geqslant 1.$$

According to Lemma 1 and Lemma 4, we have $\hat{\beta}_{n_k} \geqslant 1$ if E_k occurs. So, if we can show that

$$P(E_k,\ \text{i. o.}) = 1, \tag{3. 1}$$

then $P(\hat{\beta}_n \to 0) = 0$ and $\hat{\beta}_n$ is not strongly consistent.

To prove (3. 1), note that x_i increases with i for i large, so $\widetilde{E}_k \subset E_k$ for k large. Hence we have only to show that

$$P(\widetilde{E}_k,\ \text{i. o.}) = 1. \tag{3. 2}$$

According to the distribution of e_1, we have, for k large,

$$\begin{aligned} P(E_k) &= (P(e_1 \geqslant x_{n_k}))^{[2\log n_k/C_{n_k}]+1} \\ &= (\exp(-C_{n_k}/4))^{[2\log n_k/C_{n_k}]+1} \\ &\geqslant \exp\left(-\frac{2}{4}(\log k + \log\log k) - C_{n_k}/4\right) \\ &\geqslant \exp\left(-\frac{3}{4}(\log k + \log\log k)\right) \\ &\geqslant \exp(-\log k) = k^{-1}. \end{aligned}$$

Here we have made use of the fact that $C_{n_k} \leqslant \log n_k$, followed by (2. 5). Thus

$$\sum_{k=1}^{\infty} P(\widetilde{E}_k) = \infty. \tag{3. 3}$$

But when k is large enough, the intervals $[m_k, n_k]$ have no overlap and hence the $\widetilde{E}_k$s are independent of each other, for large k. From this and (3. 3), we obtain (3. 2). This concludes the proof of Theorem 1 for $p = 1$.

For the general case denote by a_i the number x_i defined by (2. 10), and choose the p-vectors $x_1, x_2, \cdots$ in the following way:

$$\begin{aligned} x_{ip+1} &= (a_{i+1}, 0, \cdots, 0)', \\ x_{ip+2} &= (0, a_{i+1}, 0, \cdots, 0)', \\ &\vdots \\ x_{ip+p} &= (0, 0, \cdots, 0, a_{i+1})',\ i = 0, 1, \cdots, \end{aligned}$$

and choose e_1, e_2, $\cdots$ as before. Then condition (1.6) holds, but each component of $\hat{\beta}_n$ does not converge a. s. to the corresponding component of β_0 as $n \to \infty$.

Theorem 1 is proved.

Remark This theorem, combined with the result in Chen *et al.* (1992), leaves the following question unanswered: Assuming $e(\mathscr{F})$, is the condition

$$d_n = O(1/\log n)$$

sufficient for the strong consistency of the LAD estimate $\hat{\beta}_n$ of β_0? This is the only possibility of improving the order established in Chen *et al.* (1992).

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A Tobin-type Estimate of Censored Linear Models

Abstract The paper presents a two-stage method for estimating the parameters in the censored linear model $Y_i = \max(0, \alpha_0 + X_i'\beta_0)$, $1 \leqslant i \leqslant n$. In the first stage the data are grouped in some groups and then some adjustments are made, the results are used in the latter stage to form a Tobin-type estimate. The asymptotic normality of the estimate is proved and some simulations are made.

1 Introduction

Consider the linear regression model censored from the left:

$$Y_i = \max\{\widetilde{Y}_i, 0\} = \max\{Z_i'\gamma_0 + e_i, 0\}, \ 1 \leqslant i \leqslant n, \tag{1.1}$$

where

$$Z_i = (1, X_i')', \ 1 \leqslant i \leqslant n, \ \gamma_0 = (\alpha_0, \beta_0')'. \tag{1.2}$$

The dimensionality of X_i is $d \geqslant 1$. We observe $(X_i, Y_i)'$, $1 \leqslant i \leqslant n$. Based on these observations, we try to give some estimation of γ_0, or test hypothesis on it.

The model was first considered by Tobin[1] under the assumption that $e_1, e_2, \cdots$ are iid. with a common normal distribution $N(0, \sigma^2)$. He defined the MLE of γ_0, which now is usually known as the Tobin estimator. The consistency of this estimator was established by Amemiya[2] in 1973, under the same assumption. It has also been discovered that the Tobin estimator may not be consistent when the error distribution is non-normal, this arouses the interest of studying model (1.1) under no specific assumptions on the distribution of e_i. Powell[3] is a work in this direction. He noticed the fact that $\text{med}(Y_i) = \max(Z_i'\gamma_0, 0)$ when $\text{med}(e_i) = 0$ and suggested the use of the LAD criterion in estimating γ_0.

On the other hand, Nawata in [4] proposed a method based on Least Squares. But in his scheme the Least Squares process is not used directly on the original data. In fact, he introduced a very cumbersome procedure in grouping and adjustment of the original data. After

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this he defines the "first stage estimator", also by a very complicated procedure. Then he introduces the "second stage estimator" based on the first stage estimator which is taken as the final estimate of γ_0.

Borrowing the idea of grouping and adjustment of data proposed by Nawata, the present authors advanced in [5] a method of estimating γ_0, the aim is to simplify the process of calculation and to obtain better asymptotic properties of the estimator under more reasonable conditions. There is an undesirable feature in the method that in constructing the estimator a part of the data is not used, and this may diminish its efficiency. In view of this, we propose in this paper a procedure of estimation which makes fuller use of the data. The method is still based on grouping and adjustment of the data, but the estimator resulted is of the Tobin-type.

2 The Method

Choose $\varepsilon_1 \in (0, (2d)^{-1})$, $l_n = ln^{-\varepsilon_1}$, where $l > 0$ is some chosen constnat. Partition R^d into cubes of the form

$$\{x = (x^{(1)}, \cdots, x^{(d)}) : m_i l_n \leqslant x^{(i)} < (m_i + 1) l_n,\ i = 1, \cdots, d\},$$

$$m_i = 0, \pm 1, \cdots, i = 1, \cdots, d.$$

The set of all these cubes is denoted by S_n. Choose constants $c_0 > 0$ and $\varepsilon_2 \in (1/2, 1 - d\varepsilon_1)$ and write $\{J_{n1}, \cdots, J_{nc_n}\}$ for $\{J : J \in S_n, \#(J) \geqslant c_0 n^{\varepsilon_2}\}$, where $\#(J)$ is the number of samples ($X_1, \cdots, X_n$) contained in J. Then we have $c_n \leqslant c_0^{-1} n^{1-\varepsilon_2}$. Denote $\#(J_{ni})$ by n_i, and $X_{ni}(1), \cdots, X_{ni}(n_i)$ for the X-samples included in J_{ni}. If $X_{ni}(j)$ in the original series $X_1, \cdots, X_n$ has the name X_k, then $Y_{ni}^{(0)}(j)$ means Y_k, and $e_{ni}(j)$ means e_k. From the definition of J_{ni} we have $n_i \geqslant c_0 n^{\varepsilon_2}$ for $i = 1, \cdots, c_n$. Choose constant $c_0' > 0$ and define

$$H_{ni} = \sum_{j=1}^{n_i} I(Y_{ni}^{(0)}(j) > c_0' n^{-\varepsilon_1/2}),\ H_{ni}' = \sum_{j=1}^{n_i} I(Y_{ni}^{(0)}(j) = 0),$$

where I is the indicator. Choose constants $g > 0$ and $\varepsilon_3 \in (1 - \varepsilon_1, 1)$, and write, for convenience and without loss of generality, $S_n^+ \equiv \{J_{n1}, \cdots, J_{nb}\} = \{J_{ni} : 1 \leqslant i \leqslant c_n, H_{ni} > n_i/2 + g n_i^{\varepsilon_3}\}$, $S_n^- \equiv \{J_{n,b+1}, \cdots, J_{n,b+h}\} = \{J_{ni} : 1 \leqslant i \leqslant c_n, H_{ni}' > n_i/2 + g n_i^{\varepsilon_3}\}$, when b and h all depend on n. Now we can introduce a Tobin-type likelihood function $L(\gamma, \sigma)$ as follows:

$$L(\gamma, \sigma) = \prod_{i=b+1}^{b+h} \Phi(-\sqrt{n_i} Z_{ni}' \gamma / \sigma) \prod_{i=1}^{b} \frac{\sqrt{n_i}}{\sigma} \exp\Big(-\frac{n_i}{2\sigma^2}(Y_{ni}^{(r+1)} - Z_{ni}'\gamma)^2\Big). \tag{2.1}$$

Where Φ is the distribution function of $N(0, 1)$, $Z_{ni} = (1, X_{ni}')'$ with $X_{ni} = n_i^{-1}(X_{ni}(1) + \cdots + X_{ni}(n_i))$, and ($Y_{n1}^{(r+1)}, \cdots, Y_{nb}^{(r+1)}$), with r an integer satisfying $r\varepsilon_1 \leqslant 1/2 < (r+1)\varepsilon_1$, is defined by an interative process as follows: First, let

$$Y_{ni}^{(0)} = \mathrm{med}(Y_{ni}^{(0)}(1), \cdots, Y_{ni}^{(0)}(n_i)),\ 1 \leqslant i \leqslant b.$$

Next, assume that ($Y_{n1}^{(t)}$, $\cdots$, $Y_{nb}^{(t)}$) has already been defined. Denote by $\tilde{\gamma} = (\tilde{\alpha}, \tilde{\beta}')'$ the solution of the following weighted least squares:

$$\sum_{i=1}^{b} n_i (Y_{ni}^{(t)} - Z'_{ni}\gamma)^2 = \text{minimum}. \tag{2.2}$$

Put

$$Y_{ni}^{(t+1)}(j) = Y_{ni}^{(t)}(j) - (X_{ni}(j) - X_{ni})'\tilde{\beta},\ 1 \leqslant j \leqslant n_i,\ 1 \leqslant i \leqslant b$$

and define

$$Y_{ni}^{(t+1)} = \text{med}(Y_{ni}^{(t+1)}(1), \cdots, Y_{ni}^{(t+1)}(n_i)).$$

Return to (2.1). Denote by ($\hat{\gamma}_n$, $\hat{\sigma}_n$) the maximizing point of $L(\gamma, \sigma)$, which is taken as an estimate of (γ_0, σ_0). In [5] we propose to use $\tilde{\gamma} \equiv \gamma_n = (\tilde{\alpha}_n, \tilde{\beta}'_n)'$ defined by the solution of (2.2) (with t replaced by $r+1$) as the estimate of $\gamma_0 = (\alpha_0, \beta'_0)'$, and

$$\sigma_n^2 \equiv R_n/(b-d-1) \equiv \sum_{i=1}^{b} n_i (Y_{ni}^{(r+1)} - Z'_{ni}\gamma_n)^2/(b-d-1) \tag{2.3}$$

as an estimate of σ_0^2 (for definition of σ_0^2 see Section 4). We hope that since in defining ($\hat{\gamma}_n$, $\hat{\sigma}_n$) we make use of more data as compared in the definition of (γ_n, σ_n), the former may have a better performance. The simulation results reported in the next section seems to support this hope.

3 Simulation Experiments

To get some idea about the perfomance of the proposed estimators, a small scale simulation has been done. The results are listed in the table below. The model we considered is $y = I(\alpha + \beta x + \varepsilon)$ with $\alpha = 0$ and $\beta = 1$. The data set of x is generated from the uniform distribution on the interval $[-\sqrt{3}, \sqrt{3}]$, while the data sets of the noise, ε is drawn from five different distribution:

(1) N: the standard normal distribution;

(2) L: the Laplace distribution with density function $\frac{1}{2}e^{-|x|}$;

(3) K: $\chi^2 - 0.456$ where χ^2 has the chi-squared distribution with one degree of freedom;

(4) E: $E - \log 2$, where E has the exponential distribution with density function $e^{-x}I(x > 0)$;

(5) C: the Cauchy distribution with density function $(\pi(1+x^2))^{-1}$.

We conduct seven designs. 201 replications are performed for each design and within each replication the sample size is 200. In the first five designs, the noise data are generated from the above-listed five distributions respectively. In the last two designs, labeled N_1 and N_2 in the table, the 200th noise-datum are distributed as $N(0, 3)$ and $N\left(0, \frac{1}{3}\right)$ respectively, while the remaining noise-data all come from $N(0, 1)$. Note that in these two designs the data are not iid. We consider these two designs since they may give some information concerning the behavior of the estimators in case the statistical population, from which the data are drawn, violate slightly the assumptions imposed in establishing the theoretical results. For the same reason, we consider the Cauchy distribution, for which the moment condition is not satisfied. In doing the simulation several free constants must be specified properly, which we choose as follows:

$$c_0 = 0.01,\ c_0' = 0.25,\ \varepsilon_1 = 0.225,\ \varepsilon_2 = 0.75,\ \varepsilon_3 = 0.5,\ g = 0.25.$$

Simulation Results

		N	L	K	E	C	N_1	N_2
$\alpha = 0$	α_n	0.091	0.052	0.041	0.126	0.224	0.142	0.128
$\alpha = 0$	$\hat{\alpha}_n$	0.091	0.016	0.083	0.050	0.118	0.050	0.017
$\beta = 1$	β_n	0.985	0.927	0.937	0.953	0.786	0.912	0.925
$\beta = 1$	$\hat{\beta}_n$	0.994	0.984	1.083	1.050	1.118	1.050	1.017

The numbers in the above table give the mean value of the individual simulation results. From the table it is seen that $(\hat{\alpha}_n, \hat{\beta}_n)$ is generally closer to the true value $(0, 1)$ as compared with (α_n, β_n), especially in the last two non-iid. cases. Of cause, a limited amount of simulation is not enough in giving a conclusive assertion. Besides, the choose of the free constants in the definition of the estimator is a matter deserves further investigation.

4 Large Sample Properties of $\hat{\gamma}_n$

In this section we study the asymptotic behavior of $\hat{\gamma}_n$ defined in Section 2. We shall prove the asymptotic normality of $\hat{\gamma}_n$ under a set of general conditions. First we lay down these conditions.

1° $(X_i', \tilde{Y}_i)$, $i = 1, 2, \cdots$ are iid. samples from the random vector (X', Y) assuming values in R^{d+1}.

2° $e_1, e_2, \cdots$ are iid. with a common distribution F, $\mathrm{med}(e_1) = 0$, $f(x) = F'(x)$ exists in some vicinity of 0, $f(0) > 0$ and $f'(0)$ exists. Also, $(e_1, e_2, \cdots)$ and $(X_1, X_2, \cdots)$ are independent.

3° $\mathrm{E}\,\|X\|^{2+\delta} < \infty$ for some $\delta > 0$, $P(Z'\gamma_0 > 0) > 0$ and $V_0 \equiv \mathrm{COV}(X \mid Z'\gamma_0 > 0) > 0$. Also, $\varepsilon_1 > 0$ is chosen to satisfy $\varepsilon_1 < \delta/(4 + 2\delta)$.

4° F has no singular part. If F possesses an absolute continuous part with density g, then the set of all discontinuity points of g is a closed set with Lebesgue measure 0.

Theorem 1 Under the above conditions we have

$$\sqrt{n}(\hat{\gamma}_n - \gamma_n) \to 0 \text{ in pr., as } n \to \infty. \tag{4.1}$$

In [5] it is shown that under the above conditions we have

$$\sqrt{n}(\gamma_n - \gamma_0) \xrightarrow{\mathcal{L}} N(0, \sigma_0^2 V^{-1}), \text{ as } n \to \infty, \tag{4.2}$$

with $V = \mathrm{E}(ZZ'I(Z'\gamma_0 > 0))$, and

$$\sigma_0^2 = (4f^2(0))^{-1}.$$

Combining (4.1) and (4.2), we get

Theorem 2 Under the above conditions 1°—4° we have

$$\sqrt{n}(\hat{\gamma}_n - \gamma_0) \xrightarrow{\mathcal{L}} N(0, \sigma_0^2 V^{-1}), \text{ as } n \to \infty. \tag{4.3}$$

To prove Theorem 1, we need some preparatory facts.

Lemma 1 Write $N_n = n_1 + \cdots + n_b$, where n_i and b are defined in Section 2. Then under the conditions of Theorem 1, we have

$$\lim_{n\to\infty} N_n/n = P(Z'\gamma_0 > 0), \text{ a. s.}$$

and

$$\lim_{n\to\infty} D_n \equiv \lim_{n\to\infty} \sum_{i=1}^{b} n_i Z_{ni} Z'_{ni}/N_n = \mathrm{E}(ZZ' \mid Z'\gamma_0 > 0), \text{ a. s.}$$

The first assertion is (4.9) of [5], and the second assertion can be proved likewise.

Lemma 2 Given $\varepsilon_4 < \varepsilon_3$. Under the conditions of Theorem 1 it is true with probability 1 that for n sufficiently large

$$Z'_{ni}\gamma_0 \geqslant n_i^{-1+\varepsilon_4}, \ i = 1, \cdots, b, \tag{4.4}$$

$$Z'_{ni}\gamma_0 \leqslant - n_i^{-1+\varepsilon_4}, \ i = b+1, \cdots, b+h. \tag{4.5}$$

Proof Suppose that $i \leqslant b$ but $Z'_{ni}\gamma_0 < n_i^{-1+\varepsilon_4}$. Write $Z_{ni}(j) = (1, X'_{ni}(j))'$. We have, for some $\varepsilon_5 \in (\varepsilon_4, \varepsilon_3)$, that

$$\begin{aligned} Z'_{ni}(j)\gamma_0 &< n_i^{-1+\varepsilon_4} + \|\beta_0\| \, \|X_{ni} - X_{ni}(j)\| \\ &< n_i^{-1+\varepsilon_4} + \|\beta_0\| \, l\sqrt{d}\, n^{-\varepsilon_1} \\ &< n_i^{-1+\varepsilon_5}. \end{aligned}$$

Here we use the fact that $1-\varepsilon_1 < \varepsilon_3 < 1$ and $\varepsilon_4 < \varepsilon_3$. Hence, by the definition of $S_n^{\cdot}$, we have

$$H_i \equiv \sum_{j=1}^{n_i} I(e_{ni}(j) > - n_i^{-1+\varepsilon_5}) \geqslant \frac{n_i}{2} + g n_i^{\varepsilon_3}. \tag{4.6}$$

By condition 2° of Theorem 1, it follows that there exists a constant K such that

$$p_i \equiv P(e_1 > - n_i^{-1+\varepsilon_5}) \leqslant \frac{1}{2} + K n_i^{-1+\varepsilon_5}. \tag{4.7}$$

Combining (4.6), (4.7), and noticing that $\varepsilon_3 > \varepsilon_5$, it follows that when n is sufficiently large

$$H_i/n_i - p_i \geqslant gn_i^{-1+\varepsilon_5} - Kn_i^{-1+\varepsilon_5} \geqslant gn_i^{-1-\varepsilon_3}/2. \tag{4.8}$$

Furthermore, Hoeffding's inequality gives the following estimator for probability of the event (4.8)

$$P(|H_i/n_i - p_i| \geqslant gn_i^{-1+\varepsilon_3}/2) \leqslant 2\exp(-n_i \cdot (gn_i^{-1+\varepsilon_3}/2)^2/3) = 2\exp(-g^2 n_i^{-1+2\varepsilon_3}/12).$$

Note that $-1+2\varepsilon_3 > 0$ because $\varepsilon_1 < 1/2$, so $\varepsilon_3 > 1-\varepsilon_1 > 1/2$. Together with $n_i \geqslant C_0 n^{\varepsilon_2}$, we see that there exists a constant $q > 0$ such that for n sufficiently large

$$P(|H_i/n_i - p_i| > gn_i^{-1+\varepsilon_3}/2) \leqslant \exp(-n^q) \leqslant n^{-3}. \tag{4.9}$$

Define an event

$$E_n = \{Z'_{ni}\gamma_0 < n_i^{-1+\varepsilon_4} \text{ for some } i,\ 1 \leqslant i \leqslant b\},$$

then by (4.9) we have, for n large,

$$P(E_n) \leqslant \sum_{i=1}^{b} P(Z'_{ni}\gamma_0 < n_i^{-1-\varepsilon_4}) \leqslant \sum_{i=1}^{b} P(|H_i/n_i - p_i| \geqslant gn_i^{-1+\varepsilon_3}/2)$$
$$\leqslant bn^{-3} < n^{-2}.$$

Therefore $\sum_{n=1}^{\infty} P(E_n) < \infty$ and $P(E_n \text{ i. o. }) = 0$. This proves (4.4). (4.5) can be proved in a similar fashion.

Lemma 3 If $\xi_1, \xi_2, \cdots$ are iid. random variables with $\mathrm{E}|\xi_1|^r < \infty$ for some $r > 0$, then

$$\lim_{n\to\infty} n^{-1/r} \max_{1\leqslant i\leqslant n} |\xi_i| = 0, \text{ a. s.}$$

The general case is a corollary of the special case $r = 1$, which in turn follows easily from Kolmogorov's strong law of large numbers.

Lemma 4 Under conditions of Theorem 1 we have

$$\hat{\sigma}_n = O_p(1). \tag{4.10}$$

Proof First give an estimator for $L(\gamma_n, \sigma_n)$. Since $\mathrm{E}\|X\|^{2+\delta} < \infty$, from Lemma 3 we have

$$\max(\|X_{ni}\| : b+1 \leqslant i \leqslant b+h) = o(n^{-1/(2+\delta)}), \text{ a. s.}$$

From this and (4.2), we have

$$\max(|Z'_{ni}(\gamma_n - \gamma_0)| : b+1 \leqslant i \leqslant b+h) = o_p(n^{-\delta/(4+2\delta)}), \tag{4.11}$$

we have $\varepsilon_3 > 1 - \varepsilon_1 > 1 - \delta/(4+2\delta) = (4+\delta)/(4+2\delta)$. Hence we can choose $\varepsilon_4 > (4+\delta)/(4+2\delta)$ in Lemma 2. According to Lemma 2, it is true with probability one that for n large

$$Z'_{ni}\gamma_0 \leqslant -n_i^q,\ i = b+1, \cdots, b+h, \tag{4.12}$$

where $q > -\delta/(4+2\delta)$ is a constant. Combining (4.11) and (4.12), we get

$$\lim_{n\to\infty} P(Z_{ni}'\gamma_n \leqslant -n_i^q/2,\ i=b+1,\ \cdots,\ b+h)=1.$$

This, together with (5.1), gives

$$\lim_{n\to\infty} P(Z_{ni}'\gamma_n \leqslant -n_i^q/2,\ i=b+1,\ \cdots,\ b+h;\ \sigma_n \leqslant 2\sigma_0)=1. \tag{4.13}$$

Since $n_i \geqslant c_0 n^{\varepsilon_2}$, we have

$$n_i(n_i^q)^2 = n_i^{1+2q} \equiv n_i^a \geqslant c_0^a n^{\varepsilon_2 a},\ i=b+1,\ \cdots,\ b+h.$$

Here $a=1+2q>1-2\delta/(4+2\delta)>0$. Now using the inequalities:

$$\Phi(t)\geqslant 1-(\sqrt{2\pi}t)^{-1}\exp(-t^2/2),\ t>0,$$

$$\log(1-t)\geqslant -2t,\ t>0 \text{ sufficiently small},$$

and taking $a_1\in(0,\ \varepsilon_2 a/2)$, we can derive that, when the event in the left-hand side of (4.13) occurs and n is large enough, then for any given $k>0$

$$\begin{aligned}
0 &> \log \prod_{i=b+1}^{b+h}\Phi(-\sqrt{n_i}Z_{ni}'\gamma_n/\sigma_n)\\
&\geqslant \log \prod_{i=b+1}^{b+h}\Phi(\sqrt{n_i}n_i^q/4\sigma_0) \geqslant \log \Phi^h(n^{a_1})\\
&\geqslant -2h(\sqrt{2\pi}n^{a_1})^{-1}\exp(-n^{2a_1}/2)\\
&\geqslant -n^{-k}\to 0.
\end{aligned} \tag{4.14}$$

Combining with (5.1), we obtain

$$\lim_{n\to\infty} P(L(\gamma_n,\ \sigma_n)\geqslant \sigma_n^{-b}e^{-b/2}(n_1\cdots n_b)^{1/2}e^{-1/n})=1. \tag{4.15}$$

Write $q=2/\sqrt{e}$, $q>1$. When $\sigma\geqslant 2\sigma_n$ we have

$$L(\gamma,\ \sigma)\leqslant \sigma_n^{-b}e^{-b}(n_1\cdots n_b)^{1/2}/q^b.$$

Together with (4.14), it follows that

$$\lim_{n\to\infty} P(\hat\sigma_n < 2\sigma_n)=1.$$

This, together with (5.1), gives

$$\lim_{n\to\infty} P(\hat\sigma_n < 3\sigma_0)=1. \tag{4.16}$$

This proves (4.10). The proof is completed.

Proof of Theorem 1 Given $\varepsilon>0$ and taking arbitrarily γ satisfying $\|\gamma-\gamma_n\|\geqslant \varepsilon/\sqrt{n}$, we then have

$$\begin{aligned}
\log L(\gamma,\ \hat\sigma_n) &\leqslant -b\log\hat\sigma_n + \frac{1}{2}\sum_{i=1}^{b}\log n_i - (2\hat\sigma_n^2)^{-1}\sum_{i=1}^{b} n_i(Y_{ni}^{(r+1)}-Z_{ni}'\gamma)^2\\
&= -b\log\hat\sigma_n + \frac{1}{2}\sum_{i=1}^{b}\log n_i - (2\hat\sigma_n^2)^{-1}R_n - (2\hat\sigma_n^2)^{-1}\sum_{i=1}^{b} n_i(Z_{ni}'(\gamma_n-\gamma))^2.
\end{aligned} \tag{4.17}$$

R_n is defined in (2.3). Denote by λ_n the smallest eigenvalue of D_n (see Lemma 1), and λ the smallest eigenvalue of $\mathrm{E}(ZZ' \mid Z\gamma_0 > 0)$. From Lemma 1 it follows that

$$\lim_{n\to\infty} P(\lambda_n \geqslant \lambda/2) = 1. \tag{4.18}$$

Writing $p = P(Z'\gamma_0 > 0)$, we have $p > 0$ due to condition 3° of Theorem 1. From Lemma 1, (4.17), (4.18), $\| \gamma - \gamma_n \| \geqslant \varepsilon/\sqrt{n}$, and also observing that

$$\sum_{i=1}^{b} n_i (Z'_{ni}(\gamma_n - \gamma))^2 = N_n(\gamma_n - \gamma)' D_n (\gamma_n - \gamma),$$

we then have that as $n \to \infty$,

$$\log L(\gamma,\ \hat{\sigma}_n) \leqslant -b\log \hat{\sigma}_n + \frac{1}{2}\sum_{i=1}^{b} \log n_i - (2\hat{\sigma}_n^2)^{-1} R_n - \lambda p \varepsilon^2 / 3\hat{\sigma}_n^2 \tag{4.19}$$

holds simultaneously for all γ satisfying $P(\| \gamma - \gamma_n \| \geqslant \varepsilon/\sqrt{n}) \to 1$. On the other hand, (4.14) remains true when σ_n is replaced by any $\sigma' \in (0,\ 3\sigma_0)$ and the convergence in (4.14) is also uniform for σ' lying in this interval, as long as the event in the left-hand side of (4.13) occurs. Hence, when the events in (4.13) and (4.16) occur simultaneously, we have

$$\log L(\gamma_n,\ \hat{\sigma}_n) \geqslant -b\log \hat{\sigma}_n + \frac{1}{2}\sum_{i=1}^{b} \log n_i - (2\hat{\sigma}_n^2)^{-1} R_n - o(1). \tag{4.20}$$

Summarizing (4.13), (4.16), (4.19) and (4.20), we obtain

$$\lim_{n\to\infty} P(\sup\{L(\gamma,\ \hat{\sigma}_n) : \| \gamma - \gamma_n \| \geqslant \varepsilon/\sqrt{n}\} < L(\gamma_n,\ \hat{\sigma}_n)) = 1.$$

Observing that $(\hat{\gamma}_n,\ \hat{\sigma}_n)$ maximizes $L(\gamma,\ \sigma)$, we have

$$\lim_{n\to\infty} P(\| \hat{\gamma}_n - \gamma_n \| > \varepsilon/\sqrt{n}) = 0.$$

Since $\varepsilon > 0$ is arbitrarily given, (4.1) follows, which concludes the proof of Theorem 1 and Theorem 2.

5 Large Sample Properties of $\hat{\sigma}_n$

(4.3) indicates that σ_0^2 measures the large sample precision of the estimate $\hat{\sigma}_n$, so this parameter is of some importance. In [5] it is shown that under the conditions of Theorem 1 we have

$$\sigma_n^2 \equiv R_n/(b-d-1) \to \sigma_0^2 \text{ in pr. as } n \to \infty. \tag{5.1}$$

R_n has been defined in (2.3). Also, as $n \to \infty$,

$$\sqrt{2(b-d-1)}\,\sigma_n/\sigma_0 - \sqrt{2(b-d)-3} \xrightarrow{\mathcal{L}} N(0,\ 1) \tag{5.2}$$

and

$$\sqrt{(b-d-1)/2}(\sigma_n^2/\sigma_0^2-1)\xrightarrow{\mathcal{L}}N(0,\ 1). \tag{5.3}$$

In this section we prove the following theorem:

Theorem 3 Suppose that the conditions of Theorem 1 are met and that the distribution of X is not purely atomic, then as $n\to\infty$,

$$\hat{\sigma}_n-\sigma_n\to 0,\ \text{in pr.} \tag{5.4}$$

From (5.1) and (5.4) we see that $\hat{\sigma}_n$ is a consistent estimate σ_0. Further, in the process of proof of Theorem 3 we shall prove a fact which, taken together with (5.2) and (5.3), gives the following Theorem:

Theorem 4 Under the conditions of Theorem 3 we have

$$\sqrt{2(b-d-1)}\ \hat{\sigma}_n/\sigma_0-\sqrt{2(b-d)-3}\xrightarrow{\mathcal{L}}N(0,\ 1)$$

and

$$\sqrt{(b-d-1)/2}(\hat{\sigma}_n^2/\sigma_0^2-1)\xrightarrow{\mathcal{L}}N(0,\ 1),$$

as $n\to\infty$.

Proof of Theorem 3 First prove that if the event in the left-hand side of (4.13) occurs, then for any d-vector sequence $\{t_n\}$ such that $\|t_n-\gamma_n\|=O(n^{-1/2})$ we have, as $n\to\infty$

$$0>\log\prod_{i=b+1}^{b+h}\Phi(-\sqrt{n_i}Z'_{ni}t_n/\sigma')\geqslant -n^{-k}\to 0 \tag{5.5}$$

for arbitrarily given $k>0$, and the inequality holds simultaneously for $\sigma'\in(0,\ 3\sigma_0)$. Note that we have shown that it holds when t_n is replaced by γ_n, as can be seen from (4.14). But since $\|t_n-\gamma_n\|=O(n^{-1/2})$, (4.14) still holds when γ_n is replaced by t_n. This proves (5.5). From (4.1),(4.16) and (5.5), it follows that

$$\lim_{n\to\infty}P(0>\log\prod_{i=b+1}^{b+h}\Phi(-\sqrt{n_i}Z'_{ni}\ \hat{\gamma}_n/\ \hat{\sigma}_n)\geqslant -n^{-k})=1. \tag{5.6}$$

Define

$$\tilde{\sigma}_n=\sqrt{(b-d-1)/b}\ \sigma_n,\ \tilde{\sigma}_n^2=R_n/b.$$

Then since (4.14) is still valid when σ_n is replaced by $\tilde{\sigma}_n$, hence by (5.6) we have

$$\lim_{n\to\infty}P\Big(\log\prod_{i=b+1}^{b+h}\Phi(-\sqrt{n_i}Z'_{ni}\ \hat{\gamma}_n/\ \hat{\sigma}_n)-\log\prod_{i=b+1}^{b+h}\Phi(-\sqrt{n_i}Z'_{ni}\gamma_n/\ \tilde{\sigma}_n)\leqslant n^{-k}\Big)=1. \tag{5.7}$$

Further since

$$\sum_{i=1}^{b}n_i(Y_{ni}^{(r+1)}-Z'_{ni}\ \hat{\gamma}_n)^2\geqslant\sum_{i=1}^{b}n_i(Y_{ni}^{(r+1)}-Z'_{ni}\gamma_n)^2=b\,\tilde{\sigma}_n^2,$$

we have

$$T_n\equiv\log\prod_{i=1}^{b}(\hat{\sigma}_n^{-1}\exp(-n_i(Y_{ni}^{(r+1)}-Z'_{ni}\ \hat{\gamma}_n)^2/2\hat{\sigma}_n^2))$$

$$-\log \prod_{i=1}^{b} (\tilde{\sigma}_n^{-1} \exp(-n_i(Y_m^{(r+1)} - Z'_m \gamma_n)^2 / 2\tilde{\sigma}_n^2))$$

$$\leqslant b(1 - (\tilde{\sigma}_n / \hat{\sigma}_n)^2)/2 + b\log(\tilde{\sigma}_n / \hat{\sigma}_n). \tag{5.8}$$

Noticing that

$$(1 - x^2)/2 + \log x \leqslant -(x-1)^2/2, \; x > 0.$$

From (5.8) we obtain

$$T_n \leqslant -b(\tilde{\sigma}_n / \hat{\sigma}_n - 1)^2/2. \tag{5.9}$$

Given arbitrarily $a > 0$. If $|\tilde{\sigma}_n / \hat{\sigma}_n - 1| \geqslant n^{-a}$, then (5.9) gives $T_n \leqslant -bn^{-2a}/2$. Take k in (5.7) greater than $2a$. If the event in the left-hand side of (5.7) occurs, then

$$\log L(\hat{\gamma}_n, \hat{\sigma}_n) - \log L(\gamma_n, \tilde{\sigma}_n) < 0.$$

But this is impossible in view of the definition of $(\hat{\gamma}_n, \hat{\sigma}_n)$. Hence from (5.7) it follows that

$$\log_{n\to\infty} P(|\tilde{\sigma}_n / \hat{\sigma}_n - 1| \geqslant n^{-a}) = 0.$$

So for any given constant $a > 0$ we have

$$n^a(\hat{\sigma}_n - \tilde{\sigma}_n) = O_p(1). \tag{5.10}$$

Since the distribution of X is not purely atomic, we have

$$P(\lim_{n\to\infty} b = \infty) = 1$$

and this, in view of $\tilde{\sigma}_n = \sqrt{(b-d-1)/b}\ \sigma_n$, entails

$$\tilde{\sigma}_n - \sigma_n \to 0, \text{ in pr.} \tag{5.11}$$

Now (5.4) follows from (5.10) and (5.11). Theorem 3 is proved.

Proof of Theorem 4 From (5.10) we have

$$\hat{\sigma}_n - \sigma_n - (\sqrt{(b-d-1)/b} - 1)\sigma_n = O_p(n^{-a}).$$

Or, since d is fixed and $\sigma_n = O_p(1)$ (as seen from (5.1)),

$$\hat{\sigma}_n / \sigma_n - 1 = O(b^{-1} + O_p(n^{-a})). \tag{5.12}$$

Since $b \leqslant n$, taking $a = 1$ in (5.12), we get

$$\hat{\sigma}_n / \sigma_n - 1 = O_p(b^{-1}). \tag{5.13}$$

From (5.13), and observing the fact that $b \to \infty$ a. s., Theorem 4 follows from (5.2) and (5.3) easily.

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25

Cramer-von Mises 检验的非无偏性

摘要 证明了对任意的样本量 n 和检验水平 $\alpha \in (0, 1)$, Cramer-von Mises 检验都不是无偏的.

1 主要结果

设 $X_1, \cdots, X_n$ 是从具分布 G 的总体中抽出的 iid. 样本, F 是处处连续的一维分布函数. 要检验假设 $H: G(x) = F(x), -\infty < x < \infty$. Cramer[1] 和 von Mises[2] 先后提出了检验统计量(F_n 是 $X_1, \cdots, X_n$ 的经验分布)

$$W_n^2 = \int_{-\infty}^{\infty} (F_n(x) - F(x))^2 \mathrm{d}F(x),$$

然后当 W_n^2 取大值时否定 H, 这就是周知的 Cramer-von Mises(CM)检验.

关于这个检验的早期工作, 总结在文献[3]中, 较晚一些的结果可参看文献[4], 另可参看文献[5]4.3 节, 这些工作主要考察了在原假设 H 或某种特定的对立假设(随 n 变化)下, 当样本量 $n \to \infty$ 时统计量 W_n^2 的渐近性态. 本文考察 CM 检验的一个小样本问题——无偏性问题, 结果如下:

定理 对任意的自然数 n 和检验水平 $\alpha \in (0, 1)$, 以 $W_n^2 > c_n$ 为否定域的水平 α 检验不能是无偏的.

2 定理的证明

引理 记 $A = \{x = (x_1, \cdots, x_n): 0 \leqslant x_1 \leqslant \cdots \leqslant x_n \leqslant 1\}$, 在 A 上定义函数 $f(x) = \sum_{k=1}^{n} \left(x_k - \frac{2k-1}{2n}\right)^2$, 并记 $\Delta^* = \max_{x \in A} f(x)$, 则对任意 $\Delta \in (0, \Delta^*)$, 必存在 $x = (x_1, \cdots, x_n) \in A$, 使得 $0 < x_1 < \frac{1}{2n}$, $x_1 < x_2$, 且 $f(x) = \Delta$.

证 按以下步骤:

(i) f 的最大值 Δ^* 只在$(0, \cdots, 0)$和$(1, \cdots, 1)$这两点达到. 这是因为, 若 $x = (x_1, \cdots,$

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x_n) 满足条件

$$x_1 \leqslant \cdots \leqslant x_{i-1} < x_i = x_{i+1} = \cdots = x_j < x_{j+1} \leqslant \cdots \leqslant x_n,$$

而 $x_i, \cdots, x_j$ 的公共值 y 在(0, 1)内,则可找到 ε,当 $|\varepsilon|$ 充分小,使得 $\sum_{k=i}^{j}\left(\frac{2k-1}{2n}-y\right)^2 < \sum_{k=i}^{j}\left(\frac{2k-1}{2n}-(y+\varepsilon)\right)^2$,而 $x'=(x_1, \cdots, x_{i-1}, y+\varepsilon, \cdots, y+\varepsilon, x_{j+1}, \cdots, x_n) \in A$,这时有 $f(x') > f(x)$,而 x 不能是最大值点,故若 $f(x_1, \cdots, x_n)=\Delta^*$,则每个 x_i 只能为 0 或 1. 易见只有在所有 x_i 全为 0 或全为 1 时,f 才达到最大值.

(ii) 若 $f(x)=\Delta$, $x=(0, x_2, \cdots, x_n) \in A$,则必存在 $t=(t_1, \cdots, t_n) \in A$, $f(t)=\Delta$,且 $0 < t_1 < \frac{1}{2n}$. 事实上,因 $\Delta < \Delta^*$,由(i)知 $x_2, \cdots, x_n$ 不能全为 0. 设 x 有形式

$$x_{i-1}=0,\ x_i=\cdots=x_j=y \in (0, 1),\ x_j < x_{j+1},\ j \leqslant n,$$

则可取充分小的 $\varepsilon_1 \in \left(0, \frac{1}{2n}\right)$ 及 ε_2, $|\varepsilon_2|$ 充分小,使得 $t=(\varepsilon_1, \cdots, \varepsilon_1, y+\varepsilon_2, \cdots, y+\varepsilon_2, x_{j+1}, \cdots, x_n) \in A$ 且 $f(t)=f(x)$. 若 x 有形式(0, …, 0, 1, …, 1),0 的个数记为 l,若 $l=n-1$,则存在充分小的 $\varepsilon_1 \in \left(0, \frac{1}{2n}\right)$ 及充分小的 $\varepsilon_2>0$,使得 $t=\left(\varepsilon_1, \cdots, \varepsilon_1, \frac{n-1}{n}-\varepsilon_2\right) \in A$ 而 $f(t)=f(x)$. 若 $l=1$,则因 $f(0, 1, \cdots, 1)=f(0, \cdots, 0, 1)$,而回到 $l=n-1$ 的情况. 若 $1<l<n-1$,记 $t'=(0, a, \cdots, a, 1, \cdots, 1)$,其中 $a=\frac{2l+1}{2n}$ 且 a 有 $l-1$ 个,则 $f(t')=f(x)$. 但 t' 的分量中有异于 0, 1 者,故回到前面已证的情况.

(iii) 若 $f(x)=\Delta$, $x=\left(\frac{1}{2n}, x_2, \cdots, x_n\right) \in A$,则必存在 $t=(t_1, \cdots, t_n) \in A$, $f(t)=\Delta$ 且 $0<t_1<\frac{1}{2n}$. 证明基于 $\Delta>0$,因而不能有

$$x_i=\frac{2i-1}{2n},\ i=2, \cdots, n.$$

由此出发,利用与(ii)类似的推理,细节从略.

(iv) 记 $t_1^*=\min\{x_1: x=(x_1, \cdots, x_n) \in A, f(x)=\Delta\}$,若能证明 $t_1^* \leqslant \frac{1}{2n}$,则据已证的(ii)和(iii),必存在 $y=(y_1, \cdots, y_n) \in A$,使得 $f(y)=\Delta$ 且 $0<y_1<\frac{1}{2n}$. 用反证法证 $t_1^* \leqslant \frac{1}{2n}$. 设 $t_1^* > \frac{1}{2n}$ 且 $t=(t_1^*, \cdots, t_n) \in A$,使得 $f(t)=\Delta$,则必有 $t_2<1$,因为若 $t_2=1$,则

$$\Delta=f(t)=f(t_1^*, 1, \cdots, 1)=f(0, \cdots, 0, 1-t_1^*),$$

这与 t_1^* 的定义矛盾. 现设 $t_1^* \leqslant t_2=\cdots=t_i<t_{i+1}$,因 $0<t_2<1$,可找到 $\varepsilon_1>0$ 和 ε_2,当 $|\varepsilon_2|$ 充分小,使得

$$t'=(t_1^*-\varepsilon_1, t_2+\varepsilon_2, \cdots, t_i+\varepsilon_2, t_{i+1}, \cdots, t_n) \in A,\ f(t')=\Delta, \tag{1}$$

这与 t_1^* 的定义矛盾. 若 $t_1^* = \cdots = t_i < t_{i+1}$, 对某个 i, $2 \leqslant i \leqslant n$, 记 $y = (i-1)^{-1} \sum_{k=2}^{i} \frac{2k-1}{2n}$. 若 $t_2 \geqslant y$, 则因 $t_1^* > \frac{1}{2n}$, 可找到充分小的 $\varepsilon_1 > 0$ 和 $\varepsilon_2 > 0$, 使得(1)式成立,而这与 t_1^* 的定义矛盾,若 $t_2 < y$, 则因 $\frac{1}{2n} < t_1^* = t_2 = \cdots = t_i < y$, 有

$$f\left(\frac{1}{2n}, \frac{1}{2n}, \cdots, \frac{1}{2n}, t_{i+1}, \cdots, t_n\right) > f(t_1^*, t_2, \cdots, t_n) = \Delta,$$

$$f\left(\frac{1}{2n}, y, \cdots, y, t_{i+1}, \cdots, t_n\right) < f(t_1^*, t_2, \cdots, t_n) = \Delta,$$

由此可知存在 $w \in \left(\frac{1}{2n}, y\right)$, 使得

$$f\left(\frac{1}{2n}, w, \cdots, w, t_{i+1}, \cdots, t_n\right) = \Delta,$$

而因 $\left(\frac{1}{2n}, w, \cdots, w, t_{i+1}, \cdots, t_n\right) \in A$, 这与 t_1^* 的定义及 $t_1^* > \frac{1}{2n}$ 矛盾.

(v) 现在只剩下证明:若对某个 $x = (x_1, \cdots, x_n) \in A$ 有 $f(x) = \Delta \in (0, \Delta^*)$, $0 < x_1 = x_2 < \frac{1}{2n}$, 则必可找到 $t = (t_1, \cdots, t_n)$, 使得 $f(t) = \Delta$, $0 < t_1 < \frac{1}{2n}$ 且 $t_1 < t_2$. 这个证明很简单:设 $x_1 = \cdots = x_i < x_{i+1}$. 因为 $0 < x_1 < \frac{1}{2n}$, 可找到 $\varepsilon_1 > 0$, $\varepsilon_2 > 0$ 充分小,使得 $x_1 - \varepsilon_1 > 0$, $x_i + \varepsilon_2 < x_{i+1}$ 且 $f(x_1 - \varepsilon_1, x_2 + \varepsilon_2, \cdots, x_i + \varepsilon_2, x_{i+1}, \cdots, x_n) = \Delta$. 有 $0 < x_1 - \varepsilon_1 < \frac{1}{2n}$, $x_1 - \varepsilon_1 < x_2 + \varepsilon_2$, 而 ε_1, ε_2 的取法保证了 $(x_1 - \varepsilon_1, x_2 + \varepsilon_2, \cdots, x_i + \varepsilon_2, x_{i+1}, \cdots, x_n) \in A$.

引理证毕.

定理的证明 分别以 $X_{(1)} \leqslant \cdots \leqslant X_{(n)}$ 和 $U_{(1)} \leqslant \cdots \leqslant U_{(n)}$ 记 $X_1, \cdots, X_n$ 和从 $R(0, 1)$ 中抽出的 iid. 样本 $U_1, \cdots, U_n$ 的次序统计量. 因为

$$nW_n^2 = \frac{1}{12n} + \sum_{i=1}^{n} \left(F(X_{(i)}) - \frac{2i-1}{2n}\right)^2,$$

知在原假设 H 成立时,统计量 $\widetilde{W}_n = nW_n^2 - \frac{1}{12n}$ 的分布与

$$V_n = \sum_{i=1}^{n} \left(U_{(i)} - \frac{2i-1}{2n}\right)^2 \tag{2}$$

的分布相同,故对给定显著性水平 α 时 CM 检验的否定域 $\{W_n > c_n\}$, 其 $c_n \in (0, \Delta^*)$, Δ^* 已在引理中定义. 现设 G 是一个一维连续分布, $G \neq F$. 令

$$G^{-1}(u) = \inf\{x: G(x) \geqslant u\},$$

$$\widetilde{V}_n = \sum_{i=1}^{n} \left(F(G^{-1}(U_{(i)})) - \frac{2i-1}{2n}\right)^2, \tag{3}$$

则在对立假设 G 之下,CM 检验的功效为

$$\beta(G) = P(\widetilde{V}_n > c_n).$$

取一个定义于[0, 1]的函数 g,满足以下条件:g 在[0, 1]上连续严格增; $g(0) = 0$; 在 $\left(0, \frac{1}{2n}\right)$内 $g(u) > u$; $g(u) = u$ 当 $\frac{1}{2n} \leqslant u \leqslant 1$, 且据此函数 g 定义一维分布函数

$$G(x) = g^{-1}(F(x)),$$

g^{-1}为 g 的反函数,有

$$\begin{aligned} G^{-1}(u) &= \inf\{x: G(x) \geqslant u\} = \inf\{x: g^{-1}(F(x)) \geqslant u\} \\ &= \inf\{x: F(x) \geqslant g(u)\} = F^{-1}(g(u)), \end{aligned}$$

以此代入(3)式,得

$$\widetilde{V}_n = \sum_{i=1}^{n} \left(g(U_{(i)}) - \frac{2i-1}{2n}\right)^2, \tag{4}$$

按 g 的定义易见 $\widetilde{V}_n \leqslant V_n$. 因 $c_n \in (0, \Delta^*)$, 据引理,知存在 $(t_1, \cdots, t_n) \in A$, $0 < t_1 < \frac{1}{2n}$, $t_1 < t_2$, 使得 $f(t_1, \cdots, t_n) = c_n$, A 与 f 都已在引理中定义. 由 f 的定义及函数 g 满足的条件,易知存在 $\varepsilon \in (0, t_1)$, 使得 $g(t_1 - \varepsilon) > t_1$, 并存在 $\varepsilon_1 > 0$, 使得 $t_2 - \varepsilon_1 > t_1$, 且满足:

(i) 在 $u_1 \in I_1 = [t_1 - \varepsilon, t_1 - \varepsilon/2]$, $u_i \in I_i = [t_i - \varepsilon_1, t_i]$, $2 \leqslant i \leqslant n$, 且 $(u_1, \cdots, u_n) \in A$ 时,有 $f(u_1, \cdots, u_n) > c_n$.

(ii) 在 $u_i \in I_i$, $1 \leqslant i \leqslant n$, $(u_1, \cdots, u_n) \in A$ 时, $f(g(u_1), \cdots, g(u_n)) > c_n$. 由这两个性质,及表达式(2)和(4)知

$$\begin{aligned} \beta(G) = P(\widetilde{V}_n > c_n) &\leqslant P(V_n > c_n) - P(U_{(i)} \in I_i, 1 \leqslant i \leqslant n) \\ &= \alpha - P(U_{(i)} \in I_i, 1 \leqslant i \leqslant n) < \alpha, \end{aligned}$$

即在对立假设 G 处,CM 检验的功效比水平 α 小,因而 CM 检验非无偏. 定理证毕.

3 另一种方法

此方法是基于以下的简单观察:设 X, Y 是非负随机变量, $E(X) < E(Y)$, 则必存在常数 c,使得

$$P(Y > c) > P(X > c).$$

为将这个观察用于此处的问题,定义函数

$$g(u) = \begin{cases} u + \varepsilon u(1 - 2u), & 0 \leqslant u \leqslant 1/2, \\ u, & 1/2 \leqslant u \leqslant 1, \end{cases}$$

此处 $\varepsilon > 0$ 充分小,以使 g 在[0, 1]上严格增. 定义一维分布 $G(x) = g^{-1}(F(x))$. 令

$$h(u)=\begin{cases}u(1-2u), & 0\leqslant u\leqslant 1/2,\\ 0, & 1/2\leqslant u\leqslant 1,\end{cases}$$

并定义 V_n 和 $\widetilde{V}_n$ 如前，则

$$\begin{aligned}\mathrm{E}(V_n)-\mathrm{E}(\widetilde{V}_n)&=-\varepsilon^2\sum_{i=1}^{n}\mathrm{E}(h^2(U_{(i)}))-2\varepsilon\mathrm{E}\Big(\sum_{i=1}^{n}h(U_{(i)})\Big(U_{(i)}-\frac{2i-1}{2n}\Big)\Big)\\&=-\varepsilon^2\sum_{i=1}^{n}\mathrm{E}(h(U_{(i)}))-2\varepsilon\int_0^{1/2}x(1-2x)\sum_{i=1}^{n}\Big(x-\frac{2i-1}{2n}\Big)n\binom{n-1}{i-1}\\&\quad\times x^{i-1}(1-x)^{n-i}\mathrm{d}x\\&=-\varepsilon^2\sum_{i=1}^{n}\mathrm{E}(h(U_{(i)}))+\varepsilon\int_0^{1/2}x(1-2x)^2\mathrm{d}x,\end{aligned}\tag{5}$$

这最后一个等式由下述容易证明的等式推出：

$$\sum_{i=1}^{n}\Big(x-\frac{2i-1}{2n}\Big)n\binom{n-1}{i-1}x^{i-1}(1-x)^{n-i}=2x-1.$$

由(5)式知，对充分小的 $\varepsilon>0$，有

$$\mathrm{E}(V_n)>\mathrm{E}(\widetilde{V}_n),$$

因而存在 c，使得 $P(V_n>c)>P(\widetilde{V}_n>c)$. 取检验水平 $\alpha=P(V_n>c)$ 及对立假设 G 定义如上，即得出水平 α 的 CM 检验不是无偏的.

这个结果弱于前面证明的定理，因只是肯定了对某特别选定的检验水平 α，CM 检验非无偏，而未能证明此结论对一切水平 $\alpha\in(0,1)$ 都成立(前面定理中证明了这一点). 这一方法还可以用于类似的情况. 以 χ^2 拟合优度检验为例，设原假设 $H:\{p_1^*,\cdots,p_k^*\}$. 在原假设 H 下，χ^2 统计量的数学期望为 $k-1$，而在对立假设 $\{p_1,\cdots,p_k\}$ 之下则为

$$\mathrm{E}(p_1,\cdots,p_k)=\sum_{i=1}^{k}\frac{p_i(1-p_i)}{p_i^*}+n\Big(\sum_{i=1}^{k}\frac{p_i^2}{p_i^*}-1\Big).$$

设 $p_1^*,\cdots,p_k^*$ 不全相同，为确定计，设 $p_1^*>p_2^*>0$. 取对立假设 $K:\{p_1^*+\varepsilon,p_2^*-\varepsilon,p_3^*,\cdots,p_k^*\}$，$\varepsilon>0$ 充分小. 简单计算得出：对 $p_1=p_1^*+\varepsilon$, $p_2=p_2^*-\varepsilon$，有

$$\begin{aligned}&\sum_{i=1}^{2}\frac{p_i(1-p_i)}{p_i^*}+n\sum_{i=1}^{2}\frac{p_i^2}{p_i^*}\\&\quad=\sum_{i=1}^{2}\frac{p_i^*(1-p_i^*)}{p_i^*}+n\sum_{i=1}^{2}\frac{p_i^{*2}}{p_i^*}+(1+2n)\varepsilon\Big(\frac{1}{p_1^*}-\frac{1}{p_2^*}\Big)+O(\varepsilon^2).\end{aligned}$$

因 $p_1^*>p_2^*$，可选取 $\varepsilon>0$ 充分小，使得上式右边后两项(与 ε 有关的项)之和为负，这给出

$$\mathrm{E}(p_1^*+\varepsilon,p_2^*-\varepsilon,p_3^*,\cdots,p_k^*)<\mathrm{E}(p_1^*,\cdots,p_k^*)=k-1,$$

从而证明了：当原假设 $\{p_1^*,\cdots,p_k^*\}$ 中 $p_1^*,\cdots,p_k^*$ 不全相同时，χ^2 拟合优度检验不是无偏的. 另一方面，如 Cohen 等[6]所证，在原假设是 $\{1/k,\cdots,1/k\}$ 时，χ^2 拟合优度检验的确是无

偏的.

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